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ANALYSIS AND SYNTHESIS
OF SAMPLED-DATA CONTROL
SYSTEMS

PRENTICE-HALL INTERNATIONAL SERIES
IN ELECTRICAL ENGINEERING

William L. Everitt, editor

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R

G_1

System

block diagram

H

H

H

G

G_2

G_3

G

G_1

G_2

H

-H

System

signal flow graph

Modified composite

signal flow graph

-H

-Ox2

Output transforms

$$\begin{aligned}
C(s) &= \frac{R^*(s)G(s)}{1+G^*(s)H^*(s)} \\
C^*(s) &= \\
C(z, m) &= \\
&= \frac{G^*(s)R^*(s)}{1+G^*(s)H^*(s)} \\
G(z, m)R(z) &= \frac{G(z)R(z)}{1+G(z)H(z)} \\
&= \frac{-GGG^3H)^*}{PIG_1RY} \\
C(s) &= \frac{G_2^*(s)G_3(s)G_1R^*(s)}{1+G_2^*(s)(G_1G_3H)^*} \\
GG^3 &= \\
-G(G_3H)^* &= \\
C^*(s) &= \frac{G_2Gym}{G_2(s)G_3(s)G_1R^{**}(s)} \\
1+G_2^*(s)G_1G_3H^{**}(s) &= \\
-H &= \\
G_2G_3 &= \\
H- &= \\
C(z, m) &= \\
==== &= \\
G_2(z)G_3(z, m)G, R(z) &= \frac{G_2(z)G_3(z, m)G, R(z)}{1+G_2(z)G_1G_3H(z)} \\
ox &= \\
-H &= \\
Gm &= \\
D0 &= \\
0 &= \\
Q(G_1R) &= \\
09 &= \\
-H &= \\
G &= \\
(HG &= \\
G^*(s)G(s)H(s)R^*(s) &= \\
1+GH^*(s) &= \\
C(s) &= \\
= &= \\
GR^*(s) &= \\
C^*(s) &= \\
G^*(s)R^*(s) &= \\
1+HG^*(s) &= \\
-(HG) &= \\
G(z)GH(z, m)R(z) &= \\
C(z, m) = R(z)G(z, m) &= \\
- &= \\
\infty &= \\
1+HG(z) &= \\
G_2(s)G_1R^*(s) &= \\
C(s) &= \\
1+G_2^*(s)G_2H^*(s) &= \\
-(G_2H) &= \\
X^2 &= \\
G &= \\
C^*(s) &= \\
G_2^*(s)GIR(s) &= \\
1+G_2^*(s)G_2H^*(s) &=
\end{aligned}$$

$G(z)$
 $G(z, m)$
 $G(z, m)G R(z)$
 $1+ G_a(z) G_s H(z)$

193.-
 ANALYSIS AND SYNTHESIS
 OF SAMPLED-DATA CONTROL
 SYSTEMS
 BENJAMIN C. KUO
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 University of Illinois
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C

to Margaret

PREFACE

This book has been written as a text for a course in digital and
 sampled-data control systems. However, since it was prepared
 with the needs of modern scientists and engineers in mind, it should
 also prove useful as a reference book for scientists and engineers
 working in control systems and related fields. The author has put
 much effort into preparing a book that may be studied with a
 minimum amount of guidance.

The bulk of the material presented here has been taught by the
 author since 1958 in a graduate course at the University of Illinois.
 However, with the recent advances in the theory and practice of
 sampled-data systems, more than one semester is now needed to
 cover the entire up-to-date subject. If the book is used as a text in
 a one-semester course, some of the chapters will have to be skipped
 for lack of time.

The author assumes that the reader has a knowledge of the basic
 theory of feedback control systems, although he need not have had
 prior contact with the theories of sampled- or digital-data systems.
 The reader should also have adequate training in ordinary dif-
 ferential equations, the Laplace transformation and its applica-
 tions, and the complex variables theory. In order to comprehend
 the material in Chapter 11, he would need some background on the
 theories of probability and statistics.

Both the z-transform and the state transition techniques are
 emphasized in this book. The z-transform method is straightforward

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in its application to sampled-data systems. Through the use of the
 z-transform method, many of the well-known analysis and design
 techniques of continuous-data systems can be modified and applied

to sampled-data systems. The state transition approach permits a unified study of a great variety of sampled-data systems. It utilizes the signal flow graphs and the state transition flow graphs to give a unification between computer and analytical solutions.

The book deals with the following topics: linear analysis, linear design and synthesis, statistical approach, and nonlinear systems. Chapter 1 serves as an introduction; the historical background of sampled-data systems and the importance and advantages of sampling in control systems are the highlights of this chapter. Chapter 2 gives the physical interpretation and a comprehensive mathematical description of the sampling process. Chapter 3 describes how a signal may be recovered from its sampled version and includes problems related to the process. Chapter 4 introduces the theory and basic applications of the z-transformation and presents the theorems and the limitations of the z-transforms. The analysis of sampled-data systems begins in Chapter 5. This chapter covers such topics as the block diagrams, signal flow graphs, matrix representation, and the state transition method. Chapters 6 and 7 discuss time and frequency responses of sampled-data systems. Stability studies are carried out using analytical and graphical methods. Chapters 8 and 9 are devoted to the principles and techniques of systems design. Chapter 10 deals with multirate, skip, nonuniform, and nonsynchronized samplings. Chapter 11 presents statistical analysis and design. The final chapter is on nonlinear systems; these are studied through use of the state transition, phase-plane, incremental phase-plane, and describing function methods.

Although most of the material in this text has been derived from the literature of sampled-data systems, some original ideas have been included. In a book of this nature reference cannot be made to all of the original work and contributors in the vast and active field of sampled-data systems. The author wishes to apologize to those contributors in the field whose works and names are not mentioned through either oversight or ignorance.

The author wishes to acknowledge with gratitude the encouragement and help he received from his colleagues at the Electrical Engineering Department of the University of Illinois. The author is also indebted to his graduate students, whose questions and classroom discussions have assisted in many ways in the preparation of the manuscript. The encouragement and interest of Dean W. L. Everitt, editor of this series, and Dr. E. C. Jordan, Head of the

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Electrical Engineering Department of the University of Illinois, have been invaluable in the preparation of the manuscript. Finally, the author's special thanks go to his wife Margaret, whose patience and understanding made the tedious writing of this book an enjoyable undertaking.

B.C.K.

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INTRODUCTION

1.1 WHAT ARE SAMPLED-DATA AND DIGITAL CONTROL SYSTEMS?

Within the past few years much progress has been made in the development of sampled-data and digital control systems. These systems differ from the conventional continuous-data systems in that the signal in one or more sections of the system is in the form of either a pulse train or a numerical code such as the binary code. Usually, sampled-data systems refer to a more general class of systems whose signals are in the form of pulsed data, whereas the term digital control systems implies the use of a digital computer or digital

Continuous

signal

$e(t)$

Sampler

Pulsed

data Smoothing

$ep(1)$

filter

Linear

system

$c(t)$

(a)

$ex(1)$

(b)

T $2T$ $3T$

(c)

Fig. 1-1 (a) Open-loop sampled-data system. (b) Continuous input to sampler. (c) Pulse signal output of sampler.

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1

2 INTRODUCTION

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sensing element in a system. In general, a sampled-data system is defined as one that receives information or data only at discrete instants of time. The signals in a sampled-data system may be of the form of a periodic or an aperiodic pulse train, with no information transmitted between two consecutive pulses. Figure 1-1a illustrates in a simple manner how an open-loop sampled-data system operates. A continuous input signal $e(t)$ is sampled by a sampling device, and the output of the sampler is a sequence of pulses. If the continuous input $e(t)$ is represented as shown in Fig. 1-1b, the output of the sampler, $e(t)$ is shown in Fig. 1-1c. In this case, the sampler is assumed to have a uniform sampling rate. On the basis of the signal configurations of Fig. 1-1, the sampling may be regarded as a pulse-amplitude modulator. The sampling schemes exist in many

variations; some of these are the periodic, cyclic-rate, multirate, skip, random, and pulse-width modulated samplings. The most common variations are single-rate and multirate samplings.

A digital control system is defined as a control system in which the signal at one or more points of the system is expressed in a numerical code for digital computers or digital transducers in the system. The appearance of digital signals (such as binary numbers) in part of the system requires the use of digital-to-analog as well as analog-to-digital converters. The block diagram of a typical digital control system using a digital computer is shown in Fig. 1-2. A sample-data system can also include a digital computer or a digital

Digital
code Digital

input
computer
Digital-to-
analog
converter
Continuous
output
Plant

Analog-to-
digital
converter

Fig. 1-2 A typical digital control system.

Input
Sampler
Encoder
Digital
computer
Output
Decoder
Plant

Fig. 1-3 A sampled-data system with digital computer.

data-processing unit if the pulsed signal in the system is first put into digital code for the computer and the output of the computer is decoded into pulsed-data. An example of such a system is shown in Fig. 1-3.

Although the use of sampled-data or pulsed-data in control systems

Sec. 1.1

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was recognized quite early in the history of feedback control systems, only recently has significant progress and emphasis been made in the development of the analysis and design techniques of sampled-data systems. The complexities that have arisen in the area of automatic tracking systems and the increased use of digital computers in control systems have caused considerable activity in the field of sampled-data and digital control systems.

The exact origin of the engineering adoption of the sampling concept is difficult to trace. However, a basic form of the sampled-data system was described by Gouyt some sixty years ago. Gouy

+V

Normally closed
relay contacts

Motor
driven (wheel

Relay
coil
Contact rod
Mercury

thermometer

Heating

ww

coil

Oven

Fig. 1-4 A constant temperature oven utilizing sampled-data.

devised a temperature regulation system consisting of the elements arranged as shown in Fig. 1-4. The aim of the system is to maintain a constant temperature in the oven at all times. Whenever the electrical contact rod is immersed in the mercury, current flows in the relay coil causing the relay to open and thus interrupt the heating current. Since the contact is periodically dipped into the mercury, the heating current consists of a sequence of pulses. Furthermore, since the time that the contact is immersed in the mercury

Temperature

of oven

$h(t)$

Heating

current of

Oven

$h(1)$

Pulse width is

proportional to

amplitude of

$h(t)$ at $t = nT$

$KT(k+1)T(k+2)T(k+3)T(k+4)T$

Fig. 1-5 Input and output of pulse-width sampler.

† G. Gouy, "On a Constant Temperature Oven," J. Physique, vol. 6, ser. 3, (1897)

,
pp. 479-483.

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"

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depends upon the level of the mercury and the level in turn depends upon the temperature of the oven, the pulse widths of the heating current are varied in proportion to the temperature of the oven, but the pulses all have equal amplitudes. A typical set of signals of the pulsed-data temperature control system is shown in Fig. 1-5. In contrast to the sampling scheme described in Fig. 1-1, which is called the pulse-amplitude sampling, the pulse-data of Fig. 1-5 are termed the pulse-duration or pulse width sampled-data. A pulse-amplitude sampler is essentially a linear device in the sense that inputs and outputs of the sampler are governed by the principle of superposition. However, a pulse-duration sampler is considered to be a nonlinear device since the superposition principle no longer holds between the input and the output signals. Although in this example, the merit of using pulsed-data control in regulating temperature is

T_1

Control

pulse train

T_2

S

B+

RL

A

R

B

I2

www

111

ww

카
 H
 Light
 source
 (a)
 R
 Sampler
 Hold
 +
 Amplifier
 and plant
 B
 H
 (b)

Fig. 1-6 (a) Light control system with sampled-data. (b) Block diagram of the light control system.

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not apparent, at least the system represents a classical application of the scheme of sampling in a simple control system.

Another classical example of a physical system employing sampled-data was described by Guanella, who suggested the use of sampling in electronic feedback amplifiers, electro-acoustical converters, and radio and other transmitters or receivers for the purpose of eliminating instability conditions or improving stability conditions. Fig. 1-6a shows a light control system with sampled-data which was devised by Guanella. The block diagram of the system is shown in Fig. 1-6b.

The brightness of the light source is assumed to depend upon the current or voltage supplied by the amplifier A. In addition to the reflected light, a small part of the light emerges from the source and is directed upon a photocell. A feedback signal B, proportional to the output of the photocell is subtracted from the reference signal R through the transformer T_1 . The block S is made up of the elements of the sampling device that is to sample the signal RB at periodic intervals. The circuit in S is essentially a double-triode series gate.† The triodes conduct and cut-off simultaneously and periodically in accordance with the control pulse train applied to the grids of the triodes. Therefore, the impedance between the points 1 and 2 in Fig. 1-6a varies between a small and an almost infinitely large value according to the control pulse train. The voltage on the controlling grid of the pentode is zero whenever the triodes are cut-off and is equal to R-B during the pulse duration p of the control pulse. Therefore, the block S functions as a pulse-amplitude modulator or sampler, and its input-output relations may be described by the waveforms of Fig. 1-1.

The pentode and the condenser C act as an integrating or hold device. When a voltage pulse of amplitude RB and pulser width. p is applied to the grid at point 2, a current pulse proportional to RB with equal duration is passed through the condenser C and causes a corresponding change in the voltage of C by an amount proportional to RB. Thus the voltage at the input to the amplifier A is changed by the amount $R(nT) - B(nT)$ at the n-th sampling instant $t = nT$. If the pulse signal at point 2 is represented by $e_n = K[R(nT) - B(nT)]$ at $B(nT)]$ at $t = nT$

$\sum$
 $k = 0$

then the input to A is written as e_n . Note that the hold device used here is not commonly used in the modern sampled-data control

† G. Guanella, "Electrical Oscillation Translating Systems," U.S. Patent, 2,253,976,

August 26, 1941, (Private correspondence from Guanella).

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systems. Normally, the function of a hold device is to reconstruct from the pulsed-data the original continuous signal. The pentode integrator described in the system does not fulfill this purpose. However, the present sample-and-hold scheme can be shown to improve the stability condition of the feedback system, whereas the conventional hold device, such as a zero-order hold (one that holds the amplitude of a pulse until the next occurs), used in common sampled-data systems is known to have adverse effects on system stability.

1.2 IMPORTANCE OF SAMPLING IN CONTROL SYSTEMS

The examples given in the preceding section on sampled-data systems merely illustrate the early application of sampling in feedback systems. However, in order to fully understand the merits and the reasons for using sampling in control systems, we should ask the natural question: Why sampled-data? In other words, what are the advantages and characteristics of sampling that have made sampled-data and digital control systems so important in modern control technology? To answer this question, a survey of the available class of sampled-data and digital control systems will be helpful. In general, the applications of sampled-data systems can be divided into two main categories: systems with inherent sampling and systems with intentional sampling. These systems are discussed as follows:

1. Control Systems With Inherent Sampling

In this class of systems, the signals are available only in sampled form. Examples of these systems are:

1. Radar Tracking System. The signals sent out and received by a radar tracking system are in the form of pulse trains. The scanning operation of the radar performs the function of a sampler converting both the azimuth and the elevation information to sampled-data.

2. Time Sharing Systems. Because of economic factors in the cost of information transmission, a data transmission link is often shared among several control systems. For instance, in the multichannel drift stabilization system shown in Fig. 1-7, one stabilizing amplifier is used to stabilize a group of d-c amplifiers. With the aid of a sampler, which samples the error voltage of each amplifier in turn,

† P.G. Pantazelos, "Sampled-Data Analysis of a Drift-Stabilization System," A.I. E.E.

Conference Paper No. CP 56-961, October 1956.

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Input

From other
amplifiers

Z_{in}

Z_{fb}

d-c amplifier

d-c feedback amplifier

Output

Smoothing

filter

Commutator

Stabilization

amplifier

To other

amplifiers

R

WW

Fig. 1-7 Multichannel drift stabilization system.

the stabilization amplifier is time-shared among several d-c amplifiers. Thus the cost, size, and maintenance of equipment is reduced.

3. Digital Computer in Control Systems. Digital computers have become very important as elements in control systems. There are several important reasons for this use of digital computers. First of all, digital computers are light in weight when compared to analog computers of the same capacity; and conversely, for a given weight, the capacity of digital computers is far greater than that of analog Digital

sensor

Digital

Tape or

card reader

Buffer

storage

voltage

Clamp

converter

Digital position feedback

Power

amplifier

Compensation

loop

Tach

Motor

Digital

sensor

Drill table

Fig. 1-8 Digitally-controlled drill-table positioning system.

Gear train

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Chap. 1

computers. In some cases digital computers used as compensators for control systems are preferable to the continuous type in the improvement of static and dynamic system characteristics. As an example, a digital control system used for the control of drill table position so that a number of specified holes can be drilled in a work piece is shown in Fig. 1-8. Digital computers are also used often in modern hot plate mills, with preset programming systems using devices such as plugboards or punch cards to store the sequential roll-screw-down settings required to roll an ingot to a finished plate or slab. Such digital systems simplify the mill operator's job considerably.

One also frequently finds many other applications of digital computers in control systems: fields such as guidance control for missile navigation, air-surveillance and weapon control systems, and many others.

2. Control Systems with Intentional Sampling

There is a class of system in which the sampling operation is purposely introduced. This is one of the recent trends in the design of feedback control systems. Some of the reasons for this intentional sampling are given below:

1. Sampling Improves Sensitivity. In servo applications a load must sometimes be driven by a relatively low-power-level signal. By the use of sampling in control systems, tremendous amounts of power can be controlled by sensitive control elements without excessive

power amplification. Also, the loading effect upon sensitive instruments may be minimized by the sampling operation. A classical example of this improvement in sensitivity through sampling is the chopper galvanometer described by Oldenbourg and Sartorius, the system is shown in Fig. 1-9. In this device a small signal is applied to the galvanometer coil. The chopper bar is lowered periodically and the projected needle of the galvanometer causes the load to rotate in proportion to the signal strength. The torque which is applied to the load shaft is thus determined by the chopper-bar drive rather than just by torque developed in the galvanometer coil.

† D. J. Gimpel, "Sampled-Data Systems," Control Engineering, February 1957, pp. 99-106.

C. Burdick, "Digital Computer Runs Hot Plate Mill," Control Engineering, January 1960, pp. 126-130.

\$R. C. Oldenbourg and R. Sartorius, The Dynamics of Automatic Controls, Chap. 5, publication of the American Society of Mechanical Engineers, New York.

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INTRODUCTION

Chopper-bar
-Galvanometer

To
load
Motor driven
cam

Fig. 1-9 Chopper-bar galvanometer.

2. Digital Transducers. A transducer in its broadest sense is any device which can couple unlike systems, transferring energy from one to the other and converting units. It is generally more useful for the control engineers to consider that a transducer is a device that transmits information, regardless of energy level and dimensions. A digital transducer is any coupling device that transmits data along in some form of discrete code to other unlike systems or subsystems. The most important reason for using a digital transducer is the relative immunity of digital signals to distortion by noise and nonlinearities. Furthermore, digital transducers and sensing devices, particularly rotary and linear-position sensors, are available with accuracy and resolution difficult to achieve by analog techniques.

3. Digital Coded Signals. Sampled-data in the form of digital coded signals can be stored for any length of time, transmitted over any distance, retransmitted, and detected or used as many times as necessary with no loss in accuracy. These properties permit elaborate command signal programs to be formulated and executed with extremely high accuracy.

4. Behavior. In control systems design, under certain conditions a sampled-data system with variable sampling intervals or with variable pulse widths behaves better than a continuous-data system. This is particularly true in the compensation of systems having plants with transportation lags.

+ E. J. Kompass, "What About Digital Transducers?" Control Engineering, July 1958, pp. 94-99.

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1.3 A SURVEY OF THE METHODS OF ANALYSIS AND SYNTHESIS OF SAMPLED-DATA SYSTEMS

Very little work was done on the analysis and design of sampled-data systems prior to 1950. Although early textbooks on servo-mechanisms written by MacColl (1945), James, Nichols and Phillips (1947), and Oldenbourg and Sartorius\$ (1948) all cover some of the topics of sampled-data systems, the treatments have mostly been

limited to stability analysis and the derivation of transfer functions of pulsed systems.

Since the early 1950's when digital computers were first used in control systems and when the need arose for complex automatic tracking systems, sampled-data and digital control systems have enjoyed a most fruitful period of development and research. Not only have numerous papers been published in the United States and abroad, but several books devoted solely to the subject of sampled-data and digital control systems have also become available. In this section some of the available methods in treating sampled-data and digital control systems will be indicated, and the major topics of current interest that need further investigation will be pointed out along with new problems to be tackled. In general, the studies of sampled-data and digital control systems can be put into the following categories: Linear theory, nonlinear theory, statistical theory, time varying theory, and digital control theory.

1. Linear Theory

In linear sampled-data theory, the system is assumed to consist of linear components only. A pulse-amplitude-modulated sampler is considered a linear device, since its input-output relations satisfy the superposition principle. A majority of the literature on linear theory

† L. A. MacColl, *Fundamental Theory of Servomechanisms*, Chap. 10, (Princeton, New Jersey: D. Van Nostrand Company, Inc. 1945).

H. M. James, N. B. Nichols, and R. S. Phillips, *Theory of Servomechanisms*, Chap. 5, (New York: McGraw-Hill Book Company, 1947).

§ R. C. Oldenbourg and H. Sartorius, *The Dynamics of Automatic Controls*.

|| J. R. Ragazzini and G. F. Franklin, *Sampled-Data Control Systems*, (New York: McGraw-Hill Book Company, 1958). E. I. Jury, *Sampled-Data Control Systems*, (New York: John Wiley & Sons, Inc., 1958). Y. Tsytkin, *Theory of Pulse Systems*, (Moscow: State Press for Physics and Mathematical Literature, 1959). J. T. Tou, *Digital and Sampled-Data Control Systems*, (New York: McGraw-Hill Book Company, 1959).

E. I. Jury, "Recent Advances in the Field of Sampled-Data and Digital Control Systems," *Proc. of the 1960 Conference of the International Federation of Automatic Control*, Moscow.

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assumes that the output of a sampler is a train of impulses with the strength (area) of the impulse equal to the height of the pulse at the sampling instant. This major assumption is necessary in order to simplify the mathematics involved in the analysis and design of sampled-data systems. However, the assumption is justified only if the actual pulse width of the sampler is small when compared with the dominant time constant of the system and if the sampler is followed by a hold device. In the event that the pulse width is wide, the impulse approximation can at best give a rough estimate of the actual behavior of the system, and often may even produce misleading results. However, if the sampler is not followed by a hold device or smoothing filter, the impulse-modulated approximation is not a correct description and represents an entirely different system. In fact, if the hold device is absent and if the pulse width is small so that the amplitude-modulated pulses are replaced by flat-topped pulses, the impulse approximation of these flat-topped pulses actually represents a system with loop-gain equal to $1/p$ of that of the actual system, where p is the pulse width.

Once the pulse-amplitude modulated sampler is replaced by an impulse-amplitude modulated sampler (or ideal sampler), the analysis of the system can be carried out by use of any of the following methods:

- (1) Difference equation method
- (2) Impulse response method
- (3) Frequency response method
- (4) Signal flow graphs
- (5) z-transform and modified z-transform methods.
- (6) Root locus and root contour methods
- (7) State variable method

These methods are discussed in various degrees of detail in the ensuing chapters. Although each of the above listed methods has its advantages and disadvantages, the published works indicate that the z-transform method is the most popular. The use of the z-transformation for sampled-data systems is entirely analogous to the application of the Laplace transformation to continuous-data systems. Sampled-data systems are ordinarily described by equations in the s-domain with complex coefficients such as e . The z-transformation, given by the relation $z = e^{st}$, makes possible the description of sampled-data systems by equations in z with constant coefficients, and the transfer functions become rational functions in the variable z . Therefore, most of the widely used techniques for solving linear continuous-data systems, such as the Nyquist criterion, Routh-Hurwitz criterion, root locus diagram, or Bode diagram can be modified and

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$r(t)$

$e(t)$

$e(t)$

Hold

Continuous

network

Plant

$c(t)$

$G(s)$

(a)

$r(t)$

$e(t)$

$D(s)$

Hold

Plant

$c(t)$

$G(s)$

T

Digital compensator

$D(z)$

(b)

Fig. 1-10 (a) Sampled-data control system with continuous-data compensator. (b) Sampled-data control system with digital compensator.

extended to the studies of linear sampled-data systems.

The design and synthesis of linear sampled-data systems can be divided into two general categories: continuous-data compensators and digital compensators. Typical system configurations of continuous-data compensation and digital compensation are shown in Fig. 1-10.

In general, other configurations such as feedback compensation and inner-loop compensation are possible. In control system theory usually, design refers to a procedure which uses the compensator, whether it be digital or continuous in nature, as the starting point. In other words, the designer simply has the task of finding a suitable and physically realizable compensator so that the compensated system will satisfy the specified performance criterion. In doing so, the designer usually has to rely to a great extent on his past experience in choosing the proper compensating network, and usually some

degree of trail and error is involved. Therefore, the philosophy of design of linear sampled-data control systems follows directly from the design of linear continuous-data systems, and the conventional methods such as the Bode diagram, Nyquist plot, or root locus diagram can be used.

The synthesis of linear continuous-data systems was first introduced by Aaron and Truxal. The philosophy of synthesis is such

† M. R. Aaron, "Synthesis of Feedback Control Systems by Means of Poles and Zero Location of the Closed-loop Function," Trans. A.I.E.E., 70, Part II, 1951, pp. 1439-1445.

J. G. Truxal, "Servomechanism Synthesis Through Pole-Zero Configuration," MIT Research Laboratory of Electronics Technical Report, 162, August 25, 1950; and J. G.

Truxal, Automatic Feedback Control System Synthesis, Chap. 5, New York: McGraw-Hill Book Company, 1955.

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that the over-all system transfer function C/R is first determined to satisfy the design criterion and simultaneously satisfy the physical realizability of the compensator; then the transfer function of the compensator is determined from C/R . In contrast to the design procedure, which works from the inside toward the outside, the synthesis procedure works from the outside toward the inside of the system. Synthesis procedure similar to that of Truxal's may be derived for linear sample-data systems.

In general, the following performance criteria are considered for the design and synthesis of sampled-data control systems with deterministic input:

(1) Frequency domain criteria: bandwidth, peak resonance M , gain and phase margins.

(2) Time domain criteria: For sampled-data systems, deadbeat responses can be achieved that have no overshoots and have minimum response times for aperiodic inputs such as step, velocity or parabolic. Other conventional criteria are overshoot, peak time, settling time, steady-state response, and rise time.

(3) Performance indices: Specifications of minimization of error criteria such as the integral error, mean-squared error, and integral square error.

2. Statistical Theory

In many applications of sampled-data systems (as well as continuous-data systems), the actual system inputs or disturbances are not deterministic but are random in nature. These random signals or disturbances can often be described only by their statistical properties in the form of correlation functions or power spectral density functions. Therefore, statistical analysis and design of sampled-data control systems are necessarily important.

[n(1)

$n(t)$ noise

$r(t)$

Input

Closed-loop or

open-loop sampled

data system

$c(t)$

Actual output

$M(s)$

+

or

$e(t)$

Error

Ideal

system model

$C_a(t)$

Desired output

$M_o(s)$

Fig. 1-11 Block diagram illustrating the design of optimum linear sampled-data system.

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The statistical design and synthesis of sampled-data control systems are best illustrated by the block diagram of Fig. 1-11. The main essence of the design procedure is to obtain an optimum filter (or system) that will render a minimum error between the actual and the desired output signals. The design criteria are based mainly on the minimization of the mean-square error originally suggested by Wiener. A system designed from the minimization of the mean-square error has the characteristic of emphasizing the large errors and passing over many small errors. A large class of control systems definitely exists for which minimization of the mean-square error certainly is not the ideal design criterion. However, if the minimization of the mean-square error is used as the design criterion, the random signals or disturbances are readily described by the correlation functions or the power spectral density functions. Most of the statistical studies of sampled-data systems are based on the assumption that the random processes are stationary and the systems subjected to the random signals are linear and time-invariant. However, intensified efforts on the statistical studies of nonstationary, time-varying, as well as nonlinear sampled-data systems should also be emphasized; these subjects definitely are the present and the future major topics of many investigators in the field.

3. Nonlinear Systems

The studies of nonlinear sampled-data systems can be divided into two general categories: (1) Systems with inherent nonlinearities, and (2) Systems with intentionally introduced nonlinearities. The first category includes the systems with nonlinear behavior purely due to physical limitations and imperfections of system components, while in the second category, nonlinear elements such as relays with dead zones and hysteresis are purposely inserted to improve the performance of the system. A typical relay-type sampled-data control system is shown in Fig. 1-12. Pulse-width control and amplitude quantization in sampled-data systems are also regarded as modified versions of the ordinary relay control. A pulse-width

$r(t)$

$c(t)$

-

Hold

Relay

$G(s)$

Plant

Fig. 1-12 Sampled-data control system with on-off relay.

† N. Wiener, The Extrapolation, Interpolation, and Smoothing of Stationary Time Series, New York: John Wiley & Sons, Inc., 1948.

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$e(t)$

$e(t)$

Quantizer

$-q(1)$

(a)

$-e_4$

-e3 -e2

6

g(t)

-93

+92

0

7

e₁

-92

-93

e₂

33

e₄ e(t)

(b)

Fig. 1-13 (a) Amplitude quantizer. (b) Input-output characteristic of an amplitude quantizer.

modulated signal can be thought of as the output of a relay by controlling the duration of the full positive or negative relay correction. The amplitude quantizing process is a nonlinear approximation in the sense that only a discrete set of amplitude levels are produced for any input signal. The block diagram of the amplitude quantizer and its input-output characteristics are shown in Fig. 1-13. It is apparent that the quantizer can be regarded as a relay with multiple-level corrections. In fact, if the amplitude of the input signal is small, the quantization effect becomes much more emphasized because of few levels, and the quantizer becomes an on-off relay.

The nonlinear theory of sampled-data systems has not been developed fully. The main reason for this lies in the fact that nonlinear difference and differential equations which described nonlinear sampled-data systems are usually very difficult to solve, and little has been done in the solution of these equations. The studies of nonlinear sampled-data systems have been influenced mainly by the use of the nonlinear theory of continuous-data systems. The following significant methods of analysis and design have been developed and investigated:

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(1) Phase plane method

(2) Incremental phase plane method

(3) Describing function method

(4) Discrete describing function method

(5) Difference equation method

(6) Matrix method

(7) Second method of Lyapunov

4. Time-Varying and Adaptive Systems

In most control systems the dynamic characteristics of the controlled process generally vary under actual operating conditions. This variation may be caused by normal deterioration of the system components, or drifting of parameters, or other reasons. Usually, slight variations of system parameters can be counteracted by using feedback. However, should the variations become large, the system must be considered to have time-varying characteristics. This initiated the studies of time-varying systems and adaptive systems. In sampled-data control systems, a sampler with time-varying rate may be introduced purposely for the sake of improving system performance. Adaptive sampled-data systems are increasing in practical applications. These systems are designed so that the time characteristics of the compensator or the sampling scheme may vary according to the time-varying characteristics of the controlled dynamic process. The methods of analysis of time-varying and adaptive sampled-data systems are generally of the following types: (1) matrix and

difference equation (with time-varying coefficients) and (2) statistical methods.

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2

THE SAMPLING PROCESS

2.1 INTRODUCTION

RODUC

The information received by a sampled-data control system is not continuous with respect to time but is in the form of pulses occurring at discrete instants of time. No signal is supplied at all during the periods separating the information carrying pulses. Mathematically, the pulse-data signal can be represented by a sampling operation that may be adequately described by a fictitious component called the sampling switch or sampler, as shown in Fig. 2-1a, or a pulse modu

Pulse train $p(t)$
 Carrier
 $e(t)$
 $e(t)$
 (p)
 $e(t)$
 Pulse
 modulator
 $ex(t)$
 (a)
 (b)

Fig. 2-1 Sampling process described by fictitious components.

(a) Sampler as a sampling device. (b) Pulse-modulator as a sampling device.

lator as shown in Fig. 2-1b. If the operation of the sampling switch is periodic with respect to time, that is, the sampler closes for a very short interval p at discrete periodic instants $t = 0, T, 2T, 3T, \dots$,

† W. K. Linvill, "Sampled-Data Control Systems Studied Through Comparison of Sampling with Impulse Modulation," Trans. A. I. E. E., 70, 1951, pp. 1779-1799.
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$le(t)$
 $ep(1)$
 $T+p$
 (a)
 (b)

Fig. 2-2 Typical waveforms of the input and output signals of a uniform-rate sampling device. (a) Input signal to sampler or pulse modulator. (b) Output of sampler or pulse modulator.

$nT, \dots$, the sampling is said to have a uniform rate. The waveforms of the input and the output signals of a typical uniform-rate sampling device are shown in Fig. 2-2; the input signal $e(t)$ is a continuous function of time, but the output of the sampling switch, denoted by $e(t)$, is a train of amplitude-modulated pulses of widths p occurring at the sampling instants $t = 0, T, 2T, 3T, \dots, nT, \dots$. In other words, at $t = 0$, the sampler closes for p seconds; and during this short interval $(t) = e(t)$. After $t = p$ second, the sampler opens, and the output signal is zero until the next sampling instant at $t = T$. In general, the pulses in a sampled-data system may be supplied at a variable rate, then the duration between consecutive pulses is not constant. A simple example of the variable-rate sampled-data control system is a cyclic variable-rate system in which the pulsed-data occur at a cyclic rate as shown in Fig. 2-3. Not uncommonly,

$3T$
 T
 $2T$
 $4T$
 $5T$

Fig. 2-3 Output of a cyclic-rate sampling device.

several independent sampling operations appear at different points of a sampled-data system. The sampling rates of the samplers need not be synchronous (operate at same instants of time). This type of system is generally described as a multirate sampling system. In more

† J. Tou, "Optimum Control Through Tuned Sampling," Proc. National Electronics Conference, 1959.

G. Kranc, "Compensation of an Error-Sampled System by a Multirate Controller," Trans. A. I. E. E., 76, Part II, 1957, pp. 149-159.

complicated cases the sampling operation may be entirely at random,† i. e., the interval between successive samples may be thought of as to follow some random scheme.

2.2 MATHEMATICAL ANALYSIS OF THE SAMPLING PROCESS

For the purpose of introducing the mathematical description of the sampling operation, we shall discuss only uniform-rate sampling at present. As described in the preceding section, the sampling process may be represented by a fictitious pulse-amplitude modulation process that converts a continuous signal into a train of amplitude-modulated pulses. In the uniform-rate sampling the carrier of the pulse-modulator is a unit-pulse train consisting of a train of pulses of unity height. The interval between two consecutive pulses is the sampling period T , and the pulse width p is often referred to as the sampling duration. The unit-pulse train used as the carrier of the pulse modulator $p(t)$

$$\begin{array}{ccccccc} 1 & & & & & & \\ -2T & -T-p & -T & & & & \\ T & T+P & 2T & 2T+p & & & \end{array}$$

Fig. 2-4 The unit pulse train.

is shown in Fig. 2-4. Therefore, when the input of the sampling device is a continuous function of time, the output of the device is a train of pulses whose envelope is identical to the input signal during the sampling durations.

We shall now investigate the frequency spectrum of the sampled output $e(t)$. In general, we may ask: "If the input function $e(t)$ contains no frequency component higher than ω radians per second, what frequency components will be contained in $e(t)$?" By inspection of the time response of Fig. 2-2, we shall expect to find much higher frequency components in $e(t)$ than in $e(t)$ because of the discontinuity characteristic of the pulse train. We can describe the operation of the pulse-amplitude modulator by

+ R. E. Kalman, "Analysis and Synthesis of Linear Systems Operating on Randomly Sampled-Data," (Doctoral thesis, Dept. of Electrical Engineering, Columbia University, New York, 1957).

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$$e(t) = p(t)e(t) \quad (2-1)$$

where $e(t)$ is the continuous input signal, $e(t)$ is the output signal of the modulator, and $p(t)$ denotes the unit-pulse train carrier. Since the sampling period T is assumed to be constant, $p(t)$ is a periodic function; $p(t)$ can be expanded into a Fourier series.

Hence

$$p(t) = \sum_{N=-\infty}^{\infty} C_n e^{j n \omega_s t} \quad (2-2)$$

where ω_s is the sampling frequency in radians per second and ω_s is related to the sampling period T by

$$\omega_s = 2\pi/T$$

C_n denotes the Fourier coefficients and is given by

$$C_n = \frac{1}{T} \int_0^T p(t) e^{-j n \omega_s t} dt \quad (2-3)$$

Since $p(t) = 1$ for $0 < t < p$, Eq. (2-4) becomes

$$C_2 = \frac{1}{\sqrt{2}}$$

$$e^{-jn\omega_s T}$$

From simple trigonometric identities, C_n is written

$$C_n = \frac{\sin(n\omega_s T/2)}{n\omega_s T/2}$$

Substituting C_n into Eq. (2-2), we have

$$p(t) = \sum_{n=-\infty}^{\infty} \sin(n\omega_s T/2) e^{-jn\omega_s T/2}$$

$$C_n = \frac{\sin(n\omega_s T/2)}{n\omega_s T/2}$$

The output of the sampler can now be written as

$$e(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_s t}$$

where C is given in Eq. (2-8). The Fourier transform of $e(t)$ is

By using the shifting theorem of the Fourier transform which states that

$$F[e^{j\omega_0 t} f(t)] = F(\omega - \omega_0)$$

$$E(j\omega) = \sum_{n=-\infty}^{\infty} C_n E(j\omega - jn\omega_s)$$

Equation (2-12) can also be written as

$$E(j\omega) = \sum_{n=-\infty}^{\infty} C_n E(j\omega + jn\omega_s)$$

$$C_n = \frac{\sin(n\omega_s T/2)}{n\omega_s T/2}$$

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since n is summed from $-\infty$ to $+\infty$.
Taking the limit as $n \rightarrow 0$ in Eq. (2-8), we have
 $C_0 = \lim_{n \rightarrow 0} C_n$
 $= p/T$

$n=0$

Hence from Eq. (2-12),

E^*

$$E_z(j\omega) \big|_{n=0} = -E(j\omega)$$

(2-14)

(2-15)

Equation (2-15) shows an important fact in that the frequency components contained in the original continuous function of time, $e(t)$, are still present in the output $e(t)$, except that they are multiplied by the factor p/T . For $n=0$, C_n is a complex quantity; but the magnitude of C_n may be written as

$$|C_n| = \frac{p}{T} \sin(n\omega_s p/2)$$

$T(n\omega_s p/2)$

The magnitude of $E(j\omega)$ is

$$|E^*(j\omega)| = \sum_{N=-\infty}^{\infty} |C_n| |E(j\omega + jn\omega_s)|$$

$N=-\infty$

(2-16)

(2-17)

The frequency spectrum of the periodic pulses $p(t)$ is simply a plot of the Fourier coefficient C_n as a function of ω . The amplitude plot of C_n is depicted in Fig. 2-5a. We see that the amplitude spectrum of C is not continuous, but rather it is a line spectrum at discrete frequencies of zero and $n\omega_s$, rad/sec, where n is a series of integers. The envelope of the spectrum is described by Eq. (2-16) and the amplitude of the envelope goes to zero at frequencies of $2\pi k/p$, where p is the pulse width of $p(t)$ and $k = \pm 1, \pm 2, \pm 3, \dots$. Because of the discrete property of C_n we can now write Eq. (2-17) in the following form:

$$|E^*(j\omega)| = \sum |C_n| |E(j\omega + jn\omega_s)|$$

(2-18)

In order to illustrate the magnitude spectrum of the sampler output, $E(j\omega)$, we assume that the spectrum of the continuous input time function $e(t)$ is as shown in Fig. 2-5b. By use of Eq. (2-18), the spectrum of the sampled signal is constructed in Fig. 2-5c. It is interesting to note that $|E(j\omega)|$ consists of not only the fundamental component of $E(j\omega)$ but also an infinite number of spectra called the complementary components. The n th complementary component in the output spectrum is obtained by multiplying $|E(j\omega)|$ by its corresponding Fourier coefficient $|C_n|$ and shifting it by $n\omega_s$, where $n = \pm 1, \pm 2, \pm 3, \dots$. Therefore, we may visualize the sampling device as a harmonic generator whose output contains the weighted fundamental component plus all the weighted complementary components at all frequencies separated by the sampling frequency. The

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$$-4\omega_s \quad -3\omega_s \quad -2\omega_s \quad -\omega_s$$

(a)

-

$|C_n|$

p/T

Equ 2-15

ω_s

$$2\omega_s \quad 3\omega_s \quad 4\omega_s \quad A\pi$$

$$11 |E(j\omega)|$$

(b)

$-4\omega_s \quad -3\omega_s$

$$|E^*(j\omega)|$$

p/T

$$-4\omega_s \quad -3\omega_s \quad -2\omega_s \quad -\omega_s$$

$$\omega_s \quad +\omega_s \quad 2\omega_s$$

$$3\omega_s \quad 4\omega_s$$

$$\begin{aligned}
 & (c) \\
 & -2\pi \\
 & -4\omega_s - 2\omega_s \\
 & |E^{**}(j\omega)| \\
 & 0 \\
 & 2\omega_s \quad 4\omega_s \quad 2\pi T \\
 & \text{ATT} \\
 & 1
 \end{aligned}$$

$$-5 \omega_s \quad 3 \omega_s \quad \text{έως} \quad \text{έως} \quad 3 \omega_s \quad 5 \omega_s$$

(d)

Fig. 2-5 Amplitude spectra of input and output signals of a finite pulse-width sampling device. (a) Amplitude spectrum of unit pulse train $p(t)$. (b) Amplitude spectrum of continuous input function $e(t)$ (c) Amplitude spectrum of sampler output. Sampling frequency is higher than twice the maximum signal frequency, ($\omega_s > 2\omega_c$). (d) Amplitude spectrum of sampler output. Sampling frequency is less than twice the maximum signal frequency, ($\omega_s < 2\omega_c$).

band around zero frequency still carries essentially all the information contained in the continuous input signal, but the same information also repeats along the frequency axis; the amplitude of each component is weighted by the magnitude of its corresponding Fourier coefficient C . It should be pointed out here that the frequency spectrum shown in Fig. 2-5c is sketched with the assumption that the sampling frequency ω_s , is greater than twice the highest frequency

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contained in $e(t)$; that is, $\omega_s > 2\omega_c$. If $\omega_s < 2\omega_c$ distortion in the frequency spectrum of $|E(j\omega)|$ will appear because of the overlapping of the signal components. As shown in Fig. 2-5d, if $\omega_s < 2\omega_c$, the spectrum of $|E(j\omega)|$ around zero frequency bears little resemblance to that of the original signal. Therefore, although the original signal can be recovered from the spectrum shown in Fig. 2-5c by means of a low-pass filter the complementary components of the sampler output overlap each other as shown in Fig. 2-5d, and the input signal cannot be reconstructed even by means of an ideal low-pass filter.

An alternate description of the sampled signal $e(t)$ can be obtained by means of the complex convolution method. With reference to Fig. 2-2 and Eq. (2-1), the Laplace transform of $e(t)$ is written.

or

$$E(s) = L[e^*(t)] = L[e(t)p(t)]$$

$$E(s) = E(s) * P(s)$$

(2-19)

(2-20)

where $E(s)$ and $P(s)$ are the Laplace transforms of $e(t)$ and $p(t)$, respectively. The symbol, "*", in Eq. (2-20) denotes the complex convolution operation. Considering that the unit pulse train is a sum of successively delayed positive and negative step functions, $P(s)$ can be written as

$$P(s) = \frac{1}{s} -$$

$$e^{-ps} + e^{-Ts}$$

-

$$e^{-(T+p)s} + \dots + e^{-KTs}$$

$$- (kT + p)s$$

e

+

.

1

$$(2-21)$$

Representing $P(s)$ in a closed form, we have

$$P(s) = \frac{1}{s(1 - e^{-Ts})}$$

Equation (2-20) now reads

$$E(s) = E(s) * \frac{1}{s(1 - e^{-Ts})} \quad (2-22)$$

$$E(s) = E(s) * \frac{1}{s(1 - e^{-Ts})} \quad (2-23)$$

It follows from the definitions of the complex convolution that

$$E(s) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} E(\xi) P(s - \xi) d\xi$$

$$E(s) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} E(\xi) \frac{1}{\xi(1 - e^{-T\xi})} d\xi$$

$$E(s) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} E(\xi) \frac{1}{\xi(1 - e^{-T\xi})} d\xi \quad (2-24)$$

$$E(s) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} E(\xi) \frac{1}{\xi(1 - e^{-T\xi})} d\xi \quad (2-25)$$

where ξ is a dummy variable; $\sigma_1 < c < \sigma - \sigma_2$, $\sigma > \max(1, 2, \sigma_1 + \sigma_2)$, where σ is the real part of s and σ_1 , and σ_2 , are the abscissas of convergence of $E(s)$ and $P(s)$, respectively. The path of integration of Eq.

(2-25) is along the line from $c - j\infty$ to $c + j\infty$ in the complex s -plane.

M. F. Gardner and J. L. Barnes, Transients in Linear Systems, Vol. I (New York: John Wiley & Sons, 1942), pp. 275-278.

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$$X(s) = \frac{1}{s} \sum_{n=0}^{\infty} x(nT) e^{-snT}$$

X
X
X
S
X
-plane
Poles of
 $1 - e^{-T(s-g)}$
 $(s - (1 - e^{-T(S-))})$

R_∞
X
0
X
Poles of $E()$

- jws

X
-2jws

$R_{4\infty}$

X
-3jws

X
-4jws

X
-5jws

-6jws
c-joo

Re

Fig. 2-6 Paths of contour integration in the right-half -plane and the left-half ξ -plane.

plane as shown in Fig. 2-6. The convolution integral of Eq. (2-25) may be carried out by taking the contour integration along a path formed by the c-joo to c + j ∞ line, and a semicircle of infinitely large radius going around either the right-half or the left-half -plane. The contour integral is then evaluated by means of the residues. theorem of complex variables. In other words, Eq. (2-25) may be written in the following forms:

$$\begin{aligned} & (s \\ & - \\ & 1 \\ & \xi) \\ & (1) \\ & E(s) \\ & 1 \\ & = \\ & - \\ & 2 \times j \text{ fr, } E(\xi) \\ & 2 \times j \text{ f } E(\xi) \\ & (s \\ & 1 \\ & e^{-p(s-E)} \\ & (1 - e^{-T(s-t)}) d\xi \\ & 1 \\ & - \\ & ---- \\ & e^{-p(s-E)} \\ & - \\ & \xi) (1 - e^{-T(s-\xi)}) d\xi \end{aligned}$$

(2-26)

+ R. V. Churchill, Introduction to Complex Variables and Applications (New York; McGraw Hill Book Company, 1948).

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$$\begin{aligned}
 & \frac{1}{2\pi j} \int_{\Gamma} E(s) \left(1 - e^{-T(s-\xi)} \right) ds \\
 & = \sum_{k=0}^{\infty} \frac{e^{-kT}}{k!} \frac{d^k}{ds^k} E(s) \bigg|_{s=\xi}
 \end{aligned}$$

where Γ_1 is a closed path enclosing the entire finite left half of the s -plane and Γ_2 is a closed path enclosing the right half of the s -plane. The two paths Γ_1 and Γ_2 are depicted in Fig. 2-6. In order to utilize the theorem of residues, we have to study the poles and zeros of $E(s)$ and $P(s, \xi)$. Normally, the poles of $E(s)$ are considered to be in the left half of the plane and are finite in number,

but $P(s, \xi)$ has simple poles at

$$s = \xi + j\frac{2\pi n}{T}$$

where n (integer) $< \infty$ (2-28)

where ω , is the sampling frequency in radians per second. Equation (2-28) shows that the poles of $P(s, \xi)$ are infinite in number and are located at frequency intervals of ω , ($n = 0$, and all integers) along the path of $\text{Re}(s) = \text{Re}(\xi)$ in the plane. Typical pole distributions of $E(s)$ and $P(s, \xi)$ are shown in Fig. 2-6. If the function $E(s)$ has at least one more zero than pole at ξ

$$\lim_{s \rightarrow \infty} E(s) = 0$$

that is

the second term of Eq. (2-27) is zero, since the integral along the infinite-radius semi-circle in the left half of the s -plane vanishes. Equation (2-26) then becomes

$$E(s) = \sum_{k=0}^{\infty} \frac{e^{-kT}}{k!} \frac{d^k}{ds^k} E(s) \bigg|_{s=\xi}$$

$$E(s) = \sum_{k=0}^{\infty} \frac{e^{-kT}}{k!} \frac{d^k}{ds^k} E(s) \bigg|_{s=\xi}$$

de

(2-30)

From the theorem of residues the contour integral of Eq. (2-30) is equal to the sum of the residues of the integrand at the poles enclosed by the closed path I_1 . Therefore,

Ex (s)

-

$\sum$ Residues of E()

At the poles

of E(s)

-

1

$e^{-p(s-E)}$

-

$(s - \xi)(1 - e^{-T(8 - \xi)})$

(2-31)

Equation (2-31) can be written in a more specific form if we assume that $E(\xi)$ is a rational function with only k simple poles. Then

k

E(s)

=

$\sum$

$\sum_{n=1}^k$

$N(n)$

$D'(\xi_n) (1$

1

--

$e^{-P(8 - kn)}$

$e^{-T' (s - kn) } (s - \xi_n$

(2-32)

where

$E(\xi) =$

$N(E)$

(2-33)

$D()$

$D'(n)$

$dD(\xi)$

=

(2-34)

de

t

and designates the poles of $E(\xi)$.

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Now let us evaluate $E^*(s)$ by solving the contour integral around the right-half-plane along the path I . In this case as approaches infinity, the integral along the circular portion of I' , will again vanish provided that $E(\xi)$ satisfies Eq. (2-29). Equation (2-27) now reads

1

$E^*(s) =$

$2\pi j \oint_{I'} E(\xi)$

(s

1

-

2

$e^{-p(8-5)}$

$\xi)(1 - e^{-T(s - \xi)}$

Applying the residues theorem, we have

E(s)

$$= \sum \text{Residues of } E(s) \text{ At the poles of } \frac{1 - e^{-P(s)}}{(s - \xi)(1 - e^{-P(s - \xi)})}$$

$$\alpha \xi \frac{1}{(2-35)} e^{-P(s)}$$

$$= \sum \frac{E(s)}{(s - \xi)}$$

$$(2-36)$$
 Since $P(s)$ has only simple poles located at periodic intervals along $\text{Re}(s) = \text{Re}(\xi)$ in the s -plane, Eq. (2-36) is written

$$E(s) = \sum_{n=-\infty}^{\infty} \frac{N(n)}{D'(s - \xi_n)} \cdot E(\xi_n)$$

$$(2-37)$$

$$N = -\infty$$
 where $\xi = s + jn\omega$, denotes the poles of $P(s - \xi)$. Also and

$$N(n) = 1 - e^{-P(s - \xi)} \big|_{s = s + jn\omega}$$

$$(2-38)$$

$$D'(n) = \frac{d}{ds} [(s - \xi)(1 - e^{-T(s - \xi)})];$$

$$\xi = s + jn\omega$$

$$+ jn\omega T \quad (2-39)$$
 Therefore, Eq. (2-37) becomes

$$E(s) = \sum_{n=-\infty}^{\infty} \frac{1 - e^{-jnw}, p}{jn\omega T} E(s + jn\omega)$$

$$(2-40)$$
 It is apparent that Eq. (2-40) and Eq. (2-13) are identical. The Sampling Theorem

$$(2-40)$$
 The simple physical reasoning made in the preceding section about the minimum sampling frequency for signal recovery actually answers a very basic yet important question in regard to the proper rate of sampling. In general, if sampling is used in a given system, we may

ask the following question: What are the limitations on the rate of sampling? It is easy to see that theoretically there is no upper limit on the sampling frequency. As a matter of fact, when the sampling frequency approaches infinity (if this is physically possible for a sampler), we approach the continuous case. As far as the lower limit is concerned, intuitively, we know that, if the continuous signal changes radically with respect to time, the signal's bandwidth must be relatively high, and sampling the signal at too low a rate may lose vital information on the signal between the sampling instants, and consequently, we may not be able to reconstruct the

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continuous signal from the information contained in the sampled data. From the amplitude spectra shown in Fig. 2-5, the lowest sampling frequency for possible signal reconstruction is $2\omega_c$, where ω_c is the highest frequency contained in $e(t)$. Formally, this is known as the sampling theorem. The theorem states that if a signal contains no frequency higher than ω_c radians per second, it is completely characterized by the values of the signal measured at instants of time separated by $T = \{(2\pi/\omega_c) \text{ second}\}$. Note that in practice system stability of closed-loop systems and other factors also dictate the choice of the sampling frequency and may make it necessary to sample at a rate much higher than this theoretical minimum. Furthermore, strictly speaking a band-limited signal does not exist physically in communication or control systems. All physical signals do contain components of a large frequency range. In general, because the amplitudes of the higher frequency components are greatly diminished, a band-limited signal is assumed. Therefore, in practice these facts, plus the unrealizability of an ideal low pass filter, make impossible the exact reproduction of a continuous signal from the sampled signal even if the sampling theorem is satisfied.

An interesting note on the sampling theorem should be added here. A signal can still be defined completely by sampling it at a rate less than $2\omega_c$ radians per second provided that the derivatives of the signal are known at the sampling instants as well as the function amplitude sample information. To be more specific, L. J. Fogel has proved that if a signal contains no frequency higher than ω_c radians per second, it is completely characterized by the values $e^{(k)}(nT)$, $e^{(k-1)}(nT), \dots, e'(nT)$, and $e(nT)$, ($n = 0, 1, 2, \dots$) of the signal measured at instants of time separated by $T = (k+1)(2\pi/\omega_c)$

second, where $e^{(k)}(nT) = d^k e(t)/dt^k = \omega_c^k n^k$. This means that when the values of the first derivative of $e(t)$ at $t = nT$, $e'(nT)$, for $n = 0, 1, 2, \dots$, are given in addition to the values of $e(nT)$, the maximum allowable sampling time is increased to $T = 2\pi/\omega_c$, which is twice the time required when $e(nT)$ alone is measured. The addition of each succeeding derivative allows the time between samples (sampling time) to become larger according to $T = (k+1)(2\pi/\omega_c)$, where k is the order of the highest derivative when all lower ordered derivatives are observed for each sample.

Further discussions of the signal-recovery and the sampling frequency are presented in the following sections.

R. M. Oliver, J. R. Pierce, C. E. Shannon, "The Philosophy of Pulse Code Modulation," Proc. I. R. E., 36, No. 11, November 1948, pp. 1324-1331.

L. J. Fogel, "A Note on the Sampling Theorem," Trans. on Information Theory, March 1955, pp. 47-48.

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2.3 MATHEMATICAL DESCRIPTION OF THE IDEAL

SAMPLING PROCESS-THE IDEAL SAMPLER

In the preceding section, mathematical analysis of an actual sampling process is performed by the methods of Fourier transform and complex convolution. The characteristics of the sampling operation are described in both the time domain and the frequency domain. However, exact analysis of sampled-data systems with finite pulse-width considerations is quite complex. In general, to simplify the analysis and synthesis techniques, the sampling device may be replaced by an "ideal sampler" whose output signal $e^*(t)$ contains a train of impulses. This approximation is justified if the pulse-width of the sampler output is very small when compared with the dominant time constant of the continuous signal $e(t)$, and to the sampling period.

$e(t)$
 $e^*(t)$
X
T

Fig. 2-7 Model of an ideal sampler.

A model of the ideal sampler is shown in Fig. 2-7. When the input to the ideal sampler is a continuous function $e(t)$ as shown in Fig. 2-8a, the output of the sampler, denoted by $e^*(t)$, is a train of amplitude-modulated impulses and is related to $e(t)$ by

$$e^*(t) = e(t)d(t)$$

where $d(t)$ is a unit-impulse train; that is,

$$d(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT) \dots$$

$$n = -\infty$$

$$(2-41)$$

$$(2-42)$$

where $\delta(t - nT)$ represents an impulse of unit area (or strength) occurring at time $t = nT$. Hence Eq. (2-41) can be written as

$$e^*(t) = e(t) \sum_{n=-\infty}^{\infty} \delta(t - nT) = \sum_{n=-\infty}^{\infty} e(nT)\delta(t - nT)$$

$$(2-43)$$

If it is assumed that $e(t)$ is zero for $t < 0$, Eq. (2-43) reads.

+ G. Farmanfarma, "Study of Finite Pulsed Feedback Systems," Trans. A. I. E. E., Part II, July 1958.

J. Tou, "Analysis of Sampled-Data Control Systems with Finite Sampling Duration," Proc. National Electronics Conference, 1957.

Similar to the discussion given in Sec. 2-2, the pulse-amplitude modulation may be replaced by an impulse modulator whose carrier is a train of impulses each with unity area.

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00

$$e^*(t) = \sum_{n=0}^{\infty} e(nT)\delta(t - nT)$$

$$n=0$$

$$(2-44)$$

Equation (2-44) indicates that the output of the ideal sampler is a train of impulses whose respective impulse areas (strengths) are equal to the magnitudes of the input signal at the corresponding sampling instants nT . The operation of the ideal sampler according to Eqs. (2-41) and (2-44) is illustrated in Fig. 2-8.

$e(t)t$

;)

 $0T \quad 1T \quad 11$

 (a)

 $-2T \quad 0$

 $-T$

 $e(t)$

 $2T \quad 4T \quad 6T$

 $8T \quad 10T \dots$

 $T \quad 3T \quad 5T \quad 7T \quad 9T$

 (b)

 $2T \quad 4T \quad 6T \quad 8T \quad 10T$

 $3T \quad 5T \quad 7T$

 $9T$

 (c)

Fig. 2-8 Signal waveforms illustrating the sampling operation of an ideal sampler. (a) Input signal to sampler. (b) Unit-impulse train (train of impulses with unity area). (c) Output signal of sampler.

Equation (2-44) describes the operation of the ideal sampling device only in the time domain. In order to study the frequency behavior of the sampling process, we must obtain the transform of $e^*(t)$. Different transform representations of the ideal sampler output are derived in the following explanations:

Sec. 2.3

THE SAMPLING PROCESS 31

1. The Method of Laplace Transforms

Taking the Laplace transform of $e^*(t)$ directly from Eq. (2-44), we have

$$E^*(s) = \mathcal{L}[e^*(t)] = \sum_{n=-\infty}^{\infty} e(nT) \delta(t - nT)$$

Since the Laplace transform of the unit impulse ($\delta(t)$) Eq. (2-45) becomes

$$E^*(s) = \sum_{n=-\infty}^{\infty} e(nT) e^{-s nT}$$

(2-45)

$\delta(t - nT)$ is $e^{-s nT}$

(2-46) Therefore, $E^*(s)$ represents the ordinary Laplace transform of the sampled signal $e^*(t)$. Since $e^*(t)$ describes only the values of $e(t)$ at the sampling instants, note that $E^*(s)$ carries no information concerning the behavior of the function $e(t)$ during the sampling period T . An interesting fact is that Eq. (2-46) is very similar to the defining equation of the Laplace transform of $e(t)$; that is,

$$E(s) = \int_{-\infty}^{\infty} e(t) e^{-st} dt$$

(2-47)

Therefore, $E(s)$ is the transform of a continuous time function $e(t)$, whereas $E^*(s)$ is the transform of $e^*(t)$, where $e^*(t)$ is a train of impulses modulated by $e(t)$. One should not be alarmed by the fact that $E^*(s)$ in Eq. (2-46) is an infinite series. In general, if the Laplace transform of $e(t)$ is a rational function, $E^*(s)$ can always be written in a closed form. The following examples illustrate this property of $E^*(s)$ and the general application of Eq. (2-46).

Example 2-1

Consider that a unit-step function $e(t) = u(t)$ is sampled by an ideal sampler at intervals of T seconds. From Eq. (2-44), the sampled signal (output of the sampler) is

$$e^*(t) = \sum d(t - nT)$$

The Laplace transform of $e^*(t)$ is

$$E^*(s) = \sum e^{-nts}$$

$$= \sum_{n=0}^{\infty} e^{-nts}$$

$$= \sum_{n=0}^{\infty} e^{-nts}$$

$$= \sum_{n=0}^{\infty} e^{-nts}$$

$$= \sum_{n=0}^{\infty} e^{-nts}$$

$$n = 0$$

$$(2-48)$$

$$1 +$$

$$+ e^{-2Ts} +$$

$$\text{for } e^{-ST} < 1$$

$$(2-49)$$

$$Ts$$

Example 2-2
If the input signal to an ideal sampler is $e(t) = e^{-t}$, the values of $e(t)$ at the sampling instants are described by $e(nT) = e^{-nT}$. Therefore, by use of Eq. (2-46)

$$32$$

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$$00$$

$$E^*(s) = \sum$$

$$n = 0$$

$$e^{-nT} e^{-nts} = \sum e^{-nT(s+1)}$$

$$1$$

$$=$$

$$n = 0$$

$$\text{for } e^{-ST} < 1$$

$$\cdot T(s+1)$$

$$(2-50)$$

$$2. \text{ The Method of Complex Convolution}$$

The Laplace transform of the output of the ideal sampler can also be obtained from the complex convolution method used in Sec. 2.2. After multiplying the magnitude of the pulses in the unit pulse train $p(t)$ by $1/p$ (p = pulse width) and taking the limit as p approaches zero, we have the train of unit impulses shown in Fig. 2-8b. Thus $E^*(s)$ can be obtained directly from $E(s)$; that is,

$$E^*(s) = \lim_{p \rightarrow 0} E_p(s)$$

$$p \rightarrow 0$$

$$(2-51)$$

First, let us consider Eq. (2-26). Multiplying both sides of Eq. (2-26) by $1/p$ and letting p approach zero, we have

$$E^*(s) = \lim_{p \rightarrow 0} E_p(s)$$

$$1$$

$$\lim_{p \rightarrow 0} \frac{1}{p} \int_0^p e^{-s\tau} p(\tau) d\tau = \int_0^\infty e^{-s\tau} \delta(\tau) d\tau$$

$$\lim_{p \rightarrow 0} \frac{1}{p} \int_0^p e^{-s\tau} p(\tau) d\tau$$

$$p \rightarrow 0$$

$$E)$$

$$=$$

$$1$$

$$=$$

$$e$$

$$1$$

$$- T(s - \frac{1}{T})$$

$$]$$

(2-52)

(2-53)

Since

$E^*(s)$ is written as

1

1

-

1

1

-

e

T (8

$$E^*(s) = \frac{1}{T} \int_0^T E(\xi) e^{-s\xi} d\xi = \frac{1}{T} \int_0^T E(\xi) e^{-s\xi} d\xi$$

Γ₁

-

- T(}

(2-54)

If $E(\xi)$ satisfies the condition given in Eq. (2-29), the second integral of Eq. (2-54) vanishes; and if $E(\xi)$ is a rational function having k simple poles, from the residues theorem

k

$$E^*(s) = \sum_{n=1}^k \frac{N(\xi_n)}{D'(\xi_n)}$$

N = 1

N(ξ_n)

D'(ξ_n)

1

-

- T(8 - ξ_n)

e

(2-55)

where $N(\xi)$ and $D'(\xi)$ are defined in Eqs. (2-33) and (2-34), respectively. The two expressions in Eqs. (2-46) and (2-55) are essentially the same although they are of different forms. The application of Eq. (2-55) will now be illustrated by a simple example.

Example 2-3

Consider the same signal $e(t) = e'$ used in Example 2-2. In this example, the Laplace transform of the sampled signal is to be derived from Eq. (2-55). Since the Laplace transform of $e(t)$ is $1/(s+1)$, in Eq. (2-55), $N(\xi_1) = 1$ and $D(\xi_1) = \xi + 1$. $E(\xi)$ has a simple pole at $\xi = -1$. Thus

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$$D'(\xi) =$$

=

$$dD(\xi)$$

$$\propto \xi$$

$$= 1$$

$$= -1$$

From Eq. (2-55) we have

1

$$E^*(s)$$

-

1

-

e

(2-56)

$$= \frac{1}{T} \int_0^T e^{-s\xi} d\xi$$

(2-57)

which is identical to the result obtained in Example 2-2 by using Eq. (2-46). However, the two expressions we have for $E^*(s)$ in Eqs. (2-46) and (2-55) fail to indicate the frequency domain characteristics of the sampled signal $e^*(t)$. Previously, we have shown that an actual

sampling device (finite pulse-width sampler) resembles a harmonic generator that duplicates the weighted amplitude spectrum of the input signal at integral multiples of the sampling frequency. When the finite-width pulses are approximated by impulses, we should perhaps expect the sampler to act more like a harmonic generator; and indeed it does, as for we shall show. Note that the amplitude spectrum of the output signal of the ideal sampler contains all the complementary components in addition to the primary component and that the amplitudes of all components are equal and unattenuated.

Now using Eq. (2-27), $E^*(s)$ is written as

$$E^*(s) = \lim_{p \rightarrow 0} p \int_0^p E(t) e^{-st} dt$$

$$= \lim_{p \rightarrow 0} \int_0^p E(t) e^{-st} dt$$

$$= \lim_{p \rightarrow 0} \int_0^p E(t) e^{-st} dt = \int_0^\infty E(t) e^{-st} dt$$

$$= \int_0^\infty E(t) e^{-st} dt$$

$$= \int_0^\infty E(t) e^{-st} dt$$

$$= \int_0^\infty E(t) e^{-st} dt$$

$$= \int_0^\infty E(t) e^{-st} dt$$

$$= \int_0^\infty E(t) e^{-st} dt$$

$$= \int_0^\infty E(t) e^{-st} dt$$

$$= \int_0^\infty E(t) e^{-st} dt$$

$$= \int_0^\infty E(t) e^{-st} dt$$

$$= \int_0^\infty E(t) e^{-st} dt$$

In this case since $1/(1 - e^{-sT})$ has a simple pole at infinity in the s -plane, the second integral of Eq. (2-58) along the semicircle of infinite radius in the right half of the plane may not vanish. As a matter of fact, if the degree of the denominator of $E(s)$ in s is not higher than the degree of the numerator by at least two, the integral along the semicircle may have a finite value or may converge.

even

Let us assume that $E(s)$ satisfies the following condition:

$$\lim_{t \rightarrow 0^+} E(s) = e(0^+) = \lim_{t \rightarrow 0^+} e(t)$$

not

$$(2-60)$$

In other words, the degree of the denominator of $E(s)$ is higher than that of the numerator by only one, and the initial value of the time function $e(t)$ is $e(0^+)$. We can write

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$$\lim_{s \rightarrow 0} E(s)$$

$$=$$

$$\lim_{s \rightarrow 0}$$

$$e(0^+)$$

$$00 + \{$$

$$E$$

Then, the second integral of Eq. (2-59) becomes

(2-61)

1

$2\pi j$
 $fE(s) 1$

1

1
 $-7/2$
 $\lim e(0^+)/s$

-

e
 $r(s - s) ds$

(2-62)
 $2\pi j \int_{-\infty}^{\infty} + x/2 1 e$

-

Letting

=

re
(2-63)

2,

on the semicircle of infinite radius of I, and since as $r \rightarrow \infty$,

$e^{-T(s - s)}$

> 0 , we have

1

$E(s)$

1

1
 $-T(s -$
 $E)$

e

$2\pi j \oint (t) - d = 2\pi j re$ do

$2\pi j$

$1 - x/2$

+

$= -e(0^+)/2$

(2-64)

Hence the integral along the semicircle in the right half of the s -plane is equal to $-e(0^+)/2$. The first integral of Eq. (2-59) is evaluated by means of the residues theorem and similar steps as involved in Eqs. (2-35) through (2-40). Thus

1

$2\pi j \int_{-\infty}^{\infty}$

1

00
 $E(s) = \dots E(s + jn\omega_s) \quad (2-65)$

1

$\oint -T(s - s) ds = \int_{-\infty}^{\infty} = -\infty$

e

and from Eq. (2-59) we have

00

$E^*(s) = \sum E(s + jn\omega_s) + e(0^*)$

T

$N = -\infty$

2

(2-66)

Equation (2-66) gives the Laplace transform of the output signal of an ideal sampler (impulse modulator) when the input time function $e(t)$ has a jump of $e(0^*)$ at $t = 0$. However, most of the closed-loop system transfer functions involved in control systems usually have at least two more poles than zeros, or $E(s)$ satisfies the following condition:

$\lim_{s \rightarrow 0} sE(s) = 0$

$s \rightarrow 00$

(2-67)

Then the second integral of Eq. (2-59) is zero (since $e(0^*) = 0$), and Eq. (2-66) is simply

$$E^*(s) = \sum_{n=-\infty}^{\infty} E(s + jn\omega_s)$$

(2-68)

This result can also be obtained directly from Eq. (2-40) by multiplying $E(s)$ by $1/p$ and taking the limit of the product as p approaching zero. The following example illustrates the difference in applicability between Eqs. (2-66) and (2-68).

Example 2-4

Let the input signal to an ideal sampler be a unit-step function. The Laplace transform of the sampled output $e^*(t)$ is desired. Since the input $e(t)$

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has a jump discontinuity at $t = 0$, use of Eq. (2-68) may be expected to lead to inaccurate results. This error is indicated by substituting the Laplace trans-

form of $e(t)$ into Eq. (2-68) yielding

$$E^*(s) = -\sum_{n=0}^{\infty} E(s + jn\omega_s) = \frac{1}{T} \sum_{n=0}^{\infty} \frac{1}{s + jn\omega_s}$$

which can also be written

$$E^*(s) = \sum_{n=-\infty}^{\infty} \frac{1}{T} \frac{1}{s + jn\omega_s} \quad (2-69)$$

(2-69)

$$= \frac{1}{T} \sum_{n=-\infty}^{\infty} \frac{1}{s + jn\omega_s} = \frac{1}{T} \sum_{n=-\infty}^{\infty} \frac{1}{s + jn\omega_s} \quad (57)$$

(2-70)

Only one-half of the value of $e(0^*)$ is sampled, since the first term of Eq. (2-70) is $1/2$. The correct answer of $E^*(s)$ is obtained if Eq. (2-66) is used. From Eq. (2-66) or Eq. (2-68) we see that the ideal sampler is again a harmonic generator. The sampler reproduces the spectrum of the continuous input time function $e(t)$ as well as the sidebands (complementary components) at integral multiples of the sampling frequency; and all components have the same amplitude of $1/T$.

$+1E(j\omega)$
 $IE^*(j\omega)$
 $1/T$
 $i-\omega$
 (a)
 $+Wc$
 $-2\omega_c$
 $-Wc$
 $-Wc$
 $+Wc$
 ω_c
 $2\omega_c$
 0
 (b)

Fig. 2-9 Amplitude spectra of input and output signals of an ideal sampler (impulse modulator). (a) Amplitude spectrum of continuous input function $e(t)$. (b) Amplitude spectrum of sampler output. Sampling frequency is higher than twice the maximum signal frequency, ($\omega_s > 2\omega_c$).

Assuming that the amplitude spectrum of a continuous signal is as shown in Fig. 2-9a, the corresponding amplitude spectrum of the sampled signal $e^*(t)$ when $\omega_s < 2\omega_c$ is shown in Fig. 2-9b, where ω_s is the sampling frequency in rad/sec and ω_c is the highest frequency contained in $e(t)$. Again, if the sampling frequency is less than $2\omega_c$, distortion will occur in the output spectrum because of the overlapping of the sidebands in $|E^*(j\omega)|$.

Some s-plane Properties of E^* (8)

We have shown in the preceding section that the most significant

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meaning of Eq. (2-68) is that it clearly indicates the effect of an ideal sampler in terms of the frequency domain behavior of $e(t)$ and $e^*(t)$. Two other important properties of $E^*(s)$ are also worth mentioning.

(1) $E^*(s)$ is a periodic function with period $j\omega_s$.

The property of $E^*(s)$ as a periodic function is apparent in view of Eq. (2-68) and Fig. 2-9b. Analytically, we can show this fact by substituting $s + jm\omega_s$ for s in Eq. (2-46), where m is an integer.

00

$$E^*(s + jm\omega_s) = \sum_{n=-\infty}^{\infty} e^{-n(s+jm\omega_s)T} \sum_{r=-\infty}^{\infty} e^{nT} e^{rs} \quad (2-71)$$

Therefore,

$$N = 0$$

$$E^*(s + jm\omega_s) = E^*(s)$$

$$n = 0$$

$$(2-72)$$

In other words, given any point $s = s_1$ in the s-plane, the function $E^*(s)$ has the same value at all periodic points

$$s = s_1 + jm\omega_s \quad (m$$

=

integers from $-\infty$ to $+\infty$).

$Aj\omega$

The s-plane

$$-j\omega_s/2$$

$$s_1 + j\omega_s$$

$$-j\omega_s/2$$

0

Complementary $s_1 + j2\omega_s$

strips

$$-j3\omega_s/2$$

0
 $S_1 + j\omega_s$
 $-\omega_s/2$
 Primary
 strip
 S_1
 $-\omega_s/2$
 0
 $S_1 - j\omega_s$
 $-\omega_s/2$
 0
 Complementary $S_1 - j2\omega_s$
 strips
 0
 $S_1 - j3\omega_s$
 $-\omega_s/2$

Fig. 2-10 Periodic strips in the s-plane.

This equality is illustrated in Fig. 2-10. Thus we can divide the s-plane into an infinite number of periodic strips as shown in Fig. 2-10. The strip between $-\omega_s/2$ and $+\omega_s/2$ is called the primary strip and all others occurring at higher frequencies are designated as the complementary strips. The function $E^*(s)$ has the same value at all congruent points in the various periodic strips.

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(2) If the function $E(s)$ has a pole at $s = s_1$, then, $E^*(s)$ must have poles at $s = s_1 + jm\omega_s$, ($m = \text{integers from } -\infty \text{ to } +\infty$).

The fact that this statement is true can be observed easily from Eq. (2-68). A set of hypothetical pole configuration of $E(s)$ and the corresponding $E^*(s)$ is shown in Fig. 2-11.

$j\omega_s$
 s-plane
 X
 X $j4\omega_s$
 X
 X $j3\omega_s$
 X
 Poles of $E(s)$
 X
 $j2\omega_s$
 X $j\omega_s$
 X
 X
 X
 X
 X $-j\omega_s$
 X $-j2\omega_s$
 X $-j3\omega_s$
 X $-j4\omega_s$

Fig. 2-11 Periodicity of the poles of $E^*(s)$.

2.4 SUMMARY

The nature and the operation of the sampling process have been discussed in this chapter. The sampling operation may be considered as either a fictitious sampler or a pulse-amplitude modulation process. The sampled signal contains all the sidebands (complementary components) about integral multiples of the sampling frequency. Theoretically, if the continuous signal is assumed to be band-limited, then Shannon's sampling theorem states that the necessary condition for recovery of the continuous information from the sampled signal is to have the sampler sample at least once every $(2\omega_c/\omega_s)$ second, where ω_c is the highest component frequency contained in the con-

tinuous signal. An extension of the sampling theorem is also given that allows for fewer samples, but additional information is necessary concerning the derivatives of the continuous signal at the sampling instants.

For mathematical convenience the actual sampling process may

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be approximated by the operation of an ideal sampler, whose output signal is a train of amplitude-modulated impulses. Based on this simplification, four basic forms of the Laplace transform of the sampled signal $E^*(s)$ are derived and the conditions of validity of these expressions are discussed.

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3

RECONSTRUCTION OF SAMPLED

SIGNALS

3.1 INTRODUCTION

Although the envelope of the sampled signal $e^*(t)$ corresponds to the values of the continuous signal $e(t)$ at the sampling instants, the characteristics of the two functions in the frequency domain have been shown to be quite different. In most feedback control systems employing sampled-data, the high frequency sidebands in $e^*(t)$ that resulted from the sampling operation must be removed before the signal is applied to the continuous part of the system. This step is absolutely necessary, since the continuous signal must be reconstructed from the sampled-data; moreover, signals containing high sidebands may cause excessive wear to the controlled elements. Therefore, a component which may be considered as a data-reconstruction device or a filtering device must be employed to remove the sidebands from the sampled signal $e^*(t)$ before the signal is applied to the controlled elements.

$r(t)$

The block diagram of a sampled-data feedback control system

$e(t)e^*(t)$, Digital

compensator

T

Sampler

$D(s)$

Data

reconstruction

Controlled $c(t)$,

system

Fig. 3-1 Sampled-data feedback control system with data-recon-

struction device.

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with a data-reconstruction device is shown in Fig. 3-1. The sampled signal is first processed or modified by a digital network $D^*(s)$; the output of $D^*(s)$, which is in the form of a pulse train, is then smoothed out by the data-reconstruction device before the signal is applied to the components of the controlled process.

For the purpose of studying how data reconstruction may be accomplished, let us assume that a sampler is sampling at a sampling

$AIE^*(j\omega)$

$1/T$

Filter characteristic

AAAA

$2\omega_c$

$-2\omega_c$

ω_c

-

- ω_c 0 ω_c

ω_c

Complementary

components

Primary

comp.

Complementary

components

Fig. 3-2 Reconstruction of continuous signal from sampled-data using ideal low pass filter.

frequency of $\omega_s = 2\omega_c$ rad/sec, where ω_c is the highest frequency component contained in the continuous signal being sampled. Figure 3-2 shows the amplitude spectrum of $E^*(s)$, from which it is clear that in order to duplicate the continuous signal, the sampled signal must be sent through an ideal low pass filter with an amplitude-

- ω_c

▲ Amplitude

ω_c

characteristic as shown in Fig. 3-

3. Unfortunately, an ideal filter

characteristic is physically unreali-

zable, since it can be shown that

its response begins before an input

is applied. However, even if we

could realize the ideal filter response,

we have learned in the last chapter

that perfect reproduction of the

continuous signal is still based on

the assumption that $e(t)$ is band-limited. Therefore, for all practical considerations, it is impossible to recover a continuous signal perfectly from the sampled-data. The best we can do in data-reconstruction is to try to approximate the original time function as closely as possible. Furthermore, as will be shown later in this chapter, a better approximation of the original signal requires in general, a long time delay that is undesirable in view of its adverse effect on system stabil-

Fig. 3-3 Amplitude characteristic of ideal filter.

† J. L. Bower, and P. M. Schultheiss, Introduction to the Design of Servomechanisms. (New York: John Wiley & Sons, Inc., 1958).

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ity. Consequently, the design of a data-reconstruction device usually involves a compromise between the requirements of stability and the desirability of a close approximation of the continuous signal.

3.2 DATA RECONSTRUCTION BY POLYNOMIAL EXTRAPOLATION

The problem confronting us is that we have a sequence of numbers $e(nT)$ like that shown in Fig. 2-8c and the original continuous signal $e(t)$ shown in Fig. 2-8a must be reconstructed from the information contained in this pulse sequence. Essentially, this data-reconstruction process may be regarded as an extrapolation process, since the continuous signal is to be constructed based on information available only at past sampling instants. For instance, the original function $e(t)$ between two consecutive sampling instants, nT and $(n+1)T$ is to be estimated based on the values of $e(t)$ at all preceding sampling instants of nT , $(n-1)T$, $(n-2)T$, $\dots$, 0 ; that is, $e(nT)$, $e(n-1)T$, $e(n-2)T$, $\dots$, $e(0)$. A well-known method in generating this desired approximation is based on the power series expansion of $e(t)$ in the interval between sampling instants nT and $(n+1)T$; that is

$$e(t) = e(nT) + e'(nT)(t - nT) +$$

where

$$e'(nT)(t - nT)^2 + \dots \quad (3-1)$$

$$e(t) = e(t) \text{ for } nT < t < (n+1)T \quad (3-2)$$

[In Eq. (3-1), the primes indicate the derivatives of $e(t)$ with respect to time.]

In order to evaluate the coefficients of the series given by Eq. (3-1), the derivatives of the function $e(t)$ must be obtained at the sampling instants. Since the information concerning $e(t)$ is available only at the sampling instants, these derivatives must be estimated from the impulse sequence. A simple expression involving only two data pulses gives the estimate of the first derivative of $e(t)$ at $t = nT$ as

$$e'(nT) = - \frac{e(nT) - e(n-1)T}{T}$$

An approximated value of the second derivative is given by

$$e''(nT) = - \frac{e'(nT) - e'(n-1)T}{T} \quad (3-3)$$

* J. R. Ragazzini, and L. H. Zadeh, "The Analysis of Sampled-Data Systems," Trans. A. I. E. E., 71, Part II, 1952, pp. 225-234.

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Substitution of Eq. (3-3) into Eq. (3-4) yields

$$e''(nT) = -\frac{1}{2T^2} \{e(nT) - 2e[(n-1)T] + e[(n-2)T]\} \quad (3-5)$$

From these approximated values of $e'(nT)$ and $e''(nT)$, we see that the higher the order of the derivative to be approximated, the larger will be the number of delay pulses required. In fact, we can easily

show that the number of delay pulses required to approximate the value of $e(nT)$ is $k + 1$. Thus the extrapolating device described above consists essentially of a series of time delays, and the number of delays depends on the degree of accuracy of the estimate of the time function $e(t)$. The adverse effect of time delay on the stability of feedback control systems is well known. Therefore, an attempt to utilize the higher order derivatives of $e(t)$ for the purpose of more accurate extrapolation usually meets with serious difficulties in maintaining system stability. Furthermore, a high-order extrapolator also requires complex circuitry and results in a high cost in construction. For these two reasons quite frequently only the first term of Eq. (3-1) is used in practice. A device generating only $e(nT)$ of Eq. (3-1) is normally known as the zero-order extrapolator, since the polynomial used is of the zeroth order. It is also commonly referred to as the zero-order hold device because it holds the value of the previous sample during a given sampling period. Similarly, a device that implements the first two terms of Eq. (3-1) is called a first-order hold, since the polynomial generated by this device is of the first order. Details of the zero-order hold and higher order hold devices are discussed in the following sections.

3.3 THE ZERO-ORDER HOLD

When only the first term of the power series given by Eq. (3-1) is used as shown in the last section, the extrapolator is called a zero-order hold. Then, Eq. (3-1) is simply

$$e_n(t) = e(nT) \quad (3-6)$$

The impulse response of a zero-order hold circuit is shown in Fig. 3-4. A simple diagram of a sampler with a capacitor hold is shown in Fig. 3-5. The capacitor is assumed to be charged up to the voltage $e(nT)$ at the sampling instant $t = nT$ instantaneously. (The actual rate of charging the capacitor is of course determined by the capacitance and the actual source impedance.) As the sampling switch is opened during the sampling interval T , the capacitor holds the charge until the next sampling interval. The input impedance of

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1

Unit impulse

(a)

T

(b)

Fig. 3-4 Impulse response of a zero-order hold. (a) Unit-impulse input to zero-order hold circuit. (b) Impulse response of zero-order hold.

T Sampler

To zero

impedance

source

Hold capacitor

Amplifier

Infinite

input

impedance

the amplifier shown in Fig. 3-5

is assumed to be infinite so that

there is no path for the capacitor

to discharge. The zero-order hold

circuit thus serves to stretch

the input pulses into a series of

rectangular waves of width T . In

practice, the amplifier does have a large finite input impedance; hence the actual waveform of the hold circuit output is not a train of rectangular waves but rather, a series of exponential decays with large time constant. The input and output waveforms of a zero-order hold circuit (idealized) are shown in Fig. 3-6a and Fig. 3-6b, respectively. Observe that the hold circuit output is a step approximation to the actual signal, and note also that increasing the sampling rate of the sampler tends to improve the approximation of the continuous signal.

Fig. 3-5 Simple diagram of a sampler and zero-order hold device.

(0)

$e^*(t)$

Continuous signal $e(t)$

Sampled signal $e^*(t)$

$T \quad 2T \quad 3T \quad 4T$

$5T$

67-

$h(t)$

(b)

Output signal of zero-order hold

Desired continuous signal

$2T \quad 3T \quad 4T \quad 5T \quad 6T$

Fig. 3-6 Zero-order hold operation in the time domain. (a) Input signal $e(t)$ and sampled signal $e^*(t)$. (b) Output waveform of zero-order hold circuit and continuous signal $e(t)$.

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In order to investigate the frequency response of the zero-order hold, the transfer function of the hold circuit is derived from Fig. 3-4b.

1

$G_{ho}(s)$

1

$-sT$

-

=

e

s

(3-7)

Setting $s = j\omega$ in the last equation, we have

$G_{ho}(j\omega)$

=

1

-

Equation (3-8) can be written as

$G_{ho}(j\omega)$

=

or

=

$-e$

$j\omega$

T

(3-8)

1

$2e^{-(e - j\omega T}$

$e^{-1/2} j\omega T)$

$2j\omega$

(3-9)

ω

T
 $e^{-j\omega T}$
 $(3-10)$
 $2 \sin(\omega T) e^{-j\omega T}$
 $G_{ho}(j\omega) = T \sin(\omega T)$
 Since T is the sampling period in seconds, and $T = 2\pi/\omega_s$, where ω_s is the sampling frequency in rad/sec, Eq. (3-10) becomes
 $G_{ho}(j\omega)$

$$= \frac{2\pi \sin \pi(\omega/\omega_s)}{\omega_s (\omega/\omega_s)} e^{-j\pi(\omega/\omega_s)} \quad (3-11)$$

The gain and phase characteristics of the zero-order hold are given by the following equations:

$$|G_{ho}(j\omega)|$$

$$\frac{2\pi}{\omega_s}$$

$$\sin \pi(\omega/\omega_s)$$

$$e^{-j\pi(\omega/\omega_s)}$$

$$-\pi$$

$$\omega$$

$$3\omega_c$$

$$4\omega_s$$

$$2\omega_c$$

$$\omega_c$$

$$\Delta\omega_c$$

$$2\omega_c$$

$$3\omega_c$$

$$\omega_c$$

Fig. 3-7 Gain and phase characteristics of a zero-order hold.

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$$:(\omega/\omega_s)$$

$$@s \quad \omega_s \quad \omega/\omega_s$$

$$\pi\omega$$

$$|$$

$$|G_{ho}(j\omega)|$$

$$=$$

$$2\pi \left| \sin \pi \omega / \omega_s \right|$$

and

$$\sin \pi(\omega/\omega_s)$$

$$G_{ho}(j\omega) = -\pi(\omega/\omega_s)$$

$$\pi$$

$$\left[\sin \pi(\omega/\omega_s) \right]$$

$$(3-12)$$

$$(3-13)$$

The frequency response (amplitude and phase) of the zero-order hold is shown in Fig. 3-7. This figure shows that the zero-order hold behaves essentially as a low-pass filter. However, when compared with the ideal filter characteristics of Fig. 3-3, the amplitude response of the zero-order hold is zero at ω_s , instead of cutting off sharply at $\omega_s/2$. In fact, at $\omega = \omega_s/2$, the magnitude of $G_{ho}(j\omega)$ is equal to 0.636.

We have seen from Fig. 3-6b that the effectiveness of the zero-order hold as an extrapolating device depends also on the sampling frequency ω_s ; this relationship can easily be shown from the frequency response curves. Figure 3-8a shows the amplitude spectrum of the sampler output and the frequency response of the zero-order hold when the sampling frequency is equal to the theoretical minimum ($\omega_s = 2\omega_c$). A rough sketch of the frequency response of the output

signal of the hold circuit is given in Fig. 3-8b. Observe from Fig.

$$\frac{|G_{no}(j\omega)|}{|E^*(j\omega)|}$$

$$\omega_c$$

$$2$$

$$\omega$$

$$2\omega_c$$

$$3\omega_c$$

$$4\omega_s$$

$$\omega_c$$

$$(a)$$

$$A$$

$$\omega$$

$$2\omega_c$$

$$3\omega_s$$

$$4\omega_s$$

$$\omega_c$$

$$(b)$$

$$-$$

2 ω_c . (b) Frequency

Fig. 3-8 (a) Frequency response of zero-order hold and amplitude spectrum of sampler output when ω

response of the output signal of a zero-order hold when ω

2 ω_c .

3-8b that the hold circuit output contains not only the primary component, but also a considerable number of high frequency components. Figure 3-9b indicates that the filtering property of the

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$$\omega_c$$

$$\frac{|G_{no}(j\omega)|}{|E^*(j\omega)|}$$

$$|E^*(j\omega)|$$

$$\omega_c$$

$$(a)$$

$$\omega_s$$

$$\omega_c$$

$$\omega$$

$$2\omega_c$$

$$2\omega_c$$

$$(b)$$

Fig. 3-9 (a) Frequency response of zero-order hold and amplitude spectrum of sampler output when $\omega_s > 2\omega_c$. (b) Frequency

response of the output signal of a zero-order hold when $\omega_s > 2\omega_c$.

zero-order hold is actually improved by increasing the sampling frequency.

3.4 THE FIRST-ORDER HOLD

If the first two terms of the power series given by Eq. (3-1) are used to extrapolate the time function $e(t)$ between two successive sampling instants nT and $(n+1)T$, the device is called a first-order hold. Equation (3-1) may be written as follows when its first two terms are taken.

where

$$e_n(t) = e(nT) + e'(nT)(t - nT)$$

$$e'(nT)$$

$$= \frac{e(nT) - e(n-1)T}{T}$$

$$=$$

$$T$$

$$(3-14)$$

$$(3-15)$$

Hence substituting Eq. (3-15) into Eq. (3-14) yields

$$e_n(t) = e(nT) + \frac{e(nT) - e((n-1)T)}{T} (t - nT) \quad (3-16)$$

Equation (3-16) shows that the output of the first-order hold between two consecutive sampling instants nT and $(n+1)T$ is a ramp function. The slope of the ramp function is equal to the difference in amplitudes of $e(t)$ at $t = nT$ and $t = (n-1)T$. The impulse response of the first-order hold is shown in Fig. 3-10. The impulse

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Unit impulse

2

(a)

-

T

127

(b)

1

Fig. 3-10 Impulse response of a first-order hold. (a) Unit-impulse input to first-order hold. (b) Impulse response. response of the first-order hold is derived from Eq. (3-16) by applying an impulse of unit strength at $t=0$ as input. When $n = 0$, Eq. (3-16) reads

$$e_0(t) = e(0) + \frac{e(0) - e(-T)}{T} t$$

--

T

(3-17)

For a unit-impulse input, $e(0) = 1$, and $e(T) = 0$. The impulse response for $0 < t < T$ is

$$G(t) = e_0(t) = 1 +$$

+ T

(3-18)

The impulse response for $T < t < 2T$ is obtained by setting n equal to unity in Eq. (3-16).

$$e_1(t) = e(T) + \frac{e(T) - e(0)}{T} (t - T)$$

-

-

(t - T)

Since $e(0) = 1$ and $e(T) = 0$

=

T

(3-19)

1 and $e(T) = 0$, the impulse response of the first-order hold for $T < t < 2T$ is

$$g_m(t) = e_1(t) = 1 - \frac{t - T}{T}$$

1-1

T

(3-20)

The impulse response for all other sampling intervals is zero, since $e(2T)$, $e(3T)$, . . . etc., are all equal to zero. The transfer function of the first-order hold is derived directly from the impulse response illustrated in Fig. 3-10. Therefore,

$$G_a(s) = 1 + \frac{T_s}{T} s (1 - e^{-sT})$$

(3-21)

Let us investigate how the time function of Fig. 3-6a is reproduced by the first-order extrapolation (Fig. 3-11). For the same sampling rate used in the case of Fig. 3-6, the approximated waveform shown in Fig. 3-11 does not indicate any advantage of the first-order hold over the zero-order device. As a matter of fact, the zero-order hold

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Output signal of first-order hold
Desirable continuous signal $e(t)$

Fig. 3-11 Reconstruction of a continuous signal by means of a first-order hold.

Output signal of first-order hold
Continuous signal to be reconstructed

$t = 0, T, 2T, 3T, 4T, 5T, 6T, 7T, 8T, 9T, \dots$

Fig. 3-12 Reconstruction of a continuous signal from sampled-data by means of a first-order hold. The sampling frequency is higher than that used in Fig. 3-11. actually gives better approximations in certain sampling intervals. However, Fig. 3-12 shows that, if the sampling rate is increased, the first-order hold does give a better approximation. One should not be concerned with the large error in the output of the first-order hold between sampling instants T and 0 ; this is purely because we assume that the value of $e(t)$ is zero for negative time and that it has a jump at $t=0$.

The frequency response of the first-order hold is obtained by substituting $s = j\omega$ in the transfer function in Eq. (3-21). We have

$$G_m(j\omega) = 1 + j\omega T (1 - \frac{1}{j\omega T})$$

$$(3-22)$$

From the last equation the gain and phase characteristics of the first-order hold are obtained as

$$|G_1(j\omega)| = \frac{\sin \frac{\pi \omega T}{2}}{\frac{\pi \omega T}{2}}$$

$$\angle G_1(j\omega) = -\frac{\pi \omega T}{2}$$

' $G_1(j\omega) = \tan$
 -1
 (270)
 $@s$
 W_s
 $(3-23)$
 $(3-24)$

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$G_{nt}(j\omega)$
 2π
 ω_s
 0
 $G_{ri}(j\omega)$
 ω_s
 $2\omega_s$
 $3\omega_s$
 Phase shift of zero-order hold
 Phase shift of first-order hold
 $\Delta\omega_s$
 3

Fig. 3-13 Gain and phase characteristics of a first-order hold. The frequency response curves for the first-order hold are sketched in Fig. 3-13. From the gain characteristic we see that the first-order hold cuts off at $\omega = \omega_s$, just as in the case of the zero-order hold but at a sharper rate. The fact that the amplitude curve peaks up between zero frequency and ω_s , shows that higher ripple may be contained in the output if the sampling frequency is low. In Fig. 3-13 the phase shift of the first-order hold is compared with that of the zero-order hold. Note that at low frequencies, the phase lag produced by the zero-order device exceeds that of the first-order hold but, as the frequency becomes higher, the opposite is true. However, in feedback control systems, the high frequency characteristics of the open-loop transfer function normally control the system stability; therefore, application of the first-order hold does not complicate the problem of closed-loop systems stability. Consequently, although the first-order hold in general does a better job in signal recovery than the zero-order hold, the fact that it is quite complex to construct and its adverse effect on system stability have made it seldom practical.

3.5 THE FRACTIONAL-ORDER HOLD

From the discussions given in the last two sections, we see that the amplitude characteristic of a zero-order hold drops off rapidly at low frequencies and the amplitude characteristic of a first-order hold exhibits an overshoot before cutting off sharply at $\omega = \omega_s$. An amplitude characteristic which falls between the zero-order and the

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first-order characteristics would come close to approximating the ideal filter response. Such a filter characteristic may be realized by use of a so called fractional-order hold and may be regarded as an extension or modification of the first-order hold. The output signal of a zero-order hold has a zero slope between two consecutive sampling instants, whereas that of a first-order hold has a constant slope determined by the difference in values of the two preceding samples. For a fractional-order hold the output signal between two consecutive sampling instants has a slope which is equal to k times

Unit impulse

$G_{nk}(1)$

First-order hold

2-

1

Zero-order hold

(a)

(b)

Fig. 3-14 Impulse response of a fractional-order hold. (a) Unit impulse input to fractional-order hold. (b) Impulse response. the difference in values of the two preceding samples, where k is a number between zero and unity. The impulse response of a fractional-order hold is shown in Fig. 3-14. The impulse response can be represented by a series of step and ramp functions as follows:

$$g_{hk}(t) = (1 + k)u(t) - (1 + k)u(t - T) + \frac{k}{T}(t - T)u(t - T) + \frac{k}{T}(t - 2T)u(t - 2T) + \dots$$

k

T

T

(3-25)

Taking the Laplace transform of the last equation term by term yields

$G_{hk}(s)$

1

k

-

+

$\frac{Ts^2}{1 + k}$

S

-ST

2k

-ST

e

Ts^2

k

+

$-28T$

S

$T's^2$

e

k

$-28T$

+

e

Equation (3-26) may be rearranged to read

k

$(1 - e^{-Ts})$

$G_{hk}(s) = \frac{(1 - e^{-Ts})}{Ts^2} + \frac{k}{T}(1 - e^{-Ts})$

-

(3-26)

=

$\frac{(1 - e^{-Ts})}{Ts^2} + \frac{k}{T}(1 - e^{-Ts})$

(3-27)

S

† J. R. Ragazzini, and G. F. Franklin, Sampled-Data Control Systems, (New York: McGraw Hill Book Company, Inc., 1958), pp. 40-42; also, E. I. Jury, (Sampled-Data

a

Control Systems, John Wiley & Sons, Inc. 1958), pp. 52-53.

$G_{nk}(j\omega)$

2π

$\omega\zeta$

$k=1$

$k=0.5$

$-k=0.3$

$k=0.2$

$k=0$

$G_{nk}(j\omega)$

ω

ω_s

$2\omega\zeta$

$k=0$

$k=0.2$

$k=0.3$

$k=1$

$k=0.5$

Fig. 3-15 Gain and phase characteristics of a fractional-order hold device.

Therefore, the transfer function of a fractional-order hold is

$$G_{nk}(s) = (1 - ke^{-Ts})G_{no}(s) + [G_{no}(s)]^2$$

or

$$(3-28)$$

k

$$G_{hk}(s) = (1 - ke^{-Ts})G_{no}(s) +$$

$G_{no}(s)$

$$(3-29)$$

$Ts + 1$

where $G_{ho}(s)$ and $G_h(s)$ are the transfer functions of the zero-order hold and the first-order hold, respectively. From the three preceding equations, we clearly see that when $k = 0$, $G(s) = G_{ho}(s)$; the device is identical to a zero-order hold. On the other hand, when $k = 1$, $G_{hk}(s)$ is identical to the transfer function of the first-order hold.

The frequency response in magnitude and phase of $G(j\omega)$ is plotted in Fig. 3-15 for five different values of k . It is observed from this plot that a flat response is obtained when k is approximately 0.3. In general, an optimum value of k is determined based on the frequency distribution of the input signal. A particular value of k should be selected that will result in extrapolations having the least error for this known range of frequencies.

3.6 THE EXPONENTIAL HOLD

Since the extrapolation of the sampled signal normally calls for the use of a low-pass filter, theoretically, at least, there is no reason why an ordinary low-pass passive network cannot be used. In fact,

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RECONSTRUCTION OF SAMPLED SIGNALS

if the first two terms of the power series expansion of e^s are taken, the transfer function of the zero-order hold reads.

$$G_{ra}(s) = 1 - 1(1-s) - (1-s)^2 = 1 + Ts$$

$-T's$

$G_{ho}(s)$

e

s

T

Ts

$$(3-30)$$

which is the transfer function of the R-C low-pass network shown

in Fig. 3-16. Since Eq. (3-30) represents a decaying exponential time

WW

R

Fig. 3-16 R-C passive network as hold device.

function in the time domain, this type of extrapolator is often described as the exponential hold.

A comparison of the frequency response of the zero-order and the exponential holds is made in Fig.

3-17. The reconstruction of the continuous signal $e(t)$ of Fig. 3-6a by means of the exponential hold is shown in Fig. 3-18.

$G_{no}(j\omega)$

2π

Gain characteristic of exponential hold

W_s

20.707

Gain characteristic of zero-order hold

$G_{no}(j\omega)$

$2\omega\zeta$

$3\omega\zeta$

$4\omega\zeta$

W_s

Phase characteristic of exponential hold

Phase of zero-order

hold

Fig. 3-17 Comparison of gain and phase characteristics of zero-order hold and exponential hold.

Desired continuous signal $e(t)$

Output signal of exponential hold

T

2T 3T 4T 5T

6T

Fig. 3-18 Reconstruction of $e(t)$ by an exponential hold.

If the first three terms of the power series expansion of e_r are used, we have

$G_o(s)$ T

=

$1 + (Ts/2)$

$1 + Ts + (Ts^2/2)$

(3-31)

This approximation of the zero-order hold may be realized by the R-L-C network shown in Fig. 3-19.

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Although better approximations of the zero-order hold by use of passive networks can be obtained if e_s is approximated by higher order polynomials, the physical implementations of the passive networks apparently grow more complex as the orders of the polynomials approximations are increased. Furthermore, the presence

WW

R₁

WW

R2
R3
C
L

Fig. 3-19 R-L-C passive network as hold device.

of complex poles and zeros for higher order approximations forces the use of inductive as well as capacitive and resistive elements, thus making the components rather bulky.

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4

THE Z-TRANSFORM THEORY

4.1 DEFINITION OF THE Z-TRANSFORM

In Chapter 2 the Laplace transform of a sampled time function $e^*(t)$ is shown to be

$$L[e^*(t)] = E^*(s) = \sum_{n=0}^{\infty} e(nT)e^{-s nT}$$

$$n = 0$$

(4-1)

where $e(nT)$ denotes the values of the continuous function $e(t)$ at the sampling instants. However, the expression of $E^*(s)$ normally contains the factor $e^{-s nT}$, which inevitably makes $E^*(s)$ a nonalgebraic equation. We know that when nonalgebraic equations are used difficulties in taking the inverse Laplace transform and partial fraction expansions will arise. Therefore, the introduction of a change in variable by setting

or

$$z = e^{sT}$$

$$s = \frac{1}{T} \ln z$$

$$= \ln x$$

T

(4-2)

(4-3)

is both desirable and convenient where s is the Laplace operator, T is the sampling period, and z is a complex variable defined by a complex z -plane. The variable z may be called the z -transform operator. When this substitution is made in Eq. (4-1), we have

$$E^*(s = -\frac{1}{T} \ln x) = E(x) = \sum_{n=0}^{\infty} e(nT) x^{-n}$$

$$n = 0$$

(4-4)

Therefore, the Laplace transform of the sampled function $e^*(t)$, $E^*(s)$,

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is now a function of z and is referred to as the z -transform of $e(t)$.

Hence

$E(z) = z$ -transform of $e(t) = Z[e(t)]$

where the symbol

(4-5)

is used to denote "the z -transform of." Also

$E(z) = \text{Laplace transform of } e^*(t) \text{ with } s = (1/T) \ln z$

$= E^*(s) = (1/T) \ln z$

$| s \ln$

(4-6)

Evidently, the z -transform of the time function $e(t)$ is still the same as the Laplace transform of $e^*(t)$, though disguised by the change in variable from s to z . Since the z -transform of $e(t)$ is derived from the Laplace transform of $e^*(t)$ with a change in variable, we can say in general that, in view of the relationship between $E^*(s)$ and $E(s)$ given by Eq. (2-68), any continuous function that possesses a Laplace transform also has a z -transform.

In a similar manner, the expression for $E^*(s)$ in Eq. (2-55), when z -transformed, reads

$E(x) =$

$n = 1$

$N(E_n)$

$D'(n) 1$

1

$enz-1$

(4-7)

indicating that if $E(z)$ has a finite number of simple poles, the z -transform of $e(t)$ is a rational function in z . Furthermore, the degree of the denominator of $E(z)$ in z^1 is equal to the degree of the denominator of $E(z)$ in s .

One of the disadvantages of the z -transform is that it carries information about the corresponding time function only at the sampling instants; that is, $E(z)$ defines $e^*(t)$ only. If the sampling frequency is high (at least twice as high as the highest signal frequency), the continuous signal $e(t)$, if it is band-limited, may be recovered from $e^*(t)$ by means of an ideal filter. Then the z -transform $E(z)$ carries virtually all the information contained in the Laplace transform $E(s)$. However, if nothing is known about the characteristics of $e(t)$, the inverse z -transformation is usually not unique, since the discrete time function $e^*(t)$, obtained from $E(z)$, may represent any continuous function $e(t)$ that has the same values as $e^*(t)$ at the sampling instants. In Fig. 4-1 two entirely different time functions with the

$Le_1(t)$

$ex(t)$

1

$2T \ 4T \ 6T$

$T \ 3T \ 5T$

$7T$

$ez(1)$

$e(1)$

$2T$

$4T \ 6T$

$T \ 3T \ 5T \ 7T$

Fig. 4-1 Two different time functions that have the same transforms.

$Z-$

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THE Z -TRANSFORM THEORY

same z -transform are illustrated. The only way to distinguish between these two functions, from the z -transform point of view, is to define the waveforms of the signals between sampling instants. In the past,

several methods have been developed for the evaluation of signals between sampling instants. Among these methods the submultiple sampling method and the modified z-transform methods are the most common. These methods are discussed in detail in later sections of this chapter. Regardless of the fact that the inverse z-transform is not unique and of the other limitations of the z-transform method to be brought out later, the modern design and analysis of sampled-data systems still rely heavily on the z-transform technique because of its simplicity and unique manner of application.

4.2 EVALUATION OF Z-TRANSFORMS

In general, given a time function $e(t)$, the z-transform of $e(t)$, $E(z)$, may be obtained by using the following steps:

(1) $E^*(s)$, the Laplace transform of $e^*(t)$ is obtained by use of either Eq. (2-46) or Eq. (2-55).

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(2) $E(z)$ is obtained by substituting $z = es$ in $E^*(s)$.

Example 4-1

Evaluate the z-transform of a unit impulse function $\delta(t)$. In this case assume that when an impulse is sent through an ideal sampler, the output of the sampler is still an impulse of identical strength. Thus

The Laplace transform of $e^*(t)$ is thus

Hence

Example 4-2

$e^*(t) = e(t) = \delta(t)$

$E^*(s) = 1$

$E(z) = E^*(s) | s = (1/T) \text{ in } 2$

$= 1$

(4-8)

(4-9)

(4-10)

Evaluate the z-transform of a unit-impulse train of period T .

00

-

$e(t) = \delta_r(t) = \sum \delta(t - nT)$

$n = 0$

(4-11)

E. I. Jury, Sampled-Data Control Systems (New York: John Wiley & Sons, Inc. 1958), pp. 64-72.

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Again, based on the assumption made in Example 4-1, $e^*(t)$ is identical to $e(t)$. Therefore,

$E^*(s) = \sum$

e^{-nts}

(4-12)

$n = 0$

and the z-transform of the unit-impulse train is

00

$E(z) = \sum z^{-n}$

$n = 0$

(4-13)

$n = 0$

$E(z)$ can be written in closed form as

z

-

$$E(z) = \frac{1}{1 - z^{-1}}$$

(4-14)

Example 4-3

Evaluate the z-transform of a unit-step function $u(t)$. When the step function is sampled by an ideal sampler, the output is a unit-impulse train described by Eq. (4-11). Thus the z-transform of a unit-step function is also given by Eq. (4-14). This is because the inverse z-transformation is not unique. The inverse z-transform of $E(z)$ in Eq. (4-14) can be interpreted as a unit-step function, a unit-impulse train of period T , or any continuous time function that has a value of unity at $t = nT$ ($n = 0, 1, 2, \dots$). Another difficulty in the z-transformation is that the ideal sampler cannot distinguish among a train

$e(t)$

$e(nT)$

$e(t)$

$2T \quad 4T \quad 6T$

$T \quad 3T \quad 5T$

Unit-impulse train

e

T

T

Ideal sampler

$e^*(t)$

$2T \quad 4T \quad 6T$

$3T \quad 5T$

Unit-impulse train

e

t

T

Ideal sampler

$2T \quad 4T \quad 6T$

$3T \quad 5T$

Unit-impulse train

Unit-step function

Fig. 4-2 The sampling of a unit-impulse train and a unit-step function by an ideal sampler. The output signals of the samplers are identical in both cases.

of impulses, each impulse with unit area, and a unit-step function. The sampling of the impulse train and the unit-step function by an ideal sampler is shown in Fig. 4-2. However, this difficulty does not occur in practice, since a true impulse is not found in physical systems.

Example 4-4

In this example the z-transform of the exponential function $e(t) = e^{-at}$ is

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determined by use of Eq. (2-46). The values of $e(t)$ at the sampling instants are denoted by $e(nT) = e^{-anT}$. Hence the Laplace transform of the sampled signal $e^*(t)$ is

$E^*(s)$

=

$\sum_{n=0}^{\infty}$

$n = 0$

e^{-anT}

$-nTs$

$1 + e^{-aT} + e^{-2aT} + \dots$

(4-15)

which can be written in a closed form as

$$E^*(s) =$$

$$\frac{1}{1 - e^{-(s+a)T}} \left| e^{-T} \right| < |e^{-ar}|$$

Substituting $z = e^s$ into $E^*(s)$, the z-transform of $e(t)$ is given by

$$(4-16) \quad E(z) =$$

$$\frac{1}{2}$$

$$=$$

$$-$$

$$\frac{1}{2} e^{-at-1}$$

$$\frac{2}{12} e^{-ar}$$

$$\frac{12}{2} e^{-ar}$$

Example 4-5

(4-17)

In this example the z-transform of a sine function $e(t) = \sin \omega t$ is determined by use of Eq. (2-55). In Eq. (2-55), and

W

$$E(s) = L[e(t)]$$

$$=$$

$$s^2 + \omega^2$$

$$N(s) = \omega$$

$$D'(s) = 2s$$

(4-18)

(4-19)

(4-20)

where $N(s)$ represents the numerator and $D(s)$ represents the denominator of $E(s)$. Since $E(s)$ has two finite poles, one at $s =$ the Laplace transform of the sampled function $e^*(t)$ is

$$\frac{2}{1}$$

$$E^*(s) = \sum_{n=0}^{\infty} (1 - e^{-ra-t})$$

$$=$$

$$\frac{77}{7} = 11$$

$$@$$

$$D'(s) = 2s$$

$$[$$

$$\frac{2}{j\omega}$$

$$\omega$$

$$=$$

$$\frac{2}{j\omega} \cdot 1$$

$$\frac{1}{-}$$

$$- e^{-T(8-j\omega)}$$

$$-$$

$$\frac{1}{-}$$

$$-e$$

$$-j\omega \text{ and one at } s = -j\omega,$$

$$\frac{1}{-}$$

$$-T(s + j\omega)$$

$$T$$

$$e^{-Ts} (-e^{-tw} + e^{-tor})$$

$$e$$

$$\frac{1}{2} \omega T$$

$$-2Ts$$

(4-21)

$$-e^{-Ts} (e + e^{-j\omega T}) + e^{-2Ts}$$

Equation (4-21) can be simplified to read

$$e^{-Ts} \sin \omega T$$

$$E^*(s) =$$

$$-2T's$$

$$\frac{e^{-T} \sin @ T}{(4-22)}$$

Hence the z-transform of $e(t)$ is

$$E(z) = \frac{z \sin @ T}{z-1} \frac{\sin @ T}{2z^1 \cos @ T + 1} \quad (4-23)$$

In the preceding examples the z-transforms of the time functions $e(t)$ are evaluated with the time functions as the starting point, and then the defining steps of the z-transformation are followed through. Actually, the intermediate step of finding the Laplace transform $E^*(s)$ can be eliminated by writing the z-transform directly from either Eq. (4-4) or Eq. (4-7). In this manner the z-transforms for

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other Laplace transformable functions can be determined. Thus, just as in working with the Laplace transforms of time functions, a table can be assembled for the z-transforms associated with commonly used time functions. A short table of z-transforms is given in Table 4-1, and a more extensive one is presented in Appendix A.

In using the z-transform table, remember that the inverse z-transform is not unique. One may use the table only to find the z-transform of a given time function, but given a z-transform $E(z)$, there is no unique solution for $e(t)$.

4.3 MAPPING OF THE S-PLANE INTO THE Z-PLANE

The study of the mapping of the s-plane into the z-plane by the z-transformation $z = e^{sT}$ is necessarily important. The design and analysis of continuous-data control systems often rely on the pole-zero configurations of systems transfer functions in the s-plane. Similarly, the poles and zeros of the z-transform of a system transfer function may conceivably determine the sampled response of the system. Therefore, in this section some of the important contours frequently used in the s-plane, such as the constant damping ratio

jw
s-plane
 ∞
jws
The primary
strip
 $\frac{\sigma}{\omega}$ jws
(a)

Caris
4 Im z
z-plane
z = e
s-jw
1
1
Re z
(b)

Fig. 4-3 Mapping of the primary strip in the left half of the s-plane into the z-plane by the z-transformation.

T-

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line, the constant frequency line, or the imaginary axis will be mapped into the z-plane by the transformation $z = e^s$.

The s-plane is first divided into an infinite number of periodic strips as shown in Fig. 2-18. The primary strip extends from $w =$

TABLE 4-1

TABLE OF Z TRANSFORMS

Time function

Laplace transform

Unit-impulse (t)

1

Unit step u(t)

$dr(t) = \sum_{n=0}^{\infty} \delta(t - nT)$

$n = 0$

1

14

S

1

2

14

n

1

57+1

Ts

e-at

te-at

1 - e-al

sin wt

1

s+ a

1

(s + a) 2

a

s(s + a)

w

+ w

e-at sin wt

(sa)

w

+ wi

z-transform

1

2 2 1

Z

Z-

Tz

(z - 1)2

T2z (z+1)

2(x-1)3

n

lim (-1)" " (~~~

n

an

2

e-ar

2

(2

2

2

-e-ar

Tze-ar

$$\frac{(z - e^{-ar})^2}{z(1 - e^{-ar})}$$

--

$$\frac{ze^{1)(z-e-ar}}{z \sin w T}$$

$$T)$$

$$2z \cos w T + 1$$

$$ar \sin w T$$

$$2z - 2ze^{-ar} \cos w T e$$

--

$$\cos @ T)$$

$$2x \cos w T + 1$$

$$2a T$$

$$\cos wt$$

$$e^{-at} \cos wt$$

--

$$@s$$

$$S$$

$$2(2 -$$

$$+ wa$$

$$2z$$

$$s + a$$

$$2z$$

$$(s + a)^2 + w^2$$

$$2z$$

.

$$ze^{-ar} \cos w$$

$$wT$$

$$2ze^{-ar} \cos w T + e^{-$$

$$2a T$$

$-ws/2$ to $+ws/2$; and the other complementary strips extend from $-ws/2$ to $3ws/2$, $-3ws/2$ to $-5ws/2$, . . . , for negative frequencies, and from $ws/2$ to $3ws/2$, $3ws/2$ to $5ws/2$, . . . , for positive frequencies. If we consider only the primary strip shown in Fig. 4-3a, the path described by (1)-(2)-(3)-(4)-(5)-(1) in the left half of the s-plane is mapped into a unit circle centered at the origin in the z-plane (Fig. 4-3b) by the transformation $z = e^s$. Similarly, since

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$$e^{(s+jn w,) T}$$

$$= e^{T's} e^{2x jn} = e$$

2x

$$T's$$

= 2

(4-24)

all the other left half s-plane complementary strips are also mapped into the same unit circle in the z-plane. In other words, all points in the left half of the s-plane are mapped into the region inside the unit circle in the z-plane. The points in the right half of the s-plane are thus mapped into the region outside the unit circle in the z-plane.. The loci of constant damping factor, constant frequency, and constant damping ratio in the z-plane for sampled-data systems are given in the following:

1. The Constant Damping Loci

For a constant damping factor σ , in the s-plane, the corresponding z-plane locus is a circle of radius $z = e$ centered at the origin in the z-plane as shown in Fig. 4-4.

jw

s-plane
 Im z
 Unit
 circle
 z-plane
 -02
 0
 0₁
 (a)
 Constant
 damping loci
 eat
 (b)
 Re z
 5+
 e
 Z
 ST
 2
 1
 1

Fig. 4-4 Constant damping loci in the s-plane and the z-plane.

2. The Constant Frequency Loci

For any constant frequency ω_1 in the s-plane, the corresponding z-plane locus is a straight line emanating from the origin at an angle of T rad; the angle is measured from the positive real axis (Fig. 4-5).

3. The Constant Damping Ratio Loci

For a constant damping ratio ζ , the constant ζ line in the s-plane is shown in Fig. 4-6a; the line is described by the equation,

$$s = -\zeta \omega_n + j\omega_n \tan \beta \quad (4-25)$$

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jw
 s-plane
 Im z
 z-plane
 "

jws
 jwz
 jw₁
 Re z
 0
 -jw₁
 -wy_T
 Hence

(a)
 (b)
 Fig. 4-5 Constant frequency loci in the s-plane and the z-plane.

$z = e^{sT}$
 $(-\zeta \omega_n + j\omega_n \tan \beta)T$
 Equation (4-26) can be written

where
 $2 = e^{-2\zeta \omega_n T} e^{j2\omega_n T \tan \beta}$
 $\zeta = \frac{\omega_n}{\omega_s} \tan \beta$
 $\beta = \sin^{-1} \zeta \frac{\omega_s}{\omega_n}$

= constant

(4-26)

(4-27)

(4-28)

For a given the constant path described by Eq. (4-27) is a logarithmic spiral in the z -plane except for $\theta = 0^\circ$ and 90° . The region shown shaded in Fig. 4-6a corresponds to the interior of the shaded region of Fig. 4-6b. In Fig. 4-7, the constant paths for $B = 30$ degrees are shown in both the s -plane and the z -plane. Each one-half revolution of the logarithmic spiral corresponds to the passage

- ∞

s -plane

$1/j\omega$

- $\frac{1}{2}j\omega s$

Im z

z -plane

(a)

(b)

Fig. 4-6 Constant damping ratio loci in the s -plane and the z -plane.

Re z

1

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$j\omega$

$B=30^\circ$

s -plane

- $3j\omega s$

+ $2j\omega s$

Fiws

Im z

z -plane

$\approx \approx, u$.

114

ws

fws

Re z

Ws

w=0/1

314

3

(a)

(b)

Fig. 4-7 Constant damping ratio loci for B cent) in the s -plane and the z -plane.

=

30° (50 per-

of the $-$ path in the s -plane through a change of w , along the imaginary axis.

4.4 THE INVERSE Z-TRANSFORMATION

Just as in the method of Laplace transformation, time domain

response must often be obtained from the z -transform expressions.

Since the inverse z -transformation is not unique, we must remember

that the inversion from z to time domain referred to here is the

determination of $e(t)$ or the sampled values of $e(t)$ at the sampling instants enT , but not $e(t)$, from the z -transform $E(z)$.

In general, the inverse z -transformation can be effected by one of the following methods:

(1) The Partial Fraction Method. The z -transform $E(z)$ is manip-

ulated into a sum of functions by partial fraction expansion, and the z-transform table is used to find the corresponding time functions of the expanded terms. Note that, although the z-transform tables in this book and elsewhere have the time functions tabulated in the form of continuous functions $e(t)$, the inverse transform of $E(z)$ is not necessarily equal to $e(t)$ (and most likely it is not). The inverse z-transform of $E(z)$ is simply a time function that has the same values as the corresponding $e(t)$ at the sampling instants of $t = nT$ ($n = 0, 1, 2, \dots$).

(2) The Power Series Method. The z-transform $E(z)$ is expanded into a power series in powers of z^{-1} . The coefficient of the term z^{-n} corresponds to the value of the time function $e(t)$ at the n -th sampling instant.

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(3) The Inversion Formula Method. The time function $e^*(t)$ may be obtained from $E(z)$ by the real inversion integral. The values of $e(t)$ at the sampling instants are determined by use of the following formula:

$$e(nT) = \frac{1}{2\pi j} \oint_{\Gamma} E(z) z^{n-1} dz \quad (4-29)$$

where Γ is a circle of radius $z = \rho_c$ centered at the origin of the z -plane, and ρ_c is of such a value that all the poles of $E(z)$ are enclosed by the circle.

The proofs and the illustrations of the above three methods are given in the following:

1. The Partial Fraction Method

Recall that in the analysis of continuous signals, the partial fraction expansion of a given Laplace transform $E(s)$ may be written as

$$E(s) = \frac{A}{s+a} + \frac{B}{s+b} + \frac{C}{s+c} + \dots \quad (4-30)$$

and the inverse Laplace transform of $E(s)$ yields the time function $e(t)$

$$e(t) = Ae^{-at} + Be^{-bt} + Ce^{-ct} + \dots \quad (4-31)$$

In the case of the z-transformation, it would seem at a first glance that $E(z)$ should also be expanded into a form similar to that of Eq. (4-30); that is,

$$E(z) = \frac{A}{z+a} + \frac{B}{z+b} + \frac{C}{z+c} + \dots \quad (4-32)$$

However, inverse z-transform for each term of the right-hand side of Eq. (4-32) is not found in the z-transform table. Furthermore, the z-transform of the exponential term Ae^{-at} is $Az/(z - e^{-aT})$. Therefore, if each term of the partial fraction expansion has the form of $z/(z - e^{-aT})$, $e^*(t)$ can be expressed as the sum of exponential functions. For this reason, it is more desirable to expand the function $E(z)/z$ and then to multiply each of the expanded terms by z .

Example 4-6

Given the z-transform

$$E(z) =$$

=

-

$$(1 - e^{-aT})$$

$$(\approx - 1)(\approx - e^{-aT})$$

(4-33)

where a is a constant and T is the sampling time in seconds, evaluate the inverse z-transform $e^*(t)$. The partial fraction expansion of $E(z)/z$ is

$$E(z)$$

22

=

1

~

$$1 \times -$$

1

aT

(4-34)

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Hence

Z

$$E(z) =$$

-

2

-

2

aT

(4-35)

From the z-transform table (Table 4-1) the inverse z-transform of $E(z)$ is found to be a time function whose values at the sampling instants are described by

$$e(nT) = 1 - e^{-anT}$$

-

The sampled function is given by

(4-36)

00

$$e^*(t) = \sum_{n=0}^{\infty} (1 - e^{-anT}) \delta(t - nT)$$

$n=0$

(4-37)

Nothing about the behavior of $e(t)$ in between the sampling instants can be concluded from the z-transform $E(z)$, or the sampled time function $e^*(t)$.

2. The Power Series Method

In this method $E(z)$ is expanded into a power series in powers of z^{-1} by means of long division. $E(z)$ is first written into the form of

$E(z)$

=

.

$m-1$

$$a_m z^m + a_{m-1} z^{m-1} +$$

$$+ a_1 z + a_0$$

$$b_n z^n + b_{n-1} z^{n-1} + \dots + b_1 z + b_0$$

(4-38)

Dividing the numerator by the denominator by long division yields

$E(x)$

=

$Az^{m-n}Bz^{m-n-1} +$

(4-39)

For a physically realizable system, $n > m$. Comparing Eq. (4-39)

with Eq. (4-4), we clearly see that the coefficients $A, B, \dots$ in Eq.

(4-39) correspond to the respective values of $e(nT)$ in Eq. (4-4).

Thus

Example 4-7

$A = e^{-(m-n)T}, B = e^{-(m-n-1)T}, \dots$

Determine the inverse z-transform of Eq. (4-33) by means of the power series expansion method. Equation (4-33) is written

$E(z)$

-

22

-

Dividing the denominator into the

$E(z)$:

=

-1

(1

-

$e^{-ar}) z$

$(1 + e^{-ar})z + e^{-ar}$

2

$(1 - e^{-ar})z^{-1} + (1 - e^{-2ar})z^{-2} + (1$

The inverse z-transform of $E(z)$

or

is

(4-40)

numerator yields

$2aT$

-

$e^{-3ar}) z +$

3

(4-41)

(4-42)

(4-43)

$e^{*}(t) = (1 - e^{-ar})\delta(t - T) + (1 - e^{-2ar})\delta(t - 2T) + \dots$

-

-

-

$e^{*}(t) = \sum_{n=0}^{\infty} (1 - e^{-anr})\delta(t - nT)$

$n=0$

-

This result is identical to that in Eq. (4-37) obtained by the partial fraction expansion method.

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3. The Inversion Formula Method

It is of interest to compare the defining equations of the Laplace transform and the z-transform. Given $e(t)$ as a continuous function of time that is Laplace transformable, the Laplace transform and the z-transform of $e(t)$ are respectively,

and

$E(s) = L[e(t)] = \int_0^{\infty} e(t) e^{-st} dt$

$E(x)$

$E(x) = Z[e(t)] = \sum_{n=0}^{\infty} e(nT)z^{-n}$

$n=0$

(4-44)

(4-45)

The similarity between the two equations is striking. Actually, if the sampling frequency approaches infinity, $E(z)$ reverts to $E(s)$. In a similar manner, the inverse transforms in the s and the z domains may be written as

and

1

$e(t)$

=

$2\pi j$

1

$C-j\infty$

Sotto $E(s) e' ds$

$e(nT)$

=

$2\pi j$

$\oint_C E(z) z^{n-1} dz$

1

(4-46)

(4-47)

respectively; in Eq. (4-46), the constant c denotes the abscissa of convergence of the Laplace transform, and c is so chosen that it is greater than the real parts of all the singularities of the integrand $E(s)$; in Eq. (4-47), I is a closed path (a circle) in the z -plane which encloses all the singularities of $E(z)z^{n-1}$. The proof of the inversion formula of the z -transforms is given as follows:

Since $e(nT)$ represents the values of $e(t)$ at the sampling instants, we substitute $t = nT$ in Eq. (4-46); therefore,

$e(nT)$

=

1

$\int_C E(s) e^{sn} ds$

$2\pi j \int_{c-j\infty}^{c+j\infty}$

(4-48)

where the path of integration is as shown in Fig. 4-3a. This path is seen to pass through the periodic strips of the s -plane vertically, and thus the integration of Eq. (4-48) may be broken up into a sum of integrals; each integral is performed within one of the periodic strips. Equation (4-48) can now be written

$\sum_{n=-\infty}^{\infty}$

$e(nT)$

1

$\int_{c-j\omega/2}^{c+j\omega/2} E(s) e^{sn} ds$

=

$\sum_{n=-\infty}^{\infty} E(s) e^{sn} ds$

$2\pi j \int_{c-j\omega/2}^{c+j\omega/2} E(s + jn\omega) e^{(s+jn\omega)t} ds$

$N=-\infty$

(4-49)

where $\omega_1 = 2\pi/T$. Replacing s by $s + jn\omega_1$, where n is an integer, Eq. (4-49) becomes

0

1

$(c + j\omega_1/2)$

$e(nT) =$

$2\pi j$

$\sum_{n=-\infty}^{\infty} \int_{c-j\omega_1/2}^{c+j\omega_1/2} E(s + jn\omega_1) e^{(s+jn\omega_1)t} ds$

$c - (j\omega_1/2)$

(4-50)

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or

00

$$\sum_{n=-\infty}^{\infty} e^{(nT)s} + (2) \int_{-\infty}^{\infty} E(s + jnw_1) e^{TMs} ds$$

=

2πj

n=100

c-(jw, /2)

Interchanging the summation and integration signs, we have

1 (c+(jw, /2)

e(nT)

=

$$\sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} E(s + jnw_1) e^{TMs} ds$$

2πj Jc-(jw/2)

N=1∞

Since $E^*(s) = \sum_{n=-\infty}^{\infty} E(s + jnw_1)$, Eq. (4-52) reads

T

n=-∞

(4-51)

(4-52)

e(nT)

T (c+(jw, /2)

=

$$\int_{-\infty}^{\infty} \sqrt{e^{-tw}} E^*(s) e^{rs} ds$$

c-(jw, /2)

(4-53)

When $z = e^s$ is substituted into the last equation, for the interval

$-\infty < w < +\infty$, $1/w < 1/z < w$,

$$E^*(s) = (1/T) \int_{-\infty}^{\infty} E(z) dz$$

(4-54)

ents = 2"

(4-55)

$$ds = e^s dz = z dz$$

12-1

(4-56)

Equation (4-53) now becomes

T

e(nT)

=

2πj

$$\oint_{\gamma} E(z) z^{-1} dz$$

(4-57)

or

e(nT)

1

=

$$\int_{-\infty}^{\infty} E(z) z^{-1} dz$$

n-1

(4-58)

which is the inversion formula given previously in Eq. (4-47). By use of the z-transformation the section from w , to $+w$, on the

-

s-plane

▲ jw Ac+j∞o

X

Poles of

E (s)

X

X

X

--
 j_i ωs
 Im z
 z-plane
 j ωs
 Poles of
 E(z)
 ect
 · j^{1/2}ωs
 σ
 -it ωs
 --jWs
 -j ωs
 (b)
 Ic-joo
 (a)
 Fig. 4-8 Paths of integration of the inversion formula in the s-plane and the z-plane.
 Re z

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path of integration in the s-plane is mapped into the circle of radius e^{-ar} centered at the origin in the z-plane, as shown in Fig. 4-8b.

Since $E^*(s)$ does not have singularities on or to the right of the line σ_c in the s-plane, all singularities of $E(z)$ must lie inside the circle $z = e^{-ar}$ in the z-plane.

Example 4-8

Determine the inverse z-transform of Eq. (4-33) by use of the inversion formula. From Eq. (4-58),

$$e(nT) = \frac{1}{2\pi j} \oint_C \left(\frac{1 - e^{-ar}}{z - e^{-ar}} \right)^{n+1} dz$$

n-1

dz

(4-59)

where I is a circle which encloses the poles at $z = +1$ and $z = e^{-ar}$ of $E(z)$. The contour integral of Eq. (4-59) can be evaluated using Cauchy's residue theorem. Thus

$$e(nT) = \text{Residues of } E(z)z^{n-1} \text{ at the poles of } E(z) \quad (4-60)$$

Substituting the expression of $E(z)$ into Eq. (4-60) yields.

$$e(nT) = \text{Residues of } \left(\frac{1 - e^{-ar}}{z - e^{-ar}} \right)^{n+1} \text{ at } z = 1, z = e^{-ar} \quad (4-61)$$

from which

$$e(nT) = \frac{1 - e^{-an}}{1 - e^{-ar}}$$

(4-62)

It is apparent that the same $e^*(t)$ can be obtained from $e(nT)$ as by the preceding two methods.

4.5 THEOREMS OF THE Z-TRANSFORM

Several basic theorems simplify and facilitate the applications of

the z-transformation. The derivations and illustrations of the z-transform theorems are given below.

1. Addition and Subtraction

If $e_1(t)$ and $e_2(t)$ are Laplace transformable and

$$E_1(z) = Z[e_1(t)],$$

then

Proof: By definition

Hence

$$E_2(z) = Z[e_2(t)],$$

(4-63)

$$Z[e_1(t) \pm e_2(t)] = E_1(z) \pm E_2(z)$$

$$Z[e_1(t) \pm e_2(t)] = \sum_{n=0}^{\infty} [e_1(nT) \pm e_2(nT)]z^{-n}$$

$n = 0$

$$Z[e_1(t) \pm e_2(t)] = \sum_{n=0}^{\infty} e_1(nT)z^{-n} \pm \sum_{n=0}^{\infty} e_2(nT)z^{-n}$$

(4-64)

$n=0$

$$= E_1(z) \pm E_2(z)$$

$n=0$

(4-65)

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2. Multiplication by a Constant

If $E(z)$ is the z-transform of $e(t)$, then

$$Z[ae(t)] = a Z[e(t)] = aE(z)$$

(4-66)

where a is a constant.

Proof: By definition

$n=0$

$$Z[ae(t)] = \sum_{n=0}^{\infty} ae(nT)z^{-n} = a \sum_{n=0}^{\infty} e(nT)z^{-n} = aE(z) \quad (4-67)$$

$n=0$

$n=0$

3. Real Translation

If the time function $e(t)$ is Laplace transformable and has the

z-transform $E(z)$, then

$$Z[e(t - nT)] = z^{-n}E(z)$$

(4-68)

$n-1$

and

$$Z[e(t + nT)] = z^n E(z) - \sum_{k=0}^{n-1} e(kT) z^{-k}$$

(4-69)

$k=0$

where n is a positive integer.

Proof: By definition.

$n=0$

$$Z[e(t - nT)] = \sum_{k=0}^{\infty} e(kT - nT)z^{-k}$$

which can be written as

n

$-n$

$k=0$

$n=0$

$$Z[e(t - nT)] = z^{-n} \sum_{k=0}^{\infty} e(kT) z^{-k}$$

$$= \sum_{k=0}^{\infty} e(kT - nT) z^{-(k-n)}$$

$k=0$

Since $e(t)$ is zero for $t < 0$, Eq. (4-71) becomes

$$= z^{-n} \sum_{k=0}^{\infty} e^{(kT - nT)} z^{-(k-n)}$$

$$\sum_{k=n}^{\infty} [e^{(k-n)T}] = z^{-n}$$

To prove Eq. (4-69), we write

$$\sum_{k=0}^{\infty} [e^{(k+n)T}] = \sum_{k=0}^{\infty} e^{(kT + nT)} z^{-(k+n)}$$

$$(4-70)$$

$$(4-71)$$

$$= z^{-n} E(z) \quad (4-72)$$

$$= z^{-n} \sum_{k=0}^{\infty} e^{(kT + nT)} z^{-(k+n)}$$

$$(4-73)$$

From Eq. (4-73) the reader should be able to derive Eq. (4-69) without difficulty.

Example 4-9

Using the real translation theorem, evaluate the z-transform of a unit-step function delayed by one sampling period T. From Eq. (4-68)

$$\sum_{k=0}^{\infty} [u(kT)] = z^{-1} \sum_{k=0}^{\infty} [u(kT)] = z^{-1} (z/(z-1))$$

$$\sum_{k=0}^{\infty} [u(kT)] = 1/(z-1)$$

$$(4-74)$$

$$(4-75)$$

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Example 4-10

Using the real translation theorem, evaluate the z-transform of the exponential function e^{-at} delayed by one sampling period. By using Eq. (4-68),

$$\sum_{k=0}^{\infty} [e^{-a(kT+T)}] = z^{-1} \sum_{k=0}^{\infty} [e^{-akT}]$$

The z-transform of e^{-at} is found from the z-transform table; hence

$$(4-76)$$

$$\sum_{k=0}^{\infty} [e^{-a(kT+T)}] = z^{-1} \sum_{k=0}^{\infty} [e^{-akT}]$$

$$=$$

$$=$$

$$=$$

$$= e^{-aT}$$

$$= e^{-aT}$$

$$= e^{-aT}$$

4. Complex Translation

$$(4-77)$$

The complex translation theorem states that, if $e(t)$ has the z-transform $E(z)$, then

$$\sum_{k=0}^{\infty} [e^{(kT \pm aT)}] = [E(z \pm aT)] = E(ze^{\pm aT})$$

Proof: By definition

$$\sum_{k=0}^{\infty}$$

$$\sum_{k=0}^{\infty} [e^{(kT \pm aT)}] = \sum_{k=0}^{\infty} e^{(kT \pm aT)} z^{-(k \pm 1)}$$

$$\sum_{k=0}^{\infty}$$

Let $z_1 = ze^{aT}$; then Eq. (4-79) becomes

$$\sum_{k=0}^{\infty}$$

$$\sum_{k=0}^{\infty} [e^{(kT \pm aT)}] = \sum_{k=0}^{\infty} e^{(kT \pm aT)} z_1^{-(k \pm 1)} = E(z_1)$$

Therefore,

Example 4-11

$$\sum_{k=0}^{\infty}$$

$$\sum_{k=0}^{\infty} [e^{(kT \pm aT)}] = E(ze^{\pm aT})$$

$$(4-78)$$

$$(4-79)$$

$$(4-80)$$

(4-81)
 Using the complex translation theorem, evaluate the z-transform of the function $e^{-at} \sin \omega t$. From the z-transform table (Table 4-1), the z-transform of $e^{-at} \sin \omega t$ is found to be

$$\frac{\omega z^{-1}}{z^2 - 2z \cos \omega T + 1}$$

(4-82)
 (4-83)
 Using Eq. (4-78), Eq. (4-82) may apparently be obtained by substituting ze^{-aT} for z in Eq. (4-83).

5. Initial Value Theorem
 If the function $e(t)$ has the z-transform $E(z)$ and if the limit exists, then

$$\lim_{t \rightarrow 0} e^*(t) = \lim_{z \rightarrow \infty} z E(z)$$

(4-84)
 (4-85)
 $\lim_{z \rightarrow \infty} z E(z)$

The theorem simply states that the behavior of the sampled function

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 $e^*(t)$ in the neighborhood of $t = 0$ is determined by the behavior of $E(z)$ at $z \rightarrow \infty$.

Proof: By definition of the z-transform $e(t)$,

$$E(z) = \sum_{n=0}^{\infty} e(nT) z^{-n} = e(0) + e(T)z^{-1} + e(2T)z^{-2} + \dots \quad (4-86)$$

Hence

$$\lim_{z \rightarrow \infty} z E(z) = \lim_{z \rightarrow \infty} z [e(0) + e(T)z^{-1} + e(2T)z^{-2} + \dots]$$

$$= \lim_{z \rightarrow \infty} [ze(0) + e(T) + e(2T)z^{-1} + \dots]$$

(4-87)
 6. Final Value Theorem

If the function $e(t)$ has the z-transform $E(z)$ and if the function $(1 - z^{-1})E(z)$ does not have poles on or outside the unit circle in the z-plane, then

$$\lim_{t \rightarrow \infty} e^*(t) = \lim_{z \rightarrow 1} (1 - z^{-1})E(z)$$

Proof: Let us consider two finite sequences,

$$Z_k = e(kT)z^{-k} = e(0) + e(T)z^{-1} + \dots + e(nT)z^{-n}$$

and

(4-88)

(4-89)

$$\sum_{k=0}^{n-1} e[(k-1)T]z^{-k} = e(0)x^{-1} + e(T)x^{-2} + \dots + e[(n-1)T]z^{-n}$$

k=0

(4-90)

We assume that $e(t)$ is defined only for time greater than zero so that the term $e(-T)$ that would have appeared in the last equation is zero. Now comparing Eq. (4-89) with Eq. (4-90), we see that the latter can also be written as

n

-k

$$\sum_{k=0}^{n-1} e[(k-1)T]x^{-k}$$

k=0

n-1

$$= z^{-1} \sum_{k=0}^{n-1} e(kT) z^{-k}$$

k=0

Subtracting Eq. (4-91) from Eq. (4-89) yields

n

-k

271

7-1

$$\sum_{k=0}^{n-1} e(kT) x^{-k}$$

k

k=0

$$z^{-1} \sum_{k=0}^{n-1} e(kT) z^{-k}$$

k=0

x

-k

(4-91)

(4-92)

Taking the limit of the last equation as z approaches unity, we get

7-1

77

lim

1-2

k=0

$$\left[\sum_{k=0}^{n-1} e(kT) z^{-k} \right]$$

-k

-

$$z^{-1} = c(kT) z^{-k}]$$

k=0

$$= \sum_{k=0}^{n-1} e(kT) - \sum_{k=0}^{n-1} e(kT) = e(nT)$$

=

k=0

-

k=0

(4-93)

Again, if we take the limit as n approaches infinity, Eq. (4-93) becomes

77

n-1

$$\lim_{n \rightarrow \infty} e(nT)$$

=

$$\lim_{z \rightarrow 1} \lim_{n \rightarrow \infty} \left[\sum_{k=0}^{n-1} e(kT) z^{-k} \right] \quad (4-94)$$

-1

-

1-2 $\infty+u$

$$\sum_{k=0}^{\infty} e(kT) z^{-k} \quad (4-94)$$

k=0

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Interchange the limits on the right-hand side of Eq. (4-94), and since

$$\lim_{n \rightarrow \infty} \sum_{k=0}^{n-1} z e(kT) x^{-k} = \lim_{z \rightarrow 1} \sum_{k=0}^{\infty} e(kT) x^{-k} = E(2)$$

the final-value theorem reads

$$\lim_{k \rightarrow \infty} e(nT) = \lim_{z \rightarrow 1} (1 - z^{-1}) E(z)$$

7200

2-1

(4-95)

(4-96)

Example 4-12

Given the z-transform

0.79222

E(z)

=

$$(\approx - 1)(\approx^2 - 0.416\approx + 0.208)$$

determine the final value of e(nT) by use of the final-value theorem.

Since

$$\lim_{z \rightarrow 1} (1 - z^{-1}) E(z) =$$

$$=$$

0.79222

$$0.4162 + 0.208)$$

(4-97)

(4-98)

does not possess poles on or outside the unit circle in the z-plane, the final value theorem may be applied. Thus from Eq. (4-88)

$$\lim_{n \rightarrow \infty} e(nT)$$

0.79222

$$= \lim_{z \rightarrow 1}$$

= 1

N → ∞

2-1

$$22 \cdot 0.4162 + 0.208$$

(4-99)

This result can be checked readily by expanding E(z) in powers of z^{-1} .

Expansion of E(z) by long division yields.

E(2)

$$= 0.792x + 1.12z^{-2} + 1.09123 + 1.01\approx 4 + 0.9832 - 5$$

-7

$$+ 0.9892 - 6 + 0.99\% z^{-2} +$$

(4-100)

The coefficient of this series converges rapidly to its final steady state value of unity.

7. Partial Differentiation Theorem

Let the z-transform of the function e(t, a) be denoted by E(z, a), where a is an independent variable or a constant. The z-transform of the partial derivative of e(t, a) with respect to a is given by

$$\mathcal{Z}\left\{\frac{\partial e(t, a)}{\partial a}\right\} = \frac{\partial}{\partial a} \mathcal{Z}\{e(t, a)\}$$

Proof: By definition

00

da

$$\mathcal{Z}\{e(t, a)\} = \sum_{n=0}^{\infty} e(nT, a) z^{-n}$$

N=0

(4-101)

∂

$$\sum_{n=0}^{\infty} \frac{\partial}{\partial a} e(nT, a) z^{-n} = E(x, a) \quad (4-102)$$

The following example illustrates that the z-transforms of certain types of functions may be obtained easily if the partial differential theorem is applied.

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Example 4-13

Determine the z-transform of $e(t) = te^{-at}$ by use of the partial differential theorem. The z-transform of $e(t)$ is

$$3[e(1)] = 3[te^{-at}] = 3[-13 \ 2e^{-at}]$$

From Eq. (4-101),

$$3[-23 \ --] = -32, \ 3(-1) = -3(---)]$$

$$= Tze^{-ar} \\ (\approx -e^{-ar})^2$$

$$(4-103)$$

$$(4-104)$$

4.6 THE PULSE TRANSFER FUNCTION

Thus far our examination of sampled-data systems has been confined to the discussions of the properties and mathematical representation of the sampled signals, namely, the z-transformation. The situation must be investigated when a sampled signal is applied to the input terminals of a system with linear elements.

For the linear open-loop system with a continuous signal $e(t)$ as shown in Fig. 4-9, the transfer function

$$\frac{e(t)}{c(t)} \\ \text{Input} \\ G(s) \\ E(s) \\ C(s)$$

Fig. 4-9 Linear system with continuous signal.

$$G(s) = C(s)/E(s) \quad (4-105)$$

completely describes the relationship between the continuous input and output signals. If the same system is now subjected to a sampled signal as shown in Fig. 4-10, the Laplace transform of the system output can be written as

$$C(s) = E^*(s)G(s)$$

(4-106)
 Fictitious sampler
 $c^*(1)$
 S_2
 T $C^*(s)$
 $e(t)$
 $e^*(1)$
 $c(t)$
 $G(s)$
 $E(s)$
 $TE(s)$
 $C(s)$
 S_1

Fig. 4-10 Linear system with sampled-data.

where $E^*(s)$ is the Laplace transform of the sampled signal $e^*(t)$. Since we have developed a wealth of z-transform theories and techniques in the preceding sections, it is reasonable to expect that the analysis of the system with sampled-data shown in Fig. 4-10 will be

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simplified if the z-transformation is applied. Under the present situation, the input is in sampled form; thus we can form $E(z)$ from $E^*(s)$ simply by introducing the transformation $z = e$. The z-transform of the system output is defined as $C(z)$; therefore, $C(z)$ describes $c(t)$ only at the sampling instants. In order to characterize the sampled-data system in the z-domain, we expect to derive an expression of the form

$$C(z) = G(z)E(z)$$

=

(4-107)

for example, where $C(z)$ and $E(z)$ are defined as above; but $G(z)$ at this stage can be interpreted only as the z-transform of something that apparently depends on the system transfer function $G(s)$. (As it turns out of course the function, $G(z)$, is obtained exactly by taking the z-transform of $G(s)$ or the z-transform of the impulse response $g(t)$ of the linear system G .) Therefore, the meaning of $G(z)$ in Eq. (4-107) must be determined along with the relationship between $G(z)$ and $G(s)$.

Suppose that in the system shown in Fig. 4-10 a unit-step function is applied at the input of the sampler S at $t = 0$ and the sampler is closed only at $t=0$ and then remains opened thereafter. Under this condition the output of the ideal sampler consists only of an impulse of unit area occurring at $t=0$. The output of the system, $c(t)$ in response to this unit-impulse input is simply the impulse response $g(t)$ of the system. Thus

$$c(t) = g(t) = \text{impulse response of system } G$$

(4-108)

If a fictitious sampler S , with the same sampling rate as that of S (the two samplers are assumed to be synchronized) is placed at the system output, the output of the switch S , is described by the following equation:

$$c^*(t) = g^*(t) = \sum_{n=0}^{\infty} g(nT)\delta(t - nT)$$

(4-109)

where $c(nT) = g(nT)$ is defined as the "weighting sequence" of the system. The signals, $e(t)$, $e^*(t)$, $c(t)$, and $c^*(t)$, for this case are illustrated in Fig. 4-11. The Laplace transform of $g^*(t)$ is now written

$$G^*(s) = \sum_{n=0}^{\infty} g(nT) e^{-ns}$$

$n=0$

(4-110)

and is defined as the "pulse transfer function" of the system G .

We can summarize our findings at this point as follows: When a unit-impulse function is applied to a linear system with transfer function $G(s)$, the output of the system is simply the impulse response of the system. The output of the fictitious sampler S_2 is a sequence of impulses; the area of each impulse is equal to the value of the

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$e(t)$

0

(0)

$g(1)$

0

$e^*(t)$ Impulse of unit strength

$g^*(1)$

(b)

(c)

(d)

Fig. 4-11 Signals of sampled-data system in Fig. 4-10. (a) Unit step function. (b) Sampled input, unit impulse. (c) Impulse response. (d) Weighting sequence.

impulse response $g(t)$ at the corresponding sampling instant. This impulse sequence is also called the weighting sequence of the system. The Laplace transform of the weighting sequence gives the pulse transfer function $G^*(s)$.

Once the weighting sequence of a linear system is defined by means of the principle of superposition, the output $c(t)$ and $c^*(t)$ of the system in response to any arbitrary input can be obtained. Assume that an arbitrary function $e(t)$ is applied to the system of Fig. 4-10 at $t = 0$; the sampled input signal to G is simply the sequence $e(nT)$. At time $t = nT$, the output sample $c(nT)$ is the sum of the effects of all sample $e(nT)$, $e(n-1)T$, $e(n-2)T$, . . . $e(0)$; that is, $c(nT) = \sum$ Effects of all samples, $e(nT)$, $e(n-1)T$, . . . , $e(0)$

or

-

-

-

$$c(nT) = e(0)g(nT) + e(T)g(n-1)T + e(2T)g(n-2)T$$

$$+ \dots + e(n-1)Tg(T) + e(nT)g(0)$$

(4-111)

(4-112)

Multiplying both sides of the last equation by e^{-nrs} and taking the summation from $n=0$ to $n=\infty$ yields

∞

∞

$$\sum_{n=0}^{\infty} c(nT)e^{-nrs} = \sum_{n=0}^{\infty} e(0)g(nT)e^{-nrs} + \sum_{n=0}^{\infty} e(T)g[(n-1)T]e^{-nTs}$$

-

$N=0$

$n=0$

71=0

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or

∞

+

$$\dots + \sum_{n=0}^{\infty} e[(n-1)T]g(T)e^{-nTs}$$

$$\begin{aligned}
 n=0 \\
 + \sum_{n=0}^{\infty} e(nT)g(0)e^{-nTs} \\
 N=0 \\
 \sum_{n=0}^{\infty} c(nT)e^{-nTs} = [e(0) + e(T)e^{-Ts} \\
 (4-113) \\
 + e(2T)e^{-2Ts} + \dots
 \end{aligned}$$

Equation (4-114) is written

$$\begin{aligned}
 \sum_{n=0}^{\infty} c(nT)e^{-nTs} \\
 = \\
 \text{or simply} \\
 \sum_{n=0}^{\infty} g(nT)e^{-nTs} \quad (4-114)
 \end{aligned}$$

$$\sum_{n=0}^{\infty} e(nT)e^{-nTs} \sum_{n=0}^{\infty} g(nT)e^{-nTs} \quad (4-115)$$

$$C(s) = E(s)G(s)$$

Taking the z-transform on both sides of Eq. (4-116) yields

$$\begin{aligned}
 C(z) &= E(z)G(z) \\
 (4-116) \\
 (4-117)
 \end{aligned}$$

which is the z-transform relation of the sampled-data system. $G(z)$ is often referred to as the "z-transfer function" of the linear sampled-data system.

Therefore, the z-transfer function $G(z)$ relates the sampled input $E(z)$ to the z-transform of the output $C(z)$ in the same way as the ordinary transfer function $G(s)$ does for $E(s)$ and $C(s)$. Take note, however, that the actual output of the sampled-data system, $c(t)$, is still continuous with respect to time because of the smoothing properties of system components. However, the z-transform analysis described by Eq. (4-117), which relies on the addition of the fictitious sampler S_2 , gives information on $c(t)$ only at the sampling instants. This is certainly one of the disadvantages of the z-transform method. In certain cases the loss of the information concerning response between sampling instants may not be vital; e.g., if the value of $c(t)$ does not vary appreciably between sampling instants. On the other hand, if $c(t)$ contains large-amplitude oscillations between sampling instants, the z-transform method can at best provide an approximation to the actual response and sometimes may even provide misleading results. Still another comment is that in view of Eq. (4-110) the z-transfer function $G(z)$ is derived from the impulse response $g(t)$ in exactly the same manner as $E(z)$ is derived from any signal $e(t)$. Thus $G(z)$ may be determined by any methods described in Sec. 4-2 or directly from the z-transform table. The z-transfer function relation given by Eq. (4-117) can also be derived from the relation between $E^*(s)$ and $E(s)$ stated in Eq. (2-68). The Laplace transform of the system output of the system in Fig. 4-10 is

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$$\begin{aligned}
 C(s) &= E^*(s)G(s) \\
 (4-118)
 \end{aligned}$$

From the pulse-transform on both sides of the last equation and from

the relation of Eq. (2-68), we have

$$C^*(s) = \sum_{n=-\infty}^{\infty} C(s + jn\omega_s) = \frac{1}{T} \sum_{n=-\infty}^{\infty} E^*(s + jn\omega_s) G(s + jn\omega_s)$$

In Eq. (2-72), $E^*(s)$ has been shown to be a periodic function; that is $E^*(s) = E^*(s + jn\omega_s)$, where n is an integer. Equation (4-119) can now be written as

$$C^*(s) = E^*(s) \sum_{n=-\infty}^{\infty} G(s + jn\omega_s)$$

and if we define

$$G^*(s) = \sum_{n=-\infty}^{\infty} G(s + jn\omega_s)$$

Eq. (4-120) reads

$$C^*(s) = E^*(s) G^*(s)$$

$$(4-120)$$

$$(4-121)$$

$$(4-122)$$

Equation (4-121) again shows that the definition of the pulse transfer function $G^*(s)$ is the same as that of any signal $e(t)$. In the following explanations the z -transfer functions of some simple open-loop sampled-data systems will be derived.

1. Z -Transfer Function Relations of Cascaded Elements With Sampling Switch Between Them

Figure 4-12 illustrates a sampled-data system with cascaded elements G_1 and G_2 . The two elements are separated by a second sampling switch S_2 synchronized to S_1 (the two switches have the same sampling period and operate at the same instants). The z -transform relation between the output and the input signals is derived as follows: The output signal of G_1 is

$$D(s) = G_1(s) E^*(s)$$

$$(4-123)$$

$$(4-124)$$

$$D(s) = G_2(s) D^*(s)$$

$$(4-123)$$

$$(4-124)$$

$$C(s) = G_2(s) D^*(s)$$

$$(4-123)$$

$$(4-124)$$

Fig. 4-12 Sampled-data system with cascaded elements (sampler separates the two elements). The z -transform relation between the output and the input signals is derived as follows: The output signal of G_1 is

$$D(s) = G_1(s) E^*(s)$$

and the system output is

$$C(s) = G_2(s) D^*(s)$$

$$(4-123)$$

$$(4-124)$$

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Taking the pulse transform on both sides of Eq. (4-123) yields

$$D^*(s) = G^*(s)E^*(s)$$

(4-125)

Substituting $D^*(s)$ from Eq. (4-125) into Eq. (4-124) yields

$$C(s) = G_2(s)G^*(s)E^*(s)$$

(4-126)

The pulse transform of $C(s)$ gives

$$C^*(s) = G(s)G^*(s)E^*(s)$$

(4-127)

and the z-transform of $C(s)$ is

$$C(z) = G_1(z)G_2(z)E(z)$$

(4-128)

Therefore, the z-transform of two linear elements separated by a sampler is equal to the product of the z-transforms of the two elements.

2. Z-Transform of Cascaded Elements With No Sampling Switches

Between Them

Fictitious

sampler

$c(1)$.

$C(s)$

$c(t)$

$C(s)$

$e(t)$

$E(s)$ $G(s)$

$E(s)$

$d(t)$

$D(s)$

$G_2(s)$

Fig. 4-13 Sampled-data system with cascaded elements (no sampler between the two elements).

A sampled-data system with two cascaded elements is shown in Fig. 4-13; there is no sampler between the elements. The z-transform relation of the output and the input signals is derived as follows:

The transform of the continuous output is

$$C(s) = G_1(s)G_2(s)E^*(s)$$

(4-129)

Taking the pulse transform on both sides of the last equation yields

$$C^*(s) = G_1G(s)E^*(s)$$

where $G.G(s)$ is defined as

$\emptyset$

(4-130)

$$G_1G(s) = [G_1(s)G_2(s)]^* = \frac{1}{T} \sum G_1(s + jn\omega_s)G_2(s + jn\omega_s)$$

T N118

Note that in general.

$$G.G(s) = G^*(s)G^*(s)$$

Now taking the z-transform of Eq. (4-130) yields

=

$$C(z) = G_1G_2(z)E(z)$$

(4-131)

(4-132)

(4-133)

where $G.G_2(z)$ denotes the z-transform of the product of $G_1(s)$ and

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$G(s)$. Thus the z-transform of two cascaded elements with no sampler between them is equal to the z-transform of the product of the Laplace transforms of the two elements. The following example illustrates the difference between Eq. (4-128) and Eq. (4-133).

Example 4-14

For the sampled-data systems shown in Fig. 4-12 and Fig. 4-13, given.

$G_1(s) = 1/s$ and $G_2(s) = a/(s + a)$, determine the z-transform of the output $c(t)$ when the input is a unit-step function.

For the system of Fig. 4-13 the z-transform of the output is obtained from Eq. (4-133).

$$C(z) = G_1 G_2(z) E(z) = \frac{a}{z} \frac{1}{s + a}$$

$$= \frac{z^2(1 - e^{-ar})}{(z-1)^2(z - e^{-aT})}$$

$$= \frac{z(1 - e^{-ar})}{(z-1)^2(z - e^{-aT})} \quad (4-134)$$

The output z-transform of the system in Fig. 4-12 is obtained by use of Eq. (4-128).

$$C(z) = G_1(z) G_2(z) E(z) = \frac{1}{z} \frac{a}{s + a}$$

$$= \frac{az^3 - e^{-ar}}{az(z-1)^2(z - e^{-aT})} \quad (4-135)$$

Evidently, the output z-transforms of the two systems are not alike.

4.7 LIMITATIONS OF THE Z-TRANSFORM METHOD

In the preceding sections we have seen that the z-transformation is a very convenient tool for the study of linear systems with sampled-data. However, the z-transform method has limitations; and in certain cases, care must be taken in the applications of the z-transforms.

(1) In the first place we must realize that the derivation of the z-transform method is based upon the assumption that the sampled signal is approximated by a train of impulses whose areas are equal to the input signal of the sampler at the sampling instants. This assumption is considered valid only if the sampling duration of the sampler is small when compared with the significant time constant of the system.

(2) The z-transform of the output of a linear system, $C(z)$, specifies only the values of the time function $c(t)$ at the sampling instants; $C(z)$ does not contain any information concerning the value of $c(t)$ between sampling instants. Therefore, given any $C(z)$, the inverse z-transform $c(nT)$ describes $c(t)$ only at the sampling instants $t = nT$.

$t =$

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(3) In analyzing a sampled-data system by z-transform method, the transfer function $G(s)$ must have at least two more poles than zeros (or equivalently, the impulse response of $G(s)$ must not have any jump behavior at $t = 0$). Otherwise, the system response obtained by the z-transform method is unrealistic and sometimes even incorrect.

1 ohm

Input Sampling

$e(t)$

device

1 farad

Output

$c(t)$

Fig. 4-14 An R-C integrating network. The input terminals of the R-C network are assumed to be short circuited when the sampler is opened.

In order to illustrate the points mentioned above, let us consider the R-C integrating circuit shown in Fig. 4-14. The transfer function of the circuit is $G(s) = 1/(s + 1)$, and the corresponding z-transform is $G(x) = x/(x - e^{-T})$. If the input to the sampler is a unit step function, the z-transform of the output signal is

$$C(z) = R(z) G(z) = \frac{R(z)}{z^2} G(z)$$

$$= \frac{(1 - e^{-T})(x - e^{-T})}{z^2} (\approx 1)(\approx$$

$$0.368)$$

$$(4-136)$$

where the sampler has a sampling period of one second. The value of $c(t)$ at the sampling instants are determined by expanding $C(x)$ in the inverse power of z . Therefore,

$$C(z) = 1 + 1.368z^{-1} + 1.52z^{-2} + 1.55z^{-3} + 1.56z^{-4} + \dots$$

$$c^*(1)$$

$$1.585$$

$$(4-137)$$

0
T 2T 3T 4T 5T 6T 7T

Fig. 4-15 Output signal at the sampling instants of the R-C network with sampled-data (Fig. 4-14) (Result obtained by z-transform method).

from which the sampled output $c^*(t)$ is obtained as shown in Fig. 4-15. Note that the output transform $C(z)$ specifies the values of $c(t)$ at the sampling instants only and does not give even the slightest indication of what the behavior of $c(t)$ is in between the sampling instants. As a matter of fact, if the input to the network is assumed to be an impulse train, we can show very easily by the impulse

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$c(t)$

$$1.585$$

www

T 2T 3T 4T 5T 6T 7T

$$0.585$$

t

Fig. 4-16 Actual output response of the R-C network with sampled-data shown in Fig. 4-14 (Result obtained by impulse response method or the Laplace transform method).

response method or the Laplace transform method that the actual response of the network is of the form shown in Fig. 4-16. Furthermore, the z-transform method gives the result with a jump in $c(t)$ at $t = 0$. This is because the method relies on the assumption that the sampler output is an impulse train. However, if the actual pulses of finite pulse widths are considered, this jump will not occur. As another example, consider that the capacitor in the R-C network of Fig. 4-14 is now replaced by an inductor of 1 henry. The transfer function of the R-L network is $s/(s + 1)$, and the z-transform

is
 $G(x) =$
 (4-138)
 I
 Input Sampling!
 $e(t)$
 i_{device}
 -0.368

$=$
 2
 $- 0.368$
 1 ohm
 ww
 1 henry

Output
 $c(t)$

Fig. 4-17 An R-L network with sampled-data. The input terminals of the R-L network are assumed to be short circuited when the sampler is opened.
 for $T = 1 \text{ sec.}$ For a unit step input the output z-transform is
 $C(z)$

$=$
 $-0.368z$
 $(z - 1)(\approx - 0.368)$
 $-$
 $-$
 -1

Expanding $C(z)$ into a power series of z^{-1} yields

$=$
 (4-139)
 $C(z) 0.36821 \ 0.5032z^{-2} - 0.5512z^{-3} 0.5682z^{-4} \dots$ (4-140)

The values of $c(nT)$ given by Eq. (4-140) are depicted in Fig. 4-18. The z-transform method leads to results that have the output response starting with a zero value at $t = 0$; but judging from the impulse response of the R-L network, we see that $c(t)$ must contain an impulse of strength equal to unity at $t = 0$. From the principle of

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 $c^*(t)$

T
 $2T$
 $3T \ 4T \ 5T \ 6T \ 7T \ 8T$
 -0.585

Fig. 4-18 Output signal of R-L network with sampled-data shown in Fig. 4-17. (Result obtained by z-transform method; response at the sampling instants only.)

...
 superposition, we know that there must also be unit impulses at $t = T, 2T, 3T, \dots$ in $c(t)$. The correct response when the input is a train of impulses is obtained by the impulse response method and is shown in Fig. 4-19. In the case of the previous example, whereas the z-transform method does give correct results at the sampling instants, the z-transform technique breaks down completely here.

4.8 RESPONSE OF OPEN-LOOP SAMPLED-DATA SYSTEMS BETWEEN SAMPLING INSTANTS

Although we have seen that the z-transform method is convenient in the study of open-loop systems with sampled-data, (closed-loop systems will be treated in the next chapter) z-transform analysis yields no information on system response between sampling instants. A complete description of any system response almost always

necessitates a knowledge of the waveform between sampling instants. Several methods have been developed for the evaluation of Impulses of unit area

$c(t)$ 1

-1.585

T

2T 3T 4T

5T 6T 7T 8T

0.585

m

Fig. 4-19 Actual output response of the R-L network with sampled-data in Fig. 4-17. (Result obtained by impulse response method.)

+ E. I. Jury, Sampled-Data Control Systems (New York: John Wiley & Sons, Inc., 1958), pp. 64-72.

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Fictitious

sampler

S_1

$T c(t)$

$e(t)$

$X_e^*(1)$

$G(s)$

S_o

$c(t)$

Sampler

(a)

Sampler

$e(t)$

$e(t) e'(1)$

$\bullet(n) = (x^*(n) (G(3)$

$S_o S_{no}$

T/n

Fictitious

sampler

(b)

Fictitious sampler

$S_1 T c(t)$

$c(t)$

Sp_1

$T/n c^*(t)n$

Fictitious sampler

Fig. 4-20 (a) Open-loop sampled-data system. (b) Open-loop sampled-data system with fictitious samplers for the evaluation of response between sampling instants.

response of sampled-data systems between sampling instants. Among these methods, the submultiple sampling method and the modified z-transform method are common. The state variable approach which will be introduced in chapter 5 is also quite convenient for the study of intersampling behavior of sampled systems. The analysis of open-loop systems by means of the z-transform methods is discussed below.

1. The Submultiple Sampling Method

Consider the open-loop sampled-data system shown in Fig. 4-20a.

The sampling switch S samples at uniform intervals of T seconds.

The ordinary z-transform of the output response, $C(z)$, is obtained

↑ W. K. Linvill and R. W. Sittler, "Extension of Conventional Techniques to the Design of Sampled-Data Systems," Part I, I. R. E. Convention Record, New York, 1953, pp. 99-104.

R. H. Barker, "The Pulse Transfer Function and its Application to Sampling Servo

Systems," Proc. I. E. E., London, 99, Part 4, 1952, pp. 302-317; and E.I. Jury, "Synthesis and Critical Study of Sampled-Data Control Systems," Trans. A. I. E. E., 75, Part 2, 1956, pp. 141-151.
 § R. E. Kalman and J. E. Bertram, "A Unified Approach to the Theory of Sampling Systems," Journal of Franklin Institute, 267, May 1959, pp. 405-436.

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by first taking the Laplace transform of the output of the fictitious sampler S , and then making the substitution of $z = e$. In order to recover the lost information in between the sampling instants, fictitious samplers S_{n0} and S with sampling period T/n , where $n = 1, 2, 3, \dots$, are inserted respectively between the sampler S and $G(s)$ and at the output terminal. The fictitious samplers are not physical components; they are used simply to facilitate the calculation of the system response during the sampling period.

Since the input to the fictitious sampler S_{n0} is a train of impulses of period T (spaced at T seconds apart) and since S_{n0} is sampling at a rate n times as fast as S_0 , the insertion of the switch S_{n0} does not in any way affect the original system behavior. Therefore, the system shown in Fig. 4-20b which includes the two fictitious samplers has exactly the same behavior as that of the original system in Fig. 4-20a. The continuous output $c(t)$ of the system in Fig. 4-20b may be written as a sum of impulse responses. Thus

$$c(t) = \sum_{m=0}^{\infty} e(mT)g(t - mT)$$

($m = \text{integer}$)
 (4-141)

where $g(t)$ is the impulse response of the system G . At any sampling instant $t = kT/n$ of the sampler S_{ni} , ($k = 0, 1, 2, 3, \dots$), the value of the output signal is equal to

$$c(kT) = \sum_{m=0}^{\infty} e(mT)g(kT - mT)$$

(4-142)

The output of the fictitious sampler S_n can now be written as

$$c^*(t)_n = \sum_{k=0}^{\infty} c(kT) \delta(t - kT)$$

Taking the z-transform on both sides of Eq. (4-143) yields

$$C(z)_n = Z[c^*(t)_n] = \sum_{k=0}^{\infty} c(kT) z^{-kn}$$

(4-143)

(4-144)

If we compare Eq. (4-144) with the ordinary z-transform of $c(t)$, we see that they are related by

$$C(z)_n = [C(z)]_{z=T/n}$$

$T=T/n$

$1/n$

1

(4-145)

Therefore, the z-transform of the output of sampler S_n is equal to the z-transform of $c^*(t)$ with z replaced by z and T replaced by T/n . Substituting Eq. (4-142) into Eq. (4-144), we have

∞

∞

$$C(z)_n = \sum_{k=0}^{\infty} \sum_{m=0}^{\infty} e(mT) \delta(*T$$

$$\frac{g}{mT}) \times$$

$$-k_{in}$$

$$(4-146)$$

$$k=0 \quad M=0$$

Letting (

-

n

n

m) = 1, where v is an integer, Eq. (4-146) can be written as

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Hence

where

00

-v/n

$$C(x)_n = \sum_{v=0}^{\infty} \delta(\circ 1) \cdot 2^{-3/\approx} \sum e(mT) x^{-"$$

g

n

m=0

$$C(z)_n = G(z), E(z)$$

(4-147)

(4-148)

(4-149)

00

G(z)_n =

Z \delta(VT)x-

vT -v/n

=

$$[G(x)]_2 = 21/n$$

n

T-T/n

V=0

g

Equation (4-149) implies that the response between the sampling instants can be evaluated by use of the ordinary z-transforms; the number of points on c(t) at which the values are desired determines the value of n. For instance, if in addition to the values at the normal sampling instants, t = 0, T, 2T, . . . , the values of c(t) at two other points between two consecutive sampling instants are desired, n is set equal to 3. In general, if q is the number of additional sampled values desired between two consecutive sampling instants, n = q + 1.

Example 4-15

For the system shown in Fig. 4-20a, G(s) = 1/(s + 1). The ordinary z-transform of G(s) is G(z) = 2/(\approx - e^{-7}). By means of Eq. (4-149), we have

=

-

21/n

$$G(\approx)_n = [G(2)]; = ; 1/n = 21/n - e^{-T/n}$$

T=T/n

=

21/n

1/n

e

(4-150)

for $T = 1$ sec. If in addition to the values of $c(t)$ at the regular sampling instants, we desire to know the values of $c(t)$ also at two other points (at uniform intervals) within the sampling period (one sec), we set $n = 3$ in Eq. (4-150). Thus

$$G(z) = \frac{z^3}{z^3 - 0.717} \quad (4-151)$$

It is interesting to note that the inverse z -transform of $G(z)$, yields values of the impulse response $g(t)$ not only at the actual sampling instants of $t = 0, 1, 2, \dots$ seconds, but also at t

This is done as follows: Let $z^{1/3} =$

$$= z^{1/3}, z^{2/3}, z^{4/3}, z^{5/3}, \dots$$

then $z = z^3$. Equation (4-151) now reads

$$G(z) = \frac{1}{z^3 - 0.717} = \frac{1}{z^3} + 0.717 \frac{1}{z^3} + 0.515 \frac{1}{z^3} + 0.368 \frac{1}{z^3} + \dots \quad (4-152)$$

In the last equation, the values of $g(t)$ at the fictitious sampling instants are represented by the respective coefficients of the series expansion of $G(z)^3$.

Therefore, $g(0) = 1$, $g(1) = 0.717$, $g(2) = 0.515$, $g(3) = 0.368$, $g(4) = 0.264$, $g(5) = 0.189$, $g(6) = 0.136$, etc. If the input to the system is a unit-step function, $E(z) = z/(z - 1)$. From Eq. (4-148), for $n = 3$,

$$C(z)^3 = G(z)^3 E(z)$$

Substitution of $E(z)$ and $G(z)$, into Eq. (4-153) yields

$$C(z)^3 = \frac{z^3}{(z^3 - 0.717)(z - 1)} \quad (4-154)$$

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$$\text{or} \quad C(z)^3 = \frac{z^3}{(z^3 - 0.717)(z - 1)} \quad (4-155)$$

Expanding $C(z)$, into a power series in z^{-1} yields

$$C(z)^3 = 1 + 0.717z^{-1} + 0.514z^{-2} + 1.368z^{-3} + 0.98z^{-4} + 0.703z^{-5} + 1.504z^{-6} + 1.082z^{-7} + 0.773z^{-8} + 1.55z^{-9} + \dots$$

(4-156)
The waveform of $c^*(t)$, is plotted in Fig. 4-21. In this case, the response between sampling instants may be estimated by determining two additional sampled values between consecutive sampling instants.

* (1)
1.58
Final value of $c(t)$
 $c(t)$
T
2T 3T 4T

Fig. 4-21 Output response of fictitious submultiple-sampling

sampler of the system in Example 4-15.

2. The Modified Z-Transform Method

Another interesting and powerful method which may be used to evaluate the response between sampling instants of a sampled-data system is the modified z-transform method. The method was first introduced by Barkert of England and then later was developed by Jury in this country. The method is essentially a modification of the ordinary z-transform technique obtained by inserting fictitious transportation lags (time delays) in the sampled-data system. By varying the amount of transportation lag in a prescribed manner, we may recover virtually all the information contained in the continuous signal between sampling instants.

The Delayed z-Transform In order to understand the development

R. H. Barker, "The Pulse Transfer Function and its Application to Sampling Servo Systems," Proc. I. E. E., London, 99, Part 4, 1952, pp. 302-317.

E. I. Jury, "Synthesis and Critical Study of Sampled-Data Control Systems," Trans. A. I. E. E., 75, Part 2, 1956, pp. 141-151; and E. I. Jury, "Additions to the Modified

z-transform Method," I. R. E. WESCON Convention Record, Part 4, August 21, 1957, pp. 136-156.

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of the modified z-transform, we should first study the ordinary z-transformation of impulse sequences derived from time functions delayed by non-integral multiples of the sampling period T . Recall that the z-transform of time functions delayed by integral multiples of T may be obtained easily by means of the real translation theorem stated in Sec. 4.5, Eq. (4-68). For instance, the z-transform of the time function $e(t - XT)$, where X is an integer, is simply equal to z^{-X} times the z-transform of $e(t)$. However, for a time function $e(t)$ that is delayed by kT where k is a noninteger a different treatment is necessary. Let us consider the waveform of the time function shown in Fig. 4-22a. The time function $e(t)$ is shown to be delayed by time kT where k is not an integer. This delayed time function is now sampled at $t = 0, T, 2T, \dots$, the regular sampling intervals, so that the resultant impulse train is as shown in Fig. 4-22b. (Note that this is not the same as sampling $e(t)$ at $t = kT, (k + 1)T, \dots$.) Therefore, the samples of the delayed function may be represented by

$$e^*(t - kT) = \sum_{n=0}^{\infty} e(nT - kT) \delta(t - nT)$$

(4-157)

integers $0, 1, 2, \dots$. Let us now break.

where k noninteger, and $n =$
up k into two parts, so that

$$k = \lambda + A$$

(4-158)

where λ is the largest integer less than k , and A is a positive number less than one. Substituting the relation in Eq. (4-158) into Eq. (4-157) yields

$$\bar{e}^*(t - kT) = \sum_{n=0}^{\infty} e(nT - XT - AT) \delta(t - nT)$$

-
 -
 -
 (4-159)
 Applying z-transformation to both sides of the last equation, we have

$$Z[e(t - kT)] = \sum_{n=0}^{\infty} e[nT - XT - AT]z^{-n}$$

-

$$n=0$$

 -
 -

$$\sum_{n=0}^{\infty} e[nT - AT]z^{-n}$$

 (4-160)
 Since XT is an integral multiple of T , we may regard the z-transform in Eq. (4-160) as the z-transform of $e(t - \Delta T)$ delayed by XT . Thus

-

$$Z[e(t - kT)] = \sum_{n=0}^{\infty} e(nT - AT)z^{-n}$$

 (4-161)
 When $\lambda = 0$, Eq. (4-161) becomes

-

$$\sum_{n=0}^{\infty} e(nT - AT)x^{-n}$$

 (4-162)

which is sometimes referred to as the delayed z-transform of $e(t)$ and is denoted by $E(z, A)$; that is,

$$E(z, A) = \sum_{n=0}^{\infty} e(t - AT)z^{-n}$$

 (4-163)

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The calculation of the delayed z-transform of a time function is now illustrated by an example.

Example 4-16

Derive the delayed z-transform for the time function e^{at} that is delayed by $0.5T$ second.

By use of Eq. (4-162), we have

$$E(z, \Delta = 0.5) = Z[e(t - 0.5T)] = \sum_{n=0}^{\infty} e^{-a(nT - 0.5T)} z^{-n}$$

$$= \sum_{n=0}^{\infty} e^{-0.5aT} e^{-anT} z^{-n}$$

Writing $E(z, 0.5)$ in a closed form, we have

$$E(z, 0.5) = \frac{e^{-0.5aT}}{1 - e^{-aT}z^{-1}}$$

$$(4-165)$$

$$\frac{e^{-aT/2}}{1 - e^{-aT}z^{-1}}$$

$$\frac{e^{-aT/2}}{1 - e^{-aT}z^{-1}}$$

e^{t-kT}
 $e^{*(t-kT)}$
 $0 \quad T$
 $KT \quad 4T \quad 6T \quad 8T \quad 10T$
 $3T \quad 5T \quad 7T \quad 9T$
 (b)

Fig. 4-22 (a) Delayed time function. (b) Impulse sequence of delayed time function.

Modified z-Transfer Function of Open-Loop Sampled-Data Systems

The delayed z-transform defined above may be applied to the evaluation of responses of sampled-data systems between sampling instants. Consider the open-loop system shown in Fig. 4-23a. The z-transform of $c(t)$, $C(z)$, describes the output response at the sampling instants only. Since the actual output of the system is continuous with respect to time, a modified z-transfer function may be derived to provide a relation between the sampled input and the continuous output. In

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other words, in contrast to the ordinary z-transfer function $G(z)$, which gives the relation between the sampled input $E(z)$ and the sampled output $C(z)$ as

$$C(z) = G(z)E(z)$$

(4-166)

we shall derive a modified z-transfer function $G(z, m)$ for the system such that

$$C(z, m) = G(z, m) E(z)$$

(4-167)

where m is a constant and $C(z, m)$ is called the modified z-transform of $c(t)$. By assigning proper values for m , we may obtain virtually all information of $c(t)$ from $C(z, m)$ by taking the inverse modified z-transform. We shall show how this is done in the following.

If we introduce a fictitious delay AT in the forward path of the system of Fig. 4-23a, we have the modified system shown in Fig. 4-23b. For reasons that will be clear later, the delay Δ is chosen as a fraction of the sampling period. Now consider that an impulse of unit strength is applied as input $e^*(t)$ at $t = 0$ to the system of Fig. 4-23b. The output of the system is just the impulse of the system G delayed by time AT .

$c(+)$
 $C(z)$
 $E(s) \quad E^*(s)$

$c(t)$
 $G(s)$
 $e(t)$
 $e^*(t)$
 $C(s)$

(a)
 $c(t-AT)$

$C(z, m)$
 $E(s)$
 $E(s)$

$C(s)$
 $G(s)$
 e^{-AT}

$e(t) \quad T$
 $e(t)$

$c(t) \quad L.$
 $c(t-AT)$

Fictitious time
 delay

(b)

Fig. 4-23 (a) Open-loop system with sampled data. (b) Open-loop sampled-data system with fictitious time delay.

Therefore,

$$c(t - AT) = g(t - AT)u(t - AT) \quad (4-168)$$

The sampling of this delayed impulse response by the fictitious sampler at the output gives the delayed weighting sequence: 0, $g(AT)$, $g(2T - AT)$, $g(3T - AT)$, The output signal of the fictitious sampler may be written as

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$$g^*(t - AT) = \sum_{n=0}^{\infty} g(nT - AT)\delta(t - nT) \quad (4-169)$$

Taking the z-transform on both sides of the last equation yields

$$G(z, \Delta) = \sum_{n=0}^{\infty} g(nT - AT) z^{-n} \quad (4-170)$$

$G(z, \Delta)$ is called the "modified z-transfer function" of the linear system G . When compared with the ordinary z-transfer function $G(z)$, $G(z, \Delta)$ gives values of $g(t)$ at all times when Δ is varied from zero to unity, whereas $G(z)$ defines $g(t)$ only at the sampling instants. For instance, if the sampling period is one second and if we want to know the values of $g(t)$ at $t = 0.3$ sec, 1.3 sec, 2.3 sec, simply set $A = 0.7$ sec, and the fictitious sampler will give the impulse sequence: 0, $g(0.3T)$, $g(1.3T)$, . . . $g(nT - AT)$, .

....
we
At this point we may let
 $A = 1 - m$

(4-171)
Since Δ is a number lying between 0 and 1, m is also a positive number lying between 1 and 0. With this A -to- m substitution, Eq. (4-170) becomes

$$G(z, \Delta) = \sum_{n=0}^{\infty} g(nT - T + mT)z^{-n} \quad (4-172)$$

The modified z-transform given by the last equation can be regarded as the z-transform of $g(n + m)T$ delayed by time T . Therefore,

$$G(z, m) = G(z, \Delta) |_{\Delta=1-m} = z^{-1} \sum_{n=0}^{\infty} g[(n + m)T]z^{-n} \quad (4-173)$$

=
0,
The derivation of Eq. (4-167) is similar to that of its counterpart for the ordinary z-transform given in Sec. 4.6. Assume that at t an arbitrary time function e(t) is applied to the system of Fig. 4-23b. The sampled input signal of the system is simply the impulse sequence e(0), e(T), . . . e(nT), The output of the open-loop system is denoted by c(t - AT); and the output of the fictitious sampler is the impulse sequence: c(TAT), c(2T - AT), . . . c(nT - AT), At any time t = nT the output sample c(nT - AT) is equal to the sum of the effects of all samples e(n - 1)T, e(n - 2)T, . . . e(0): that is

$$c(nT - \Delta T) = e(0)g(nT - \Delta T) + e(T)g[(n - 1)T - AT] + e(2T)g[(n - 2)T - \Delta T] + \dots + e[(n - 1)T]g[(T - AT)] \quad (4-174)$$

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Multiplying both sides of Eq. (4-174) by z⁻ⁿ and taking the summation from n = 0 to n = ∞, we have

$$\sum_{n=0}^{\infty} c(nT - AT)z^{-n} = \sum_{n=0}^{\infty} e(nT)x^{-n} \sum_{m=0}^{\infty} g(nT - AT)x^{-m} \quad (4-175)$$

or simply

where m = 1 - Δ and

n=0

n=0

$$C(z, m) = E(z)G(x, m) \quad (4-176)$$

c(t)

c*(t)

C(T) C(27)

t

T 27 37 4T 57 6T

7T 8T 9T 10T 11T

(0)

c(1)

c(t-AT).

c(2T-4T)

C(T-AT)

T

2T 37 4T 57 6T

(b)

c(t).

c*(t-AT)

7T

8T

9T 10T

11T

T 27 3T 4T 5T 6T 7T

8T 9T 10T 11T

(c)

Fig. 4-24 Samples of delayed time functions. (a) c(t) and c*(t); or c*(t - AT), (m = 1, Δ = 0). (b) c(t) and c*(t - AT) when m = 1 and Δ = 0. (c) c(t) and c*(t - ΔT) when m = 0 or Δ = 1.

-

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$$C(z, m) = \sum_{n=0}^{\infty} c(nT - AT) z^{-n}$$

$$= z^{-1} \sum_{n=0}^{\infty} c[(n + m)T] z^{-n}$$

(4-177)
 $C(z, m)$ is defined as the modified z-transform of $c(t)$ and is denoted by $3m[c(t)]$. Therefore,

$3m[c(t)] = \text{modified z-transform of } c(t)$

$= \text{z-transform of } c(t - AT)$

$$= z^{-1} Z[c(t + mT)]$$

(4-178)

When the value of m (or A) is varied from zero to unity, the fictitious sampler of the system given in Fig. 4-23b provides values of $c(t)$ at virtually all times. This process is illustrated in Fig. 4-24. Note that, when $m = 1$ ($A = 0$), the modified z-transform should revert to the ordinary z-transform, and therefore, we can write

$$C(z, 1) = C(z)$$

(4-179)

However, Eq. (4-179) is valid only if the response $c(t)$ does not have any jump discontinuity at $t=0$. When $m = 0$ ($A = 1$), the function $c(t)$ is delayed by one sampling period T . Thus

$$C(z, 0) = z^{-1} \sum_{n=0}^{\infty} c(nT) z^{-n} = z^{-1} C(z)$$

(4-180)

The following example illustrates how the modified z-transform of a function $g(t)$ may be determined by use of Eq. (4-173).

Example 4-17

Determine the modified z-transform of $g(t) = e^{-t}$. As a first step we write

$$g(n + m)T = e^{-(n+m)T}$$

The modified z-transform of $g(t)$ is obtained from Eq. (4-173). Thus

$$G(z, m) = z^{-1} \sum_{n=0}^{\infty} e^{-(n+m)T} z^{-n}$$

Expanding $G(z, m)$ in the last equation, we have

(4-181)

(4-182)

$$G(z, m) = z^{-1}(e^{-mT} + z^{-1}e^{-(m+1)T} + z^{-2}e^{-(m+2)T} + \dots)$$

(4-183)

which can also be put a closed form:

$$G(z, m) = z^{-1} e^{-mT} \frac{1 - e^{-T}}{1 - z^{-1}e^{-T}}$$

$$= z^{-1} e^{-mT} \frac{1 - e^{-T}}{1 - z^{-1}e^{-T}}$$

$$= z^{-1} e^{-mT} \frac{1 - e^{-T}}{1 - z^{-1}e^{-T}}$$

$$= z^{-1} e^{-mT} \frac{1 - e^{-T}}{1 - z^{-1}e^{-T}}$$

$$= z^{-1} e^{-mT} \frac{1 - e^{-T}}{1 - z^{-1}e^{-T}}$$

$$= z^{-1} e^{-mT} \frac{1 - e^{-T}}{1 - z^{-1}e^{-T}}$$

$$= z^{-1} e^{-mT} \frac{1 - e^{-T}}{1 - z^{-1}e^{-T}}$$

$$= z^{-1} e^{-mT} \frac{1 - e^{-T}}{1 - z^{-1}e^{-T}}$$

$$= z^{-1} e^{-mT} \frac{1 - e^{-T}}{1 - z^{-1}e^{-T}}$$

$$= z^{-1} e^{-mT} \frac{1 - e^{-T}}{1 - z^{-1}e^{-T}}$$

$$= z^{-1} e^{-mT} \frac{1 - e^{-T}}{1 - z^{-1}e^{-T}}$$

$$= z^{-1} e^{-mT} \frac{1 - e^{-T}}{1 - z^{-1}e^{-T}}$$

$$= z^{-1} e^{-mT} \frac{1 - e^{-T}}{1 - z^{-1}e^{-T}}$$

$$= z^{-1} e^{-mT} \frac{1 - e^{-T}}{1 - z^{-1}e^{-T}}$$

$$= z^{-1} e^{-mT} \frac{1 - e^{-T}}{1 - z^{-1}e^{-T}}$$

$$= z^{-1} e^{-mT} \frac{1 - e^{-T}}{1 - z^{-1}e^{-T}}$$

$$= z^{-1} e^{-mT} \frac{1 - e^{-T}}{1 - z^{-1}e^{-T}}$$

$$= z^{-1} e^{-mT} \frac{1 - e^{-T}}{1 - z^{-1}e^{-T}}$$

delayed function $c(t - AT)$. Also, the modified z-transfer function $G(z, m)$ of any linear system with impulse response $g(t)$ is defined as the summation of impulses that are samples of the delayed impulse response $g(t - AT)$. The modified z-transform may also be defined by use of the complex convolution integral which we have used in Chapter 2 to derive the ordinary z-transform. We recall that the definition of the ordinary z-transform of $c(t)$ may be written as

$$C(z) = L[c(t) \times \delta_r(t)]_{z=e^{Ts}} \quad (4-185)$$

where $\delta_r(t)$ is a train of impulses each with unit area. In a similar manner, the modified z-transform of $c(t)$ is written as

$$C(z, A) = L[c(t - AT) \times \delta_r(t)]_{z=e^{Ts}} \quad (4-186)$$

where $0 < A < 1$.

From the theory of complex convolution Eq. (4-186) can be expressed as

$$C(z, A) = \{L[c(t - AT)] * L[\delta_r(t)]\}_{z=e^{Ts}}$$

$$C(z, A) = [e^{-ATs} C(s) * A_r(s)]_{z=e^{Ts}} \quad (4-187)$$

$$(4-188)$$

$$(4-189)$$

$$1$$

$$-T's$$

$$(4-190)$$

$$L[c(t - AT)] = e^{-ATs} C(s)$$

$$L[\delta_r(t)] = A_r(s)$$

$$=$$

$$1$$

Therefore, the convolution integral reads

$$-$$

$$1$$

$$C(z, A) =$$

$$= [$$

$$1$$

$$rc + j\infty$$

$$2\pi j =$$

$$1c - j\infty$$

$$C(s) e^{-$$

$$-$$

$$ATE$$

$$1$$

$$-T(s - s)$$

$$\alpha s$$

$$-$$

$$z=e^{Ts}$$

$$(4-191)$$

$$1$$

$$2$$

The integral above can be evaluated by integrating along the line from $c - j\infty$ to $c + j\infty$ and the semicircle of infinite radius located either in the right half or the left half of the s -plane. The possible contours of integration are identical to those shown in Fig. 2-6. This contour integration method will provide correct results for $C(z, A)$ only if the integration around the large semicircle of T , or T_2 is zero. Since the contour in the right-half s -plane encircles the poles of $1/(1 - e^{-T(-)})$ which are infinite in number, the integral of Eq. (4-191) can best be evaluated by integrating along the closed contour in the left-half plane. However, one difficulty is still encountered

here the term e^{-T} has a pole at ∞ which means that there is a pole on the contour of integration. To ensure that the integral converges on the infinite semicircle in the left half of the plane, a new parameter, m , must be introduced such that

$$A = 1.$$

(4-192)

Now replacing A by $1/m$ in Eq. (4-188), we can show that Eq. (4-191) becomes

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or

$$C(z, m) e^{-ST}.$$

$$\frac{1}{C+j\omega} = \frac{e^{-ST}}{2\pi j} \int_{C-j\infty}^{C+j\infty} C(s) e^{sm} ds$$

$$= \frac{e^{-ST}}{2\pi j} \int_{C-j\infty}^{C+j\infty} C(s) e^{sm} ds$$

$$= \frac{e^{-ST}}{2\pi j} \int_{C-j\infty}^{C+j\infty} C(s) e^{sm} ds$$

$$= \frac{e^{-ST}}{2\pi j} \int_{C-j\infty}^{C+j\infty} C(s) e^{sm} ds$$

$$= \frac{e^{-ST}}{2\pi j} \int_{C-j\infty}^{C+j\infty} C(s) e^{sm} ds$$

$$= \frac{e^{-ST}}{2\pi j} \int_{C-j\infty}^{C+j\infty} C(s) e^{sm} ds$$

$$= \frac{e^{-ST}}{2\pi j} \int_{C-j\infty}^{C+j\infty} C(s) e^{sm} ds$$

$$= \frac{e^{-ST}}{2\pi j} \int_{C-j\infty}^{C+j\infty} C(s) e^{sm} ds$$

$$C(z, m) = e^{-ST} \frac{1}{2\pi j} \int_{C-j\infty}^{C+j\infty} C(s) e^{sm} ds$$

$$= \frac{e^{-ST}}{2\pi j} \int_{C-j\infty}^{C+j\infty} C(s) e^{sm} ds$$

$$= \frac{e^{-ST}}{2\pi j} \int_{C-j\infty}^{C+j\infty} C(s) e^{sm} ds$$

$$= \frac{e^{-ST}}{2\pi j} \int_{C-j\infty}^{C+j\infty} C(s) e^{sm} ds$$

$$= \frac{e^{-ST}}{2\pi j} \int_{C-j\infty}^{C+j\infty} C(s) e^{sm} ds$$

$$= \frac{e^{-ST}}{2\pi j} \int_{C-j\infty}^{C+j\infty} C(s) e^{sm} ds$$

since $C(s)$ vanishes on the infinite semicircle in the left half of the plane. Applying the residue theorem, Eq. (4-194) is reduced to

$$C(z, m) = e^{-ST} \sum \text{Residues of } C(s) e^{sm} \text{ at the poles of } C(s)$$

$$= \frac{e^{-ST}}{2\pi j} \int_{C-j\infty}^{C+j\infty} C(s) e^{sm} ds$$

$$= \frac{e^{-ST}}{2\pi j} \int_{C-j\infty}^{C+j\infty} C(s) e^{sm} ds$$

$$= \frac{e^{-ST}}{2\pi j} \int_{C-j\infty}^{C+j\infty} C(s) e^{sm} ds$$

$$= \frac{e^{-ST}}{2\pi j} \int_{C-j\infty}^{C+j\infty} C(s) e^{sm} ds$$

$$= \frac{e^{-ST}}{2\pi j} \int_{C-j\infty}^{C+j\infty} C(s) e^{sm} ds$$

$$C(z, m) = \frac{1}{z^1} x^{-1}$$

$$= \frac{1}{z^1} x^{-1}$$

$$= \frac{1}{z^1} x^{-1}$$

$$= \frac{1}{z^1} x^{-1}$$

$$= \frac{1}{z^1} x^{-1}$$

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$$= \frac{1}{z^1} x^{-1}$$

$$= \frac{1}{z^1} x^{-1}$$

$$= \frac{1}{z^1} x^{-1}$$

If $C(s)$ is a rational function with k simple poles, we may represent $C(s)$ as the quotient of two polynomials; thus

$$C(s) = \frac{N(s)}{D(s)}$$

$$= \frac{N(s)}{D(s)}$$

$$= \frac{N(s)}{D(s)}$$

$$= \frac{N(s)}{D(s)}$$

$$= \frac{N(s)}{D(s)}$$

$$= \frac{N(s)}{D(s)}$$

$$= \frac{N(s)}{D(s)}$$

$$C(x, m) = x^{1/2} D(S) I$$

2

n=

$$D'(Sn) = dD(S) I$$

de

and s, designates the poles of C(S).

1

(4-197)

-

emrs

ez-1

(4-198)

(4-199)

Still another alternate form of the modified z-transform can be obtained by performing the contour integration in the right half of the plane along the semicircle. The closed path encircles the singularities of $1/(1 - e^{-T(-)})$. In this case, the notation need not be changed from A to 1 m, since e^{-AT} does not have poles on the right-half -plane contour. Therefore, the modified z-transform of c(t) given by Eq. (4-191) is now written as

-

1

1

$$C(z, A) = 2; \int_0^{\infty} C(s) e^{-\frac{1}{2}x-n} d\hat{s} \dots n$$

-ATE

1

$$T(s-5) de$$

de

|z=eTs

(4-200)

Equation (4-200) is a valid representation of the modified z-transform defined by Eq. (4-191) if the integration along the semicircle in the right half of the $\hat{s}$ -plane is convergent. Observe that in Eq. (4-200) the pole of $1/(1 - e^{-T(-)})$ on the semicircle in the right half $\hat{s}$ -plane is cancelled by the zero of e^{-T} which is at $+\infty$. This result implies that the integral along the semicircle will converge as long as C(s) satisfies the condition:

-

$$\lim_{s \rightarrow \infty} C(s) = 0$$

84 ∞

(4-201)

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In other words c(t) may possess a jump (but no impulse) at $t = 0.t$

Applying the residue theorem, we reduce the integral in Eq. (4-194) to

ΔTE

$$C(z, A) = - \text{Residues of } C(s) e^{-ATE}.$$

-

-

1

at the poles of $1/1 - e^{-T(-)}$,

1

-

$$- T(s - \hat{s})$$

e

$\alpha \hat{s}$

eTs

(4-202)

Since $1/1 - e^{-T(-)}$ has simple poles at $s = s_{\pm jn\omega_s}$, Eq. (4-196) may

$\hat{s}$

be written as

$$C(z, A) = -$$

$-\sum_{n=-\infty}^{\infty}$

$$N(S_n)$$

$$N=-8$$

In Chap. 2 we saw that

$$N(S_n)/D'(S_n) = -1/T$$

$$(4-203)$$

=ets

$$(4-204)$$

Substitution of Eq. (4-204) into Eq. (4-203) yields

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or

$$C^*(x, \Delta) = \sum_{n=-\infty}^{\infty} C(s + jn\omega) e^{-\alpha(s + jn\omega)}$$

$$T \quad 1=1\infty$$

00

$$C^*(s, A) = \sum_{n=-\infty}^{\infty} C(s + jn\omega) e^{-A(s + jn\omega)}$$

n=-∞

Substituting $A = 1/T$ into Eq. (4-206), we have

- -

00

T

$$(4-205)$$

$$(4-206)$$

$$C^*(s, m) = - \sum_{n=-\infty}^{\infty} C(s + jn\omega) e^{-(1-m)(s + jn\omega)}$$

$$(4-207)$$

which may be regarded as an alternative form of the modified z-transform of $c(t)$. In a later chapter we shall see that Eq. (4-207) may be conveniently applied to the statistical analysis and design of sampled-data systems.

The Inverse Modified z-Transform The major contribution of the modified z-transform is that it gives information on the actual outputs of linear sampled-data systems. The modified z-transform of $c(t)$, $C(z, m)$, contains all the information of $c(t)$ between sampling instants. The process of the inverse modified z-transform is denoted by $3'$, and may be effected by means of either the power series expansion method or the inversion formula.

† Note that in the case of the z-transform formula given in Eq. (2-68), the time function must not have any jump discontinuity at $t=0$. We have seen in Chapter 2 that if the time function has a jump at $t = 0$, the z-transform of the time function

$c(t)$ evaluated by taking the contour integral around the right half of the s -plane path

provides only one half of the value of $c(0+)$. In the case of the modified z-transform,

Eq. (4-200) leads to the correct answer even if $c(t)$ jumps at $t = 0$ simply because

the time function is first shifted to the right by a period of AT , where Δ is between $n = 0$

and 1; then the shifted function $c(t - AT)$ is sampled by the fictitious sampler at the

regular sampling instants: $0, T, 2T, 3T, \dots$. Therefore, unless $\Delta = 0$ or $\Delta = 1$, the

initial value of $c(t)$, $c(0+)$ is never sampled.

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(a) **The Power Series Method.** In this method the actual output response $c(t)$ is obtained by expanding $C(z, m)$ into a power series in z by long division. Just as in the z-transformation, if $C(z, m)$ is written as a ratio of two polynomials in z as

$$C(x, m) = \frac{a_0(m)x^0 + a_1(m)x^1 + \dots + a_{k-1}(m)x^{k-1} + a_k(m)x^k}{b_0(m)x^0 + b_1(m)x^1 + \dots + b_{k-1}(m)x^{k-1} + b_k(m)x^k}$$

(4-208)

it can be expanded into a power series in z^{-1} by dividing the denominator into the numerator using long division. Therefore, Eq. (4-208) is written as

$$C(z, m) = \frac{A(m)z^2 + B(m)z^{k-j-1} + \dots}{1}$$

(4-209)

Comparing Eq. (4-209) with Eq. (4-177), we see that the coefficient $A(m)$ corresponds to $c(j - k + m - 1)T$; $B(m)$ corresponds to $c(j - k + 1 + m - 1)T$; etc. When m varies continuously between zero and unity, the response $c(t)$ at all times is given by the coefficients $A(m)$, $B(m)$, ... The details of the method are illustrated by the following examples.

Example 4-18

The time function $g(t) = e^{-rt}$ given in Example 4-17 is to be reconstructed from its modified z -transform by use of the power series expansion method. The modified z -transform of $g(t)$ is given by Eq. (4-184).

$$G(z, m) = \frac{e^{-mT}}{1 - e^{-rT}z^{-1}}$$

Dividing the denominator into the numerator yields

$$G(z, m) = e^{-mrT}z^{-1} + e^{-(m+1)T}z^{-2} + e^{-(m+2)T}z^{-3} + \dots + e^{-(m+n)T}z^{-(n+1)} + \dots$$

(4-210)

(4-211)

The time function $g(t)$ during the first sampling period is given by the coefficient e^{-mr} of the z^{-1} term in Eq. (4-211). When $m = 0$ ($A = 1$), the value of $g(t)$ at $t = 0$ is obtained; and when $m = 1$ ($A = 0$), $g(T^-)$ is obtained. The values of n and m prescribed for the beginning and the end of each succeeding sampling period are tabulated as follows:

1st sampling period:

$$n = 1, m = 0, g(0^+) = g(T, 0) = e^{-mrT} |_{m=0}$$

$$n = 1, m = 1, g(T^-) = g(T, 1) = e^{-mT} |_{m=1} = e^{-T}$$

2nd sampling period:

$$n = 2, m =$$

$$\begin{aligned} n &= 2, m = 0, g(T) = g(2T, 0) \\ n &= 2, m = 1, g(2T^-) = g(2T, 1) \end{aligned}$$

$$\begin{aligned} &= e^{-(m+1)T} |_{m=0} \\ &= e^{-T} \\ &= e^{-(m+1)T} |_{m=1} = e^{-2T} \end{aligned}$$

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nth sampling period:

$$n = n,$$

$$n = n,$$

$$m =$$

$$m =$$

$$= e^{-(n-1)T}$$

$$0, g[(n-1)T^*] = g(nT, 0) = e^{-(m+n-1)T} \mid m=0$$

$$= 1, g(nT^-) = g(nT, 1) = e^{-(m+n-1)T} \mid m+1 = e^{-nr}$$

The response between any two consecutive sampling instants is determined by assigning values for m continuously from zero to unity. The total time function $g(nT, m)$ which is the inverse modified z-transform of $G(z, m)$ is shown in Fig. 4-25.

$$g(t)$$

$$e^{-mT}$$

$$m=0$$

$$1$$

$$e^{-(m+1)T}$$

$$g(nT, m)$$

$$e^{-(m+2)T}$$

$$0$$

$$m=1$$

$$m=0$$

$$m=1$$

$$m=0$$

$$m=1$$

$$m=0$$

$$T$$

$$2T$$

$$3T$$

$$4T$$

$$5T$$

Fig. 4-25 Inverse modified z-transform of $G(z, m)$

Example 4-19

$$e^{-mr}$$

$$Z-$$

The sampled-data system shown in Fig. 4-14 is to be analyzed by the modified z-transform method. The output response of the system shown in Fig. 4-16 is obtained by means of the submultiple-sampling method in Example 4-15. A block diagram of the system with the fictitious time delay for the application of the modified z-transform is depicted in Fig. 4-26. From Example 4-18 the modified z-transform of the system transfer function is obtained as

$$mT$$

$$G(z, m)$$

$$=$$

$$\sim$$

$$e$$

$$(4-212)$$

The modified z-transform of the output response is given by Eq. (4-176).

$$C(z, m) = E(z)G(z, m)$$

and for a unit-step function input $C(z, m)$ becomes

$$-mT$$

$$C(z, m)$$

$$e$$

$$=$$

$$-$$

$$2 \quad 1 \quad z \quad e$$

$$(4-213)$$

$$(4-214)$$

Expanding Eq. (4-214) by long division yields the following power series:

$$C(z, m)$$

$$T = e^{-mT-1} + e^{-mT} (1 + e^{-T}) x^2 + e^{-mr}(1 + e^{-T} + e^{-2T})x - 3$$

$$e$$

$$+$$

$$\bullet$$

$$\bullet$$

$$+e^{-mr}$$

$$-nT$$

$$-$$

$$+$$

$$(4-215)$$

The output responses $c(t)$ at the beginning and the end of various sampling periods are tabulated as follows:

1st sampling period:

$$n = 1, m = 0,$$

$$c(0) = e^{-mT} \mid m$$

$$m=0=$$

$$: 1$$

$$n =$$

$$1, m = 1,$$

$$c(T) = e^{-mT} \mid m=1 = e^{-r}$$

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$$e^{*}(t)$$

$$\frac{1}{s} e^{*}(t) X^{-1}(1) G(s) = + + +$$

$$1$$

$$-$$

$$AT=(1-m)T$$

$$s+1$$

Fictitious delay

Fictitious sampler

$$C(z, m)$$

$$c(nT, m)$$

$$c(t)$$

Fig. 4-26 Open-loop sampled-data system with fictitious delay to evaluate response between sampling instants by the modified z-transform method.

2nd sampling period:

$$n = 2, m = 0,$$

$$c(T+) = e^{-mr}(1 + e^{-T}) \mid m=0 = 1 + e^{-T}$$

$$n = 2, m = 1,$$

$$c(2T-) = e^{-mr}(1 + e^{-T}) \mid m=1 = e^{-T}(1 + e^{-T})$$

nth sampling period:

$$n = n, m = 0, c(n-1)T+$$

$$=e^{-mr}$$

$$1 - e$$

$$1 - e$$

$$m=0$$

$$n = n,$$

$$m = 1, c(nT-) = e$$

$$-mr1$$

$$-e$$

$$nT$$

$$-$$

$$m=1$$

$$=$$

$$=$$

$$1 - e^{-nr}$$

$$1 - e^{-T}$$

-
 1 e-T
 The output response c(t) between any two consecutive sampling instants is determined by assigning values of m continuously from zero to unity. For instance, the response during the first sampling period is obtained from e-mr when m is varied from zero to unity. The response during the second sampling period is obtained from e-mr(1 + e-7) when m is varied from zero to unity. The output response c(t) thus obtained is sketched in Fig. 4-27. The Inversion Formula Method Like the z-transform method, the inverse modified z-transform can also be obtained by means of the inversion integral

$$c(t) = \frac{e^{-mT}(1+e^{-T})}{e^{-mT}(1+e^{-T}+e^{-2T})} c(nT, m)$$

m=0
 m=0
 1
 m=0
 e-m
 www
 m=1
 m=1
 m=1
 o
 T
 27 37 4T 5T 6T
 77

Fig. 4-27 Output response obtained by the modified z-transform method for the sampled-data system of Fig. 4-26.

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$$c(nT, m) = \frac{1}{2\pi j} \oint_C C(z, m) z^{n-1} dz \quad (4-216)$$

where Γ is a closed path in the z-plane that encloses all the singularities of $C(z, m)z^{n-1}$. We can show that when $m = 1$, the inversion integral for the modified z-transform reverts to that of the z-transform. During the sampling period between $t = (n - 1)T$ and $t = nT$,

- the value of $c(n - 1)T+$ is obtained from $c(nT, m)$ by assigning zero value for m; the value of $c(nT-)$ is obtained by substituting $m = 1$ in $c(nT, m)$.

Example 4-20

The same system given in Example 4-19 is analyzed by use of the inversion formula for modified z-transform. Substituting the expression of $C(z, m)$ from Eq. (4-214) into Eq. (4-216), we have

$$c(nT, m) =$$

$$\frac{1}{2} \sum_{j=0}^{\infty} x_j f \quad e^{-mT} \quad e^{-1} \quad - \quad 5^k - 1$$

dz
(4-217)
where Γ may be a circle in the z -plane centered at the origin; the circle should be large enough to enclose the poles of the integrand of Eq. (4-217) 1 and $z = e$. Applying the residue theorem to Eq. (4-217) yields

at $x =$
 $c(nT, m) = e^{m}$
 $\sum$ Residues of
 z^*

$(\approx - 1) (\approx - e^{-1})$
at the poles $z = 1, z = e^{-1}$.

Equation (4-218) can be further reduced to
(4-218)

$c(nT, m) = e$

$-mT$
(4-219)

1

e
 nT
 e
(4-219)

We see that this result agrees with that obtained in Example 4-19.

4.9 THEOREM of the MODIFIED Z-TRANSFORMS

Several basic theorems similar to the z -transforms may simplify and facilitate the application of the modified z -transform method to the studies of sampled-data systems. The derivations and applications of some of these theorems are as follows:

1. Real Translation If the modified z -transform of the time function $e(t)$ is $E(z, m)$, then

$\sum_m e(t \pm nT) = z^{\pm 1} E(z, m)$

(4-220)

where n is an integer. This theorem is actually the same as the delayed z -transform discussed in Sec. 4.8 where the delayed z -transform is given by Eq. (4-161). Therefore, no additional proof is necessary here.

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2. Complex Translation The complex translation theorem states

that if $c(t)$ has the modified z -transform $E(z, m)$, then

$\sum_m [e^{aT} c(t)] = \sum_m [E(z \pm a)] = e^{\pm aT} E(z, m)$ (4-221)

Proof: By definition

$\sum_m [e^{aT} c(t)] = \sum_m E(z \pm a)$

$=$
 $=$
 $L[e^{aT} f_a(t) e(t - aT) \delta(t)] s = (1/T) \ln z$

$L[z^p] = a(nT - 27) e(nT - \backslash T) \delta(t - nT)] \cdot (1/1) \ln z$

$n=0$

Since $A \neq 1$, Eq. (4-222) becomes.

$=$
 $\sum_m [e^{aT} c(t)] = f[e$

$f_a m T$

or

00

770

(4-222)

$\sum_m [e^{aT} c(t)] = f[e^{aT} c(t)] \dots$

$$\sum_{n=0}^{\infty} e^{a(n+m)T} z^{n+m} = e^{amT} z^m \sum_{n=0}^{\infty} e^{anT} z^n = e^{amT} z^m \frac{1}{1 - ze^{aT}}$$

Multiplication of the right-hand side of Eq. (4-224) by e^{-ar} yields

$$\sum_{n=0}^{\infty} e^{[(n+m)T](ze^{aT})^{-n}} = e^{-ar(m-1)} \sum_{n=0}^{\infty} e^{[(n+m)T](ze^{aT})^{-n}} \quad (4-225)$$

which is then reduced to the following form:

$$\sum_{n=0}^{\infty} e^{[(n+m)T](ze^{aT})^{-n}} = e^{-ar(m-1)} \sum_{n=0}^{\infty} e^{[(n+m)T](ze^{aT})^{-n}} \quad (4-226)$$

$$\sum_{n=0}^{\infty} e^{[(n+m)T](ze^{aT})^{-n}} = \sum_{n=0}^{\infty} E(s \pm a) = e^{-ar(m-1)} E(ze^{aT}),$$

Example 4-21

Given $e(t)$ and $E(s)$ as $e(t) = t$ and $E(s) = 1/s^2$, respectively, evaluate the modified z-transform of $e^{-a}e(t)$.

By use of the complex translation theorem stated in Eq. (4-221),

$$\sum_{n=0}^{\infty} e^{-(s+a)^2} = \sum_{n=0}^{\infty} e^{-ar(m-1)} E(ze^{aT}, m) \quad (4-228)$$

From Appendix A, the modified z-transform of $e(t)$ is found as

$$\sum_{n=0}^{\infty} (1) = \sum_{n=0}^{\infty} (12)$$

hence

$$\sum_{n=0}^{\infty} (t) = \sum_{n=0}^{\infty} (32) = [$$

Equation (4-230) is simplified to

$$\sum_{n=0}^{\infty} e^{-ar(m-1)} E(ze^{aT}, m) = \sum_{n=0}^{\infty} e^{-ar(m-1)} E(ze^{aT}, m) \quad (4-229)$$

$$\sum_{n=0}^{\infty} e^{-ar(m-1)} E(ze^{aT}, m) = \sum_{n=0}^{\infty} e^{-ar(m-1)} E(ze^{aT}, m) \quad (4-230)$$

$$\sum_{n=0}^{\infty} (t) = \sum_{n=0}^{\infty} (12)$$

$$\sum_{n=0}^{\infty} e^{-ar(m-1)} E(ze^{aT}, m)$$

$$\sum_{n=0}^{\infty} e^{-ar(m-1)} E(ze^{aT}, m)$$

$$\sum_{n=0}^{\infty} e^{-ar(m-1)} E(ze^{aT}, m)$$

$$\sum_{n=0}^{\infty} e^{-ar(m-1)} E(ze^{aT}, m)$$

$$\sum_{n=0}^{\infty} e^{-ar(m-1)} E(ze^{aT}, m)$$

$$\sum_{n=0}^{\infty} e^{-ar(m-1)} E(ze^{aT}, m)$$

~

e-ar)?

(4-231)

A check on the modified z-transform table in Appendix A shows that this is the correct answer for the modified z-transform of e-att.

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3. Initial Value Theorem The initial value theorem states that if the time function e(t) has the modified z-transform E(z, m), the initial value of e(t) may be determined by the following relation:

$$\lim_{m \rightarrow 0} e(nT, m) = \lim_{m \rightarrow 0} zE(z, m)$$

lim e(t)

=

n-0

m=0

28400

(4-232)

7-.0*

if the limit exists.

m=0

Proof: The proof of this theorem actually follows directly from the initial value theorem for the ordinary z-transforms. By definition the modified z-transform of e(t) is

-1

00

$$E(z, m) = z^{-1} \sum_{n=0}^{\infty} e(nT + mT) z^{-n}$$

n=0

-n

(4-233)

Multiplying both sides of the last equation by z and taking the limit as m approaches zero, we have

$$\lim_{m \rightarrow 0} z E(z, m) = \sum_{n=0}^{\infty} e(nT) z^{-1}$$

m=0

7=0

(4-234)

which is recognized as the ordinary z-transform of e(t). Therefore, from Eq. (4-85), the relation given in Eq. (4-232) is obtained.

Example 4-22

To illustrate the application of the initial value theorem for modified z-transforms, the initial value of the time response of the system in Example 4-18 will be determined. In Example 4-18 the modified z-transform of the system impulse response g(t) is

-m T

e

$$G(z, m)$$

=

2 -e

The value of g(t) at t =

-

0 is given by Eq. (4-232) as

-MT

$$\lim_{m \rightarrow 0} g(t)$$

=

$$\lim_{m \rightarrow 0} zG(z, m)$$

ze

=

L-.0*

lim

m=0 2

1
 (4-235)
 (4-236)
 $m \rightarrow 0$
 $z \rightarrow 00$
 $z \rightarrow 00$

In a similar manner, the initial value of the system output $c(t)$ in Example 4-19 is obtained by use of Eq. (4-232). Thus

$$\lim_{t \rightarrow 0^+} c(t)$$

$$m = 0$$

$$\lim z$$

$$\frac{c}{z}$$

$$\frac{e^{-mr}}{1}$$

$$\frac{1}{1}$$

$$\frac{1}{z}$$

$$\frac{2100}{(4-237)}$$

4. Final Value Theorem

The final value theorem for the modified z -transform states that if the time function $e(t)$ has the modified z -transform $E(z, m)$ and if the function $[(z - 1)/2]E(z, m)$ possesses no poles on or outside of the unit circle in the z -plane, then

$$\lim_{n \rightarrow \infty} e(nT, m)$$

$$= \lim_{z \rightarrow 1} (z - 1) E(z, m)$$

$$(4-238)$$

$$\lim_{z \rightarrow 1} (z - 1) E(z, m)$$

Proof: The proof of this theorem can be effected in much the

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same way as that of the final value theorem for the ordinary z -transforms. Therefore, it is left as an exercise at the end of this chapter.

The final value theorems for the z -transforms and the modified z -transforms normally will give identical results to the final value of a time function $c(t)$ if there are no ripples or hidden oscillations between sampling instants in $e(t)$. In general, if ripples between sampling instants or discontinuities at the sampling instants exist, the z -transform method is inadequate and usually will bring misleading results. The final value theorem stated in Sec. 4.5 by Eq. (4-88) could conceivably give the final value of $e(t)$ only at the sampling instants. However, if the final value theorem for the modified z -transform is used where hidden oscillation exists, the final value of $e(t)$, represented by

$$\lim_{n \rightarrow \infty} e(nT, m)$$

$$= \lim_{z \rightarrow 1} (z - 1) E(z, m)$$

$$(4-239)$$

will be a function of m ; and as the value of m is varied from 0 to 1 (inclusive), the final value of $e(t)$ during one sampling period is determined.

Example 4-23

The final values of the time response of the systems in Examples 4-18 and 4-19 are evaluated by use of Eq. (4-238). From Fig. 4-25 we see that the inverse modified z-transform of $G(z, m) = e^{-mr}/(x - e^{-7})$ does not possess any ripple or discontinuity between or at the sampling instants. Therefore, we should expect that Eqs. (4-88) and (4-238) will both provide identical results for $g(t)$ as t tends to infinity. Using Eq. (4-238), we have

$$\lim_{t \rightarrow \infty} g(t) = \lim_{n \rightarrow \infty} g(nT, m) \\ = \lim_{n \rightarrow \infty} (z^{-1})^n = 0 \quad (4-240)$$

We can easily see that Eq. (4-88) will give the same result as Eq. (4-240). From Fig. 4-27 we see that the system output response of the system in Example 4-19 has discontinuities at the sampling instants. It has been shown previously that the final value of $c(t)$ is between 0.585 and 1.585. In the steady state $c(t)$ jumps from 0.585 to 1.585 at the sampling instants. Therefore, it is essential that the final value theorem for the modified z-transforms be used in this case. From Eq. (4-238), the final value of $c(t)$ is

$$\lim_{t \rightarrow \infty} c(t) = \lim_{n \rightarrow \infty} c(nT, m) = \lim_{m \rightarrow 1} (z^{-1})^m = 0$$

$$(4-241)$$

and is a function of m . The fact that the final value of $c(t)$ is a function of m is an indication that ripples or hidden oscillations exist between sampling instants in $c(t)$. It is easy to see that by assigning values for m from 0 to unity in Eq. (4-241), the steady state response of $c(t)$ during one sampling period is obtained.

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5

BLOCK DIAGRAM, SIGNAL FLOW GRAPH, AND MATRIX

REPRESENTATION OF SAMPLED-DATA SYSTEMS

5.1 BLOCK DIAGRAM ANALYSIS AND TRANSFER FUNCTIONS OF CLOSED-LOOP SAMPLED-DATA SYSTEMS

In the preceding chapter the block diagrams of simple open-loop systems with sampled-data have been shown, and they are analyzed by means of the z-transform method. We have seen that when the sampling operation is represented by a sampling switch (or an impulse modulator), the block diagram concept, used so effectively in the analysis of continuous-data systems, can also be adapted to the studies and analysis of sampled-data systems.

In short, a block diagram is simply a graphic shorthand notation representing the interaction between elements of a physical system. Like the block diagrams that lead to the transfer functions or output transforms of continuous signal systems, the transfer functions or output transforms for sampled-data systems can also be evaluated with the help of block diagram algebra. Furthermore, for sampled-data systems the process of transfer function evaluation may be effected in either the s- or the z-domain; the transfer function, for instance, may be represented by either the pulse transfer function $G^*(s)$ or the z-transfer function $G(z)$.

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With Cascaded Elements

For open-loop systems with cascaded elements, the case where synchronized samplers exist between elements must be distinguished from the case where the elements are connected directly together. For instance, the two systems shown in Fig. 5-1a and Fig. 5-1b be-

Fictitious

sampler

$c(1)$

$T^* C^*(s)$

$e(t) e^*(1)$

$C(z)$

$G_1(s)$

$G_2(s)$

$c(t)$

$C(s)$

Synchronized

samplers

(a)

Fictitious

$c^*(1)$

sampler

$C^*(s)$

$C(z)$

$c(t)$

$C(s)$

$e(t)$

$e^*(1)$

$$G_1(s)$$

$$G_2(s)$$

(b)

Fig. 5-1 Open-loop sampled-data systems with cascaded elements.

(a) Cascaded elements with sampler in between elements. (b)

Cascaded elements connected directly together.

have in entirely different ways and thus need separate treatment.

In Sec. 4-6 we have seen that the Laplace transform, the pulse transform, and the z-transform of the output response of the system in Fig. 5-1a are respectively

and

G^*

$$C(s) = G_1(s) G_2(s) E^*(s)$$

=

$$C^*(s) = G^*(s) G^*(s) E^*(s)$$

$$C(z) = G_1(z) G_2(z) E(z)$$

(5-1)

(5-2)

(5-3)

From Eqs. (5-2) and (5-3) the pulse transfer function of the open-loop system is simply

$$C^*(s)$$

$$E^*(s)$$

=

$$G^*(s) G^*(s)$$

and the z-transform of Eq. (5-4) is

$$C(z)$$

$$= G_1(z) G_2(z)$$

$$E(z)$$

(5-4)

(5-5)

Therefore, for cascaded elements with synchronized samplers between

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all elements, the over-all pulse-(or z-) transfer function of the open-loop system is generally equal to the product of the pulse-(or z-) transfer functions of the individual elements. This is not the case for the system shown in Fig. 5-1b. The transfer functions derived in Sec. 4-6 show that the Laplace transform of the continuous output signal is

$$C(s) = G_1(s) G_2(s) E^*(s)$$

(5-6)

The pulse transform of $c(t)$ or the Laplace transform of $c^*(t)$ is given by

=

$$C^*(s) = G_1(s) G_2(s) E^*(s)$$

$$G_1 G^*(s) E^*(s)$$

where $G_1 G^*(s)$ is defined as

$$G_1 G^*(s) = G_1 G^\dagger(s) = \sum_{n=0}^{\infty} G_1(s + jn\omega_s) G_2(s + jn\omega_s)$$

(5-7)

T

$N = \infty$

and the z-transform of $c(t)$ is

(5-8)

(5-9)

$$C(z) = G_1 G_2(z) E(z)$$

=

where $G_1 G_2(z)$ denotes the z-transform of $G_1 G^*(s)$ or $G_1(s) G_2(s)$.

Closed-Loop Sampled-Data Systems

The derivation of transfer functions and output response transforms of closed-loop sampled-data systems follows the same method

as that used for the open-loop systems. Because of possible variations in the location of the sampler, the transfer function for even a simple single-loop system with sampled-data, unlike that of a continuous-data system, depends entirely on where the sampler (or samplers) is (are) located. A single-loop sampled-data system with error-sampling is shown in Fig. 5-2. The transfer function of the system and the output response transform are derived as follows:

From the definition of the Laplace transform, the transform of

$c(t)$ is

Fictitious

sampler

$c^*(t)$

$T C^*(s)$

$r(t)$

$R(s)$

$e(t)$

$e^*(t)$

$G(s)$

$E(s)$

$\uparrow E^*(s)$

$c(t)$

$C(s)$

$H(s)$

Fig. 5-2 Closed-loop sampled-data system.

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$C(s) = G(s)E^*(s)$

The Laplace transform of the continuous error signal $e(t)$ is

$E(s) = R(s) - C(s)H(s)$

—

Substituting $C(s)$ in Eq. (5-10) into Eq. (5-11) yields

$E(s) = R(s) - H(s)G(s)E^*(s)$

Taking the pulse transform of Eq. (5-12) yields

10

01

$E^*(s) = R^*(s) - [H(s)G(s)E^*(s)]^*$

—

$E^*(s) = R^*(s) - [H(s)G(s)]^*E^*(s)$

$= R^*(s) - HG^*(s)E^*(s)$

(5-10)

(5-11)

(5-12)

(5-13)

(5-14)

Solving for $E^*(s)$ in Eq. (5-14), we have the pulse transform

of $e(t)$

$E^*(s)$

=

$R^*(s)$

$1 + HG^*(s)$

(5-15)

The Laplace transform of the continuous output response is obtained

by substituting $E^*(s)$ from Eq. (5-15) into Eq. (5-10). Hence

$C(s)$

=

$G(s) R^*(s)$

$1 + HG^*(s)$

(5-16)

The pulse transform of $c(t)$ or the Laplace transform of $c^*(t)$ is

obtained by taking the pulse transform of Eq. (5-16), thus

$C^*(s) = G^*(s)E^*(s)$

=

$$\frac{G^*(s)}{1 + HG^*(s)} = \frac{R^*(s)}{C^*(s)} \quad (5-17)$$

The z-transform of $c(t)$ is obtained from the last equation simply by changing variables of $s =$

$$T \ln z \text{ yielding} \\ C(z)$$

$$= \frac{R(z)}{1 + HG(z)} \quad (5-18)$$

As another example of the derivation of the closed-loop transfer function of a sampled-data system, consider the system shown in Fictitious sampler

$$\begin{aligned} & T \quad C^*(s) \\ & C(z) \\ & r(t) \\ & e(t) \\ & G(s) \\ & R(s) \\ & E(s) \\ & c(t) \\ & C(s) \\ & c(1) \\ & H(s) \\ & C(s) \end{aligned}$$

Fig. 5-3 Closed-loop sampled-data system.

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Fig. 5-3. In this system the output signal $c(t)$ is sampled by a sampler, and the sampled signal is fed back through the element $H(s)$ to the input. The signal at the output of $H(s)$ is continuous with respect to time. The output transforms: $C(s)$, $C^*(s)$, and $C(z)$ are derived as follows. In this case, a so-called transfer function for the closed-loop system cannot be derived because the reference input R is combined with the transfer function $G(s)$ of the forward path. The Laplace transform of $c(t)$ is

$$C(s) = E(s)G(s)$$

and the Laplace transform of the error signal $e(t)$ is

$$E(s) = R(s) - H(s)C^*(s) \quad (5-19)$$

$$(5-20)$$

$$(5-21)$$

$$(5-22)$$

Substituting Eq. (5-20) into Eq. (5-19) yields

$$C(s) = R(s)G(s) - G(s)H(s)C^*(s)$$

Taking the pulse transform on both sides of Eq. (5-21) yields

$$C^*(s) = GR^*(s) - HG^*(s)C^*(s)$$

where

$$GR^*(s) = RG^*(s) \quad (5-23)$$

$$(5-23)$$

$$= [G(s)R(s)]^*$$

Solving for $C^*(s)$ from Eq. (5-22), we have

$$C^*(s) =$$

$$\frac{GR^*(s)}{1 + HG^*(s)} \quad (5-24)$$

which is the pulse transform of $c(t)$. We see that the pulse transfer function in the form $C^*(s)/R^*(s)$ cannot be obtained in this case, since $GR^*(s)$ should be considered as one single expression. In order to obtain the continuous output $C(s)$, we substitute $C^*(s)$ into Eq. (5-21). Therefore,

$$\begin{aligned} &= \frac{C(s) G(s) R(s)}{H(s)GR^*(s)} \\ &= \frac{G(s)}{1 + HG^*(s)} \end{aligned}$$

From Eq. (5-24) the z-transform of $c(t)$ is obtained as

$$\begin{aligned} &C(z) \\ &= \frac{GR(z)}{1 + HG(z)} \end{aligned} \quad \begin{array}{l} (5-25) \\ (5-26) \end{array}$$

Since $GR(z)$ denotes the z-transform of $G(s)R(s)$, the z-transfer function $C(z)/R(z)$ cannot be obtained for this system. In a similar manner, the output transforms or transfer functions (if they exist) for some simple sampled-data system configurations are derived and are shown in Table 5-1, which is printed on the inside of the front and back cover. (In Table 5-1, the same results are also obtained by use of the signal flow graph method.)

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Output Transforms of More Complex System Configurations

In general, the labor involved in the derivation of the output transforms of sampled-data systems increases with the increase in complexity of the systems configurations. Unfortunately, the block diagram reduction technique devised for continuous feedback control systems cannot be efficiently applied to sampled-data systems simply because of the mixing of sampled and continuous signals in the latter. A simple example will illustrate this difficulty. In Fig. 5-4a two parallel paths are shown that may be thought of as part of the block diagram of a continuous-data system. We know that the two paths can be combined and reduced to a single block as shown in Fig. 5-4b. However, the two channels of the block diagram shown in Fig. 5-5 cannot be combined into one, since one path has sampled data and the other has a continuous signal.

As an illustration of the derivation of the output transforms of sampled-data systems with complex configurations, let us consider the block diagram shown in Fig. 5-6. With any experience in the

$$\begin{aligned} &G_1(s) \\ &E(s) \\ &C(s) \\ &E(s) \\ &C(s) \\ &G_1(s) + G_2(s) \\ &G_2(s) \end{aligned}$$

(a)
(b)

Fig. 5-4 Parallel paths with continuous data.

$$\begin{aligned} &E^*(s) G(s) \\ &E(s) \\ &+ \\ &C(s) \end{aligned}$$

$E(s)$
 $G_2(s)$
 Fig. 5-5 Parallel paths with continuous and sampled data.

$E^*(s)$
 $D^*(s)$
 $G_1(s)$
 Fictitious
 sampler
 $C(s)$
 Synchronized
 samplers

$C(z)$
 $c^*(1)$
 $R(s)$
 $G_2(s)$
 $E(s)$
 $E(s)$
 $c(1)$
 $C(s)$

Fig. 5-6 Block diagram of closed-loop sampled-data system.

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manipulation of block diagrams one will intuitively write down the following two equations directly from the system configuration:

The Laplace transform of the continuous output is

$$C(s) = G_2(s)E(s) + G_1(s)G_2(s)D^*(s)E^*(s)$$

and the Laplace transform of the continuous error signal is

$$E(s) = R(s) - C(s)$$

(5-27)

(5-28)

However, from here on the next step seems to be uncertain. We see that there are at least two alternatives: one is to take the pulse transform of both sides of Eq. (5-27); the other is to take the pulse transform of both sides of Eq. (5-28) first. A closer look reveals that since $E(s)$ and $E^*(s)$ are eventually to be eliminated from Eq. (5-27), the pulse transform of $C(s)$ should not be taken at this stage, since taking it will make the first term on the right hand side of Eq. (5-27) read $G_2E^*(s)$ and then $E^*(s)$ could not be eliminated from Eq. (5-27). Therefore, the logical step is to take the pulse transform of $E(s)$ in Eq. (5-28) and then substitute the resulting equation into Eq. (5-27). We have,

$$E^*(s) = R^*(s) - C^*(s)$$

Substitution of Eq. (5-29) into Eq. (5-27) yields

$$C(s) = G_2(s)R(s) - G_2(s)C(s)$$

Solving for $C(s)$ yields

$$C(s) = \frac{G_1(s)G_2(s)D^*(s)[R^*(s) - C^*(s)]}{1 + G_2(s)}$$

(5-29)

(5-30)

$C(s)$

=

$G_2(s)$

$1 + G_2(s)$

$R(s)$

+

$G_2(s)$

$1 + G_2(s)$

$G_1(s)D^*(s)[R^*(s) - C^*(s)]$

-

(5-31)

Now taking the pulse transform on both sides of Eq. (5-31), we have

$$C^*(s) = \frac{1}{1 + G_2(s)} \left[\frac{G_1(s)}{R(s)} + \frac{G_2(s)}{R(s)} \right] D^*(s)$$

$$[G_2(s)G_1(s)] D^*(s) [R^*(s) - C^*(s)]$$

Solving for $C^*(s)$ in the last equation yields

$$\frac{1 + G_2(s)}{[G_2(s)G_1(s)] D^*(s) R^*(s)}$$

$$+ \frac{1}{1 + G_2(s)}$$

$$C^*(s) = \frac{1 + G_2(s)}{1 + G_2(s)} D^*(s)$$

(5-32)

(5-33)

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The Laplace transform of $c(t)$ is obtained by substituting $C^*(s)$ from Eq. (5-33) into Eq. (5-31). After simplification, $C(s)$ reads

$$C(s) = \frac{G_2(s)}{1 + G_2(s)} R(s) + \frac{G_2(s)G_1(s)}{1 + G_2(s)} D^*(s)$$

$$R^*(s) - \frac{G_2(s)R(s)}{1 + G_2(s)}$$

This expression for $C(s)$ can be simplified still further if the term:

$$\frac{G_2(s)R(s)}{R^*(s) - [G_2(s)R(s)]}$$

is written in an equivalent form. Taking the inverse transform of the last term, we have

$$R(s) - \frac{G_2(s)R(s)}{1 + G_2(s)} R(s)$$

Substitution of the pulse transform of $R(s)/[1 + G_2(s)]$ into Eq.

(5-34) yields

$$C(s) = \frac{G_2(s)R(s)}{1 + G_2(s)} + \frac{G_2(s)G_1(s)}{1 + G_2(s)}$$

$$D^*(s) \\ R(s) \\ 1 + G_2(s) \\ (5-36) \\ 1 + \\ G_2(s)G_1(s) \\ 1 + G_2(s)] * D^*(s)$$

From the examples given above, we see that because the simultaneous equations of a sampled-data system contain both sampled and unsampled forms of the same variable, an analytic solution of the equations is quite difficult to obtain. In general, no definite rule exists that will lead to the optimum procedure of taking the pulse transform and making proper substitution of variables. Even an experienced analyst might find himself enmeshed in a confusion of tedious algebraic manipulation. In the following section the signal flow graph technique of solving sampled-data systems problems is introduced. As we shall see, when the signal flow graph of a sampled-data system is constructed, the output transforms (or the transforms of any system variable) as a function of the input signal can virtually be written by mere inspection from the signal flow graph.

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5.2 SIGNAL FLOW GRAPHS OF SAMPLED-DATA SYSTEMS

We have seen that system transfer functions of continuous-data systems can be derived from either the signal flow graph or the block diagram simply by use of Mason's gain formula. Now we shall investigate how the signal flow graph technique can be extended to the analysis and portrayal of systems with sampled-data. However, any attempt to apply Mason's gain formula directly to the block diagram or equivalent signal flow graph of a system with sampled-data will encounter immediate difficulty. As is evident, the main difficulty stems from the fact that sampling switches exist in sampled-data systems; since we cannot replace the samplers by their equivalent transfer functions, the conventional signal flow graph algebra simply does not know how to handle a flow graph with sampling switches. In a special case, if all the signals in a sampled-data system are sampled, then all the transferred quantities of the systems are starred and the general gain formula can still be applied directly. For instance, let us consider the sampled-data system shown in Fig. 5-7. The signal flow graph of the system is constructed in Fig. 5-8. Since only sampled signals exist throughout the system, the sampling switches are omitted from the signal flow graph. Now applying the gain formula to the signal flow graph with only sampled variables, the closed-loop transfer function of the system is obtained as

$$R^* \\ RA \\ E \\ C \\ G^*(s) \\ H^*(s)$$

Fig. 5-7 An all-sampled-data system.

Fig. 5-8 Equivalent signal flow graph of the sampled-data system shown in Fig. 5-7.

† R. Ash, W. H. Kim, and G. M. Kranc, "A General Flow Graph Technique for the Solution of Multiloop Sampled Systems," Trans. of A. S. M. E., Journal of Basic Engineering, June 1960, pp. 360-370.
J. M. Salzer, "Signal Flow Reductions in Sampled Data Systems," I. R. E., WESCON Convention Record, Part 4, 1957, pp. 166-170.

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$$\begin{aligned} C^*(s) \\ R^*(s) \\ = \\ G^*(s) \\ 1+G^*(s)H^*(s) \end{aligned}$$

(5-37)

We can conclude now that Mason's gain formula can be applied only to signal flow graphs (or block diagrams) that contain nodes either representing all continuous signals or all sampled signals but not mixed signals.

However, the majority of the sampled-data systems contain both sampled and continuous signals; therefore, the samplers cannot be omitted simply. A signal flow graph method must be devised to take care of systems with sampled as well as continuous data.

Next, we shall introduce a method of determining the output transforms (or transfer functions) of sampled-data systems by use of Mason's gain formula. The method involves the following steps: (1) With the system block diagram as the starting point, construct the "original signal flow graph" of the system. The original flow graph is defined as merely the equivalent flow graph of the system block diagram. For instance, Fig. 5-8 gives the original flow graph of the block diagram shown in Fig. 5-7; the original flow graph of the system in Fig. 5-2 is drawn in Fig. 5-9.

$$\begin{aligned} 1 \\ R \\ X_1 = E \\ X_2 \\ G \\ X_3 = C \\ -H \end{aligned}$$

Fig. 5-9 Original signal flow graph of the system in Fig. 5-2.

(2) The second step involves the construction of the "sampled signal flow graph" from information obtained from the original signal flow graph. This step needs some careful discussion. Let us consider the flow graph shown in Fig. 5-9. Note that the flow graph still contains the sampling switch so that the gain formula cannot be applied directly. However, the following set of equations is written from the flow graph:

$$\begin{aligned} X_1 &= ER - HX_3 \\ X_2 &= E^* X^* \\ X_2 \end{aligned}$$

$$\begin{aligned} = \\ X_1 = C = GX_2 \end{aligned}$$

(5-38)
(5-39)
(5-40)

where for simplicity $E^*(s)$ is denoted by E^* , $R(s)$ by R , $H(s)$ by H and $G(s)$ by G . X_1 , X_2 , and X , are Laplace transforms of flow graph variables at nodes 1, 2, and 3, respectively. Substituting Eq. (5-39) into Eq. (5-40), and then substituting Eq. (5-40) into Eq. (5-38), we have the following equations:

$$X_1 = HGX^* + R$$

$$\begin{aligned} - \\ - \\ (5-41) \end{aligned}$$

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$$X_2 = X^*$$

X_2

(5-42)

X_3

$$X_1 = GX^* =$$

$$= GX_2$$

(5-43)

=

Careful examination of Eqs. (5-41), (5-42), and (5-43) reveals that the same equations can be obtained if the sampler in the original flow graph of Fig. 5-9 is deleted and if an artificial signal source of signal strength X , $X = E^*$ is applied at node X_2 , the output node of the sampler. (Now the system flow graph has two separate inputs.) If now we take the pulse transform on both sides of Eqs.

(5-41) through (5-43), we have

$$X^* = -(HG)^*X^* + R^*$$

-

$$X^* = X^*$$

$$X^* = G^*X^*$$

(5-44)

(5-45)

(5-46)

Since the last three equations contain only sampled variables, the signal flow graph constructed from these equations is called the sampled signal flow graph of the system in Fig. 5-2 and is shown in Fig. 5-10. Note that the sampled flow graph has only the sampled variables: R^* , X^* , X^* , X^* and C^* .

-(HG)

0

X_2

Fig. 5-10 Sampled flow graph of the system in Fig. 5-2.

Now we can apply Mason's gain formula directly to the sampled flow graph shown in Fig. 5-10 for the evaluation of the sampled output $C^*(s)$ or $C(z)$. Once the sampled flow graph is drawn, all other dependent variables X , E^* , and X^* (in this case, $X = E^*$ = X) can also be expressed as functions of the input R^* by use of the gain formula. Therefore, from Fig. 5-10

$$C^*(s)$$

=

$$G^*(s)$$

$$1 + HG^*(s)$$

$$R^*(s)$$

1

and

$$E^*(s)$$

=

$$HG^*(s) R^*(s)$$

$$\text{where } HG^*(s) = (HG)^*.$$

$$1 + HG^*(s)$$

(5-47)

(5-48)

In general, if a multiloop sampled-data system contains several samplers (synchronized), then by deleting the samplers and adding artificial signal sources at the corresponding output nodes of the samplers, a set of equations similar to that of Eqs. (5-41) through (5-43) can be written. For instance, if the node variables of the

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original flow graph are represented by X_n ($n = 0, 1, 2, \dots$), then the following equation may be written for the variable X , at node j :

$$X_1 = \sum_k a_{1k} X_k +$$

$$\sum_k b_{kj} R_k \quad (5-49)$$

where it is assumed that a sampler is located between node i and node j and that signal is transmitted from node i to node j ; therefore, X is the sampled signal which flows to node j from the sampler following node i . a represents the transmission between node i and node j ; b denotes the transmission from node k to node j ; R is the signal applied at node k which flows to node j . Taking the pulse transform on both sides of Eq. (5-49) yields

$$X_j = \sum_i a_{ij} X_i + \sum_k (b_{kj} R_k)^*$$

where all the variables are now in sampled form.

$$(5-50)$$

This step can generally be omitted in writing Eq. (5-49). Once the samplers in the original flow graph are replaced by corresponding input signals, Eq. (5-50) can be written directly from the flow graph by applying Mason's gain formula and taking the pulse transform of the variables at the same time. For instance, Eqs. (5-44) through (5-46) can be written directly from Fig. 5-9 by applying the gain formula and taking the pulse transform at nodes X_1 , X_2 , and X , simultaneously.

(3) In the previous step we have demonstrated how the sampled variables of a sampled-data system can be expressed as functions of the inputs by use of the sampled signal flow graph and Mason's gain formula. However, the sampled flow graph does not contain the unsampled variables X_1 , X_2 , Since the actual output responses of most sampled-data systems are of the continuous type, the expressions of continuous responses are often desired. For this operation we shall define the "composite signal flow graph." A composite signal flow graph is a composition of the original flow graph and the sampled flow graph of a sampled-data system; the sampling switches in the original flow graph are deleted and are replaced by their corresponding artificial input sources. If on the original flow graph a sampler is located between node i and node j and the signal is transmitted from i to j , the sampling operation is described by drawing a branch of unity gain from node X^* of the sampled flow graph to node X , on the original flow graph. From the composite signal flow graph, all the sampled and the unsampled output variables can be determined as functions of the input signals by means of Mason's gain formula. Actually, once the composite signal flow graph is constructed, a separate sampled flow graph need not be constructed,

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since the former graph will contain all the information on the unsampled variables.

Let us use the simple sampled-data system shown in Fig. 5-2 to illustrate the construction of the composite flow graph introduced above. The original signal flow graph of the system is shown in Fig. 5-9, and the sampled signal flow graph is already obtained as shown in Fig. 5-10. As is apparent, the variables in the two flow graphs are related by Eq. (5-39); that is $X_2 = X^*$. Therefore, the composite flow graph is constructed by drawing a branch with unity gain from node X of the sampled flow graph to node X_2 on the original flow graph. The composite flow graph is shown in Fig. 5-11. Now by use of Mason's formula the continuous output response of the system is written

$$\begin{aligned} C(s) \\ -(HG)^* \\ = \end{aligned}$$

$$\frac{G(s)}{1 + HG^*(s)} R^*(s)$$

$$G$$

$$1$$

$$G$$

$$R$$

$$E = X_1$$

$$X_2$$

$$X_3 = C$$

$$-H$$

$$(5-51)$$

Fig. 5-11 Composite signal flow graph of the sampled-data system shown in Fig. 5-2.

which agrees with the result obtained previously in Eq. (5-16). The unsampled input $R(s)$ does not appear in the expression for $C(s)$ since there is no forward path from R to C in the composite flow graph. In a similar manner, the continuous signal $E(s)$ is obtained from the composite flow graph,

$$E(s) = \frac{R(s)}{G(s)H(s)R^*(s)}$$

$$= \frac{1}{1 + HG^*(s)} \frac{1}{1 + HG^*(5)} R(s)$$

$$(5-52)$$

As is easy to see, the sampled variables $C^*(s)$ and $E^*(s)$ given in Eqs. (5-47) and (5-48), respectively, can also be derived directly from the composite flow graph simply by use of the gain formula. Example 5-1.

In this example the sampled and the unsampled output variables of the

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$$117$$

$$X_2 = X_1 \cdot D \cdot G$$

$$1$$

$$G_2$$

$$1$$

$$X_1 = E$$

$$X_3$$

$$X_q = C$$

Fig. 5-12 Original signal flow graph of the sampled-data system shown in Fig. 5-6.

sampled-data system shown in Fig. 5-6 will be evaluated by use of the composite signal flow graph method. Following the steps outlined above:

(1) The original signal flow graph of the system is constructed in Fig. 5-12.

(2) From the original flow graph, the following set of equations is written in accordance with the form of Eq. (5-49). (As mentioned previously, equations can and usually should be written directly in the form of Eq. (5-50) by applying Mason's formula and taking the pulse transform simultaneously. But as an illustration of the method of the composite flow graph, the originally suggested steps are followed):

$$E = X_1 = \frac{D \cdot G G^2}{1 + G_2} X^* + \frac{X^*}{1 + G_2}$$

$$X_2 = X^*$$

$$R$$

$$1 + G_2$$

$$X,$$

$$\begin{aligned}
&= \\
&D^*G_1 \\
&1+ G_2 \\
&R \\
&X^* + \\
&1 + G_2 \\
&D^*G_1G_2 \\
&G.R \\
&X_1 = \\
&1+ G_2 \\
&X^* + \\
&1 + G_2 \\
&(5-53) \\
&(5-54) \\
&(5-55) \\
&(5-56)
\end{aligned}$$

For the sake of simplicity the functions of s symbols are omitted in the above equations; that is, X , denotes $X_1(s)$, D^* denotes $D^*(s)$, etc. Taking the pulse transform on both sides of the above four equations yields

$$\begin{aligned}
&X^* = \\
&= \\
&- \\
&GG_2 \\
&\cdot D^* (G + G_4) * X^* + (1 - RG_2) * \\
&X^* = X^* \\
&X_3^* = D^* \\
&1+ G_2 \\
&(1 - C_{12}) * X^* + \\
&1+ G \\
&R \\
&+^2G.)^* \\
&X + + (1 + G_2) \\
&(GR)^* \\
&*
\end{aligned}$$

$$\begin{aligned}
&X_i = D (GG)^*x; + \\
&X^* \\
&+G_2 \\
&(5-57) \\
&(5-58) \\
&(5-59) \\
&(5-60)
\end{aligned}$$

From the above equations with sampled variables, the sampled signal flow graph of the system is drawn in Fig. 5-13; the composite flow graph of the system is drawn in Fig. 5-14.

Now by use of Mason's gain formula, the sampled and the unsampled output variables of the system are determined from the composite flow graph:

$$\begin{aligned}
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&1 \\
&R \\
&(1 - AG_2)^* \\
&R \\
&(TAGZ \\
&G_1 \\
&1+G_2 \\
&0x2 \\
&G_1G_2 \\
&D \\
&1+ G_2'
\end{aligned}$$

$G_2 R$
 $G_1 G_2$
 $-D^*(1+G_2)$
 Fig. 5-13 Sampled signal flow graph of the sampled-data system shown in Fig. 5-6.

G_1
 1
 $1+G_2$
 $G_2 R$
 $G_1 G_2$
 $G_1 G_2$
 $1+G_2$
 X_2
 $D^* G_1$
 G_2
 1
 X_α
 $X_4 = C$
 X_3
 $E = X_1$
 Fig. 5-14 Composite signal flow graph of the sampled-data system shown in Fig. 5-6.

C^*
 $=$
 $X = (1 + G R_2$
 $*$
 $1 +$
 $D^* (G, G_{12}) *$
 $+$
 R
 $*$
 $(5-61)$
 $1 + D^* (G G_2) * (1 + G_2$
 $1 + G_2 /$
 which can be simplified to the following form:

C^*
 $=$
 $*$
 R
 D^*
 $(G R)^* + [(G R) + (1 + C) *] D + (G G)''$

Because of the relationship,

$1 + D^*$
 $R^* = R G_2$
 Eq. (5-62) becomes.

$1 + G$
 $(G, G_{12}) *$
 $(1 + R G C,)^* + (1 + R G,)^*$
 $1 +$
 $*$

(5-62)

(5-63)

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C^*
 $(G R) * + D^* (G, G_+,) *$

$=$

$:)^*$

R^*

$1 + D^*$

(GG)*

which agrees with the result obtained previously in Eq. (5-33).

(5-64)

The unsampled output signal $C(s)$ is obtained from the composite flow graph when Mason's gain formula is applied. Thus

$C =$

$RG_2 \frac{1}{1+D^*}$

$\left[\frac{G_1 G_2}{1+G_2} \right] + (RG_2) * D^* G_2$

(5-65)

$\frac{1}{1+G_2} + D^* (1G_2 G_2)^* + G_2 D^* (G_1 G_2)^*$

$\frac{1}{1+G_2}$

+ Equation (5-65) is simplified to the following form:

R

$\frac{1}{1+G_2}$

D

C

$G.R$

$\frac{1}{1+G_2}$

+

$G_1 G_2$

$\frac{1}{1+G_2}$

$D \cdot (G)^*$

X

$\frac{1}{1+D(GC)^*}$

$*$

$\frac{1}{1+G_2}$

(5-66)

which agrees with the result obtained previously in Eq. (5-36).

Example 5-2

As another illustrative example, let us consider the multisampler system shown in Fig. 5-15. The original flow graph of the system is drawn in Fig. 5-16.

$R(s)$

$E(s)$

$G_1(s)$

$G_2(s)$

$H(s)$

T

R_0

Fig. 5-15 A multisampler sampled-data system.

G_1

$X_2 = X^*$

X_3

$-H$

$C(s)$

G_2

X_5

$X_5 = C$

Fig. 5-16 Original signal flow graph of the system in Fig. 5-15.

Note that in the signal flow graph of Fig. 5-16, variables X_2 and X_3 are assigned to the inputs of the two samplers. Therefore, the following equations can be written directly from Fig. 5-16 by the use of Mason's gain formula (also taking the pulse transforms at the same time):

$X = (G_1 G_2)^* X^* + (G_2 H)^* X^* + R^*$

$X^* = X$

(5-67)

(5-68)

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 $-(G, G_2)$
 R^*
 G

$(G_2H)^*$
 (G, G_2)
 $-H$

X_5
 $X=C^*$
 $-(G_2H)$
 Fig. 5-17 Sampled flow graph of the system shown in Fig. 5-15.

$-(G_1G_2)$
 $(G_2H)^*$

G_1
 X
 $(G_1G_2)^*$
 1
 $-(G_2H)^*$
 $-H$

G_1
 G_2
 1

X_2
 X_3
 X_5
 $X_5=C$
 -1

Fig. 5-18 Composite signal flow graph of the system shown in Fig. 5-15.

$$X^* = G^*X^* - H^*X^*$$

$$X^* = X^*$$

$$X = (G, G_2)^*X^* - (G_2H)^*X^*$$

(5-69)

(5-70)

(5-71)

These equations with sampled signals above are now represented by the sampled signal flow graph shown in Fig. 5-17. Next, we eliminate the two samplers from the original flow graph and connect nodes X and X^* of the sampled flow graph to nodes X , and X , on the original flow graph, respectively, by unity gain branches; we have the composite flow graph as shown in Fig. 5-18. The sampled and the unsampled outputs of the system are obtained from the composite flow graph using Mason's formula. Therefore,

$$C^* = X^*$$

$$= \frac{(G, G)^*R^*}{1 + (G_1G_2)^* + (G_2H)^* - (G_1G_2)^*(G_2H)^* + (G_1G_2)^*(G_2H)^*}$$

which is then simplified to

$$(G, G_2)^*R^*$$

$$C^* = \frac{1 + (G_1G_2)^* + (G_2H)^*}{(5-72)}$$

$$(5-73)$$

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The continuous output is

$$C = X_5$$

$$= G_1G_2[1 + (G_2H)^*] - (G_1G_2)^*G_2H$$

$$1 + (G_1 G_2)^* + (G_2 H)^* \\ R^*$$

(5-74)

5.3 MODIFIED Z-TRANSFORM OF OUTPUTS OF CLOSED-LOOP

SAMPLED-DATA SYSTEMS

The modified z-transfer function of an open-loop sampled-data system is discussed in Sec. 4-8. In order to describe the response between sampling instants, a fictitious time delay AT is inserted in the open-loop system as shown in Fig. 5-19.

$$E(s) +$$

$$E^*(s)$$

$$G(s)$$

T

Fictitious

sampler

$$C(z, m)$$

T

$-AT$

e

$$C(s)$$

Fictitious time

delay

Fig. 5-19 Open-loop system with sampled-data and with the fictitious time delay.

Without the fictitious delay the Laplace transform of the output response of the system is

$$C(s) = G(s)E^*(s)$$

and the ordinary pulse transform of $c(t)$ is

or

$$C^*(s) = G^*(s)E^*(s)$$

=

$$C(z) G(z) E(z)$$

(5-75)

(5-76)

(5-77)

With the fictitious delay inserted in the open-loop system, the modified pulse transform of the delayed output is

where

=

$$C(s) G(s) E^*(s)$$

00

(5-78)

$$G_m(s) = \mathcal{L} [g(t - AT)] = \int_0^\infty \sum_{n=0}^{\infty} g(nT - kT) \mathcal{E}(t - nT)$$

-

$n=0$

-

-

(5-79)

and a similar expression defines $C(s)$. The modified z-transform of the system output is therefore,

where

$$C(z, m) = G(z, m)E(z)$$

$$G(z, m) = [G^*(s)]_{s=(1/T) \ln z}$$

(5-80)

(5-81)

Note that the modified z-transform operation does not affect the term: $E^*(s)$, in Eq. (5-78). Therefore, the modified z-transform of Eq. (5-75) is given by Eq. (5-80), but not by $C(z, m) = G(z, m)E(z, m)$.

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In the analysis of a closed-loop sampled-data system by use of the modified z-transform method, a fictitious time advance of AT must be inserted in the feedback path as shown in Fig. 5-20; then

Fictitious
sampler
 $C(z, m)$

T

$C(s)$
 $R(s)$
 $E(s)$ $E^*(s)$
 $G(s)$
 $-AT$

e

Fictitious time
delay

$H(s)$
 $+AT$

e

Fictitious time
advance

Fig. 5-20 Fictitious time delay and time advance in a closed-loop sampled-data system for modified z-transform analysis.

the actual signal in the system is not affected by the fictitious delay in the forward path. From the modified system block diagram of Fig. 5-20, the modified z-transform of the system output $c(t)$ is derived as follows:

$$C(s) = G(s)E^*(s)e^{-ATS} \quad (5-82)$$

Taking the modified z-transform on both sides of Eq. (5-82) yields

$$C(z, m) = G(z, m)E(z) \quad (5-83)$$

Substituting the relationship of

$$(5-84)$$

$$E(x)$$

$=$

$$R(z) \frac{1}{1+GH(z)}$$

into Eq. (5-83), we get

$$C(z, m) =$$

$$G(z, m)$$

$$\frac{1}{1+GH(z)}$$

$$R(z)$$

$$(5-85)$$

From the above derivations we see that the modified z-transform of a time function $c(t)$ can be obtained directly from the Laplace transform $C(s)$ once $C(s)$ is given. This means that if $C(s)$ is the output of a sampled-data system, the modified z-transform of the output can be obtained directly from $C(s)$ once $C(s)$ is determined from the composite signal flow graph of the system. Since in sampled-data systems $C(s)$ is usually of the form (see Eq. 5-75):

$$C(s) = A(s) B^*(s)$$

$=$

$$(5-86)$$

where $A(s)$ denotes a continuous function and $B^*(s)$ denotes a discrete function, the modified z-transform of $c(t)$ is written directly from Eq. (5-86),

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$$C(z, m) = A(z, m)B(z) \quad (5-87)$$

where $A(z, m)$ is the modified z-transform of $A(s)$ and $B(z)$ is

merely the ordinary z-transform of $B(s)$ or $B^*(s)$.
 As an illustrative example, the modified z-transform of the output of the system in Example 5-2 is determined in the following:
 The Laplace transform of $c(t)$ is given in Eq. (5-74). Applying the modified z-transform operation to Eq. (5-74) in the same manner as from Eq. (5-86) to (5-87), we have

$$C(z, m) = \frac{G_1 G_2(z, m)[1 + G_2 H(z)] - G_1 G_2(z) G_2 H_2(z, m)}{1 + G_1 G_2(z) + G_2 H(z)} R(z) \quad (5-88)$$

In a more formal approach the modified z-transform of any signal in a sampled-data system can be obtained by simply modifying the composite flow graph as described in the following and then applying Mason's gain formula. Starting with Eq. (5-49), if the modified z-transform of the variable X , is desired, we simply take the modified pulse transform on both sides of Eq. (5-49) to get

$$X^*m = \sum_i (a_{ij}) X^* + \sum_k (b_{kj} R_x) m \quad (5-89)$$

where (a) and (b_{jR_x}) denote the modified pulse transforms of $a)$ and b_{jR_k} , respectively. Actually, Eq. (5-89) can always be written directly from the original flow graph of the sampled-data system by applying Mason's formula and taking the modified z-transform simultaneously.

Once Eq. (5-89) is written, the composite flow graph of the system is modified by adding nodes and branches on the graph corresponding to the portrayal of Eq. (5-89); then, when Mason's formula is applied to the modified flow graph, the modified pulse transform (or modified z-transform) of X , is obtained.

Example 5-3

As an illustration of the construction of the modified composite flow graph of a multiloop, multisampler system, let us consider the system in Example 5-2 (Fig. 5-15). The set of equations written from the original signal flow graph in Fig. 5-16 is repeated as follows:

$$\begin{aligned} X_1 &= X_2 \\ X_2 &= X^* \\ G_1 G_2 X^* + H G_2 X^* + R \\ X_1 &= G_1 X - H X^* \\ X_1 &= X^* \\ X_1 &= G_1 G_2 X - G_2 H X^* \end{aligned} \quad \begin{aligned} (5-90) \\ (5-91) \\ (5-92) \\ (5-93) \\ (5-94) \end{aligned}$$

The pulse transforms of the above five equations are given in Eqs. (5-67)

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through (5-71); based on these equations, the sampled and the composite flow graphs of the systems are constructed in Figs. 5-17 and 5-18, respectively. In order to determine the modified z-transform of the system output $c(t)$, we take the modified pulse transform of both sides of Eq. (5-94) (since $X_1 = C$), thus

$$X^*m$$

$$C$$

$$=$$

$$(G_1 G_2) * X^* - (G_2 H) * X^*$$

X_{5m}

$$(5-95)$$

$$-(6, 62)$$

$$(G, G_2 m$$

1

$$(G_2 H) *$$

$$-- (G_2 H)$$

G

$$(G, G_2)$$

G₁

$$X_2$$

$$X_1$$

$$-H$$

$$-H$$

$$-(G_2 H) *$$

1

$$G_2$$

-1

Fig. 5-21 The modified composite signal flow graph of the system in Fig. 5-15.

5m

*

m

The modified composite flow graph of the system is now drawn in Fig. 5-21; a node at some convenient point representing the variable $X = C$ is added to the flow graph, and branches with branch gains of (G, G_2) and $-(G_2 H) *$ are connected from nodes X and X^* respectively to X . The modified pulse transform of $c(t)$ is obtained directly from Fig. 5-21 by use of Mason's gain formula. Therefore,

C*

m

-

$$(G, G_2) [1 + (G_2 H) *] - (G, G_2) * (G_2 H) * R^*$$

$$1 + (G, G_2) * + (G_2 H) *$$

and the modified z-transform of $c(t)$ is

$$C(z, m) =$$

-

$$GG(z, m) [1 + G.H(\approx)] G, G, (z) GH(\approx, m)$$

$$1 + G_1 G_2(\approx) + G_2 H(z)$$

$$(5-96)$$

R(≈)

$$(5-97)$$

Clearly this result agrees with that obtained previously in Eq. (5-88).

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5.4 TRANSMISSION MATRIX OF SAMPLED-DATA SYSTEMS+

The fact that a sampled-data system deals with sequences of numbers rather than with continuous time functions makes the use of matrix representation convenient for analysis and synthesis. We have seen that the matrix method is particularly useful in the analysis and design of sampled-data systems with time-varying components as well as systems with nonlinear elements.

In this section the matrix representation of linear time-invariant sampled-data systems will be discussed. In later chapters, the matrix method will be extended to the studies of time varying and nonlinear sampled-data systems.

e(t)

(1)

$$G(s)$$

$$E(s) E(s)$$

T
Fictitious
sampler
 $c^*(t)$
 $C^*(s)$
 $C(z)$
 $c(t)$
 $C(s)$

Fig. 5-22 Open-loop sampled-data system.

Consider the open-loop sampled-data system shown in Fig. 5-22. The system consists of a linear time-invariant plant preceded by a sampler and followed by a fictitious sampler; the samplers are assumed to have a sampling period of T sec and are synchronized. It follows from the discussion on the weighting sequence given in Sec. 4.6 that the continuous output of the system is obtained through the use of the superposition principle; that is,

$$c(t) = \sum_{k=0}^{\infty} g(t - kT)e(kT) \quad (5-98)$$

where $g(tkT)$ is the impulse response of the plant G due to an impulse occurring at $t = kT$.

At the sampling instants, the output of the systems is

$$c(nT) = \sum_{k=0}^{\infty} g(nT - kT)e(kT) \quad (5-99)$$

For a physical system initially at rest, we may normally assume that the response of the system is zero prior to the application of an output. Hence,

B. Friedland, "A Technique for the Analysis of Time Varying Sampled-Data Systems," A. I. E. E. Trans. 75, Part 1956, pp. 407-414.

E. Kinnen, and J.T. Tou, "Analysis of Nonlinear Sampled-Data Control Systems," A. I. E. E. Trans. 75, Part II, 1960, pp. 386-390.

Sun, C. H., "Matrix Solution of Nonlinear Sampled-Data Control Systems," Ph. D Dissertation, University of Illinois, Urbana, Illinois, 1961.

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$g(nTkT) = 0$ for $n < k$

(5-100)

The upper limit of Eq. (5-99) can be changed from ∞ to n as indicated below:

$$c(nT) = \sum_{k=0}^n g(nT - kT)e(kT) \quad (5-101)$$

(5-101)

In general, Eq. (5-101) may be written as a matrix equation:

C_k

$= GKE_k$

(5-102)

where

$c(0)$

$c(T)$

C_0

$C_k = c(2T)$

$= C_2$

نت
 $= \text{col } [c_0, c_1, c_2, \dots, c_n]$
 where $c_j = c(jT)$
 (5-103)
 and
 $c(nT)$
 $e(0)$
 e_0
 $e(T)$
 e_1
 $E_x = e(2T)$
 $= e_2$
 $=$
 $\text{col } [e_0, e_1, e_2, \dots, e_n]$
 (5-104)
 G_k

$=$
 $=$
 where $e, e(jT)$
 $e(nT)$
 g_0
 $0 \ 0$
 0
 0
 g_2
 $50 \ 50$
 $50 \ 50$
 g_0
 0
 0
 0
 g_0
 0
 0
 (5-105)
 g_3
 g_2
 50
 g_0
 0
 $g_n \ g_{n-1} \ g_{n-2} \ g_{n-3}$
 g_0

where $g_j = g(jT) = g(nT_kT)$. G is defined as the transmission matrix for G .
 Note that for the transmission matrix G the element at the j -th row and the p -th column is physically the contribution to the j -th sampled output ($j = 0, 1, 2, \dots, n$) due to a unit impulse input at the p -th ($p = 0, 1, 2, \dots, n$) sampling instant. Since an output contribution exists only after an input is applied but not before it, this

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accounts for the fact that all the elements of the matrix G above the principal diagonal are equal to zero. This is characteristic of the transmission matrix of a physical sampled-data system. As explained in Sec. 4.4, Eq. (4-39), when $G(z)$, the z -transform of $g(t)$, is expanded into a power series in powers of z^{-1} , the coefficient for z^{-1} is the value of the impulse response at $t = kT$, or g_k . Thus, the transmission matrix may be obtained from the z -transform of the transmission function $G(z)$ by the following steps:

(1) Given $G(z)$, expand $G(z)$ into a power series in powers of z^{-1} ; that is,

$$G(z) = g_0 + g_1 z^{-1} + g_2 z^{-2} + g_3 z^{-3} + \dots + g_k z^{-k} + \dots$$

(5-106)

(2) Write the transmission matrix starting with g_0 at the principal diagonal, and working toward the left with g_1 , g_2 , and g_3 , consecutively, as shown below:

$$\begin{pmatrix} g_0 & 0 & 0 & 0 \\ g_1 & g_0 & 0 & 0 \\ g_2 & g_1 & g_0 & 0 \\ g_3 & g_2 & g_1 & g_0 \\ \vdots & \vdots & \vdots & \vdots \end{pmatrix}$$

(3) Fill in with zeros for all the elements above the principal diagonal.

As an illustrative example, the transmission matrix of a discrete integrator is determined as follows:

$$G(z) = \frac{1}{1 - z^{-1}}$$

$$= \frac{1}{1 - z^{-1}} = 1 + z^{-1} + z^{-2} + z^{-3} + \dots$$

(5-107)

Since all the coefficients in the expansion of $G(z)$ are equal to 1, the transmission matrix for G is simply

$$G_k = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}$$

(5-108)

$$G_k = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}$$

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Chap. 5

BLOCK DIAGRAM, SIGNAL FLOW GRAPH, AND MATRIX REPRESENTATION Transmission Matrices of Parallel and Series Combinations of Components

Since in general, sampled-data control systems consist of a number of components interconnected directly or through sampling switches, the transmission matrices of parallel and series combinations of system components must be investigated. A parallel combination of two components is shown in Fig. 5-23a. By the principle of super-

$c^*(1)$

(1)

A

$$_e(t)X_e^*(1)$$

$c(1)$

$$_e(1)X_e^*(1)$$

G
-c(t)
T
B
(a)
(b)

Fig. 5-23 (a) Parallel combination of components. (b) Trans-

mission matrix $G_k = A_k + B_k$.
position, we can see that the transmission matrix for the parallel
combination is simply equal to the sum of the transmission matrices
of the two components; that is

=
 $G_k = A_k + B_k$
Also, we know that
(5-109)
(5-110)

$G(z) = A(z) + B(z)$
Thus the transmission matrix equation and the z-transform equation
satisfy the same rule of linear combination.

A series combination of two elements is shown in Fig. 5-24a.
Let the input and output signals of the two elements be as designated
in Fig. 5-24a. Then, the output of the series combination is given
by the following matrix equations:

$C_k = B_k M_k = B_k A_k E_k$
(5-111)
 $c^*(1)$

$e(t)$
 $* (1)$
 $m(1)$
 $m^*(1)$
 $e(t)$
A
B
X
(1)
G
c (1)
T
c(1)
T
(a)
(b)

Fig. 5-24 (a) Series combination of components. (b) Transmis-

sion matrix $G_k = B_k A_k$
Equation (5-111) implies that the transmission matrix of the series
combination of two elements A and B connected through a sampling
switch, is equal to the product of the transmission matrices of the
two elements. Similarly, the z-transform of the system output is

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$C(z) = B(z) A(z) E(z)$
=
(5-112)

Note that Eqs. (5-111) and (5-112) are identical in form just as
Eqs. (5-109) and (5-110) are. Note especially that since matrix
multiplication is not commutative, BA generally does not equal
 AB . However, the multiplication of $B(z)$ and $A(z)$ in the z-trans-
form equation of Eq. (5-112) is commutative.

Transmission Matrix of a Feedback Control System

Using the properties of the combination of components developed
above, we are now able to determine the transmission matrix for a

feedback control system with sampled-data. The transmission matrix for the simple feedback system with sampled-data shown in Fig. 5-25 is derived as an example. In matrix notation the sampled error signal is written as:

Digital
-compensator-
Plant

$r(1)$
 $e(t)$
 $D(s)$
 $G(s)$
- $c(t)$

where

Fig. 5-25 Unity feedback control system with sampled-data.

$E_k R_k C_k$

C_k

$E_x = \text{col } [e_0, e_1, e_2, \dots, e_n]$

$R_k \text{ col } [r_0, r_1, r_2, \dots, r_n]$

$=$
 $C_k \text{ col } [C_0, C_1, C_2, \dots, C_n]$

$=$
(5-113)

(5-114)

(5-115)

(5-116)

By use of the matrix equation in Eq. (5-111), the system output is found to be

$C_k G_k D_k E_k$

(5-117)

where D denotes the transmission matrix for the discrete compensator and G_k represents the transmission matrix for the plant, which includes the zero-order hold.

Substitution of Eq. (5-117) into Eq. (5-113) yields

E_k

$=$

R_k

-

$G_k D_k E_k$

Solving for E_x in the last equation, we have

$E_k (1 + G_k D_k) = R_k$

$(1 + G_k D_k)^{-1} R_k$

(5-118)

(5-119)

provided that $(1 + G_k D_k)$ is nonsingular. Now substituting Eq.

(5-119) into Eq. (5-117) yields

Letting

$C_k G_k D_k (1 + G_k D_k)^{-1} R_k$

M_k

$=$

$G_k D_k (1 + G_k D_k)^{-1}$

(5-120)

(5-121)

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where M denotes the closed-loop transmission matrix of the system, then

$C_k = M_k R_k$

$M_k^{-1} (1 + G_k D_k)^{-1}$

It can also be shown easily that

(5-122)

(5-123)

The above form for M is frequently convenient for computation of system responses.

It is clear that Eq. (5-121) corresponds to the z-transform of the output of the closed-loop system; that is

$$C(z) = \frac{G(z)D(z)}{1+G(z)D(z)} R(z)$$

(5-124)

5.5 THE STATE-VARIABLE APPROACH

In general, system analysis and synthesis may be carried out by two major approaches. One approach is the use of the conventional transfer functions, block diagrams, or signal flow graphs. The other method, which has gained popularity among modern engineers, is the state-transition or state-variable approach. In the state-transition approach the system is characterized by a set of first-order differential or difference equations describing the state variables of the system. Each component of the system is decomposed into the basic mathematical elements describing it. The system is then analyzed or designed by solving the set of first-order equations, rather than a single higher-order equation or a set of higher-order equations.

The state-transition method has at least the following advantages as compared to the block diagram or transfer function approach:

(1) By use of vector-matrix notation a set of first-order differential equations may be represented by a single expression. This simplifies the mathematical notation.

(2) Obtaining the solution of a set of first-order differential equations is much easier than solving a high-order differential equation.

(3) The initial conditions of the system are readily given by

↑ T. R. Bashkow, "The A Matrix, a New Network Description," I. R. E. Trans. on Circuit Theory, Vol. CT-4, Sept. 1957, pp. 117-120; R. E. Kalman, J. E. Bertram,

"A Unified Approach to the Theory of Sampling Systems," Journal of Franklin Institute, 267, May 1959, pp. 405-436. E. O. Gilbert, "A Method for the Symbolic Representation and Analysis of Linear Periodic Feedback Systems," A. I. E. E. Trans.

Part 2, 79, pp. 512-523, 1960.

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the state-transition equations. The calculation of the initial conditions by conventional techniques can be tedious.

(4) The state-transition approach allows a unified study of sampled-data systems. This includes uniform sampling, multirate sampling, cyclic sampling, as well as random sampling and finite pulse-width considerations.

(5) The state-transition method can be applied to certain types of nonlinear systems and time-varying systems.

State Variables And State-Transition Equations of Continuous-Data Systems

For an n-th order linear system with continuous-data, the system dynamics may be represented by the following set of n equations:

$$dx/dt = f(x_1, x_2, \dots, x_n, r_1, T_2, \dots, r_m)$$

where $x_1, x_2, x_3,$

$\vdots$
($i = 1, 2, 3, \dots, n$)

(5-125)

..., $\hat{x}_n$ are the set of n system variables which are

defined as state variables; $r_1, r_2, r_3, \dots, r_m$ are the m inputs of the system; $f_1, f_2, \dots, f_n$ represent functional relations. Often, the output of the system is designated as one of the state variables, although this designation is not essential. The set of first-order equations may be described by an analog computer diagram or a signal flow graph. In essence, the n equations may be written as

$$\frac{dx}{dt} = \sum_{j=1}^m a_{ij}(t) x_j(t) + \sum_{k=1}^m b_{ik}(t) r_k(t)$$

($i = 1, 2, 3, \dots, n$)
(5-126)

where a_{ij} is a constant denoting the gain of the path connected between the output of the j -th integrator and the input of the i -th integrator; b_{ik} is the gain between the k -th input r and the input of the i -th integrator. A typical signal flow graph portraying the summing of signals (state variables and inputs) at the input node of the i -th integrator is illustrated in Fig. 5-26. The s^{-1} operator denotes the integration operation. Note that the convention used here is not in exact agreement with the practical analog computer notation in that amplification and integration are not accompanied by a sign change.

In vector-matrix notation the set of first-order linear equations of Eq. (5-125) may be written as

↑ If Q of the system outputs y is not included in the state variables, these outputs

must be written as functions of the state variables and the inputs; that is

$$Y_q(t) = g_q(X_1, X_2, \dots, X_n, T_1, T_2, \dots, T_m), \quad q = 1, 2, \dots, Q.$$

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4(1)

-1

a_{in}

O_{ii}

a_{ij}

Outputs of

integrators

Fig. 5-26 Analog signal flow graph simulating the equation

=

dt

dx/dt

$= Ax + Br$

where x is a column matrix for the state variables; that is,

and

$x = \text{col}[x_1(t) \ x_2(t)$

$x_3(t) \ \dots \ x_n(t)]$

$r = \text{col} [r_1(t)$

$r_2(t)$

$r_3(t)$

$r_s(t) \ \dots r_m(t)]$

(5-127)

A is an $n \times n$ square matrix whose elements are defined by a_{ij} ; B

is an $n \times m$ matrix.

The matrix equation of Eq. (5-127) gives a detailed description of the transition of state of the system's state variables x from $x_1(t)$ to $x_1(t) + dx_1(t)$ when time is changed from t to $t + dt$. Since dt is an infinitesimal time change, the transition of a continuous-data system takes place in a continuous fashion with respect to

time.
 Finally, a quantitative description of the state transition of the system may be obtained by solving the matrix equation of Eq. (5-127). Taking the Laplace transform on both sides of Eq. (5-127), we have

$$sX(s) = AX(s) + BR(s) + x(t_0)$$

Solving for $X(s)$ yields

$$X(s) = (sI - A)^{-1}x(t_0) + (sI - A)^{-1}BR(s)$$

(5-128)
 (5-129)

where $x(t)$ denotes the initial condition of $x(t)$ evaluated at $t = t_0$. Since A is a constant matrix, the inverse Laplace transform of $(sI - A)^{-1}$ is simply

$$[(sI - A)^{-1}] = p(t - t_0) = e^{(t - t_0)A}$$

--
 The inverse Laplace transform of $X(s)$ in Eq. (5-129) is

$$x(t) = p(t - t_0)x(t_0) + \int_{t_0}^t p(t - \tau)Br(\tau)d\tau$$

-(5-130)
 (5-131)
 where the second term on the right-hand side of the equation is
 The inverse Laplace transform of $1/(s - a)$ is $e^{(t - t_0)a}$

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the convolution integral. Now, the equations represented by the vector-matrix of Eq. (5-131) are defined as the state-transition equations; $(t - t_0)$ is sometimes called the state-transition matrix of the dynamic system.

Although the use of vector matrices in the state-variable approach leads to simplified notations, the matrix method is not necessarily the simplest way of arriving at the state-transition equations. The problem is considerably simplified if the solution is carried out using the signal flow graph approach. The state-transition flow graph is defined as the analog computer simulation flow graph with the initial conditions (t_0) , $X_2(t_0)$, $\dots$, $X_n(t_0)$ as input signals applied at the corresponding output nodes of the integrators. The state-transition equations of the system can then be determined from the transition flow graph merely by applying Mason's formula and taking the inverse Laplace transform.

To illustrate the application of the state-variable approach to continuous-data systems, the following example is given.

Example 5-4

Consider a continuous-data system with open-loop transfer function

We let

and

=

$\frac{1}{s+1}$

=

$\frac{5-2}{1+s}$

$C(s)$

$R(s) = \frac{s(s+1)}{s^2+1}$

$C(s) = \frac{s^{-2}X(s)}{R(s)} = s^{-1}X_2(s)$

$R(s) = (1 + s^{-1})X(s)$

Equation (5-134) can also be written as

$X(s) = s^{-1}X_2(s) + R(s)$

=

-

(5-132)

(5-133)

(5-134)

(5-135)

From these equations the signal flow graph which represents the analog computer simulation of the system is constructed in Fig. 5-27. The state variables are designated as x and x_2 , where x is also the system output c .

x_2

1

S

$X_1 = c$

-1

Fig. 5-27 Signal flow graph representing the analog computer simulation of a second-order feedback control system.

From the signal flow graph the following two first-order differential equations are written in accordance with Eq. (5-126):

$dx/dt = x_2(t)$

$dx_2/dt = -x_2(t) + r(t)$

Therefore,

A

=

-1

1

(5-136)

(5-137) ✓

B

-0

R

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Thus

For a unit-step input

-

1

1

S

$(SI - A)^{-1} =$

$s(s + 1)$

1

(5-138)

0

$s + 1$

1

$p(t - t_0)$

$1 - e^{-(1-t_0)}$

=

$e^{-(1-t_0)}$

(5-139)

t

-

$t_0 - 1 + e^{-(1-t_0)}$

$1 - e^{-(1-t_0)}$

(5-140)

$\int_{t_0}^t Br(T) dt = [$

Therefore, the state-transition equations of the system are given by the matrix equation:

$x_1(t)$

$x_2(t)$.

= [

$1 - e^{-(t-t_0)} \quad x_1(t_0) \quad t - t_0 - 1 + e^{-(1-t_0)}$

$e^{-(1-t_0)}$

X

$$+ \frac{x_2(t_0)}{1 - e^{-(1-t_0)}} \quad (5-141)$$

The problem is considerably simplified if the solution is carried out using the signal flow graph method. When the initial conditions $x_1(t_0)$ and $x_2(t_0)$ are applied to the state-variable nodes x_1 and x_2 , respectively, we have the state-transition flow graph as shown in Fig. 5-28.

$$R(s)$$

$$S$$

$$OX_1(s)$$

$$X_2(s)$$

Fig. 5-28 State transition flow graph of a second-order feedback control system.

The state-transition equations in the Laplace transform domain are readily written by applying Mason's formula to the transition flow graph. Thus

$$X_1(s)$$

$$=$$

$$+$$

$$S$$

$$x_1(t_0) \frac{x(t_0)}{s(s+1)}$$

$$+$$

$$R(s)$$

$$s(s+1)$$

$$X_2(s)$$

$$x_2(t_0)$$

$$R(s)$$

$$=$$

$$+$$

$$s+1$$

$$s+1$$

$$(5-142)$$

$$(5-143)$$

Taking the inverse Laplace transform of Eqs. (5-142) and (5-143) yields (r is a unit-step input).

$$x_1(t) = x_1(t_0) + x_2(t_0)[1 - e^{-(1-t_0)}] + t - t_0 + e^{-(1-t_0)}$$

$$x(t) = x(t_0)e^{-(t-t_0)} + 1 - e^{-(1-t_0)}$$

$$(5-144)$$

$$(5-145)$$

These equations are in exact agreement with the result in Eq. (5-141).
State- Variables and State-Transition Equations of Discrete-Data Systems
In contrast to continuous-data systems, which are described by

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differential equations, discrete-data systems are generally described by difference equations. For an n -th order linear system with discrete-data, the system dynamic may be represented by the following set of first-order difference equations:

$$x[(k+1)T] = \sum_{j=1}^m d_{1,j} x_j(kT) + \sum_{p=1}^n C_s r_p(kT)$$

$$j=1$$

$$p=1$$

$$(i = 1, 2, 3, \dots, n)$$

$$(5-146)$$

where $X_1, X_2,$

are the state variables, and kT denotes the sampling instants. In general, the outputs of the system may be expressed as functions of x , and r , whether or not the outputs are

$n=0$

-

where the second term on the right-hand side of the equation is recognized as the convolution summation. Comparison of Eq. (5-153) with Eq. (5-131) shows that the solutions for discrete-data and continuous-data systems are analogous.

Transition Equations of Sampled-Data Systems

In most sampled-data systems the signals found in the systems are continuous as well as discrete. Thus, ordinarily a sampled-data system containing discrete and continuous elements should be described by sets of difference equations and differential equations. In general, we may classify the dynamic elements in a sampled-data system into the following three categories:

(1) continuous dynamic elements - described by differential equations

(2) discrete dynamic elements - described by difference equations

(3) sample and hold elements-described by discrete state-transition equations.

The state variables and transition equations of the first two categories have been discussed in the preceding sections. Namely, we can write the set of differential equations in Eq. (5-126) for the continuous elements and the set of difference equations of Eqs. (5-146) and (5-147) for the discrete elements. Usually, sample-and-hold elements are connected between discrete and continuous elements. We shall

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assume that sampling and hold are one combined operation. Thus, the output of a sampled-and-hold element between two consecutive sampling instants is a step function. If the input and the output of the i -th sampled-and-hold element are denoted by $y(t)$ and $h(t)$, respectively, we can write.

$$h(kT+) = y(kT)$$

where kT really means $kT + T$ for $0 < T < T$.

(5-154)

In general, the input to a sampled-and-hold element is a linear combination of the system's state variable and the system inputs. Equation (5-154) can be written as

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$$h(kT) = \sum_{j=1}^m S_{1,j} x_j(kT) + \sum_{i=1}^n V_{i1} u_i(kT)$$

where $S_{1,j}$ and V_{i1} are constants.

$$u = 1$$

(5-155)

The state-transition equations of the complete sampled-data system can be obtained from the combined solutions of Eqs. (5-155), (5-147), (5-146), and (5-126). In fact, the state-transition equations can be written directly from the composition of the signal flow graphs portraying Eqs. (5-128), (5-146), (5-147), and (5-155).

To illustrate the analysis of a sampled-data system by the state-transition approach, the following examples are given:

Example 5-5

Consider the second-order sampled-data system shown in Fig. 5-30. The continuous-data part of the system is considered in Ex. 5-4. Therefore, from Eq. (5-141), the state-transition equations of the continuous elements in matrix form are written as

Zero-order

hold

$$G(s) =$$

1

$$s(s+1)$$

C

The solution of this problem may be simplified if the signal flow graph technique is used. The transition flow graph of the continuous-data elements is repeated in Fig. 5-31a. By use of Eq. (5-157), the state-transition flow

1
S
1
S
H(s)
X₂(s)
X₁(s)
r(10)
n(%)
51
H(s)
-
(a)
x₂(10)
1x₁(10)
-1
X₂(s)
X₁(s)
(b)

Fig. 5-31 (a) State transition flow graph of continuous-data elements. (b) State transition flow graph of the sampled-data system in Example 5-5.

graph of the sampled-data system is shown in Fig. 5-31b. The state-transition equations of the system are now obtained from the transition flow graphs applying Mason's formula. We have

1
1
S
X(s) = (-36 + 1)] + (6) + 3(8 + 1) %2(%)
[1 (s
5(s x₂(to)
1
5(5
+55 (8 + 1) * (%)
5²(s
(5-160)

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X₁(5)
=
-
1
1
+
5(s + 1) x₁ (to) + 5 + 1 x₂(to) +55 + 1) r (to)
Taking the inverse Laplace transform of the last two equations yields

$$x_1(t) = [2-(t - t_0) - e^{-(t-t_0)}]x_1(t_0) + [1 - e^{-(t-t_0)}]X_2(t_0)$$

$$+ [(t - t_0) - 1 + e^{-(t-t_0)}]r(t_0)$$

(5-161)
(5-162)

$x(t) = - [1 - e^{-(t-t_0)}] x_1(t_0) + e^{-(t-t_0)} x_2(t_0) + [1 - e^{-(t-t_0)}] r(t_0)$ (5-163)
 These two equations are identical to those given by Eq. (5-158).

Example 5-6

As another example of illustrating the application of the state-variable approach to sampled-data systems, we shall consider the sampled-data system shown in Fig. 5-32. Given that

$D(z)$.

$D(s)$

Zero-order hold

$G(s)$

Fig. 5-32 Sampled-data system of Example 5-6.

$G(s) = \frac{1}{s^2 + a_1 s + a_0}$

and $D(z) = \frac{1}{1 + b_1 z^{-1}}$

the transition equations of the system are desired.

The transition flow graph of the continuous-data elements is already shown in Fig. 5-28. The flow graph for the discrete dynamic elements must be constructed. The transfer function for the discrete elements can be written as

$X_1(z)$

$E(z) = \frac{1}{s^2 + a_1 s + a_0} (a_0 + a_1 z^{-1}) X(z)$

$(1 + b_1 z^{-1}) X(z)$ (5-164)

Letting $X_3(z) = (a_0 + a_1 z^{-1}) X(z)$ (5-165)

and $E(z) = (1 + b_1 z^{-1}) X(z)$ (5-166)

we have from Eq. (5-166) $X(z) = E(z) - b_1 z^{-1} X(z)$ (5-167)

Taking the inverse z-transform of Eqs. (5-165) and (5-167) yields

$x_3(kT) = a_0 x(kT) + a_1 x[(k-1)T]$ (5-168)

$x(kT) = e(kT) - b_1 x[(k-1)T]$ (5-169)

Letting $x[(k-1)T] = x(kT)$, the last two equations become

$x_3(kT) = a_0 [e(kT) - b_1 x(kT)] + a_1 x_1(kT)$

$= a_0 e(kT) + (a_1 - a_0 b_1) x(kT)$
 $x[(k+1)T] = e(kT) - b_1 x(kT)$

(5-170)
 (5-171)

Equations (5-170) and (5-171) are the difference equations describing the

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*(KT)

-b₁

$\delta(a, -a, b,)$

$e(kT)\alpha$

$Ox(k+1)T$

$Dxz(kT)$

$XSKT)$

-1

%

(a)

-b₁

(a, raob,)

$r(kT)$

1

$e(kT)$

$X4(k+1)T$

$X3(kT)$

$(kT) h(t) H(s)$

$X2(s) X1(s)$

$X1(s)$

%

(b)

Fig. 5-33 (a) Signal flow graph portraying the difference equations of the discrete elements. (b) The transition signal flow graph of the sampled-data system in Example 5-6.

discrete-data elements of the system. Note that the equations are of the form of Eqs. (5-146) and (5-147), respectively.

The signal flow graph which portrays the two difference equations is drawn in Fig. 5-33a. This discrete transition flow graph must be linked with the transition flow graph of the continuous-data elements. This linking is accomplished through the transition equation of the sampled-and-hold element. From Fig. 5-32, the transition equation of the sampled-and-hold element is

Also,

$h(t) = x3(to)$

$e(to) = e(kT) = r(to) - x1(to) = r(kT) - x1(to)$

(5-172)

(5-173)

The transition signal flow graph of the entire sampled-data system is the composition of the flow graphs of Figs. 5-28 and 5-33a through the use of Eqs. (5-172) and (5-173). The complete transition flow graph of the system is shown in Fig. 5-33b. Using Mason's gain formula, the transition equations are written directly in the Laplace domain. [r , $x_1(to)$, $x_2(to)$, and $x_3(kT)$ are the input nodes.]

-

ao

$|x_1(to)$

1

$= [\frac{1}{s} - \frac{36}{s+1}] + (46) + 5 + 1) + 2(4)$

$X, (s) (3)$

$(s+1)$

S

α_1

+

a) ab) $x(kT)$

$s^2 (s + 1)$

+

$$\frac{1}{s(s+1)} x_2(t_0) + \frac{1}{s(s+1)} r(kT) \quad (5-174)$$

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 $X_2(s)$

$$\begin{aligned} &= \frac{1}{s(s+1)} x_1(t_0) + \frac{1}{s(s+1)} x_2(t_0) \\ &+ \frac{1}{s(s+1)} x_3(kT) + \frac{1}{s(s+1)} r(kT) \quad (5-175) \\ &h(t) = x_3(kT) \\ &= -ax_1(t_0) + (a_1 - ab_1)x_1(kT) + a_0 r(kT) \quad (5-176) \\ &x(k+1)T = -x_1(t_0) - b_1x_1(kT) + r(kT) \end{aligned}$$

(5-177)
Taking the inverse Laplace transform in Eqs. (5-174) and (5-175) and letting $t_0 = kT$, the transition equations are obtained as follows:

$$\begin{aligned} x_1[(k+1)T] &= [1 - a_0(T - 1 + e^{-1})]x_1(kT) + (1 - e^{-1})x_2(kT) \\ &+ (a_1 - ab_1)(T - 1 + e^{-1})x_1(kT) \\ &+ a(T-1+e^{-T})r(kT) \end{aligned}$$

$$\begin{aligned} x_2[(k+1)T] &= -a_0(1 - e^{-1})x_1(kT) + e^{-1}x_2(kT) \\ &+ (a_1 - ab_1)(1 - e^{-1})x_1(kT) \\ &+ a(1-e)r(kT) \quad (5-178) \\ x_3(kT) &= -ax_1(kT) + (a_1 - ab_1)x_1(kT) + a_0 r(kT) \quad (5-179) \end{aligned}$$

$$(5-180)$$

$$(5-181)$$

Note that since the state variable $x_3(kT)$ does not appear in the transition equations of Eqs. (5-178), (5-179), and (5-181), Eq. (5-180) is not needed in obtaining the solution.

Response of Sampled-Data Systems Between Sampling Instants

The state-variable method may be readily applied to the evaluation

of system response between sampling instants. From the studies in the previous sections, the state-transition equations of a sampled-data system may be written in the vector-matrix form as

$$x(t) = Y(t - t_0)x(t_0) + \int_{t_0}^t \phi(t - \tau)r(\tau)d\tau$$

(5-182)

For uniform-rate sampling, if only the response at the sampling instants is desired, we let $t = kT + T$, and since $t_0 = kT$, $t - t_0 = T$. Equation (5-182) reads

(5-183)

$$x[(k + 1)T] = Y(T)x(kT) + \int_{kT}^{(k+1)T} \phi(T - \tau)r(\tau)d\tau$$

However, if the response at $t = kT + AT$, where $AT < T$, is desired, we simply substitute $t = kT + AT$ and $t_0 = kT$ into Eq. (5-182). We have

$$x(kT + AT) = Y(AT)x(kT) + \int_{kT}^{kT + AT} \phi(AT - \tau)r(\tau)d\tau \quad (5-184)$$

For example, the response of the system in Example 5-5 at $t = kT$ and $t = kT + AT$ is given by the following transition equations:

$$x_1(kT + AT) = (2 - e^{-AT})x_1(kT)$$

$$+ (1 - e^{-AT})x_2(kT) + (AT - 1 + e^{-AT})r(kT) \quad (5-185)$$

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$$x(kT) = \begin{bmatrix} 2 - e^{-1} \\ 1 - e^{-1} \end{bmatrix} x_1(kT) + \begin{bmatrix} 1 - e^{-1} \\ AT - 1 + e^{-1} \end{bmatrix} r(kT)$$

$$+ (1 - e^{-1})x_2(kT) + (AT - 1 + e^{-1})r(kT) \quad (5-186)$$

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TIME RESPONSE OF SAMPLED-DATA CONTROL SYSTEMS

6.1 INTRODUCTION

Like the analysis of systems with continuous data, the study of the transient response of sampled-data systems is extremely important. In general, the output response of a sampled-data system can be separated into two parts: the steady-state response and the transient response. Since all physical systems contain inertia and friction, the output response before the steady-state is reached does not follow the reference input signal at all times. The deviation between the actual output and the desired output is termed the error signal or, generally, the actuating signal. Again, as in the study of continuous-data systems, the transient behavior of a sampled-data system is described by such terms as overshoot, rise time, delay time, settling time, damping ratio, damping factor, natural undamped frequency, etc. The definitions of some of these terms are given in Fig. 6-1. The transient response also indicates the relative stability of the system. The steady-state error of a system is defined as the error between input and output signals when steady state is reached. Although theoretically a steady-state-error-free system exists, all physical systems do in practice have a certain amount of error in the steady-state operation because of inherent properties of feedback and friction.

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c(1)4

Unit-step

input

Maximum overshoot

1.05

1.0-

0.95

0.90-

0.5

0.1

1

max

delay

T_s

settling

time

time

-T, rise time

Fig. 6-1 Time response of sampled-data system to unit-step input showing the time domain performance criteria.

A brief outline of the subjects covered in this chapter follows:

(1) Time Response of Systems with Typical Test Input Signals.

Unfortunately, the actual input excitations of feedback control systems are not known ahead of time so that conventional analysis and design techniques require the assumption of some basic test signals such as the step, ramp, parabolic, and sinusoidal functions as system inputs. In studies of transient response the step response is of importance because, as shown in Fig. 6-1, the performance criteria in the time domain are all defined with respect to a step displacement type of input. In the next section, we shall investigate the

transient behavior of a simple linear closed-loop system with and without sampling. As was mentioned in Chapter 1, the main advantage of sampling is to gain sensitivity and to achieve time sharing of equipments. However, just as in any servo application, we should expect to pay some kind of penalty for the desirable feature we gain. Therefore, in the first part of this chapter, we shall compare the performances of a continuous-data system and the same system with sampling. The effect of variation of sampling frequency upon transient response and system stability will also be investigated.

(2) Stability Analysis.

Various methods of testing system stability will be discussed. In continuous-data systems the most useful methods have been the Routh-Hurwitz criterion, the Nyquist criterion, and the root locus plot. We shall see in this chapter, how these methods can be extended to the studies of sampled-data control systems. The Nyquist criterion, however, will be covered in Chapter 7.

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(3) The z-Plane Technique.

We know that the poles and zeros of the transfer function in the s-plane completely define the performance of a continuous-data system. However, in the studies of sampled-data systems, concentration is centered on the z-plane. By means of the root locus plot in the z-plane, we shall see how the pole-zero configurations affect the time response of sampled-data systems.

(4) Steady-State Errors.

By means of the final-value theorem for z-transforms, the steady-state error of a sampled-data system can be evaluated for a specific input signal. However, the z-transform method specifies the error only at the sampling instants. For systems with no hidden oscillations between sampling instants, z-transform analysis is quite adequate.

6.2 COMPARISON OF TRANSIENT RESPONSES OF CONTINUOUS-DATA AND SAMPLED-DATA SYSTEMS

Let us consider the simple continuous-data system shown in Fig. 6-2. For a unit-step displacement input, the output transform of the system is

$$C(s) = \frac{K}{s(s^2 + s + K)} \quad (6-1)$$

A 50 per cent damping system (damping ratio = 0.5) is obtained when K is equal to unity. Assuming zero initial conditions, the unit-step response of the system is obtained by taking the inverse Laplace transform of Eq. (6-1); thus

$$c(t) = 11.157e^{-0.5t} \sin(0.866t + 60^\circ) \quad (6-2)$$

The response $c(t)$ is plotted in Fig. 6-5 as a function of time. We see from the plot that the maximum overshoot of the system is approximately 16 percent.

Now, let us consider that error sampling is employed in the system of Fig. 6-2. The same system now with sampled-data in the error

$$\begin{aligned} R(s) \\ E(s) \\ 1/G(s) = \\ K \\ s(s+1) \\ K=1 \\ -C(s) \\ H(s) = 1 \end{aligned}$$

Fig. 6-2 Closed-loop system with continuous-data.

$$C(z)$$

$$R(s)$$

$$R(z) +$$

$$E(s) \quad E^*(s)$$

$$1$$

$$C(s)$$

$$G(s) =$$

$$T \quad E(z)$$

$$s(s+1)$$

$$H(s) = 1$$

Fig. 6-3 Closed-loop system with sampled-data.

channel is shown in Fig. 6-3. The output transform of such a system. configuration is given in the last chapter in Eq. (5-17) as

$$C^*(s) =$$

$$G^*(s)$$

$$1 + HG^*(s)$$

$$R^*(s)$$

and the z-transform of the output is given by Eq. (5-18) as

$$C(x) =$$

$$G(z)$$

$$1 + HG(z)$$

$$R(z)$$

$$(6-3)$$

$$(6-4)$$

For the transfer functions previously chosen for the continuous system, the z-transforms of $G(s)$ are found in Table 4-1. Therefore,

$$G(x) =$$

$$1$$

$$=$$

$$z(1 - e^{-T})$$

$$s(s+1) \quad (\approx -1)(x - e^{-T})$$

$$(6-5)$$

Since $H(s) = 1$, $HG(z)$ is also given by Eq. (6-5). For a unit-step displacement input,

$$R(z)$$

$$Z$$

$$=$$

$$-$$

$$2 \quad 1$$

Substituting Eqs. (6-5) and (6-6) into Eq. (6-4) yields

$$C(z) =$$

$$22(1$$

$$z^2(1 - e^{-T})$$

$$-$$

$$-T$$

$$(z - 1)(z - 2 \text{ zenten})$$

$$(6-6)$$

$$(6-7)$$

Let us assume that the sampling period T is one second. Eq. (6-7) becomes

$$0.63222$$

$$C(z)$$

$$=$$

$$(\approx -1)(2^2 - 0.736x + 0.368)$$

$$(6-8)$$

By means of long division, Eq. (6-8) is now expanded into a power series in z^{-1} :

$$C(z) = 0.632z^{-1} + 1.0962z^{-2} + 1.205x^{-3} + 1.12\%^{-1}$$

+1.01425 + 0.982-6 +
The inverse z-transform of C(z) is

-
-
(6-9)

$$\ddot{c}^*(t) = 0.6328(t - T) + 1.0968(t - 2T) + 1.2058(t - 3T),$$

$$+1.128(t - 4T) + 1.0148(t - 5T)$$

$$+0.988(6T) + \dots$$

(6-10)

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The output sequence $c(nT)$ is drawn in Fig. 6-5 together with the output response of the continuous-data system. The continuous output response of the sampled-data system is estimated by connecting the discrete output sequence by a smooth curve. Strictly speaking, the modified z-transform should be used to determine the response between sampling instants. However, in this simple system, the approximation by a smooth curve made here is quite good since the sampling frequency is substantially higher than the natural frequency of the system.

A comparison of the sampled-data system response with the response of the continuous-data system shows that the sampling operation has caused a decrease in rise time, but the overshoot is increased. The responses also show that the sampled system has faster oscillation and a slightly shorter settling time. In general, the effect of sampling is that of reducing the system stability. However, sampling of the error signal does improve the stability of systems with transportation lags.

In the above investigation we have arbitrarily chosen K to have a value of unity. When the value of K is increased to 5, the effect on the transient response of the sampled-data system is interesting. If the sampling period is still one second, the output z-transform for K = 5 becomes

$$C(x) = \frac{3.16x^2}{(x - 1)(x^2 -$$

$$z + 0.368)$$

(6-11)

which is expanded into the following series in z^{-1} :

$$C(z) = 3.16z^{-1} + 6.32z^{-2} + 8.32z^{-3}$$

$$+ 9.17z^{-4} + 9.24z^{-5} + \dots$$

(6-12)

If enough terms are carried out in the above series, we can show that the series actually diverges, which means that the system is unstable. (As will be proved in the next section, the marginal value of K for system stability when T = 1 sec in this case is 4.32.) This again shows that the effect of sampling generally is to reduce the margin of stability. We recall a well known fact for the second-order system with continuous-data as shown in Fig. 6-2: the system is always stable for all positive values of K. However, with sampling operation in the error channel, even a second-order system may become unstable for a relatively low value of K.

A zero-order hold is now inserted in the forward path of the system in Fig. 6-3. The sampled-data system with zero-order hold is shown in Fig. 6-4. The output z-transform C(z) of the system is derived as follows: The closed-loop pulse transfer function is

T

 $C(z)$ $R(s)$ $R(z)+$ $E(s)$ $E^*(s)$ 1-6-ST $C(s)$

1

 $G(s)=$

S

 $E(z)$ $s(s+1)$ $G_{ro}(s)$

1.6

 $c(t)$ $c^*(t)$

1.4

 $\bar{s}$ $H(s)=1$

Fig. 6-4 Sampled-data system with zero-order hold.

1.2

1

Second order system with sampled-data
and zero-order holdSecond order system with sampled -data
and no hold device

0.8

0.6

0.4

0.2

Second order system with continuous-data

(166)

い

1

0

1

2

3

4

5 6 7 8 9 10 11 12 13 14

Time (sec)

15 16 17

Fig. 6-5 System outputs of systems in Figs. 6-2, 6-3, and 6-4.

 $C^*(s)$

=

 $G_O G^*(s)$ $R^*(s) 1 + G_O G^*(s)$ Taking the z-transform on both sides of Eq. (6-13) yields
where

=

 $C(x) G_{ho}G(z)$ $R(z)$ $G_{re}G(x) = 3 [(1 -$

S

 $1 + G_O G(z)$ $= (1-z)^3$

ST

K

 $s(s +$

) ((41))]

K

(1)]

s² (s

1

-

(6-13)

(6-14)

= K(1 - x -

-

)

3

1

(+ + 1)

1

(6-15)

s

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From the z-transform table we have

GhoG(z) = K(1

-

ざりして

Tz

2

- 1)2

-

Z

1

-

Z

+

-

(6-16)

For T = 1 sec, and K = 1, Eq. (6-16) is simplified to the following form:

(0.3682 + 0.264)

GhoG(z) =

-

(≈ 1)(≈ 0.368)

-

(6-17)

Substituting Eq. (6-17) into Eq. (6-14) when R(z) = x/(x - 1), the z-transform of the output response is

C(z)

=

-

z(0.3682 + 0.264)

z - 1)(x² z + 0.632)

(6-18)

By means of long division the output z-transform C(z) is expanded into a power series in z⁻¹. Thus

C(z) = 0.368z + 1.00z² + 1.40z³ + 1.40x + 1.15% - 5

-6 + 0.80 z - 7

-8

9

+ 0.90z

+ 0.862 + 0.97 z + 1.05 x 10

+ 1.06z⁻¹¹ + 1.01z⁻¹ 12 + 0.96x - 13 +

(6-19)

The inverse z-transform of $C(z)$ is obtained from Eq. (6-19) in the usual manner and is plotted in Fig. 6-5. A comparison of the output responses of the sampled-data systems with and without hold circuit shows that the zero-order hold causes a longer delay time and higher peak overshoot (over 40 percent) in the step response. With the zero-order hold, it also takes a longer time for the transient response to settle. The trend towards higher overshoot and oscillation in the system with hold device is expected, since the addition of the hold circuit is equivalent to adding phase lag in the system. and since it is well known that phase lag usually has an adverse effect on the stability of feedback control systems.

6.3 STABILITY OF SAMPLED-DATA SYSTEMS

In the studies of linear feedback control systems with continuous signals, we have always kept in mind that the most important performance criterion of all is over-all system stability. If a system is unstable, then to speak of its transient or steady-state response would be insignificant, since the system is virtually useless unless it is first stabilized by some means of compensation. Based on the information we have obtained from the illustrative examples given in the last section, we shall expect the problem of stability to be more acute in control systems with sampled-data, since the sampling operation usually undermines system stability.

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In words, the stability of a linear continuous-data system is defined as: A system is stable if for any bounded input, the corresponding output is also bounded. Logically, this same definition should also be carried over to linear systems with sampled-data, especially since most sampled-data systems do have output responses that are continuous functions of time. However, the analysis and design of sampled-data systems have relied to a great extent on the z-transform technique, which provides information on responses only at the sampling instants in the form of $c^*(t)$ or $c(nT)$. Therefore, the stability tests of sampled-data systems are usually conducted with respect to the output pulse sequence rather than with the actual output signal. However, the stability analysis based on the z-transform method will lead to erroneous results if the system response contains hidden oscillations of the form shown in Fig. 6-6. In this case, the pulse sequence occurring at the sampling instants converges, but the actual

$c(1)$
 $c(nT)$
AAAAA
5T 6T
8T

Fig. 6-6 Hidden oscillation in output response of sampled-data system.

$C(1)$
 $C(z)$
 $R(s)$
 $r(1)$
 $C(s)$
 $R(s)$
 $R^*(s)$
 $c(1)$
 $G(s)$
 $G(s)$

$c(1)$
 $r(t)$ T
 (1)
 $C(s)$
 (a)
 (b)

Fig. 6-7 Continuous- and sampled-data systems.

Usually, a perfect integrator or velocity servomechanism is considered as an exception to this definition.

ex-

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time response is unbounded. In such a case, modified z-transform or the submultiple-sampling method should be used.

Let us now consider the continuous and the sampled-data systems shown in Fig. 6-7. The stability of the continuous system is completely determined by the location of the poles of $G(s)$ in the complex s-plane. For a stable system all the poles of $G(s)$ must lie in the left-half s-plane; if at least one of the poles of $G(s)$ is found in the right-half of the s-plane, the system is unstable. This can readily be seen from the inverse Laplace transform of $G(s)$; a pole of $G(s)$ with positive real part corresponds to a rising exponential function in the time domain.

If we assume that for the sampled-data system in Fig. 6-7b there are no hidden oscillations in the system and thus that z-transform method is applicable, the system is said to be stable if the output sequence $c(nT)$ is bounded for a bounded input. Therefore, for a bounded $c(nT)$, the corresponding actual response $c(t)$ (whatever form $c(t)$ may take) must also be bounded, and its Laplace transform $C(s)$ must have poles only in the left-half s-plane.

As was shown in Sec. 4-5, the z-transformation maps the left-half of the s-plane into the interior of the unit circle in the z-plane. Therefore, the z-transform of the response $c(t)$ must have all its poles located inside the unit circle in the z-plane. Since $C(z)$ is a product of $R(z)$ and $G(z)$, we can now say that the stability of the sampled-data system depends entirely on the pole location of the z-transfer function $G(z)$ because the input is assumed to be bounded. For a stable system, the poles of $G(z)$ must not be found in the region outside the unit circle in the z-plane. The regions of stability in the s-plane and the z-plane are depicted in Fig. 6-8. In the s-plane the regions of stability and instability are separated by the imaginary axis, whereas in the z-plane they are separated by the unit circle $z = 1$. Poles of $G(z)$ of the system in Fig. 6-7b that are on the unit circle in the z-plane will give rise to sustained oscillation in the output response.

Let us now consider a closed-loop sampled-data system of the form shown in Fig. 6-9. Based on the discussions above, the closed-loop system will be stable if the poles of the closed-loop z-transfer function $C(z)/R(z)$ are all located inside the unit circle in the z-plane. In other words, the roots of the characteristic equation.

$$1 + HG(z) = 0$$

(6-20)

must all lie inside the unit circle. Or, if z represents the roots of Eq. (6-20), then the closed-loop sampled-data system is stable if and only if all the z_i 's satisfy

$$|z_i| < 1$$

(6-21)

200

$j\omega$
 Im z
 s-plane
 Stable
 region
 Unit circle
 Unstable
 region
 Stable
 region
 Stable
 region
 Unstable
 region
 Unstable
 region
 z-plane
 Unstable
 region
 Re z
 Unstable
 region

Fig. 6-8 Regions of stability in the s-plane and the z-plane.

$C(z)$
 $R(s)$
 $E(s)$
 $E^*(s)$
 $C(s)$
 $G(s)$
 $G(z)$

Fig. 6-9 Simple closed-loop sampled-data system.

At this time, we shall investigate the possible ways of testing the stability of a sampled-data control system. We recall that for continuous-data feedback control systems, system stability may be investigated by use of one of the following five well-known methods:

(1) The Routh-Hurwitz Criterion: Determines whether any roots of the characteristic equation lie in the right-half of the s-plane.

(2) The Nyquist Criterion: Investigates whether or not the open loop transfer function plot encircles the $(-1, j0)$ point in the G-plane and hence determines the difference between the number of poles and zeros of the closed-loop transfer function that are in the right-half s-plane.

(3) The Bode plot: This may be used to determine whether or not any real frequency exists at which the open-loop gain is greater than 0 db when the phase shift is greater than 180 deg.

(4) Decibels of G as a Function of Phase Shift of G : This is a plot of the magnitude of G in db versus the phase shift of G . From this plot the absolute stability as well as the relative stability (measured by phase and gain margins) of the closed-loop system may be

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investigated. By superimposing this plot upon the Nichols chart, the frequency response of the closed-loop system may be obtained.

(5) The Root Locus Plot: The root locus diagram indicates graphically the location of the roots of the characteristic equation when the open-loop gain K of the system is varied.

We shall see that the stability analysis of sampled-data systems does not depend on the derivation of a wealth of new techniques at all. As a matter of fact, with some modifications each of the techniques mentioned above can be used to study the stability of sampled-data control systems. The only new techniques that are introduced

$$\begin{vmatrix} a_k & \bar{a}_n & \bar{a}_{n-1} & 0 & 10 & 0 & 0 & \bar{a}_n & 0 & 0 \\ \bar{a}_{n-2} & \bar{a}_{n-1} & a_n & 0 & 1800 & 18 & 18 & \bar{a}_{n-k+1} & \bar{a}_{n-k+2} & \bar{a}_{n-k+3} \\ \bar{a}_n & 0 & 0 & \bar{a}_0 & \end{vmatrix}$$

where $k = 1, 2, 3, \dots, n$, and a is the conjugate of a_k . Note that when all the coefficients of the polynomial are real, the determinant Δ is symmetrical with respect to its principal diagonal. The stability criterion can also be stated as: A sampled-data system is stable when all its characteristic equation roots lie inside the unit circle in the z -plane provided that the coefficients of the equation satisfy the following conditions:

$a_k < 0$ for k odd
 $a_k > 0$ for k even)
 (6-24)

However, if the conditions stated in Eq. (6-24) are not satisfied, then the characteristic equation has at least one root that falls outside the unit circle. Unlike the Routh-Hurwitz criterion, the Schur-Cohn test does not indicate in any definite manner how many roots are outside the unit circle: nor does it tell if any of the roots are located on the unit circle.

Example 6-1

by
 Consider that the characteristic equation of a sampled-data system is given
 (6-25)

equation are determined as follows:

$$(2z - 1)(z + 2)$$

$$= 2z^2 + 3z - 2 = 0$$

The Schur-Cohn determinants of this

$$\begin{vmatrix} a_0 & a_n \\ a_n & \bar{a}_0 \end{vmatrix}$$

$$= \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} = 21 - 1 = 0$$

Immediately, we know that Eq. (6-25) has at least one root located outside the unit circle in the z -plane, since A , is not less than zero.

The determinant A , is written

$$\begin{vmatrix} a_0 & 0 \\ a_n & a_{n-1} \end{vmatrix}$$

$$= \begin{vmatrix} 2 & 0 \\ 0 & 1 \end{vmatrix} = 2$$

$$\begin{array}{r} a_n \\ 3 \\ - \\ A_2 \\ \bar{a}_n \\ 0 \\ \bar{a}_0 \quad \bar{a}^1 \\ 2 \\ \bar{a}_{n-1} \quad \bar{a}_n \quad 0 \\ \bar{a}_0 \\ 3 \\ 202 \\ 323 \quad 2 \\ - \quad 2 \\ --- \\ 0 \quad - \quad 2 \\ 202 \quad 0 \\ 144 \quad (6-27) \end{array}$$

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Thus, the sequence of Schur-Cohn determinants reads: 1, 0, 144, which

has only one change in sign (where for the second-order system to be stable, there should be two changes in sign). From the given characteristic equation in Eq. (6-25), we know that the roots are at $z = 1/2$ and $z = 2$; the latter is located outside the unit circle.

Example 6-2

by

Consider that the characteristic equation of a sampled-data system is given $(\approx + 1)(\approx + 0.5)(\approx + 2)$

$$\approx^3 + 3.5\approx^2 + 3.5\approx + 1 = 0 \quad (6-28)$$

From the last equation we know that one root is inside the unit circle, one is right on the unit circle, and one is outside the circle. The Schur-Cohn determinants of this equation are determined as follows:

$$\Delta_1 = 0 \quad (6-29)$$

$$\Delta_2 = \begin{array}{r} 1 \\ 0 \quad 1 \\ 3.5 \\ 3.5 \\ 1 \\ 0 \\ 1 \\ 1 \\ D_2 \\ = \\ 0 \\ (6-30) \end{array}$$

$$\begin{array}{r} 1 \\ 0 \\ 1 \\ 3.5 \\ 3.5 \\ 1 \\ 0 \quad 1 \\ 1 \\ 0 \\ 0 \end{array}$$

1
 3.5
 3.5
 3.5 1
 0 0
 1
 3.5
 3.5 3.5 1
 0
 0
 1
 A3
 = 0
 (6-31)
 1
 0
 0
 1
 3.5
 3.5
 3.5 1
 0 0 1 3.5
 1
 3.5 3.5 1 0 0

Thus, the sequence of Schur-Cohn determinants reads: 1, 0, 0, 0, and the stability criterion is not met. However the test says only that not all of the roots of the characteristic equation are inside the unit circle. No conclusion can be made on how many roots are in the region outside the unit circle or on the unit circle.

From these two examples, we see that the application of the Schur-Cohn stability criterion is not nearly as convenient as the Routh-Hurwitz test ordinarily used for continuous-data systems. First of all, for higher order systems the use of the Schur-Cohn test relies on the evaluation of higher order determinants. There does not seem to be any simple way of tabulating the equation coefficients so that stability condition can be investigated without manipulating with the determinants. Certainly, the attractiveness of the Routh-Hurwitz criterion is due in part to the fact that the equation coefficients of the continuous-data system can be tabulated into the form of the Routh tabulation and then stability condition is obtained simply by manipulation of the coefficients of the tabulation. Furthermore, the

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Routh's test will give the number of roots that are in the right-half of the s-plane and also will tell whether any root exists on the imaginary axis, whereas the Schur-Cohn test tells only whether all the roots are inside the unit circle in the z-plane.

6.5 JURY'S STABILITY TEST

A stability test which is simpler than the Schur-Cohn test procedure is devised by Jury and Blanchard. The method is presented below without proof.

Referring to the polynomial of Eq. (6-22) with $a_n > 0$, the first step in Jury's stability test is the formation of the following table using the coefficients of the polynomial:

n-k
 Row
 ~°
 z¹
 z²
 2n-1
 z"

1
ao
αv
a2
an-k
An-1
an
2
an
An-1
An-2
Ak
ai
8
ao
3
bo
9
b1
b2
bn-k
bn-1
45
4
bn-i bn-
bn-3
bk
●
bo
5
Co
C1
C2
Cn-2
Co
6
Cn-2 Cn-3 Cn-4
●
2n
-
5 Po
Pi P2 P3
2n
-
4 P3 P2
Pi
Po
2n
-
3 qo qi
92

Note that the elements of row $2k + 2$ consist of the coefficients of the row $2k + 1$ written in reverse order. The elements in the table are defined as

ao
An-k
bk
-
an
ak
Co
Cn-2-k

d_k
 $C_{n-2} \quad C_k$
 q_0
 $P_0 \quad P_3$
 $P_3 \quad P_0$
 b_0
 $C_k =$
 b_{n-1-k}
 $b_{n-1} \quad b_k$
 $P_0 \quad P_i$

q_2
 $P_3 \quad P_2$
 The necessary and sufficient conditions for the polynomial $F(z)$ to have no roots on and outside the unit circle in the z -plane are:
 $F(1) > 0$
 $F(1) < 0$
 n even
 n odd

E. I. Jury, and J. Blanchard, I. R. E. Proc., 49, pp. 1947-48.

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$|a_0| < a_n$
 $|b_0| b_{n-1}$
 $|C_0$

C_{n-2}
 $d_0 > d_{n-3}$

$(n-1)$ constraints
 Example 6-3

The polynomial given by Eq. (6-28) is to be tested by Jury's method.
 We have $F(1) = 9 > 0$, but $F(1) = 0$, which violates the condition that
 for $n =$

$= 3$ $F(-1)$ must be less than zero. Therefore, the polynomial has at
 least one root lying on or outside the unit circle.

Example 6-4

Consider the polynomial
 $F(z) = z^4 - 1.368z^3 + 0.4z^2 + 0.08z + 0.002 = 0$
 The first step of the stability test finds that
 $F(1) = 0.114 > 0$ and $F(-1) = 2.69$

0
 $(6-32)$
 Since $n = 4$ is even, these conditions satisfy the $F(1) > 0$ and $F(-1) > 0$
 requirements for stability. Next, we tabulate the coefficients in the following
 manner:

Row
 1
 23
 2
 3
 -
 0.002
 1
 0.08
 - 1.368
 0.4
 - 1.368
 1
 0.4
 0.08

0.002
 1
 1.368
 -0.399
 0.083
 4
 · 0.083
 -
 - 0.399
 1.368
 - 1
 5
 0.993
 -
 - 1.385
 0.512

From the above tabulation, we have

$$1001 = 0.002 < 1$$

1601

$$|a_0| < a$$

$$= 1 > 0.083$$

$$|b_0| |b_3|$$

$$= 0.993 \cdot 0.512$$

$$|c_0| > |C_2|$$

Since all these constraints are satisfied, the system represented by the polynomial as its characteristic equation is stable.

6.6 EXTENSION OF ROUTH-HURWITZ CRITERION

TO SAMPLED-DATA SYSTEMS

Any attempt to apply the conventional Routh-Hurwitz test to the study of stability of sampled-data systems meets with immediate difficulty. If we try to apply the test directly to the characteristic

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equation of a sampled-data system in the s-domain, we see that the equation is of the form $1 + HG^*(s) = 0$ [or $1 + G^*(s)H^*(s)$, as the case may be] which is normally a transcendental function of s; therefore, the Routh's test breaks down completely. Trying to apply the Routh's test to the characteristic equation in the z-domain also fails, since the stability boundary in the z-plane is no longer the imaginary axis but rather is the unit circle $z = 1$. However, if we map the interior of the unit circle in the z-plane onto the left-half plane of another complex variable plane by some kind of transfor-

4jw

=

▲ Im z

∞

36

s-plane

4

The primary strip

w - plane

jww

W =

-j1

z=esT

S

0

z-plane

r-plane

jwr

8

Fig. 6-10 The mapping of the primary strip in the s-plane into the interior of the unit circle in the z-plane and the mapping of the z-plane unit circle to the r-plane and the w-plane through the bilinear transformations.

-j1

Re z

1

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s-plane

jw

z-plane

Im z

Stable

region

Unstable

region

Stable region

Re z

Stable

region

Unstable

region

Unstable region

juw

jw,

w-plane

r-plane

Stable

region

Unstable

region

Stable

Unstable

region

region

0w

Stable

region

Unstable

Stable

Unstable

region

region

region

Fig. 6-11 Regions of stability in the s-plane, the z-plane, the w-plane, and the r-plane.

mation, the Routh-Hurwitz criterion may be applied again directly to the equation in this new variable.

We shall investigate the bilinear transformation

$z =$

$aw + b$

$cw + d$

(6-33)

where w is a complex variable; that is, $w = \sigma + j\omega$. In Eq. (6-33)

if we set $a = b = 1$, $c = 1$, and d

$=$

1, we have

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$$\frac{z-1}{z+1} = w$$

(6-34)

We shall see that Eq. (6-34) maps the interior of the unit circle in the z -plane onto the left-half of the w -plane. For the purpose of identification of this process, we shall call Eq. (6-34) the "w-transformation." However, if in Eq. (6-33) $a = b = 1$ but $c = 1$ and $d = -1$, the bilinear transformation also maps the inside of the z -plane unit circle into the left-half of the w -plane. To distinguish this case from the previous one, we use another complex variable, r , so that

$$\frac{z-1}{z+1} = r$$

(6-35)

which is called the "r-transformation." The mappings of the primary strip in the s -plane into the unit circle of the z -plane and then from z to the r and w planes by the r and w transformations are shown in Fig. 6-10. If all the periodic strips in the s -plane are considered, the z -transformation maps the entire left-half of the s -plane into the interior of the unit circle in the z -plane; the r or the w transformation then maps the inside of the z -plane unit circle onto the entire left-half of the r or the w plane. This is illustrated in Fig. 6-11. As far as the application of the Routh-Hurwitz criterion is concerned, either the w - or the r -transformation may be used; we can easily see that no real advantage is gained by choosing one transformation over the other. Both transformations will convert the original characteristic equation in z into a quotient of two polynomials in either r or w of the same order. In order to be consistent, the r -transformation is used here exclusively as the medium for the application of the Routh-Hurwitz criterion to sampled-data systems. As will be shown later in the discussion on frequency domain design of sampled-data systems, for certain types of systems the use of the r -transformation may lead to simpler solutions, whereas for other types of systems the design may be more straightforward if the w transformation is used.

The determination of stability of sampled-data systems by use of the r -transformation is illustrated by the following examples:

Example 6-5

Consider the characteristic equation given by Eq. (6-28), which is repeated in the following:

$$(\approx + 1)(\approx + 0.5)(\approx + 2)$$

$$= z^3 + 3.522z^2 + 3.5z + 1 = 0$$

(6-36)

Substituting the r -transformation of Eq. (6-35) into the last equation, we have

$$\frac{z-1}{z+1} = r$$

$$\left(\frac{r-1}{r+1}\right)^3 + 3.5\left(\frac{r-1}{r+1}\right)^2 + 3.5\left(\frac{r-1}{r+1}\right) + 1 = 0$$

(6-37)

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which is simplified to

$$r = 0$$

(6-38)

In this case the roots of Eq. (6-38) can be obtained by mere inspection, and

thus the Routh's test need not be applied. The roots of Eq. (6-38) are $r = 0$, $r = -1$, and $r = 3$. The roots of the characteristic equation in the z -plane and the r -plane are sketched in Fig. 6-12.

z -plane

Stable

root

Im z

jur

r -plane

1

Re z

2

Stable

Unstable

Unstable

roots

root

roots

Fig. 6-12 Roots of Eqs. (6-36) and (6-38) in the z -plane and the r -plane, respectively.

Example 6-6

Let the open-loop transfer function of a unity-feedback control system with sampled-error signal be

22.57

$G(s)$

=

$s^2 (s + 1)$

Let the sampling period be one second; the z -transform of $G(s)$ is

(6-39)

$G(z) =$

22.57 $\approx (0.3632 + j0.264)$

$(\approx -1)^2 (\approx -0.368)$

(6-40)

The characteristic equation of the system is given by the numerator of which is

$1 + G(z) = 0$

23 + 5.9422 + 7.7z

0.368

-

0

(6-41)

(6-42)

Substituting the r -transformation of Eq. (6-35) into Eq. (6-42) yields

3

2

$(+1) + 5.94$

94 $(\approx +1)^2 + 7.7(1) - 0.368 = 0$

r

The last equation is simplified to

$14.27r^3 + 2.3r^2 + 11.74r + 3.13 = 0$

-

(6-43)

(6-44)

By inspection we see that Eq. (6-44) has a negative coefficient, and since the necessary conditions for a polynomial to have no root with positive real part are that all its coefficients must be of the same sign and none of the coefficients may vanish, we know that Eq. (6-44) must have at least one root in the right-half of the r -plane. This fact means that Eq. (6-42) must have at least one root outside the unit circle in the z -plane and that the system is

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unstable. But just for the purpose of illustrating the application of the Routh-Hurwitz criterion, we make the following Routh's tabulation using the coefficients of Eq. (6-44):

$$\begin{array}{r} \text{d03} \\ 7-2 \\ 14.27 \\ 2.3 \\ 11.74 \\ 3.13 \\ - \end{array}$$

$$\begin{array}{r} \text{pol} \\ - 27 \ 44.6 \\ = 31.1 \\ 2.3 \\ 0 \\ 7-0 \\ 3.13 \end{array}$$

Since there are two changes of sign in the first column of the last tabulation, the characteristic equation in Eq. (6-44) has two roots located in the right-half of the r -plane. This corresponds to having two roots outside the unit circle in the z -plane for Eq. (6-42). This result can be checked by solving Eq. (6-42); the three roots are at $z = -2$, $z = -3.97$, and $z = 0.032$. When mapped into the r -plane, the three corresponding roots are at $r = \infty$, $r = 0.6$, 1.07 . The z -plane roots and the r -plane roots are illustrated in

$$\begin{array}{r} \text{and } r = \\ \text{Fig. 6-13.} \end{array}$$

$$\begin{array}{r} \text{Im } z \\ j\omega \\ z\text{-plane} \\ r\text{-plane} \\ \text{Unit circle} \\ 0.33 \\ * \end{array}$$

$$\begin{array}{r} - 3.97 \\ 0.032 \\ \text{Re } z \\ * \end{array}$$

$$\begin{array}{r} -1.07 \\ 16.6 \\ \text{Unstable} \\ \text{roots} \\ \text{Stable} \\ \text{root} \\ \text{Stable} \\ \text{root} \\ \text{Unstable} \\ \text{roots} \end{array}$$

Fig. 6-13 The roots of the characteristic equation of Example 6-6 sketched in the z -plane and the r -plane.

Example 6-7

In this example the Routh-Hurwitz criterion is applied to the sampled-data system in Fig. 6-3. As has been shown in Sec. 6.2, for a sampling period of one second, the system is stable for $K = 1$, but is unstable for $K = 5$. The marginal value of K for a stable operation in this case is apparently between 1 and 5. The Routh's test will be used here to investigate the variation of the marginal value of K as the sampling period T is varied. The characteristic equation of the system with K and T unspecified is written

$$z^2 + [K(1 - e^{-1}) - (1 + e^{-T})]z + e^{-T} = 0$$

-e
 By use of the r-transformation, Eq. (6-45) is converted to
 0
 (6-45)

$$K r^2 + 2r + \frac{2(1+e^{-T})}{1-e^{-T}}$$

 -
 - K = 0
 (6-46)

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The Routh's tabulation of the last equation is as follows:

7.2
 K
 2
 7-0

$$2(1 + \frac{K}{2(1 + e^{-T})})$$

 1-e-T
 -
 - K
 1

The Routh-Hurwitz criterion says that a system is stable if all the elements in the first column of the Routh's tabulation are positive. Therefore, for the system to be stable, the following conditions are desired:

and
 $K > 0$
 $K < 2(1 + e^{-T})$
 1-e-T
 (6-47)

The conditions of stability given by Eq. (6-47) are plotted in Fig. 6-14; the unstable values of K and T are indicated by the shaded region. One point of special interest is that when T is equal to one second, the marginal value of K for stability is 4.32. When the sampling period is increased (slower sampling rate), the plot in Fig. 6-14 shows that the marginal value of K is decreased, thus proving that in this system sampling does reduce system stability.

K
 8.17
 Unstable region
 Stable
 region

4.32
 2.63
 2.21

T
 1
 2
 3
 T (sec)

Fig. 6-14 Marginal values of K for stability as a function of the sampling period T of the system shown in Fig. 6-3.

As a closing remark to this section, we can say that the stability study of sampled-data systems by use of the bilinear r-transformation and the Routh-Hurwitz criterion is quite straightforward in principle. In transforming from the z to the r domain, the degree of the polynomial is not increased (it may actually be reduced if the orig-

inal polynomial has roots at $z = 1$). However, in the case of high-order systems, the work involved in simplifying the polynomial in r before applying the Routh's test is quite tedious.

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6.7 EFFECT OF POLE-ZERO CONFIGURATION OF

$C^*(s)$ IN THE S-PLANE AND $C(z)$ IN THE Z-PLANE

UPON SYSTEM TRANSIENT BEHAVIOR

We know that the Routh-Hurwitz criterion tests only the absolute stability of a system. However, if a system is stable, the criterion does not indicate how stable the system is. Additional information about the time response of a stable sampled-data system at the sampling instants must be obtained from the location of the poles of $C^*(s)$ in the left-half of the s-plane or the poles of $C(z)$ inside the unit circle in the z-plane. A knowledge of the pole-zero locations in the s-plane or the z-plane enables the designer to predict the transient behavior of sampled-data systems.

From the studies of continuous-data systems, the correlation between the locations of the poles and zeros of $C(s)$ in the s-plane and the transient response of $c(t)$ is well known. For instance, complex conjugate poles of $C(s)$ in the left-half s-plane give rise to sinusoidally damped responses; poles on the negative real axis correspond to monotonically decaying responses; and complex conjugate poles on the imaginary axis will give rise to undamped constant-amplitude sinusoidal oscillations. All poles in the right-half s-plane correspond to unstable responses. Since the z-transformation is merely a mapping process, and the mapping of the s-plane into the z-plane has been demonstrated in Sec. 4-3, the correlation between the pole-zero location of $C(z)$ in the z-plane and the transient behavior of the sampled-data system at the sampling instants can be established simply by investigation of the correspondence of the s-plane and the z-plane pole-zero configurations. In regard to the pole locations of $C^*(s)$ in the s-plane, we shall consider the following cases:

1. Poles on the Real Axis in the s-Plane

The periodic property of $C^*(s)$ has been established previously in Chapter 2; that is, the poles of $C^*(s)$ that are located in the primary strip of the s-plane are also duplicated in the complementary strips. Therefore, in this case, poles of $C^*(s)$ on the real axis must also be accompanied by complementary poles located on the horizontal axes of $s \pm jn\omega_s$, where $n = 1, 2, 3, \dots$. The negative real axis, together with its complementary axes of $s \pm jn\omega_s$ ($n = 1, 2, 3, \dots$) in the s-plane, when mapped by the z-transformation correspond to the positive real axis inside the unit circle in the z-plane. Similarly, the positive real axis and its complementary axes

=
=

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of $s \pm jn\omega_s$, ($n = 1, 2, 3, \dots$) in the s-plane are mapped into the portion of the positive real axis outside the unit circle in the z-plane. These can be shown by substituting $s \pm jn\omega_s$ into

$z = e^{st}$. Therefore,

Complementary

strip

Primary

strip -02

$z = e^{(\pm \sigma \pm jn \omega_s)T}$

=

etor

$A_{j\omega}$
 s -plane
 $j\omega_s$
 $j\omega_s/2$
 $-1-j\omega_s/2$
 Complementary
 $-j\omega_s$
 $*$

strip
 (a)
 z -plane
 Unit circle
 Im
 (6-48)
 $\pi\omega_s$
 Re

الناسا
 (b)

Fig. 6-15 (a) Transient responses corresponding to various pole locations of $C^*(s)$ in the s -plane (real poles only). (b) Transient sequence corresponding to various pole locations of $C(x)$ in the z -plane.

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For region inside the left-half s -plane, Eq. (6-48) gives

$z = e$

(6-49)

whose magnitude is always less than unity, and thus corresponds to the region inside the unit circle in the z -plane. For the region in the right-half of the s -plane, Eq. (6-48) becomes

$z = e^{+\sigma T}$

(6-50)

which always has a value greater than unity. Thus the right-half of the s -plane corresponds to the exterior of the unit circle in the z -plane. A typical set of transient behaviors due to poles of $G^*(s)$ on the real axis in the s -plane is illustrated in Fig. 6-15a. The corresponding transient behaviors described by the z -plane poles on the positive real axis are shown in Fig. 6-15b. Consequently, we can conclude that a uniform variation in system transient response comes from a positive real pole of $C(z)$ in the z -plane. Furthermore, if the pole is inside the unit circle, the transient dies out monotonically or the output sequence $C(nT)$ decays uniformly; and if the pole is outside the unit circle, the transient increases in magnitude monotonically and the output sequence is divergent.

2. A Pair of Complex Conjugate Poles in the Primary Strip of the s -Plane

Consider a pair of complex conjugate poles at $s = \sigma \pm j\omega$, in

$\sigma_1 -$

the s -plane, where $\omega_1 < \omega_s/2$. The corresponding poles in the z -plane are at

$\sigma_1 \pm j\omega_1$

(6-51)

which are complex conjugate poles inside the unit circle in the z -plane. The oscillatory transient responses due to complex conjugate poles in the s -plane are shown in Fig. 6-16a. In this case, since it is assumed that the sampling frequency ω_s is greater than twice the frequency of the complex s -plane poles, $2\omega_1$, the output sequence defined by the z -plane poles will also be oscillatory as shown in Fig. 6-16b. Notice that when the s -plane poles are on the imaginary axis, at $s = j\omega$, where $\omega < \omega_s/2$, the envelope of the output sequence is

sinusoidal with frequency ω_1 .

=

3. Complex Conjugate Poles on the Boundaries of the Periodic Strips in the s-Plane

-

σ_1

Now let us consider a pair of complex poles located at $s = -\sigma_1 \pm j\omega_1/2$, where ω_1 is the sampling frequency. The corresponding z-plane poles are at

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jw

s-plane

Complementary

strip

jws

2

jws/2

3

Primary

strip

X1

A

X

Complementary

strip

(a)

-jws/2

-jws

KA

M

3

Im z

z-plane

Re z

1

Unit circle

(b)

Fig. 6-16 (a) Transient responses corresponding to various pole locations of $C^*(s)$ in the s-plane (complex conjugate poles only).

(b) Transient response sequence corresponding to various pole locations of $C(z)$ in the z-plane.

t

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A

4jw

jus

s-plane

A

jus

MA

T

-02

-01

(a)

jws

Im z
z-plane
t
Unit circle
MA
1

Re z
(b)
Fig. 6-17 (a) Transient responses corresponding to various pole locations of $C^*(s)$ in the s-plane (complex conjugate poles on the boundaries between periodic strips). (b) Transient response sequence corresponding to various pole locations of $C(z)$ in the z-plane.

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$2 = e^{-17}$
(6-52)
These are a pair of double poles located on the negative real axis inside the unit circle of the z-plane. Since the s-plane poles are complex, the envelope of the output sequence will be oscillatory. However, in this special case, assume that $w_1 = w_s/2$; exactly one sample will be in each one-half period of the envelope of the output sequence. Therefore, the pulses in the output sequence will occur in alternating (positive and negative) magnitudes as shown in Fig. 6-17b.

A significant point concerning the above mentioned case is that the sampling frequency w , is at a minimum specified by Shannon's sampling theorem. Apparently, if the sampling is decreased from the present value so that $w_1 > w_s/2$, the sampled-data will not be able to describe the original signal completely, and the z-transform method becomes unreliable.

The results of the above discussion concerning the correspondence of the transient behavior and the pole locations in the s-plane and the z-plane for sampled systems are tabulated in Table 6-1.

TABLE 6-1
CORRESPONDENCE OF THE TRANSIENT BEHAVIOR OF SAMPLED-DATA SYSTEMS AND THE POLE LOCATIONS OF $C^*(s)$ IN THE s-PLANE AND $C(z)$ IN THE z-PLANE

Location of poles of $C^*(s)$ in the primary strip of the s-plane	Location of poles of $C(z)$ in the z-plane
(1) Poles in the left-half s-plane	Poles inside the unit circle in the z-plane
Poles on the negative real axis in the s-plane	Poles on the positive real axis in the z-plane
Complex conjugate poles in the s-plane	Complex conjugate poles
Complex conjugate poles on the boundaries between periodic strips	
(2) Poles on the imaginary axis in the s-plane	
(3) Poles in the right-half s-plane	

inside the unit circle in the z-plane
 Double poles on the negative real axis inside the unit-circle in the z-plane
 Poles on the unit circle in the z-plane

Poles outside the unit circle in the z-plane

Transient behavior

of $c^*(t)$

Stable response

Monotonically decaying

output sequence

Positively damped oscillatory output sequence

Decaying output sequence

with alternating (positive and negative magnitudes)

Sustained oscillatory output sequence with constant

amplitude

Unstable response

† A damped sinusoid is not a periodic function, but the maxima and the minima

of the function occur at periodic intervals of T_i

$= \frac{2\pi\omega_i}{T_i}$

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6.8 EFFECT OF THE POLE-ZERO CONFIGURATIONS

IN THE Z-PLANE UPON THE MAXIMUM

OVERSHOOT AND PEAK TIME OF TRANSIENT

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In the preceding section we have obtained the relationship between the characteristic equation roots in the z-plane and the transient response of a second-order sampled-data system. When the characteristic equation has complex roots inside the unit circle in the z-plane, the corresponding output response of the system will be oscillatory with positive damping. In general, the complex roots that are inside and are closest to the unit circle give rise to more oscillatory responses (less damped). It remains for us to determine the relationships between the pole-zero configuration of the closed-loop transfer function and the maximum overshoot and peak time (time at which maximum overshoot occurs) of a sampled-data system.

For a simple second-order system with continuous-data, if the closed-loop transfer function is given by

$\frac{C(s)}{R(s)}$

$= \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$

(6-53)

then the maximum overshoot and the peak time are given by

and

$\frac{1}{\omega_n \sqrt{1-\zeta^2}}$

$\frac{\pi}{\omega_n \sqrt{1-\zeta^2}}$

cus, shoot & max

$\frac{1}{\omega_n \sqrt{1-\zeta^2}}$

$\frac{\pi}{\omega_n \sqrt{1-\zeta^2}}$

$\frac{1}{\omega_n \sqrt{1-\zeta^2}}$

$\frac{\pi}{\omega_n \sqrt{1-\zeta^2}}$

$$1 + \exp$$

$$- \text{peach lime I}$$

$$\pi$$

$$=$$

$$\max$$

$$\omega_n \sqrt{1}$$

$$2$$

(6-54)

(6-55)

For systems of the order higher than two, no simple relations can be obtained between $C_{\max}$, c_{Tax} , and the pole-zero configuration.

However, if a system is characterized by only a pair of dominant poles (poles of C/R that dominate the transient behavior) described by Eq. (6-53) and if the other poles and zeros are far to the left in the complex s -plane, the effects of these other poles and zeros on the transient response are negligible; c_{inax} and Tax in this case can still be approximated by using Eqs. (6-54) and (6-55).

For a sampled-data system, the problem of establishing the overshoot and peak-time-pole-zero configuration is more complex. This is because normally when the z -transform method is used, the output response is described only at the sampling instants. We shall select the following model as the closed-loop z -transfer function of a second-order system (B. C. Kuo, Automatic Control Systems, (Englewood Cliffs, N. J.: Prentice-Hall, Inc., 1962)).

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order feedback control system with sampled-data (the sampler is in the forward path):

where

$$\frac{C(z)}{R(z)}$$

$$=$$

$$\frac{K(z - z_1)}{(z - p_1)(z - p_1)}$$

$$=$$

$$\frac{K}{(z - p_1)(z - p_1)}$$

$$(1 - p_1)(1 - p_1)/(1 - \approx 1)$$

(6-56)

(6-57)

and p is the complex conjugate of p_1 . If the system does not have a hold circuit following the sampler, the zero of Eq. (6-56) will be located at the origin in the z -plane, whereas if the system has a zero-order hold, the zero may lie at any place on the real axis in the z -plane.

When the system is subject to a unit step function input, the output transform $C(z)$ is given by

$$C(z)$$

$$=$$

$$zK(z - z_1)$$

(6-58)

$$(\approx - 1)(\approx - p_1)(z - P_1)$$

(6-58)

The output response at the sampling instants is obtained by applying the inversion formula of Eq. (4-47). Thus

$$c(nT) = \frac{2\pi}{273} \sum_{j=1}^J Kz(221) (z_1)(z_{p1})(z_{p1})$$

where I is a contour which encloses all the singularities of the integrand. Applying Cauchy's residue theorem, the output sequence $c(nT)$ is written

$$c(nT) = 1 + 2 \frac{K(p_{121})}{p_1^n \cos(\theta_1 + \phi_1)} \quad (6-60)$$

where $\phi_1 = \arg p_1$ (argument of p_1)

$$C(1) = \frac{C_{\max}}{\max c(nT)}$$

OT 2T 3T 4T 5T 6T 7T 8T 9T 10T 11T 12T 13T

Fig. 6-18 Continuous time response and output sequence of sampled-data control system.

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$$\theta_1(p_{121}) = \arg(p_1 - 21) - \arg(p_1 - 1) - \pi/2 \quad (6-62)$$

Since, as illustrated in Fig. 6-18, $c(nT)$ describes the values of the continuous response $c(t)$ only at the sampling instants, the problem of evaluating $C_{\max}(t)$ from the information contained in $c(nT)$ is a difficult one. The likelihood that $c_{\max}(t) = \max c(nT)$ is very small, implying that the maximum overshoot happens to occur exactly at a sampling instant. The determination of the maximum overshoot and the peak time by use of $c(nT)$ has been discussed in the literature. However, in these earlier methods, we must assume that only two dominant poles of $C/R(z)$ contribute significantly to the overshoot and peak time; all other closed-loop poles and zeros are considered to be very near to the origin of the z -plane. The method used here is similar to the work of D. P. Lindorff; the maximum overshoot and the peak time are computed by considering not only the dominant quadratic poles but also the influence of a closed-loop zero which may be located at any point on the real axis in the z -plane.

The poles and zeros of Eq. (6-56) are now depicted in Fig. 6-19. From the geometrical location of these poles and zeros, we define the angle α as shown in Fig. 6-19, and α is given by

$\alpha = \arg(p_1 - z_1) - \arg(p_1 - 1) + \pi/2$
 By use of Eq. (6-62) the angle α becomes

α

α

α

α

(6-63)
 (6-64)

Also, we can readily show that the following relationship holds between α and the closed-loop pole-zero locations:

$\sec \alpha = 2$

$K(p_1 - z_1)$

$(p_1 - 1)(p_1 - P_1)$

(6-65)

where K is given in Eq. (6-57). Substituting Eqs. (6-64) and (6-65) into Eq. (6-60), the system output sequence is written

$c(nT) = 1 + \sec \alpha x^n \cos(n\beta, \pm \alpha - n)$

Letting $t = nT$, then

and

n

$|P_1|^{1/T} = e^{-5\omega n}$

$\beta = \arg p_1 = \omega T = \omega_2 \sqrt{1 - g^2 T}$

(6-66)
 (6-67)

(6-68)

Therefore, a continuous function that passes through the points of $c(nT)$ is

† J. T. Tou, Digital and Sampled-data Control Systems, (New York: McGraw-Hill Book Co., Inc., 1959), pp. 327-336. E. I. Jury, Sampled-data Control Systems, (New York: John Wiley & Sons, Inc., 1958), pp. 132-142.

D. P. Lindorff, "Application of Pole-zero Concepts to Design of Sampled-Data Systems," I. R. E. Transactions on Automatic Control, Dec. 1959, pp. 173-184.

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-1

Im

$|P_1| = e^{-\sigma T}$

$\alpha \pi \theta_2 - \gamma$

$\alpha = \theta_2 - \theta_3 +$

EN

121

$\theta_3 = \angle P_1 - 1$

z-plane

P_1

θ_2

θ_1

θ_1

θ_1

θ_1

θ_1

θ_1

θ_1

z-plane
 $\alpha = \gamma - \theta_2$
 $-Z_1$
 $\alpha = -(\theta_2 - \theta_3 +)$
 $\theta_2 = \angle P_1 - Z_1$
 $\theta_3 = \angle P_1 - 1$
 P_1
 θ_3
 Re
 α
 θ_r
 1
 X

Fig. 6-19 Geometrical representation of α for a second-order sampled-data system with closed-loop transfer function given by Eq. (6-56).

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$$c(t) = 1 + \sec \alpha \exp(-\zeta \omega_n t) \cos(\omega_d t \pm \alpha - \pi) \quad (6-69)$$

The overshoot of $c(t)$ will be used to characterize the actual overshoot of the step response of the sampled-data system whose closed-loop transfer function is given by Eq. (6-56). The overshoot of the response in Eq. (6-69) is determined in the usual manner; that is, by taking the first derivative of $c(t)$ with respect to time and setting dc/dt equal to zero. The steps described above leads to the following relationship:

$$\tan(\omega_d t \pm \alpha - \pi) =$$

$$\frac{\pi}{\sqrt{1-\zeta^2}} \quad (6-70)$$

Since the time t solved from the last equation is the peak time $T_{\max}$, we have

$$T_{\max} = \frac{\pi}{\omega_d \sqrt{1-\zeta^2}} \quad (6-71)$$

Substituting Eq. (6-71) into Eq. (6-69) and simplifying, the maximum response is written

$$C_{\max} = 1 + \frac{\sec \alpha}{\sqrt{1-\zeta^2}} \exp\left(-\zeta \omega_n T_{\max}\right) \quad (6-72)$$

Hence, the maximum overshoot of the system response is

$$\text{Overshoot} = \frac{\sqrt{1 - \xi^2}}{\xi} \exp\left(\frac{-\pi}{\tan^{-1} \frac{\sqrt{1 - \xi^2}}{\xi}}\right) \quad (6-72)$$

We can readily see from Eq. (6-73) that the maximum overshoot is uniquely specified by the damping ratio and the angle α (see Fig. 6-19).

In Fig. 6-20 the percent overshoot given by Eq. (6-73) is plotted as a function of α . The angle α and its sign are determined from the pole-zero location of the closed-loop transfer function as described in Fig. 6-19. The value of α can be calculated by use of Eq. (4-26) or (4-27); that is, once the complex poles (P_1, ξ_1) of C/R are given, we have

$$P_1 \tan \alpha = \xi_1 \quad (6-74)$$

In the above expression, since p_1 and ξ_1 are obtained from the location of P_i , α is computed readily. The damping ratio of the complex roots is determined from

$$\xi = \sin \alpha \quad (6-75)$$

Or

$$\xi = \frac{1}{T_{\max}} \quad (6-76)$$

$$\xi = \frac{1}{T_{\max}} \tan^{-1} \frac{Q}{\sqrt{1 - Q^2}}$$

$$\xi = \frac{1}{T_{\max}} \tan^{-1} \frac{Q}{\sqrt{1 - Q^2}} \quad (6-76)$$

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Percent overshoot

3
1,000

500

5=0

100

C=0.1

=0.2

50

X=0.3

30

$\zeta = 0.4$
 $\zeta = 0.5$
 $\zeta = 0.6$
 $\zeta = 0.7$
 $\zeta = 0.8$
 α degrees
 -90 -60 -40 -20 0 +20 +40 +60 +90

Fig. 6-20 Percent overshoot of second-order sampled-data control systems.

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The curves of peak time (normalized in $T_{max} P1/T$) are plotted in Fig. 6-21 as functions of α and ζ . Note here that the time function of Eq. (6-69) closely approximates the actual time response of the system only if the sampling frequency of the sampler is sufficiently high. In general, for the method used here to be correct, the sampling frequency should be equal to at least twice the natural frequency of the system. In other words, the complex poles of C/R must be located roughly in the

250°
 220°
 200°
 180°
 160°
 140°
 120°
 100°
 80°
 60°
 40°
 20°
 0°
 $\zeta = 0$
 $\zeta \leq 0.1$
 $T_{max} P1$
 T
 $\zeta = 0.2$
 $\zeta = 0.3$
 $\zeta = 0.4$
 $\zeta = 0.5$
 $\zeta = 0.6$
 $\zeta = 0.7$
 $\zeta = 0.8$
 α
 -90° -70° -50° -30° -10° 0° +10° +30° +50° +70° +90°

Fig. 6-21 Peak time of step response of second-order sampled-data systems versus α .

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A

first and the fourth quadrants in the z -plane. Otherwise, the output sequence $c(nT)$ does not contain sufficient information about the actual time response; consequently, the inverse transform of $C(z)$ produces a time function which will not be an accurate approxima-

tion of the true output response.

The curves of percent overshoot and peak time given in Figs. 6-20 and 6-21 can also be used to approximate the overshoot and peak time of sampled-data systems with orders higher than two. In general, the method described above is applicable to systems having two complex dominant poles and a zero located at any point on the real axis; all other poles and zeros are very near to the origin of the z-plane.

Let us now consider an illustrative example using this method.

Im
 -0.717
 $\alpha = 13^\circ$
 z-plane
 $|p_1| = 0.8$
 $\phi = 51^\circ$
 $p_1 = 0.5 + j0.617j$
 $|p_1| = 0.6$
 $\alpha = 0^\circ$
 $-p_1 = 0.368 + j0.48j$
 $\phi_1 = 53^\circ$
 $0.2 \ 0.4 \ 0.6 \ 0.8 \ 1.0$
 p_1
 p_1
 System without hold:
 $C(z)$
 $=$
 $0.632z$
 $R(z) = \frac{z^2 - 0.736z + 0.368}{z^2 - 0.736z + 0.368}$
 System with zero-order hold:
 (poles: $0.368 + j0.48j$)
 (zero: 0)
 $C(x)$
 $R(z)$
 $0.3682 + 0.264$
 (poles: $0.5 + j0.617j$)
 $=$
 Re
 $2 + 0.632$ (zero: -0.717)

Fig. 6-22 Pole-zero locations of second-order sampled-data control system in Figs. 6-3 and 6-4.

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Consider the simple error-sampled-data system of Fig. 6-3. The overshoot and peak time of the system estimated from the z-transform method used in Sec. 6-2 are 20 percent and 2.9 sec, respectively. (see Fig. 6-5). From the geometrical location of the poles and zeros of C/R depicted in Fig. 6-22, we have $p_1 = 0.6$, $\alpha = 0^\circ$, and $\phi_1 = 53^\circ$. By use of Eqs. (6-74) and (6-75), the damping ratio of the quadratic poles is found to be 0.485. Therefore, from Fig. 6-20, we found the overshoot to be 20 percent; and from Fig. 6-21, the peak time is approximately equal to 2.85 sec.

$=$
 1

In a similar manner, the overshoot and peak time of the second-order sampled-data system with zero-order hold of Fig. 6-4 are calculated with reference to the pole-zero location of the closed-loop transfer function depicted in Fig. 6-22. The overshoot and peak time of the system with zero-order hold are found to be 48 percent and 3.5 sec, respectively. ($p = 0.8$, $\alpha = 13^\circ$, $\phi_1 = 51^\circ$, $y = 0.244$.) These

quantities check closely with those shown in Fig. 6-5.

=
.
—

6.9 ROOT LOCUS METHOD FOR SAMPLED-DATA SYSTEMS

The root locus technique has been quite useful and popular in the analysis and design of linear control systems with continuous-data. The root locus diagram of a continuous-data system is essentially a plot of the roots of the characteristic equation as a function of the loop gain K (K varies from 0 to ∞). In the preceding section, we have shown that the stability and transient behavior of sampled-data systems depend entirely upon the poles (and zeros, for transient response) of a closed-loop pulse-transfer function $C^*(s)/R^*(s)$ or the closed-loop z -transfer function $C(z)/R(z)$. Therefore, if the conventional root locus technique can be conveniently adapted to sampled-data system, the transient behavior and other important properties of these systems may conceivably be obtained with ease from a plot of the poles of $C^*(s)/R^*(s)$ or the poles of $C(z)/R(z)$ with the system loop gain K as a variable parameter.

Let us first consider the possibility of constructing the root locus plot of a sampled-data system in the s -plane. For the simple closed-loop system shown in Fig. 6-23, the closed-loop pulse-transfer function is given by

$C^*(s)$

—

$$\frac{G^*(s)}{R^*(s)} \frac{1}{1 + HG^*(s)}$$

(6-77)

† W. R. Evans, "Graphical Analysis of Control Systems," Trans. A. I. E. E. 67, 1948, pp. 547-551.

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T

$C(z)$
 $C^*(s)$
 $R(s)$
 $E(s)$
 X
 $E^*(s)$
 $G(s)$
 $C(s)$
 $H(s)$

Fig. 6-23 Simple closed-loop system with sampled-data.

The characteristic equation of the system is

$$1 + HG^*(s) = 0$$

(6-78)

Equation (6-78) is generally a transcendental function of s , and since the conventional root locus technique applies only to algebraic equations, the difficulty involved in this case is apparent. The main idea

$\Delta i\omega$
 $j\omega$
 s -plane
 s -plane
 $j2\omega_s$
 $K=0$
 $K=0$
 $0\pi\omega_s$
 $\infty 4K$
 $j\omega_s$
 $K=0$ X

$*K=0$
 iz WS
 $\infty 48$
 $K=0$
 $K=0$
 -1
 0
 -1
 $-j$ = WS.
 $\infty 4K$
 K_{∞}
 $-jws$
 $K=0$ X
 $K=0$
 $\infty 4K$
 K_{∞}
 $-j2ws$
 $K=0$ X
 $K=0$
 (0)
 (b)

$$G(s) = K/s(s + 1), H(s) = 1$$

Fig. 6-24 Pole-zero configuration of $HG^*(s)$ and the root locus diagram in the s -plane of the sampled-data system in Fig. 6-3.
 (a) Poles of $HG^*(s)$. (b) Root loci.

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of the root locus is that the roots of $1 + HG^*(s) = 0$ are plotted from the knowledge of the poles and zeros of $HG^*(s)$. However, we know that the poles and zeros of $HG^*(s)$ are infinite in number. As an example, consider the system shown in Fig. 6-3. The pole-zero configuration of $HG^*(s)$ is depicted in Fig. 6-24a. We can show that the root loci of the characteristic equation take the form of Fig. 6-24b when the forward gain K is varied from 0 to ∞ . Clearly, for more complex pole-zero configurations for $HG^*(s)$, the construction of the root loci in the s -plane will be more difficult because of the periodic property of $HG^*(s)$. However, the closed-loop z -transfer function $C(z)/R(z)$ is a rational function of z , and the characteristic equation of the system in Fig. 6-23 in the z -domain is an algebraic equation in z ,

$$1 + HG(z) = 0$$

(6-79)

Therefore, the construction of the z -plane root loci should be quite straightforward since rules of construction of the conventional root locus diagrams are directly applicable to the z -plane loci. In essence, the construction of the z -plane root loci is to find the points that satisfy the following conditions:

(1)

(2)

$$|HG(x)| = 1$$

$$|HG(z)|$$

=

$$180^\circ + k \times 360^\circ$$

TABLE 6-2

(6-80)

(6-81)

RULES OF CONSTRUCTION OF ROOT LOCUS DIAGRAMS

(1) Starting points

($K = 0$)

(2) Ending points

($K \rightarrow \infty$)

(3) Number of separate root loci

(4) Symmetry of root loci

(5) Asymptotes of root loci as $s \rightarrow \infty$

The root loci start at the poles of $HG(z)$. (The poles include both those in the finite z -plane and at infinity.)

The root loci end at the zeros of $HG(z)$. (The zeros include both in the finite z -plane and at infinity.)

where

Number of loci $N = Z$ if $Z > P$

Number of loci $N = P$ if $Z < P$

P = number of finite poles of $HG(z)$

Z = number of finite zeros of $HG(z)$.

=

The root loci of systems with rational transfer functions are symmetrical with respect to the real axis of the z -plane.

For large values of s , the root loci are asymptotic to straight lines with angles determined by

$(2k + 1)\pi$

$P - Z$

where $k = 0, 1, 2,$

$k \leq \frac{P-Z}{2}$

up to (but does not include)

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(6) Intersection of the asymptotes

(7) Root loci on the real axis

(8) Angles of departure and arrival

(9) Intersection of the root loci with the unit circle

(a) The intersection of the asymptotes lies only on the real axis in the z -plane.

(b) The point of intersection of the asymptotes on the real axis is given by

$\sigma =$

=

-

(b_{11})

$\sum \text{Poles of } HG(z) - \sum \text{zeros of } HG(z)$

$P - Z$

On a given section of the real axis in the z -plane, root loci may be found in the section only if the total number of poles and zeros of $HG(z)$ to the right of the section is odd.

The angle of departure of the root locus from a pole (or the angle of arrival at a zero) of $HG(z)$, can be determined by assuming a point z_1 which is very close to the pole (or zero) and is on the root locus associated with the pole (or zero), and applying the following equation:

m

-

$m+n$

$$\sum_{i=1}^m \frac{1}{z + p_i} - \sum_{j=1}^n \frac{1}{z + q_j} =$$

=

i=1

j=1

$$(2k + 1)\pi$$

The values of K at the crossing points on the unit circle in the z-plane made by the root loci may be obtained by use of the modified Routh-Hurwitz criterion.

(10) Breakaway points of If the characteristic equation of a sampled-data system the root loci is written as

$$K = f(x)$$

all breakaway points on the root loci are determined by solving for the roots of

(11) Calculation of K on the root loci

$$\frac{dK}{dz}$$

$$= 0$$

The value of K corresponding to any point on the root loci in the z-plane may be determined graphically from the following equation:

1

$$K =$$

$$|HG(\approx 1)|$$

2, . . . all integers.

where k = 0, 1,

If we write HG(z) in a standard form as

$$HG(z) =$$

$$\frac{K(z + p_1)(z + p_2) \dots (z + p_m)}{(z + q_1)(z + q_2) \dots (z + q_n)}$$

$$(6-82)$$

$$(z + p_1)(z + p_2) \dots (z + p_m)$$

$$(z + q_1)(z + q_2) \dots (z + q_n)$$

Then, the two conditions given in Eqs. (6-80) and (6-81) become

$$|HG(z)| =$$

$$=$$

$$m$$

$$K \prod_{i=1}^m |z + p_i|$$

$$i=1$$

$$m+n$$

$$\prod_{j=1}^{m+n} |z + q_j|$$

$$=$$

$$1$$

$$(6-83)$$

$$j=1$$

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and

$$m$$

$$m + n$$

$$|HG(x)| =$$

$$\sum_{i=1}^m \frac{1}{z + p_i} - \sum_{j=1}^{m+n} \frac{1}{z + q_j}$$

$$2=1$$

$$-$$

$$\sum_{j=1}^{m+n} \frac{1}{z + q_j}$$

$$= (2k + 1)\pi$$

$$(6-84)$$

$$j=1$$

Consequently, given the pole-zero configuration of HG(z), the con-

struction of the root locus diagram in the z-domain involves the following two steps:

(1) A search for the points in the z-plane that satisfy the condition given by Eq. (6-84).

(2) The value of K at a given point z, on the root loci is determined from Eq. (6-83).

As mentioned above, the rules of construction developed from the properties of the root loci for continuous-data systems are still applicable in the z-plane; these rules are tabulated in Table 6-2. The proofs of the rules are not included here; they may be found in most standard textbooks on feedback control systems.†

Example 6-8

In this illustrative example, the root loci of the sampled-data system shown in Fig. 6-3 will be constructed in the z-plane. The open-loop z-transfer function of the system is given by Eq. (6-5) and is now repeated as follows:

$G(z)$:

$$= \frac{K(1 - e^{-T})}{(z - 1)(z - e^{-1})} \quad (6-85)$$

Since the system has unity-feedback, $H(s) = 1$, and the characteristic equation is

$$1 + G(z) = 0 \quad (6-86)$$

The root loci of the system are the plots of the roots of Eq. (6-86) when K is varied from 0 to ∞ . Let us consider that the sampling period T is one second, then, Eq. (6-85) reads

$$G(z) = \frac{0.632Kz}{(z - 1)(z - 0.368)} \quad (6-87)$$

= 1, one of
- ∞ . From
+ 1 and between
From rule (10),

Equation (6-87) has two poles at $z = 1$, $z = 0.368$, and a zero at $z = 0$. The root loci in the z-plane can be constructed on the basis of the rules given in Table 6-2. From rule (1), the loci start from $z = 0.368$ and $z = 1$; the loci terminates at $z = 0$, and the other locus terminates at $z = -1$. From rule (7), the parts of the real axis between $z = 0.368$ and $z = 1$ and between $z = -1$ and $z = 0$ constitute sections of the loci.

the breakaway points (there are two) are located at $z = 0.62$. However, we can show that for this two-pole and one zero configuration, the complex conjugate section of the root loci is a circle. This is proved as follows:

-
0.62 and $z =$
Let

$$z = x + jy \quad (6-88)$$

† B. C. Kuo, Automatic Control Systems, Chapter 8, (Englewood Cliffs, N. J.: Prentice-Hall Inc., 1962).

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Then, Eq. (6-87) becomes

$$G(z) = \frac{0.632K(x + jy)}{(x + jy - 1)(x + jy - 0.368)}$$

$$= \frac{[(x - 1) + jy][(x - 0.368) + jy]}{(6-89)}$$

On the root loci, z must satisfy Eq. (6-84); therefore,

$$\frac{1}{G(z)} = \tan^{-1} (2)$$

$$= \frac{\tan^{-1} y(2x - 1.368)}{(x - 1)(x - 0.368)}$$

$$= \frac{y^2}{(2k + 1)\pi}$$

$$\tan/G(z) = \frac{1}{1+}$$

$$\text{Hence}$$

$$x$$

Taking the tangent of both sides of the last equation yields

$$\frac{y}{x} = \frac{y(2x - 1.368)}{(x - 1)(x - 0.368) - y^2}$$

$$= \frac{y(2x - 1.368)}{(x - 1)(x - 0.368) - y^2}$$

$$= \frac{y^2}{(6-91)}$$

$$= \frac{y^2}{(x - 1)(x - 0.368) - y^2}$$

Simplifying, we have

$$x^2 + y^2 = 0.368$$

which is an equation of a circle with the center at the origin of the z -plane, and the radius is 0.62.

The complete root loci of the system for $T = 1$ sec are drawn in Fig.

6-25. The intersection of one of the loci with the unit circle is at $z = -1$; the corresponding value of K at the point is equal to 4.32.

Im
z-plane
K=2.18
K=0.244
∞ K K=4.32
K=4.1

K4∞
K=0
K=0
Re
-1
-0.62
0.368 0.62

1
 $K=2.18$
 Unit circle
 $K2(1 - e^{-T})$
 $G(x)$

=
 $T = 1 \text{ sec}$
 $(x - 1)(z - e^{-T})$

Fig. 6-25 Root locus plot of sampled-data system in Fig. 6-3.

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Im
 z-plane
 Critical valu

$\infty K K=2.63$

$K=1.35$

-1

$K=2.16$

-0.368

$K=0$

$K+\infty 0.135$

$K=0.463$

0.368

$K=0$

Re

$K=1.35$

Unit circle

$G(x)$

=
 $K2(1 - e^{-T})$

$(x - 1)(z - e^{-T})$

$T = 2 \text{ sec}$

Fig. 6-26 Root locus plot of sampled-data system in Fig. 6-3.

In a similar manner, the root loci of the system for T

scribed by

$x^2 + y^2$

$= 0.135$

=

2 sec are de-

(6-94)

and the loci are depicted in Fig. 6-26. The critical value of K for system stability when $T = 2 \text{ sec}$ is found (from Fig. 6-14) to be 2.63 as compared with 4.32 for $T 1 \text{ sec}$. Notice that when the sampling period of the system is increased from one to two seconds, the stability margin is decreased; however, the circular section of the loci has a smaller radius. This is usually a misleading feature of the z -plane root loci, since one would think that normally the stability of systems is measured by the closeness of the root loci with the unit circle. The fact is that the loci in Fig. 6-25 and Fig. 6-26 seem to be of the same form, but the values of K at various points on the two plots are not in correspondence.

By the same analytical method used above, we can show that the complex conjugate section of the root loci of this system is generally described by the following equation:

$x^2 + y^2 = e^{-T}$

(6-95)

Relating the z -plane loci to the s -plane loci shown in Fig. 6-24 can be very interesting. Evidently, since the z -plane is the conformal mapping of the s -plane through the z -transformation $z = e^{sT}$, the z -plane root loci of Fig. 6-25 are conformal to the s -plane loci in Fig. 6-24.

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Example 6-9

Plot the root locus diagrams of the sampled-data system with zero-order hold as shown in Fig. 6-4. The open-loop transfer function of the system is given by Eq. (6-16) and is now simplified to

$$G_oG(z) = K[(T+e^{-T})z - Te^{-T} + 1 - e^{-T}]$$

.

-

$$(\approx - 1)(\approx - e^{-T})$$

(6-96)

Equation (6-96) can also be written as

$$G_oG(z) = K(T - 1 + e^{-T}).$$

(z+a)

$$\approx - 1)(\approx - e^{-T})$$

(6-97)

where

1

$$a =$$

(6-98)

-T

$$e - Te^{-T}$$

$$T - 1 + e^{-T}$$

The open-loop z-transfer function $G_oG(z)$ has poles at $z = 1$, $z = e^{-T}$ and a zero at $z = a$. The root loci in the z-plane must start from $z =$

1 and $z =$

e^{-T} , and one of the loci terminates at $z = -a$, whereas the other locus termi-

Im

z-plane

$$K=15$$

$$K \rightarrow \infty$$

$$-0.717$$

0

Unit circle

Root locus

$$K=2.43$$

$$K=0.196$$

$$K=0$$

$$K=0$$

*

Re

$$0.368$$

$$0.65$$

1

$$K=2.43$$

$$G_oG(z)$$

=

$$K[(T+e^{-T})z - Te^{-T} + 1]$$

-

$$(z - 1)(z - e^{-T})$$

$$e^{-T}]$$

$$T = 1 \text{ sec}$$

Fig. 6-27 Root locus plot of the sampled-data system with zero-order hold in Fig. 6-4.

TABLE 6-3 ROOT LOCI OF SOME SIMPLE 2-TRANSFER FUNCTIONS

Characteristic of loci

One pole

Straight line from the pole to minus infinity on the real axis of the z-plane

$K=0$

Root locus plot

$K=0$

One pole and one zero

Straight line from the pole to the zero on the real axis in the z-plane

$K=0$

$K=0$

-1

$K=0$

Two real poles

Straight line between the poles on the real axis and a vertical line bisecting the real loci

$K=0$

$K=0$

$K=0$

Two complex conjugate Vertical straight lines from the poles

two poles

* $K=0$

$K=0$

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le-zero configuration

of HG(z)

TABLE 6-3 (Continued)

Characteristic of loci

Root locus plot

Two poles and one

zero.

(real poles)

Straight line segments on the real axis, and a circle described by

zero: $z = 1$;

pole: $P_1 = \mu_1$

$r =$

$(x + a)^2 + y^2 = r^2$

$a = \{$

$[\mu_1\mu_2 - \zeta(\mu_1 + \mu_2) + \zeta^2]\}$

-

θ

Two poles and one

zero.

(complex conjugate

Straight line segments on the real axis, and circular segments de-

scribed by

poles)

zero: $z = 1$

pole: $p_1 = \mu_1 + jv$

$T =$

$P_2 = \mu_1 - jv$

$$(x + a)^2 + y^2$$

ξ

$$a =$$

-

$$r^{-2}$$

$$(\mu_1^2 + v^2 - 2\mu_1\mu_2 + \xi^2)\}$$

Two poles and two
zeros.

(real poles and real
zeros)

$$\text{zero: } Z_1 = 0$$

$$Z_2 = \xi$$

$$\text{pole: } P_1$$

=

$$\mu_1$$

$$P_2 = \mu_2$$

Two poles and two
zeros.

(complex or real)

$$\text{zero: } z_1 = 1 + jn$$

$$\text{pole: } P_1$$

=

-

$n?$

$$M_1 + ju$$

$$P_2 = M_2 - jv$$

Straight line segments on real
axis, and circular segments de-
scribed by

$$a =$$

$$r^2 = a^2$$

$$(x + a)^2 + y^2 = r^2$$

$$\mu_1\mu_2 - \xi_1\xi_2$$

$$(\mu_1 + \mu_2) - (\xi_1 + \xi_2)$$

$$(\mu_1 + \mu_2)\xi_1\xi_2 - (\xi_1 + \xi_2) M_1 M_2$$

$$+ (\mu_1 + \mu_2) - (\xi_1 + \xi_2)$$

Straight line segments on real
axis, and circular segments de-
scribed by

$$(x + a)^2 + y^2 = r^2$$

$$(P_{12} + v^2)$$

-

$$a =$$

$$(\mu_1 + \mu_2) -$$

$$r^2 = a^2 +$$

-

$$(\xi_1\xi_2 + n^2)$$

$$(1 + 2)$$

$$(\mu_1 + \mu_2) (\xi_1\xi_2 + n^2)$$

$$(\mu_1 + \mu_2) - (\xi_1 + \xi_2)$$

$$(\xi_1 + \xi_2) (M_1 M_2 + v^2)$$

$$(\mu_1 + \mu_2) - (\xi_1 + \xi_2)$$

X

X

X

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K=1.45

Im

z-plane

$K=3.7$
 -1.5
 $K=0$
 -0.524
 0
 $K=0$
 $*$
 0.135
 $K=0.157$
 $K=0$
 Re
 0.48
 Root locus
 $K=1.45$
 Unit circle
 $K[(T - 1 + e^{-T})z - Te^{-T} + 1 - e^{-T}]$
 $\text{GroG}(z) =$
 $-$
 $T = 2 \text{ sec}$
 $(\approx -1)(x - e^{-T})$
 $-$

Fig. 6-28 Root locus plot of the sampled-data system with zero-order hold in Fig. 6-4.

$-$
 $-$
 nates at ∞ . The real part of the root loci lies in the section between $z = e^{-T}$ and $z = +1$ and between minus infinity and $z = a$ on the real axis. The complex conjugate section of the loci is obtained by substituting $\approx = X + jy$ and Eq. (6-97) into the phase angle requirement of root loci in Eq. (6-84). After simplification, the result is
 $(x + a)^2 + y^2 = a^2 + a + (a + 1)e^{-T}$
 $-$
 (6-99)
 $a, 0$ on
 $= 1$
 This last equation represents a circle with the center located at $(-a, 0)$ on the real axis, and the radius of the circle is $[a^2 + a + (a + 1)e^{-T}]^{1/2}$. The root loci of the sampled-data system with zero-order hold for $T = 1 \text{ sec}$ and $T = 2 \text{ sec}$ are constructed in Fig. 6-27 and Fig. 6-28, respectively. Notice that for the system with zero-order hold, the root loci may intersect the unit circle at complex conjugate points rather than on the real axis as in the case when the hold circuit is absent. The critical value of K for stability is 2.43 for $T = 1 \text{ sec}$, and it is 1.45 for $T = 2 \text{ sec}$.
 $-$

$-$
 From these two examples, we see that the construction of the root locus diagram for a sampled-data system in the z -plane is similar to that of the continuous case, except that the interpretation on system stability is different since the stable region is now inside the unit circle.

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When investigating the transient behavior from the z -plane root loci, the constant damping, the constant damping ratio, and the constant frequency loci in the z -plane must be referred to. The mapping of these loci from the s -plane to the z -plane has been given in Sec. 4.3; the application of these loci with the root locus plots will be discussed in the next section.

By use of the method illustrated in Examples 6-8 and 6-9, the root loci of any sampled-data system with rational $HG(z)$ may be constructed. Just as in continuous-data systems, the Spirulet can also be used as an aid in the cut-and-try method of plotting the root

loci in the z-plane. The root loci of some simple z-transfer functions. are given in Table 6-34.

6.10 TRANSIENT RESPONSE AND RELATIVE STABILITY FROM ROOT LOCUS PLOTS

Once the root locus diagram of a given sampled-data system is constructed in the z-plane, the absolute stability as well as the relative stability of the system can be observed from the root loci. Absolute stability requires that all roots of the characteristic equation lie inside the unit circle in the z-plane. Relative stability, on the other hand, is also studied in terms of the locations of the roots of the characteristic equation. The questions that concern us are: If the system is stable, how stable and how good is it? Does its transient behavior satisfy the designer's specifications? Specifically, we want to obtain information of some performance criteria such as the settling time, maximum overshoot, and damping ratio of the system. In general, if the realization of certain minimum settling time and specific damping ratio is of interest, the behavior of the root loci with respect to the constant damping factor and the constant damping ratio loci in the z-plane should be investigated. These loci and the constant frequency loci were given in Sec. 4-3. The constant damping loci are a family of concentric circles centered at the origin of the z-plane; the radius of the circle corresponding to a given damping factor σ is $e^{-\sigma T}$, where T is the sampling period in seconds. The constant frequency loci are straight lines emanating from the origin of the z-plane at angles of ωT radians measured from the positive real axis. These loci and their s-plane counterparts are shown.

† Available from the Spirule Company, 9728 El Venado, Whittier, Calif.
M. Mori, "Root-Locus Method of Pulse Transfer Function for Sampled-Data Control Systems," I. R. E. Trans. on Automatic Control, PGAC-3, November 1957, p. 13-20.

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in Fig. 4-4 and 4-5. If the design of a certain sampled-data system requires a certain minimum settling time, then all the characteristic equation roots of the system must lie to the left of the constant damping locus $s = \sigma$ in the s-plane, and correspondingly in the z-plane all roots must lie inside the circle with radius $e^{-\sigma T}$ (Fig. 6-29). For a constant damping ratio, the constant loci in the z-plane are shown to be a family of logarithmic spirals, except for $\zeta = 0$ and $\zeta = 1$. Typical constant loci for $\zeta = 0.5$ are shown in Fig. 6-30 for the s-plane and the z-plane. However, if a certain damping ratio is specified, all the characteristic equation roots must lie to the left of the constant damping ratio line in the s-plane (if this damping ratio is 50 percent, the roots must not be found in the

jw
s-plane
s-plane
Im
z-plane
Unit circle

Fig. 6-29 Constant damping loci in the s-plane and the z-plane.

jw
1232 jus
30°
jws
30°
Zjws
- $\xi\omega\zeta$
-jws
Im

$\omega\zeta$
 z-plane
 Re
 $\sigma=0$
 1
 Re
 Unit circle
 $-3/j\omega$
 Fig. 6-30 Constant damping ratio in the s-plane and the z-plane
 for $\zeta = 0.5$.
 -

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shaded region in Fig. 6-30). Correspondingly, no characteristic equation roots must be found to the right of the logarithmic spiral of the given in the z-plane. As is shown in Fig. 6-30, since as the frequency is increased, the magnitude of the spiral decreases logarithmically and its phase angle increases linearly, and eventually the shaded region in the z-plane is to cover the entire interior of the unit circle. At this point, it would seem that the damping ratio requirement for sampled-data systems could never be realized. For-
 $j\omega$ s-plane
 --

$\omega\zeta$
 $j\omega$
 Im
 z-plane
 Insignificant
 poles
 30°
 $\omega\zeta$
 $\frac{1}{2} j\omega$
 X
 $\sigma=0$
 Re
 Primary
 strip
 Insignificant
 poles
 $-\frac{1}{2} j\omega$
 -1ω
 Unit circle
 30°
 $-j\omega$
 $-\frac{2}{3} j\omega$
 (a)
 Insignificant
 $\frac{1}{2} j\omega$
 Im
 poles
 30°
 s-plane
 z-plane
 X
 X
 X
 Primary
 strip
 X
 X
 -
 - $j\omega$

---jws
 30°
 $\omega\zeta$
 Re
 MIT.
 1

$\omega\zeta$
 -ws
 - 3 jws
 (b)

Fig. 6-31 (a) Constant damping ratio loci in the s-plane and the z-plane for $\zeta = 0.5$ when only the primary strip in the s-plane is significant. (b) Constant damping ratio loci in the s-plane and the z-plane for $\zeta = 0.5$ when only the primary strip and the two strips between $j\omega$, are significant.

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 $3=60^\circ$

s = plane
 45°
 $20^\circ \ 10^\circ \ 3^\circ$
 $30^\circ \ 15^\circ \ 5^\circ$
 $90^\circ - \beta$
 (0)

$3=0^\circ \ C = 0$
 $B=3^\circ = 0.052-$

Im
 $ws/4$
 $B=5^\circ = 0.087-$
 $B=10^\circ = 0.174$
 $B=15^\circ \ 0.2597$
 $-8 = 20^\circ = 0.342$
 $-B = 30^\circ = 0.5$
 $-8 = 45^\circ = 0.707$
 $-3 = 60^\circ \ C = 0.866$

$ws/2$
 $w=0$
 $8 = 90\% = 1.0$

Fig. 6-32 (a) The constant
 (b)

loci in the s-plane for the periodic strip of 0 to $j\omega_s/2$. (b) The constant loci in the z-plane corresponding to the loci in the s-plane in (a).

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Unfortunately, most servo systems have components that act as low-pass filter. Therefore, if the forward path transfer function of a sampled-data system introduces attenuation at high frequencies so that the high-frequency poles in the complementary strips become insignificant, only a portion of the logarithmic spiral need be considered. For instance, if the forward path transfer function introduces rapid attenuation at frequencies higher than $w/2$ so that only poles in the primary strip of the s-plane are significant, to realize a damping ratio of $\gamma = 0.5$, the characteristic equation roots must be found inside the unshaded region of Fig. 6-31a. If rapid attenuation is introduced by the forward path elements at frequencies higher than w , then only the poles in the strips between $+j\omega_s$ and $j\omega$, in the s-plane need be considered; the corresponding region in the z-plane is the unshaded area inside the unit circle as shown in Fig. 6-31b. In Fig.

6-32 the constant loci for various values of

-

are plotted in the

∞ KK=15

Im

z-plane

Root locus

K = 2.43

Constant

damping factor

locus

$\sigma = 0.332$

Constant

damping ratio

= 0.5

K = ∞ 0

+0.717

K = 0.49

Constant frequency

w=0.558

rad/sec

32°

K=0

| x=0

K=

1.0

0.368

0.65

K = 0.196

Unit circle

T = 1 sec

Fig. 6-33 Root locus plot of the sampled-data system with zero-order hold shown in Fig. 6-4.

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s-plane and the z-plane for the positive half of the primary strip.

The loci for frequency range of 0 to $-\omega_s/2$ are just the mirror image of the curves shown with the mirror placed on the real z-axis.

The relative stability of a sampled-data system can now be investigated by superimposing the constant, σ , and a loci on the system root loci plot. Let us consider the system shown in Fig. 6-4.

The root locus plot of the system is again constructed in Fig. 6-33 for T = 1 sec. Inspection of the root locus plot shows that the root locus intersects the $\zeta = 0.5$ locus at the point that corresponds to K = 0.49. Therefore, if the forward gain of the system is 0.49, the damping ratio of the second-order system is equal to 50 percent.

This answer is significant, since in this particular system the forward transfer function between the sampler and the output provides sufficient attenuation at frequencies higher than $\omega_s/2 = \pi$, so that only the constant locus in the primary strip is necessary. By use of the constant damping and the constant frequency loci, the damping factor and of the system when K = 0.49 are found to be 0.332 sec⁻¹ and 0.558 rad/sec, respectively.

6.11 SYSTEM RESPONSE BETWEEN SAMPLING INSTANTS

Thus far, the z-transform method has been used to evaluate the output responses of sampled-data systems at sampling instants only. In Sec. 6.2, the output sequence of a simple second-order system with sampled-data is obtained by the z-transform method, and the output responses are sketched in Fig. 6-5 for the system operating with and without a zero-order hold. However, the z-transform analysis does not yield any information on the response between sampling instants.

Therefore, in this particular system the continuous output response is estimated by connecting the output samples by a smooth curve. In general, in a well-behaved system if the sampling frequency is high as compared with the natural frequency of the system, this estimation of the continuous response from the sampled data is reasonably accurate. But for a certain class of systems, oscillation may exist in the response between sampling instants; then a complete description of the actual system response would certainly require the knowledge of the response waveform between sampling instants as well. Also, in evaluating the maximum overshoot of a time response, the maximum value given by the sampled sequence is usually less than the actual value, unless the maximum overshoot happens to fall right on a sampling instant. In order to calculate the maximum

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overshoot more accurately, we again have to recover the signal between sampling instants.

In Chapter 4 the submultiple sampling method and the modified z-transform method are used to evaluate the response between sampling instants for open-loop sampled-data systems. These methods will now be extended to the studies of closed-loop systems with sampled-data.

1. The Submultiple Sampling Method

The application of the submultiple sampling method to the evaluation of response between sampling instants of a closed-loop system with sampled-data is illustrated in Fig. 6-34. In this single-loop system fictitious samplers S_{n0} and S_n with sampling period T/n where $n = 1, 2, 3, \dots$ are inserted between S . (the regular sampler with sampling period T) and $G(s)$ and at the output terminal, respectively. The fictitious samplers are not physical components; they are used simply to facilitate the calculation of system response during the sampling periods. Since the input to the fictitious sampler S_{n0} is a train of impulses of period T , and S_{n0} is sampling at a rate n times as fast as S_0 , the insertion of the switch S_{n0} does not in any way affect the original system behavior. Therefore, the system shown in Fig. 6-34 behaves exactly as if the two fictitious samplers were absent.

$r(t)$
 $e(t) \times$
 $X_{e^{*}(1)} \times e^{*}(1),$
 $E(s) \rightarrow S_0 \rightarrow E(z) \rightarrow S_{n0} \rightarrow E(z)$
 $B(s)$
 $T \rightarrow c(1)$
 $T \rightarrow c^{*}(1)$
 $G(s)$
 $c(1)$
 $H(s)$

Fig. 6-34 Closed-loop sampled-data system with fictitious samplers for the evaluation of response between sampling instants.

At the output terminal of the system shown in Fig. 6-34, the actual continuous output is denoted by $c(t)$; the ordinary z-transform of the output $C(z)$ is obtained from the output of the sampler S_1 . But if additional information on $c(t)$ in between the ordinary samp-

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ling instants is desired, the output of S_n is used. It follows from Eq. (4-148) that the output of S may be written as

$C(z)_n = G(z)_n E(z)$
 (6-100)

where $E(z)$ is the ordinary z -transform of $e(t)$, and $G(z)$, is defined in Eq. (4-149) as

$$G(z) = \frac{G(s)}{s} \bigg|_{s=\frac{1}{z}}$$

$$T = T/n$$

The output transform of the sampler S_n is

$$(6-101)$$

$$E(z) = R(z) - B(z) = R(z) - HG(z) E(z)$$

$$(6-102)$$

Solving for $E(z)$ in the last equation yields

$$E(z)$$

$$=$$

$$R(z)$$

$$1 + HG(z)$$

$$(6-103)$$

Substitution of Eq. (6-103) into Eq. (6-100) yields

$$C(z) = G(z)E(z) + HG(z)E(z)$$

$$R(z)$$

$$(6-104)$$

Therefore, the transform of the response at discrete instants between sampling instants can be evaluated from the ordinary z -transforms.

The inverse z -transform of $C(z)$ gives the output sequence $c(kT/n)$

for $k = 0, 1, 2, 3, \dots$, and $nq+1$ where q is the number of additional sampled values desired between any two consecutive sampling instants.

Example 6-10

Evaluate the output response at two additional instants (uniform) during each sampling period for the sampled-data system shown in Fig. 6-3. The sampling period is one sec, $K = 1$, and the input signal is a unit step function. The open-loop transfer function of the system is given by Eq. (6-5).

$$G(z)$$

$$=$$

$$(1 - e^{-T})$$

$$(z - 1)(z - e^{-1})$$

$$(6-105)$$

Using Eq. (6-101), we have from the last equation

$$G(z) =$$

$$\frac{1 - e^{-T/n}}{z^{1/n} - 1}$$

$$(6-106)$$

$$=$$

$$=$$

$$(6-106)$$

Since T is one sec and since, for two additional samples between sampling instants, $n = 2 + 1$,

$$n = 3$$

$$=$$

$$3$$

Substituting the value of n into Eq. (6-106), we have

$$\text{or}$$

$$z^{1/3}(1 - e^{-1/3})$$

$$=$$

$$(z^{1/3} - 1)(z - e^{-1/3})$$

$$(6-107)$$

$$G(z)$$

$$=$$

$$(6-108)$$

$$0.293z^{1/3}$$

$$G(z)$$

$$=$$

$$(6-109)$$

$$(z^{1/3} - 1)(z^{1/3} - 0.717)$$

Now substitution of Eq. (6-109) into Eq. (6-104) yields

$$C(z)^3 = \frac{0.29321}{z(z - 0.368)} (\approx 1/3 - 1)(\approx 1/3 - 0.717) (\approx -1)(\approx -0.368) + 0.632 \approx$$

(6-110)

Letting

we have

$$23 = 21/3$$

$$233 = 2$$

(6-111)

(6-112)

Substituting Eq. (6-111) and (6-112) into Eq. (6-110), we have

$$C(z)^3 =$$

or

$$C(z) =$$

$$0.293(0.368)$$

$$(\approx 3 - 1)(\approx 3 - 0.717)(\approx 3 - 1)(\approx 3 - 0.368) + 0.632z^3$$

$$0.2932 \quad 0.1082$$

-

(6-113)

-

$$23 - 1.717z^3 + 0.717z^5 - 0.73623 + 1.26\approx 3 - 0.527\approx 3 + 0.368\approx 3 - 0.632\approx 3 + 0.264$$

By means of long division Eq. (6-114) is written

(6-114)

$$1.6$$

$$c(t)$$

$$1.4$$

$$1.2$$

$$1.0$$

$$0.8$$

$$0.6$$

$$0.4$$

$$0.2$$

•

$$0$$

$$0$$

$$1$$

$$2$$

$$3$$

$$4 \quad 5 \quad 6$$

Time (sec)

$$7 \quad 8$$

$$9$$

$$10$$

Fig. 6-35 System output between sampling instants of the system in Example 6-8.

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2

$$C(z)^3 = 0.293zz^1 + 0.503zz^2 + 0.648z^5^3 + 0.86z^3' + 1.01zz^3 + 1.10zz^6$$

$$+1.1427 +1.16258 +1.20\approx 5^\circ + 1.132510+ 1.125'' + \dots$$

(6-115)

In the last equation the coefficient of corresponds to the value of $c(t)$ at $t = kT/3$ for $k = 0, 1, 2, 3, \dots$. Notice that the coefficients of $z^9, z^{23}, z^{30}, z^{37}$,

$\dots$ agree with the output sequence given by Eq. (6-9) (slight discrepancies are due to slide rule inaccuracy); the other samples of Eq. (6-115) give the

value of $c(t)$ between sampling instants. The output response of the system, based on the information contained in $C(z)$, is depicted in Fig. 6-35.

2. The Modified z-Transform Method

The complete time response of a closed-loop sampled-data system can also be obtained by use of the modified z-transform technique discussed in Sec. 4.8. The technique of determining the output transform of a closed-loop system with sampled-data in terms of the modified z-transform has been discussed in Sec. 5.3. For the single-loop system shown in Fig. 6-36, the modified z-transform of the system output is given by Eq. (5-85) as

Fictitious

sampler

T $C(z, m)$

$C(s)$

$R(s)$

$E(s)$

$E^*(s)$

$G(s)$

$e^{-(1-m)T}$

Fictitious

time delay

$H(s)$

$e^{(1-m)T}$

Fictitious

time advance

Fig. 6-36 Fictitious delay and advance in closed-loop sampled-data system for the evaluation of the modified z-transform of output response.

$C(z, m) =$

$G(z, m)$

$1 + HG(z)$

$R(z)$

(6-116)

where $G(z, m)$ is defined as the modified z-transfer function of the forward path elements. The output modified z-transform for more complex system configurations may be derived by use of the signal flow graph technique described in Sec. 5.3.

As illustrated in Sec. 4.8, the inverse modified z-transform may be performed either by the power series expansion method (Eq. 4-208 and Eq. 4-209) or the inversion formula method (Eq. 4-216).

To illustrate the determination of the complete system response

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by the modified z-transform method, the following illustrative example is given.

Example 6-11

Compute the complete output response of the closed-loop system of Fig. 6-3 when the system is subjected to a unit step function input. The gain constant K is unity, and the sampling period is one sec.

The forward path transfer function of the system is

-

1

$G(s) = 5(8 + 1)$

$s(s)$

(6-117)

From the table of modified z-transforms, the modified z-transform of $G(s)$ is obtained as

$G(z, m)$

$z(1 - e^{-mr}) + (e^{-m^0} - e^{-1})$

=

$(\approx - 1)(z - e^{-T})$

For $T = 1$ sec, Eq. (6-118) becomes.

$$\begin{aligned} &= \\ (6-118) \\ G(z, m) &= \\ &= \frac{z(1 - e^{-m}) + (e^{-m} - 0.368)}{(\approx - 1)(\approx - 0.368)} \end{aligned}$$

(6-119)
The z-transform of $G(s)$, $G(z)$, is ordinarily found from the z-transform table, but it is interesting to note that $G(z)$ is also obtainable from the modified z-transform $G(z, m)$ by setting m equal to unity (see Eq. 4-179).

Therefore,

$$0.632z \\ G(z) = G(z, 1)$$

$$\begin{aligned} &= \\ &= \frac{z}{(\approx - 1)(\approx - 0.368)} \end{aligned}$$

(6-120)

which checks with the result from Eq. (6-105).

From Eq. (6-116) the output modified z-transform is

$$\begin{aligned} &= \frac{z(1 - e^{-m}) + (e^{-m} - 0.368)}{(\approx - 1)(\approx - 0.368)} \end{aligned}$$

$$0.632z$$

$$C(z, m)$$

$$\begin{aligned} &= \\ &= \frac{1 +}{(\approx - 1)(\approx} \end{aligned}$$

$$\begin{aligned} &= \\ &= \frac{0.368)}{R(z)} \end{aligned}$$

$$\begin{aligned} &= \\ &= \frac{0.368)}{R(z)} \end{aligned}$$

For a unit step input and after simplification, Eq. (6-121) becomes

$$C(z, m) =$$

$$\frac{2(1 - e^{-m}) + (e^{-m} - 0.368)}{(\approx - 1)(\approx^2 - 0.736\approx + 0.368)}$$

$$\begin{aligned} &= \\ &= \frac{1}{(\approx - 1)(\approx^2 - 0.736\approx + 0.368)} \end{aligned}$$

$$\begin{aligned} &= \\ &= \frac{1}{(\approx - 1)(\approx^2 - 0.736\approx + 0.368)} \end{aligned}$$

$$(6-121)$$

(6-122)
The complete output response of the system is obtained by first taking the inverse modified z-transform of Eq. (6-122) and then letting m take on various values between 0 and 1. First, let us use the power series expansion method. Expanding $C(z, m)$ of Eq. (6-122) into a power series in z^{-1} by dividing the denominator into the numerator using long division, we have

$$C(z, m)$$

When $m =$

$$\begin{aligned} &= \\ &= \frac{(1 - e^{-m})z^{-1} + (1.368 - 0.736e^{-m})z^{-2} + (1.266 - 0.174e^{-m})z^{-3}}{+ (1.058 + 0.143e^{-m})z^{-4} + (0.935 + 0.25e^{-m})z^{-5} + (0.92 - 0.16e^{-m})z^{-6} + \dots} \end{aligned}$$

$$\begin{aligned} &= \\ &= \frac{(1 - e^{-m})z^{-1} + (1.368 - 0.736e^{-m})z^{-2} + (1.266 - 0.174e^{-m})z^{-3}}{+ (1.058 + 0.143e^{-m})z^{-4} + (0.935 + 0.25e^{-m})z^{-5} + (0.92 - 0.16e^{-m})z^{-6} + \dots} \end{aligned}$$

(6-123)
1, $C(z, m)$ is reverted to $C(z)$; therefore,

$$\begin{aligned} &= \\ &= \frac{1}{(\approx - 1)(\approx^2 - 0.736\approx + 0.368)} \end{aligned}$$

which checks closely with Eq. (6-10).

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The output response $c(t)$ at all times is given by the coefficients of Eq. (6-123) when m is varied between zero and unity continuously. The correspondence of the coefficients and output response is tabulated in the following:

Coefficient of z^{-1} $(1 - e^{-m}) = c(1 + m - 1)T$

Coefficient of z^{-2}

$$(1.368 - 0.736e^{-m}) = c(2 + m - 1)T$$

$$(1.266 - 0.174e^{-m}) = c(3 + m - 1)T$$

$$(1.058 + 0.143e^{-m}) = c(4 + m - 1)T$$

Coefficient of z^{-3}

Coefficient of z^{-4}

Coefficient of z^{-5}

Coefficient of z^{-6}

$$(0.935 + 0.25e^{-m}) = c(5 + m - 1)T$$

$$(0.92 + 0.16e^{-m}) = c(6 + m - 1)T$$

If we arbitrarily set $m = 1/3$ and $m = 2/3$, the output response at two points between sampling instants may be obtained. The results are tabulated as follows:

$c(0.333)$

$c(1.333)$

$c(2.333)$

0.283

0.84

1.10

$c(3.333)$

0.956

$c(4.333)$

$c(5.333)$

1.11

1.03

$c(0.667)$

0.487

$c(1.667)$

1.00

$c(2.667)$

1.17

$c(3.667)$

1.12

$c(4.667)$

$c(5.667)$

1.06

1.00

These sampled values together with the response at the actual sampling instants are sketched in Fig. 6-37.

Let us now perform the inverse modified z -transform of Eq. (6-122) by means of the inversion formula. Substituting Eq. (6-122) into the inversion formula of Eq. (4-216), we have

1

-

-

$$c(nT, m) = 27; \oint [(1 - e^{-m})z^{-1} + (e^{-m} - 0.368)],$$

$$(x-1)(x^2 = 0.7362 + 0.368)$$

-

$n-1 \oint dz$

(6-125)

where I is a closed path in the z -plane which encloses all the singularities

of $C(z, m)^{-1}$,
 Factoring the denominator of the integrand of Eq. (6-125) yields
 $[(1 - e^{-m}) + (e^{-m} - 0.368)] =$
 $- e^{-m} z$

$$c(nT, m) = 27; \quad (z - 1)(x - 0.368 - j0.482) (\approx 0.368 + j0.482)$$

From the residues theorem,

$\frac{dz}{(6-126)}$
 $c(nT, m)$ Residues of $C(z, m)^{-1}$ at the poles of $C(z, m)$. (6-127)
 The residue of $C(z, m)^{-1}$ at $z = 1$ is computed as

$$\frac{(1 - e^{-m}) + (e^{-m} - 0.368)}{(0.632 - j0.482)(0.632 + j0.482)}$$

The residue of $C(z, m)z^{-1}$ at $z = 0.358 + j0.482$ is

$$\frac{1}{(6-128)} \frac{[(1 - e^{-m})(0.368 + j0.482) + (e^{-m} - 0.368)]}{(-0.632 - j0.482)(j0.964)}$$

(6-129)
 which is simplified to

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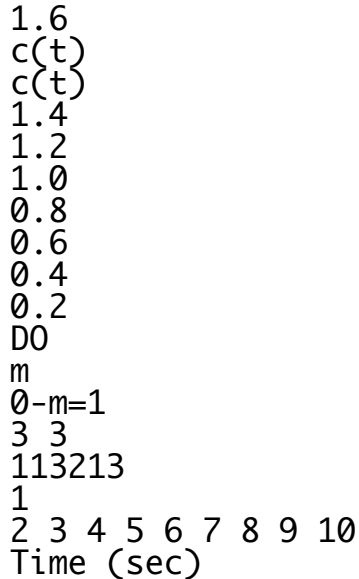


Fig. 6-37 System output response between sampling instants of system in Example 6-11.

$$- 1.31(0.606)^n [0.632e^{-m} + j0.482(1 - e^{-m})]e^{-j(n-1)52.6^\circ}$$

(6-130)
 In the same manner the residue of $C(z, m)z^{-1}$ at $z = 0.368 - j0.482$ is
 $- 1.31(0.606)^n [0.632e^{-m} - j0.482(1 - e^{-m})]e^{-j(n-1)52.6^\circ}$

Therefore, the inverse modified z-transform of $C(z, m)$ is

$$c(nT, m) = 1 - 1.31(0.606)^n \{ [0.632e^{-m} + j0.482(1 - e^{-m})]e^{-j(n-1)52.6^\circ} + [0.632e^{-m} - j0.482(1 - e^{-m})]e^{-j(n-1)52.6^\circ} \}$$

- Equation (6-132) can be reduced to the following form:

$$c(nT, m) = 1 - 1.31(0.606)^n [1.264e^{-m} \cos (n - 1)52.6^\circ - 0.964(1 - e^{-m}) \sin (n - 1)52.6^\circ]$$

(6-131)

(6-132)

(6-133)

When the value of m is varied between zero and unity, the last equation describes the complete output response of the system when the input is a unit step function. For $m = 1$, this equation describes the response at the sampling instants only. Thus,

$$c(nT, 1) = c(nT) = 1 - 1.31(0.606)^n [0.465 \cos (n - 1)52.6^\circ - 0.6091 \sin (n - 1)52.6^\circ]$$

(6-134)

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When proper values for n and m are substituted into Eq. (6-133), the complete output response of the system can be shown to be identical to the response shown in Fig. 6-37.

Thus far in this section we have demonstrated how the response between sampling instants of a closed-loop sampled-data system can be evaluated by means of the submultiple sampling and the modified z -transform methods. The particular system chosen in the illustrative examples, however, is of such a nature that the z -transform method is quite valid in the calculation of the system response. But in general if the system response between sampling instants is not well behaved, it may be necessary to apply the two methods discussed above so that the behavior of the system response between sampling instants may be computed. In order to demonstrate the real usefulness of the modified z -transform and some oscillatory properties of system response between sampling instants, the following example is given.

Example 6-12

The open-loop transfer function of the unity-feedback system with sampled-data shown in Fig. 6-38 is given by

$$G(s) = \frac{27.8(s + 1)}{s(s^2 + 6s + 48.6)}$$

(6-135)†

The sampling period is one second. Since the natural frequency of the system is equal to 6.96 rad/sec (48.6) and the sampling frequency is 6.28 rad/sec, the z -transform method should not be used in this problem. The modified z -transform will be used here to compute the complete unit-step response of the system.

$$R(s) = E(s) \times E^*(s),$$

$$G(s)$$

$$C(s)$$

$$H(s) = 1$$

Fig. 6-38 Sampled-data system in Example 6-12.

The modified z -transform of $G(s)$ is obtained by use of the expression given in Eq. (4-196) or directly from the modified z -transform table. From Eq. (4-196), we have

$$G(z, m) = z^{-1} \text{Residues of } G(s)$$

at the poles of $G(z)$ (6-136)

3

Or

$$G(z, m) = \phi^{-1}$$

$$N(s) \text{ ems. } T$$

$n=1$

$$D'(s_n) = 1$$

(6-137)

+ E. I. Jury, "Synthesis and Critical Study of Sampled-Data Control Systems," Trans.

A. I. E. E., 75, Part 2, 1956, pp. 141-151.

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where

$$N(s)/D(s)G(s)$$

and s_i denotes the poles of $G(s)$.

(6-138)

The poles of $G(s)$ are at $s = 0$, $s =$

$$-3 + j6.28, \text{ and } s =$$

$$-3 - j6.28;$$

therefore, Eq. (6-137) gives

$$G(z, m) = 2^{-1} [9$$

27.8

$$9 + (6.28) \frac{1}{z}$$

+

-

$\frac{z}{z^2}$

+

$$27.8(2 + j2\pi)$$

$$- 2(6.28)^2 \frac{1}{z^2} + j12\pi \frac{1}{z}$$

$$27.8(-2j2\pi) \text{ emT}(-3-j2x)$$

$$- 2(6.28)^2 + j12\pi \frac{1}{z}$$

For $T = 1$ second, Eq. (6-139) is simplified to

1

$$G(z, m) = 0.573$$

[

2

-

$\frac{z}{z^2}$

-

$$e^{-3m}$$

$$0.05$$

$$(\cos 2\pi m$$

-

$$- 7.27 \sin 2\pi m)$$

1

$$\text{emT}(-3+j2x)$$

$$- (3 + j2x) T$$

--

$$2-1e^{-3-}$$

$$-3-j2x) T$$

(6-139)

(6-140)

The z -transform of $G(s)$ is readily obtained from Eq. (6-140) by letting

$m = 1$. Therefore,

$$0.573$$

$$G(z) = G(z, 1)$$

=

$\frac{z}{z^2}$

-

0.0284
 1 2- 0.05
 (6-141)
 Substituting Eq. (6-140) and Eq. (6-141) into Eq. (6-116) yields the modified z-transform of the output of the system as
 $C(z, m) = \frac{0.573 - 0.505z + 0.05z^2}{(z - 1)(z^2 - 0.505z + 0.05)}$

The output response at the sampling instants is obtained from
 $C(z) = C(z, 1) = \frac{(z - 1)(z^2 - 0.505z + 0.05)}{(z - 1)(z^2 - 0.505z + 0.05)}$
 (6-142)
 By long division, Eq. (6-143) is expanded into a power series in z^{-1} .
 $C(z) = 0.545z^{-1} + 0.821z^{-2} + 0.933z^{-3} + 0.976z^{-4} + 0.994z^{-5} + \dots$
 (6-144)
 The output samples at the sampling instants of $c(nT)$ are now plotted in Fig. 6-39.

Fig. 6-39 Response at sampling instants of the system in Example 6-12.

Time (sec)

Fig. 6-39 Response at sampling instants of the system in Example 6-12.

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However, in this particular system, the exact output response cannot be approximated by connecting the output sequence $c(nT)$ with a smooth curve. Because of the low sampling frequency chosen as compared with the natural frequency of the system and because of the oscillatory nature of the system, we do not expect that the response between sampling instants will be well-behaved. To evaluate the intersampling response, we shall take the inverse transform of $C(z, m)$. Substituting Eq. (6-142) into the inversion formula of Eq. (4-216) yields

$$c(nT, m) = \frac{1}{2\pi j} \oint_{\gamma} \frac{(0.573 - 0.505z + 0.05z^2)}{(z - 1)(z^2 - 0.505z + 0.05)} z^{n-1} dz$$

(6-145)

Factoring the denominator of the integrand of Eq. (6-145) and applying the residues theorem, we have

$c(nT, m)$

$$= \sum \text{Residues of} \\ \frac{0.573 [(z - 0.05) - (\approx 1)e^{-3m}(\cos 2\pi m - 7.27 \sin 2\pi m)]}{(z - 1)(z - 0.37)(\approx -0.135)}$$

(6-146)

at the poles $z = 1$, $z = 0.37$, and $z = 0.135$.

Computing the residues at the three poles of $C(z, m)$, the inverse z-transform of $C(z, m)$ is finally reduced to the following expression:

$$c(nT, m) = 1 + 2.44[(0.135)^n - (0.37)^n] (\cos 2\pi m - 7.27 \sin 2\pi m)e^{-3m} + (0.24)(0.135)^n - (1.24)(0.37)^n$$

(6-147)

Determining the maximum and the minimum of the response described by Eq. (6-147) is of interest. This determination is made by differentiating $c(nT, m)$ with respect to m and setting the result equal to zero; that is,

$$\frac{\partial c(nT, m)}{\partial m} = 0$$

(6-148)

From the last equation, the following relation is obtained:

m
max
= function of n

(6-149)

and since n is always an integer and the value of m should lie between zero and unity, the maximum point can be located by a few cut-and-try steps.

Substituting Eq. (6-147) into Eq. (6-148) yields

$$\frac{\partial c(nT, m)}{\partial m}$$

$$\frac{\partial}{\partial m}$$

$$= 2ne^{-3m}(\sin 2\pi m - 7.27 \cos 2\pi m) - 3e^{-3m}(\cos 2\pi m - 7.27 \sin 2\pi m) = 0$$

(6-150)

Eq. (6-150) is reduced to

$$15.3 \sin 2\pi m$$

$$- 48.6 \cos 2\pi m$$

$$= 0$$

(6-151)

from which we have

$$\tan 2\pi m$$

$$= 3.18$$

(6-152)

Equation (6-152) is satisfied when

$$2\pi m = 1.26, 1.26 + \pi, 1.26 + 2\pi,$$

(6-153)

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However, since m is between 0 and 1, only the first two values given in the last equation are significant. Therefore

and

$$2\pi m = 1.26 \text{ or } 4.4$$

$$m = 0.2 \text{ or } 0.7$$

(6-154)

(6-155)

We can actually show that when $m = 0.2$, the response is at a maximum and that when $m = 0.7$, the response is at its minimum. Furthermore, since $m_{\max}$ and $m_{\min}$, as determined above, are not functions of n , a maximum point and a minimum point will occur in each sampling period. Therefore, we can conclude that

Occurs at $m = 0.2 + n$ for $n = 1, 2, 3,$

$$\dot{c}(nT, m) \text{ max}$$

(6-156)

However, the maxima in all sampling periods are not all of the same value, unless the response is of the form of a sustained oscillation with constant amplitude. Also,

$$c(nT, m) \text{ min occurs at } m = 0.7n \text{ for } n = 1, 2, 3, \dots \quad (6-157)$$

Similarly, the minima in all the sampling periods are not of the same value.

The values of $c(nT, m)$ are calculated and tabulated in Table 6-4 for $n = 1, 2, 3, 4$; and various values of m . The complete output response of the system is depicted in Fig. 6-40. Clearly the actual response between sampling instants is altogether different from the smooth curve predicted from the z -transform approximation. This difference verifies our original prediction that the z -transform method will give misleading results in this particular system. The oscillation between sampling instants, as observed in this system, not indicated by the z -transform solution is often referred to as the "hidden oscillation." In practice, the hidden oscillation between sampling instants may be positively damped (as in the example just given), undamped, or negatively damped (unstable, or hidden instability). A significant point in the system of this example is that the poles of $C(z)$ and the poles of $C(z, m)$ are identical. This indicates that the stability of the system response at the sampling instant

and in between the sampling instants is the same. However, in one rare class of systems the modified z -transform of the system output may have more poles than $C(z)$. In this case, if all the poles of $C(z)$ be located inside the unit circle of the z -plane (indicating that the response at the sampling instant

is a stable one) but one or more poles of $C(z, m)$ be located outside the unit circle, the response between the sampling instants would actually be unstable. This type of instability is sometimes referred to as the "hidden instability," since, like the hidden oscillation phenomenon, it is not indicated by the ordinary z -transform method.

+ R. H. Barker, "The Pulse Transfer Function and its Application to Sampling Servo Systems," Proc. I. E. E., 99, Part 4, December 1952, pp. 302-317. E. I. Jury,

"Hidden Oscillations in Sampled-Data Control Systems," Trans. A. I. E. E., 76, Part 2, January 1957, pp. 391-394.

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3

$$c^*(t)$$

$$c(t)$$

$$-c(t)$$

2

1

c(1)
 1
 2
 3
 4
 5
 Time (sec)
 Fig. 6-40 Complete step response of the system in Example 6-12.
 TABLE 6-4
 SYSTEM RESPONSE OF THE SYSTEM IN EX. 6-12

m	c(1, m)	c(2, m)	c(3, m)	c(4, m)
0.2				
0.216				
0.3				
0.5				
0.7				
0.716				
0.85				
1.0				
2.72				
2.63				
2.24				
0.7				
0.094				
0.116				
0.287				
0.545				
1.92				
1.87				
1.67				
0.9				
0.592				
0.603				
0.69				
0.821				
1.36				
1.28				
0.96				
0.84				
0.844				
0.93				
1.14				
1.14				
1.11				
0.98				
0.94				
0.94				
0.95				
0.97				

6.12 STEADY-STATE RESPONSE STEADY-STATE ERROR ANALYSIS

In the preceding sections, only the transient response of sampled-data systems is discussed. However, just as in systems with continuous signals, the steady-state response is also of great importance in the analysis and design of sampled-data systems. The steady-state error of a linear feedback control system with unity-feedback (with or without sampled-data) is defined as the difference between the reference input $r(t)$ and the actual output $c(t)$

when steady state is reached; that is

Steady-state error = $\lim_{t \rightarrow \infty} e(t)$

$\lim_{t \rightarrow \infty}$

$$= \lim_{t \rightarrow \infty} [r(t) - c(t)] \quad (6-158)$$

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The entire study of the steady-state error of sampled-data system is based on the final-value theorem for z-transforms. If the system is of such a nature that the z-transform method is valid so that $e(t)$ can be characterized by its values at the sampling instants, the steady-state error at the sampling instants is described by

$$\lim_{t \rightarrow \infty} e^*(t) = \lim_{n \rightarrow \infty} e(nT)$$

$\lim_{n \rightarrow \infty}$

$\lim_{z \rightarrow 1}$

(6-159)

From the final-value theorem of the z-transformation, we have

$$\lim_{t \rightarrow \infty} e^*(t)$$

$\lim_{z \rightarrow 1}$

$\lim_{z \rightarrow 1}$

$$\lim_{z \rightarrow 1} (z-1)E(z)$$

$\lim_{z \rightarrow 1}$

where $E(z)$ is the z-transform of $e(t)$.

(6-160)

In cases where the system response cannot be adequately described by the samples at the sampling instants (such as if hidden oscillation exists), the steady-state error between sampling instants can be described by use of the final-value theorem for modified z-transforms. Therefore, the actual steady-state error is

$$\lim_{t \rightarrow \infty} e(t)$$

$\lim_{t \rightarrow \infty}$

$$\lim_{L \rightarrow \infty} e(nT, m)$$

$\lim_{L \rightarrow \infty}$

$\lim_{N \rightarrow \infty}$

$\lim_{z \rightarrow 1}$

$$\lim_{z \rightarrow 1} (z-1)E(z, m) \quad (6-161)$$

$\lim_{z \rightarrow 1}$

where $E(z, m)$ is the modified z-transform of $e(t)$.

In the studies of linear feedback control systems with continuous-data, it is known that the steady-state error of a given system depends on the type of reference input applied to the system and also, to a great extent, on the nature and characteristics of the system itself. We shall show that these properties are carried over to feedback control systems with sampled-data. Therefore, the concept of the error constants and the error coefficients can be conveniently extended from the continuous-data systems to the sampled-data systems. The error constants are a direct measure of the steady-state error of a system when typical test signals such as a step, ramp, or acceleration function is applied. The definitions and the derivations of the error constants are given as follows:

Referring to the unity-feedback system shown in Fig. 6-41, the

z-transform of the error signal is written

$$E(z) = R(z) - C(z)$$

$\lim_{z \rightarrow 1}$

$R(z)$

$$1 + G(z)$$

(6-162)

For a sampled-data system (error-sampled) with non-unity feedback, the error signal is defined as the difference between the reference input and the feedback signal.

Thus, the error transform is given by

$$\frac{E(z)}{R(z)} = \frac{1}{1 + G(z)H(z)}$$

$$= \frac{R(z)}{1 + HG(z)}$$

The discussion given in this section on steady-state errors can be applied to nonunity feedback systems if $G(z)$ is replaced by $G(z)H(z)$ or $HG(z)$.

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Substituting Eq. (6-162) into Eq. (6-160), the steady-state error at the sampling instants is described by

$$e = \lim_{t \rightarrow \infty} e^*(t)$$

$$= \lim_{z \rightarrow 1} \frac{z-1}{z} \frac{R(z)}{1 + G(z)}$$

(6-163)

This last equation is valid only if $1 + G(z)$ does not have any zeros located on or outside the unit circle in the z -plane. Equation (6-163) indicates that the steady-state error of a sampled-data system with unity-feedback depends on the reference input signal $R(z)$, and the open-loop transfer function $G(z)$. The computation of the steady-state error is simplified if some basic facts and relations concerning the type of reference input, the system type, and the error are established. We shall consider only three basic types of test signals: step displacement, ramp, and acceleration.

$$\frac{R(s)}{E(s)} = \frac{E^*(s)}{G(s)}$$

$$- C(s)$$

Fig. 6-41 Sampled-data control system with unity-feedback.

1. Steady-State Error of Sampled-Data Systems with Step Displacement Input

Displacement Input

If the reference input of the system of Fig. 6-41 is a step displacement of magnitude R , the z -transform of $r(t)$ is given by

$$R(z) = \frac{R}{z-1}$$

Substituting Eq. (6-164) into Eq. (6-163) yields

$$e_{ss} = \lim_{z \rightarrow 1} \frac{R}{z-1} \frac{z-1}{z} \frac{1}{1 + G(z)}$$

If we let

$$K_p = \lim_{z \rightarrow 1} G(z)$$

(6-164)

(6-165)

(6-166)

and define K, as the "positional error constant," Eq. (6-165) becomes

$$e_{ss} = \frac{R}{1 + K_D}$$

Solving for K, in the last equation, we have

$$K_D = \frac{R}{e_{ss}} - 1$$

Since R is the amplitude of the step displacement input, which is also the desired amplitude of the steady-state output response, Eq. (6-168) may be written as

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Positional error constant K,

$$K_D = \frac{\text{Desired output}}{\text{Allowable } e}$$

(6-169)

Equation (6-169) is quite useful in the design of feedback control systems with sampled-data when the reference input is considered as a step displacement. Once the desired output positional response and its allowable steady-state error are specified, the corresponding K, of the system is determined from Eq. (6-169); and by using Eq. (6-166), the required behavior of G(z) at z = 1 is also determined.

2. Steady-State Error of Sampled-Data Systems with Ramp Function Input

If the input to the sampled-data system in Fig. 6-41 is a ramp function, $r(t) = Rt$, where R is in displacement/sec, the z-transform of $r(t)$ is given by

$$R(z) = \frac{RTz}{(z-1)^2}$$

Substituting Eq. (6-170) into Eq. (6-163) yields

$$e_{ss} = \lim_{z \rightarrow 1} (z-1) \left[\frac{R(z)}{1 + G(z)} \right]$$

Equation (6-171) can be written as

$$e_{ss} = \lim_{z \rightarrow 1} \frac{R(z)}{1 + G(z)}$$

(6-171)

(6-172)

Now, let the velocity error constant be defined as

$$K_v = \lim_{z \rightarrow 1} (z-1) G(z)$$

then, Eq. (6-172) becomes

$$\begin{aligned}
 & \text{or} \\
 & K_o \\
 & = \\
 & R \\
 & = \\
 & - \lim_{z \rightarrow 1} [(x - 1)G(z)] \\
 & \text{ex} \\
 & - \\
 & R \\
 & K_v \\
 & \dots
 \end{aligned}$$

Desired output velocity
 ess Steady-state error in displacement
 (6-173)
 (6-174)
 (sec⁻¹) (6-175)

The steady-state error given in Eq. (6-174) is often referred to as the velocity error. However, we should keep in mind that the velocity error thus defined is not an error in velocity, but rather an error in displacement due to a velocity (ramp) function input of the form of $r(t) = Rt$. Figure 6-42 further clarifies this point by showing a typical ramp input and the output response of a system at sampling instants lags behind the input by an error in displacement when steady state is reached.

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$c(t)$
 ess = ess
 Reference input,
 $r(t) = Rt$
 $2T \quad 3T$
 $4T \quad 5T$
 $6T \quad 7T \quad 8T \quad \dots$

Output sequence
 $c(nT)$

Fig. 6-42 Output sequence of sampled-data system with ramp input.

3. Steady-State Error of Sampled-Data Systems With Acceleration Function Input

Consider now that an acceleration input $r(t) = Rt^2/2$ is applied to the system in Fig. 6-41. The z-transform of $r(t)$ is

$$\begin{aligned}
 R(x) &= \\
 & T^2 z(x + 1) \\
 & 2(x-1)^3
 \end{aligned}$$

Therefore, the steady-state error at sampling instants is

$$\begin{aligned}
 & R(z+1) \\
 & 2 \rightarrow i (\approx -1)^2 [1 + G(x)]
 \end{aligned}$$

-
 T^2
 ess =
 $\lim$
 which can be written as
 es

$$\begin{aligned}
 & = \\
 & - \\
 & 2-1 \\
 & \lim (z = D'G(z)) \\
 & T^2 \\
 & R \\
 & 2
 \end{aligned}$$

Ka

Now, let the acceleration error constant be defined as

$$K_1 = - \lim_{z \rightarrow 1} [(z - 1)^2 G(z)]$$

(6-176)

Then, Eq. (6-177) becomes

$$e_{ss} = R/K_a$$

(6-177)

(6-178)

(6-179)

(6-180)

(6-181)

The acceleration error constant also has the significance of

Ka

=

R

=

Desired output acceleration

ess Steady-state error in displacement

(6-181)

In a similar manner, the steady-state error given by Eq. (6-180) is called the acceleration error, which is an error in displacement due to an acceleration function input.

From the foregoing discussions, we see that when the reference input of a unity-feedback sampled-data system is of the step displacement, ramp, or acceleration type, the steady-state error at the sample

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ling instants due to each input depends on the error constants Kp, K_v, and K_a, respectively. The error constants are summarized as follows:

(1) Positional error constant $K = \lim_{z \rightarrow 1} G(z)$

(6-182)

2-1

(2) Velocity error constant $K_1 = \lim_{z \rightarrow 1} (z - 1)G(z)$

-

(6-183)

T21

(3) Acceleration error constant

$$K_a = - \lim_{z \rightarrow 1} [(z - 1)^2 G(z)]$$

2-1

(6-184)

Steady-State Error Between Sampling Instants

In general, the steady-state error analysis given in the preceding section is valid only for the error at the sampling instants. However, when a system has a zero steady-state error at the sampling instants, the error at other times is not necessarily also zero. A complete description of the steady-state error may be made by use of the modified z-transform. In Fig. (6-161) we see that the actual steady-state error of a sampled-data system may be written as

$$\lim_{L \rightarrow \infty} e(t) = e_{ss} = \lim_{m \rightarrow \infty} e(nT, m) = \lim_{m \rightarrow \infty} (12') E(z, m) \quad (6-185)$$

L→∞

71-400

2-1

where $E(z, m)$ is the modified z-transform of $e(t)$. However, in most practical cases, the limit of $E(z, m)$ as z approaches the limit of unity, is independent of m , therefore, Eq. (6-185) and Eq. (6-161) will yield identical results under this condition. As an illustration, we may try to evaluate the steady-state error of the system in Example 6-12. Since the system has hidden oscillation, the modified

z-transform technique was used in the example to compute the output step response. The modified z-transform of the error signal is written as

$$E(z, m) = R(z, m) - C(z, m) \quad (6-186)$$

For a unit-step displacement input, the modified z-transform of $r(t)$ is

$$R(z, m) = 1/(z - 1) \quad (6-187)$$

Substitution of Eq.

(6-187) and Eq.

(6-142) into Eq. (6-186) and

Eq. (6-185) yields

ess

$$= \lim_{m \rightarrow \infty} e(nT, m)$$

$$= \lim_{m \rightarrow \infty} (1 - z^{-1})$$

$$7-0$$

$$0 < m < \infty$$

$$2-1$$

$$0 < m < 1$$

$$1$$

$$-$$

$$1)$$

$$=$$

$$\lim_{m \rightarrow \infty} 0.573(1 - z^{-1})$$

$$2-1$$

$$0 < m < 1$$

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$$Z$$

$$(2$$

$$-$$

$$X$$

$$-0.05)$$

$$-$$

$$(z$$

$$-$$

$$22$$

$$-$$

$$1)e^{-31}$$

$$\text{Hence}$$

$$\text{ess} =$$

$$1 - \lim_{m \rightarrow \infty} 0.573-$$

$$-$$

$$2-1$$

$$0 < m < \infty$$

$$-3m (\cos 2\pi m$$

$$0.5052 + 0.05$$

$$(\approx 0.05)$$

$$---$$

$$- 0.505z + 0.05$$

$$-$$

$$7.27 \sin 2\pi m)$$

$$(6-188)$$

$$=$$

$$0$$

(6-189)

and the steady-state error of the system due to a unit step function input is zero. Since Eq. (6-189) is not a function of m , it indicates that the hidden oscillation is not carried into the steady-state and that the correct result on the steady-state error of the system may be obtained by use of the z -transforms.

Steady-State Error From Closed-loop Pole-Zero

Configurations

The relation between the error constants K_p , K_v , and K_a and the poles and zeros of the closed-loop transfer function $C(z)/R(z)$ provides useful information when analysis and synthesis of sampled-data systems are effected in the z -plane utilizing the pole-zero concept. Consider that the closed-loop transfer function relating to Fig.

6-41 is denoted by

$$\frac{C(z)}{R(z)} = \frac{K \prod_{i=1}^n (z - z_i)}{\prod_{j=1}^m (z - p_j)} \quad (6-190)$$

$$= \frac{K \prod_{i=1}^n (z - z_i)}{\prod_{j=1}^m (z - p_j)}$$

$$= \frac{G(z)}{1+G(z)}$$

The relationships between the error constants K_p , K_v , and K_a and the poles and zeros p_i and z_i are derived as follows:

(1) K_p Taking the limit as z approaching unity in Eq. (6-190), we have

$$\lim_{z \rightarrow 1} \frac{C(z)}{R(z)} = \lim_{z \rightarrow 1} \frac{G(z)}{1+G(z)}$$

$$= \frac{K_p}{1+K_p}$$

Thus, the positional error constant given in terms of the closed-loop poles and zeros is

$$K_p = \lim_{z \rightarrow 1} \frac{C(z)}{R(z)} = \frac{1 - C(1)}{1 - R(1)}$$

(6-192)

(2) K.: For a unit-magnitude ramp-function input, the z-transform of the error signal of the system in Fig. 6-41 is

$E(z) =$

$-\frac{1}{Tz}$

$$[1 - C(z)/R(z)] = \frac{1}{Tz} [1 - C(z)]$$

(6-193)

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Applying the final-value theorem to the last equation, we have

$$e^* = \lim_{z \rightarrow 1} (1 - z) E(z)$$

$$= \lim_{z \rightarrow 1} (1 - z) \frac{1}{Tz} [1 - C(z)]$$

(6-194)

For a finite nonzero velocity error, $1 - C(z)/R(z)$ must have a zero at $z = 1$.

or the steady-state error in Eq. (6-194) becomes infinite or zero. Thus, Eq. (6-194) can be written as

$$e^* = \lim_{z \rightarrow 1} (1 - z) \frac{1}{Tz} [1 - C(z)]$$

$$= \lim_{z \rightarrow 1} \frac{d}{dz} \left[\frac{1 - C(z)}{R(z)} \right]$$

(6-195) Since we have assumed that the system has a finite velocity error, the steady-state error sequence of the system corresponding to a step displacement input must equal zero; that is,

$$\lim_{z \rightarrow 1} C(z)/R(z) = 1$$

Equation (6-195) can now be written as

or

$$\lim_{z \rightarrow 1} \frac{d}{dz} \left[\frac{1 - C(z)}{R(z)} \right]$$

=

=

-

$$\lim_{z \rightarrow 1} \frac{d}{dz} \left[\frac{1 - C(z)}{R(z)} \right]$$

$$= \lim_{z \rightarrow 1} \frac{d}{dz} \left[\frac{1 - C(z)}{R(z)} \right]$$

$$\lim_{z \rightarrow 1} \frac{d}{dz} \left[\frac{1 - C(z)}{R(z)} \right]$$

Substitution of Eq. (6-190) into Eq. (6-198) yields

(6-196)

(6-197)

(6-198)

$\lim_{z \rightarrow 1} \{d[\ln K + \ln(x - z_1) + \ln(x - z_2) + \dots]$

$\frac{1}{z}$

$=$

$\frac{K \& T}{z-1}$

$\frac{dz}{dz}$

$-$

$+ \ln(z - z_m) - \ln(z - p_1) - \ln(z - p_2)$

$- \frac{1}{z}$

$- \ln(x - p_2)]\}$

$-$

$-$

$-$

(6-199)

and it follows that

$\frac{1}{z}$

$=$

$\frac{K \& T}{z}$

$\frac{1}{z}$

$(1$

$-$

$p_1)$

$-$

M

$\frac{1}{z}$

$i=1$

$-$

$z_i)$

(6-200)

(3) Ka: For an acceleration input, $r(t) = t^2/2$; the steady-state

error sequence is given by Eq. (6-177). Or

$e_{ss} = \lim_{z \rightarrow 1} (z + 1) [1 - 0] =$

$\frac{T^2}{2} \lim_{z \rightarrow 1} \{d[C(z)/R(z)]\}$

$-$

$\frac{T^2}{2}$

$\frac{2}{z-1}$

$-$

$\frac{C(z)}{R(z)}$

$\frac{1}{z}$

$R(z).$

$-$

$\cdot$

$\cdot$

Ka

(6-201)

For a finite nonzero acceleration error, $1 C(z)/R(z)$ must have a

double zero at $z = 1$; otherwise, the steady-state error in Eq. (6-201)

is infinite or zero. Taking the second derivative with respect to z in

both the numerator and the denominator of Eq. (6-201), we have

e'_{ss}

$\frac{1}{z}$

Ka

$=$

$\frac{T^2}{2} \lim_{z \rightarrow 1} \{d[C(z)/R(z)]\}$

$\frac{2}{z-1}$

$\frac{dz}{dz}$

(6-202)

In terms of the logarithmic derivative, the following relation is rec-

ognized:

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d^2

d^2

$C(z)$

In

$\frac{d^2}{dz^2} R(z)$

d^2

$[C(z)/R(z)]$

d

2

$C(x)/R(z)$

dz

=

$C(z)/R(z)$

$C(z)R(z)$

It follows now that Eq. (6-202) is written as

d

1

=

Ka

$\frac{d^2}{dz^2} C(z)$

$T^2 \lim_{z \rightarrow 1} \frac{1}{dz^2} \frac{C(z)}{R(z)}$

$\frac{1}{2-1} \frac{d^2}{dz^2} R(z)$

$8]$

=

dz

$[\frac{1}{2} [C(z)/R(z)]]]$

$C(z)/R(z)$

(6-203)

(6-204)

Differentiating the right-hand side of Eq. (6-199) with respect to z and taking the limit as z approaches unity, Eq. (6-204) is written.

1

1

=

Ka

T^2

$(1$

$-$

1

$21) 2$

$-$

n

$(1$

$-$

1

$P)$

(6-205)

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● FREQUENCY RESPONSE OF SAMPLED-DATA SYSTEMS 7.1 INTRODUCTION

In the preceding chapter time response of sampled-data systems has been studied through the use of the z-transform or the modified z-transform technique. The stability of closed-loop sampled-data systems is investigated by application of the Schur-Cohn criterion or the modified version of the Routh-Hurwitz criterion (the r-transformation) or Jury's criterion to the z-transform characteristic equations. However, among the many techniques available for the analysis and design of feedback control systems, the frequency response method has proved quite popular. The basic features of the frequency response method are that the description of the system is given in terms of its response to sinusoidally varying input signals and that the sinusoidal steady-state behavior of the system is obtained by replacing the Laplace operator s by $j\omega$ in the system transfer function. As is well known, the output response for linear systems with continuous-data contains only the same frequency component as the sinusoidal input signal, and only in systems with nonlinear elements may harmonics or subharmonics of the input signal appear in the system output. However, for linear systems with sampled-data, the sampler acts as a harmonic generator so that the system output in general may contain higher harmonics of the input. The main difficulty in studying sampled-data systems by the frequency response method,

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therefore, stems from the fact that these high frequency components in the system usually make the construction of the frequency loci plots more difficult.

The study of sampled-data systems in the frequency domain essentially relies on the extension of all the existing techniques devised for the analysis of continuous-data systems. Some of the well-known methods are:

1. The Nyquist Diagram

From the Nyquist diagram of the open-loop system transfer function, the stability of the closed-loop system can be investigated. The behavior of the Nyquist plot with respect to the $(-1, 0)$ point in the open-loop transfer function plane gives information on the absolute stability as well as the relative stability of the system.

Frequency response characteristics of the system can also be obtained directly from the Nyquist plot.

2. The Bode Diagram (Gain and Phase versus Log of Frequency)

The Bode diagram is a plot of the amplitude and phase angle of the open-loop transfer function as a function of frequency (or log of frequency) and may be used to investigate the absolute and relative stabilities of the closed-loop system.

3. The Gain-Phase Plot

The gain-phase plot of the open-loop transfer function is used to determine system stability by investigating the amplitude in db versus the phase angle of the open-loop transfer function.

In the following sections, we shall see how the above mentioned conventional methods may all be extended to aid the analysis and design of sampled-data control systems.

7.2 THE NYQUIST PLOT

The Nyquist criterion is essentially a graphical study of the stability of a closed-loop system. We have established in the preceding chapter that the stability of linear control systems with continuous- or sampled-data is determined by the location of the roots of the characteristic equations. For a continuous-data system, all the roots of the characteristic equation must lie in the left-half of the s-plane,

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whereas for a sampled-data system, the roots of the characteristic equation must be found inside the unit circle $|z| = 1$ in the z-plane for system stability. Let us consider the single-loop feedback systems shown in Fig. 7-1 (a) and (b). For the continuous-data system, the closed-loop transfer function is

$$\frac{R(s)}{E(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

Fig. 7-1 Block diagrams of continuous- and sampled-data systems.

(a) Continuous-data control systems. (b) Sampled-data control systems.

$$\frac{R(s)}{E(s)} = \frac{G(s)}{1 + G(s)H(s)} \quad (7-1)$$

and the closed-loop transfer function of the sampled-data system is

$$\frac{R^*(s)}{E^*(s)} = \frac{G^*(s)}{1 + HG^*(s)} \quad (7-2)$$

Eq. (7-1) makes clear that the relative and absolute stability of the continuous-data system may be studied by constructing the Nyquist plot (or Bode plot or gain-phase plot) for $G(j\omega)H(j\omega)$. Logically, the same procedure may be extended to sampled-data systems by constructing the Nyquist plot for $HG^*(j\omega)$ or $HG(z)$. Because of the difference in the criterion on pole-zero locations. in the s-plane and the z-plane, the Nyquist criterion must be restated.

for sampled-data systems.*

The Nyquist path originally defined in the s -plane for continuous-data systems is shown in Fig. 7-2a. The path is described by four separate sections and virtually encloses the entire right-half of the S -plane with the exception of the origin and points on the imaginary axis where the roots of the characteristic equation may locate. The corresponding Nyquist path in the z -plane mapped by the z -transformation is shown in Fig. 7-2b. The small semicircles at $z = +1$ + It is expected that the reader is familiar with the conventional Nyquist criterion

so that extension is made here only to the stability analysis of sampled-data systems.

The reader may refer to any standard text book on feedback control systems, such as

Automatic Control Systems by B. C. Kuo, Englewood, N. J., Prentice-Hall Inc., 1962,

for details of the derivation and applications of the conventional Nyquist criterion.

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$j\omega$

$+\infty$

7

s -plane

(IV)

(I)

$j\omega$

8

Ro

6

75

7

4

-(II)

(IV)

4

- ∞

8

15

(III)

- ∞

Im z

2-

plane

(III)

8/(1)

2-

Re z

3 (11)

3(11)

(a)

(b)

Fig. 7-2 (a) The Nyquist path in the s -plane. (b) The Nyquist path in the z -plane.

and other points on the path correspond to the small indentations on the s -plane Nyquist path. The large circle of infinite radius in the z -plane corresponds to the large semicircle (Sec. IV) in the s -plane. The Nyquist path in the z -plane is not shown as a continuous closed path because, corresponding to point (5) on the path in the s -plane, the z -plane path jumps abruptly from the unit circle to the

infinite-radius circle and at point (7) the z-plane path jumps abruptly back from the infinite circle to the unit circle. The exact place where the jump occurs is indeterminate. However, the area that is enclosed by the z-plane Nyquist path is the region between the two circles. Later we shall see that though this Nyquist path appears to be discontinuous, no difficulty results in the interpretation of the stability to the Nyquist diagram.

The procedure and statement of the Nyquist criterion for sampled-data systems are given as follows:

(1) The Nyquist path in the z-plane is defined as shown in Fig. 7-2b.

(2) The Nyquist plot of $HG^*(s)$ or $HG(z)$ is constructed. This is a plot of $HG^*(s)$ or $HG(z)$ when z takes on the values along the Nyquist path in the z-plane (or s takes on all values along the path in the s-plane).

(3) The Nyquist criterion states that for a stable closed-loop sampled-data system, the Nyquist plot for $HG(z)$ must encircle the

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$(-1, 0)$ point of the $HG(z)$ plane as many times as the number of poles of $HG(z)$ that are outside the Nyquist path in the z-plane, and the encirclement (if there is any), must be made in the clockwise direction.

encirclement

In other words,

where

(MOJA +)

The Dove? [It $HQ(2)$] outside unit corell

$N =$

$N = Z - P$ noe poes [1++16(2)] (7-3)

$(\pm)N =$ number of encirclement of the $(-1, 0)$ point made by the $HG(z)$ Nyquist plot in the $HG(z)$ plane. A positive sign for N corresponds

Z

P :

$=$

$=$

to counterclockwise (positive) encirclement, and a negative sign for N means clockwise (negative) encirclement.

number of zeros of $1 + HG(z)$ that are outside the unit circle (or enclosed by the Nyquist path) in the z-plane.

number of poles of $1 + HG(z)$ that are outside the unit circle (or enclosed by the Nyquist path) in the z-plane. (The poles of $1 + HG(z)$ are the same as the poles of $HG(z)$).

Therefore, for a stable closed-loop system, Z must be zero; hence,

N

$=$

$-$

$- P$

(7-4)

Usually, if $HG(z)$ does not the z-plane, the system is

The Nyquist criterion for stability is then simplified to

have poles outside the unit circle in said to be open-loop stable, and $P = 0$.

$N = 0$ }

(7-5)

which means that if the open-loop system is stable, the Nyquist plot of $HG(z)$ should not encircle (or enclose)† the $(-1, 0)$ point at all to ensure that the closed-loop system is also stable.

As the above discussion indicates, the application of the Nyquist criterion to sampled-data systems is essentially an investigation of

the behavior of the Nyquist plot of $HG(z)$ with respect to the $(-1, 0)$ point. Therefore, the major work in the application of the Nyquist criterion is to construct the Nyquist diagram for $HG(z)$; once this is done, the stability condition of the closed-loop system is determined by mere inspection of the diagram. However, the plotting of the Nyquist diagram for sampled-data systems is complicated by the fact that the pulse-transfer function $HG^*(s)$ is an infinite summation of functions, due to the sampling operation. The transfer locus can no longer be sketched for a sampled-data system by "Enclosed" means encircled in the counterclockwise direction. The area that is enclosed by a closed-path is always to the left of the path when one travels along the path in the prescribed direction.

In this case, if $HG(z)$ encircles the $(-1, 0)$ point at all, the direction must be counterclockwise; and since $N = Z =$ positive number, the point will be enclosed. § Transfer locus usually is considered only as a portion of the complete Nyquist diagram. In the s -plane, the transfer locus of $G(s)H(s)$ corresponds only to sections I and III on the Nyquist path in the s -plane.

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mere inspection of the transfer function, at least not as simply as in the continuous-data case, where the transfer locus may usually be sketched by inspection of the form of the transfer function $G(s)H(s)$ of the system.

Three methods for the construction of the transfer locus of $HG(z)$ are discussed in the following:

1. Two-Term Approximation Method

From Eq. (4-121), $HG^*(j\omega)$ is related to $G(j\omega)H(j\omega)$ by $HG^*(j\omega) = + \sum_{n=0}^{\infty} H(j\omega + jn\omega_s)G(j\omega + jn\omega_s)$

(7-6)

which is an infinite sum of functions. Therefore, it seems that an exact plot of $HG^*(j\omega)$ in any form would require the consideration of an infinite number of terms from Eq. (7-6). However, we can get a closer look at the problem by expanding Eq. (7-6) into the following series:

$$HG^*(j\omega) = +[G(j\omega)H(j\omega) + G(j\omega + j\omega_s)H(j\omega + j\omega_s) + G(j\omega + j2\omega_s)H(j\omega + j2\omega_s) + \dots]$$

$$(7-7) \quad + G(j\omega - j\omega_s)H(j\omega - j\omega_s) + G(j\omega + j2\omega_s)H(j\omega + j2\omega_s) + G(j\omega - j2\omega_s)H(j\omega - j2\omega_s) + \dots$$

In the majority of feedback control systems, the magnitude of $G(j\omega)H(j\omega)$ decreases rapidly as the frequency ω increases, with the result that the higher frequency components in $HG^*(j\omega)$ are not significant. Therefore, it is usually adequate to approximate $HG^*(j\omega)$ by just the first few significant terms of the infinite series. As an example, let the loop transfer function of a simple sampled-data system be

$$G(s)H(s) = \frac{1.57}{s(s+1)}$$

(7-8)

and the sampling frequency ω_s is 4 rad/sec. As derived from Eq. (7-7), the pulse transfer function of the sampled-data system is

$$HG^*(j\omega) = [G(j\omega)H(j\omega) + G(j\omega + j4)H(j\omega + j4) + G(j\omega - j4)H(j\omega - j4) + \dots]$$

$$+G(jw - j4)H(jw - j4) + G(jw + j8)H(jw + j8) + \dots]$$
 For $\omega = 1 \text{ rad/sec}$, Eq. (7-9) becomes

$$HG^*(j1) = - [G(j1)H(j1) + G(-j3)H(-j3) + G(j5)H(j5) + G(-j7)H(-j7) + \dots]$$
 (7-9)
 (7-10)

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51
 1.5
 B
 $HG(z) = \text{plane}$
 W_r
 1.0
 0.8
 0.5
 -1.5
 -1.0
 -0.5
 10.5
 1.0
 1.5
 |
 -0.5
 1.25
 0.8
 -1.0
 1.0
 0.8
 -1.5+
 10
 C

Fig. 7-3 Transfer loci for $HG(x)$
 $K = 1.57, T$

=
 =
 A Exact plot
 B-Two-term approx.
 C - $G(jw)H(jw)$
 $0.792Kx$
 $(\approx 1)(\approx 0.208)$
 1.57 sec.

But from the plot of $G(jw)H(jw)$ shown in Fig. 7-3, the magnitudes of $G(j4)H(j4)$, $G(j5)H(j5)$, $\dots$, and all other higher frequency components are very small; hence, $HG^*(j1)$ can be approximated by just the first two terms of the infinite series given in Eq. (7-10).

Similarly, for $\omega = 2 \text{ rad/sec}$,

$$HG^*(j2) = +[G(j2)H(j2) + G(-j2)H(-j2)$$

$$+G(j6)H(j6) + G(-j6)H(-j6) + \dots]$$

$$\approx [G(j2)H(j2) + G(-j2)H(-j2)]$$
 (7-11)

By taking the two most significant terms of each infinite series of $HG^*(jw)$ for any given frequency ω an estimate of $HG^*(j\omega_1)$ may be obtained. The accuracy of the estimation of course depends

on how well the high frequency components are attenuated in the system. Therefore,

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$$HG^*(j3) \approx -[G(j3)H(j3) + G(-j1)H(-j1)]$$

$$HG^*(j4) = -[G(j4)H(j4) + G(-j0)H(-j0)]$$

(7-12)

(7-13)

$$HG^*(j5) = -[G(j1)H(j1) + G(-j3)H(-j3)] \quad (7-14)$$

The plot of $HG^*(j\omega)$ using the two-term approximation method is shown in Fig. 7-3. As may be seen from this $HG^*(j\omega)$ plot, the value of $HG^*(j\omega/2)$ is real; at frequencies above $\omega/2$, the transfer locus rises into the upper half plane and reaches infinity at $\omega = \omega_s$. For $0 < \omega < \omega_s/2$, the $HG^*(j\omega)$ plot is an exact mirror image of the locus in the upper half plane. Note that the transfer locus for $G(j\omega)H(j\omega)$ that corresponds to frequency range from $-\infty$ to $+\infty$ is single-valued, whereas the locus for $HG^*(j\omega)$ repeats itself for every sampling frequency; that is, the plot of $HG^*(j\omega)$ for $0 < \omega < \omega_s$, is identical to that for frequencies between ω , and $2\omega_s$, $2\omega_s$, and $3\omega_s$, . . . , etc. This contrast is expected, since we have established the fact that $HG^*(j\omega)$ is a periodic function with period equal to ω_s .

@s

2. The z-Transform Method

An exact diagram for $HG^*(j\omega)$ can be drawn by considering $HG(z)$ instead of $HG^*(s)$. For the same system given in Eq. (7-8), the z-transfer function is given by

$$HG(z) = \frac{0.792z}{z^2 - 1.57z + 0.208}$$

$$HG(z) = \frac{0.792z}{(z - 1)(z - 0.208)}$$

$$HG(z) = \frac{0.792z}{(z - 1)(z - 0.208)}$$

$$HG(z) = \frac{0.792z}{(z - 1)(z - 0.208)}$$

(7-15)

From the s-z-plane mapping properties, the points on the unit circle in the z-plane and the points on the $j\omega$ axis of the s-plane are joined by the following relations:

$$z = 1/90^\circ$$

@=

$$\omega_s/4$$

$$z =$$

$$1/180^\circ$$

@=

$$2\omega_s/4$$

$$z = 1/270^\circ$$

$$\omega = 3\omega_s/4$$

$$z = 1/360^\circ$$

$$\omega = n\omega_s$$

where n is an integer. Therefore, for $G(s)H(s)$ given by Eq. (7-8),

$$HG^*(j1) = HG(1 \angle 90^\circ)$$

$$=$$

$$=$$

$$0.863/-147^\circ$$

$$0.792/90^\circ$$

$$1.57 \angle 90^\circ - 1)(\angle 90^\circ - 0.208)$$

(7-16)

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$$0.792/180^\circ$$

$$HG^*(j2) = HG(1 \angle 180^\circ)$$

$$\begin{aligned}
&= 1.57 \\
&(2/180^\circ)(1.280 / 180^\circ) \\
&= 0.515/180^\circ \\
&(7-17) \\
&0.792/270^\circ \\
&HG^*(j3) = HG(1 / 270^\circ) = 1.57 \\
&(1.414/-135^\circ)(1.02 / - 102^\circ) \\
&= 0.863 / 147^\circ \\
&(7-18) \\
&HG^*(j4) = HG(/ 360^\circ) = 1.57 \\
&0.792/360^\circ \\
&(0/360^\circ)(0.792 / 360^\circ) \\
&= \infty \quad (7-19)
\end{aligned}$$

From these data, the exact plot of $HG^*(j\omega)$ is constructed in Fig 7-3.

A comparison of the transfer locus plots obtained from the two preceding methods shows that there is a discrepancy between the results. Individually the high frequency terms in $HG^*(j\omega)$ are not significant, but the accumulated error is still large. Furthermore, the approximated plot shows what purports to be a safer relative stability margin than the actual plot and would endanger stability of the system in design.

3. The Bilinear Transformation Method

A still more convenient method in the construction of the transfer locus for $HG^*(j\omega)$ is to utilize the bilinear w- or r-transformation introduced in Sec. 6-6. Again, as in the extension of the Routh-Hurwitz criterion, either the r- or the w-transformation may be used. Let us use the r-transformation for the present illustration. The r-transformation is characterized by

$$\begin{aligned}
2 &= \\
r + 1 \\
r - 1
\end{aligned}$$

Substituting Eq. (7-20) into Eq. (7-15) yields

$$\begin{aligned}
HG(z) &= \\
2 &= \\
r + 1 \\
r - 1 \\
+1) \\
&= 1.57 \\
&0.792(r + 1)(r - 1) \\
&2(0.792r + 1.208) \\
&(7-20) \\
&(7-21)
\end{aligned}$$

Now let $r = j\omega\tau$, where ω , is the imaginary part of r . Then, Eq.

(7-21) becomes

$$\begin{aligned}
HG^*(j\omega) &= \\
&1.57 (j\omega\tau + 1)(j\omega\tau - 1) \\
&- \\
&2 \\
&(j\omega\tau + 1.525) \\
&(7-22)
\end{aligned}$$

where $HG^*(j\omega\tau)$ means $HG(z = +1)$, but not $HG^*(s = j\omega\tau)$.

The amplitude and phase angle of $HG^*(j\omega\tau)$ when ω , is varied from $-\infty$ to $+\infty$ are tabulated in the following:

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$z=1$

1

$\angle = +90^\circ$

3

$= -90^\circ$

(a)

z plane

z plane

$z=1+\epsilon e^3$

θ

R_∞

2

22

$= 0^\circ$

$z = \text{Rejo.}$

(b)

Fig. 7-4 (a) Section II of the Nyquist path in the z-plane. (b) Section IV of the Nyquist path in the z-plane.

0τ

$HG^*(j\omega r)$

0τ

$HG^*(j\omega r)$

18

$\infty / -90^\circ$

0

$0.515 / -180^\circ$

-1

$0.863 / -147^\circ$

+1

$0.863 / 147^\circ$

$+\infty$

$\infty / 90^\circ$

The transfer locus of $HG^*(j\omega r)$ obtained by means of the r-transformation method is identical to that of $HG^*(j\omega)$ shown in Fig.

7-3 and obtained by the z-transform method. Similarly, we can show that, if the w-transformation of $z = (1+w)/(1-w)$ is used, the same plot of $HG^*(j\omega)$ would be obtained.

The complete Nyquist plot of $HG(z)$ is obtained by considering all the four sections of the Nyquist path in the z-plane. This involves the addition of the $HG(z)$ plot that corresponds to Sec. II and IV to the transfer loci of $HG(z)$ shown in Fig. 7-3.

An enlarged view of the small indentation of Sec. II of the Nyquist path is shown in Fig. 7-4a. The points on this section may be represented by

$z = 1 + \epsilon e^3$

(7-23)

where ϵ tends to zero. As the tip of the small vector shown in Fig. 7-4a is moved from point (1) to point (3) along the small semicircle, $\angle$ is varied from $+90^\circ$ to -90° . Substitution of Eq. (7-23) into

Eq. (7-15) yields

$HG(z) = 1.57 \times$

$0.792(1+\epsilon e^3)$

$\epsilon e (0.792 + \epsilon e^3)$

Since ϵ tends to zero as a limit, Eq. (7-24) is written as

(7-24)

$HG(z) = 1.57$

$0.792 \quad 1.57$

$=$

$\epsilon e \quad 0.792 \quad \epsilon$

(7-25)

Therefore, the magnitude of $HG(z)$ on Sec. II of the Nyquist

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path is infinite (this is usually true for physical systems); and as is varied from $+90^\circ$ through -180° (clockwise direction), the $HG(z)$ plot according to Eq. (7-25) will rotate through $+180^\circ$ (counterclockwise) about the origin of the $HG(z)$ -plane.

The Nyquist plot of $HG(z)$ for Sec. IV on the Nyquist path may be drawn in the similar manner as that for Sec. II. Referring to Fig. 7-4b, the points on this infinite circle may be represented by

$z = Re^{j\theta}$

(7-26)

where R tends to infinity. As we travel from any point on this large circle and back to the same point in the prescribed direction, θ is varied through $+360^\circ$ (counterclockwise direction). Substitution of Eq. (7-26) into Eq. (7-15) yields

$HG(z) = 1.57$

$0.792 Re$

$(Re-1)(Re$

$1)(Re^* - 0.208)$

Thus, the Nyquist plot for $HG(z)$

corresponding to Sec. IV of

the Nyquist path should wrap

around the origin - 360° (clock- $N = 0$

wise) with zero magnitude. The

complete Nyquist plot for $HG(z)$

of Eq. (7-15) is now shown in

Fig. 7-5.

z

$=$

$=1.57 \ 0.792$

R

(7-27)

Im

$HG(z) = plane$

R_∞

Re

$N = 0$

$P \ 0$

ZAN

$= 0$

$=$

Fig. 7-5 Nyquist plot for $HG(z)$

$0.792Kz$

$K = 1.57, T = 1.57$

$(\approx 1)(x - 0.208)$

sec. The closed-loop system is stable.

The complete Nyquist diagram

may now be used to study the

stability of the open-loop and the

closed-loop sampled-data system.

The stability information obtained

from the $HG(z)$ plot is actually

two-fold. From the origin of the

$HG(z)$ -plane we may investigate

the stability of the function $HG(z)$ itself, although in this case we

know that $HG(z)$ is stable since it does not have any pole outside

the unit circle in the z -plane. In certain cases, the stability of $HG(z)$

should be determined before the stability of the closed-loop system

is analyzed. We may check the stability of $HG(z)$ by investigating

the net number of encirclement of the origin made by the Nyquist

plot of $HG(z)$. Since $HG(z)$ made one clockwise and one counter-clockwise encirclement of the origin, the net encirclement is zero; therefore,

$$\begin{aligned} \text{or} \\ N &= P - Z = 0 \\ P &= Z \\ (7-28) \\ (7-29) \end{aligned}$$

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where in this case,

N = the number of encirclement of the origin made by $HG(z)$.

P the number of poles of $HG(z)$ that are outside the unit

circle.

Z

=

circle.

the number of zeros of $HG(z)$ that are outside the unit

circle.

This analysis of the $HG(z)$ plot with respect to the origin in the $HG(z)$ is merely an interesting check of the Nyquist criterion since P and Z are both equal to zero in this case as may be shown by inspection of Eq. (7-15).

The stability of the closed-loop sampled-data system is now determined from the number of encirclements of the $(-1, j0)$ point made by the $HG(z)$ plot. From Fig. 7-5 we see that the $(-1, j0)$ point is not encircled by the $HG(z)$ plot at all; therefore, N (with respect to the -1 point now) is equal to zero. But since P is zero (the poles of $1 + HG(z)$ are the same as the poles of $HG(z)$), we have

=

$$Z = 0$$

This means that the closed-loop system is also stable (for $K = 1.57$ and $T = 1.57$ sec). However, if the value of K is increased to 3.05 or greater, the $HG(z)$ plot will pass through or encircle the $(-1, j0)$ point once; the encirclement is made in the counterclockwise direction. Hence,

N

$$Z - P = 1$$

$$(7-30)$$

Since P is zero, we have $Z = 1$, meaning that the characteristic

Im

$HG(z) = \text{plane}$

$R \rightarrow \infty$

Re

-1

$$N = 1$$

$$N = 0$$

$$P = 0$$

$$P = 0$$

$$Z = 1$$

$$Z = 0$$

Fig. 7-6 Nyquist plot for $HG(z)$:

=

($\approx$

The closed-loop system is unstable.

$0.792Kz$

$1)(\approx 0.208)$

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equation of the system has one root outside the unit circle in the z-plane and that the closed-loop system is unstable. The Nyquist plot of $HG(z)$ for $K = 5$ is constructed in Fig. 7-6. Normally, if the open-loop system is stable, only the unit circle portion of the Nyquist path in the z-plane need be considered because under this condition the open-loop transfer function $HG(z)$ does not have any pole outside the unit circle and $P = 0$. Usually the zeros of $HG(z)$ are determined by inspection of the transfer function; thus the behavior of $HG(z)$ with respect to the origin of the $HG(z)$ plane need not be investigated. Since the poles of $HG(z)$ are identical to the poles of $1 + HG(z)$, N must equal zero for the closed-loop system to be stable. Therefore, if $P = 0$, the Nyquist criterion may be stated as:

If a sampled-data system is open-loop stable, the closed-loop system is also stable if the Nyquist plot of $HG(z)$ that corresponds to Sec. I, II, and III (the unit circle) on the Nyquist path in the z-plane does not enclose the $(-1, j0)$ point.

As another example, we may construct the Nyquist plot for the same system discussed above but with a zero-order hold inserted in the forward path. For this second-order sampled-data system with zero-order hold, the open-loop z-transfer function is readily given by Eq. (6-96).

$$K[(T - 1 + e^{-T})z - Te^{-T} + 1 - e^{-T}]$$

$$(7-31)$$

$$GLOG(2)$$

$$(\approx -1)(z - e^{-1})$$

For $K = 1.57$ and $T = 1.57$ sec,
 $GOG(z)$

$$1.57 (0.778z + 0.465)$$

$$(21)(\approx 0.208)$$

By means of the r-transformation, Eq. (7-32) is written as

$$GOG(j\omega_r)$$

$$= 1.23 (j\omega_r + 0.25)(j\omega_r + 1.525)$$

$$(7-32)$$

$$(7-33)$$

From either Eq. (7-32) or Eq. (7-33), the Nyquist diagram for $GOG(z)$ is plotted as shown in Fig. 7-7. Since the poles of $G_1OG(z)$ are all located inside the unit circle in the z-plane, the open-loop system is stable; only the Nyquist plot that corresponds to Sec. I of the Nyquist path need be plotted (Sec. III gives the mirror image of Sec. I). However, for completeness, the entire Nyquist plot of $GOG(z)$ is shown in Fig. 7-7. For the value of K chosen (1.57), the Nyquist plot for $GOG(z)$ does not enclose the $(-1, j0)$ point, and the closed-loop system with zero-order hold is stable. However, if the value of K is increased so that the $GOG(z)$ plot now passes through the negative real axis at a point to the left of the -1 point,

W_r
 1.5
 1.0
 0.8
 -1.0 -0.8
 1.0
 0.5
 $HG(z) = \text{plane}$
 1.5
 0.5-
 -0.5
 -0.8
 -1.0
 -1.5 B

=
 Fig. 7-7 Nyquist diagrams for sampled-data system. $G(s)H(s) = \frac{K}{s(s+1)}$

$K = 1.57$, $T = 1.57$ sec, curve A with zero-order hold,
 curve B without hold device.

the critical point $(-1, j0)$ is encircled twice in the counterclockwise direction (enclosed), and the closed-loop system is unstable. In this case, $P = 0$, and $N = 2$; hence $Z = 2$; the closed-loop system transfer function has two poles outside the unit-circle in the z -plane.

A comparison of the relative stability of the basic sampled-data system with and without zero-order hold is made by drawing the Nyquist diagrams of the two cases on the same set of coordinates (see Fig. 7-7). The system without hold device definitely has a safer stability margin. The gain margin and phase margin of the systems can be determined in the usual manner.

Typical Nyquist plots for standard $HG(z)$ (or $G(z)H(z)$) functions are shown in Table 7.1 at the end of this chapter.

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Relationships Between Frequencies

From the discussions given in the above sections on the construction of the transfer locus of $HG(z)$ or $HG^*(s)$, we can see that there are definite relationships among the frequencies w , w_r , w_w , and w_s , where w_s is the sampling frequency and w_r , w_w , and w_o are respectively the imaginary parts of r , w , and s . We shall now derive the frequency relationships as follows:

For the r -transformation,

$$z = \frac{r+1}{r-1}$$

$$r = \frac{z+1}{z-1}$$

$$z = \frac{r+1}{r-1}$$

$$(7-34)$$

and for real frequencies,

$$z = e^{jwT}$$

$$(7-35)$$

and

$$r = jw,$$

$$(7-36)$$

Hence Eq. (7-34) becomes

$$jw = \frac{z+1}{z-1}$$

$$ewr = \frac{z+1}{z-1}$$

$$jwT = \frac{z+1}{z-1}$$

$$-1$$

$$+1$$

$$1$$

$$(7-37)$$

By means of simple trigonometric manipulations, w , is determined

from Eq. (7-37) as

w_r

--

=

$\cot$
 $(\frac{\omega}{\omega_c})$

2

from which, if we solve for w ,

(7-38)

@=

$= 2/2 \cot^{-1}(-$

ω_c

-

$\cot^{-1}(-w_r)$

(7-39)

π

For the w -transformation,

Therefore,

-

ω_w

and

$w =$

$\text{Im}(w)$

-

$Z = 1$

$2+1$

=

$W = s$

W

π

1

0τ

=

$= ---$

1

r

(7-40)

$\tan$

(7)

(7-41)

(7-42)

$\tan(w_w)$

From Eq. (7-39) and Eq. (7-42), the relationships between w_r , w_w , w , and w_c are established and are illustrated in Fig. 7-8.

7.3 THE BODE DIAGRAM

The Bode diagram can also be constructed for a sampled-data system by use of the bilinear transformations. If the r -transformation is used, the Bode plot of an open-loop transfer function $HG(z)$ is

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-5

-

4

-3

-2

0

10

IN

2

3

4

5

30
20
10
Gain margin
- 5.77
-10
- 20
-30
-40
1
0.01 0.02 0.05 0.1 0.2
0.5
1.0
2
270°
180°
90°
\$.s.
0°
-90°
-180°
wr 1.525

Magnitude of HG* (jw)
52
10
20
50
100
Phase margin
Phase shift of HG*(jw,)
0.01 0.02
0.05 0.1 0.2
0.5
1.0 2
5 10 20
50
100
wp

Fig. 7-9 Bode diagram of sampled-data system with $G(s)H(s) = K/s(s + 1)$, $K = 1.57$ and $T = 1.57$ sec.

=
the r-transformation the Bode diagram for sampled-data feedback control systems is constructed in much the same way as that for the continuous-data systems. The Bode diagram has at least one attractive feature over the Nyquist plot in the polar plane in that it may be constructed by use of the asymptotes. We shall see later that the

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30
20
10
db
-10
-13.9
- 20
-30
-40
Gain margin
Magnitude of GnoG* (jw)
1

0.01 0.02 0.05 0.1 0.2

1

0.25

Wr

270°

180°

90°

0°

90°

180°

1

1

0.5 1.0 2

1.525

5 10 20 50 100

Phase margin

Phase shift of $G_o G^* (j\omega)$

0.01 0.02 0.05 0.1 0.2

0.5

1.0

2

5 10 20

50

100

ω_p

Fig. 7-10 Bode diagram of sampled-data system with $G(s)H(s)$

=

= $K/s(s + 1)$ and zero-order hold, $K = 1.57$, T

=

= 1.57 sec.

Bode plot may also be conveniently used for the design and synthesis of compensation of sampled-data systems.

7.4 THE GAIN-PHASE PLOT

The gain-phase diagram for an open-loop transfer function $G(z)$ is a plot of the magnitude of $G(z)$ as a function of the phase shift

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of $G(z)$, with frequency as a varying parameter on the locus. The gain-phase plot of $G(z)$ may be used together with the Nichols chart to determine the frequency response of the closed-loop system. For a basic sampled-data system with unity-feedback, the magnification of the closed-loop system is given by

$M =$

=

$\frac{C(z)}{R(z)} \frac{G(z)}{1 + G(z)}$

(7-44)

where M is a function of complex variables. The magnitude and phase shift of M , when plotted against frequency, represent the frequency response of the closed-loop system. From the theory of continuous-data systems, we know that the peak value of $M(M_p)$ usually gives a qualitative measure of the relative stability of the system. The optimum value of M , for most control systems should lie within 1.1 and 1.5.

Let us now use the same system given in the preceding sections to illustrate the construction of the gain-phase plot for $G(z)$. Substitution of Eq. (7-22) into Eq. (7-44) yields

16

$M = 2\text{db}$

12

ω

4.4
 $M = 3\text{db}$
 8
 db
 $G(z)$
 3.4
 $K = 1.57$
 4

$M = 4\text{db}$
 1.8
 1.25
 1.0
 0.5
 -4
 $w_r = 0$
 -8
 270°
 -240° -220° -200°
 -180°
 Phase shift (degrees)
 Nichols chart
 (constant M loci)
 -160°
 -140°
 -120°

Fig. 7-11 Gain-phase plot for sampled-data system with $G(s) =$

K
 $K = 1.57$, $T = 1.57$ sec.
 $s(s + 1)$
 -90°

H. M. James, N. B. Nichols, and R. S. Phillips, Theory of Servomechanisms, (New York: McGraw-Hill Book Co., 1947).

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$M =$
 $-K(w + 1)$
 $3.05j2w$, $K(w; + 1)$
 (7-45)

where in this case K is not yet specified. Examination of the last equation shows that M can be greater than unity. Furthermore, if the real and imaginary parts of the denominator of M are made zero at the same frequency w_r , the value of M becomes infinite.

This will occur when

$w/$
 -
 3.05
 K
 -

$1 = 0$
 (7-46)

Therefore, M is infinite when w , is zero and when K is 3.05, and the closed-loop system becomes unstable. These values of w , and K check with the marginal value of K and or obtained from either the Nyquist plot or the Bode diagrams presented earlier.

A plot of $G(z)$ in gain-phase is now constructed in Fig. 7-11.

The plot is usually obtained by transferring the data from the Bode plot of $G(z)$. The Nichols chart is also superimposed on the gain-phase plot to give information on closed-loop system frequency response. It is seen that for $K = 1.57$, the peak resonance M , is equal to 1.58, and the resonant frequency w_{rp} is 0.97 (equivalent to

$\omega = 3 \text{ rad/sec}$). From the gain-phase plot and the Nichols chart, the closed-loop frequency response (M versus ω ,) is plotted in Fig. 7-12.

3.0
2.5
2.0
 $M_p = 1.58$
1.5
1.0
0.5
0
-3 -2
-1
1
2 3
4
 ω_p

Fig. 7-12 Magnification curve (M versus ω ,) of sampled-data system with $G(s) = K/s(s + 1)$, K

-
1.57 and T
-
1.57 sec.

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7.5 MODIFIED Z-TRANSFORM ANALYSIS

The frequency response analysis of sampled-data systems conducted in the preceding sections is based on the z-transform methods. For systems with well behaved responses between sampling instants, the z-transform method is of course quite adequate. But a general study

30
20
7.12.
5.5
10
 $m = 0.2$
 $m = 0.3$
0
-7.75-
-11.7-
-16.4-
10
 $m = 1.0$
 $m = 0.5$
Magnitude of $G(z, m)$
-20
 $m = 0.7$
-30
-40
0.01
0.1
1.0
 ω_p
10
100
360°
270°
Phase shift of $G(z, m)$
 $m = 1.0$
180°
 $m = 0.7$
90°

$m = 0.5$
 0°
 $m = 0.3$
 $m = 0.2$
 -90°
 -180°
 0.01
 0.1
 1.0
 W_r
 10

Fig. 7-13 Bode diagram of sampled-data system with $G(s) = \frac{27.8(s+1)}{s(s^2+6s+48.6)}$, $T = 1$ sec and modified z-transform is used;
 $m = 0.2, 0.3, 0.5, 0.7$ and 1 .
 100

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of all systems, or of systems with hidden oscillations, must employ the modified z-transform method. In this section, we shall show how the modified z-transform method may be used to analyze sampled-data systems in the frequency domain. Let us consider the sampled-data system illustrated in Example 6-12. We know that the output response of the system due to a unit step function input possesses hidden oscillations between sampling instants. The z-transform analysis in this case predicted a monotonically increasing time response, whereas the actual response oscillates violently in between sampling instants. Clearly, if the z-transform method is carried over to the frequency domain analysis of the system, similar misleading information will result. However, we shall show that by using the modified z-transform method, salient characteristics of the system frequency response and relative stability may be accurately anticipated. For a sampled-data system with unity-feedback, the "magnification" of the closed-loop system may in general be defined as

W_r
 3.0
 2.0
 $1+G$
 $G(z)$ -plane
 1.0
 0.7
 0.3
 0
 $-1.0 \ -0.8 \ -0.6 \ -0.4$
 -0.3
 $0.2 \ 0.4 \ 0.6 \ 0.8 \ 1.0$
 -0.7
 -1.0
 -2.0
 -3.0

=
 Fig. 7-14 Nyquist diagram of sampled-data system with $G(s) = \frac{27.8(s+1)}{s(s^2+6s+48.6)}$ and $T = 1$ sec. from z-transform data.
 =

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$$M(m) = C(x, m) = G(x, m)$$

$$R(z) = \frac{1}{1 + G(z)}$$

(7-47)

where $M(m)$ is a function of m and frequency. Since the numerator of Eq. (7-47) contains the modified z -transform of $G(s)$ and since the denominator has the ordinary z -transform, the Nichols chart cannot be used for the determination of $M(m)$. However, if $G(z, m)$ is constructed in the Bode diagram for several values of m ($0 < m < 1$) and if the plot of $G(z)$ is made in the polar coordinates, the plot of $M(m)$ versus frequency may be constructed without much difficulty. The Bode plots of $G(z, m)$ are drawn in Fig. 7-13 for $m = 0.2, 0.3, 0.7$, and 1.0 ; and the transfer locus for $G(z)$ is plotted in Fig. 7-14. From these plots the magnification curves for the various values of m are drawn in Fig. 7-15. For $m = 0.2$, the sampled output $c(t)$

$$c(nT, m)$$

$$C(t)$$

2-

0.2

1.2

2.2

3.2

4.2

Output step response

2

1

$$M_p = 3.08$$

T

1

1

1

-6 -5 -4

-3 -2 -1

1 2

3

4

5 6

ω_p

$M(m)$ versus frequency

Fig. 7-15a Magnification curve and the corresponding time response from modified z -transform with $m = 0.2$ of the sampled-data system with $G(s)$

=

$$27.8(s + 1)$$

$$s(s^2 + 6s + 48.6)'$$

$$T = 1 \text{ sec.}$$

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3

2

$$c(t)$$

3.3

4.3

0.3

1.3

2.3

Output step response

3-

2

1

$M_p = 2.55$

L

1

1

-6 -5 -4 -3 -2 -1

0 1

2

3

4

5

6

W_r

$M(m)$ versus frequency

Fig. 7-15b Magnification curve and the corresponding time response from modified z-transform with $m = 0.3$ of the sampled-data system with $G(s)$

$27.8(s + 1)$

$s(s^2 + 6s + 48.6)$

$T = 1$ sec.

gives the peak overshoots of the hidden oscillation during each sampling period, and the corresponding $M(m)$ versus ω plot indicates a peak resonance M_p of 3.08. For other different values of m the corresponding $M(m)$ curves are all different. For $m = 0.7$ and 1.0 , the time responses at sampling instants erroneously predict an overdamped system, and the corresponding $M(m)$ curves exhibit no resonant peaks at all.

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$c(t)$

3.

$c(nT, m)$

$c(t)$

2-

0.7

1.7

t

2.7

3.7

Output step response

21

1-

0.206

1

-5 -4 -3 -2 -1

0

2

3

4

5

6

ω_p

$M(m)$ versus frequency

Fig. 7-15c Magnification curve and the corresponding time response from modified z-transform with $m = 0.7$ of the system with sampled-data and $G(s) =$

$27.8(s + 1)$

$s(s^2 + 6s + 48.6)$

T = 1 sec.

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$c(t)$

$c(nt, m) c(t)$

$2-$

T

2T

3T

4T

Output step response

2

1

0.352

1

-6 -5 -4 -3 -2 -1

0

1

2

3

4 5 6 7

r

M(m) versus frequency

Fig. 7-15d Magnification curve and the corresponding time response

from modified z-transform with m =

1.0 (same as z-transform)

of the sampled-data system with G(s):

T = 1 sec.

=

$27.8(s + 1)$

$s(s^2 + 6s + 48.6)$

and

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TABLE 7-1

$G(s) = K/s$

$G(x)$

=

Kz

Z-

- 1

$K(r + 1)$

$G(r)$

=

Im

w $G(jw)$

Im
0
 $W = \infty$
Re
8-3
 $G(jw)$
 $w_1 = 0$
0.5K Re
Wr
3340
 $G(s) = K/s^2$
 $G(z)$
=
KTz
 $(x-1)^2$
 $G(r)$
-
 $KT (r^2 - 1)$
2
-
Im
 $G(jw)$
 $w_1 = 0$
Im
 $G(jw)$
++
Re
-KT
0
Re

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K
 $G(s) = 5a (a > 0)$
 $s+a$
TABLE 7-1 (continued)
 $G(z)$
=
z-
Kz
 e^{-ar}
 $G(r)$
=
(A
 $K(r + 1)$
 $(1 - e^{-aT}) (r + A)$
 $1+e^{-ar}$
 $= 1- e^{-ar}$
Im
 $G(jw)$
Im
+ Wr
-ω
843
+W
340
Re
ωp
∞
@r=

$$\begin{aligned}
&K \\
&A(1-e) \\
&W_r \\
&K \\
&1-e^{-T} \\
&G(s) \\
&G(r) \\
&= \\
&= \\
&K \\
&(s+a)(s+b) \\
&G(x) \\
&= \\
&Kz(e^{-ar}-e^{-bT}) \\
&- \\
&(b-a)(z-e^{-aT})(z-e^{-bT}) \\
&K(r^2-1)(e^{-ar}-e^{-bT}) \\
&(1-e^{-ar})(1-e^{-bT})(r+A)(r+B)(b-a) \\
&- \\
&(A=1+e^{-aT}) \\
&1+ \\
&(B=1+e^{-bT}=3.7) \\
&- \\
&I_m \\
&G(j\omega) \\
&-\omega \\
&W \\
&\infty \\
&+W \\
&I_m \\
&G(j\omega) \\
&+W_r \\
&ab \\
&18 \\
&\omega=0 \\
&Re \\
&-K(e^{-at}-e^{-bT}) \\
&(b-a)(1+e^{-aT})(1+e^{-bT}) \\
&-W_r \\
&\omega_p \\
&Re \\
&K(e^{-at}-e^{-bT}) \\
&(b-a)(1-e^{-aT})(1-e^{-bT})
\end{aligned}$$

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 G(s)

$$\begin{aligned}
&= \\
&K \\
&s(s+a) \\
&TABLE\ 7-1\ (continued) \\
&G(x) \\
&= \\
&K(1-e^{-aT})z \\
&a(z-1)(z-e^{-aT}) \\
&G(r) \\
&= \\
&K(r^2-1) \\
&2a(r+A) \\
&(A=1+e^{-aT}) \\
&3-8 \\
&+W
\end{aligned}$$

K
 Re
 Im
 $G(j\omega)$
 $G(j\omega)$
 $+Wr$
 $wr=0$
 $0_{\infty} = +\infty 0$
 $2aA$
 $-wr$
 Im
 K
 $G(s) = 5^2 (s + a)$
 040
 $+W$
 $G(z)$
 $G(r)$
 -
 =
 KTz
 $K(1 - e^{-aT})z$
 -
 $a(z - 1)^2 - a^2(z - 1)(x - e^{-aT})$
 -
 $KT(r^2 - 1) r + (TaA - 2)/Ta$
 $4a$
 $r + A$
 Im
 $1 + e^{-ar}$
 $A =$
 Im
 $G(j\omega)$
 W
 ∞
 ∞
 $+00 + wr$
 $+00 + wr$
 $@_1 = 0.$
 $(4 > 247)$
 $+ wr$
 $+wr$
 ω_p
 $G(j\omega)$
 $-KT (A - 2/aT)$
 $4aT$
 $\rightarrow 8$
 Wr
 $(A < 27)$

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TABLE 7-1 (continued)

$G(s)$

=

K

$s (s + a)^2$

$K2(1.$

e^{-ar}

$G(z)$

=

$aTe^{-ar})(x +$

$G(r) = K' (r^2 - 1) (r + C)$

$$\frac{(r + A)^2}{K' K} \frac{1 - e^{-ar}}{1 - e^{-ar}} = \frac{aTe^{-ar}}{2a(1 - e^{-ar})} \frac{A}{1 + e^{-ar}} \frac{1}{1 - e^{-ar}} \frac{+\infty}{Wr} \frac{-C > 0}{+ Wr} \frac{+\infty 0}{-Wr} \frac{e^{-2aTaTe^{-ar}}}{1 - e^{-ar} aTe^{-ar}} \frac{(\approx - 1)(z - e^{-ar})^2}{1 - 2aTe^{-ar}} \frac{C}{=}$$

$$C < 0$$

$$Im$$

$$w, = /A^2 - 2AC$$

$$Wr$$

$$wr$$

$$-$$

$$e^{-ar} - e^{-27}$$

$$\frac{1 - e^{-ar} + e^{-2ar}}{W}$$

$$Im$$

$$G(jw)$$

$$+W$$

$$G(ju)$$

$$843$$

$$Wr$$

$$= 0$$

$$Re$$

$$-K'C$$

$$42$$

$$Re$$

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DESIGN AND COMPENSATION OF SAMPLED-DATA SYSTEMS BY CONTINUOUS NETWORKS

8.1 INTRODUCTION

Generally all performance specifications given on the steady-state and transient responses of a feedback control system cannot be satisfied simultaneously given only the basic fixed elements over which the engineer has little or no control. Although the adjustment of the loop gain would seem to be a simple way of changing the system performance, this adjustment is very seldom satisfactory as a means of improving the system, except perhaps in very trivial situations. For instance, if the system loop gain is increased to reduce the steady-state error, the transient response usually will become more oscillatory, or the system may even become unstable. Therefore, in control system engineering, compensating devices must frequently be introduced whose function is not only to provide absolute stability

but more important to meet specifications called for on the steady-state as well as the transient performances. Basically, the principle of compensation of sampled-data systems is similar to that of systems with continuous-data. Regardless of whether the sampling operation is inherent or purposely introduced into a system, the output step response of a sampled-data system is usually continuous with respect to time and is characterized by the usual terms of specifications, such as percent overshoot, response time, rise

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time, delay time, steady-state error, etc. Therefore, the system should be compensated so that all these quantities lie within the specified margins. However, due to the sampling operation, the design of the compensation networks for sampled-data systems does present more problems and difficulties than for systems without sampling. The possibility of using computers as compensating elements has also made the design of sampled-data systems more versatile and interesting. Whenever the preliminary studies made on a certain sampled-data system reveal that the over-all system performance does not satisfy the given specification, compensation should be employed. Theoretically, compensation of sampled-data control systems may be incorporated in any suitable manner, but the following methods are the most common.

1. Cascade Compensation

Compensating elements are inserted in the forward path of a sampled-data system. The following two schemes may be used:

Compensating
elements

Controlled
system

$R(s)$
 $E(s)$ $E^*(s)$

T

$G_e(s)$

$G_1(s)$

$C(s)$

Continuous
signal

Feedback
elements

$H(s)$

(a)

Synchronized
samplers

Controlled
system

$R(s)$

$E(s)$

$D(s)$

$G_1(s)$

$\cdot C(s)$

Digital
controller

$H(s)$

(b)

Fig. 8-1 Cascade compensation of sampled-data systems. (a) Continuous network compensation. (b) Digital network compensation.

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(a) Compensation by Continuous Networks. In this method continuous-data networks or devices such as those used in continuous-data control systems are connected in series with the elements of the controlled system (Fig. 8-1a).

(b) Compensation by Digital Networks. In this method, digital controllers or networks are employed as compensating elements. The input and output signals of a digital controller are all in pulsed form. A digital network may be a network with synchronized samplers located at the input and output terminals of the network, or in general, it can be a special-purpose digital computer or a portion of a general-purpose digital computer. A sampled-data control system with cascaded digital controller is illustrated in Fig. 8-1b.†

2. Feedback Compensation

This includes inserting compensating elements in the feedback path, or forming additional minor feedback loops and other more complex measures such as the conditional feedback. A sampled-data system with conditional feedback is illustrated in Fig. 8-2.

$R(s)$
 $H_1(s)$

T

$H_2(s)$

$D^*(s)$

$IN(s)$

+

$G(s)$

$C(s)$

+

Fig. 8-2 Sampled-data system with conditional feedback.

In this case, it is also appropriate to regard the system as a continuous-data system with digital controller or digital computer compensation.

G. Lang & J. M. Ham, "Conditional Feedback Systems-A New Approach to Feedback Control," A. I. E. E. Trans. Part 2, Vol. 74, 1955, pp. 152-161. D. R. Katt,

"Conditional Feedback Systems Applied to Stabilizing a Missile in Pitch Attitude

WESCON Convention Record, 1957, pp. 171-175.

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3. Multirate Sampling Compensation

The performance of certain sampled-data systems may be improved by use of a multirate controller which has samplers at its input and output; the samplers do not operate with the same sampling rate.

In general, the multirate controller may be located in either the forward or the feedback path. A typical control system with multirate controller is illustrated in Fig. 8-3.

Multirate Controller

$R(s)$

$E(s)$

$D^*(s)$

S

$G_1(s)$

$-C(s)$

T/n

T

$H(s)$

Fig. 8-3 Sampled-data with multirate controller.

As far as the design techniques are concerned, we recall that the design of systems with continuous-data is usually carried out either

in the frequency domain or the time domain (s-plane). This is also true for the compensation and design of sampled-data systems, in view of the fact that we have extended most of the conventional methods such as the Bode plot, Nyquist plot, root locus diagram, etc., to the sampled-data case. In terms of the Bode plot and the Nyquist plot, the compensation of sampled-data systems essentially involves the reshaping of the original system locus until the desired values for M , phase margin, gain margin, and bandwidth are obtained. In terms of the root locus diagram, we are working with virtually the pole-zero configuration of the closed-loop transfer function that gives information on the frequency response as well as the time response. By adding poles and zeros at proper places in the z-plane, the root loci of a sampled-data system may be reshaped so that the corresponding system has improved behavior. In the present chapter, only the compensation with continuous networks will be discussed. Digital compensation and other techniques will be covered in Chapter 9.

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8.2 CASCADE COMPENSATION BY CONTINUOUS-DATA NETWORKS

A sampled-data control system with a cascade continuous-data compensating network $G_c(s)$ is shown in Fig. 8-4. Since the compensating network is connected directly to the zero-order hold and the controlled system, the z-transform of the open-loop transfer function is

$$C(z)/E(z) = Z[G_o(s) G_c(s) G_1(s)] = G_o G_c G_1(z) \quad (8-1)$$

The main objective of continuous-data compensation is to select a realizable network with transfer function $G_c(s)$, which will cause the sampled-data system to perform satisfactorily when inserted in it as shown in Fig. 8-4. However, as Eq. (8-1) shows, the compen-

Zero-order Compensating Controlled

hold

network

system

$R(s)$

$E(s)$

T_s

$1-e$

$H(s)$

$G_c(s)$

$G_1(s)$

$-C(s)$

S

Fig. 8-4 Sampled-data system with continuous-data compensation. sating network function $G_c(z)$ cannot be derived and separated from the over-all expression $G_o G_c G_1(z)$: hence the influence of various forms of $G_c(s)$ on the over-all system behavior is not apparent. Therefore, the effort of selecting a suitable $G_c(s)$ usually involves the construction of the transfer locus of $G_o G_c G_1(z)$ (if the frequency domain method is used) each time a new compensating network is suggested. Perhaps only after the construction of quite a few $G_o G_c G_1(z)$ plots can an intelligent choice of $G_c(s)$ be made. For this reason, although cascade compensation by means of continuous-data networks may seem to be the simplest method to carry out, the design technique is far from being as straightforward as its counterpart in continuous-data control systems.

Several approximation methods in regard to the design of continuous-data compensation as shown in Fig. 8-4 have been devised in the literature, but none can be regarded as really satisfactory.

+ W. K. Linvill, "Sampled-Data Control Systems Studied Through Comparison of Sampling with Amplitude Modulation," Trans. A. I. E. E., 70, Part 2, 1951, pp. 1779-1799.

1788. E. I. Jury, Sampled-Data Control Systems (New York: John Wiley & Sons, Inc. 1958), Chapter 6.

‡ J. R. Ragazzini & G. E. Franklin, Sampled-data Control Systems (New York: McGraw-Hill Book Company, Inc., 1958, pp. 123-129.

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Either the approximation is too rough, or the amount of work involved in the design is still overwhelming. For the purpose of illustration, the continuous-system approximation method: is described in the following.

In the continuous-system approximation method, the sampler and the zero-order hold are approximated by a pure time delay equal to one-half the sampling period of the system. The approximating system is then a continuous system with time delay, and the conventional design techniques may be used.

Let us consider the system shown in Fig. 8-4. The transfer function of the zero-order hold has been given in Eq. (3-10) as

$$G_{re}(j\omega) = T \sin(\omega T/2) e^{-j\omega T/2}$$

ωT

(8-2)

By definition the open-loop pulse-transfer function of the system is given by

$$G_1 G_c G_T(j\omega) = \sum_{n=0}^{\infty} G_a(j\omega + jn\omega_s) G_c(j\omega + jn\omega_s) G_T(j\omega + jn\omega_s)$$

Substitution of Eq. (8-2) into Eq. (8-3) yields

$$G_1 G_c G_T(j\omega) = \sum_{n=0}^{\infty} T \sin\left(\frac{(\omega + jn\omega_s)T}{2}\right) e^{-j(\omega + jn\omega_s)T/2} G_c(j\omega + jn\omega_s) G_1(j\omega + jn\omega_s)$$

$\sin$

$N=18$

$\sin$

$\omega + n\omega_s$

$\frac{T}{2}$

$e^{-j(\omega + n\omega_s)T/2}$

T

$\omega + n\omega_s$

$\frac{T}{2}$

$$\times G_c(j\omega + jn\omega_s) G_1(j\omega + jn\omega_s)$$

(8-3)

(8-4)

If the controlled system $G_c(j\omega)$ has low pass characteristics (which is true for most servo systems) and if at low frequencies $\sin \frac{(\omega + n\omega_s)T}{2} \approx \frac{(\omega + n\omega_s)T}{2}$ equals about one, we can approximate $G_0 G_c G_T^*(j\omega)$ by just the first term of its series expansion. Hence

$$G_0 G_c G_T(j\omega) \approx e^{-j\omega T/2} G_c(j\omega) G_1(j\omega)$$

(8-5)

Therefore, the sampler and the zero-order hold are approximated by a time lag of one-half the sampling period T , and the approximating

$R(s)$

$E(s)$

$e^{-j\omega T/2}$

$H(s)$

$G_c(s)$

$G_1(s)$

$C(s)$

Fig. 8-5 Continuous-system approximation of the sampled-data system shown in Fig. 8-4.

In general, this approximation is questionable, since otherwise we can apply this method to all sampled-data systems and there would be no need to devise the z-transform technique or any other efforts for the studies of sampled-data systems.

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A linear continuous-data system is as shown in Fig. 8-5. The real significance of this pure time-delay approximation is now depicted by the sketch of a hypothetical signal at the output of the zero-order hold as shown in Fig. 8-6. The hold circuit output, which is a staircase wave, may be roughly approximated by a smooth curve passing through the center of each step during a sampling period. This is of course equivalent to delaying the sampler input by one-half the sampling period.

$h(t) \uparrow$
 $e(t)$
 $, h(t)$
 $e(t)$
 Approximated $h(t)$

6T

7T 8T

0 T 2T 3T 4T 5T

Fig. 8-6 Waveforms showing the approximation of the sampler and the zero-order hold by a pure time delay of $T/2$.

Example 8-1

To illustrate the application of the continuous-system approximation method described above, the transfer function of the controlled system of the sampled-data control system in Fig. 8-4 is given as

$$G_1(s) = K/[s(s + 1)] \quad (8-6)$$

where $K = 1.57$ and T

=

1.57 sec. The approximating continuous model of the uncompensated system (open-loop) is represented by

$$G_1(j\omega) = e^{-j\omega T/2} G_1(j\omega)$$

Substituting Eq. (8-6) into Eq. (8-7), we have

$$G_0(j\omega) = e^{-j\omega T/2} K/[j\omega(j\omega + 1)] \quad (8-7)$$

(8-8)

The Bode plot of $G_0(j\omega)$ is now constructed in Fig. 8-7 for the specified values of K and T . From the plot we see that the phase margin of the uncompensated system is approximately 0 degree. Note that the exact plot of $G_0(j\omega)$ of this system shown in Fig. 7-10 gives a phase margin of approximately 15 degrees. As can be shown, the continuous-system approximation described by Eq. (8-5) usually gives a pessimistic representation of the system stability. Furthermore, the phase shift curve of $G_0(j\omega)$ is discontinuous at $\omega = \omega_s/4$ rad/sec, since the term $\sin(\omega T)/T$ changes sign at the sampling frequency.

=

Let us assume that the specification calls for a phase margin of at least 45 degrees. We may, at this point, investigate the possibility of using either a phase-lead or phase-lag network for the continuous compensation. Because of the rapid increase in phase lag at frequencies above 1 rad/sec, a phase-lead

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network may not be effective in improving the phase margin of this system.

(As is well known, phase-lead compensation would increase the gain crossover frequency and therefore would make the addition of more phase lead less effective if the phase shift of the uncompensated system decreases very rapidly in the vicinity of the new gain crossover.) A phase-lag compensation would seem to be more appropriate for the compensation of this system.

A typical phase-lag network is given by the following transfer function:

$$G_0(s) = \frac{1 + aTs}{1 + Ts}$$

(8-9)

60
db
40
20
0

Magnitude curve of G_0 $G_1(j\omega)$
of uncompensated system

-20
-40

Magnitude curve of
1 phase-lag network

Magnitude curve of $G_0 G_c G_1$
of compensated system

-60
-80

ω_c

-100

0.01

0.02 0.05 0.1 0.2 0.5 1.0 2

0.016 0.04

5

10 20

50 100

4

@rad/sec

-90°

-180°

Phase shift

Phase

margin 45°

of $G_0 G_c G_1(j\omega)$

of compensated

system

-270°

-360°

Phase shift of $1/(s+1)s$

Phase margin of

uncompensated system = 0°

Phase shift of uncompensated

system

-Phase shift is

discontinuous here

0.01 0.02 0.05 0.1

0.2

0.5 1.0 2

5

10 20

50

100

4

@rad/sec

Fig. 8-7 Bode plots of sampled-data system in example 8-1. $G(s)$

$Ks(s+1)$, $K = 1.57$, $T = 1.57$ sec. Phase-lag compensation.

=

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where a and γ are constants and a is always less than unity. From the Bode plot of the uncompensated system, we see that the phase margin of 45 degrees may be realized if the gain crossover frequency is moved from the present value (approximately 1 rad/sec) to a frequency of 0.5 rad/sec. Since the phase lag of the compensating network will also have some effect on the final phase shift curve of the compensated system, as a standard practice the new gain crossover should be selected at a frequency somewhat lower than 0.5 rad/sec, say $\omega = 0.4$ rad/sec. From the Bode plot in Fig. 8-7, the gain of $G_1G^*(j\omega)$ at this new gain crossover is about 8 db. Thus the phase-lag network must produce 8 db of attenuation at the new gain crossover frequency. Hence

=

- 8 db

Solving for a from the last equation yields

$$a = 10^{-8/20}$$

$$= 10^{-0.4} =$$

0.4

(8-10)

(8-11)

This fixes the distance between the two corner frequencies of the lag network. To determine the exact location of the two corners, it is necessary to locate the upper corner frequency $1/aT$ at such a place that the phase lag contributed by the lag network is negligible at the new gain crossover. This is done by placing the upper corner at a frequency which is equal to $1/10$ times the new gain crossover frequency. Therefore,

$\frac{1}{aT}$

$$\omega = 0.4$$

$$aT = \frac{1}{10 \times 0.4}$$

=

$$0.04 \text{ rad/sec}$$

(8-12)

and the lower corner frequency of the lag network is determined readily from

$\frac{1}{T}$

=

$$0.04a$$

=

$$0.016 \text{ rad/sec}$$

(8-13)

Substituting the values of a and γ determined from above into Eq. (8-9), the transfer function of the phase-lag compensating network is.

$$G_c(s) = \frac{1 + 25s}{1 + 62.5s}$$

$$(8-14)$$

The magnitude and phase shift plots of the compensated system transfer function $G_{ho} G G^*(j\omega)$ are sketched in Fig. 8-7. The phase margin of the compensated system is equal to about 45 degrees as required.

8.3 CASCADE COMPENSATION BY BILINEAR

TRANSFORMATION METHOD†

The reasons for the rather wide spread use of frequency domain design techniques for continuous-data systems are quite obvious. First, the method guarantees at the outset of the design that the compensating network is of a realizable form (preferably of the RC type); and second, successive application of the cut-and-try process

† M. P. Pastel and G. J. Thaler, "Sampled-Data Design By Log Gain Diagrams," I. R. E. Transactions on Automatic Control, Vol. AC-4, No. 2, Nov. 1959, pp. 192-197.

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can be quickly made in the Bode diagram and the Nichols chart. We have demonstrated in the last section that for the design of sampled-data systems the above mentioned advantages are preserved only if certain approximation schemes are adopted. Without approximation the Bode diagram method cannot be employed, and furthermore, directly connecting a continuous compensating network into the sampled-data system requires complete recalculation of the system z -transfer function.

In the preceding chapters the bilinear transformation $z = (r + 1)/(1)$ has been used to convert z -transform function $G(z)$ of a sampled-data system into a new function $G(r)$, which is a rational function in variable r . This bilinear r -transformation maps the unit circle in the z -plane onto the imaginary axis in the r -plane. Therefore, the Bode diagram of $G(r)$ can be drawn in log of magnitude and phase against frequency. Then, by use of the Nichols chart and correlation information available from continuous-data system, the dynamic behavior of the closed-loop sampled-data system can be predicted from the $G(r)$ plot and the Nichols chart.

In this section we shall present a straightforward method of designing sampled-data control systems with continuous-data network utilizing the r -transformation. The method does not rely on any approximation of the sampling operation and has all the advantages of the frequency design of continuous-data systems.

The block diagram of the system under consideration is shown in Fig. 8-8, in which $G_1(s)$ and $G_c(s)$ are the transfer functions of

$R(s)$

$E(s)$

$G_{ho}(s)$

Zero-order

hold

$G_c(s)$

Compensating

network

$G_1(s)$

Controlled

system

$C(s)$

Fig. 8-8 Sampled-data control system with continuous-data compensation.

the controlled system (fixed elements) and the compensating network, respectively. The proposed steps of design are as follows:

(1) Determine the r -transform function $G_o G_i(r)$ of the system.

without compensation by substituting $z = (r + 1)/(r - 1)$ into $G_{ho} G_1(z)$.

(2) Construct the Bode diagram for $G_{ho} G_1(r)$ and transfer the log magnitude and phase versus ω , loci to the Nichols chart. Predict the dynamic behavior of the uncompensated system by evaluating the phase margin, gain margin, peak resonance, resonant frequency, and bandwidth from the Nichols chart.

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(3) Should the system need improvement, a compensating function $G_c(r)$ of the form of $(1 + a_1 r)/(1 + T_1 r)$ or $(1 + a_1 r)(1 + b_1 r)/(1 + t_1 r)(1 + T_1 r)$ or any other more complex type, if necessary, is

multiplied to $G_o G_i(r)$ to reshape the transfer locus of the uncompensated system. These compensating functions in the r -domain, in general, can be classified as phase-lead, phase-lag, or lead-lag configurations. But they may have no direct significant correlation of phase-lead or phase-lag in the s -plane. The purpose of identifying $G(r)$ in the r -domain is purely for the use of the knowledge of continuous-data systems compensation in regard to the effects of phase-lead and phase-lag networks on the system transfer locus. The transfer function of the actual compensating network in s , $G_c(s)$ is determined from taking the inverse transform of $G_c(r)$ $G_O G_1(r)$, or $G_c(s) G_o(s) G_1(s) = \text{inverse transform of } [G(r)G_o G_i(r)]$ (8-15). Since $G_o(s)G(s)$ is known, $G_c(s)$ is readily obtained from the last equation.

However, the physical realizability of $G_c(s)$ must always be kept in mind. Furthermore, if possible, $G_c(s)$ should be realized by RC elements. In order for $G_c(s)$ to be RC realizable, the following requirements are necessary:

(a) The poles of $G_c(s)$ must lie in the left-half of the s -plane and must be simple and real.

(b) The number of poles of $G_c(s)$ must be greater than or at least equal to the number of zeros of $G_c(s)$.

The zeros of $G_c(s)$ may lie anywhere in the s -plane. Therefore, the poles of $G_c(s)$ must be restricted to the negative real axis of the s -plane. Since the poles of $G_c(s)$ are generated by the poles of $G'(r)$, it follows that $G(r)$ must have only simple poles to the left of the -1 point on the real axis in the r -plane (see Fig. 8-9).

(4) Once $G(r)G_o G_i(r)$ is determined, take the inverse transform to obtain $G_c(s) G_o(s) G_1(s)$. It should be noted that since

s -plane
 r -plane
 Poles of
 $G_c(s)$
 Poles of
 $G'(r)$

Fig. 8-9 Locations of the poles of $G_c(s)$ and $G(r)$ in the s -plane and the r -plane for RC realizability.

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$$G(r)G_o G_1(r) = [G_c G_o G_1(2)] \quad 2 = (+1)/(7-1)$$

$$G_c(r) = 3[G_c(s)] := (r+1)/(r-1)$$

(8-16)

(8-17)

The z -transform of the uncompensated open-loop transfer function can be written

$$G_O G_1(2) (1$$

=

-

$$(-1)^3 [G_1(s)]$$

In terms of the r -transforms, we have

$$G_O G_1(r) = \{(1 - 2^{-1})^3 [G_1(s)]\}$$

=

2

$r+$

13

$$G_1(s)$$

$$z = (r+1)/(r-1)$$

s

$$| z = (r + 1), \quad ' (r - 1)$$

(8-18)

(8-19)

After compensation, the r-transform function of the open-loop system is

$$Ge(r) GhoG(r) = G10 GeG1(r)$$

$$= \frac{13}{2} [G.(s) G_1(s)].$$

$$\} z = \{r + 1), (r- 1)$$

$$or$$

$$3[G_1(s) G_1(s)]$$

$$r+ 1$$

$r+1Ge(r) G_10G_1(r)$

$$= \frac{12}{2} = (r+ 1), (r-1)$$

$$(8-20)$$

$$(8-21)$$

The function $G.(s)G_1(s)/s$ is found from Eq. (8-21) by making the partial fraction expansion of $(r + 1) G.(r) GoG_1(r)/2$ and finding the corresponding pairs of functions in s . A table of the Laplace-z-r transform functions is given in Table 8-1 for this purpose. Investigating the transform pairs of $G(s)$ given in Table 8-1, we see that when $G(s)$ has only one pole at $s = 0$, $G(r)$ has a term $(r + 1)$ in the numerator. For instance, the r-transform of the function $1/s$ is $(r+1)/2$. When $G(s)$ has two poles at $s = 0$, the corresponding $G(r)$ has the term $(r + 1)(r - 1)$ in the numerator. For a function with three poles at $s = 0$, the corresponding r-transform function has $r(r + 1)(r - 1)$ in the numerator. These facts suggest that when making the partial fraction expansion of $G(r)GG(r)$, the number of poles that the function $G_1(s)/s$ has at $s = 0$ must be investigated. If $G(s)/s$ has only one pole at $s = 0$, $G_1(s)$ does not have any pole at $s = 0$; the partial fraction expansion of $Ge(r) GoG(r)$ is performed, since the $(r+ 1)$ term that is needed to correspond to the poles of $G_1(s)/s$ at $s = 0$ is already available at the right-hand side of Eq. (8-21). However, if $G(s)/s$ has two poles at $s = 0$, $Ge(r) GoG_1(r)/(r1)$ must be expanded, since the r-transform of $1/s^2$ is $Tr 1)(r+1)/4$.

(5) Once $Ge(s) G1(s)$ is determined from Eq. (8-21), $Ge(s)$ is readily obtained. However, because of the lack of correspondence between the number of s-plane zeros and the r-plane zeros, the function $G.(s)$ obtained from the previous steps will in many cases have,

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TABLE 8-1

Laplace

transform

$G(s)$

z-transform

$G(z)$

$1/5$

2

Z

r-transform

$G(r)$

$r+ 1$

12

1135

113

Tz

$(z - 1)2$

$T^2 x(x+1)$
 $2(x-1)^3$
 $\frac{1}{s+a}$
 $\frac{1}{z}$
 $2-e^{-ar}$
 $\frac{1}{a^2}$
 $\frac{s+a}{T(r+1)(r-1)}$
 $\frac{4}{T^2 r(r+1)(r-1)}$
 $\frac{8}{r+1}$
 $(1-e^{-ar})^2 (r$
 $r+1$
 $-$
 $1+e^{-ar}$
 $1-e^{-ar}$
 $(r+1)(r-1)Te^{-ar}$
 Tze^{-ar}
 $(z-e^{-ar})^2$
 $(1-e^{-ar}$
 $-e^{-ar})^2 r +$
 $+e^{-ar}$
 e^{-ar}
 a
 $s(s+a)$
 a
 $(1-e^{-ar})z$
 $(z-1)(z-e^{-ar})$
 $s^2 (s+a)$
 Tz
 $(\approx -1)^2$
 $\frac{w}{s^2 + w^2}$
 $\frac{w}{(s+a)^2 + w^2}$
 $\frac{(1-e^{-ar})}{a(z-1)(z-e^{-aT})}$
 $-$
 $z \sin wT$
 $2z \cos wT + 1$
 $ze^{-ar} \sin wT$
 $2ze^{-ar} \cos T + e^{-2aT}$
 $(r+1)(r-1)$
 $2(r+1)$
 $1+e^{-ar}$
 e^{-ar}
 $T(r+1)(r-1)$
 $\frac{4}{(r+1)(r-1)}$
 $\frac{1+e^{-ar}}{2ar + e^{-aT}}$
 $\frac{(r+1)(r-1) \sin wT}{2[(r^2+1) - (r^2-1) \cos wT]}$
 $-$
 $\frac{(r+1)(r-1) e^{-ar} \sin wT}{(r+1)^2 - 2(r+1)(r-1)}$
 $e^{-ar} \cos wT + (r-1)e^{-2ar}$
 more zeros than poles. This can be circumvented by adding one or

more remote poles to $G_e(s)$ on the negative real axis in the s -plane. These additional poles will have negligible effect on the over-all system response.

A numerical example is given in the following to illustrate the presented method of design.

Example 8-2

Consider that the sampled-data system shown in Fig. 8-8 has a sampling period of one second and a transfer function for the controlled system

$$G_1(s) = \frac{1}{s(s+1)(s+2)}$$

The system is without compensation at the moment.

The z -transform of the zero-order hold and $G_1(s)$ is (8-22)

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$$G_{ho}G_1(z) = (1 - \frac{1}{z}) \frac{1}{s^2} \left[\frac{5^2 (5+1) (s+2)}{0.33x^2 + 0.692 + 0.078} \right]$$

-

$$G_{ho}G_1(z) = \frac{0.25}{(z-1)(z+0.368)(z-0.135)}$$

(8-23)

$$G_{ho}G_1(r) = \frac{0.25}{(r-1)(r+0.784)(r-0.326)}$$

The corresponding r -transform function is then.

$$G_{ho}G_1(r) = \frac{0.25}{(r-1)(r+0.784)(r-0.326)}$$

(8-24)

The Bode plot for $G_{ho}G_1(r)$ is shown in Fig. 8-10. The magnitude in db and phase of $G_{ho}G_1(r)$ are transferred from the Bode plot to the Nichols chart

80

60

40

20

Gain crossover

db 0

-20

-40

-60

-80

0.1

1

Magnitude curve $G_{ho}G_1$
(uncompensated system)

New gain
crossover

Magnitude curve $G_e G_{ho}G_1$
(compensated system)

0.326

1.0

0.784

3 6 10

2.16

ω_p

60 100

107

1,000

360°

270°
 Phase
 margin 22°
 180°
 Phase
 margin 45°
 Phase shift
 (uncompensated system)
 90°
 0°
 Phase shift
 (compensated system)

-90°
 -180°

0.1
 10

ω_p
 100

1,000

Fig. 8-10 Bode diagram of the uncompensated and the compensated system of Example 8-2.

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28
 $M = 1.15$ -
 $-M = 2.0$

24
 $M = 1.3$

20
 $M = 1.6$
 $M = 2.5$
 $-M = 4.0$

16
 12
 8
 4

db
 0
 2.8

の

10

10

@

8

-4 Uncompensated

3

-8

system

$G_{ho}G_1(j\omega_r)$

1.8

2.8

-12

1.8

Compensated system

$G_{el}j\omega, G_{ho}G_1(j\omega)$

1.2

-16

$\omega.$

-20

1.2
 0
 -24
 -28
 -32
 260° 240° 220° 200° 180° 160°
 140° 120° 100° 90°

Phase shift

Fig. 8-11 Nichols chart for the system in Example 8-2.

as shown in Fig. 8-11. From this Nichols chart the following quantities on the performance of the system are measured:

Peak resonance

Phase margin

Gain margin

Mp

= 4

Φ. M.

=

22 deg.

G. M.

=

3 db

Clearly the uncompensated system has a rather poor margin of stability.

Let us now assume that the desired phase margin of the system is to be greater than or equal to 45 deg, and M, 1.5. The steady-state accuracy of the system should be maintained at the original level. In other words, the velocity error constant K, must not be affected by the compensation. By referring to the relationship between s and r as shown in Fig. 7-8, we can see that in order to preserve the zero frequency behavior of the open-loop transfer function, the function $G_e(r)G_oG_c(r)$ must equal $G_oG_c(r)$ in magnitude as r approaches infinity, since as $s \rightarrow 0$, $r \rightarrow \infty$. Therefore, the magnitude curve of $G_oG_c(r)$ shown in the Bode plot of Fig. 8-10 must not be affected by the compensating network at ω , ∞ , or the constant K, requirement would

simply imply that

$\lim_{r \rightarrow \infty} G_e(r) = 1$

= 1

(8-25)

748

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Inspection of Fig. 8-10 reveals that a phase margin of 45 deg may be realized if the gain crossover frequency (ω_c) is shifted from the present value 3) 10 0, 5. Since the high frequency (ω) gain of $G_c(s)$ must be kept unchanged, a phase-lag model for $G_c(s)$ would definitely shift the gain crossover toward a lower frequency, thus further degenerating the stability of the system. Therefore, a phase-lead model is more appropriate for $G_c(s)$. Let

(a > 1)

$\omega_c = a\omega_r$

$G_e(r) =$

(8-26)

The Bode plot of $G(r)$ is depicted in Fig. 8-12 in magnitude and phase versus ω . The objective of the phase-lead model is to provide ample attenuation to $G_c(j\omega_c)$ near the gain crossover while not affecting the high frequency behavior, and the effect on the phase shift curve is to be kept at a minimum. Since the phase shift of the phase-lead compensator when added to the original phase shift curve would have an effect of reducing the phase margin of the system, a safety factor should be included when selecting the new gain crossover. Therefore, instead of using $\omega_c = 5$, let us select $\omega_c = 6$ as the new gain crossover of the compensated system. As Fig. 8-10 shows, the magnitude of $G_c(j\omega_c)$ at 6 is equal to 5 db. This suggests that the phase-lead compensator must provide 5 db of attenuation at this frequency

. Therefore,
 $20 \log 10 a' = 5 \text{ db}$
 Solving for a from the last equation yields $a = 1.78$.

(8-27)
 Having determined the value of a, only the selection of a proper value for T is necessary to complete the design. Referring to the phase shift curve of the lead network shown in Fig. 8-12, we see that if the lower corner ($1/\alpha T$) of the network is placed far above the new gain crossover = 6, the phase shift of $G(j\omega)$ will not be appreciably affected by the compensation.

16' (jw)

db

90°

[G(jw)

0

20 log10

1/0T

1/T

0-1

sin m =

0+1

φ

Fig. 8-12 Bode plot of the phase-lead model for G(r)

11+ arr

altor

Wr

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Usually, the lower corner $1/\alpha T$ of the compensator is placed at a frequency (,) about one decade above ω ; that is,

$1/\alpha T$

= 10ω

= ≈ 60

Therefore, the compensating network is described by

(8-28)

$1 + 0.0167r$

$G_e(r)$

$1.781 + 0.0094r$

(8-29)

The open-loop transfer function of the compensated system is now

$G_e(r)G_oG_1(r)$

=

-

$0.25 (r - 1)(r + 0.784)(r - 0.326)(1 + 0.0167r)$

$1.78 (r + 2.16) (r + 1.31)(1 + 0.0094r)$

(8-30)

The magnitude and phase shift of the compensated system function $G_e(r)G_oG_1(r)$ are now plotted in the Bode plot and the Nichols chart in Figs. 8-10 and 8-11, respectively. The peak resonance of the compensated system is equal to 1.3, and the phase margin is approximately 45 degrees. Consequently, the system as compensated is considered satisfactory.

The corresponding Laplace transform of the compensating network is.

determined as follows:

Since $G(s)/s$ has two poles at $s = 0$, in order to determine $G_e(s)$, the partial fraction expansion of the function $G'(r)G_oG_1(r)/(r-1)$ is to be evaluated. Thus from Eq. (8-30)

$G_e(r)G_oG_1(r)$

$(r-1)$

= $0.25 \frac{1}{r-1}$

[1

2.2

$$r + 2.16$$

+

$$0.55$$

$$r + 1.31$$

$$48.4$$

$$r + 107$$

$$(8-31)$$

or

$$Ge(r)GoG_1(r) = 0.25(r$$

2.2

$$r + 2.16$$

Substitution of Eq. (8-32) into Eq. (8-21) yields

$$1) 1$$

-

+

$$0.55$$

$$r + 1.31$$

$$48.4$$

$$r + 107$$

$$(8-32)$$

3

$$Ge(s)G(s)$$

$$(r + 1)(r - 1)$$

--

$$= 0.25$$

S

2

$$| z = (r + 1)/(r - 1)$$

2.2

$$0.55$$

$$48.4$$

1

+

$$r + 2.16 \quad r + 1.31 \quad r + 107$$

$$(8-33)$$

The corresponding Laplace transform of each of the terms on the right-hand side of Eq. (8-33) is found from Table 8-1. Therefore,

or

$$Ge(s)G,(s)$$

S

$$0.5$$

-

$$Ge(s)G,(s)$$

-

S

$$0.55$$

$$s(s + 1)$$

+

$$0.138$$

$$s(s + 2)$$

$$0.02 \quad 23s \quad 36s^2$$

-

$$12.1$$

-

$$5(5$$

$$(8-34)$$

$$(s + 0.02)$$

1253

$$(8-35)$$

$$s^2 (s + 1)(s + 2) (s + 0.02)$$

This leads to a compensation function

$$Ge(s)$$

$$= \frac{0.02 - 23s + 36s^2 + 123}{(s + 0.02)(8-36)}$$

In order for $G_e(s)$ to be RC realizable, two remote poles must be added to $G_e(s)$, say at $s = -40$ and $s = -100$. Thus

$$G_o(s) = \frac{12s^3 + 0.02 - 23s + 36s^2}{(s + 0.02)(s + 40)(s + 100)} \quad (8-37)$$

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The realization of the compensating network, whose transfer function is given by Eq. (8-37), can be carried out by any standard method of synthesis of RC two ports described in the literature of network synthesis.†

Note that the bilinear transformation $z = (1 + w)/(1 - w)$ can also be used for the proposed method of design of sampled-data control systems with continuous-data compensation. Whereas there is no distinct advantage in using either the r or the w transformation, the latter is considered as an alternative design technique only. Since for a fixed value of z , r and w are related by

$$r = 1/w \quad (8-38)$$

the method using the w -transformation can be regarded as a "dual" of the r -transform method described above.

We can easily verify the statement that in order for the compensating function $G_e(s)$ to be RC realizable, $G(w)$ must have only simple poles that lie between 0 and -1 on the negative real axis in the w -plane. The zeros of $G'(w)$ can lie at any place in the w -plane.

Similar to the r -transform method, the open-loop transfer function of the uncompensated system after being w -transformed is

$$G_o G_1(w) = \{(1 - x^{-1})^3 [G.(s)]\},$$

$$= \frac{2w + 31}{G_1(s)} \quad z = \frac{1+w}{1-w} \quad 1 + \frac{w}{s} = \frac{1+w}{1-w} \quad (8-39)$$

After compensation the w -transform of the open-loop system is

$$G_e(w) = G_h G_1(w)$$

$$= G_h G_c G_1(w)$$

$$= \frac{2w + 1}{s} \text{ from which we have } \frac{3[G.(s) G_1(s)]}{s}$$

$$= \frac{z}{3[G_e(s) G_1(s)]} \quad z = \frac{1+w}{1-w} \quad \therefore \frac{1+w}{1-w} = \frac{3[G_e(s) G_1(s)]}{s} \quad (8-40)$$

$$w + 1 [G: (w) G G_1 (w)] \quad (8-41)$$

2w

The function $G_e(s)G_1(s)/s$ is found from Eq. (8-41) by taking the partial fraction expansion of the right-hand side of the equation and then finding the corresponding pairs of functions in s . A table of transforms relating s , z , and w for some common transfer functions similar to those of Table 8-1 can be derived if necessary. As can be easily shown, the following partial fraction expansions of functions must be made corresponding to the number of poles of

$G_1(s)/s$ at $s = 0$:

See for example, N. Balabanian, Network Synthesis (Englewood Cliffs, N. J.: Prentice-Hall, Inc., 1958), Chap. 7.

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$G(s)/s$ has one pole at $s = 0$

expand $G(w)G_hG_1(w)$

$G_1(s)/s$ has two poles at $s = 0$

expand

$G_1(s)/s$ has three poles at $s = 0$ expand

$G_e(w)G_hG_1(w)$

$1-w$

$G_e(w)G_hG_1(w)$

$(1-w)(1+w)$

The remaining steps of the design procedure follow in exactly the same way as that described for the r -transform method.

80

60

40

Magnitude curve of G_hG , (w)

(uncompensated system)

20

Gain crossover

db 0

Magnitude curve of

$G_c(w)G_hG_1(w)$

(compensated system)

New gain

-20

crossover

-40

1

-600

-80

0.0094

1 0.46

0.001

0.01

0.1

의

0.0167

ww

0.2 0.33

1.0 3.05

0.76 1.27

10

0°

-90°

Phase shift

-180°

(compensated system)

Phase

margin 45°
 -270°
 -360°
 0.001
 0.01
Phase
margin 22°
Phase shift
(uncompensated system)
 0.1

ww
 1.0
Fig. 8-13 Bode diagram of the uncompensated and the compensated systems of Example 8-2. (The w-transformation is used).
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As an illustration, the design problem given in Example 8-2 is reworked using the w-transformation. Substituting z

into Eq. (8-23) yields
Go $G_1(w)$
 $= \epsilon 0.25$

$-\$
 $-\$
 $=$
 $(1+w)(1-w)$
(8-42)
 $(1-w)(1+0.784w)(10.326w)$
 $w(1+2.16w)(1+1.31w)$

The Bode plot for $G_0G_1(w)$ is shown in Fig. 8-13, and the plot in the Nichols chart is clearly identical to that of Fig. 8-11. The values of ww on the Nichols chart plot are obtained from the corresponding or values through the relation

$=$
 $w\omega$
 $1/wr$
(8-43)

As inspection of the Bode plot in Fig. 8-13 shows, a phase-lead model for $G(w)$ is not suitable for the compensation of the system, since it would simply shift the gain crossover to a higher frequency (ww) and thus reduce the phase margin still further. A phase-lag model for $G'(w)$ is more appropriate; that is,

$G_e(w)$
 $=$
 $1 + ATW$
 $1 + \tau w$
 $(a < 1)$
(8-44)

The selection of the values of a and τ simply follows the conventional method of designing phase-lag compensation.† The essence of phase-lag compensation is to move the gain crossover frequency to a lower frequency where the phase margin is as specified. For the same specifications given in Example 8-2 and from Fig. 8-13, we can see that a phase margin of 45° may be obtained if the gain crossover is moved from ww 0.33 to ww 0.2 (these values for ww correspond to w, 3.0 and w, 5 , respectively). In fact, if we substitute $r = 1/w$ into Eq. (8-29) and simplify, we have the satisfactory compensating model for $G'(w)$. Therefore,

$$G(w) = [G(r)]_{r=1/w}$$

$$\frac{1 + 60w}{1 + 107w} \quad (8-45)$$

The Bode plot of the compensated system function $G(w)$ $G_{10}G_1(w)$ is now shown in Fig. 8-13. Without doubt, the two compensated systems described by Figs. 8-11 and 8-13 are identical. Also, by use of Eq. (8-45) and $G(w)$ $G_0G_1(w)$, the same transfer function for $G_e(s)$ is obtained as in Eq. (8-37).

Thus, the r -transform $z = (r + 1)/(r - 1)$ and the w -transform $z = (1 + w)/(1 - w)$ may lead to the same (or similar) results for $G_e(s)$ in a given problem. Since the solution of $G_e(s)$ for a given design problem is not unique, independent use of the r and the w † B. C. Kuo, Automatic Control Systems (Englewood Cliffs, N. J.: Prentice-Hall, Inc., 1962), Chap. 9.

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transform methods is unlikely to lead to identical results for $G_e(s)$ as illustrated above. But for a fixed set of design specifications, the results obtained by the two methods should be equally valid. As a summary to this section, some pertinent facts about the continuous-data compensation of sampled-data systems using r or w transformation are summarized as follows:

(1) The r -transform method.

(a) For RC realizability of $G(s)$, $G_e(r)$ must have only simple poles to the left of the -1 point on the real axis in the r -plane. The zeros of $G(r)$ can lie anywhere in the r -plane.

(b) Usually, a phase-lead model for $G(r)$ is more desirable.

(c) In order to determine $G_e(s)$, a specific function $(1, 1/(r+1)$ or $1/(r+1)(r-1)$) must be multiplied to $G_e(r)$ $G_{10}G(r)$ corresponding to the number of poles of $G_1(s)/s$ at $s = 0$ before the partial fraction expansion is taken.

(2) The w -transform method.

(a) For RC realizability of $G_e(s)$, $G'(w)$ must have only simple poles that lie between 0 and 1 on the real axis in the w -plane. The zeros of $G(w)$ can lie anywhere in the w -plane.

(b) Usually, a phase-lag model for $G(w)$ is more desirable.

(c) In order to determine $G_e(s)$, a specific function $(1, 1/(1-w)$, $1/(1-w)(1+w)$) must be multiplied to $G'(w)$ $G_{10}G_1(w)$, corresponding to the number of poles of $G_1(s)/s$ at $s = 0$ before taking the partial fraction expansion.

8.4 FEEDBACK COMPENSATION WITH CONTINUOUS-DATA NETWORKS

Compensating networks for sampled-data control systems can also be introduced into feedback paths. In general, the compensators may be of the continuous-data or the digital-data types. However, only feedback compensation with continuous-data networks will be considered in this chapter.

Two simple forms of feedback-compensated systems with sampled-data are depicted in Fig. 8-14. In more complex situations of course compensations may be introduced in the forward path as well as the feedback paths.

As can be shown, the two feedback-compensated systems with

continuous-data networks in Fig. 8-14 may be represented by equivalent sampled-data systems with digital compensators located in the

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forward paths only. Therefore, designing the feedback compensations with continuous-data networks is essentially equivalent to finding the cascade-digital-compensators in the equivalent systems.

Let us consider the system shown in Fig. 8-14a. The z-transform of the output response is given by

$$\frac{C(z)}{R(z)} = \frac{G_o G_1(z)}{1 + G_H(z) + G_r G_1(z)} \quad (8-46)$$

Multiplying the numerator and the denominator of Eq. (8-46) by a function $G(z)$ yields

$$\frac{C(z)}{R(z)} = \frac{G_o G_1(z) G_e(z)}{G_e(z) + G_e(z) G_1 H(z) + G_e(z) G_m G_1(z)} \quad (8-47)$$

Now consider the sampled-data system with cascade digital compensation shown in Fig. 8-15. The z-transform of the system output is given by

$$\frac{C(z)}{R(z)} = \frac{G_e(z) G_o G_1(z)}{1 + G_e(z) G_o G_1(z)} \quad (8-48)$$

If the systems of Figs. 8-14a and 8-15 were to be equivalent, we would equate Eq. (8-47) to Eq. (8-48). After simplification we have

$$\begin{aligned} & \frac{R(s)}{R(s)} \\ & \frac{H(s)}{H(s)} \\ & \frac{G_o(s)}{G_o(s)} \\ & \frac{G_1(s)}{G_1(s)} \\ & \frac{-C(s)}{-C(s)} \\ & \frac{G_o(s)}{G_o(s)} \\ & \frac{G_1(s)}{G_1(s)} \\ & \frac{-C(s)}{-C(s)} \end{aligned}$$

(b)
Fig. 8-14 Sampled-data control systems with feedback compensation (continuous-data networks).

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$$\begin{aligned} & \frac{R(s)}{R(s)} \\ & \frac{G_c(s)}{G_c(s)} \\ & \frac{G_h(s)}{G_h(s)} \\ & \text{Digital} \\ & \text{compensator} \\ & \frac{G_1(s)}{G_1(s)} \\ & \frac{C(s)}{C(s)} \end{aligned}$$

Fig. 8-15 Sampled-data system with cascade digital compensation.

$$G_e(z) + G_e(z) G_o H(z) = 1$$

Solving for $G_o H(z)$ in the last equation, we have

$$\begin{aligned} & G_o H(z) \\ & - \\ & 1 - G_e(z) \\ & G_e(z) \end{aligned}$$

(8-49)

(8-50)

Taking into account the transfer function of the zero-order hold,
Eq. (8-50) leads to

$$3[H(s)]$$

=

1

-

z^{-1}

$$1 - G_e(z)$$

$$G_e(z)$$

(8-51)

The inverse transform (z to s) of Eq. (8-51) gives the transfer
function $H(s)$ of the feedback compensating network, implying that
once $G_e(s)$ is determined for the system in Fig. 8-15, the function
 $H(s)$ of the feedback compensated system in Fig. 8-14a is obtained
by use of Eq. (8-51).

However, there is still the physical realizability of $H(s)$ to be
considered. In other words, what are the restrictions on $G_e(z)$ so
that $H(s)$ is always a physically realizable transfer function?

Evidently, the realizability of $H(s)$ (and of $H(s)/s$) relies entirely
upon the physical realizability of the z -transform of $H(s)/s$, and vice
versa. For a transfer function to be realizable physically, the system
represented by the transfer function must not produce an output
prior to the application of an input. This statement is true for
continuous-data as well as sampled-data systems. For the output to
occur at or after the application of an input, the series expansion
of $3[H(s)/s]$ should have no positive powers in z . Since $3[H(s)/s]$
can usually be expressed in terms of the ratio of a power series in
 z^{-1} , the requirement for physical realizability is met with the follow-
ing form:

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- k

$$A_k z^{-k}$$

$$3 \sum_{k=0}^m [H(s)] z^{-k}$$

$k=0$

m

$$b_k z^{-k}$$

$k=0$

(8-52)

where $b_0 \neq 0$. Substituting Eq. (8-52) into Eq. (8-51) and solving
for $G_e(z)$, we have

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$$G_c(z)$$

=

m

m

$$\sum_{k=0}^m b_k z^{-k}$$

$k=0$

n

-

$$\sum_{k=0}^M b_k z^{-k} + (1 - z^{-2}) \sum_{k=0}^n a_k z^{-k}$$

$k=0$

M

$$\sum_{k=0}^M b_k z^{-k}$$

$k=0$

$k=0$

$k=0$

$$C_k z^{-k}$$

(8-53)

where $b_0 \neq 0$.

Therefore, the physical realizability condition of $H(s)$ requires that $G(z)$ be of the form of Eq. (8-53) with $b_0 \neq 0$.

If $H(s)$ is to be a realizable RC transfer function, then all the poles of $H(s)$ must be simple and must lie on the negative real axis of the s -plane, excluding the origin and infinity. The zeros of $H(s)$ can lie at any place in the s -plane. Therefore, $H(s)/s$ can be expanded into the following form by partial fraction expansion:

$$H(s) = \frac{A_0}{s} + \sum_{k=1}^m \frac{A_k}{s + s_k}$$

(8-54)

where A_0 and A_k are constants and where s_k ($k = 1, 2, 3, \dots, m$) are simple negative poles. The z -transform of $H(s)/s$ is therefore

$3[H(s)]$

$$= \frac{A_0}{1 - z^{-1}} + \sum_{k=1}^m \frac{A_k}{1 - e^{s_k T} z^{-1}}$$

-

e

(8-55)

which has simple poles inside the unit circle with only one pole at

$z = 1$

$z = 1$

In order to expand $3[H(s)/s]$ into the form of Eq. (8-55), $G(z)$ must have an equal number of zeros and poles as Eq. (8-51) implies. Furthermore, comparing Eqs. (8-52) and (8-53), we see that the poles of $[H(s)/s]$ are generated from the zeros of $G(z)$. Therefore, the RC realizability of $H(s)$ requires in addition that $G(z)$ be of the form of Eq. (8-53); $G(z)$ must have an equal number of zeros and poles; and the zeros of $G(z)$ must be simple and lie inside the unit circle in the z -plane. There are no restrictions on the location of the poles of $G(z)$, and therefore, an RC transfer function for $H(s)$ can be derived from even an unstable $G(z)$ that satisfies all the above mentioned requirements but has a pole or poles outside the unit circle.

To illustrate the realization of a digital compensator by the continuous-data feedback compensator of Fig. 8-14a, the following example is given.

Example 8-3

A digital compensator with transfer function $G(z)$ given below is shown

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in Fig. 8-16a. Determine the transfer function $H(s)$ of the equivalent continuous data feedback structure in Fig. 8-16b.

$$G(z) = \frac{(1 - 0.5z^{-1})(1 - 0.8z^{-1})}{(1 - 0.2z^{-1})(1 - 0.1z^{-1})}$$

-

(8-56)

Since $G(z)$ has an equal number of poles and zeros and since all the zeros are simple and lie inside the unit circle $|z| = 1$, $H(s)$ can be realized by an RC network.

Substituting Eq. (8-56) into Eq. (8-51) and simplifying, we get

$$\frac{H(s)}{s}$$

Expanding by partial fraction expansion, Eq. (8-57) becomes

$$\frac{-z^{-1} + 0.382}{z^2 - 2}$$

$$= \frac{1}{1 - z^{-1}} (1 - 0.2z^{-1})(1 - 0.1z^{-1})$$

$$= \frac{H(s)}{2.25} - 0.86$$

$$\frac{1}{s} - \frac{1}{s - 1} + \frac{1}{s - 0.2}$$

$$= \frac{3.11}{s} - \frac{0.12}{s - 1} - \frac{0.12}{s - 0.2}$$

The pairs of transforms corresponding to the right-hand side of the last equation are found from the z-transform table. Thus

$$\frac{H(s)}{s} = -0.86$$

$$= \frac{1}{s} - \frac{1}{s - 1} + \frac{1}{s - 0.2}$$

$$\frac{H(s)}{s} = \frac{3.11}{s} - \frac{3.53}{s + 1.61} + \frac{3.18}{s + 2.3}$$

The negative sign for $H(s)$ means that positive feedback should be used in the system in Fig. 8-16b. The sampled-data system with feedback compensation given in Fig. 8-14b can also be approximated by the cascade-digital compensation configuration of Fig. 8-15. Evidently, for the two systems in Figs. 8-14b and 8-15 to be equivalent, the following relationship must be true:

$$G_c(s) = G_c(z)$$

$$\frac{H(s)}{G_n(s)}$$

Fig. 8-16 Digital compensator and equivalent feedback compensator with continuous-data network.

$$H(s)G_1(s)$$

$$\frac{1}{1 - G(z)}$$

$$G(z)$$

(8-61)
Since the left-side of Eq. (8-61) contains $G_1(s)$ as well as $H(s)$, the requirements on $G(z)$ for an RC realizable $H(s)$ are necessarily more complicated than those listed in the previous case. However, for $3[H(s)G(z)/s]$ to be physically realizable, $G(z)$ must have an equal number of poles and zeros. Also, the simple poles of $H(s)$ in the left-half of the s -plane must be generated by the simple zeros of $G(z)$ inside the unit circle $|z| = 1$. In addition to these restrictions, $G(z)$ must contain additional zeros that correspond to the poles of $G_1(s)$.

Usually, the function $H(s)$ obtained from Eq. (8-61) may contain more zeros than poles, but remote poles may be added to make $H(s)$ RC realizable.

Example 8-4

Consider that the controlled process of the sampled-data control system in Fig. 8-15 has the transfer function

$$G_1(s) = \frac{1}{s(s + 0.694)}$$

(8-62)
The sampling time is one second. The system is considered to be satisfactorily compensated by a cascade-digital compensator (Fig. 8-15) with transfer function

$$G_c(z) = \frac{(1 - z^{-1})(1 - 0.52z^{-1})}{(10.12z^{-1})}$$

Note that $G(z)$ contains zeros at $z =$
(8-63)

1 and $z = 0.5$ which correspond to the poles of $G_1(s)$ at $s = 0$ and $s = -0.694$, respectively. We assume that the design of the digital compensation is flexible enough so that the inclusion of these two zeros in $G(z)$ is permissible.

Substituting Eq. (8-63) into Eq. (8-61) and simplifying yields

$$H(s)G_1(s) = \frac{1.62z^{-1} - 0.82z^{-2} + 0.12z^{-3}}{(1 - z^{-1})^2(1 - 0.2z^{-1})(1 - 0.5z^{-1})}$$

Expansion of Eq. (8-64) by partial fraction expansion yields

$$H(s)G_1(s) = \frac{2.252z^{-1}}{(8-64)}$$

$$= \frac{1}{(1 - z^{-1})^2}$$

$$= \frac{1.31}{1 - z^{-1}} + \frac{0.02}{1 - 0.5z^{-1}}$$

$$+ \frac{2z^{-1}}{1 - 0.2z^{-1}} + \frac{1 - 0.52z^{-1}}{(8-65)}$$

The pairs of Laplace transforms corresponding to the right-hand side of the last equation are found from the z -transform table. Thus

$H(s)G(s)$ 2.25

1.31

=
2
S
S
=

0.02

+

1.33

$s + 1.61$ $s + 0.694$

$1.38s^2 + 4.7s + 2.5$

$s^2(s + 0.694)(s + 1.61)$

Solving for $H(s)$ from Eq. (8-66), we have

(8-66)

$H(s)$

$1.38(s + 2.74)(s + 0.66)$

-

$(s + 1.61)$

In order that $H(s)$ is RC realizable, a remote pole at say $s =$
added to $H(s)$. Therefore,

(8-67)

50, must be

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$H(s)$

=

$1.38(s + 2.74)(s + 0.66)$

$(s + 1.61)(s + 50)$

(8-68)

This section has outlined methods for using feedback compensation of sampled-data systems with continuous-data networks. In designing the feedback compensating network, an equivalent cascade-digital compensation model may be used. Since the transfer function of the digital compensator $G(z)$ is isolated from the transfer function of the controlled process $G_i(z)$ (see Eq. (8-48)), various conventional design techniques such as the root locus and the Bode diagram methods can be used immediately, without approximation in designing $G(z)$. However, detailed discussions and illustrations of the design of digital compensations will be given in the next chapter. Note that the feedback structure depicted in Fig. 8-16 may also serve as a realization of a digital compensator utilizing RC networks.

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9

DESIGN AND SYNTHESIS BY DIGITAL CONTROLLERS

9.1 INTRODUCTION

In the preceding chapter, compensation of sampled-data control systems is accomplished by the insertion of continuous-data networks in the forward or the feedback paths of the systems. Since the continuous-data network is connected directly to the controlled process, the z-transform of the product of the two transfer functions is a single function. Therefore, analytically it is difficult to evaluate the effects of various continuous networks on the over-all performance of the system. Usually, in order to reduce the design effort, approximation methods such as the time-delay approximation described in the last chapter have to be used. A design method using the bilinear r-transformation or the w-transformation was introduced. The method has the advantage of giving exact solutions without using approximations.

Alternately, compensation of a sampled-data system can also be accomplished by introducing pulsed-data networks, or digital controllers into the system. The process, therefore, is referred to as the digital compensation of sampled-data systems.

Although continuous-data networks are simple to implement and economical to build, the digital controller is in general more versatile, and thus better system performance can be achieved by means of digital compensation. Furthermore, digital compensation may be

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$R(s)$

$R(s)$

$R(s)$

$E(s)$

$D(s)$

$G_0(s)$

$G_1(s)$

$C(s)$

to

$D^*(s)$

Zero-order Controlled

hold

process

Digital controller

(a)

$D(s)$

Digital controller

$G_0(s)$

(b)

$G_1(s)$

Controlled

process

$C(s)$

$C(s)$

$G_1(s)$

$H(s)$
 $G_h(s)$
 $D(s)$
(c)

Fig. 9-1 Control systems with digital compensations. (a) Cascade-digital compensation. (b) Bypass digital compensation. (c) Feed-back digital compensation. applied to continuous-data systems as well as sampled-data systems. Several possible configurations of the digital compensation of sampled-data and continuous-data control systems are depicted in Fig. 9-1.

Figure 9-1a shows that a digital controller is introduced in the forward path of a feedback control system. The system could be considered as a sampled-data system or a continuous-data system with digital compensation. If the uncompensated system is a sampled-data system, $D(s)$ and the sampler following $D(s)$ represent the digital compensator unit. If the uncompensated system is considered a continuous-data system, $D(s)$, the two synchronized switches and the zero-order hold $G_h(s)$ represent the digital compensator unit.

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The system shown in Fig. 9-1b can be considered as a continuous-data system with bypass digital compensation. The actuating signal $E(s)$ to the controlled process is equal to the sum of the continuous error $E(s)$ and the sampled error $E_2(s)$. Figure 9-1c illustrates the insertion of a digital controller in the feedback path of a continuous-data system.

The conventional design techniques utilizing such tools as the Nyquist plot, the Bode diagram, and the root locus plot, can be applied readily to the design of digital compensation. Let us investigate the sampled-data system shown in Fig. 9-1a. The closed-loop z-transfer function of the system is written

$$C(z) D(z) G_h G_1(z) = \frac{R(z)}{1 + D(z) G_h G_1(z)} \quad (9-1)$$

In Eq. (9-1) since $D(z) G_h G_1(z)$ is the product of the pulse transfer function $D(z)$ of the digital controller and the z-transform of $G_h G_1(s)$, the variation of $D(z)$ can be controlled independently. Therefore, in terms of the frequency loci, the reshaping of the locus of $G_h G_1(z)$ by use of various forms of $D(z)$ can be studied in a straightforward manner, and the selection of the desired $D(z)$ is easily accomplished.

Likewise, selection of the desired $D(z)$ can be effected by observing the effect of the adjustment and variation of the poles and zeros of $D(z)$ upon the root locus of the characteristic equation. Under certain circumstances, zeros may be introduced to $D(z)$ to cancel the unwanted poles of $G_h G_1(z)$, and new poles of $D(z)$ are added at appropriate places to improve the system performance. The first part of this chapter is devoted to the use of conventional techniques for the design of digital compensation of feedback control systems with an emphasis on the Bode plot and the root locus methods.

9.2 THE DIGITAL CONTROLLER

Before entering into the design of digital compensation, we shall discuss the construction and the realizability of the digital controller, the fundamental ingredient of a control system with digital compensation. The block diagram of a digital controller is shown in Fig. 9-2.

With reference to Fig. 9-2, the transfer function of the digital

controller is defined as

$$\frac{E_2(z)}{D(z)} = \frac{E_1(z)}{(9-2)}$$

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X.

$$e_1(t) \quad e^*(1)$$

$$E_1(s)$$

$$e_2(1)$$

$$e(1)$$

$$D(s)$$

$$TE^*(s)$$

$$E_2(s)$$

$$E_2(s)$$

$$D(z)$$

Digital controller

Fig. 9-2 The block diagram of a digital controller.

The input $e(t)$ to the digital controller is in the form of discrete samples of $e(t)$. The digital controller performs certain linear operations on the input samples and delivers a modified pulse sequence $e^*(t)$ to the output.

There are many ways of implementing a digital controller. A digital controller may contain a passive network preceded and followed by samplers. The digital controller may also be a special-purpose digital computer or a general-purpose computer. This approach enables one to realize certain functional operations that are not obtainable with passive elements.

The a priori requirement in the design of the digital controller $D(z)$ is that it be physically realizable. The effort would be worthless if the design results in a transfer function for $D(z)$ which cannot be realized by physical elements, passive or active.

The condition of physical realizability of any system implies that no output signal of the system will appear before an input signal is applied. If the transfer function $D(z)$ of the digital controller is expressed as a ratio of two polynomials in z

$$D(z) = \frac{a_n z^n + a_{n-1} z^{n-1} + \dots + a_0}{b_m z^m + b_{m-1} z^{m-1} + \dots + b_0}$$

$$+ a_0$$

$$+ b_0$$

$$(9-3)$$

Expanding $D(z)$ in a power series in z^{-1} , the coefficients of the series represent the values of the weighting sequence of the digital controller. The coefficient of the z^0 term corresponds to the value of the weighting sequence at $t = kT$, where k is an integer. Clearly, for the digital controller to be physically realizable, the power series expansion of $D(z)$ must not contain any positive power in z . Any positive power in z in the series expansion of $D(z)$ simply indicates "prediction" or simply that the output signal precedes the input signal. Therefore, for $D(z)$ to be physically realizable, the highest power of the denominator must be equal to or greater than that of the numerator in Eq. (9-3), or simply $m \geq n$.

Quite often $D(z)$ has equal number of poles and zeros and is written as

$$D(z)$$

$$=$$

$$-1$$

$$a_0 + a_1 z^{-1} +$$

$$b_0 + b_1 z^{-1} +$$

$$-1$$

$$\cdot$$

$$+anz$$

$$+bmz^1$$

$$n$$

$$(9-4)$$

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where n and m are any positive integers. In this case, the denominator must not contain any factor of z' . Therefore, the condition for $D(z)$ to be realizable is that $b_0 \neq 0$ in Eq. (9-4).

The realization of the transfer function $D(z)$ of a digital controller by physical networks or devices is quite flexible and may take many forms. The most common measures are the realization by pulsed-data RC networks and digital computers.

1. Realization of $D(z)$ by Pulsed-data RC Networks

The use of a pulsed-data RC network to realize the transfer function $D(z)$ of a digital controller follows directly from the discussion given in Sec. 8-4. As was shown in Sec. 8-4, the design of a feedback compensation with continuous-data networks can be carried out in an equivalent system with series digital compensation. In essence, the two systems shown in Fig. 8-16 are equivalent to each other if the transfer functions are related by

$$D(z)$$

$$=$$

$$1 + G_0 H(z)$$

$$(9-5)$$

The problem in this section of course involves the determination of $H(s)$, the transfer function of the feedback pulsed-data network, once $D(z)$ is given.

A digital controller can also be realized with a series pulsed-data network consisting of a zero-order hold and an RC network preceded and followed by a sampler. In more complicated situations a combination of the series and the feedback pulsed-data configurations may lead to the realization of more complex digital controllers.

These three possibilities of realizing a digital controller are identified and discussed below as the series pulsed-data networks, the feedback pulsed-data networks, and the series-feedback pulsed-data networks.

Series Pulsed-data Networks. The series pulsed-data network which is used for the realization of the pulse transfer function of a digital controller is depicted in Fig. 9-3. The transfer functions of the two systems in Fig. 9-3 are related by

$$D(z) = G_0 G_c(z)$$

$$(9-6)$$

Since the transfer function of the zero-order hold is $(1 - e^{-T})/s$, we obtained from Eq. (9-6)

$$D(z)$$

$$=$$

$$\frac{1}{z} [G_c(s)]$$

$$(9-7)$$

$$D(z)$$

$$=$$

$$\frac{1}{z} [G_c(s)]$$

$$(9-7)$$

J. Sklansky, "Pulsed RC Networks for Sampled-data Systems," I. R. E., Convention Record, Part 2, 1956, pp. 81-99.

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Thus, given a specific $D(z)$, the transfer function $G_c(s)$ of the

pulse-data network can be determined from Eq. (9-7) by taking the inverse (z to s) transform.

If $G_e(s)$ is to be an RC realizable transfer function, all the poles of $G_e(s)$ must be simple and lie on the negative real axis of the s -plane with the exception of the origin and infinity. The zeros of $G_e(s)$ can be located anywhere in the s -plane. Therefore, $G_e(s)/s$ can be expanded into the following form by partial fraction expansion:

$$\frac{G_e(s)}{s} = \frac{A_0}{s} + \sum_{k=1}^m \frac{A_k}{s + s_k}$$

where A_0 and A_k are constants and s_k ($k = 1, 2, 3, \dots$) are simple negative real poles. The z -transform of Eq. (9-3) is

$$\frac{G_e(s)}{s} = \frac{A_0}{s} + \sum_{k=1}^m \frac{A_k}{s + s_k}$$

$$\frac{G_e(s)}{s} = \frac{A_0}{s} + \sum_{k=1}^m \frac{A_k}{s + s_k}$$

(9-8)
m) are

(9-9)
which has simple positive real poles inside the unit circle $|z| = 1$, with only one pole at $z = 1$. Comparing Eq. (9-9) with Eq. (9-7), we see that in order for $G_e(s)$ to represent an RC network, the pulse transfer function $D(z)$ must have the following properties:

- (i) The number of poles of $D(z)$ must be equal to or greater than the number of zeros of $D(z)$.
- (ii) The zeros of $D(z)$ are arbitrary in location.
- (iii) The poles of $D(z)$ must be simple, real and positive, and lie inside the unit circle $|z| = 1$ in the z -plane.

As an illustration, an example is presented below.

$e(1)$
 $e^*(1)$
 $e_z(1)$
 $e(1)$
 $D(s)$
 $G_{ro}(s)$
 Zero-order
 hold
 $\cdot D(z) \cdot$
 (a)
 $e,(t)$
 $e(1)$
 $G_{ho}(s)$
 $G_c(s)$
 $e_2(1) e_z(1)$
 $G_{no}(s)$
 T
 Zero-order Passive
 Zero-order

hold
hold
network
·D(z).

(b)

Fig. 9-3 Realization of pulse transfer function of the digital controller by a series pulse-data network. (a) Digital controller. (b) Equivalent series pulse-data network.

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Example 9-1

Given the pulse transfer function of a digital controller

D(z)

=

$$\frac{1 - 0.52z^{-1}}{1 - 0.2z^{-1}}$$

7

2 (9-10)

The problem is to realize D(z) by the series pulsed-data network of Fig. 9-3b. Since D(z) meets the three requirements of RC realizability listed above, it can be realized by the series pulse-data network. Substitution of Eq. (9-10) into Eq. (9-7) yields

(10.52-1)

$$(1 - 0.52z^{-1}) (1 - 0.2z^{-1})$$

Expanding the last equation into partial fractions, we have

$$\frac{1}{s} [G_1(s)] = (1$$

$$g[2] =$$

-

$$0.625$$

1 2

$$0.375$$

+

$$\frac{1}{s} - 0.2z^{-1}$$

(9-11)

(9-12)

The corresponding Laplace transforms of the right side of Eq. (9-12) are found from the z-transform table. Thus

and

$$G_e(s)$$

S

=

$$0.625$$

+

S

$$0.375$$

$$\frac{s + 1.61}{s + 1}$$

S+ 1

=

$$\frac{s(s + 1.61)}{s(s + 1)}$$

$$G_e(s)$$

=

$$\frac{s+1}{s+1.61}$$

$$\frac{s+1}{s+1.61}$$

(9-13)

(9-14)

An RC network realizing $G_e(s)$ is shown in Fig. 9-4a, and the pulse-data network for the realization of D(z) is given in Fig. 9-4b.

(b) Feedback Pulse-data Networks. As was shown in Sec. 8-4, the feedback structure with zero-order hold and the RC network shown in Fig. 9-5b is equivalent to the digital controller of Fig. 9-5a. The

$R_1 = 10^2$
 $R_2 = 1.64 \times 10^5$
 $C = 10 \mu\text{f}$

(a)
 $e(t)$
 $e(1)$
 $G(s)$
 $G(s) (1) A$
 $e(1)$

Zero-order
 hold

RC network

$D(z)$

(b)

Fig. 9-4 Pulse-data network realization of $D(z)$

RC network realizing $G(s)$.

1
 $0.5z^{-1}$

=

1

$0.2z^{-1}$ (a)

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$e(t)$

$D(s)$

$G(s)$

$e_2(1)$

Zero-order

hold

$-D(z)$.

(a)

$e(t)$

$G(s)$

$e_2(t)$

Zero-order

hold

$H(s)$

RC network

(b)

Fig. 9-5 Realization of pulse transfer function of the digital controller by a feedback pulse-data network. (a) Digital controller.

(b) Equivalent feedback pulse-data network.

transfer functions of the two systems are related by

or

=

1

$1 + G(s) H(z)$

$D(z)$

$3[H(s)] =$

1

-1

$-$

z

$[1, D(z)]$

$D(z)$

1

$1 -$

(9-15)

(9-16)

In order to realize $H(s)$ by an RC network, the pulse transfer

function $D(z)$ must have the following properties:

(i) $D(z)$ must have the same number of poles and zeros.

(ii) The poles of $D(z)$ are arbitrary.

(iii) The zeros of $D(z)$ must be simple, real, positive, and lie inside the unit circle of the z -plane.

The justifications to these restrictions have been discussed in detail in Sec. 8.4 and therefore are not repeated here.

Note that in the feedback structure, restrictions are placed on the location of the zeros rather than on the poles of $D(z)$. In fact, $H(s)$ is RC realizable even if $D(z)$ has poles outside the unit circle. This characteristic of the feedback structure is not unexpected since it is well known that any feedback system, even one with only passive elements, may be stable or unstable under various operating conditions.

An example illustrating the realization of a pulse-transfer function by the feedback pulse-data network is given in Example 8-3.

(c) Series-feedback Pulse-data Networks. Both the series network and the feedback pulse-data network discussed above contain restrictions in the realization of pulse transfer functions. In the series structure the zeros of $D(z)$ are arbitrary, whereas in the feedback

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structure the poles of $D(z)$ are arbitrary. This implies that certain combinations of the series and the feedback configurations may relieve all the restrictions on the poles and zeros $D(z)$ with the exception of course that $D(z)$ must be physically realizable. Many such combinations are conceivable, but let us consider the simple feedback-series pulse-data network shown in Fig. 9-6. Sklansky has

$e, (t)$
 $D(s)$
 $ez (1)$
 $e, (1)$
 $H(s)$
 $\cdot D(z)-$
 (a)
 $G_{no}(s)$
 $G_c(s)$
 $ez (1)$
 Zero-order
 hold
 RC network
 T
 RC network

(b)
 Fig. 9-6 Realization of pulse transfer function of the digital controller by a feedback-series pulse-data network. (a) Digital controller. (b) Equivalent feedback-series pulse-data network.

shown that any physically realizable pulse transfer function may be realized by the pulsed-data network shown in Fig. 9-6b, where $H(s)$ and $G_c(s)$ represent RC networks.

The over-all z -transfer function of the feedback-series structure is given by

$$E_2(z) = \frac{GoGe(z)}{E_1(z) + GoH(z)} \quad (9-17)$$

If the two systems in Fig. 9-6 are equivalent to each other, we have

$$E_2(z) = E_1(z)$$

$$= D(x)$$

$$= \frac{G_{ho} G_c(z)}{1 + G_H(z)}$$

$$(9-18)$$

If the pulse transfer function $D(z)$ has complex conjugate poles and zeros and poles and zeros that are on or outside the unit circle, $D(z)$ can be realized neither by the series structure nor by the feed-back structure of the pulse-data network with RC elements. Under this condition we can write $D(z)$ as the product of two pulse transfer functions $D_1(z)$ and $D_2(z)$; that is,

+ Sklansky, op. cit.

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< where
and
pals complex allstate Roks.
 $D(z)$

$$= D_1(z)$$

$$= D_s(z) D_c(z)$$

$$\frac{1}{1 + G_{ho} H(z)}$$

$$D_2(z) G_{ho} G_c(z)$$

=
つ
(9-19) 1
(9-20)
(9-21)
 $D_r(z)$ and $D_c(z)$ are identical to the transfer functions of the feedback pulse-data network and the series pulse-data network in Eq. (9-15) and (9-6), respectively. Therefore, $D(z)$ must have all the properties of a pulse transfer function that can be realized by the feedback pulse-data network with RC elements. This means that $D_r(z)$ should contain all the poles of $D(z)$ that are complex and all the unstable poles. The zeros of $D(z)$ must be simple, real, and positive, and lie inside the unit circle. The number of poles and zeros of $D(z)$ must be equal. Similarly, $D_c(z)$ must satisfy all the requirements in order to be realized by a series pulse-data network with RC elements. Thus $D_c(z)$ should contain all the zeros of $D(z)$ that are complex or on or outside the unit circle. Since it is assumed that $D(z)$ is physically realizable and that $D_c(z)$ has an equal number of poles and zeros, $D_r(z)$ certainly does not have more zeros than poles.

To illustrate the formulations of $D_r(z)$ and $D_c(z)$ from a specific pulse transfer function $D(z)$, let us consider the realization of the following pulse transfer function by a pulse-data network:

$$D(z) = \frac{\text{realizable zeros}}{\text{unrealizable zeros}}$$

$$= \frac{(1 - 0.5z^{-1})(1 - 0.6z^{-1})(1 - 0.8z^{-1})(1 + 0.2z^{-1})(1 - z^{-1} + z^{-2})}{(1 - 0.2z^{-1})(1 - 0.37z^{-1})(1 - 0.9z^{-1})(1 - z^{-1})(1 - 0.52z^{-1} + z^{-2})}$$

+
-

-
 - realizable poles
 $2 = 0 - 2$
 $7 = 0.37$
 z
 unrealizable poles
 1
 (9-22)
 where the zeros that are real, positive, and simple, and lie inside the unit circle are referred to as the realizable zeros and the complex or non-positive (including $2 = 0$) or multiple-order or unstable (including those on the unit circle) zeros are referred to as the unrealizable zeros. Same references are made for the poles of $D(z)$. The pulse transfer functions $D_1(z)$ and $D_s(z)$ are formulated as follows:
 $D_1(z) =$
 -
 $D_s(z) =$
 -
 Realizable zeros of $D(z)$
 Unrealizable poles of $D(z)$
 $(1 + 0.52z^{-1})(1 - 0.621z^{-1})(1 - 0.82z^{-1})$
 1
 -
 $(1 - z^{-1})(1 - 0.5z^{-1} + z^{-2})$
 Unrealizable
 zeros of $D(z)$
 Realizable poles of $D(z)$
 1
 $(1 + 0.221z^{-1})(1 - z^{-1} + z^{-2})$
 -
 (9-23)
 0
 -
 $(1 - 0.2x^{-1})(1 - 0.37z^{-1})(1 - 0.9z^{-1})$
 (9-24)

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$D_1(z)$ satisfies the conditions of realizability by a feedback pulse-data network with RC elements, and $D_s(z)$ is realizable by a series pulse-data network with RC elements. Cascading the two pulse-data networks, we have the feedback-series structure shown in Fig. 9-6. However, the example given above is fabricated in such a manner that the pulse transfer functions $D_1(z)$ and $D_s(z)$ that are formulated from $D(z)$ by properly assigning the realizable and the unrealizable poles and zeros of $D(z)$ to the two functions are directly realizable by the feedback and the series structure, respectively. Careful inspection of the pulse transfer function given in Eq. (9-22) reveals that $D_1(z)$ and $D_s(z)$ may be directly realizable by their corresponding pulse-data networks only if the following conditions are met:

- (i) The number of realizable poles of $D(z)$ = the number of unrealizable zeros of $D(z)$.
- (ii) The number of realizable zeros of $D(z)$ = the number of unrealizable poles of $D(z)$.

As another illustrative example let us consider the following pulse transfer function:

realizable
 zero

+

$$\begin{aligned}
 & -1 \\
 & + \\
 & (9-25) \\
 & (1 - 0.5z^{-1})(1 + 0.2z^{-1})(1 - z^{-1} + z^{-2}) \\
 & \text{unrealizable zeros} \\
 & D(z) \\
 & = \\
 & (1 - 0.2z^{-1})(1
 \end{aligned}$$

$$\begin{aligned}
 & - \\
 & - \\
 & z^{-1})(1 - 0.5z^{-1} + z^{-2}) \\
 & K \\
 & \text{realizable} \\
 & \rightarrow \\
 & \text{pole} \\
 & \text{unrealizable poles}
 \end{aligned}$$

-1
 Evidently, $D(z)$ is physically realizable, although it represents an unstable system. According to the rules given in this section, the poles and zeros of $D(z)$ are assigned to $D_1(z)$ and $D_2(z)$ as follows:

$$\begin{aligned}
 & D_1(z) = \\
 & \text{Realizable zeros of } D(z) \\
 & \text{Unrealizable poles of } D(z) \\
 & = \\
 & 1
 \end{aligned}$$

$$\begin{aligned}
 & - \\
 & D_2(z) = \\
 & \text{Unrealizable zeros of } D(z) \\
 & \text{Realizable poles of } D(z)
 \end{aligned}$$

$$\begin{aligned}
 & = \\
 & z \\
 & (1 - 0.5z^{-1}) \\
 & z^{-1})(1 - 0.5z^{-1} + z^{-2})
 \end{aligned}$$

$$\begin{aligned}
 & -1 \\
 & (9-26) \\
 & (1 + 0.2z^{-1})(1 - z^{-1} + z^{-2}) \\
 & (10.2z^{-1}) \\
 & (9-27)
 \end{aligned}$$

Since $D_1(z)$ of Eq. (9-26) has two zeros at $z = 0$ that must be referred to as unrealizable zeros, it cannot be realized by a feedback pulsed-data network. Similarly, the $D_2(z)$ in Eq. (9-27) has two poles at $z = 0$ and is not realizable by a feedback structure. This difficulty apparently stems from the fact that $D(z)$ has one realizable zero and three unrealizable poles and one realizable pole but three unrealizable zeros. Under this condition the numerator and the denominator of $D(z)$ must be multiplied by $(1 - az^{-1})(1 - bz^{-1})$, where a and b are real numbers, $0 < a < 1$, $0 < b < 1$, and $a \neq b$;

$$\begin{aligned}
 & - \\
 & - \\
 & (1 - z^{-1})(1 - 0.5z^{-1} + z^{-2}) \\
 & - \\
 & D_s(z) \\
 & (10.52)(1 - az^{-1})(1 - bz^{-1}) \\
 & - \\
 & =
 \end{aligned}$$

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also, a and b are not equal to any other pole and zero of $D(z)$. Then, $D_1(z)$ and $D_2(z)$ may now be written as

$$D_1(z) = \frac{-1}{(1+0.22)(1 - z^{-1} + z^{-2})} \approx (1 - 0.2z^{-1})(1 - az^{-1})(1 - bz^{-1})$$

(9-28)
(9-29)

Now we see that $D_1(z)$ can be realized by a feedback pulse-data network and $D_2(z)$ can be realized by a series pulsed-data network. The over-all pulse transfer function $D(z)$ may then be realized by the feedback-series structure of Fig. 9-6b.

2. Realization of $D(z)$ by Digital Computers

The implementation of a pulse transfer function $D(z)$ can be accomplished with a general-purpose or a special-purpose digital computer; the latter is sometimes also referred to as a pulsed-data processing unit. Because of the advantages in computing speed, storage capacity, and flexibility, the use of digital computers in control systems has become increasingly important. Furthermore, due to the discrete property of the signals in a digital controller, the realization of a pulse transfer function by a digital computer seems to be quite natural.

In general, a pulse transfer function $D(z)$ can be realized through a program of a general-purpose digital computer. The following three methods of programming have been discussed in the literature,[†] direct programming, cascade programming, and parallel programming. In the pulse-data processing unit method, a special-purpose digital computer may be built which is capable of performing specific arithmetic manipulations of multiplication, storage, addition, and subtraction, etc. Digital and analog devices may also be combined in a pulse-data processing unit. For instance, the storage operation may be implemented with either analog or digital devices. If digital elements are used, the stored samples could be retained in coded forms on registers, magnetic tapes, or drums. If analog elements are used, data could be stored in the form of some analog quantities such as current, voltage, or mechanical displacement.

The above mentioned methods of implementing a pulse transfer function are discussed in the following paragraphs.

[†] A. R. Bergen, J. R. Ragazzini, "Sampled-data Processing Techniques For Feedback Control Systems," Trans. A. I. E. E., Part 2, Nov. 1954, pp. 236-244.

[‡] J. M. Salzer, "Frequency Analysis of Digital Computers Operating in Real Time, Proc. I. R. E., Vol. 42, No. 2, 1954, pp. 457-466.

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(a) Direct Programming. A physically realizable pulse transfer function $D(z)$ of a digital controller may be expressed in the general form

$$D(z) = \frac{E_2(z)}{E_1(z)} = \frac{a_0 + a_1z^{-1} + a_2z^{-2} + \dots + a_nz^{-n}}{b_0 + b_1z^{-1} + b_2z^{-2} + \dots + b_mz^{-m}}$$

(9-30)

where b_0, m and n are positive integers, and $E(z)$ and $E_1(z)$ represent the z -transforms of the output and the input signal of the digital controller, respectively. Cross-multiplying in Eq. (9-30) and taking the inverse z -transform yields

$$\sum_{k=1}^m b_{k-1} e(t) + z \sum_{k=1}^m b_{k-1} e(t - kT) = z \sum_{k=1}^m a_{k-1} e(t - kT)$$

Solving for $e(t)$ from the last equation yields

$$e(t) = \frac{1}{b_0} \left[\sum_{k=1}^m a_{k-1} e(t - kT) - \sum_{k=1}^m b_{k-1} e(t - kT) \right]$$

$$(9-31)$$

$$(9-32)$$

The present value of the output $e^*(t)$ depends on the present and the past information of the input and also on the past information of the input signal as well. To implement the digital computer program, two basic mathematical operations are required. The first operation is the data storage. Past samples on the output and the input signal must be stored in order that they may be used in the computation of the output sample $e(t)$. The second operation involves arithmetic manipulations including multiplication of the stored output and the input samples by constants $b_0, \dots, b_{m-1}$, addition and subtraction of samples.

and
If we let

$$x^*(t) = \sum_{k=0}^m b_k e(t - kT)$$

$$y^*(t) = \sum_{k=0}^n b_k e(t - kT)$$

$$e(t) = x^*(t) - y^*(t)$$

The block diagram representation of the direct digital programming to obtain $x^*(t)$ from $e^*(t)$ is shown in Fig. 9-7a, and the direct digital program for the implementation of the relationship between $e(t)$ and $y^*(t)$ is depicted in Fig. 9-7b. The rectangular blocks in the figures indicate the time delays which represent the storage operation. The weights, $a_0/b_0, a_1/b_0, \dots, a_m/b_0$, and $b_1/b_0, b_2/b_0, \dots, b_n/b_0$, represent the weighting which are applied to the various input and output samples. The total sampled output $e^*(t)$ is obtained by subtracting $y^*(t)$ from $x^*(t)$. Thus, the block diagram of the digital program which gives the realization of $D(z)$ is shown in Fig. 9-7c.

T
 $e^{(1)}$
 $e^{*}(1-T)$
 Storage
 T
 Storage
 T
 $e^{(1-2T)}$
 $e^{*}(1-mT)$
 Storage
 00
 bo
 (1)
 (1)
 (a)
 $1b^3$
 b^2
 bo
 bo
 ez
 (1)
 Storage
 T
 Storage
 Storage
 Storage
 $e^{(1-T)}$
 $e^{(1-27)}$
 T
 T
 T
 (b)
 Input
 $e_1^{(1)}$
 Program for $x^{*}(t)$
 $ex(t)$
 Program for $y^{*}(t)$
 $fy^{*}(1)$

am
 bo
 815

(c)

Fig. 9-7 Block diagram for direct digital programming of $D(z)$.

(a) Direct digital programming for $x^{*}(t)$ of Eq. (9-33). (b)

Direct digital programming for $y^{*}(t)$ of Eq. (9-34).

(b) Cascade Programming. A complex pulse transfer function.

$D(z)$ may be written as the product of a number of simple pulse transfer functions each realizable by a simple digital program. The digital programming of $D(z)$ may then be represented by the series of cascaded digital programs of the simple pulse transfer functions. Writing Eq. (9-30) in factored form, we have

$y^{*}(1)$

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$D(z) =$

$E_2(z)$

$E_1(z)$

$=$

m

$$a_0 \prod_{k=1}^m (1 + c_k z^{-1})$$

k=1

72

$$b_0 \prod_{k=1}^n (1 + d_k z^{-1})$$

k=1

(9-36)

where c and d are the zeros and the poles of D(z), respectively. For n>m, D(z) is written as the product of n simple pulse transfer functions

$$D(z) = \prod_{k=1}^n D_k(z)$$

k=1

(9-37)

where

$$D_k(z)$$

$$a_k (1 + c_{1k} z^{-1})$$

-

$$(k = 1)$$

(9-38)

$$b_k (1 + d_{1k} z^{-1})$$

$$D_k(z)$$

$$1 + c_{k2} z^{-1}$$

-

$$(k$$

$$= 2, 3,$$

m)

(9-39)

$$1 + d_{k2} z^{-1}$$

1

and

$$D_k(z)$$

=

$$(k = m + 1,$$

n)

(9-40)

$$1 + d_{k2} z^{-1}$$

$$e^{(1)}$$

$$e^{(1)}$$

$$D(z)$$

(a)

$$e^{*(1)}$$

$$D_1(z)$$

$$D_2(z)$$

$$D_z(z)$$

$$D_a(z)$$

$$D_n(z)$$

$$e^{(1)}$$

(b)

Storage

T

Input

Storage

T

Ck

Output

(c)

Fig. 9-8 Cascade programming of pulse transfer function. (a)

Digital programming. (b) Cascaded digital programming. (c)

Digital program of $D(z) = (1 + c_1 z^{-1}) / (1 + d_1 z^{-1})$.

Thus the digital program of D(z) shown in Fig. 9-8a can be represented by the series connection of n simple digital programs as shown in Fig. 9-8b. The schematic diagram of the digital program of D(z), (k = 2, 3, . . ., m), is illustrated in Fig. 9-8c.

Parallel Programming. As an alternative method to cascade programming, a pulse transfer function may be realized by the parallel

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combination of a number of simple digital programs. This type of programming involves taking the partial fraction expansion of $D(z)$. Thus Eq. (9-30) is written.

$D(z)$

=

$\sum_{k=1}^n$

$\frac{A_k}{z - p_k}$

(9-41)

where it is assumed that $D(z)$ has only simple poles. (If $D(z)$ has multiple poles, the method of parallel programming is of course still applicable.) $D(z)$ may also be written as

when

$\sum_{k=1}^n$

$D(z) = \sum_{k=1}^n D_k(z)$

$D_k(z)$

=

$\sum_{k=1}^n$

$\frac{A_k}{z - p_k}$

$D_k(z) = \frac{A_k}{z - p_k}$

(9-42)

(9-43)

$D(z)$ can be realized by the n simple digital programs for $D_1(z)$, $D_2(z)$, . . . , $D_n(z)$, operating in parallel as shown in Fig. 9-9b. The pulse transfer function $D(z)$ is realized by the simple digital program depicted in Fig. 9-9c.

$D(z)$

$D_1(z)$

$D_2(z)$

$D_n(z)$

$D_1(z)$

$D_2(z)$

$D_n(z)$

$D_1(z)$

$D_2(z)$

$D_n(z)$

$D_1(z)$

$D_2(z)$

$D_n(z)$

$D_1(z)$

$D_2(z)$

$D_n(z)$

$D_1(z)$

$D_2(z)$

$D_n(z)$

$D_1(z)$

$D_2(z)$

$D_n(z)$

$D_1(z)$

$D_2(z)$

$D_n(z)$

$D_1(z)$

$D_2(z)$

$D_n(z)$

$D_1(z)$

$D_2(z)$

$D_n(z)$

Fig. 9-9 (a) Digital programming. (b) Parallel programming. (c)

Digital program of $D_1(z)$ b. $1 - dz^{-1}$

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(d) Pulsed-data Processing Units or Special-purpose Computers. In the method of realizing a pulse transfer function by digital programming, a program of a general-purpose digital computer is written so that when the machine reads this program, it will carry out the mathematical operations described by the specific pulse transfer function. When different pulse transfer functions are being realized,

the same digital computer may be used with variations in the programming. On the other hand, a special-purpose computer, sometimes also referred to as a pulsed-data processing unit, may be built specially for the realization of a specific pulse transfer function $D(z)$. However, a pulsed-data processing unit is good only for the realization of a certain pulse transfer function. Separate computers must be built to realize different pulse transfer functions.

The principle of realization of $D(z)$ by a pulsed-data processing unit is similar to that of digital programming described in the preceding sections. In essence the data processing unit should be capable of performing specific arithmetic manipulations such as multiplication, time delays, and addition and subtraction. In a data processing unit these arithmetic manipulations may be implemented by transmission lines for time delays, potentiometers and summing amplifiers for multiplication, weighting, and subtraction and addition. These operations need not be carried out by digital elements only. Sometimes digital as well as analog devices may be used in the implementation of a pulsed-data processing unit.

To illustrate the realization of a pulse transfer function by a pulsed-data processing unit, the following simple example is considered.

Example 9-2

Consider the pulse transfer function

$$D(z) =$$

$$\frac{(1 - 0.62z^{-1})}{(1 - 0.342z^{-1})}$$

$$= \frac{E_a(z)}{E_1(z)} \quad (9-44)$$

which evidently is physically realizable. Cross multiplying in Eq. (9-44) and taking the inverse z-transform yields

$$e(t) - 0.6 e(t - T) = e(t) - 0.34 e(t - T)$$

$$(9-45)$$

where T is the sampling period in seconds. Solving for $e^*(t)$ in the last equation, we have

$$e^*(t) = e(t) - 0.6 e(t - T) + 0.34 e(t - T) \quad (9-46)$$

The block diagram of the computer for the realization of Eq. (9-46) is shown in Fig. 9-10.

Note that the block diagram of the data processing unit in Fig. 9-10 is identical to that of the digital programming of the same $D(z)$. However, the

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$e^*(1)$

T

(0.6)

Delay line

pot.

(0.34)

pot.

Delay line

T

$e_2(1)$

T

Fig. 9-10 Realization of pulse-transfer function by a pulse-data

processing unit.

difference is that in Fig. 9-10, the elements of the block diagram represent actual physical components, whereas in the digital programming the diagram represents the sequence of arithmetic manipulations of a general-purpose computer. In the realization of the pulse transfer function of Eq. (9-44), the data processing unit requires two delay lines, two potentiometers, and one summing amplifier.

9.3 DIGITAL COMPENSATION DESIGN THROUGH BILINEAR TRANSFORMATION

In Sec. 8.3 the bilinear transformations $z = (r+1)/(r-1)$ and $z = (1+w)/(1-w)$ were used for the Bode diagram design of continuous-data network compensation of sampled-data systems. In a similar fashion the extension of the Bode diagram method to the design of digital compensation may be easily carried out if any one of the bilinear transformations is utilized. In fact, the Bode diagram design of digital compensation in the r or the w domain is more straightforward than that of the continuous network compensation, since the transfer function of the digital controller can be adjusted independently without affecting the transfer functions of the other parts of the system. Therefore, the effects of the adjustment and variation of the poles and zeros of $D(w)$ (or $D(r)$) upon the overall system behavior are easily observed.

With reference to the system shown in Fig. 9-1a, the design procedure of the digital compensation is outlined as follows:

(1) Assuming that the w -transformation is used, determine the w -transform function $G_0G_1(w)$ of the uncompensated system by substituting $z = (1+w)/(1-w)$ into $G_1G_1(z)$.

(2) Construct the Bode diagram for $G_0G_1(w)$ in magnitude versus ω and phase shift versus ω . Transfer the plots in the Bode diagram to the Nichols chart if necessary. Predict the dynamic behavior of the uncompensated system by finding the phase margin,

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gain margin, resonant frequency, and bandwidth from the Bode plot and the Nichols chart.

(3) Should the system need compensation, the open-loop transfer function of the system with digital compensation becomes $D(w)G_0G_1(w)$. A proper digital controller $D(w)$ is to be selected so that the desired system performance specifications are satisfied. The selection of $D(w)$ simply follows the conventional technique of compensating continuous-data systems, which involves a trial-and-error procedure and is to some extent based on the experience and the imagination of the designer.

The physical realizability of $D(z)$ requires that it have at least as many poles as zeros or that $m \geq n$ in Eq. (9-3). To determine the restrictions on $D(w)$ so that the corresponding $D(z)$ is always realizable, let us consider that $D(w)$ is described by the form

$$D(w) = \frac{c_0 + c_1 w + \dots + c_p w^p}{d_0 + d_1 w + \dots + d_q w^q} \quad (9-47)$$

where c and d are constant coefficients and p and q are positive integers. Substituting $w = (z-1)/(z+1)$ into Eq. (9-47) and simplifying, we have

$D(z)$

$$= \frac{a p z^3 + a p - 1 z p - 1 + b a z + b q - 1 z^4 - 1 + a_0}{(x + 1)^{\alpha - D}}$$

(9-48)

where a and b are related to c and d , respectively. Equation (9-48) indicates that $D(z)$ has the same number of poles and zeros regardless of the relative magnitudes of p and q . This means that the physical realizability condition of $D(z)$ does not affect the condition of the number of zeros and poles of $D(w)$. Thus in Eq. (9-47) p and q can be chosen arbitrarily and the corresponding $D(z)$ will always have the same number of poles and zeros. However, if it is desired that the digital controller is stable, then all the poles of $D(w)$ must lie in the left-half of the w -plane (corresponding to the inside of the unit circle in the z -plane), and in addition $D(w)$ must not have more zeros than poles. From Eq. (9-48) note that if $p > q$, $D(z)$ will have $(p - q)$ poles at $z = 1$ that correspond to sustained oscillatory responses. However, an unstable pulse transfer function is considered realizable. In certain cases the use of unstable digital controllers is even desirable.

(4) Once $D(w)$ is determined, $D(z)$ is obtained by substituting $w = (x - 1)/(x + 1)$ into $D(w)$. The final step in the design involves the realization of $D(z)$ by one of the methods discussed in Sec. 9.2. To illustrate the procedures of design presented, an example follows.

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$R(s)$

$E(s)$

$E(s)$

$D(s)$

$G_{no}(s)$

$G_1(s)$

$C(s)$

E_{sit}

n_3

Digital controller $D(z)$

Fig. 9-11 Sampled-data control system with digital compensation.

Example 9-3

Consider the sampled-data feedback control system shown in Fig. 9-11.

The fixed parts of the system are described by the transfer function

K

$G_1(s)$

$s(1 + 0.1s)(1 + 0.5s)$

(9-49)

The sampling period is 0.5 sec. Design a digital controller for the system to meet the following specifications:

The velocity error constant $K_v \geq 1.5$

The phase margin $\phi_m \geq 50^\circ$

The resonant peak $M_r \leq 1.3$

The open-loop transfer function of the uncompensated system is given by

$1 e^{-Ts}$

K

$G_{ho}(s) G_1(s)$

=

$s(1 + 0.1s)(1 + 0.5s)$
 The z-transform of $G_h(s) G_1(s)$ with $T = 0.5$ sec. is
 (9-50)

or
 $G_h G_1(z)$

$$= \frac{K(0.13x^2 + 0.177z + 0.0092)}{z(z-1)(20.350 - 0.054)}$$

$G_h G_1(z) =$
 $0.13K(z + 1.31)(x + 0.054)$
 $z(z-1)(x - 0.368)$

Applying the final-value theorem to Eq. (9-52), we found that for K, the gain constant K must equal 3.

(9-51)
 (9-52)

Substitution of $z = (1+w)/(1-w)$ and $K = 3$ into Eq. (9-52) yields
 $G_h G_1(w) =$

$$= \frac{0.75(1-w)(1+0.9w)(1-0.134w)}{w(1+w)(1+2.17w)}$$
 (9-53)

The Bode plot of $G_h G_1(w)$ is plotted in Fig. 9-12. Note from the Bode diagram that for K

3 the uncompensated system is on the verge of becoming unstable. (It can be shown that marginal value of K for stability is equal to 3.3.) The phase margin of the system is approximately zero deg. To obtain a phase margin of 50 deg while maintaining the velocity constant at 1.5, a phase-lag model is suggested for $D(w)$, since phase-lead compensation is ineffective under the circumstances.

Let the transfer function of $D(w)$ be of the form

$1 + ATW$
 $D(w)$

$=$
 $1 + TW$
 (9-54)

To realize a phase margin of 50 deg, the gain crossover of the system should be shifted from ω_w

0.6 to ω_w

0.2 , provided that the phase shift at this new gain crossover is not affected very much by the phase-lag com-

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80
 60
 40
 20
 db

Magnitude curve of $G_h G_1(w)$
 (uncompensated system)

0
 -20

Magnitude curve of
 $D(w) G_h G_1(w)$
 (compensated system)-

-40
 1
 -60
 I
 7.46
 -80
 0.001
 0.005 0.01 0.02
 0.1
 0.462 1.0
 10
 ww
 0°
 -90°
 -180°
 -270°
 -360°
 0.001
 0.01
 Phase shift
 (compensated)
 0.1
 ww
 50°
 Phase shift (uncompensated)
 1.0
 10

Fig. 9-12 Bode diagram of sampled-data system with digital compensation in Example 9-3.

=
 pensation. The Bode plot indicates that the open-loop gain of the uncompensated system at $\omega_w 0.2$ is approximately +12 db. Therefore, to make 0.2 the new gain crossover, the phase-lag model of $D(w)$ must provide 12 db of attenuation at this frequency (ω_w) and thus determine the distance between the two corner frequencies of $D(w)$; therefore,

$$\omega_w$$

$$\omega_w$$

$$a = 10^{-12/20} = 0.25$$

(9-55)

Choosing the upper corner frequency of $D(w)$ at one decade below $= 0.2$, we have

Thus

$$1/a\tau$$

$$1/T$$

=

$$0.02$$

$$0.005$$

(9-56)

(9-57)

The digital controller is then described by the w -transfer function

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$$1 + 50w$$

$$D(w)$$

=

$$1 + 100w$$

(9-58)

The open-loop transfer function of the compensated system is given by

$$0.75(1+50w)(1-w)(1+0.9w)(1-0.134w)$$

$$w(1+100w) (1+w)(+2.17w)$$

$$D(w)G_oG_1(w) =$$

(9-59)

The Bode plot of the compensated system is now plotted in Fig. 9-12. Note that the phase margin is now approximately equal to 50 deg and is satisfactory.

The magnitudes and phase shifts of $G_oG_1(w)$ and $D(w)G_oG_1(w)$ are plotted in the Nichols chart as shown in Fig. 9-13. From these curves and the Nichols chart, the following quantities on the system's performance are measured and comparisons are made between the compensated and the uncompensated system.

40

0.01-

30

(Uncompensated)

0.01

$G_oG_1(w)$

0.02

20

$M=1.12$

0.3

1.2

1.3

db 10

2.0

0.02

0.06

3.0

0.5

4.0

0.6

1.0

0.9

0.5

-10

ww

T0.6

ww

-20

1.0

8448

-30

-260-240

-220 -200

0.2

0.4 $M=0.707$

(Compensated)

$D(w)G_oG_1(w)$

-180 -160

-140 -120

-100-90

Phase shift (deg)

Fig. 9-13 Nichols chart and the plots of the open-loop transfer functions for the compensated and the uncompensated system in Example 9-3.

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Uncompensated

system

almost infinite

Compensated
system

1.2

Resonant peak M_p

Resonant frequency $\omega_r(\omega)$

0.6 (2.1 rad/sec)

0.2 (0.8 rad/sec)

•

Phase margin ϕ_m

$\approx 0^\circ$

50 deg

Gain margin G_m

≈ 0 db

12 db

Bandwidth BW (rad/sec)

0.9 (3 rad/sec)

0.4 (1.5 rad/sec)

The final steps in the design involve the determination of $D(z)$ from $D(\omega)$ and the realization of $D(z)$ by any one of the methods described in Sec. 9-2. Substituting $w \approx -1/(1 + j\omega T)$ into Eq. (9-58) yields

==

3.75.1062

ω

$G_c(s)$

Zero-order

hold

1.25-106.0

$G_{no}(s)$

L

10 μ

$G_c(s) -$

10 μ f

(S)H

pot.

0.25

ω

$-D(z) \cdot$

(a)

$G_{ro}(s)$

1.22-106

$H(s)$

(b)

0.24

T

pot.

Delay line

(c)

$\cdot D(z) -$

T

0.99

T

pot.

Delay line

$\rightarrow$

Summing

amplifier

$D(z) = 0.25(x - 0.96)/(z - 0.99)$

-

Fig. 9-14 Various ways of realizing the digital controller in the system of Example 9-3.

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$$D(z) = 0.25$$

2

$$0.96$$

$$0.99$$

(9-60)

By use of the techniques described in Sec. 9-2, the pulse transfer function of the digital controller $D(z)$ is realized by a series pulsed-data network, a feedback pulsed-data network, and a pulsed-data processing unit.

First, for a series pulsed-data network, we substitute $D(z)$ into Eq. (9-7) which leads to

$$G_e(s)$$

$$1 + 12.5s$$

=

$$1 + 50s$$

(9-61)

The RC network which realizes $G_e(s)$ and the series pulsed-data network for the realization of $D(z)$ are shown in Fig. 9-14a.

For the feedback pulsed-data network realization, we substitute $D(z)$ from Eq. (9-60) into Eq. (9-16). This leads to

$$H(s)$$

=

$$3s$$

$$s + 0.082$$

(9-62)

The RC network for the realization of $H(s)$ and the feedback pulsed-data network which realizes $D(z)$ are shown in Fig. 9-14b.

Finally, the block diagram of the pulsed-data processing unit realization of $D(z)$ is depicted in Fig. 9-14c. The diagram can also be regarded as the schematic diagram of the direct programming of a digital computer.

9.4 DESIGN IN THE Z-PLANE USING THE ROOT LOCUS

In Sec. 6.7 the root locus technique has been extended to the analysis of sampled-data feedback control systems. Note that preferably the root locus plots for sampled-data systems should be plotted in the z -plane and that their properties are quite similar to those of the root locus for continuous-data systems in the s -plane. Therefore, the application of the root locus method to the analysis of sampled-data control systems is quite straightforward. Once the root loci of the characteristic equation are plotted in the z -plane, much knowledge concerning the transient response of the system can be gathered by observing the location of the roots for the particular loop gain K .

On the other hand, the design of sampled-data control systems can also be carried out in the z -plane using the root locus. In essence, the design of a sampled-data system in the z -plane involves the search for a set of characteristic equation roots that give rise to a satisfactory over-all system performance. In terms of the root locus design, these desirable characteristic equation roots are found by reshaping the root loci of the original system through adjustment of the loop gain and the use of compensation networks. The most

elementary problem in root locus design is the determination of the loop gain to yield suitable relative stability. The loop gain of the system can be adjusted to give appropriate dynamic performance as measured by the position of the complex poles with respect to the constant damping ratio curves inside the unit circle. However, no simple rules are available for the determination of appropriate compensation networks from the root locus diagram. In most design problems the effects on the characteristic equation roots when the loop

gain K , as well as other parameters, is varied must be studied. Therefore, the design in the z -plane with root locus usually involves a certain amount of trial-and-error. Usually, some knowledge on the effect of adding or shifting of open-loop poles and zeros on the shape of a root locus is helpful on the part of the designer. The use of the root contour method is also illustrated in this chapter when a sampled-data system has variable parameters other than K . In the design of continuous-data control systems, usually, the design may fall into one of the following categories: (1) phase-lead compensation, (2) phase-lag compensation, and (3) cancellation compensation. Other more complex schemes of course do exist, but knowing the effects of these three basic types of compensation usually gives a designer much insight into the design problem. With reference to the design of sampled-data systems by the Bode diagram, the basic forms of the compensating network $D(w)$ or $D(r)$ have also been classified as phase-lead and phase-lag; and certainly cancellation compensation can always be used whenever and wherever feasible. Therefore, in the following paragraphs, the design of compensation for sampled-data systems is also described as phase-lead, phase-lag, or cancellation.

Phase-Lead Compensation

A simple phase-lead network model in the w -domain is described by the transfer function

$$D(w) = \frac{1 + ATw}{1 + \tau w}$$

$$(a > 1) \quad (9-63)$$

where a is a constant greater than or equal to zero. The $D(w)$ in Eq. (9-63) is referred to as a phase-lead transfer function, since in terms of the Bode diagram it produces positive phase shift that may be added to the system phase shift in the vicinity of the gain crossover frequency (ω_w) to increase the phase margin. However, the B. C. Kuo, Automatic Control Systems, (Englewood Cliffs, N. J.: Prentice-Hall, Inc., 1962), Chapter 8.

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use of phase-lead compensation is not recommended under all circumstances. In fact, it has been shown that the use of phase-lead compensation is unlikely to bring satisfactory results when the uncompensated system has the following properties:

- (1) when the transfer function of the controlled process $G_1(w)$ has two poles that are close together and the corner frequencies are located near the gain crossover frequency (ω_w) in the Bode plot
- (2) when the transfer function of the controlled process $G_1(w)$ has a pair of complex conjugate poles located near the gain crossover

Both the above cases will cause the phase shift curve of $G_1(w)$ to vary rapidly near 180 degrees.

- (3) when the uncompensated system is initially unstable or lightly damped.

For these same reasons phase-lead compensations were not recommended in the illustrative examples on the design of sampled-data systems (Example 9-3, for instance).

With reference to Eq. (9-63), the pole-zero configuration of $D(w)$ is shown in Fig. 9-15a. Note that the pole and the zero of $D(w)$

ω_w
w-plane

Unit
circle
-1/T
-1/07
1-T
1+T
(a)
(b)
z-plane
+1
1-07
1+0T

Fig. 9-15 Pole-zero configurations of phase-lead network.

always lie on the negative real axis in the w-plane with the zero. to the right of the pole. Substitution of $w = (z - 1)/(z+1)$ into Eq. (9-63) yields

$$D(z) = \frac{(a + 1)z + 1}{1 + (a + 1)Tz} = \frac{(1 + aT)z + 1}{1 + (1 + aT)Tz} \quad (9-64)$$

Since a and T are both positive numbers and since $a > 1$, the pole and the zero of $D(z)$ always lie on the real axis on or inside the unit circle in the z-plane; the zero is always to the right of the pole. + B. C. Kuo, Automatic Control Systems, (Englewood Cliffs: Prentice-Hall, Inc., 1962), Chapter 9.

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pole. A typical set of pole-zero configuration of $D(z)$ is shown in Fig. 9-15b.

Having established the pole-zero properties of the phase-lead model of $D(z)$, we shall now use the following illustrative example to investigate its effects on the reshaping of the root loci.

Example 9-4

A sampled-data feedback control system with digital compensation is shown in Fig. 9-16. The controlled process of the system is described by the transfer function

$\frac{R(s)}{E(s)} = X$
 $\frac{D(s)}{G_0(s)G_1(s)}$
Zero-order Controlled
hold
 $\frac{C(s)}{\text{process}}$
Digital controller $D(z)$

Fig. 9-16 Sampled-data control system in Example 9-4.

$$G_1(s) =$$

$$= \frac{K}{s(s+1)} \quad (9-65)$$

The sampling period is one second. The open-loop transfer function of the system without compensation is

$$G_0G_1(z) = \frac{K}{(z-1)(z-0.368)}$$

(9-66)

The root locus diagram of the uncompensated system is plotted in Fig. 9-17. Note that the complex conjugate part of the root loci is a circle with center

720-368

z-plane

K=2.43

8-1

K-

717

K=Q

0

IK=0

+0.368-

K=2.43

T = 1 sec

Fig. 9-17 Root locus diagram of the system in Example 9-4

without compensation.

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at $z = -0.717$ and a radius of 1.37. The closed-loop system becomes unstable for all values of K greater than 2.43. Let us assume that K is set at this marginal value so that the two characteristic equation roots are on the unit circle as shown in Fig. 9-17.

Suppose that the transfer function $D(z)$ of the digital controller is of the form

$D(z) = ($

$(az + 1)$

az

$T+1$

$2 + (1 - AT)/(1 + AT)$

$\approx + (1 - T)/(1 + T)$

where for phase-lead compensation, $a > 1$ and $\infty > T > 0$.

(9-67)

The constant factor $(aT + 1)/(T + 1)$ in $D(z)$ is necessary, since the insertion of the digital controller should not effect the velocity error constant

K_v , while improving the stability of the system. In other words, $D(z)$ must satisfy the condition

$\lim_{z \rightarrow \infty} D(z)$

$=$

1

2-1

(9-68)

The design problem now essentially involves the determination of appropriate values for a and T so that the system is stabilized. However, at this point, how the values of a and T should be chosen is not clear. Although we know from the properties of root loci that an added open-loop zero has the effect of pulling the root loci toward it, whereas an additional open-loop pole has a tendency to push the loci away, no simple ways exist for telling which combinations are the most effective for stabilizing the system because of the unlimited number of possible combinations of the pole and zero of $D(z)$.

Several sets of values of a and T are used in the following to illustrate the effects of the phase-lead compensation. As a first trial, let $a = 6.06$ and $T = 0.165$. The transfer function of the digital controller reads

$D(z) = 1.72$

$\approx$

$z + 0.717$

and the open-loop transfer function of the compensated system is

$0.64K \approx$

$$D(z) = \frac{G(z)}{(z - 1)(z - 0.368)}$$

The root loci of the compensated system are shown in Fig. 9-18 as loci (2). Note that although the complex conjugate portion of the loci lie inside the unit circle, we should not jump to the conclusion that the phase-lead compensated system is more stable. As a matter of fact, for $K = 2.43$ one of the roots of the characteristic equation is on the negative real axis outside the unit circle, and the system is unstable. This simply shows that for the values chosen for a and T , the compensated system is worse than the original system. Four other sets of values for a and T are tried, and the corresponding root loci of the compensated systems are plotted in Fig. 9-18. (Only the positive complex conjugate parts of the root loci are shown.) The characteristic equation roots of the compensated systems when $K = 2.43$ are indicated on the loci. The pole and zero locations of $D(z)$ of the various compensations are tabulated in Table 9-1. From the root locus diagram, we see that among the five compensations only when $a = 2$, $T = 0.4$, and $a = 3$, $T = 0.1$

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z-plane

Uncompensated (1)

(5)

$$K = 2.43$$

(6)

$$K = 2.43$$

(4)

(3)

$$-0.717$$

$$0.368$$

$$K=2.43$$

(2)

(2)

(1) Uncompensated

$$(2) = 0.165, a = 6.06$$

$$(3) T = 0.1, a = 3$$

$$(4) T = 0.4, a = 2$$

$$(5) T=1, a=2$$

Unit circle

$$(6) \sigma = 3, \alpha = 3$$

$aT +$

$$D(x) = \frac{(a+1)}{2 + (1 - AT)/(1 + AT)}$$

$$\approx \frac{(1 - T)}{(1 + T)}$$

Fig. 9-18 Root loci of the uncompensated and compensated systems for Example 9-4.

result in stable systems. But then the damping ratios of the systems are less than 10 percent, and from Fig. 6-20 we can predict that the overshoots of the step responses would exceed 70 percent, a circumstance which is unacceptable under almost any circumstances.

Note that the system is improved, though only slightly, when the value of a is relatively small. However, if the value of T is too small, the pole of $D(z)$ will move very close to the 1 point; and since one root locus starts at this pole and ends at $-\infty$, the compensated system will be very lightly damped or unstable. On the other hand, the root loci in Fig. 9-18 show that for large values of the phase-lead compensation also results in unstable systems.

The general ineffectiveness of the phase-lead compensation is anticipated in this problem, since the original system is on the verge of instability and this situation is one for which phase-lead compensation is not recommended.

In the next section we shall see that a phase-lag compensation is more satisfactory for improving the stability of this system. We can also show that if the loop gain K is reduced from the value of 2.43, the effectiveness of the phase-lead compensation is improved; however, this improvement is usually at the expense of higher bandwidth and resonant frequency.

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TABLE 9-1

pole and zero of $D(z) = (aT + 1) =$

T

a

$z + (1 - ar)/(1 + aT)$

$\approx + (1 - T)/(1 + T)$

Zero of $D(z)$

-0.538

Pole of $D(z)$

-0.818

0.1

3

0.165

6.06

0

-0.717

0.4

2

-0.111

-0.375

1.0

2

+0.333

0

3.0

3

+0.80

+0.50

Phase-Lag Compensation

A second possibility for simple compensation is available in the phase lag model. In the w -domain, a simple phase-lag model is given by the transfer function

$D(w)$

$=$

$1 + ATW$

$1 + TW$

(9-71)

where $0 < a < 1$ and $0 < T < \infty$.

In the design method using the Bode diagram, the attenuation property of $D(w)$ at high frequency (ww) is utilized to move the gain crossover of the uncompensated system to a lower point where the desired phase margin is measured. So that a significant amount of phase lag will be contributed near the new gain crossover, the upper corner frequency $1/T$ of $D(w)$ should be placed far below the

Aww

w -plane

$-1/07$

$-1/7$

z -plane

Unit circle.

HE

$1+T$

$$\frac{1 - 0.7}{1 + 0.7} \\ (a) \\ (a) D(w)$$

$$= \frac{(b) \quad 1 + a\tau\omega}{(a < 1) \quad 1 + \tau\omega} \\ \frac{a\tau + 1}{z + (1 - aT)/(1 + a\tau)} \\ (b) D(z) = \frac{1 + 0.7}{z + (1 - 0.7)/(1 + 0.7)}$$

Fig. 9-19 Pole-zero configurations of phase-lag network.

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new gain crossover. Therefore, in the phase-lag compensation the value of γ is very large.

T

Since $D(w)$ in Eq. (9-71) is identical to the transfer function for the phase-lead compensator only when $a < 1$ for phase-lag, the z -transform of Eq. (9-71) is also expressed by Eq. (9-67) with $a < 1$. The pole-zero configuration of the phase-lag $D(w)$ and $D(z)$ are depicted in Fig. 9-19. Note that since a is less than unity, the pole always lies to the right of the zero.

The same sampled-data control system used for Example 9-4 is used in the following illustrative example to show the effects of the phase-lag compensation in the root locus diagram.

Example 9-5

Consider the same system given in Example 9-4 with $K = 2.43$. The system

(3)

(2)

Uncompensated

(1)

$K = 2.431$

-0.717

0

z -plane

(14)

$K = 2.43$

-X-

0.368

Unit circle

(1) Uncompensated

(2) $\sigma = 0.1$, $a = 0.5$

(3) $T = 0.4$, $a = 0.5$

(4) $T = 1.0$, $a = 0.5$

$D(z)$

=

$$\frac{(1 + AT)^2 + (1 - AT)}{(1 + AT)}$$

$$T + 1 \approx \frac{(1 - T)}{(1 + T)}$$

Fig. 9-20 Root loci of the uncompensated and the phase-lag compensated system for Example 9-5.

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is now to be stabilized by means of the simple phase-lag compensator of Eq. (9-71).

First, we shall investigate the effects of the phase-lag compensation on

the root loci when small values of T are chosen. With $a = 0.5$, the root loci of the system with the phase-lag compensation are plotted for $\tau = 0.1, 0.4$, and 1.0 , as shown in Fig. 9-20. (Only the positive complex conjugate parts of the loci are shown.) For small values of T , the phase-lag compensation has made the system unstable. This simply verifies the conclusion we obtained from the Bode diagram method that, for the phase-lag compensation to be effective, the value of T should be relatively large.

Let us assume that the design specification requires the damping ratio of the complex closed-loop poles to be approximately 60 percent. Referring to locus (1) in Fig. 9-21, which is the root locus of the original system, we note that the complex closed-loop poles have a damping ratio of 60 percent when the loop gain K is equal to 0.5 . In essence, the phase-lag compensation can be regarded as a means of increasing the velocity error constant K_v by a ratio of $4.86 (=2.43/0.5)$, while keeping the complex closed-loop poles relatively unaffected. Since T is to be very large and since a is less than unity, the pole

Uncompensated
(1)
-0.717
1
z-plane
(2)
(3)
 $K=2.43$
 $K=2.43$
 $=0.6 \quad K=0.5$

(2)
0 0.368
0.98
0.905

Unit circle

(1) Uncompensated
(2) $\sigma = 100, a = 0.2$
(3) $T=50, \alpha=0.2$

$$D(x) = \frac{(a+1)^2}{(1-r)^2} \frac{(1-r)^2}{(1+ar)^2} \frac{(1-r)^2}{(1+ar)^2}$$

Fig. 9-21 Root loci of the uncompensated and the phase-lag compensated system for Example 9-5.

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and zero of $D(z)$ will appear as an integrating dipole near the point $z = +1$. Therefore, the complex conjugate parts of the original root loci are not affected significantly by the addition of the integrating dipole, since from points on these loci the dipole and the pole of $G_h G(z)$ at $z = 1$ appear as a simple pole. In order to increase the velocity error constant by a factor of 4.86 (from $K = 0.5$ to $K = 2.43$), the constant ratio a of the integrating dipole should be chosen to be at least $1/4.86$, preferably $1/5$, to allow the dipole to contribute a slight phase lag near the new gain crossover. Thus we let $a = 0.2$, and is chosen to be 100 . As will be shown, the exact value of T is not critical. Substituting these values of a and T into Eq. (9-67) yields the transfer function of the phase-lag controller

$$D(z) = 0.191$$

$\approx$
2
- 0.905
0.98

The open-loop transfer function of the compensated system is

$$D(z) G_{hoG_1}(z) =$$

$$=$$

$$---$$

$$0.07K(z - 0.905) (\approx + 0.717)$$

$$(1)(\approx 0.368) (\approx$$

$$=$$

$$0.98)$$

$$(9-72)$$

$$(9-73)$$

In Fig. 9-21 the root loci of the phase-lag compensated system are shown as loci (2). Note that the complex roots for $K = 2.43$ on the compensated loci lie very close to the roots for $K = 0.5$ on the uncompensated loci. As a matter of fact, for the values of a and τ chosen, the addition of the integrating dipole has affected the shape of the original root loci only slightly except for the small loop of loci around the dipole.

The root loci of the compensated system when $a = 0.2$ and $\tau = 50$ are also plotted in Fig. 9-21 (loci 3). For $K = 2.43$, the complex characteristic equation roots lie very close to those of loci (2). This shows that the precise location of the dipole is not critical, as long as it is close to the $z = +1$ point.

The Root Contour Method

The root contour diagram of a closed-loop system is the plot of the roots of the characteristic equation when any system parameter other than the open-loop gain K is varying. Apparently, the root contour diagram is quite useful in the design of feedback control systems, since it can be used to indicate the effects of variations in the poles and zeros of the open-loop transfer function. By use of the root contour plot, the effects of phase-lead and phase-lag compensations may be clearly indicated, and appropriate values of the parameters of the controllers can be easily determined.

As an illustrative example to the root contour method, let us consider the sampled-data system given in Example 9-3 and Example 9-4. Since the phase-lead and the phase-lag controllers have basically the same transfer function, but with different ranges for a , the open-loop transfer function of the sampled-data system with either compensation is given by

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$$D(z) G_{hoG_1}(z)$$

$$=$$

$$\frac{(x + 1) + a\tau(x-1)}{(2 + 1) + \tau(\approx - 1)}$$

$$\times$$

$$0.368K (z + 0.717)$$

$$(x-1)(\approx 0.368)$$

$$(9-74)$$

When $K = 2.43$, the characteristic equation of the compensated system is written

$$(x + 1)[(x - 1)(\approx 0.368) + 0.89] (z + 0.717)]$$

$$+ \tau(x - 1)^2 (z - 0.368) + 0.89a\tau (z - 1)(\approx + 0.717) = 0 \quad (9-75)$$

Equation (9-75) is considered to have two variable parameters in a and τ . First, we let $a = 0$, but consider that it is varied from 0 to ∞ . Then, Eq. (9-75) is reduced to

$$(x + 1)[(\approx - 1)(\approx - 0.368) + 0.89(\approx + 0.717)]$$

$$+ T(x - 1)^2(\approx - 0.368) = 0$$

$$K=2.43$$

$$(9-76)$$

z-plane

Root contour

$$(a = 0)$$

$$T=0.1$$

$$T=0.4$$

$$T=0$$

$$T=1$$

$$T=3$$

Root locus of
uncompensated system

$$T=10$$

$$T=100$$

$$T=40$$

$$T=0.4$$

$$T=40$$

$$T=0.1$$

$$T=1$$

$$848$$

$$T=100$$

$$848$$

$$ett$$

$$0$$

$$0.368$$

$$+1$$

$$T=3$$

$$T=10$$

$$\infty$$

Root contour

$$(a = 0)$$

Unit circle

$$K=2.43$$

Root contour

$$(a = 0)$$

X-

poles of $G(z)$

zeros of $G(z)$

$G(z)$

=

$$7(z - 1)^2(z - 0.368)$$

$$(\approx + 1)[(\approx - 1)(\approx - 0.368) + 0.89(z - 0.717)]$$

-

Fig. 9-22 Root contours of the sampled-data control system in
Example 9-3, with a as the variable parameter, $K = 2.43$ and
 $a = 0$.

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Dividing both sides of Eq. (9-76) by the terms that do not contain
 T , we have

$$1 +$$

-

$$T(x - 1)^2(z - 0.368)$$

$$(\approx + 1)[(\approx - 1)(z - 0.368) + 0.89(\approx + 0.717)]$$

Since Eq. (9-77) can be written into the form of

T

$$1 + G(z) = 0$$

=

0

(9-77)

(9-78)

with appearing in $G(z)$ as a multiplying factor, the conventional root locus method can be applied to the plotting of the root contours of Eq. (9-76) when γ is varied from 0 to ∞ . A further investigation of Eq. (9-77) reveals that the denominator of the equivalent $G(z)$ is identical to the characteristic equation of the original system multiplied by a factor $(z+1)$. Thus, the poles of $G(z)$ are found at the points where $K = 2.43$ on the original loci of the uncompensated system and at $z = -1$. The zeros of $G(z)$ are obtained readily from Eq. (9-77) at $z = 1, 1$, and 0.368 ; these zeros correspond to the ending points of the root contours. Now the pole-zero configuration of $G(z)$ is depicted in Fig. 9-22, and the root contours when γ is varied from 0 to ∞ are plotted accordingly. Note that the starting points of the root contours lie at the points of the original root loci where $K = 2.43$.

Now consider that a as well as γ , is varied from 0 to ∞ . Dividing through Eq. (9-75) by the terms that do not contain a yields

$$\frac{1 + 0.89aT(21)(x + 0.717)}{(\approx + 1)[(\approx - 1)(\approx - 0.368) + 0.89(\approx + 0.717)]}$$

which is of the form

$$1 + T(x - 1)^2(z - 0.368)$$

$$1 + G'(z) = 0$$

with a appearing as a multiplying factor.

=

= 0

(9-79)

(9-30)

Again, we notice that the denominator of $G'(z)$ is identical to the left side of Eq. (9-76), whose roots when γ is varied are represented by the root contours of Fig. 9-22. This implies that the root contours of Eq. (9-75) start at points on the root contours of Fig. 9-22 where a is variable but has any fixed positive value. The ending points of the root contours are at the zeros of $G'(z)$, namely, ≈ 1 and $2 = 0.717$.

$z =$

z

The root contours of Eq. (9-75) with a as the varying parameter are plotted in Fig. 9-23 for various fixed values of γ . (Only the positive complex conjugate parts of the contours are shown.) Note that all the contours pass through the two complex roots of the original system ($K = 2.43$); and the corresponding value of a at

1

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Root contours

$1 < 0$

$T = \text{constant}$

(lead compensation)

z -plane

$T=1$

$a=1$

$T=0 \quad T=0.1$

Root contours

$0 < a < 1$

$T = \text{constant}$

(lag compensation)

Root contour

($a=0$)

$x_7=3$

$K=2.43$

$T=0.4$

Ideal

lead compensation

$D(z)=(2+1)+0.7(2-1)$

$(z+1)$

e^{-x}

-0.717

$XT=10$

$T=40$

$XT=100$

844

+1

Root locus of

uncompensated system

$G'(x) =$

$0.89a(x-1)(x+0.717)$

poles of $G'(z)$

zeros of $G'(z)$

$(z+1)[(x-1)(z-0.368)+0.89(\approx+0.717)]+7(z-1)^2(z-0.368)$

Fig. 9-23 Root contours of the sampled-data control system in

Example 9-3. a and 7 are both variable; $K = 2.43$.

the points is unity; since for $a = 1$, $D(w)$ is unity, the system is without compensation.

The effects of the phase-lead and the phase-lag compensation are clearly observed from the root contour plots of Fig. 9-23. For $a > 1$, the compensation is phase-lead. The general ineffectiveness of the phase-lead compensation for this problem is clearly illustrated by the root contours. For large values of 7 , the root contours for $a > 1$ lie mostly outside the unit circle. Although for very small values of the root contours for $a > 1$ may lie inside the unit circle, the corresponding real pole of $G'(z)$ is very close to the 1 point (see Fig. 9-22); and since one of the root contours extends from this pole to along the negative real axis, the system is not much improved.

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On the other hand, when $a < 1$, the compensation is phase-lag.

The root contours again indicate clearly that the system is unstable for small values of T . But the system can be stabilized at will if a large value of T is used.

Cancellation Compensation

On a theoretical basis at least, it is always possible to use a digital compensator with poles and zeros that can cancel the undesirable poles and zeros of the controlled process transfer function, and new open-loop poles and zeros can be inserted at more advantageous locations in the z -plane to satisfy the design specifications. The cancellation compensation suggested above is often used in systems that have complex poles in the controlled process, since in such cases simple phase-lead compensation is known to be ineffective. However, the cancellation compensation does not always furnish a completely satisfactory solution to a design problem. Quite often, the resulting compensator may involve excessive complexity; and if the undesirable complex poles of the controlled process are near the unit circle in the z -plane, inexact cancellation can often result in a z -plane

z-plane

ON

0

(a)

QX

+1

Root

locus

Unit circle

Root

locus

(b)

z-plane

P₁

1+1

+1

Unit circle

(c)

Unit circle

Fig. 9-24 Root locus diagrams illustrating the effects of inexact cancellation of undesirable poles.

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conditionally stable system. As a simple example to the effect of inexact cancellation, let us refer to the root locus diagrams given in Fig. 9-24. In Fig. 9-24a the pole-zero configuration of the open-loop transfer function of a hypothetical sampled-data system is shown. A digital controller with the transfer function

-

-

$$D(z) = K_1$$

$$(\approx -p_1)(z - P_2)$$

$$(z_a)(z_b)$$

(9-81)

is devised to cancel the complex poles at p_1 and p , and to add new poles at $z = a$ and $z = b$. Figure 9-24b shows a particular case of inexact cancellation, which occurs when the complex zeros of the digital controller are not exactly equal to the poles p and p_2 of the controlled process. In this case, the inexact cancellation does not have any adverse effect on the performance of the compensated system, since the closed-loop dipole near p_1 and p as a result of the inexact cancellation, does not contribute significantly to either the transient or the frequency response.

On the other hand, if the poles and zeros of the inexact cancellation are arranged as shown in Fig. 9-24c, parts of the root loci between the uncanceled pole and zero pairs lie outside the unit circle. Then the system may be conditionally stable or even unstable.

9.5 SYNTHESIS IN THE Z-PLANE

Thus far, the frequency domain and the root locus method of design of the digital compensation of sampled-data control systems have been discussed. The principal disadvantage of these methods of design lies in the extent of the cut-and-try effort imposed on the designer. In proceeding from specifications to the final system, the designer must base his decisions on his past experiences and on the nature of the problem and select an appropriate compensator to modify the open-loop system in such a way that the closed-loop system is satisfactory. If as the result of the designer's first rough guess, the specifications are not met by the system, he must keep on modifying the open-loop system until the closed-loop system has a satisfactory performance.

A logical synthesis procedure should be devised to enable the

designer to proceed from specifications to the final system in a straightforward manner. A well-known method for the synthesis of linear continuous-data control systems using pole-zero configurations in the s-plane was first set forth by Truxal. The philosophy of
 † J. G. Truxal, Automatic Feedback Control System Synthesis, (New York: McGraw-Hill Book Company, 1955), Chapter 5.

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Truxal's synthesis will be extended to the synthesis of sampled-data control systems in this section.

The synthesis procedure essentially involves the following three steps:

- (1) The closed-loop z-transfer function of the final system is determined from the specifications and the fixed parts of the system.
- (2) The corresponding z-transfer functions of the open-loop system and the digital controller are found.
- (3) The digital controller is synthesized.

In contrast to the conventional design methods, the synthesis procedure outlined above enables the designer to work with the over-all system function, so that in a sense the system is forced to meet the specifications.

Determination of the Closed-Loop Transfer Function From the Specifications and the Fixed Parts of the System

The first step in the synthesis involves the determination of the closed-loop transfer function of the sampled-data system from the specifications and the fixed elements of the system. Although, in general, specifications in control systems may take a wide variety of forms, any but the most commonly used ones are difficult to consider. Therefore, for the proposed synthesis procedure only the following time domain specifications are imposed :

damping ratio of the control poles

overshoot

peak time

velocity error constant

The first three specify the transient response of the system, whereas the last one defines the steady-state performance of the system to a ramp input.

The basic model of the over-all system is shown by the block diagram in Fig. 9-25. The system is characterized by two basic components: the digital controller and the controlled process that includes the fixed elements of the system. The closed-loop transfer function of the system in Fig. 9-25 is given by

$R(s)$

$E(s)$

$C(s)$

$D(s)$

T

$G_o(s)G_1(s)$

$G_o(s)$

Fixed elements

Digital Controller

Fig. 9-25 Simple sampled-data feedback control system with digital compensation.

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$M(z)$

=

$C(z)$

$R(z)$

$$D(z) = \frac{G_p(z)}{1 + D(z)G_1(z)} \quad (9-82)$$

where $G_1(z)$ is the z-transform of $G_0(s) G_1(s)$ and is assumed to be given at the outset of the synthesis.

In general, the selection of the desired closed-loop function $M(z)$ is governed by both the specifications and the nature of $G_1(z)$. Evidently, the $M(z)$ selected by the designer not only must satisfy all the specifications but also should lead to a transfer function $D(z)$ for the digital controller which is physically realizable. It has been shown in Sec. 9.2 that for $D(z)$ to be physically realizable the degree of the numerator in z must not exceed that of the denominator.

Solving for $D(z)$ in Eq. (9-82) in terms of $M(z)$ and $G_1(z)$ yields

$$D(z) = \frac{M(z)}{1 - M(z)G_1(z)} \quad (9-83)$$

If $M(z)$ is written as the ratio of polynomials $p(z)/q(z)$ and $G_1(z)$ as $n(z)/d(z)$, then Eq. (9-83) becomes

$$D(z) = \frac{p(z)}{q(z) - p(z)n(z)} \quad (9-84)$$

As the last equation demonstrates, to ensure that $D(z)$ is a physically realizable pulse transfer function (numerator order equals or is less than the order of the denominator), the excess of poles over zeros for $M(z)$ must at least equal the excess of poles over zeros for $G_1(z)$. For instance, if $G_1(z)$ has two finite poles and one finite zero, an excess of poles over zero of one, the selected configuration for $M(z)$ must contain at least one more pole than zeros.

If a stable digital controller is desired, then the poles of $D(z)$ must lie inside the unit circle with the exception of a simple pole at $z = 1$. An unstable $D(z)$ of course may still be realizable, but if the synthesis can be accomplished with a stable digital controller, certainly an unstable $D(z)$ should be avoided because it will correspond to an unstable open-loop system.

With reference to Eq. (9-84), we notice that the stability of $D(z)$ requires that the polynomial $q(z) - p(z)n(z)$ have no zero on or outside the unit circle with the exception of a possible simple zero at $z = 1$. Furthermore, if the fixed elements $G_1(z)$ has zeros outside the unit circle, $M(z)$ must contain the same zeros, or these zeros will become the unstable poles of $D(z)$.

The second factor governing the choice of $M(z)$ is the set of performance specifications. These specifications have been listed previously as the damping ratio, maximum overshoot, peak time,

configurations of the closed-loop transfer function of a second-order sampled-data system and the maximum overshoot, damping ratio, and peak time have been established in Sec. 6.7. Figures 6-20 and 6-21 give the percent overshoot and the peak time, respectively, in terms of the locations of the closed-loop poles and zero of the second-order system. Also, in Sec. 6-11 the relationships between the error constants and the closed-loop pole-zero configurations were derived. These relationships and the plots that were obtained previously are basic and of great importance in regard to the synthesis procedure proposed here.

The pole-zero configurations considered in detail in the following discussion are all characterized by the fact that the principal characteristics of the transient response of the system are fixed and controlled by a pair of complex conjugate poles p_1 and p_2 and a real zero z_1 as shown in Fig. 9-26. All other poles are placed either

z-plane

Control poles

Control

zero

z_1

$\cdot x + x \cdot$

$P_4 \ P_3$

z_2

z_2

Dipole

$+1$

P_5

Unit circle

Fig. 9-26 Pole-zero configuration of $M(z)$.

so close to the origin in the z-plane (such as p_1 and p_2) or so close to a zero (as p_5 is close to z_2) that they form a dipole and contribute insignificantly to the over-all transient response. With only the basic control poles P_1 , P_2 , and the zero z_1 , $M(z)$ is of the form:

where

$$M(z) = \frac{K(z - z_1)(z - z_2)}{(z - p_1)(z - p_2)}$$

$M(z)$

$C(z)$

$=$

$-$

K

$$(1 - p_1)(1 - p_2)$$

$-$

$$(1 - z_1)$$

(9-85)

(9-86)

The constant K is so chosen that for a unit step function input the steady-state error at the sampling instants is zero.

The closed-loop transfer function given by Eq. (9-85) can be used only to synthesize a system with a $G_c(z)$ which has only one

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more pole than it has zeros. If the number of poles of $G_c(z)$ exceeds the number of zeros by more than one, additional poles must be inserted in $M(z)$ to make $D(z)$ physically realizable. By locating these additional poles near the origin in the z-plane, their contribution to the transient response can be made negligible.

Consider that $M(z)$ is given by the basic structure of Eq. (9-85) and that the pole-zero configuration of $M(z)$ is illustrated in Fig.

9-27. Substitution of Eq. (9-85) into Eq. (9-83) yields

z-plane

ON

Ipi

Kav

+1

Unit circle

D(z)

Fig. 9-27 Basic pole-zero configuration of M(z).

K(z-z₁)

1

=

X

-

(z - p₁)(z - p₂) - K(z

-

z₁)

GALE

(9-87)

Since K is given by Eq. (9-86) and

p₁p₂ = |p|²

p₁ + p₂ = 2|p|cos θ &

p₂

(9-88)

(9-89)

where p is the magnitude of p₁ (or p₂) and θ is the angle of p₁ (see Fig. 9-27), we can show that D(z) can be written as

D(z)

=

1 - 2p cos θ + |p|²

1 - z₁

X

2-

-1) [z

1) [z -

-

z₁z₂

1

(9-90)

z₁(1 - 2|p|cos θ + |p|²) + |p|²]G₁(z)

-

1.

-

z₁

If G₁(z) has at least one more pole than it has zeros, then D(z) is physically realizable. Furthermore, D(z) is stable if G₁(z) has no zero on or outside the unit circle and if the following condition is satisfied:

|z₁|(1 - 2|p|cos θ + |p|²) + |p|² < 1

(9-91)

1 - z₁

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N

30

20

15

10

5

2

1.5
 1.0
 =0.5
 --0.707
 777
 Unstable regions
 0.5
 C=0.707
 = 0.5
 0.2
 0.15
 0.1
 0° 10° 20° 30° 40° 50° 60° 70° 80° 90°
 φ

Fig. 9-28 Plots of the unstable regions of z_1 given by Eq. (9-91). The condition of Eq. (9-91) is drawn in Fig. 9-28 for $\gamma = 0.5$ and 0.707 to facilitate the choice of z_1 and that will result in a stable $D(z)$. Note that $D(z)$ is always stable if z_1 is negative. If $G_1(z)$ contains a simple zero outside the unit circle, the closed-loop zero z_1 should be chosen as this zero; and if this z_1 falls outside of the unstable region of Fig. 9-28 for the specified values of γ and ϕ , $D(z)$ will be stable. However, if the value of z_1 , chosen according to the zero of $G_1(z)$ outside the unit circle, falls inside the unstable region of Fig. 9-28, $D(z)$ will be unstable. But in general there is a certain amount of freedom available in the selection of γ and ϕ , so that z_1 may still be located outside the unstable region by properly choosing γ and ϕ .

The expression relating the velocity error constant to the closed-

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loop poles and zeros in the z -plane has been derived in Sec. 6.11 and is repeated below:

$$K_v = \lim_{z \rightarrow 1} (z-1) \left[\frac{1}{T} \sum_{i=1}^m \frac{z_i}{z_i - 1} \right]$$

$$= \frac{1}{T} \sum_{i=1}^m \frac{z_i}{z_i - 1}$$

$$(9-92)$$

where T is the sampling period, p_i and z_i are the closed-loop poles and zeros, respectively. For the basic pole-zero configuration of Fig. 9-27, Eq. (9-92) is written

$$K_v = \frac{1}{T} \left[\frac{1}{p_1} + \frac{1}{p_2} + \dots + \frac{1}{p_m} - \frac{1}{z_1} - \frac{1}{z_2} - \dots - \frac{1}{z_n} \right] \quad (9-93)$$

Substitution of Eqs. (9-88) and (9-89) into Eq. (9-93) yields

Kv

2. = T (

1

22 pcos

- 2 pcos + |p|² 1 21

(9-94)

An efficient method for increasing the velocity error constant without affecting the transient response appreciably is to add a dipole near $z = +1$. To increase K , the zero at z must be located to the right of the pole at p in order that $1/(1 - p_1)$ may be less than $1/(1 - z_1)$. The zero and the pole are placed sufficiently close to each other so that the residue of the function $C(z)/z$ at the pole P is negligible. With the insertion of the dipole, $M(z)$ reads

$$M(z) = \frac{C(z)}{K(1 - p_1)(z - z_1)(z - z_2)(z - z_3) \dots (1 - z_1)(z - p_1)(z - p_2)(z - p_3) \dots}$$

The corresponding relation for K is

-

-

(9-95)

1

=T

Ko

1

-

2-2p cosp

2|p|cos & + |p|²

1

+

1

Pi

1

30

or

Kv

X = T [i

22p cosp

Zi

Pi

- 2 pcos + |p|²

1

-

21/1

-

1

1

-

21

Pi

--

--

1

1 Z1

-

1

21

(9-96)

(9-97)

The effects of the addition of the dipole near $z = 1$ on the transient response and the error coefficient K , will now be investigated. For a unit step displacement input the residue of $C(z)/z$ at the pole p is given by

$$\lim_{z \rightarrow p} (z - p) M(z) \times (z - p)$$

Substituting Eq. (9-95) into the last equation, we have

$$\lim_{z \rightarrow p_1} (z - p_1) \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)(p_i - z_i)} \\ = \frac{K_p}{(1 - p_2)(1 - p_i)(p_1 - z_1)(p_i - z_i)}$$

$$\lim_{z \rightarrow z_1} (z - z_1) \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)(p_i - z_i)} \\ = \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_i - z_i)}$$

(9-98)

(9-99)

If p_1 and z_1 are much closer to the $z = 1$ point than the closed-loop

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poles and zero are in the original configuration, the residue is approximately

or

$$\lim_{z \rightarrow p_i} (z - p_i) \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)(p_i - z_i)} \\ = \frac{K_p}{(1 - p_1)(1 - p_2)(p_1 - z_1)(p_i - z_i)}$$

$$\lim_{z \rightarrow z_i} (z - z_i) \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)(p_i - z_i)} \\ = \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)}$$

$$\lim_{z \rightarrow p_i} (z - p_i) \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)(p_i - z_i)} \\ = \frac{K_p}{(1 - p_1)(1 - p_2)(p_1 - z_1)(p_i - z_i)}$$

$$\lim_{z \rightarrow z_i} (z - z_i) \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)(p_i - z_i)} \\ = \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)}$$

$$\lim_{z \rightarrow p_i} (z - p_i) \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)(p_i - z_i)} \\ = \frac{K_p}{(1 - p_1)(1 - p_2)(p_1 - z_1)(p_i - z_i)}$$

$$\lim_{z \rightarrow z_i} (z - z_i) \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)(p_i - z_i)} \\ = \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)}$$

$$\lim_{z \rightarrow p_i} (z - p_i) \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)(p_i - z_i)} \\ = \frac{K_p}{(1 - p_1)(1 - p_2)(p_1 - z_1)(p_i - z_i)}$$

$$\lim_{z \rightarrow z_i} (z - z_i) \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)(p_i - z_i)} \\ = \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)}$$

$$\lim_{z \rightarrow p_i} (z - p_i) \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)(p_i - z_i)} \\ = \frac{K_p}{(1 - p_1)(1 - p_2)(p_1 - z_1)(p_i - z_i)}$$

$$\lim_{z \rightarrow z_i} (z - z_i) \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)(p_i - z_i)} \\ = \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)}$$

$$\lim_{z \rightarrow p_i} (z - p_i) \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)(p_i - z_i)} \\ = \frac{K_p}{(1 - p_1)(1 - p_2)(p_1 - z_1)(p_i - z_i)}$$

$$\lim_{z \rightarrow z_i} (z - z_i) \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)(p_i - z_i)} \\ = \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)}$$

$$\lim_{z \rightarrow p_i} (z - p_i) \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)(p_i - z_i)} \\ = \frac{K_p}{(1 - p_1)(1 - p_2)(p_1 - z_1)(p_i - z_i)}$$

$$\lim_{z \rightarrow z_i} (z - z_i) \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)(p_i - z_i)} \\ = \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)}$$

$$\lim_{z \rightarrow p_i} (z - p_i) \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)(p_i - z_i)} \\ = \frac{K_p}{(1 - p_1)(1 - p_2)(p_1 - z_1)(p_i - z_i)}$$

$$\lim_{z \rightarrow z_i} (z - z_i) \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)(p_i - z_i)} \\ = \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)}$$

$$\lim_{z \rightarrow p_i} (z - p_i) \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)(p_i - z_i)} \\ = \frac{K_p}{(1 - p_1)(1 - p_2)(p_1 - z_1)(p_i - z_i)}$$

$$\lim_{z \rightarrow z_i} (z - z_i) \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)(p_i - z_i)} \\ = \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)}$$

$$\lim_{z \rightarrow p_i} (z - p_i) \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)(p_i - z_i)} \\ = \frac{K_p}{(1 - p_1)(1 - p_2)(p_1 - z_1)(p_i - z_i)}$$

$$\lim_{z \rightarrow z_i} (z - z_i) \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)(p_i - z_i)} \\ = \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)}$$

$$\lim_{z \rightarrow p_i} (z - p_i) \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)(p_i - z_i)} \\ = \frac{K_p}{(1 - p_1)(1 - p_2)(p_1 - z_1)(p_i - z_i)}$$

$$\lim_{z \rightarrow z_i} (z - z_i) \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)(p_i - z_i)} \\ = \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)}$$

$$\lim_{z \rightarrow p_i} (z - p_i) \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)(p_i - z_i)} \\ = \frac{K_p}{(1 - p_1)(1 - p_2)(p_1 - z_1)(p_i - z_i)}$$

$$\lim_{z \rightarrow z_i} (z - z_i) \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)(p_i - z_i)} \\ = \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)}$$

$$\lim_{z \rightarrow p_i} (z - p_i) \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)(p_i - z_i)} \\ = \frac{K_p}{(1 - p_1)(1 - p_2)(p_1 - z_1)(p_i - z_i)}$$

$$\lim_{z \rightarrow z_i} (z - z_i) \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)(p_i - z_i)} \\ = \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)}$$

$$\lim_{z \rightarrow p_i} (z - p_i) \frac{K_p}{(1 - p_1)(1 - p_2)(1 - p_i)(p_1 - z_1)(p_i - z_i)} \\ = \frac{K_p}{(1 - p_1)(1 - p_2)(p_1 - z_1)(p_i - z_i)}$$

The sampling period is 0.1 second. The magnitude and angle of the complex poles of C/R are $p = 0.352$ and 59.2° , respectively. The velocity error constant of the original system, computed from Eq. (9-94), is 9.6 sec^{-1} . Let us assume that the velocity error constant is to be increased to 20 by the addition of a dipole near $z = 1$. The added transient term resulting from the addition of the dipole is to have an initial amplitude less than 5 percent. With the addition of the dipole near $z = 1$, the resulting closed-loop transfer function is written

$$C(z) = \frac{0.822(1 - p)(z - z_1)}{R(z)(1 - z)(z^2 - 0.362z + 0.1)(z - p_1)}$$

(9-103)

For a unit step input the residue of $C(z)/z$ at the pole p is approximately

$$= \frac{0.822(-1 + z_1)}{1 - p}$$

z_i

(9-104)

If the transient step response is not to be affected by the dipole by more than 5 percent, we set

$$\frac{1 - p_i}{1 - z_i} = 1.05$$

Also, with the dipole K , is given by Eq. (9-97). Thus

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K_v

$$= \frac{1}{9.6}$$

$$= \frac{1}{z_i p_i}$$

$$0.1(2z_1 - 21)(1 - p_i)$$

For K , 20, Eq. (9-106) reads

$$= \frac{1}{z_i p_i}$$

z_i

$$= (-0.54)$$

p_i

(9-106)

(9-107)

Solving Eqs. (9-105) and (9-107) simultaneously, the pole and zero positions are found to be

$$p_1 = 0.9028$$

$$z_i =$$

$$0.9074$$

(9-108)

(9-109)

The final pole-zero configuration is shown in Fig. 9-29, and the closed-loop transfer function is

$$\frac{C(z)}{R(z)} = \frac{0.363(0.1)(x - 0.9074)}{(z^2 - 0.36z + 0.1)(z - 0.9028)}$$

z-plane

(9-110)

$$0.18 + j0.31$$

$$0.9074$$

$$0.9028$$

$$0.18 - j0.31$$

Unit circle

Fig. 9-29 Pole-zero configuration of $C(z)/R(z)$ for example of using dipole to increase K .

As an aid to the synthesis procedure, the first terms on the right side of Eqs. (9-94) and (9-97) are plotted in Fig. 9-30 for $\alpha = 0.5$ and $\alpha = 0.707$.

Note that the method of increasing K by adding a dipole near $z = 1$ to the closed-loop transfer function is the well-known phase-lag compensation. This correspondence is easily identified if we plot the inverse root locus of the example given above. In Fig. 9-31 the sketch of the inverse root locus of this example is based on the pole-zero configuration in Fig. 9-29. The open-loop poles and zeros are found on the inverse root locus plot. An integrating dipole is located near $z = 1$ with the pole to the right of the zero. An alternate method of increasing K , is to add a single zero to

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9

8

6

2

$$\alpha = 0.707$$

$$\alpha = 0.5$$

$$15^\circ$$

$$30^\circ$$

$$45^\circ \quad 60^\circ \quad 75^\circ$$

$$90^\circ$$

Fig. 9-30 Plot of

$$\alpha = 0.707$$

$$2 - 2\alpha \cos \theta$$

$$1 - 2\alpha \cos \theta + \alpha^2$$

versus θ for $\alpha = 0.5$ and

z-plane

Closed-loop

poles

Open-loop and

Open-loop

poles

Unit circle

closed-loop zeros

Fig. 9-31 Inverse root locus plot of the example of using dipole to increase K . The open-loop zero-pole configuration corresponds to integral control.

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$C(z)/R(z)$. As a matter of fact, the addition of a zero to C/R corresponds to the phase-lead compensation discussed previously. For instance, consider that in the preceding example of increasing K_v , a zero is to be added at $z = 22$. In order to make $D(z)$ physically realizable, the addition of z must be accompanied by a pole p located very near the origin. Thus the velocity error constants is given by

$$\frac{1}{1 - 0.1} = \frac{1}{0.9} = 1.111$$

$K_v = 20 \times 9.6 \times 1.111 = 213.3$

(9-111)

If p , is located very near the origin in the z -plane, the second term on the right side of Eq. (9-111) is approximately 0.1. Eq. (9-111) becomes

$$\frac{1}{1 - 0.1} = \frac{1}{0.9} = 1.111$$

$K_v = 20 \times 9.6 \times 1.111 = 213.3$

(9-112)

Solving for z , from the last equation yields

$$z = 0.35$$

(9-113)

The pole-zero configuration of the final $C(z)/R(z)$ is shown in Fig. 9-32a. The correspondence of this scheme of increasing K , with the phase-lead compensation is clearly shown by the inverse root locus plot of Fig. 9-32b.

The bandwidth of the closed-loop system is well known to increase in the case of the phase-lead compensation, whereas the phase-lag compensation usually decreases the bandwidth. The increase in band-

z -plane
 z -plane
 Pole very
 near the
 $0.18 + j0.31$
 origin
 Closed-loop
 poles
 $10.1 \pm j0.35$
 $0.18 - j0.31$
 (a)
 Unit circle
 Open-loop
 poles
 Unit circle
 Open-loop and
 closed-loop zeros
 (b)

Fig. 9-32 (a) Pole-zero configuration of $C(z)/R(z)$ for example of using a simple pole to increase K . (b) Inverse root locus plot of the example of using a simple zero to increase K . The open-loop pole-zero configuration corresponds to derivative control.

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width is particularly objectionable in sampled-data control systems, since the passage of high frequency harmonics generated by the sampler may cause hidden oscillations in the output response. Also, because of the inherently low relative stability or instability of a sampled-data system, a phase-lead compensation seldom gives satisfactory compensation results. The addition of a simple zero to $C(z)/R(z)$ (accompanied by a pole near the origin) at any place on the real axis other than the origin or near the origin has the effect of causing larger overshoot in the step response and wider bandwidth. The synthesis is complicated by the fact that the transient response is no longer controlled just by the zero z , and the basic control poles p_1 and p_2 . Therefore, with the addition of the zero the design curves of Figs. 6-20 and 6-21, which are prepared on the basis of the simple P_1, P_2, z , configuration, are no longer valid for the prediction of the transient response and have to be modified. In view of the reasons given above, the method of adding a zero for increasing K , is not generally preferred in the synthesis of a sampled-data control system. The dipole method is considered to be more effective and straightforward.

Example of the Synthesis Method

The method of synthesis is now illustrated by the following examples. The examples are chosen for the purpose of facilitating the understanding of the basic elements of the design procedure.

Example 9-6

The fixed parts of a sampled-data system with zero-order hold is described by the transfer function

$$G_p(s) = \frac{(1 - e^{-Ts})}{s^2 (s + 1)}$$

(9-114)

The sampling period is 0.1 second.

Design a digital controller to realize the following specifications:

- (a) $\zeta = 0.707$
- (b) $T_{max} \leq 0.3$ sec.
- (c) Overshoot ≤ 10 per cent.
- (d) $K, 5 \text{ sec}^{-1}$.

The selection of a suitable $C(z)/R(z)$ is described by the following steps:

- (1) The z -transfer function of the fixed parts of the system is given by

$$G_p(z) = \frac{0.005(\approx 0.9)}{(1 - \approx 0.9)}$$

$$\frac{0.905}{(9-115)}$$

Since $G_p(z)$ has one more pole than zero, the basic configuration of $C(z)/R(z)$ given by Eq. (9-85) can be used. Therefore,

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$$C(z)$$

$$(1 - p)(1 - p_2) (\approx - \approx 1)$$

$$R(z) = (1 - z_1)(\approx - p_1) (\approx - f_2)$$

$$(9-116)$$

and the condition of the physical realizability of the digital controller $D(z)$

is satisfied.

(2) Since the zero of $G(z)$ is located inside the unit circle, the zero z_1 can be chosen freely.

(3) As Fig. 6-20 shows for 0.707 and an overshoot of 10 percent, the angle α must lie in the range between -80° and $+18^\circ$, where α is an angle determined by the pole-zero configuration of Eq. (9-116) and is illustrated in Fig. 6-19.

max

=

0.3

(4) For α at the extremes of the range ($-80^\circ \leq \alpha \leq +18^\circ$) and T sec, determine the corresponding range for using either Eq. (6-67) or Fig. 6-21, where is the angle of p_1 .

Therefore, when $\alpha = -80^\circ$,

and

$T_{\max} / T = 215^\circ$

=

$0.1 \times 215^\circ$

0.3

=

71°

When $\alpha = +18^\circ$,

$T_{\max} / T = 120^\circ$

and

$0.1 \times 120^\circ$

0.3

=

40°

(9-117)

(9-118)

(9-119)

(9-120)

(5) Selecting $\alpha = 0^\circ$ from Fig. 6-21 or Eq. (6-76), we have

For $T_{\max}$

$T_{\max} / T =$

=

135°

0.3 sec, Eq. (9-121) gives $> 45^\circ$.

(9-121)

In order that the bandwidth of the system may not be too high, the angle should not be too large. Therefore, we choose $= 50^\circ$. From Eq. (4-27),

p is determined as

$|p|$

=

-85

$\exp 57.3 \sqrt{1 - 5^2}$

= $\exp$

2

- 50

57.3

-

0.42

(9-122)

0° , and $p = 0.42$, z_1 is found to be equal to

For 50° , α

0.0737.

=

When the values of p , α , and z_1 are substituted into Eq. (9-49), the velocity error constant is

which is satisfactory.

K_v

-

8.2 sec⁻¹

(9-123)

Based on the above design, the closed-loop transfer function of the synthesized system is

$C(z)$

$0.688(z - 0.0737)$

=

$R(z) = 22(0.54\% + 0.178)$

(9-124)

The pole-zero configuration of $C(z)/R(z)$ is illustrated in Fig. 9-33.

(6) When the values of p , α , and z , are substituted into Eq. (9-90), the transfer function of the digital controller is given by

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$p = 0.42$

$\alpha = 0^\circ$

z -plane

$0.27 + j0.32$

0.0737

1

$0.27 - j0.32$

Unit circle

Hence

Fig. 9-33 Pole-zero configuration of the closed-loop transfer function $C(z)/R(z)$ for the system designed in Example 9-6.

$D(z) =$

$D(z)$

=

$0.688(z - 0.0737)$

-

1

X

$1)(z - 0.23) G_p(z)$

-

138(z

$0.0737)(z$

$(z - 0.23)(z - 0.905)$

$0.9)$

(9-125)

(9-126)

As we can see, $D(z)$ is physically realizable and can be realized by any one of the methods described in Sec. 9.2.

=

For $\alpha = 0^\circ$ and 0.707 , the maximum overshoot of the step response of the system as predicted from Fig. 6-20 is 6.6 percent. The peak time, computed from Fig. 6-21, is 0.27 sec. The actual step response of the system at the sampling instants is determined by substituting $R(z) = 2/(1 - z^{-2})$ into Eq. (9-124). Therefore,

$C(z)$

=

$0.688(z - 0.0737)$

$(z^2 - 1)(z^2 - 0.54z + 0.178)$

Expanding the last equation by long division yields.

$C(z)$

-

$= 0.688z^{-1} + 1.009z^{-2} + 1.058z^{-3} + 1.027z^{-4} + 2z^{-5} + 0.992z^{-6}$

$\dot{c}(nT)$
 $\dot{c}(nT)$

The step response of the system at the sampling instants is plotted in Fig. 9-34. Note that $c(nT)_{\max}$ is equal to 1.058 at $nT = 0.3$ sec, and corresponds to a maximum overshoot of 5.8 percent and a peak time of 0.3 sec. We should not expect that the z-transform solution would give the exact maximum overshoot and peak time of the continuous output. However, in this case the z-transform solutions are quite close to the actual quantities. The following tabulation gives a comparison of the results and the specified quantities:

	Specified	Designed
$T_{\max}$	0.3 sec	0.27 sec
Overshoot	<10%	6.6%
K_v	≥ 5	8.2

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1.2
 1.0
 0.8
 0.6
 0.4
 0.2
 1.095
 1.058

|
 Step response of system
 designed in Ex.9-6 ($K_v=5$)
 Step response of system
 with dipole compensation
 in Ex.9-7 ($K_v = 20$)

0
 0.1
 0.2
 0.3 0.4
 0.5
 0.6
 0.7
 0.8

Fig. 9-34 Step responses of systems designed in Examples 9-6 and 9-7.

Example 9-7

Consider that the same $G(z)$ and the same specifications of the last example are used except that the bandwidth of the system is to be kept approximately the same as the previous system and K_v is increased to 20 sec^{-1} . In order to increase K_v , without appreciably affecting the bandwidth, a dipole may be added near $z = +1$ to the closed-loop transfer function $C(z)/R(z)$ selected for the last example. The final system function is now

$C(z)$
 $R(z)$

$$0.688(1 - p) (\approx - 0.0737) (\approx - z_1) \\ (1 - \approx z_2)(\approx^2 - 0.54z + 0.178)(\approx P_i) \\ (9-129)$$

Since the maximum overshoot is specified as less than 10 percent and since the basic pole-zero configuration of C/R without the dipole corresponds to a system with an overshoot of 6.6 percent, the added transient term must, as a result of the addition of the dipole, have an amplitude less than 3 percent. Thus we set

$$\frac{1}{P_i} \\ = 1.03 \\ \frac{1}{Z_i} \\ (9-130)$$

From Eq. (9-97) the velocity error constant is written

$$\frac{1}{P_i} \\ -0.1 \\ K_z = 20.82 \\ \text{Solving Eqs. (9-130) and (9-131) simultaneously yields}$$

$$\frac{1}{Z_i} \\ ((\approx + 1) (p)) \\ (9-131)$$

$$\frac{1}{P_i} \\ Z_1 = \\ = 0.9595 \\ (9-132)$$

and

$$\frac{1}{P_i} \\ = \\ = 0.9583 \\ (9-133)$$

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The final system transfer function is

$$\frac{C(z)}{R(z)} \\ ****$$

$$0.9595) \\ 0.7086(z - 0.0737) (z \\ (\approx^2 - 0.54\approx + 0.178)(z - 0.9583) \\ (9-134)$$

For a unit step function input C(z) is expanded into a power series by long division.

$$\frac{3}{C(z)} = 0.7086z^{-1} + 1.04x^2 + 1.095\approx^{-3} + 1.06\approx^{-} + 1.02z^{-5} \\ + 0.99\approx^{-6} + 0.96\approx^{-7} + 0.93z^{-8} + \\ (9-135)$$

The step response of the system at the sampling instants is now plotted in Fig. 9-34. Note that the peak time is approximately 0.3 sec and that the maximum overshoot is 9.5 percent. All these quantities satisfy the specifications.

By substituting Eq. (9-134) into Eq. (9-83), we obtain the transfer function

of the digital controller.

$$D(z) = \frac{142(\approx -0.0737)(\approx -0.96)(\approx -0.905)}{(\approx +0.983)(\approx +0.223)(\approx -0.9)}$$

Clearly $D(z)$ is stable and physically realizable.

(9-136)

The system designed in this example has a velocity constant four times greater than that of the system designed in the previous example. When the step responses of the two systems are compared, the overshoot of the dipole compensated system has only about 3 percent higher overshoot. However, the transient response of the dipole compensated system contains a long tail of small amplitude. This long-duration tail is anticipated in all systems with integral compensation, since the added transient term due to the dipole has a time constant of $1/\rho t$, which is long compared to the original time constants of the system.

In this section, an attempt has been made to present an extension of Truxal's synthesis method of the continuous-data systems to the synthesis of sampled-data systems. The synthesis procedure is aimed toward reducing the complexity of the problem to the point where the dynamic characteristics of the system can be described by two control poles and one zero in the z -plane. The synthesis procedure essentially involves a selection of these control poles and zero to meet the performance specifications and a physically realizable digital compensator. The compensator need not be stable, although stability is preferable. Additional poles may be added to $C(z)/R(z)$ near the origin of the z -plane if the fixed part of the system has an excess of poles over zeros of more than one. However, the realizability of $D(z)$ in sampled-data systems is not nearly as critical as in the continuous-data case, since a third order continuous-data plant with hold, say

$G_p(s)$

=

$$\frac{(1 - e^{-Ts})}{s(s+1)(s+2)}$$

(9-137)

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corresponds to a z -transfer function $G(z)$ with three poles and two zeros, producing an excess of poles over zeros of only one. A dipole may be added near $z = +1$ in the z -plane to increase the velocity error constant without appreciably affecting the bandwidth and the transient overshoot of the designed system.

9.6 DESIGN OF SAMPLED DATA SYSTEMS WITH

"MINIMAL" OR "DEADBEAT" RESPONSES+

In the preceding sections the design and synthesis of sampled-data control systems have relied to a great extent on the extension of the techniques of synthesis and design of continuous-data control systems. The criteria for the frequency-domain design of sampled-data control systems are generally expressed in the form of the phase and gain margins, resonant peak, and bandwidth. The design may be viewed as the shaping of the frequency locus of the system to obtain satisfactory relative stability. In the synthesis of sampled-data systems, the criteria of performance are expressed in the form of time domain specifications, such as the overshoot, peak time, and damping ratio.

Because of the flexibility in the application and realizability of a digital controller, a sampled-data system can be designed with minimum settling time and zero steady-state error at the sampling instants when a specific test signal such as a step, ramp, or parabolic is applied. With reference to the system configuration of Fig. 9-35, $R(z)$

$E(z)$
 $D(s)$
 $G(s)$
 $C(s)$
 $G_1(s)$
 $D(z)$
 $G_p(z)$

Fig. 9-35 Single-loop sampled-data feedback control system.

a physically realizable digital controller $D(z)$ can be designed so that for a particular test input the outputs at the sampling instants are as specified. In essence the design criteria are as follows:

(1) The system must have zero steady-state error at the sampling instants for the specified input test signal.

(2) The rise time should equal a minimum number of sampling periods.

(3) The settling time, measured at the sampling instants, should be finite.

† J. E. Bertram, "Factors in the Design of Digital Controllers for Sampled-data Feedback Systems," Trans. A. I. E. E., vol. 75, Part 2, July 1956, pp. 151-159.

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(4) $D(z)$, $G(z)$, and $C(z)/R(z)$ must all be physically realizable.

A response that satisfies all these properties is referred to as a "minimal" or "deadbeat" response.

The design of the "minimal" response system in a sense has the same philosophy as the synthesis method described in the preceding section. The design procedure starts with the selection of an over-all system function $M(z)$ meeting the specifications of minimum rise time, finite settling time, and zero steady-state error and simultaneously resulting in a physically realizable controller. To obtain such complete control over the transient behavior of the system requires specification of both the zeros and poles of the over-all system function $M(z)$. However, in view of the four design criteria listed above and of the selection of the over-all minimal response function, $M(z)$ for a given input signal is likely to have restrictions. These restrictions are found by considering the criteria and realizability conditions in the following:

Steady-State Error and Finite Settling Time

As has been shown previously, the transfer function of the digital controller in the system of Fig. 9-35 is given by

$$D(z) = \frac{M(z)}{1 - M(z)G(z)} \quad (9-138)$$

where $M(z)$ denotes the over-all system function $C(z)/R(z)$. We shall show that in order to obtain zero steady-state error in response to test signals of the form $r(t) = t^m$, constraints must be imposed on the function $1/M(z)$ which appears in the denominator of $D(z)$. The z-transform of the error signal of the system is written as

$$E(x) = R(x)[1 - M(z)] \quad (9-139)$$

Applying the final-value theorem to the last equation, the steady-state error is given by

$$\lim_{N \rightarrow \infty} e(nT) = \lim_{z \rightarrow 1} (1 - z^{-1})R(z)[1 - M(z)] \quad (9-140)$$

If one of the requirements is that the steady-state error be zero,

we equate Eq. (9-140) to zero.

$$\lim_{n \rightarrow \infty} e(nT) = \lim_{z \rightarrow 1} (1 - z^{-1})R(z)[1 - M(z)] = 0$$

$z=1$

(9-141)

From the z-transform table, we see that for the type of input considered (tm), the z-transform is of the form

$R(z)$

$=$

$A(z)$

$(1 - z^{-1})^n$

(9-142)

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where $n = m + 1$ and that $A(z)$ is a polynomial in z^{-1} with no zeros at $z = 1$. Substitution of Eq. (9-142) into Eq. (9-141) yields

$$\lim_{n \rightarrow \infty} e(nT) = \lim_{z \rightarrow 1} (1 - z^{-1})^n A(z) [1 - M(z)] = 0 \quad (9-143)$$

$=$
 7000

$(1 - z^{-1})^n$

72

$z^{-1}) A(z) [1 - M(z)] = 0$ (9-143)

$z=1$

-

-

Since $A(z)$ does not have zeros at $z = 1$, the necessary condition for the steady-state error to be zero in Eq. (9-143) is that $1 - M(z)$ contain the factor $(1 - z^{-1})^n$ which will cancel the same factor in the denominator. Thus $1 - M(z)$ must have the form

-

$$1 - M(z) = (1 - z^{-1})^n F(z)$$

-

1

(9-144)

where $F(z)$ is a polynomial (to be chosen) in z^{-1} with no zeros at $z = 1$. Solving for $M(z)$ in Eq. (9-144), we have

or

-

$$M(z) = 1 - (1 - z^{-1})^n F(z)$$

$=$

$M(z)$

$=$

-

-

$$1 - (1 - z^{-1})^n F(z)$$

$z=1$

(9-145)

(9-146)

Since $F(z)$ is assumed to be a polynomial in z^{-1} , it has only poles at $z = 0$. Note that the characteristic equation of the system with zero steady-state error is of the form

$$1 - M(z) = 0$$

(9-147)+

where p is an integer greater or equal to n . Therefore, in order to have the steady-state error die out within a finite time interval when the input signal is t^n , the characteristic equation of the system should have the form described by Eq. (9-147).

Physical Realizability Considerations

The limitations which are imposed on the forms of $G(z)$ and

$D(z)$ by their physical realizability also restrict the selection of

the over-all system function $M(z)$. In general, physically realizable $D(z)$ and $G_p(z)$ can be expressed by the following expressions:

$$D(z) = \frac{d_n z^n + d_{n+1} z^{n+1} + \dots + d_{n+k} z^{n+k}}{1 + d_{n+1} z^{-1} + \dots + d_{n+k} z^{-k}}$$

(9-148)

and

$$G_p(z) = \frac{g_p z^p + g_{p+1} z^{p+1} + \dots + g_{p+k} z^{p+k}}{1 + g_{p+1} z^{-1} + \dots + g_{p+k} z^{-k}}$$

(9-149)

where $n \geq 0$ and $p \geq 0$. Substituting the last two equations into the closed-loop transfer function $M(z)$ yields

In general $F(z)$ need not be a polynomial; it can be a rational function in z^{-1} as long as it has no poles or zeros at $z = 1$, and no poles outside the unit circle. Then,

the characteristic equation of the system with zero steady-state error is of the form

$$z^3 F_a(z) = 0, \text{ where } F_a(z) \text{ denotes the denominator polynomial of } F(z).$$

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$M(z)$

=

=

$C(z)$

$R(z)$

$$\frac{(d_n z^n + d_{n+1} z^{n+1} + \dots + d_{n+k} z^{n+k}) (g_p z^p + g_{p+1} z^{p+1} + \dots + g_{p+k} z^{p+k})}{1 + (d_{n+1} z^{-1} + d_{n+2} z^{-2} + \dots + d_{n+k} z^{-k}) (g_{p+1} z^{-1} + g_{p+2} z^{-2} + \dots + g_{p+k} z^{-k})}$$

(9-150)

$M(z)$ can be expanded into a power series in z^{-1} . Therefore,

$$M(z) = m_m z^{-m} + m_{m+1} z^{-(m+1)} + \dots$$

where $m = n + p$.

$-m-1$

.

(9-151)

The above discussion simply verifies that if $G_p(z)$ and $D(z)$ are physically realizable, the over-all system function is also physically realizable. More important, Eq. (9-151) indicates that to ensure that

$D(z)$ is physically realizable, the lowest power of the series expansion of $M(z)$ in z must be at least as large as the lowest power in z^{-1}

in the series expansion of $G_p(z)$. Comparing Eq. (9-151) with Eq. (9-145), we note that the selection of $F(z)$ is not entirely arbitrary.

In order that the system may have a deadbeat response and simultaneously result in a physically realizable $D(z)$, $F(z)$ should be so chosen that $M(z)$ does not contain the constant term 1; this means that $F(z)$ should contain the constant term 1.

As an illustration, consider that the controlled process of the sampled-data system of Fig. 9-35 is described by the z -transfer function

$$G_p(z) = 1 - \alpha z^{-1} - \alpha^2 z^{-2} - \dots - \alpha^{N-1} z^{-(N-1)}$$

(9-152)

Note that the impulse sequence of the controlled process given by the last equation has a time lag equal to two sampling instants.

Let us assume that a digital controller is to be designed so that

the step response of the system is of the minimal type.
 For a unit step input, $R(z) = 1/(1 - z^{-1})$; the function $1 - M(z)$ must have the form

•

-

$$1 - M(z) = (1 - z^{-1}) F(z)$$

(9-153)

However, we shall show that the fastest response that can be obtained from this system is a step response which settles to zero steady-state error in two sampling periods. The fastest step response that can be obtained from this system is when

$$M(z) = z^{-2}$$

2

(9-154)

since the highest power in z^{-1} of the series expansion of $M(z)$ should be greater or equal to the highest power in z^{-1} of the series expansion of $G(z)$. We shall show that selection $M(z) = z^{-2}$ as the over-all system function will result in a digital controller that

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is physically unrealizable. Substitution of $M(z) = z^{-2}$ and Eq. (9-152) into Eq. (9-138) yields

$$D(z) =$$

z^{-1}

1

$$= \frac{1}{z^2}$$

2

z

1

2

-

x

2

=

1

-

z

z^{-2}

- 2

(9-155)

which evidently is not a physically realizable pulse transfer function. For $M(z) = z^{-2}$, the transfer function of the digital controller is

$$D(z) =$$

-

1

2

-

z

1

2

(9-156)

which is realizable.

In this case, the function $F(z)$ is given by

$$F(z) =$$

$$\frac{1 - M(z)}{1 - z^{-1}}$$

1

-

-

$$1 + z^{-1}$$

1

(9-157)

Therefore, the restriction on the selection of $F(z)$ when $G_c(z)$ has a time lag has been demonstrated by this example. In general, we can show that if $G_c(z)$ has either time lag or zeros on or outside the unit circle, $F(z)$ cannot be chosen arbitrarily. The case when $G_c(z)$ has zeros on or outside the circle will be discussed later. The example just given also illustrated the constraints imposed on the possible form of $M(z)$ when $G_c(z)$ has a time lag. However, as will be shown in the following paragraphs, $G_c(z)$ has poles on or outside the unit circle, additional restriction on $M(z)$ is imposed

Input function

TABLE 9-2

 $r(t)$ Step $u(t)$

-

Ramp t $R(z)$ $1 - M(z)$

1

2

 $(1 - x^{-1}) F(x)$

 Tz^{-1} $(1 - x^{-1})^2$ $(1 - z^{-1})^2 F(z)$ Parabolic t^2

-

 $(1 - x^{-1})^3 F(z)$ $T^2 z^{-1} (1 + x^{-1})$ $(1 - z^{-1})^3$

-

-

in that $M(z)$ should also contain these unstable poles of $G_c(z)$.

Tabulated in Table 9-2 are the expressions of $1/M(z)$ necessary for a sampled-data system to have a minimal type of output in response to a step, ramp, and parabolic input.

=

If the controlled process $G_c(z)$ has no time lag and all its zeros are inside the unit circle, then $F(z)$ can be chosen to be unity. With $F(s) = 1$, from Table 9-2, the over-all system functions of systems with minimal outputs in response to step, ramp, and parabolic inputs are determined and given as follows:

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Step input

Ramp input

 $M(x) = z^{-1}$

1

2

(9-158)

 $M(z)$ $= \frac{z-1}{z^2-1}$

2

-

(9-159)

 $M(z) = \frac{3z-1-3z^2}{z^2-1}$

2

(9-160)

Parabolic input

Note that when the system is designed to yield a minimal response at the application of a step input, the transient settles in one sampling period. When the system is designed on the basis of a ramp input, the transient settles in two sampling periods, and so on. In general, for an input of the form t^m , the minimum settling time of minimal response system is $m + 1$ sampling periods. The following example illustrates the design of a sampled-data system with minimal response.

Example 9-8

Consider that the controlled process, including the hold device, of the system of Fig. 9-35 is described by

$$G_p(s) = \frac{1 - e^{-Ts}}{s^2 (s + 1)} \quad (9-161)$$

The sampling period is 0.1 second. Design a digital controller so that the system has a minimal response at the sampling instants when the input is a unit step function. The transfer function of the system is purposely chosen to be the same as that of the system in Example 9-7 so that the techniques and philosophies of the present minimal design may be compared with the synthesis method presented in Sec. 9.5

The z-transform of $G_p(s)$ is

$$G_p(z) = \frac{0.0052z^{-1}(1 - 2z^{-1})(1 - 0.92z^{-1})}{(1 - 0.9052z^{-1})} \quad (9-162)$$

Since $G_p(z)$ does not have a time lag and since all its zeros are inside the unit circle in the z-plane, $F(z)$ is chosen to be 1. From Table 9-2, hence

$$1 - M(z) = 1 - z^{-1}$$

$$M(z) = 1 - z^{-1}$$

For a unit step input, the z-transform of the system output is

$$C(z) = \frac{1 - z^{-1}}{1 - 0.9052z^{-1}}$$

$$= 0.9052 + 0.0948z^{-1} + 0.0052z^{-2} + \dots$$

The step response at the sampling instants is shown in Fig. 9-36.

$$c(nT) = u(1)$$

$$0.1 \quad 0.2 \quad 0.3 \quad 0.4 \quad 0.5 \quad 0.6 \quad 0.7 \quad 0.8 \quad 0.9$$

Fig. 9-36 Output response at the sampling instants of the sampled-data system designed in Example 9-8.

$$(9-163)$$

$$(9-164)$$

$$(9-165)$$

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Substitution of $M(z)$ and $G_p(z)$ into Eq. (9-138) yields

$$D(z) = \frac{1 - 0.9052z^{-1}}{1 - z^{-1}}$$

0.9052-1

0.92-1

(9-166)

which is physically realizable.

In the compensated system although the transient response settles to zero error in one sampling period in response to a step input, the performance of the system must be investigated when inputs other than a unit step are applied. Using Eq. (9-139), the error transform for a unit ramp input is written

$$E(z) = R(z)[1 - M(z)]$$

=

0.12-1

$$(1 - z^{-1})^2 (1 - z^{-1})$$

Hence

$E(z)$

0.12-1

=

=

1

0.1(x1+x

-

(9-167)

+ + .)

.

.

(9-168)

Step input

$r(t)$

$c(nT)$

Step input

$r(t)$ $c(nT)$

Step input

$C(NT)$ $r(1)$

0T 2T 4T 6T

0T 2T 4T 6T

0T 2T 4T 6T

$-e(nT)$

$e(nT)$

$e(nT)$

T 2T 3T

2T

3T

(a)

(b)

-2-

(c)

Ramp input

5r

5r

Ramp input

Ramp input

4 $r(1)$.

4-

(10) $\int y(t) dt$

$c(nT)$

3 $C(NT)$

0T 2T 4T 6T

0T 2T 4T

6T

0T 2T 4T 6T

$e(nT)$

to

$e(nT)$
 $Le(nT)$
 OT 27 4T 6T
 OT 27
 37
 (d)
 (e)
 (f)
 $r(t)$
 Parabolic
 input
 8 $r(1)$
 Parabolic
 input
 Parabolic
 8 $r(t)$ input
 6-
 4
 $c(nT)$
 $c(nT)$
 2F
 OT 27 4T
 6T
 OT 27 47 67
 OT 27 4T
 67
 6t $e(nT)$
 6t
 $e(nT)$
 $e(nT)$
 라
 2
 2
 OT 27 4T 6T
 OT 27
 4T 67
 07 27 37
 (g)
 (h)
 (i)

Figure 9-37

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The steady-state error is not zero when a ramp input is applied to the system designed to yield a minimal output in response to a step input. Similarly, for a unit parabolic input t^2 the error transform is

$$E(z) = 0.01(z^{-1} + 3z^{-2} + 5z^{-3} + 7z^{-4} + 9z^{-5} + \dots)$$

Evidently, the magnitude of the error sequence increases with time.
(9-169)

Referring to the computations above, we may conclude that the designed system has a minimal response at the sampling instants only when the input is a step function. When the system is subjected to an input other than a step function, the response is not only not minimal but is not acceptable by any standards. This characteristic, however, is typical of the minimal response design. The designed system may be said to be "tuned" for a particular test signal, and it does not respond satisfactorily to any other type of input. Figure 9-37 illustrates the responses of the minimal response systems designed on the basis of the step, ramp, and acceleration

inputs. In each case the effect of having only minimal response for a particular input is clearly shown. For instance, Fig. 9-37b shows that the step response of a system which is designed to have a minimal response to a ramp input has an overshoot of 100 percent.

Comments on the Minimal Response Design

Both the minimal response design and the z-plane synthesis method rely heavily on the cancellation of the undesirable poles and zeros of $G(z)$. In this respect the philosophies of the two methods are quite similar. In the design for minimal response $D(z)$ is selected with poles and zeros cancelling the unwanted poles and zeros of $G(z)$ so that all the poles of the over-all system function lie at the origin in the z-plane. In the synthesis method described in Sec. 9.5, $D(z)$ is designed to cancel the undesirable poles and zeros of $G(z)$, and at the same time, poles and zeros of $D(z)$ are inserted at more appropriate places so that the over-all system satisfies the specified performance criteria.

Since the minimal design can result in a system whose output follows the input exactly after a minimum number of sampling periods with no overshoot and since the design procedure is straightforward, the minimal design would, at first glance, seem to be a far superior method than the z-plane synthesis and any of the other methods described previously. However, the minimal response design has at least the following disadvantages:

(1) The design refers to the response at only the sampling

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instants. A minimal response at the sampling instants does not necessarily assure a minimal response at all times.

(2) The design is optimum only for the specific input for which it is intended. The designed system does not provide satisfactory performance to other inputs.

(3) From the point of view of sensitivity, the multiple order pole of the closed-loop transfer function at the origin (or at any other point in the z-plane) is undesirable. It will be shown that the sensitivity of the multiple order pole due to a change in any system parameter k is infinite.^t

The sensitivity of the roots of the characteristic equation of a sampled-data system with respect to the incremental change of a system parameter k is defined as

$$S_1^k = \frac{dz}{dk} \frac{z}{z}$$

$$= \frac{dz}{dk} \frac{z}{z}$$

$$(9-170)$$

Assuming that the characteristic equation is a polynomial with constant coefficient, we can write it as

$$P(z) + kQ(z)$$

$$= 0$$

$$(9-171)$$

where k is a system parameter, such as the open-loop gain, $P(z)$ and $Q(z)$ are polynomials which do not contain k . Consider that k is varied by an increment Δk . Eq. (9-171) becomes

$$P(z) + (k + \Delta k)Q(z) = 0$$

Dividing both sides of Eq. (9-172) by $P(z) + kQ(z)$ yields

$$(9-172)$$

$$\frac{1 + \Delta k Q(z)}{P(z) + kQ(z)}$$

$$P(z) + kQ(z)$$

$$= 0$$

which can be written as

(9-173)

where

$$1 + A_k B(z) = 0$$

$$B(z)$$

$$=$$

$$Q(z)$$

$$P(z) + kQ(z)$$

(9-174)

(9-175)

Notice that the denominator of $B(z)$ contains the original characteristic equation. Expanding $B(z)$ by partial fraction expansion, in the neighborhood of a characteristic equation root z of multiplicity n , $B(z)$ is approximated by

† G. Franklin, discussion of paper by S. F. Schmidt, "Application of Continuous System Design Concepts to the Design of Sampled-data Systems," Trans. A. I. E. E.

Part 2 May 1959, pp. 78-79.

J. G. Truxal, I. M. Horowitz, "Sensitivity Considerations in Active Network Synthesis," Proceedings of the Second Midwest Symposium on Circuit Theory, East Lansing, Mich., December 1956. F. F. Kuo, "Pole-zero Sensitivity in Network Functions,"

I. R. E., Transactions on Circuit Theory, Vol. CT-5, December 1958, pp. 372-373.

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$$B(x)$$

$$=$$

$$A_1$$

$$(2-21)"$$

$$-$$

$$A_i$$

$$(Az)"$$

Substitution of Eq. (9-176) into Eq. (9-174) yields

$$1 +$$

$$A_k A$$

$$(Az)"$$

$$=$$

$$0$$

from which, we obtain

(9-176)

(9-177)

$$A_k - (Az)" - 1$$

$$=$$

$$Az$$

$$A_1$$

(9-178)

Taking the limit on both sides of Eq. (9-178) as Δk and Az approach zero, we have

$$\lim_{\Delta k \rightarrow 0} (A_k) = dk$$

$$-(Az)" - 1$$

$$= \lim_{\Delta k \rightarrow 0} A_1$$

$$= 0$$

$$Az \rightarrow 0$$

$$A_k \rightarrow 0$$

$$Az \rightarrow 0$$

(9-179)

Thus at a multiple order root of the characteristic equation, either

$dk/dz = 0$ or S is infinite. In other words, the minimal response design which places all poles of $C(z) R(z)$ at the origin of the z -plane has the system operating at a point of infinite sensitivity of pole position to parameter variations. Furthermore, Eq. (9-179) shows that the larger the number of poles placed at a single point, the more sensitive will be the pole locations to parameter changes. The derivations given above simply show that a minimal response is extremely sensitive to parameter changes in practical systems. Therefore, in the design of any system, if consideration is placed on the sensitivity of response to parameter changes, the use of the minimal response design must be precluded.

Design of Systems with Plant Poles and Zeros Outside the Unit Circle
 We know that the minimal response design of sampled-data control systems is achieved through cancellation of the undesired poles and zeros of the transfer function of the controlled process $G(z)$. Unfortunately, because of parameter variations and inaccuracy in the description of a physical system by a transfer function, perfect cancellation of poles and zeros are impossible in any practical system. It has been shown previously that an imperfect cancellation of either a zero or a pole of $G(z)$ outside the unit circle by the pole and zero of $D(z)$ may lead to a system which is unstable. For this reason the design procedures discussed previously have been limited

† R. H. Barker, "The Theory of Pulse-Monitored Servo and Their use for Prediction," Report No. 1046, Signals Research and Development Establishment, Christchurch, England, Nov. 1950.

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to only those systems with $G(z)$ which have poles and zeros inside the unit circle. For a system with a $G(z)$ that has poles and zeros outside the unit circle, a separate treatment is necessary. Let us first illustrate the effect of imperfect cancellation of poles and zeros of $G(z)$ that lie outside the unit circle. Consider that the controlled process is described by

$$G(z) = (z - z_0) A(z)$$

where z_0 is a zero and p , a pole on or outside the unit circle in the z -plane; $A(z)$ represents a rational function with all poles and zeros inside the unit circle. Assuming that a digital controller is designed with poles and zeros that will cancel the zero z_0 and pole p of $G(z)$ and some poles and zeros of $A(z)$, $D(z)$ is given by

$$D(z) = (z - z_a) A_d(z)$$

where z_a and p_a are the zero and pole which are to cancel p , and z_0 , respectively. Substitution of Eqs. (9-180) and (9-181) into the over-all system function yields

$$M(z) = \frac{D(z)G(z)}{1 + D(z)G(z)}$$

=
 $\frac{z_d}{z - p_a}$

Zd

-

-

$$A.(2) \quad A_c(z) = \frac{1 + (2 - 2^4)z^{-1}}{(2 - 2;)} A.(2) \quad A_a(z)$$

-

.

The characteristic equation of the system is

-

-

$$(z - p_a)(z - p_o) + (z - z_a)(1 - z^{-1})A(z)A_a(z) = 0 \quad (9-183)$$

Let us assume that imperfect cancellation occurs between z , and p_a , so that

but p_o

$$p_o = p_a + A_z,$$

Substituting Eq. (9-184) into Eq. (9-183) yields

-

-

$$(9-184) \quad (z - p_a)(z - p_o) + (z - z_a)(1 - p_a - A_z)A(z)A_a(z) = 0$$

(z

(9-185) Collecting terms and dividing through the last equation by the terms that do not contain A_z , we have

1+

-

$$-A_z, A(z)A_d(z)(z - z_a)$$

(z

--

$$(1 - p_a)(1 - p_o) + (1 - z_a)(1 - p_a)A(z)A_a(z)$$

-

-

Since $p_o = p_a + A_z$, Eq. (9-186) is simplified to

1 +

$$-A_z, A(z)A_a(z)$$

d

$$(1 - p_a)[1 + A(z)A_a(z)]$$

=

0

$$(9-186) \quad (A_z, 0)$$

$$= 0 \quad (A_z, 0) \quad (9-187)$$

The zeros of the last equation are the roots of the characteristic

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equation. Using the root locus theory we can see that, if A_z is considered as the variable parameter, one of the root loci of the characteristic equation starts at $z = p_a$ which is located outside the unit circle. Therefore, for small values of A_z , the root will be very close to p_a , and the closed-loop system is unstable.

If imperfect cancellation exists between p , and z_a as well as between p_a and z , we let

$$p_o = z_a + A_p.$$

$$(9-188)$$

In addition to the relationship of Eq. (9-184). Substituting Eq. (9-188) into Eq. (9-185) yields

-

-

-

$$(z - p_a)(z - z_a - A_p) + (z - z_a)(1 - p_a - A_z)A(z)A_a(z) = 0$$

(9-189)

By rearranging the terms, and then dividing through both sides of Eq. (9-189) by the terms which do not contain A_p , we have

$$-A_p(z - p_a)$$

-

$$1 +$$

$$(x - z_e)(1 +$$

$$-A_2, A, (2) A_d(2)$$

-

$$\approx -p_a[1 + A, (z) A_a(2)]$$

=

$$0$$

(9-190)

-

The zeros of the last equation are the roots of the characteristic equation when imperfect cancellations of the pole p , and the zero z , of $G, (z)$ are made by the pole and zero of $D(z)$. Observe also that the denominator of the second term on the left side of Eq. (9-190) is identical to the left side of Eq. (9-187) times $(z - z_a)$. Therefore, when A_p is considered a variable parameter, one of the root loci of Eq. (9-190) starts on the root locus of Eq. (9-187) which begins at $z = P.$, and one locus starts at $z = z_a$. Since both A_p , and A_z , are assumed to be small, the two roots of Eq. (9-190) on the two root loci starting at z , and p , are located very close to the starting points, and the system is unstable. Of course, if z , and P_o are located outside but very close to, or on the unit circle, imperfect cancellation of z , and p , may result in a system which is conditionally stable. On the other hand, if z , and p , are located just inside the unit circle, imperfect cancellation of the pole and zero may still lead to an unstable system.

The conclusion drawn from this discussion is that if a controlled system of a sampled-data control system has a pulse transfer function which contains zeros and poles which lie on or outside the unit circle in the z -plane, no attempt is to be made to cancel these poles and zeros with the poles and zeros of a digital controller, since imperfect cancellations may cause the system to be unstable. This simply imposes additional constraints on the selection of the over-all system transfer function for the minimal response design. Rewriting Eq. (9-138) yields

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$$D(z)$$

-

-

$$M(z)$$

$$1 - M(z) \wedge G_p(z)$$

$$1$$

$$X$$

(9-191)

If $G, (z)$ has poles and zeros that lie on or outside the unit circle in the z -plane, Eq. (9-191) can be written as

$$M(2)$$

-

$$II (1 - piz^{-1})$$

$$i$$

$$1$$

$$D(z)$$

=

$$1 - M(2) * II (1$$

$$X$$

(9-192)

$212^{-1})$

$A_1(z)$
where p and z represent the poles and zeros, respectively, of $G_1(z)$ that lie on or outside the unit circle. $A_1(z)$ is a rational function in z^{-1} with poles and zeros only inside the unit circle. Since $D(z)$ is not allowed to contain the poles p_1 and the zeros z , with reference to Eq. (9-192), we see that these poles and zeros of $G_1(z)$ must be cancelled by the poles and zeros of $M(z)$ and $1 - M(z)$.

For a response with finite settling time, $M(z)$ and $1 - M(z)$ are polynomials in z^{-1} . Therefore, to prevent $D(z)$ from containing the poles and zeros of $G_1(z)$ that are on or outside the unit circle, $M(z)$ must contain the terms $\prod (1 - p_i z^{-1})$ and $1 - M(z)$ must

contain $\prod (1 - p_i z^{-1})$. In general, $M(z)$ and $1 - M(z)$ have the following forms:

$$M(z) = \prod_{i=1}^n (1 - p_i z^{-1}) \prod_{j=1}^m (1 - z_j^{-1})$$

$$212^{-1})(\prod_{i=1}^m (1 - p_i z^{-1}) + \prod_{j=1}^m (1 - z_j^{-1}) + \dots) \quad (9-193)$$

where m should equal or be greater than the lowest power in z^{-1} in the series expansion of $G_1(z)$.

$$1 - M(z)$$

$$= \prod_{i=1}^n (1 - p_i z^{-1}) \prod_{j=1}^m (1 - z_j^{-1}) (1 + a_1 z^{-1} + a_2 z^{-2} + \dots)$$

(9-194)

where n is an integer which depends on the order of the input test signal. Note that if some of the poles of $G_1(z)$ are at $z =$

1 , then

$1 - M(z)$ is raised to the power of (121) equal to either the order of the input signal poles or the order of the poles of $G_1(z)$ at $z = 1$, whichever is higher.

To illustrate the design of a system which has poles and zeros of $G_1(z)$ outside or on the unit circle, the following numerical example is given.

Example 9-9

For the system of Fig. 9-35, the transfer function of the controlled process is given by

$-T's$

$$G_p(s) = G_o(s) G_1(s)$$

$$s^3(1+0.04s)$$

The sampling period is 0.05 second.

(9-195)

The problem is to design a digital controller that will produce an output

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with zero steady-state error and minimum finite settling time in response to a ramp input. The z-transform of $G(s)$ is

$$G(z) = \frac{3.92 \times 10^{-11}(1 + 2.78z^{-1})(1 + 0.2z^{-1})}{(1 - z^{-1})^2(1 - 0.286z^{-1})} \quad (9-196)$$

which has one zero outside the unit circle at $z = -2.78$ and two poles on the unit circle at $z = +1$.

In order to obtain a response with finite settling time, $M(z)$ must be a polynomial in z^{-1} and should contain the zero of $G(z)$ at $z = -2.78$. Thus $M(z)$ is of the form

$$M(z) = (1 + 2.78z^{-1})(M_1z^{-1} + M_2z^{-2}) \quad (9-197)$$

The lowest order of $M(z)$ in z^{-1} is unity, since the lowest order in z^{-1} in the series expansion of $G(z)$ is unity. For a unit ramp input, $1 - M(z)$ should contain a factor $(1 - z^{-1})^2$ and also the poles of $G(z)$ that are on or outside the unit circle. In this case, since $G(z)$ has two poles at $z = +1$, having the term $(1 - z^{-1})^2$ in $1 - M(z)$ would be sufficient to satisfy both requirements. Hence $1 - M(z)$ is of the form.

$$1 - M(z) =$$

$$(1 - z^{-1})^2 F(z) = (1 - z^{-1})^2(1 + a_1z^{-1})$$

where in order for the

$$(9-198)$$

response to be deadbeat and simultaneously result in a physically realizable $D(z)$, $1 - M(z)$ must contain the constant term 1. Note that $M(z)$ in Eq. (9-197) is chosen to be a polynomial in z^{-1} of order three, which means that the transient response will settle in three sampling periods, the shortest possible settling time for this system. Referring to Eq. (9-159), we see that if $G(z)$ does not have poles or zeros on or outside the unit circle, the minimum settling time for a ramp input is two sampling periods. Because of the zero of $G(z)$ at $z = -2.78$, $F(z) = 1$ can no longer be chosen. Hence the settling time is increased by one sampling period. In general, the more zeros and poles of $G(z)$ which lie outside the unit circle, the longer will be the finite settling time.

Now, with three unknowns and three independent equations, M_1 , M_2 , and a_1 can be solved from Eq. (9-197) and Eq. (9-198). Substitution of Eq. (9-197) into Eq. (9-198) yields the following equations:

$$M_1z^{-1} + (2.78M_1 + M_2)z^{-2} + 2.78M_2z^{-3} = (2 - a_1)z^{-1} + 2a_1z^{-2} - a_1z^{-3} \quad (9-199)$$

$$1)z^{-2} - a_1z^{-3}$$

$$(9-199)$$

Equating the coefficients of the corresponding terms of both sides of the last equation, we have

$$M_1 = 2 - a_1 \quad (9-200)$$

$$2.78M_1 + M_2 = 2a_1 - 1 \quad (9-201)$$

$$2.78M_2 - a_1 = 0 \quad (9-202)$$

Solving the last three equations yields the coefficients

$$M_1 = 0.723$$

$$(9-203)$$

$$M_2$$

$$a_1 =$$

$$-0.46$$

$$1.277$$

$$(9-204)$$

$$(9-205)$$

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Hence, the over-all system transfer function is

$$M(z) = 0.723z^{-1} + 1.554z^{-2}$$

$$1.277z^{-3}$$

For a unit ramp input, the z-transform of the output is

$$(9-206)$$

$$C(z) = M(z)$$

$$Tz^{-1}$$

$$(1-z^{-1})$$

$$= 0.05 (0.723z^{-2} + 3z^{-3} + 4z^{-4} + 5z^{-5} + \dots)$$

$$(9-207)$$

The output response at the sampling instants is shown in Fig. 9-38a. When a unit step input is applied, the output z-transform is

$$1$$

$$-$$

$$C(z) = M(z) (1 - z^{-1})$$

$$=$$

$$0.723z + 2.277z^{-2} + 2z^{-3} + 2z^{-4} +$$

$$(9-208)$$

$$2$$

$$z^{-5} z^{-6} z^{-7}$$

$$8-$$

$$7-$$

$$6H$$

$$5-$$

$$r(t)$$

$$2$$

$$2.277$$

$$c(nT)$$

$$27 \quad 37$$

$$4T$$

$$5T \quad 6T$$

$$7T$$

$$\text{Time } (T = 0.05 \text{ sec})$$

$$(a)$$

$$r(1)$$

$$c(nT)$$

$$27 \quad 3T \quad 4T \quad 5T \quad 6T$$

$$7T$$

$$\text{Time } (T = 0.05 \text{ sec})$$

$$(b)$$

$$8T \quad 9T$$

$$10T$$

$$8T \quad 9T \quad 10T$$

Fig. 9-38 Ramp and step responses of the system in Example 9-9. (a) Ramp response. (b) Step response.

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The step response of the system has an overshoot of over 100 percent. (Fig. 9-38b).

Substitution of Eqs. (9-206) and (9-196) into Eq. (9-191) yields the pulse transfer function of the digital controller

$$D(x) = \frac{1840(10.286z^{-1})(1 - 0.6362z^{-1})}{(1 + 0.2z^{-1})(1 + 1.277z^{-1})} \quad (9-209)$$

Note that although $D(z)$ has a pole outside the unit circle in the z -plane, it is still physically realizable.

9.7 DESIGN USING THE STALENESS WEIGHTING FACTOR

In the preceding section we have seen that sampled-data systems which are designed for finite settling time and zero steady-state error for a particular test input usually do not provide satisfactory performances when other types of inputs are applied. For instance, the system in Example 9-9 is designed to respond with finite settling time and zero steady-state error when subjected to a ramp input.

However, the step response of the designed system exhibits a highly undesirable peak overshoot of 128 percent. A system that is extremely sensitive to the form of its input signal is seldom acceptable in practice, since it is very unlikely that the system will be subject to only the specific kind of input signal for which it is designed.

Therefore, a design procedure should be developed to lead to a system that will respond satisfactorily not just to one signal but to a number of different test signals. A compromise may be reached among the responses of a system to several different test signals by introducing a "staleness-weighting factor" into the over-all system transfer function. If $M(z)$ denotes the over-all system function that provides a minimal response (zero steady-state error and minimum finite settling time) for a specific test signal, the staleness-weighting factor is introduced into the system in the following manner:

Insertion of the factor $(1 - cz^{-1})$ into the denominator of

$1 - M(z)$ yields

$$\frac{1 - M(z)}{1 - M(z)}$$

$$= \frac{1 - M(z)}{(1 - cz^{-1})(1 - M(z))} \quad (9-210)$$

where c is the staleness-weighting factor. For the over-all system to be stable, c should satisfy the relation $-1 < c < +1$.

When the staleness-weighting factor is introduced into the denominator of $1 - M(z)$, the characteristic equation of the compensated

† J. E. Bertram, "Factors in the Design of Digital Controllers for Sampled-data Feedback Systems," Trans. A. I. E. E., 75, Part 2 July 1956, pp. 151-159.

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system is no longer of the form $2^n = 0$ and the error of the system does not settle to zero in a finite period of time. However, the addition of the staleness-weighting factor does not improve the infinite sensitivity of the system response to parameter variations, since the characteristic equation still contains a multiple order pole at $z = 0$. With a proper choice of the value of c , the peak overshoot of the step response can be reduced when the system is designed for a ramp input, and the transient response can be reduced to a tolerable amount within reasonable time interval.

The use of the staleness-weighting factor in the design of sampled-data systems is illustrated by the following numerical example.

Example 9-10

Consider the sampled-data system designed in Example 9-8. The controlled process, including the hold device, is described by the transfer function $G(z)$

$$G(z) = \frac{0.005(10.92-1)}{(121)(1-0.9052z^{-1})}$$

The sampling period is 0.1 second. A digital controller must be designed so that the ramp response has zero steady-state error at the sampling instants, and at the same time a compromise between the overshoot of the step response and the settling time may be reached. According to the previous design rules for a ramp input hence

$$1 - M(z) = (1 - z^{-1})^2$$

which is the minimal over-all system function for a ramp input. The step response and the ramp response of the compensated system have been illustrated in Figs. 9-37b and 9-37e, respectively. Note that while the ramp response is described as minimal, the step response exhibits a peak overshoot of 100 percent.

Now applying the staleness-weighting factor to $1 - M(z)$ in accordance with Eq. (9-210) yields

$$1 - M_c(z)$$

$$= \frac{(1 - x^{-1})^2}{1 - cz^{-1}}$$

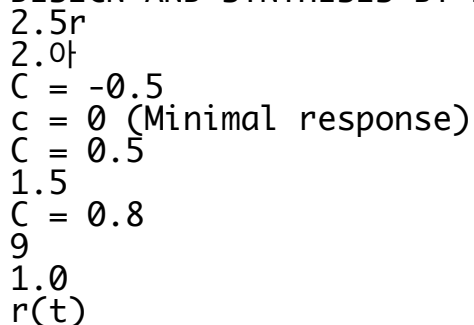
Solving for $M_c(z)$ in Eq. (9-214), we have

$$M_c(z) = \frac{(2 - c)z^{-1}}{1 - cz^{-1}}$$

$$(9-215)$$

Referring to Eq. (9-215), we see that for negative values of c greater than -1, the introduction of the staleness-weighting factor actually increases the peak overshoot of the step response. The optimum value of c would seem to be slightly less than +1, since this value would produce the least amount of overshoot in the step response. However, we shall show that the overshoot can be reduced only at the expense of longer settling time. To illustrate the effect of the staleness-weighting factor and to assist the selection of an optimum

value of c , the step and the ramp responses of the system at the sampling



0.5
 T
 $2T \ 3T$
 $4T$
 $5T \ 6T \ 7T$
 $8T \ 9T \ 10T$
 Time (sec)
 6
 5
 $c(nT)$
 4
 $c = -0.5$
 3
 $2 \ r(t)$
 $--$
 (a)
 $-C = 0.8$
 $C = 0.5$
 $c = 0$ (Minimal response)
 T
 $2T \ 3T$
 $4T$
 $5T$
 $6T$
 $7T \ 8T \ 9T \ 10T$
 Time (sec)
 (b)
 $\frac{1}{s}$
 $\frac{1}{s^2}$

Fig. 9-39 Step and ramp responses of the system in Example 9-10. (a) Unit step responses. (b) Ramp responses. c values are plotted in Figs. 9-39a and 9-39b, respectively, for several values of c between 1 and +1. Observe that for $c = 0.8$, the peak overshoot in the step response is only 20 percent but the transient in the step and the ramp responses settle very slowly. On the other hand, for negative or small values of c , whereas the ramp response may be satisfactory, the overshoot in the step response is too large to be acceptable. A satisfactory compromise

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 between overshoot and settling time may be reached when c is in the neighborhood of 0.5.

∞
 In the literature, a more quantitative method for the selection of a desirable value of c has been devised to yield a compromise between the step and the ramp responses. The method is based on the minimization of the sum of the squared error $\sum [e(mT)]^2$ which is simply the sum of the squares of the coefficients of the following equation when it is written as a series or polynomial in z^{-1} :

$$E(x) = R(x)[1 - M(z)]$$

(9-216)

Unfortunately, the minimum values of the sums of the squared errors of a ramp response and a step response often occur at opposite ends of the permissible range of the staleness-weighting factor c . In any given system, a minimum of the squared error can usually not be found that will result in optimum performance for both the step and the ramp inputs.

9.8 COMBINATION OF THE Z-PLANE SYNTHESIS
 AND THE MINIMAL RESPONSE DESIGN

Sampled-data control systems designed by the minimal response method described in the preceding sections, whether or not the staleness-weighting factor is used, all contain the following disadvantages:

- (1) The design is conducted on the basis of the response at the sampling instants with the result that intersampling ripples may exist in the output response.
- (2) The characteristic equation of the system has multiple order roots at $z = 0$, a condition which corresponds to that of the system operating at a point with infinite sensitivity to parameter variations.
- (3) Even with the staleness-weighting factor, a system which will perform satisfactorily in response to various types of inputs is difficult to design.

In this section the z -plane synthesis method described in Sec. 9.5 will be combined with the design procedure for finite settling time and zero steady-state error. In essence, a pair of complex control poles are inserted in the over-all system transfer function for the control over the transient response; and simultaneously, the system must also have zero steady-state error for a ramp input. The domi-

† J. E. Bertram, op. cit.

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nant poles of $M(z)$ are selected to lie in the first and the fourth quadrants of the z -plane so that hidden oscillations do not exist in the output response.

The basic concept and procedure of the proposed design technique can best be explained by the following example.

Example 9-11

The sampled-data control system under consideration has the block diagram shown in Fig. 9-35. The fixed part of the system is described by the transfer function

$$G(z) = \frac{0.005z^{-1}(1 - 0.9z^{-1})}{(1-z)(1 - 0.9052z^{-1})}$$

(9-217)

The sampling period is 0.1 second. Note that the system is identical to those used in Examples 9-6, 9-8, and 9-10. The problem is to select a digital compensator which will enable the system to meet the following set of design criteria:

- (1) The steady-state error of the system in response to a ramp function input should be zero ($K_v = \infty$).
- (2) The overshoot of the step response should be as small as possible and the rise time and the settling time as short as possible.
- (3) The damping ratio of the dominant control poles of $C(z)/R(z)$ should equal 0.707.

Since $G(z)$ has one more pole than zero, to insure that $D(z)$ is physically realizable, the over-all system function $M(z)$ must have at least one more pole than zero. To satisfy the zero steady-state error requirement on the ramp response, $1 - M(z)$ should contain the factor $(1-z)$. This also satisfies the requirement that $1 - M(z)$ must include the poles of $G(z)$ which lie on or outside the unit circle in the z -plane. The poles and zeros of $G(z)$ inside the unit circle are cancelled by the poles and zeros of $D(z)$. Therefore, $D(z)$ is of the form

$$D(z) = \frac{1}{X(z)}$$

$$G_2(z) = \frac{1 - M(z)}{K_a} \wedge G_1(z)$$

$$(1 - 0.905z^{-1})(1 + a_1z^{-1})(1 + a_2z^{-1})$$

$$(10.9\%)z^{-1}(1 - z^{-1})(1 + b_1z^{-1})(1 + b_2z^{-1}) \dots$$

(9-218)
 where K_a is a constant; a_1 , a_2 , and b_1 , b_2 , are the zeros and poles, respectively, that are added to $D(z)$ to improve the performance of the system. For $D(z)$ to be physically realizable the order of the denominator of $D(z)$ must be at least as high as that of the numerator. In general, the least possible number of poles and zeros should be used in $D(z)$ so that the implementation of the digital controller is the simplest.
 + Because of the zero steady-state error requirement imposed on the ramp response and the possibility of $G_1(z)$ having zeros outside the unit circle, the zeros of $M(z)$ cannot be chosen arbitrarily. Therefore, the designer does not have complete control over the peak overshoot of the step response as he did in the synthesis method of Sec. 9.5.

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Taking the product of $D(z)$ and $G_1(z)$, the open-loop transfer function of the compensated system is written as

$$D(z)G_1(z) =$$

$$\frac{0.005Kz^{-1}(1 + a_1z^{-1})(1 + a_2z^{-1})}{(1 - z^{-1})^2(1 + b_1z^{-1})(1 + b_2z^{-1})}$$

(9-219)

The minimum order in z of the system is two. In other words, the simplest form of $D(z)G_1(z)$ is

$$D(z)G_1(z) = \frac{0.005Kz^{-1}(1 + a_1z^{-1})}{(1 - z^{-1})^2}$$

$$(9-220)$$

(9-220)

which has two unknowns in K_a and a_1 . The characteristic equation of the system is the numerator of $1 + D(z)G_1(z)$ and is equal to the sum of the numerator and the denominator of Eq. (9-220). Therefore, the characteristic equation reads

$$(11)z^2 + 0.005Kaz(1 + a_1z^{-1}) = 0$$

$$1 + (0.005Ka - 2)z^{-1} + (1 + 0.005Kaa_1)z^{-2} = 0$$

(9-221)

(9-222)

Let us use the characteristic equation of the system of Example 9-6.

The characteristic equation roots are at $z = -0.27 + j0.32$ and $z = -0.27 - j0.32$

$$\text{and the equation is}$$

$$10.542z^{-1} + 0.1782z^{-2}$$

$$= 0$$

(9-223)

Equating the coefficients of equal powers of z^1 in Eqs. (9-222) and (9-223) and solving for K_a and a yields

$K_d = 292$

=

$a_i - 0.563$

Therefore, the pulse transfer function of the digital controller is

$D(z)$

=

$292(1 - 0.563z^{-1})(1 - 0.905z^{-1})$

$(1 - z^{-1})(1 - 0.9z^{-1})$

The over-all system transfer function is

(9-224)

(9-225)

$M(z)$

$C(z) 1.462 - 1(1 -$

-

$0.563z - 1)$

=

=

$R(z) 10.54z^2 - 1 + 0.178z - 2$

(9-226)

The step and the ramp responses of the system with the designed digital compensation are plotted in Fig. 9-40. The step response exhibits a peak overshoot of 46 percent. The ramp response gets to within one percent of its final value in about four sampling periods. These responses compare favorably with the minimal step and ramp responses of the same system shown in Fig. 9-39. Even with the staleness-weighting factor design the overshoot of the step response in Fig. 9-39 for $c = 0.5$ is comparable to that of the step response in the present design, but the settling time in the present system is faster.

More significant is the fact that the characteristic equation of the system designed with the present method does not possess any multiple order root. Therefore, the sensitivity of the system to parameter variations is definitely improved.

The peak overshoot of the designed system can be decreased still further by reducing the resonant frequency of the system; however this reduction

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1.5

0.5

5

$r(t)$

Characteristic Eq: $1 - 0.54z^{-1} + 0.178z^{-2}$

Characteristic Eq: $1 - 1.4z^{-1} + 0.54z^{-2}$

+

27

37 4T 57 6T 77 8T 97 10T

Time (sec)

(a)

4

$C(AT)$

3T

$r(t)$

Characteristic Eq: $1 - 0.54z^{-1} + 0.178z^{-2}$

Characteristic Eq: $1 - 1.4z^{-1} + 0.54z^{-2}$

2T 3T

4T

57 6T

Time (sec)

(b)

77

8T 97 IOT

Fig. 9-40 Step and ramp responses of the system in Example

9-11. (a) Step responses. (b) Ramp responses.

would cause a longer rise time and settling time. For instance, if the characteristic equation roots are chosen to be $z = 0.7j0.225$ and $z = 0.7 - j0.225$, a change which will yield a damping ratio of 0.707, the characteristic equation becomes

$$1 - 1.42z^{-1} + 0.54z^{-2} = 0$$

(9-227)

Equating the coefficients of equal powers of z^{-1} in Eqs. (9-222) and (9-227) and solving for Ka and a , yields

 Ka

$$= 120$$

The over-all system transfer function is

(9-228)

$$a_1 =$$

$$-0.767$$

$$-1$$

$$0.62 (10.7672 - 1)$$

 $M(2)$

$$=$$

$$1 - 1.4z^{-1} + 0.54z^{-2}$$

(9-229)

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The step and the ramp responses are also plotted in Fig. 9-40. The overshoot in the step response is reduced to 30 percent. However, the peak time, rise time, and the settling time are increased considerably.

A compromise is usually necessary in selecting the location of the control poles of the system in order to yield reasonable results for the overshoot and response time.

9.9 SYNTHESIS BY THE STATE VARIABLE AND STATE TRANSITION APPROACH

The state variable and state transition concepts introduced in Chapter 5 as a general approach to the analysis of linear discrete-data control systems can be used for the purpose of synthesis. We shall see that with the help of the variable gain concept, the state transition method provides a simple and systematic synthesis procedure for linear (as well as nonlinear) discrete-data control systems. The development of the synthesis procedure makes use of the transition flow graph and equations. Digital compensation is designed for the linear system to have deadbeat performance. In Chapter 12 the same technique is extended to synthesize nonlinear control for the compensation of linear and nonlinear systems. The synthesis method using the state variables concept is essentially a numerical method. Once the synthesis procedure and equations are set up, the computation can be carried out simply by a digital computer. The method has one important advantage over the minimal response design covered in Sec. 9.6 in that the state transition approach insures that the deadbeat response does not have intersampling ripples.

Consider the sampled-data control system shown in Fig. 9-41.

The variables x_1 , x_2 , x_p , are the state variables of the controlled process; $h(t)$ is the output of the zero-order hold, and $e(t)$ is the

...

$$-h^*(1)$$

Digital controller

$e(t)$
 $X_1 = C$
 $D(s)$
 $z.o.h.$
 $G_1(s)$
 T

$h(t)$
 4444
 $D(z)$
 $X_p \quad X_3 \quad X_2$

Fig. 9-41 Control system with digital controller.

† J. T. Tou, B. Vadhanaphuti, "Optimum Control of Nonlinear Discrete-Data Systems," A. I. E. E. Trans., Part 2, September 1961, pp. 166-171.

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actuating signal. Since the output of the digital controller is a train of impulses, its values at the sampling instants are equal to the output of the zero-order hold at the sampling instants. Therefore, the z-transfer function of the digital controller can be written as

$$D(z) = H(z)E(z)$$

where

(9-230)

$$H(x) = h(0+) + h(T+)z^{-1} + h(2T+)z^{-2} + \dots + h(nT+)z^{-n}$$

and

(9-231)

$$E(z) = e(0+) + e(T+)z^{-1} + e(2T+)z^{-2} + \dots$$

$$+ \dots + e(nT+)z^{-n}$$

(9-232)

In the state variable approach, since only one term in each of Eqs. (9-231) and (9-232) is of interest during one sampling period, the digital controller may be replaced by a variable gain amplifier with gain $K(kT)$ which will have different values during different sampling periods. $K(kT)$ is given by

$$K(kT) = h(kT+)/e(kT+)$$

(9-233)

This way the digital controller is represented by the variable gain amplifier with gain $K(kT)$ on the state transition flow graph during the time interval $kT < t \leq (k+1)T$.

The synthesis procedure and technique are outlined as follows:

(1) Draw the state transition flow graph of the system with the digital controller represented by the variable gain amplifier $K(kT)$. From the transition flow graph, the transition equations are written in the following vector-matrix form:

$$x(t) = y(t - t_0, K_k)x(t_0) + O(t - t_0, K_x)r(t_0)$$

where for simplicity $K(kT)$ is written as K_k .

(9-234)

(2) For a deadbeat response the system error must be zero for $t > nT$, where n is the smallest possible positive integer. This condition is realized if

and

$$x_1(nT) = r(nT)$$

$$x_2(nT) = x_3(nT)$$

=

=

$$x(nT) = 0$$

=

(9-235)

(9-236)

Since $x(nT)$, $x_2(nT)$, ... are functions of K_k , for $k = 0, 1, 2, \dots, n$,

the variable gain K for $k = 0, 1, 2, \dots, n$, can be determined from Eqs. (9-235) and (9-236).

(3) Compute $h(0+), h(T+), \dots, \dots, h(nT+)$, using the K 's determined in the last step, Eq. (9-233), and the relationship $e(kT+) = r(kT) - x_1(kT)$ (9-237)

(4) Determine the transfer function $D(z)$ of the digital controller using Eqs. (9-230), (9-231), and (9-232).

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The synthesis procedure is illustrated by the following numerical examples.

Example 9-12

Consider that in the sampled-data system of Fig. 9-41. $G_c(s) = 1/s(s + 1)$, $r(t) =$ unit step function, $T = 0.1$ second, and all initial conditions are zero. A digital controller is to be designed for the system to exhibit a deadbeat response. Note that the problem is identical to that stated in Example 9-8 except that in the previous example the deadbeat response is realized only at the sampling instants. In fact, the so-called deadbeat response exhibits overshoots and ripples during sampling periods, and the design should actually be termed insignificant. However, with the state transition approach, inter-sampling ripple is certain not to occur in the deadbeat response.

The state transition flow graph of the system is constructed as shown in Fig. 9-42. Note that the digital controller is represented by a variable gain amplifier with gain $K(t_0)$ connected between the nodes $e(t)$ and $h(t)$. The transition equations are written directly from the flow graph.

$r(t_0)$
 $K(t_0)$
 $e(t)$
 $h(t)$ $H(s)$
 -1

$x_1(t_0)$
 $(x_2(5))$
 $x_1(5)$

Fig. 9-42 Transition flow graph of control system with digital compensation. (Example 9-12).

$x_1(t)$
 $=$
 $-$
 $-$
 $-$
 $\{1 - K(t_0)[(t - t_0) - 1 + e^{-(t-t_0)}]\}x_1(t_0)$
 $+ [1 - e^{-(t-t_0)}]x_2(t_0) + K(t_0)[(t - t_0) - 1 + e^{-(t-t_0)}]r(t_0)$
 $-$
 $-$

$x(t) = -K(t_0)[1 - e^{-(t-t_0)}]x_1(t_0) + e^{-(t-t_0)}x_2(t_0)$

For t_0

read

$1 =$
 $+ K(t_0)[1 - e^{-(t-t_0)}]r(t_0)$
 $-$

(9-238)

(9-239)

$kT, t = (k + 1)T, T = 0.1$ second, $r(kT) = 1$, the last two equations

$x_1[(k + 1)T] = (1 - 0.005K)x_1(kT) + 0.095x_2(kT) + 0.005Kk$ (9-240)

$x_2[(k + 1)T] = -0.095Kx_1(kT) + 0.905x_2(kT) + 0.905Kk$ (9-241)

-

We are now ready to incorporate the conditions for deadbeat response listed in Eqs. (9-235) and (9-236). Substituting $k = 0, x_1(0) = 0$, and $x_2(0) = 0$, into Eqs. (9-240) and (9-241), we have

and

$$\begin{aligned} x_1(T) &= 0.005K, \\ x_2(T) &= 0.095K, \end{aligned} \quad \begin{aligned} (9-242) \\ (9-243) \end{aligned}$$

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If the output response is to reach its steady state value in one sampling period with zero error, it is necessary that $x_1(T) = 1$ and $x_2(T) = 0$. In view of Eqs. (9-242) and (9-243), we see that this balance cannot be achieved. Next, we try k_1 . Substitute k_1 and the values of $x_1(T)$ and $x_2(T)$ into Eqs. (9-240) and (9-241), and if deadbeat response is to occur after two sampling periods, we have

$$\begin{aligned} x_1(2T) &= 0.014K, \quad 0.000025K, K, + 0.005K_1 = 1 \\ x_2(2T) &= -0.000475K, K_1 + 0.086K_0 + 0.095K_1 = 0 \end{aligned}$$

Solving for K , and K_1 , from the last two equations yields K_0

$$\begin{aligned} -203. \text{ Therefore,} \\ e(T+) &= r(T) - x_1(T) = 1 - 0.005(105.6) = 0.472 \end{aligned}$$

$$e(0+) = r(0) - x_1(0) = 1$$

$$\begin{aligned} h(T+) &= Ke(T+) = -203(0.972) = -197 \\ h(0+) &= Koe(0+) = 105.6 \end{aligned} \quad \begin{aligned} (9-244) \\ (9-245) \end{aligned}$$

$$105.6 \text{ and}$$

$$\begin{aligned} (9-246) \\ (9-247) \\ (9-248) \\ (9-249) \end{aligned}$$

Substituting the last four equations into Eq. (9-230) yields the transfer function of the digital controller.

$$D(z) = \frac{h(0+) + h(T+)z^{-1}}{e(0+) + e(T+)z^{-1}}$$

$$= \frac{105.6 + 197z^{-1}}{1 + 0.472z^{-1}}$$

$$\begin{aligned} (9-250) \end{aligned}$$

The compensated system has a unit step response without intersampling ripples like that shown in Fig. 9-43. However, the digital compensation thus

0 7 27 374757677787 97

Fig. 9-43 Deadbeat response of the system in Example 9-12.

designed is optimum only when a step input is applied to the system. The system may not perform satisfactorily when any other basic test signal such as a ramp or a parabolic function is applied.

In this section we have described the synthesis of control systems

with digital controllers by means of the state transition technique. The synthesis procedure involves four major steps and is very simple and straightforward. The synthesis method makes use of the variable gain amplifier representation of the unknown controller. The content of the controller is entirely general. Once the optimum K is determined for $k = 0, 1, 2, 3, \dots, n$, the controller could actually contain linear elements, nonlinear elements, and time-invariant as well as time-variant elements. The nature of the state transition design assures that the output response of the synthesized system is free

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from intersampling ripples (provided that the sampling frequency is high enough). However, the design is optimum only for the specific input for which it is intended. The designed system does not provide satisfactory performance for a wide class of inputs.

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10

MULTIRATE AND SKIP SAMPLING

The sampled-data systems considered in the preceding chapters have one feature in common; they contain a number of samplers which operate in synchronism with the same sampling rate. Such systems are usually described as single-rate sampled-data systems. In general, there are systems which contain two or more samplers operating with different sampling rates; the sampling may or may not be synchronized. Systems having a sampler which operates with a nonuniform aperiodic or cyclic-variable rate are also common; and are called cyclic-rate sampled-data systems. The multirate or cyclic-rate sampled-data systems may also be regarded as skip-sampled systems, since the sampling schemes can be thought of as the result of periodically omitting (or skipping) samples from a uniform sampling operation.

In order to distinguish it from single-rate sampled systems a multirate sampled-data system is defined here as a system containing synchronized switches that do not operate with the same sampling

rate.

In many instances, the practical application of multirate sampled systems arises from actual physical existence of such systems. This may occur, for instance, in the case of flight-control systems when a data link computer operates at a rate different from the speed of the radar antenna. In addition to the above mentioned case, situa-

G. M. Kranc, "Input-Output Analysis of Multirate Feedback Systems," I. R. E. Transactions on Automatic Control, November 1957, pp. 21-28.

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tions may arise where multirate sampling must be incorporated intentionally to improve the performance of a given system. A digital controller operating at a higher rate than the basic sampling rate of the system is an example of this sort of situation. In a later section we shall show that systems with properly designed multirate digital compensators have better performance than those with single-rate sampling. The skip or cyclic-rate sampling technique may also be employed to obtain a novel type of controller with which deadbeat performance can be achieved.

10.1 MULTIRATE SAMPLING

Open-Loop Systems

The block diagrams of typical open-loop systems with multirate sampling are shown in Fig. 10.1. In the systems of Figs. 10-1a and 10-1b, the samplers are assumed to be synchronized, but one sampler is operating N times faster than the other. In Fig. 10-1c the two synchronized samplers operate with rates of $1/T_1$ and $1/T_2$, respectively. Usually, in order to simplify the mathematics of analysis, it is necessary to assume that N is an integer and that T_1 and T_2 are commensurable.

1

Consider the multirate system shown in Fig. 10-1a, and assume that N is an integer. The output signal of the system is sampled at a rate N times faster than the sampling rate at the input. The output of the multirate sampler with sampling rate T/N is written as

$e(t)e^{(1)}$

$T \ S_1$

$c(+)$ X

$G(s)$

$(1)N$

$T/N \ S_N$

(a)

$e(r)e^{(f)N}$

$c(1)$

$G(s)$

$c(t)$

$T/N \ S_N$

$T S_1$

(b)

$e^{(N_1)}$

$c(1)$

$e(1)$

$G(s)$

T_1

S_{N_1}

T_2

S_{N_2}

(N_2)

(c)

Fig. 10-1 Open-loop systems with multirate sampling.

† G. M. Kranc, "Compensation of an Error-Sampled System by a Multirate Controller," Trans. A. I. E. E., 77, July 1957, pp. 149-159.

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$e(t)$

$e^*(t)$,

T

$2T$

$3T$

$4T$

$c(t)$

$(N$

$(N=3)$

$co(1)$

$7/3$

T

$2T$

$T \ 4T \ 5T$

$2T \ 7T \ 8T$

3

$6/00$

$3T$

1979

(0)

E/M

$4T$

T

$2T$

$3T$

$4T$

$c_2(1)$ at

$4T$

345

$2T$

$5T$

3

$15/m$

175

$7T$

$10T$

137

(b)

olm

$8T$

$11T$

3

$c^*(t) = c(t) + c^*(t) + c^*(t)$

Fig. 10-2 (a) Typical signals at various points of the multirate sampled-data system of Fig. 10-1a. (b) Skip-sampled components of $c^*(t)$ shown in (a).

00

kT

$c^*(t) \times = \sum_{k=0}^{\infty} c(kT) \delta(t - kT)$

$k=0$

-

(10-1)

which is essentially a unit-impulse train of frequency NT , amplitude-modulated by $c(t)$. As an illustration, a typical set of signals that appears at various points of the system is depicted in Fig. 10-2a for $N = 3$. By pure physical reasoning, the output signal $c^*(t)$ x_for

$N = 3$ can be regarded as the sum of three skip-sampled components,

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$c^*(t)$, $c(t)$, and $c_r(t)$, as shown in Fig. 10-2b. In general, the skip-sampled component $c(t)$ ($n = 0, 1, 2, \dots, N-1$) is formed from $c^*(t)$ by taking every N -th sample starting with the n -th, where $n = 0, 1, 2, \dots, N-1$. From Fig. 10-2b we see that a skip-sampled component of a multirate sampled signal $c^*(t)$ can be obtained by sampling $c(t)$ with a "skip-sampled" sampler which consists of a time advance, a time delay, and a basic sampler S , whose sampling period is T . The skip-sampled sampler that gives the n -th component of $c^*(t)$ is shown in Fig. 10-3.

N

N

$c(t)$

$e^{j2\pi nTs}/N$

$C_h(t)$

$e^{-j2\pi nTs}/N$

$T S_1$

Time delay

Time advance

Fig. 10-3 A skip-sampled sampler.

A sampler with multirate sampling rate T/N can now be decomposed into a parallel combination of N skip-sampled samplers, each with appropriate delay and advance elements as shown in Fig. 10-4. The equivalent block diagram of the system of Fig. 10-1a with sampler decomposition is shown in Fig. 10-5. Since now all samplers of the skip-sampled sampler model have the same sampling rate, it

$c^*(1)N$

$c(t)$

SN

T/N

equivalent to

$e^{j2\pi nTs}/N$

$e^{-j2\pi nTs}/N$

$C_1(1)$

$c_0(t)$

$c(t)$

$e^{j2\pi nTs}/N$

$e^{-j2\pi nTs}/N$

$C_2(1)$

+

$c^*(t)$

$e^{j2\pi nTs}/N$

$e^{-j2\pi nTs}/N$

$c_3(1)$

$e^{j2\pi nTs}/N$

$C_K(1)$

*

$(C_{N-1}(1)$

$(N-1)Ts$

$e^{j2\pi nTs}/N$

$e^{-j2\pi nTs}/N$

$(C_1(1)$

$(N-1)Ts$

$e^{j2\pi nTs}/N$

Fig. 10-4 Decomposition of multirate sampler into N skip-sampled samplers.

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is conceivable that ordinary z-transform technique can be used to analyze the system. From Fig. 10-4, the output of the multirate sampled system is written.

$$c^*(t)_n = \sum_{k=0}^{N-1} c^*(t) \quad (10-1)$$

(10-2) where $c(t)$ represents the n -th ($n = 0, 1, 2, \dots, N-1$) skip-sampled component and

$$c^{\#}(t) = \sum_{k=0}^{N-1} c(kT + nT) \delta(t - kT - nT) \quad (10-3)$$

$$c^{\#}(t) = \sum_{k=0}^{N-1} c(kT + nT) \delta(t - kT - nT)$$

Substituting Eq. (10-3) into Eq. (10-2), we have

$$c^*(t)_x = \sum_{n=0}^{N-1} \sum_{k=0}^{N-1} c(kT + nT) \delta(t - kT - nT) \quad (10-4)$$

The continuous output of $G(s)$ is obtained from the superposition principle; that is

Thus

$$c(t) = \sum_{m=0}^{N-1} e(mT)g(t - mT) \quad (10-5)$$

$$c(kT + n) = \sum_{m=0}^{N-1} e(mT)g(kT + n - mT) \quad (10-6)$$

Substitution of Eq. (10-6) into Eq. (10-4) yields

$$c^*(t)_x = \sum_{n=0}^{N-1} \sum_{k=0}^{N-1} \sum_{m=0}^{N-1} e(mT)g(kT + n - mT) \delta(t - kT - nT) \quad (10-7)$$

Taking the Laplace transform on both sides of Eq. (10-7), we have

$$C^*(s)_N = \sum_{n=0}^{N-1} \sum_{k=0}^{N-1} \sum_{m=0}^{N-1} e(mT)g(kT + n - mT) \delta(t - kT - nT) \quad (10-8)$$

$$\sum_{n=0}^{nT} \sum_{k=0}^{N-1} g_k kT +$$

$$- (k + n/N) T_s$$

$$\sum_{m=0}^{M-1} e^{j(mT - nTs)} \quad (10-8)$$

$$c_i(t)$$

$$c_o(t)$$

$$e^{Ts/N}$$

$$e^{-Ts/N}$$

$$e(t) \times e^*(t)$$

$$c(t)$$

$$G(s)$$

$$e^{2Ts/N}$$

$$e^{-2Ts/N}$$

$$c_2(1)$$

$$(1)^N$$

$$T_s$$

$$CIZIN$$

$$e^{kTs/N}$$

$$e^{-kTs/N}$$

$$C_K(1)$$

$$(N-1)Ts$$

$$X$$

$$C_{N-1}(1)$$

$$-(N-1)Ts$$

$$e$$

$$N$$

$$e$$

$$N$$

Fig. 10-5 Equivalent skip-sampled system of the multirate sampled system of Fig. 10-1a.

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or

$N-1$

$\sum$

(10-9)

$$C^*(s) N = [G(s) e^{nTs/N}]^* E^*(s) e^{-nTs/N}$$

$n=0$

The z-transform of $c^*(t)x$ then reads

$N-1$

$-n/N$

$$C(z)N = \sum_{n=0}^{N-1} Z[G(s) e^{nTs/N}] E(z) z^{-n/N}$$

N

$72=0$

(10-10)

Equation (10-10) gives an important relationship for the analysis of the multirate sampling system of Fig. 10-1a; the equation represents the z-transform of the output of the multirate sampler in terms of the products of the ordinary z-transforms of $E(s)$ and $e^{-nts}/NG(s)$. The z-transform of $e^{-Ts}/NG(s)$ can be expressed in terms of the modified z-transform of $G(s)$. From the modified z-transform theory, we have

00

$$3m[G(s)] = G(z, m) = 2$$

$$\sum_{k=0}^{N-1} g[(kT + mT)] x^{-k} \quad 0 < m < 1$$

$k=0$

(10-11)

Multiplying both sides of Eq. (10-11) by $z^{-n/N}$ and replacing m by n/N , we have

$$\sum_{k=0}^{n/N} z^{1-(n/N-k)} G(z, \sim) = \geq 8(kT + nT) \quad (10-12)$$

$$Z^{1-N/N} G(z, \sim) = Z[ew^{TM}w \sim G(s)]z^{-n/N} \quad 0 < n/N < 1$$

Thus
Now Eq. (10-10) can be written as

$$\sum_{n=1}^{N-1} x'(n/N) \quad 0 < n/N \leq 1 \quad (10-13)$$

$$C(x)x = G(x)E(x) + \sum_{n=1}^{N-1} z^{1-(n/N)} G(2, 1) E(z) \quad (10-14)$$

Equation (10-14) is more convenient to use than Eq. (10-10), since in it $G(z, n/N)$ is readily determined from the modified z -transform table with m replaced by n/N .

An alternate form for $C(z)$ can be obtained if we let $(v/N) = k + (n/N)$. Then

$$\sum_{v=0}^{N-1} \sum_{k=0}^{v/N} 8(kT + T)e^{-\{k+(n/N)\}T's}$$

$$= \sum_{n=0}^g \sum_{k=0}^{n/N} 8(1)T$$

$$Z^{-UTS/N} = G(z)N \quad (10-15)$$

and
where
(10-16)
(10-17)

$$C(z) = G(2)x E(z)$$

$$G(z)x = [G(z)]^{2=2-1/N}$$

N
 $T=T/N$
Equation (10-16) gives the z -transform of the output of the multirate sampled system of Fig. 10-1a as the product of $E(z)$ and $G(2)N$ where the latter is obtained from the ordinary z -transform $G(z)$ by replacing z and T by z^{-} and T/N , respectively.
 $-1/N$

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Example 10-1

Determine the z -transform of the output of the system shown in Fig. 10-6. Using Eq. (10-14), we find the z -transform of the system output for $N = 3$ to be

$$C(\approx), = G(\approx) E(\approx) + \sum_{7=1} z^{1-(\sim^3)} G(\approx, 7) E(z)$$

3

(10-18)

From the table of z-transforms and modified z-transforms, the following functions are obtained :

G(z)

=

(1-e)

(1)(z-e-T)

(10-19)

e(t)

X

e*(t)

1

G(s)=-

C(1) XC²(FINS₁

s(s+1)

TIN SN

Fig. 10-6 Multirate sampled-data system of Example 10-1.

1

and

G(z, m)

=

Thus

C(z)³ = E(z)

E(2) [(z

-

+

e

e-27/3

2-e

which is simplified to

C(2)³ =

-

(1/3 - 1) (2 1/3 - e-T³) E(2)

.

e-mr

e

(1 e-T)

-

- 1)(1 - e-7)

♦

(10-20)

1

+

-

e-T/3

1 z-e

22/3

(10-21)

(1-e-T/3) 1/3

(10-22)

.

For any given input E(z) the values of the output response of the system at the sampling instants kT/3, (k = 0, 1, 2, ...) are determined by first letting 23 = 213 and then by expanding C(z), by long division into a power series in powers of z⁻¹; and c(kT/3) is given by the coefficient of the z* term. This procedure is essentially similar to the submultiple sampling method used for the determination of response between sampling instants of a sampled-data system.

We shall now show that the output z-transform of the system can also be obtained from Eq. (10-17); that is

$$G(z)^3 = \frac{(1 - e^{-T})z}{1 - e^{-Tz}} = \frac{(1 - e^{-T/n})z}{1 - e^{-Tz/n}}$$

Evidently, this equation will lead to the same result for $C(z)$ as that given by Eq. (10-22).

For more complex open-loop systems with multirate sampling (with the basic sampler preceding the multirate rate sampler or samplers), methods similar to that described above can be applied. The z -transforms of the outputs of the systems shown in Figs. 10-7a and 10-7b are respectively

$$C(z) = G_1(z)G_2(z)E(z)$$

$$C(z) = G_1(z)G_2(z)NE(z)$$

$$(10-24)$$

$$(10-25)$$

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$$e(t)$$

$$e^*(t)$$

$$G_1(s)$$

$$G_2(s)$$

$$X$$

$$TIN$$

$$TIN$$

$$(a)$$

$$e(t)$$

$$X$$

$$c(1)N$$

$$G_1(s)$$

$$G_2(s)$$

$$TIN$$

$$(b)$$

Fig. 10-7 Block diagrams of open-loop multirate sampled systems.

The previous section is devoted to the study of the multirate sampling system of Fig. 10-1a through the use of the weighting sequence and the z -transform methods. The purpose of going through these seemingly complicated derivations of $C(z)$ is to obtain a good understanding of the characteristics of a multirate sampling system. Actually, a simplified analysis of the system can be made by referring to the equivalent block diagram with skip-sampled samplers and applying the signal flow graph technique. As has been shown previously, the signal flow graph method can be used effectively in the study of multiloop sampled-data systems. The concept of the sampled-flow graphs shall now be extended to sampled-data systems with multirate samplers.

Let us consider the open-loop multirate system shown in Fig. 10-1b. Since the input signal is sampled with a rate of N/T and

since the output of the system is sampled with a rate of $1/T$, the analysis of the system is necessarily different from that described for the system of Fig. 10-1a. The equivalent skip-sampled block diagram of the system is depicted in Fig. 10-8a, and the equivalent signal flow graph with skip-sampled samplers is constructed in Fig. 10-8b. From this signal flow graph, the pulse transform of $c(t)$ is written directly+

$$C^* = \sum_{n=0}^{N-1} (G e^{-nTs/N}) * X^* = \sum_{n=0}^{N-1} (G e^{-nTs/N}) * (E e^{j\omega nTs/N}) \quad (10-26)$$

where for simplicity, $C^*(s)$ is denoted by its shorthand notation C^* , $G(s)$ by G , etc.

The z-transform of $c(t)$ is

$$C(z) = \sum_{n=0}^{N-1} [e^{-rs/N} G(s)] Z[ents/N E(s)] \quad (10-27)$$

† Actually, $C(z)$ can be written directly from the block diagram of Fig. 10-8a. The introduction of the signal flow graph method here is intended to lay the foundation for future work when multiloop systems with multirate sampling are studied.

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$$\begin{aligned} & eTs/N \\ & e^{-Ts/N} \\ & e_1(1) e(1) \\ & e(t) \\ & e2Ts/N \\ & e^{-2Ts/N} \\ & e2(1) \\ & + \\ & G(s) \\ & [C(+)] X_{S_1} c^* (1) \\ & T S_1 C(z) \\ & -ekTs/N \\ & --- e-KTSIN ex(t) \\ & (N-1)Ts \\ & e \\ & N \\ & e \\ & -(N-1)Ts \\ & N \\ & (a) \\ & * \\ & eN-1(1) \\ & 1 \\ & eTs/N \\ & e^{-TS/N} \\ & oxa \\ & 1 \\ & G \\ & E0 \\ & e2Ts/N \\ & e^{-2Ts/N} \\ & E \\ & OX2 \end{aligned}$$

CTC
 e^{kTs}/N
 e^{-kTs}/N
 $--0Xk$
 $-(N-1)Ts$
 e^{-N}
 $(N-1)Ts$
 e^{-N}
 $0XN-1$
 $XN-1$

(b)

Since

and

Fig. 10-8 (a) Equivalent skip-sampled system of the multirate sampled system of Fig. 10-1b. (b) Signal flow graph of the equivalent block diagram of (a).

$E(s) = zE(z, n/N) \quad 0 < n/N \leq 1$

$-nTs/N$

$ents/NG(s) = G(z, 1 - n/N) \quad 0 < n/N < 1$

Equation (10-27) now becomes

$N-1$

(10-28)

(10-29)+

$C(z) = E(z)G(z) + \sum_{n=1}^{N-1} zE(z, n/N)G(z, 1 - n/N) \quad (10-30)$

As another possible multirate sampling scheme, let us consider the open-loop system of Fig. 10-1c, in which the two samplers are sampling at rates of $1/T_1$ and $1/T_2$ samples per second, respectively.

If we assume that T_1 and T_2 are commensurable, i. e., the ratio of T_1 to T_2 is a rational number, we can let $T_1 = T/N_1$, and

1

2

2

+ Note that $[ents/NG(s)] = 2G(z, n/N)$ for $n \leq N$ but $[e^{-nTs}/NG(s)] = 2G(2, -n/N)$.

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$Co(1)$

esT/N_1

e^{-sT}/N_2

e^{-sT}/N_2

(1)

e^{2sT}/N_1

e^{-2sT}/N_1

$+ G(s)$

e^{2sT}/N_2

e^{-2sT}/N_2

$+ eksT/N_1$

X

$-e^{-ksT}/N_1$

$eksT/N_2$

e

$(N_1-1)Ts$

$-(N-1)Ts$

N_1

e

$$\begin{aligned}
& N_1 \\
& (0) \\
& e \\
& N_2 \\
& X \\
& -e^{-ksT/N_2} \\
& (N_2-1)TSX-N_2-11TS \\
& -N_2-1(1 \\
& e \\
& N_2 \\
& 0\% \\
& eSTIN_2- \\
& e-STIN_2 \\
& ST/NOX_1 \\
& STINA \\
& G_1 \\
& e^{2ST/N_1} \\
& e^{-2ST/N_1} \\
& e^{2sT/N_2} e^{-2ST/N_2} \\
& E \\
& E \\
& 0X2 \quad X2 \\
& 2012 \quad 201 \\
& CN2 \\
& CN2 \\
& ekst/N_1 \quad e^{-ksT/N_1} \\
& ekst/N_2 \\
& e^{-ksT/N_2} \\
& 0XK \\
& (N-1)ST \\
& -(N_1-1)ST \\
& e \\
& (N_2-1)ST \\
& e \\
& N_1 \\
& e \\
& NA \\
& e \\
& N_2 \\
& -(N_2-1)ST \\
& N_2 \\
& XN_1-1 \quad XN_1- \\
& YN_2-1 \quad YN_2-1 \\
& (b)
\end{aligned}$$

Fig. 10-9 (a) Equivalent block diagram with skip-sampled samplers of the multirate sampled system of Fig. 10-1c. (b) Signal flow graph of the equivalent block diagram of (a).

=
1
2
T = T/N₂, where T is the least common multiple of T₁ and T₂, and N₁ and N₂ are integers. In general, if a sampled-data system contains switches that operate with periods of T₁, T₂, . . . , T_n, a number T and a set of integers N₁, N₂, . . . , N_n can always be found such that T₁ = T/N₁, T₂ = T/N₂, T₃ = T/N₃, . . . , T_n = T/N_n, provided that the ratios of T₁ to T₂ to T₃ to . . . T_n are rational numbers. As would follow, the two samplers of the system under study can be replaced by equivalent skip-sampled samplers. The equivalent block diagram of the system is shown in Fig. 10-9a, and the signal flow graph of this equivalent block diagram is shown in Fig. 10-9b. The z-transform of the system output is determined in the following manner:

1
2

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CNN,

$$= \sum_{n=0}^{N_1-1} e^{-nTs/N_1} Y^* (e^{nTs/N_1} G e^{-kTs/N_1})^* X_i \quad (10-31)$$

But
 Y^*

$$= \sum_{k=0}^{N_1-1} (e^{nTs/N_1} G e^{-kTs/N_1})^* (e^{kTs/N_1} E)^* \quad (10-32)$$

Therefore,
 C^2

$$= \sum_{n=0}^{N_2-1} \sum_{k=0}^{N_1-1} e^{-nTs/N_2} (e^{nTs/N_2} G e^{-kTs/N_1})^* (e^{kTs/N_1} E)^* \quad (10-33)$$

or
 $C(z)N_1 = \sum_{n=0}^{N_2-1} \sum_{k=0}^{N_1-1} Z(e^{nTs/N_2} G e^{-kTs/N_1}) Z(e^{kTs/N_1} E) \quad (10-34)$

Example 10-2

Determine the z-transform of the output of the multirate sampled-data system shown in Fig. 10-10a. Assuming that $T_1 = T/N_1$ and $T_2 = T/N_2$, where $T = 6$ sec, $N_1 = 3$, and $N_2 = 2$, the z-transform of the system output according to Eq. (10-34) is

$$C(z)_2 = \sum_{n=0}^{2-1/2} \sum_{k=0}^2 Z(e^{nTs/2} G(s) e^{-kTw/3}) Z(ATV/3 E(s)) \quad (10-35)$$

Expanding the right-hand side of the last equation, we have
 $C(z)_2 = G(2)E(\approx) + Z[G(s) e^{-Ts/3}] Z[e^{Ts/3} E(s)] + Z[e^{-2Ts/3} G(s)] Z[e^{2Ts/3} E(s)]$
 $+ Z[e^{Ts/3/2} G(s)] Z[e^{-1/2} + 2^{-1/2} Z[e^{Ts/3} G(s)] Z[e^{Ts/3} E(s)]$
 $+ 2^{-1/2} Z[e^{-Ts/3} G(s)] Z[e^{2Ts/3} E(s)]$
From Eqs. (10-28) and (10-29), Eq. (10-36) is written as

(10-36)

$$C(z)^2 = G(z) E(z) + zG(z, 2/3) E(z, 1/3) + zG(z, 1/3) E(z, 2/3) + z^{1/2}G(z, 1/2) E(z) + z^{3/2}G(z, 1/6) E(z, 1/3) + z^{1/2}G(z, 5/6) E(z, 2/3)$$

(10-37)

For a unit step input $E(z, m) = 1/(z - 1)$, and substituting the z-transform and the modified z-transform of $G(s)$ into Eq. (10-37), we have

$$C(z) =$$

$$= \frac{1}{z^2 - e^{-0.5}}$$

$$= \frac{(z - 1)(z - e^{-0.5}) [z^2 + (e^{-0.5} + e^{-1})z + e^{-0.75} + (e^{-1} + e^{-1.5})z^3]}{z^2 + 0.1533z + 0.0067z/2 + 0.41823z^2}$$

(10-38)

If we let $z = z^{1/2}$, then z

$$=$$

$z^{1/2}$, then z

$$C(z) =$$

$$= \frac{z^2 + 0.1533z^2 + 0.41823z + 0.0067zy}{z^2 + 0.0025z + 0.0025}$$

(10-39)

which is expanded into the power series

$$C(z) = 1.0418z + 1.1562z^2 + 0.4247z^3 + 1.1562z^4 + 0.4252z^5 + \dots$$

(10-40)

The waveform of $c(t)$ and the signal from the output sampler, $c^*(t)$, are sketched in Fig. 10-10b.

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$$e(t) \times e^* (t)$$

$$T_1 = 2 \text{ sec}$$

$$G(s)$$

$$G(s) = 5s + 1$$

$$c(1)$$

$$X$$

$$c(1)^3,$$

$$T_2 = 3 \text{ sec}$$

(a)

$$c(1)$$

$$(1)^3$$

$$c(1)^3$$

$$1.585$$

$$1.155 \quad 0.425 \quad 1.156$$

$$0.418$$

$$-c(t)$$

$$1$$

$$0$$

$$2 \quad 3 \quad 4 \quad 6$$

$$8910 \quad 12$$

$$14$$

(b)

Fig. 10-10 (a) Multirate sampled-data system for Example 10-2.

(b) Unit step response of the system in Example 10-2.

When a zero-order hold or a transportation lag is located in an

-H
eTs/N2

$(e^{-sT/2G})^*$
 $-H$
 (b)
 C
 1
 $Y_0 - (est/2H)^*$
 $(est/2R)^*$
 1
 $**$
 $0lest/2R)^*$
 $(e^{-sT/2G})^*$
 $Y_0 - (est/2H)$
 $-H$
 $e^{-ST/2}$
 G
 OC
 (c)

$N_1 = 2, N_2 = 1$

Fig. 10-12 (a) Signal flow graph. (b) Sampled flow graph. (c) Composite flow graph of multirate sampled system of Fig. 10-11.

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$N_1 - 1$

$Y^* =$

$m=1$

$(e^{kTs/N} : G e^{-mTs/N_1})^* X^* \quad (k = 1, 2, \dots, N_2 - 1) \quad (10-47)$

$Y^* C^*$

$=$

$"$

$(10-48)$

Using these equations, we can construct a sampled-flow graph that portrays the functional relations between the variables $(e^{nTs/N} : R)^*$, X , Y , and C^* , where $n = 0, 1, 2, \dots, N_1 - 1$, and $k = 1, 2, \dots, N_2 - 1$.

-

As an illustrative example, let $N_1 = 2$ and $N_2 = 1$ in the system of Fig. 10-11. The signal flow graph model with skip-sampled samplers

$||$
 $C(z)$
 $R(s)$
 $D(s)$
 $G(s)$
 TIN
 $C(s)$
 (a)
 $N-1$
 $-nTs/NG(s)]$
 $C(z) =$
 $=$
 $R(z) \sum_{n=0}^{N-1} [e^{nTs/ND(s)}] Z[e^{-nTs/NG(s)}]$
 $C(z, m) =$
 $1 + \sum_{n=0}^{N-1} Z[e^{nTs/ND(s)}] Z[e^{-nTs/NG(s)}]$
 $R(z) \sum_{n=0}^{N-1} [e^{-nTs/NG(s)}] Z[e^{nTs/ND(s)}]$
 $1 + \sum_{n=0}^{N-1} Z[e^{nTs/ND(s)}] Z[e^{-nTs/NG(s)}]$

$$\begin{aligned}
& T \quad C(z) \\
& R(s) \\
& D(s) \\
& G(s) \\
& T_{IN} \\
& C(s) \\
& C(2) \\
& = \\
& C(z, m) \\
& = \\
& N-1 \\
& (b) \\
& G(z) \sum_{n=0}^{N-1} [e^{-nts/ND(s)}] Z[e^{1TM}s/N R(s)] \\
& N-1 \\
& 1 + \sum_{n=0}^{N-1} [e^{-nts/ND(s)}] Z[eurs/NG(s)] \\
& N-1 \\
& G(z, m) [ers/N D(s)] Z[e^{1rs}/N R(s)] \\
& N-1 \\
& n=0 \\
& 1 + \sum_{n=0}^{N-1} [e^{-*ND(s)}] \sum_{n=0}^{N-1} [e^{TG(s)}] \\
& 71=0
\end{aligned}$$

Fig. 10-13 z-transforms and modified z-transforms of outputs of some simple multirate sampled feedback control systems.

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$$\begin{aligned}
& R(s) \\
& C(s) \\
& G(s) \\
& + \\
& T_{IN} \\
& C(z) \\
& = \\
& N-1 \\
& H(s) \\
& (c) \\
& \sum_{n=0}^{N-1} [ens/NG(s)] [ers/NR(s)] \\
& N-1 \\
& 1 + \sum_{n=0}^{N-1} [e^{-nts/NG(s)}] Z[ents/N H(s)] \\
& C(z, m) \\
& = \\
& n=0 \\
& N-1 \\
& n=0 \\
& N-1 \\
& -nrs/NG(s)] Z[enTs/N R(s)] \\
& 1 + \sum_{n=0}^{N-1} [e^{-nts/NG(s)}] Z[e^{TM}s/N H(s)] \\
& 20
\end{aligned}$$

Fig. 10-13 (continued)

is depicted in Fig. 10-12a. The following equations are written from the signal flow graph:

$$X^* R^* H^* Y^*$$

$$X^* = (e^{sT}, {}^2R)^* - (e^{T/2H})^* Y^*$$

$$C^* = G^* X^* + (e^{-ST}, {}^2G)^* X^*$$

$$Y^* = C^*$$

$$(10-49)$$

$$(10-50)$$

$$(10-51)$$

(10-52)

Based on these equations, the sampled-flow graph of the system is now depicted in Fig. 10-12b, and the starred transform of the system output is written directly as

$$C^*(z) = \frac{G^*R^* + (e^{-ST/2}G)^* (e^{3T/2}R)^*}{1 + G^*H^* + (e^{-ST/2}G)^* (e^{3T/2}H)^*}$$

The z-transform of the output is

$$C(z) = \frac{G(z)H(z) + Z[e^{-3T/2}G(s)] Z[e^{3T/2}H(s)]}{1 + G(z)H(z) + Z[e^{-3T/2}G(s)] Z[e^{3T/2}H(s)]}$$

(10-53)

(10-54)

The continuous output $C(s)$ and its modified z-transform are determined from the composite-flow graph. The composite-flow graph is obtained by combining the flow graphs of Figs. 10-12a and 10-12b (In this case, only the significant portion of the original flow graph is included: see Fig. 10-12c).

From the composite flow graph shown in Fig. 10-12c, we obtain

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-ST/2

$$1 + G^*H^* + (e^{-ST/2}G)^* (e^{3T/2}H)^*$$

$C(s)$

$$= \frac{R^* \{G[1 + (e^{-ST/2}G)^* (e^{3T/2}H)^*] - (e^{-ST/2}G)(e^{3T/2}H)^*G^*\}}{1 + G^*H^* + (e^{-ST/2}G)^* (e^{3T/2}H)^*}$$

+

(10-55)

$C(z, m)$

$$= \frac{(e^{3T/2}R)^* [(e^{-ST/2}G)(1 + G^*H^*) - GH^* (e^{-ST/2}G)^*]}{1 + G^*H^* + (e^{-ST/2}G)^* (e^{3T/2}H)^*}$$

$$R(z) \{G(z, m) [1 + Z(e^{-ST/2}G) Z(e^{3T/2}H)] - G(z) Z(e^{-ST/2}G) Z(e^{3T/2}H)\}$$

$$1 + G(z)H(z) + Z(e^{-ST/2}G) Z(e^{3T/2}H)$$

+

-ST/2

$$3(e^{3T/2}R) \{3m(e^{-ST/2}G)[1 + G(z)H(z)] - G(z, m) H(z) Z(e^{-ST/2}G)\}$$

$$1 + G(z)H(z) + Z(e^{-ST/2}G) Z(e^{3T/2}H)$$

(10-56)

By similar methods the z-transforms and modified z-transforms of some other simple closed-loop systems with multirate sampling are given in Fig. 10-13.

We shall now study the performance of a multirate sampled-data system and compare it with that of a single-rate sampled system. A numerical example is given below.

Example 10-3

The block diagram of a multirate sampled-data feedback control system is shown in Fig. 10-14. The sampling period of the low speed sampler is one second; the high speed sampler samples N times as fast, N being an integer. Each sampler is followed by a zero-order hold. The transfer function of the combination of the first zero-order hold and the compensator is

$D(s)$

=

$$\frac{1 - e^{-st}}{s(s + 1)}$$

(T = 1 sec)

(10-57)

Since the zero-order hold following the high-speed sampler holds the sampled signal only for T/N second, the transfer function for this hold and the plant is

$G(s)$

= K

\$2

(10-58)

From Fig. 10-13a, the closed-loop z-transfer function of the system is determined

$C(z)$

$R(z)$

$$= \sum_{n=0}^{N-1} [e^{nsT}/ND(s)] [e^{-nsT}/NG(s)]$$

$$1 + \sum_{n=0}^{N-1} Z[e^{nsT}/ND(s)] Z[e^{-nsT}/NG(s)]$$

(10-59)

$n=0$

$R(s)$

$E(s)$

Zero-order

$Dc(s)$

LX

$C(s)$

hold

Zero-order

hold

$G_o(s)$

TIN

$D(s)$

$G(s)$

Fig. 10-14 Block diagram of multirate sampled-data system in Example 10-3.

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The following z-transforms are determined:

Z

1

-

e-

$$D(z) = 3 [35 + 1]$$

$$3[+] =$$

-

KIN

$$G(z) = 3[$$

$$3[1]=KIN$$

-8

-

$$3[ents/ND(s)] = 23[1].$$

$$z^m 5(s+1) \quad m=n/N$$

2-1

-

0.632

- 0.368

(10-60)

(10-61)

$$z(1 - e^{-n/N}) + e^{-n/N}$$

$$(z - 0.368)$$

$$\begin{aligned}
& - \\
& 0.368 \\
& = \\
& KZT \\
& 3m[13] \\
& H \\
& - \\
& m=1-(n/N) - K3mm-1-(n+1)/N \\
& K/N \\
& = \\
& 2-1 \\
& e^{-ns/N} \\
& - e^{-s/N} \\
& 3 [e^{TnTxNG(1)}] = 3 [K \alpha^{**}(1 - e)] \\
& (10-62) \\
& (10-63)
\end{aligned}$$

The characteristic equation of the system is obtained by setting the denominator of Eq. (10-59) equal to zero. Therefore,

$$\begin{aligned}
& N-1 \\
& 1 + \sum Z[ents/ND(s)] Z[e^{-nrs}/NG(s)] = 0 \\
& (10-64) \\
& 72=0
\end{aligned}$$

which is reduced to

$$\begin{aligned}
& 1+D(z)G(z) = 0 \\
& (10-65)
\end{aligned}$$

when $N = 1$, and the system becomes a single-rate sampled system.

Substituting Eqs. (10-60) through (10-63) into Eq. (10-64), we have

$$\begin{aligned}
& 1 + \\
& 0.632K/N \\
& (\approx - 1)(z - 0.368)
\end{aligned}$$

$$\begin{aligned}
& - \\
& - \\
& N-1 \\
& + \\
& : (1 - e^{-n/N}) + e^{-n/N} \\
& (z - 0.368)
\end{aligned}$$

$$\begin{aligned}
& - \\
& X \\
& ---- \\
& 0.368 K/N \\
& (\approx - 1)
\end{aligned}$$

$$\begin{aligned}
& = \\
& 0 \\
& (10-66) \\
& 22 - (1.368 - K + \\
& K + \\
& 0.632K \\
& N(1 - e^{-1/N} \\
& Z + \\
& 0.632K \\
& N(1-e-/N) \\
& + 0.368(1 - K) (10-67)
\end{aligned}$$

Equation (10-66) is reduced to

Applying the Schur-Cohn or the modified Routh stability criterion, we obtain the maximum allowable gain for stability from the following conditions:

$$\begin{aligned}
& (1) \\
& (2) \\
& (3) \\
& K > 0 \\
& 0.632 + \\
& (0.36
\end{aligned}$$

-

$$0.632$$

$$N(1 - e^{-1/N})$$

$$K > 0$$

$$2.736 - (1.36$$

$$1.264$$

$$N(1-e)K > 0$$

$$(10-68)$$

$$(10-69)$$

$$(10-70)$$

When various positive values of N (in this case, N can take on non-integral values greater than one) are substituted into Eqs. (10-69) and (10-70), we find that the stability of the system is completely determined by Eqs. (10-68) and (10-69), and the condition stated in Eq. (10-70) is superfluous. Thus the maximum allowable gains for various values of N are determined from

$$K <$$

$$0.632$$

$$(10-71)$$

$$-$$

$$0.632$$

$$N(1 - e^{-1/N})$$

$$0.368$$

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4
3
Kmax
2
1
1
2 3 4
5 6 7 8
N
9 10

Fig. 10-15 Maximum value of K for stability versus N for the system in Example 10-3.

and a plot of the maximum K versus N is drawn in Fig. 10-15.

Several pertinent facts on the performance of this multirate sampled-data system can be drawn from Fig. 10-15. When $N = 1$, the system has single-rate sampling, and the maximum allowable gain for system stability is equal to unity. Also the high-speed sampler tends to make the system more stable.

10.2 VARIABLE-RATE SAMPLING

In one class of sampled-data systems the samplers do not operate with a uniform rate; that is, the durations between consecutive sampling pulses are not constant. Such systems are called variable-rate sampled-data systems. Although the sampling scheme of the variable-rate samplers may have various forms, the following discussion is limited to systems containing samplers that operate with periodically variable rates; this type of system is described as the cyclic-variable rate sampled-data system. As will be shown later, cyclic-variable rate sampling offers many useful features such as the improvement of the performance of a sampled-data system and the time-sharing of equipment in discrete-data systems.

The block diagram of a typical cyclic-variable rate sampled-data system is shown in Fig. 10-16a. The sampler S operates with a constant sampling rate of $1/T$ samples per second, whereas the sampler S_e operates with a cyclic-variable rate at instants $t = 0, T/N_1, T/N_2, T/N_3, \dots, T/N_n; T, (1 + 1/N_1)T, (1 + 1/N_2)T, (1 + 1/N_3)T, (1 + 1/N)T; 2T, (2 + 1/N_1)T, (2 + 1/N_2)T, \dots, (2 + 1/N_n)T; 3T,$ where $N_1 > N_2 > N_3 > \dots > N_n > 1$. The sampling functions of S and S_e are described in Fig. 10-16b. An examination of Fig. 10-16b reveals that the cyclic-rate sampler S_e can be replaced by a

parallel combination of the equivalent skip-sampled samplers used in preceding sections. However, unlike the case of multirate sampling, the delays of the skip-sampled components here are not uniform but are determined by the values of $N_1, N_2, N_3, \dots, N_n$. For instance, the k th skip-sampled sampler consists of a basic sampler with sampling

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Cyclic-variable

rate sampler

$R(s)$

$OE(s)X$

$E(s)$

S

Zero-order

hold

$Dc(s)$

Zero-order

$C(s)$

Sc hold

$Gc(s)$

$Do(s)$

0

T T

N_1 N_2

$\frac{1}{N_1}$

N_3

$\frac{1}{N_3}$

(a)

$\cdot G_k(s) \cdot$

$k=0,1,2,\dots,n$

T

27

T

$(1+)$

$\frac{1}{T}$

$)T(1+ IT 2T (2 + 1/1)T$

$(2+$

t

$(1 + 1) T$

N_2

(b)

Fig. 10-16 (a) Block diagram of cyclic-rate sampled-data system.

(b) Top: Sampling function of sampler S . Bottom: Sampling

function of cyclic sampler Sc .

X_0

T

eTs/N_1

X_1

$e-Ts/N$

T

ets/N_2 X_2 X_2 $e-TS/N_2$

Sc

(a)

eTs/N_k

eTs/N_n

$e-Ts/N_k$

$e-Ts/N_n$

T

(b)

Fig. 10-17 (a) Cyclic-variable rate sampler. (b) Equivalent skip-

sampled flow graph model of a cyclic-variable rate sampler.

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T_s

period T ; the sampler is preceded by a time advance unit with transfer function e^N and followed by a time delay unit with transfer function $e^{-T/N}$. The equivalent skip-sampled flow graph model of a cyclic-variable rate sampler is now constructed in Fig. 10-17, and the equivalent flow graph of the sampled-data system of Fig. 10-16a is drawn in Fig. 10-18. The transfer function of the combination of the zero-order hold following the single-rate sampler S and $D(s)$ is designated as $D_o(s)$; therefore,

D_o

oxo

G_o

$e^{-T_s/N_1} G_1$

$e^{T_s/N_1} D_o$

$ox,$

$e^{-T_s/N_2} G_2$

$e^{T_s/N_2} D_o$

OX_2

X_2

C

$@T_s/N_k G$

$e^{T_s/N_k} D_o$

OX

$e^{-T_s/N_n} G_o$

$e^{T_s/N_n} D_o$

-1

Fig. 10-18 Equivalent signal flow graph with skip-sampled samplers of sampled-data system of Fig. 10-16a.

$-T's$

$D_o(s) = 1 - e^{-TM} D(s)$

S

$.$

(10-72)

However, since the zero-order hold holds a sampled pulse signal until the next pulse comes along, the transfer functions of the zero-order holds that follow all the equivalent samplers in Fig. 10-18 are necessarily different. For instance, the transfer function of the zero-order hold that follows the k th skip-sampled sampler is

$1 - \exp(-T_s (N \dots -))$

1

1

N

$+1$

(10-73)

for k

$=$

$2, 3,$

$n - 1.$

$\dots$

S

Therefore, with reference to Fig. 10-18, we have

$G_o(s)$

1

$-T_s' N_i$

e

$=$

$G(s)$

S
(10-74)

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$R(s)$

$E(s)$

Zero-order

hold

$G(s)$

$C(s)$

S_c

Cyclic-rate

sampler

(a)

K

$S+1$

$\begin{matrix} | & | & || & | & \downarrow \downarrow \\ T & T & (1+1)7 & 27 \end{matrix}$

e

TS/N

(b)

$(2+1)7 \quad 37 \quad (3+1)T$

e^{-TS/NG_1}

Go

$G(s) = 1 - e^{-TS/N}$

TS/N

$-G(s)$

S

$G_1(s) =$

$1 - e^{-Ts(1-4)}$

$-G(s)$

S

(c)

Go

G

$-(e$

$lets/NG)$

(e

$-TS/NG_1$

$-le$

TS/NGC

$TS/NR)$

$Gets/N$

Go

(d)

Fig. 10-19 (a) Cyclic-variable rate sampled-data system for Example 10-4. (b) Sampling function of cyclic-variable rate sampler. (c) Equivalent signal flow graph of cyclic-variable sampled-data system with skip-sampled samplers. (d) Composite signal flow graph of the cyclic-rate sampled-data system of (a).

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$G_1(s)$

=

1

-

-

$\exp[-T_3(N, -N)] \quad 6(3)$

T_s
 S
 $\frac{1}{N}$
 $G_n(s)$
 1
 $-$
 $-$
 $-T_s[1-(1/N)]$
 $\cdot G(s)$
 S
 $G(s)$
 $(10-75)$
 $(10-76)$

It follows that the z-transform and the modified z-transform of the system output are given by

$C(x) =$
 and
 $C(z, m)$
 $=$
 respectively.

$$G_o(z)D_o(z) + \sum_{k=1}^N Z[e^{j\omega T/N} D_o(s)] \{e^{-j\omega T/N} G_x(s)\}$$

$$1 + G_o(z) D_o(z) + \sum_{k=1}^N Z[e^{j\omega T/N} D_o(s)] Z[e^{-j\omega T/N} G_x(s)]$$

$$D_o(z)G_o(z, m) + \sum_{k=1}^N Z[e^{j\omega T/N} D_o(s)] Z[e^{-j\omega T/N} G_x(s)]$$

$$1 + G_o(x)D_o(x) + \sum_{k=1}^N Z[e^{j\omega T/N} D_o(s)] \{e^{-j\omega T/N} G_x(s)\}$$

$R(z)$
 $(10-77)$

$R(x)$
 $(10-78)$

In order to illustrate the technique of analysis and the characteristics of sampled-data systems with cyclic-variable rate samplers, two numerical examples are given below.

Example 10-4

Consider the simple cyclic-variable rate sampled-data system shown in Fig. 10-19a. Assume that the sampler operates at instants of $t = 0, T/N, T, (1 + 1/N)T, 2T, (2 + 1/N)T, 3T, \dots, (k + 1/N)T, (k + 1)T, \dots$, where N is a positive number greater than unity and T is equal to one second.

The sampling function of the sampler is depicted in Fig. 10-19b.

The purposes of this example are to determine the optimum value of N for which the system is most stable and to show the effects of the variations of N on the system performance.

The equivalent signal flow graph of the system is now depicted in Fig. 10-19c. The cyclic-rate sampler is decomposed into two basic samplers operating in synchronism with a sampling period of one second. The equivalent sampler in the first branch is not accompanied by any advance and delay units; the sampler in the second branch is preceded by an advance unit with transfer function $e^{Ts/N}$ and followed by a delay unit with transfer function $e^{-Ts/N}$. Since the sampler in the second branch operates at the instants $t = T/N, (1 + 1/N)T, (2 + 1/N)T, \dots, (k + 1/N)T, \dots$, the transfer functions of the zero-order holds in the two branches are different and are given by

$$1 - e^{-Ts/N}$$

S
 $(10-79)$

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and

1

$$e^{-Ts[1-(1/N)]}$$

S

$$(10-80)$$

respectively. The hold device in the first branch holds the output of the basic sampler for only the duration of T/N second, whereas the hold device of the second branch holds the corresponding basic sampler output for the remaining time $(1 - 1/N)T$ during the interval of T seconds. Therefore, with reference to Fig. 10-19c,

and

-

$$G_o(s)$$

$$1 - TS/N$$

=

$$G(s)$$

S

1

--

$$-Ts[1-(1/N)]$$

$$G_1(s)$$

=

$$G(s)$$

S

$$(10-81)$$

$$(10-82)$$

The z-transform and the modified z-transform of the system output can be determined by use of the composite signal flow graph technique. In Fig. 10-19c if we regard R , X^* , X^* as input node variables and X_o , X_1 , C as output node variables and write the starred transforms of the output variables as functions of the input variables, we have

$$X = R^* - G_o^* X^* - (e^{-T/N} G)^* X^*$$

-

$$X^* G^*$$

$$(10-83)$$

$$X^* = (e^{T^3/NR})^* - (e^{Ts/NG_o})^* X^* - G^* X^*$$

$$(10-84)$$

$$C^* = G^* X + (e^{-7s/NG_o})^* X^*$$

$$(10-85)$$

$$C = GX^* + (e^{-T/NG_o})^* X^*$$

$$(10-86)$$

The composite signal flow graph of the system is constructed from these four equations as shown in Fig. 10-19d. The z-transform and the modified z-transform of the system output are readily determined from the composite signal graph as

$$C(z)$$

and

$$C(z, m)$$

=

=

+ -

$$\{G_o(\approx)[1 + G_1(\approx)] - Z[e^{TM^3/NG_o(s)}] Z[e^{-TM^3/NG_1(s)}]\} R(z)$$

+

-

$$\frac{Z[e^{TM^3/NR(s)}] \{[1 + G_o(\approx)] Z[e^{TM^3/NG_1(s)}] - Z[e^{TM^3/NG_o(s)}] G_o(z)\}}{1 + G_o(\approx) + G_1(\approx) - Z[e^{TM^3/NG_o(s)}] Z[e^{-TM^3/NG_1(s)}] + G_o(\approx) G_1(\approx)}$$

$$(10-87)$$

$$\{[1 + G_1(\approx)] R(\approx) - Z[e^{TM^3/NR(s)}] Z[e^{-Ts/NG_1(s)}]\} G_o(z, m)$$

N

$$1 + G_o(\approx) + G_1(\approx) - Z[e^{TM^3/NG_o(s)}] Z[e^{-TM^3/NG_1(s)}] + G_o(z) G_1(\approx)$$

+

-

$$\{[1 + G_o(z)] Z[e^{TM^3/NR(s)}] - 3[es/Go(s) (R\approx)]\} m[e^{-TM^3/NG_1(5)}]$$

$$1 + G_0(z) + G_1(z) - Z[e^{Ts} N G_0(s)] Z[e^{-Ts} / N G_1(s)] + G_0(z) G_1(z) \quad (10-88)$$

The characteristic equation of the system is obtained by setting the denominator of Eq. (10-87) or Eq. (10-88) equal to zero. Thus

$$1 + G_0(z) + G_1(z) - Z[e^{Ts} N G_0(s)] Z[e^{-Ts} / N G_1(s)] + G_0(z) G_1(z) = 0 \quad (10-89)$$

Making use of the relationships of Eqs. (10-41) through (10-44), we obtain the following z-transforms of the system:

$$\begin{aligned} e^{-Ts} N \\ \frac{e^{-Ts}}{s} N \\ G_0(z) &= 3 \left[\frac{1}{s} \right] \\ \left[\frac{1}{s} \right] &= 3 \left[K \frac{s(s+1)}{s(s+1)} \right] \\ &= 0.368 (e^{-1/N} - 0.368) \\ &\approx 0.368 \end{aligned} \quad (10-90)$$

$$\begin{aligned} &376 \\ &\text{Chap. 10} \\ &\text{MULTIRATE AND SKIP SAMPLING} \\ &e^{-Ts} [1 - (1/N)] \end{aligned}$$

$$\begin{aligned} e \\ G_1(z) &= 3 \left[\frac{1}{s} \right] \\ \left[\frac{1}{s} \right] &= 3 \left[K \frac{s(s+1)}{s(s+1)} \right] \\ &= K (1 - e^{-Ts} [1 - (1/N)]) \\ &= K (e^{-1/N} - 0.368) \end{aligned}$$

$$\begin{aligned} &= 3 \left[\frac{e^{-Ts} N G_0(s)}{s(s+1)} \right] = 3 \left[K \frac{e^{-Ts}}{s(s+1)} \right] \\ &= 3 \left[\frac{e^{-Ts} N G_0(s)}{s(s+1)} \right] = 3 \left[K \frac{e^{-Ts}}{s(s+1)} \right] \end{aligned}$$

$$\begin{aligned} &3 [e^{-Ts} * G_0(s)] = [e^{-Ts}]. \\ &\text{For a unit-step displacement input,} \\ &\left[\frac{1}{s} \right] \end{aligned}$$

$$\begin{aligned} &1) \\ &\left[\frac{1}{s} \right] \\ &K \frac{s(s+1)}{s(s+1)} \\ &0.368 \\ &(10-91) \end{aligned}$$

$$\begin{aligned} &= \\ &z - 0.368 \\ &(10-92) \end{aligned}$$

$$K(1 - e^{-1/N})$$

$$\begin{aligned} &= \\ &2 - 0.368 \\ &(10-93) \end{aligned}$$

$$\begin{aligned} &\text{and} \\ &R(z) = 1 \end{aligned}$$

$$\begin{aligned} &= \\ &3 [e^{-Ts} / N R(s)] \end{aligned}$$

$$\begin{aligned} &= \\ &z=1 \end{aligned}$$

(10-94)

(10-95)

Substituting Eqs. (10-90) through (10-93) into Eq. (10-89) and simplify.

ing, we obtain the characteristic equation

$$(\approx 0.368) [\approx - (1.368K + 0.736K + 0.368)$$

-

$$+ (0.368e^{+e1/N}) (K + 1)K] = 0$$

(10-96)

The characteristic equation has a fixed root at +0.368 and a variable root whose location depends on the values of N and K. For the closed-loop cyclic-rate sampled-data system to be stable, the roots of the characteristic equation must lie inside the unit circle of the z-plane. Therefore, for stability the values

of K and N must satisfy the following condition:

$$|(1.368K^2 + 0.736K + 0.368) - (0.368e^{1/N} + e^{-1/N})(K + 1)K| < 1 \quad (10-97)$$

Separating the terms containing K from the terms containing N, the above condition is met when

and

$$0.368e^N + e^{-1/N} >$$

$$0.368(1 + K) + (K - 1)$$

K

$$0.368e/N + e^{-1/N}$$

$$0.368(1+K)^2 + K^2 + 1$$

$$(K + 1)K$$

(10-98)

(10-99)

The stability boundary curves for the system are plotted from Eqs. (10-98) and (10-99) in Fig. 10-20. Observe that when the sampler operates at a uniform rate of one sample per second ($T = 1$ sec), the maximum gain for stability is equal to 2.16; when the rate of the uniform sampling is increased to two samples per second ($T = 0.5$ sec or $N = 2$), the marginal value of K for stability is 4.1. The effect of the cyclic sampling is clearly demonstrated by the stability boundary curves. For both large and small values of N, the closed-loop system with cyclic sampling is stable even for very high gains. For instance, when $1/N$ equals 0.1 or 0.9, the system is stable for values of K between 31.75 and 35.1 (of course the sensitivity of the system stability with respect to the variation in system gain K is poor). If we substitute Eqs. (10-90) through (10-95) into Eq. (10-87) and simplify, the z-transform of the system output is obtained as

† J. T. Tou, "Multiple Variable-rate Discrete-data Control Systems," Proceedings of

International Automatic Control Conference, Butterworth Publications, Moscow, 1960.

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30

25

20

K

15

10

5

Unstable region

Unstable region

Kmax Values of K for deadbeat performance

Kmin

Unstable

region

Kmin

Stable region

Kmin

$\frac{0}{-5}$

0

0.1

0.2 0.3

C(z)

=

'Unstable region

0.4 0.5 0.6 0.7 0.8 0.9 1.0

1/N

Fig. 10-20 Stability boundary curves for the cyclic-rate sampled-data system in Example 10-4.

$Kz[(0.368e^{1/N} + e^{-1/N} - 1.368)K + 0.632]$

- -

$(\approx -1)[\approx - (1.368K^2 + 0.736K + 0.368) + (0.368e^{1/N} + e^{-1/N})(K^2 + K)]$

(10-100)

If deadbeat performance is to be achieved, the characteristic equation of the system should be of the form: "0. Therefore, to achieve deadbeat performance, the root of the characteristic equation must be located at the origin in the z-plane; that is,

-

$1.368K^2 + 0.736K + 0.368 (0.368e^{1/N} + e^{-1/N}) (K^2 + K) = 0$ (10-101)

The values of K for deadbeat responses when 1/N takes on various values between zero and unity are plotted in Fig. 10-20. For a given value of N

V/4

NIN

0 I

T

2T

3T

4T t

N

T

system of Example 10-4.

N

2T

3T

Fig. 10-21 Two cyclic sampling schemes of the sampled-data

4T

t

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(except for $N = 2$), two possible values of K will give rise to a deadbeat performance. However, the analysis of the system thus far made is based on the z-transformation, and a deadbeat response described at the basic sampling instants $t = 0, T, 2T, \dots$ does not necessarily mean that the actual continuous output of the system is deadbeat. We shall show that although replacing $1/N$ by $(11/N)$ in Eq. (10-100) does not alter the z-transform of the system and thus the system outputs for the two cyclic sampling schemes of Fig. 10-21 at the basic sampling instants of $t = 0, T, 2T, \dots$ are identical, the actual continuous outputs of the two cases are not the same. We shall use Eq. (10-88), the modified z-transform of the system output, to determine the system response in between the basic sampling instants. In order to use Eq. (10-88), the modified z-transforms of $G_o(s)$ and $e^{-TS/NG}(s)$ must be determined. From the definition of the modified z-transform, we have

-

$G_o(\approx, m) = 3m [K(1 - e^{-1/N})]$

2

Ts

$$K(e^{-TS}/N)$$

$$K(1 - e^{-Ts}/N)$$

$$\frac{e^{-Ts}}{s(s+1)}$$

$$\bullet (10-102)$$

$$\frac{K}{s^2} [s^2 + 3(s + 1)]$$

$$\frac{3m}{s} [e^{-TV}/NG, 3m]$$

$$\frac{K(e^{-TUN}}{s(s+1)}$$

$$K(e^{-Ts}/N e^{-Ts})$$

$$\frac{s(s+1)}{(10-103)}$$

$$J_u[e^{T_{TM}^*G}, (3)] = 3 \ll [K(erty * = 9'')] = = 3 [K (ot;)]''$$

The evaluation of the two modified z-transform functions given above is simple when the values of the output response at $t = (k+1/N)T$ for $k = 0, 1, 2, \dots$ are desired. For this case, we set $m = 1/N$, and Eqs. (10-102) and (10-103) become

$$\frac{0.368}{s(s+1)}$$

$$Go(\approx, m) = 2^{-1}$$

$$\frac{3}{s} [K($$

$$\frac{3}{s} [K(1$$

$$K(e^{-T}/N)$$

$$\frac{s(s+1)}{K(1 - e^{-V}/N)}$$

$$\frac{1}{s} Z[ets/Go(s)]$$

$$\frac{3m}{s} [e^{-T}/NG_1(s)]$$

$$\frac{-Ts}{s} [1 -$$

$$\frac{(10-104)}{K(e^{-1}/N - 0.368)}$$

$$\frac{(z - 0.368)}{(10-105)}$$

Substituting Eq. (10-104) and Eq. (10-105) into Eq. (10-88), we have the modified z-transform of the system output ($m = 1/N$) for deadbeat performance.

$$C(z, 1/N) =$$

$$\frac{K}{(\approx - 1)(\approx - 0.368)}$$

$$\frac{[z - 0.368(1 + K)]}{-K + K(e^{1/N-1} + e^{-1}/N)](1 - e^{-1}/N) + [\approx(1 - K) + zKe^{-1/4}/N]$$

$$\frac{-0.368(K + 1) + Ke^{-1}/N}{\approx^{-1}(e^{-1}/N - 0.368)}$$

After simplification, the last equation is reduced to

$$\frac{(10-106)}{C(z, 1/N) =}$$

$$\frac{K}{1) [(1 - e^{-1/3}) + 2^{-1}]}$$

$$\frac{1}{1}$$

$$\frac{1}{K+1} + 1] (10-107)$$

By long division the modified z-transform of the system output at the sampling instants $t = (k + 1/N)T$, ($k = 0, 1, 2, \dots$) is

$$C(z, 1/N) = K(1 - e^{-1/N}) z^{-1} +$$

$$\frac{K}{K+1} z^{-2} + \frac{K}{K+1} z^{-3} + \dots$$

$$(10-108)$$

Using Eq. (10-101) or Eq. (10-22), we can predict by the z-transform method that deadbeat performance of the system is possible for $K = 4$ when

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$1/N$ equals 0.223 or 0.777. Substituting $1/N = 0.223$ and $K = 4$ into Eq. (10-108), we have

$$C(z, 0.223) = 0.821 z^{-1} + 0.82 z^{-2} + 0.82 z^{-3} + \dots$$

which is a deadbeat response. However, for $1/N$

(10-108) becomes.

(10-109)

0.777 and $K = 4$, Eq.

$$C(z, 0.777)$$

$$= 1.2621 z^{-1} + 0.82 z^{-2} + 0.82 z^{-3} + \dots$$

$$(10-110)$$

Since the response now has an overshoot of 1.26, it is not deadbeat. For $K = 10$ and $1/N = 0.0953$ the deadbeat response of the system is described by the following modified z-transform:

$$1/N=0.223$$

$$1/N=0.777$$

Sampling functions

$$c(1)$$

$$2.16$$

$$2$$

$$K=4$$

$$m=1/N=0.777$$

$$0.8$$

$$m=1/N=0.223$$

$$0.223$$

$$0.777 \quad 1$$

$$2$$

$$c(1)$$

$$7$$

$$6$$

$$5$$

$$4$$

$$3$$

$$2$$

$$6$$

$$m=1/N=0.9047$$

$$m=1/N=0.0953$$

$K=10$
 0.91
 $1/N=0.0953$
 $1/N=0.9047$
 $0 \ 0.0953$
 $0.9047 \ 1$
 2

Fig. 10-22 Output step responses of cyclic-rate sampled-data system of Example 10-4.

31
31

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 But for K

$$C(z, 0.0953) = 0.91z^{-1} + 0.91z^{-2} + 0.91z^{-3} +$$

(10-111)

10 and $1/N = 0.9047$, the modified z-transform of the system output for $m = 0.9047$ is

$$C(z, 0.9047) = 6z^{-1} + 0.91z^{-2} + 0.91z^{-3} +$$

(10-112)

and the maximum overshoot is 600 percent. The output responses of the system for the two cases calculated above are depicted in Fig. 10-22. As can be shown in a similar manner, true deadbeat response for this system with cyclic sampling is possible only for $0 < 1/N < 0.5$.

Example 10-5

Let us consider the feedback control system with cyclic sampling shown in Fig. 10-16a. We assume that the sampler operates with the same cyclic scheme as that shown in Fig. 10-19b and that the sampling period is one second. As will be demonstrated in this example, an optimum value of $1/N$ exists at which the maximum allowable gain for system stability reaches its peak value. When $1/N$ is increased toward unity or is decreased toward zero, the maximum allowable gain for stability is decreased.

The transfer functions of the controller $D(s)$ and the fixed elements of the system are given by

and
 $D(s)$

$$G(s) = \frac{1}{s+1} = K/s^2$$

(10-113)

(10-114)

respectively. With reference to Eq. (10-77), the z-transform of the system output is given by

$$C(z) = \frac{Go(z) Do(z) + [eTsN Do(s)] [e^{-T/NG_1(s)}]}{1 + Do(z)Go(z) + Z[e^{Ts} N Do(s)] Z[e^{-TM} NG_1(s)]}$$

in which the following z-transforms are given by

$$D_o(z) = \frac{1}{z-1}$$

$$Go(z) = \frac{1 - e^{-Ts}}{Ts(s+1)}$$

$$\begin{aligned}
 &= \\
 &= \\
 &= \\
 &3[K(1-p)] \\
 &1 \\
 &(1-e^{-Ts}) \\
 &- \\
 &R(s) \quad (10-115) \\
 &- \\
 &0.632 \\
 &z \quad 0.368 \\
 &(10-116) \\
 &K \\
 &(\approx -1)N \\
 &(10-117) \\
 &\diamond - 0.368 \\
 &) + 6-1/x \\
 &e^{-1/N} \quad 0.368 \\
 &(10-118) \\
 &(10-119) \\
 &3 [c^{TM*} * Do(s)] = 3 [c^{TM*} * (1+T)] = (1 - e^{-1/N}) \\
 &e \\
 &3[e^{-**G}, (4)] = 3[Ke^{-7x} (1 - e^{-(1)})] K(1-1/N) \\
 &2T
 \end{aligned}$$

Substituting Eqs. (10-116) through (10-119) into Eq. (10-115) yields the z-transform of the system output as

$$\begin{aligned}
 &C(z) = \\
 &K[(1 - e^{-1/N})(1-e)x + (1-e) + e^{-0.368} \\
 &N \\
 &K \\
 &N \\
 &'+[K(1 - 1/N)(1 - e^{-1/N}) - 1.368]z + \sqrt{(1 - e^{-1/N}) + 0.368(1 - K) + Ke^{-1/N}} \\
 &- \\
 &-R(s) \\
 &(10-120)
 \end{aligned}$$

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The characteristic equation of the system is

$$z^2 + K(1 - 1/N)(1 - e^{-1/N})z - 1.368 + e^{-1/N} = 0$$

Applying the Schur-Cohn stability criterion to the last equation, we obtain the following inequalities:

$$\begin{aligned}
 &(1) \\
 &(2) \\
 &(3) \\
 &0.632 \\
 &K < \\
 &1/N(1 - \\
 &- \\
 &e^{-1/N}) - 0.368 + e^{-1/N} \\
 &K > 0 \\
 &-1.368 \\
 &K >
 \end{aligned}$$

$$1/N(1 - e^{-\gamma})$$

0.684

(10-122)

(10-123)

(10-124)

Since the right-hand side of Eq. (10-124) is always negative for $0 < 1/N \leq 1$, the maximum and the minimum allowable gains for stability are given by Eqs. (10-122) and (10-123), respectively. The stability boundaries for this system are plotted in Fig. 10-23. Observe that from the stability point of

Unstable region

1.6

1.44

1.4

Kmax

1.2

1.0

K 0.8

0.6-

Stable region

0.4-

0.2

Kmin

0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0

1/N

Unstable region

Fig. 10-23 Stability boundary curves for the cyclic-rate sampled-

data system of Example 10-5.

view, an optimum value of $1/N$ is equal to 0.44, since the maximum allowable gain reaches a peak of 1.44 under this condition. For a uniform sampling with sampling period of one second ($N = 1$), the maximum allowable gain for stability is 1.0.

For this system to have deadbeat performance, the characteristic equation of the system must be of the form $z^2 = 0$. Therefore, in Eq. (10-121), the coefficients of z and z^0 must vanish simultaneously with

$$K(11/N)(1 - e^{-\gamma/N}) - 1.368 = 0$$

(10-125)

K

and

-

N

$$(1 - e^{-\gamma}) + 0.368(1 - K) + Ke^{-\gamma/2} = 0$$

(10-126)

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However, we can show that Eqs. (10-125) and (10-126) cannot simultaneously be equal to zero for $0 < 1/N < 1$; hence deadbeat performance cannot be obtained for this system with the specific cyclic sampling function used.

10.3 STATE VARIABLES AND TRANSITION EQUATIONS

OF MULTIRATE AND SKIP-SAMPLED SYSTEMS

The state variable and state transition methods introduced in Sec. 5.6 can readily be used to study sampled-data systems with multirate- or variable-rate sampling. We shall demonstrate that the state transition equations of a system with multirate- or variable-rate sampling can be written out directly from the transition flow graph in much the same way as those for a system with uniform-rate sampling.

1. Cyclic Sampling

...9

1

1

The operation of a cyclic-rate sampler is described by the pulse-

trains shown in Fig. 10-16b or Fig. 10-19b. The sampling operation of the cyclic sampler is assumed to take place at the instants $t = kT, kT/N_1, kT/N_2, kT/N_3, \dots, kT/N_n, (k+1)T, T + kT/N_1, \dots$ or $t = kT, kT + T_1, kT + T_1 + T_2, kT + T_1 + T_2 + T_3, \dots, (k+1)T, (k+1)T + T_1, \dots$; where the N 's are positive numbers greater than one, $T_1, T_2, \dots$ are all less than T . Referring to Eq. (5-182) we may write the transition equation of a uniform-rate sampling system during the time interval to $t < t_0 + T$, where $0 < T < T$, as

$$x(t) = (t - t_0) x(t_0) + \int_{t_0}^t r(\tau) d\tau$$

(10-127)

Since Eq. (10-127) is valid for any t between 0 and T , it may be applied to the cyclic-rate sampling system. For instance, at $t = kT + T_1$, Eq. (10-127) becomes

$$x[(kT + T_1)] = Y(T_1)x(kT) + \int_{kT}^{kT+T_1} r(\tau) d\tau$$

(10-128)

Similarly, at the very next sampling instant $t = kT + T_1 + T_2$; Eq. (10-127) reads

$$x[(kT + T_1 + T_2)] = Y(T_2)x(kT + T_1) + \int_{kT+T_1}^{kT+T_1+T_2} r(\tau) d\tau$$

1
2

(10-129)

In general, if $T = T_1 + T_2 + T_3 + \dots + T_q$, at $t = kT + T_1 + T$:

$T_{q-1} < (k+1)T$, Eq. (10-127) is written

$$x(kT + T_1 + T) = Y(T)x(kT + T_1 + T_2 + \dots + T_{q-1}) + \int_{kT+T_1+T_2+\dots+T_{q-1}}^{kT+T_1+T} r(\tau) d\tau$$

(10-130)

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and at $t = kT + T$ we have

$$x[(k+1)T] = Y(T)x(kT + T_1 + T_2 + \dots + T_{q-1}) + \int_{kT+T_1+T_2+\dots+T_{q-1}}^{(k+1)T} r(\tau) d\tau$$

1

By substituting Eq. (10-128) into Eq. (10-129) and so on, we see

that Eq. (10-131) can be written as

$$x[(k+1)T] = Y(T)Y(T_1)\dots Y(T_{q-1})x(kT) + Y(T)Y(T_1)\dots Y(T_{q-1})\int_{kT}^{(k+1)T} r(\tau) d\tau$$

Example 10-6

$$+ Y(T)Y(T_1)\dots Y(T_{q-1})\int_{kT}^{(k+1)T} r(\tau) d\tau$$

$$= Y(T)Y(T_1)\dots Y(T_{q-1})\left[\int_{kT}^{kT+T_1} r(\tau) d\tau + Y(T_1)\int_{kT+T_1}^{kT+T_1+T_2} r(\tau) d\tau + \dots + Y(T_1)\dots Y(T_{q-1})\int_{kT+T_1+\dots+T_{q-1}}^{(k+1)T} r(\tau) d\tau \right]$$

1
2

$$+ Y(T)Y(T_1)\dots Y(T_{q-1})\int_{kT}^{(k+1)T} r(\tau) d\tau$$

2

(10-132)

Consider the system shown in Fig. 10-19. Assuming that $G(s) = 1/s(s+1)$, the sampler closes at instants $t = kT, kT + T_1, (k+1)T, (k+1)T + T_1, \dots$, where $0 < T_1 < T$.

The state transition flow graph of the linear elements described by $G(s)$ has been shown in Fig. 5-31a. During the time interval $kT < t \leq kT + T_1$, the transition flow graph of the over-all system is shown in Fig. 5-31b. The

$$x_1(t) = \{e^{-(t-T_1)}\}x_1(kT) + \{[2 - (t - T_1) - e^{-(t-T_1)}](1 - e^{-T_1}) + e^{-[1-e^{-T_1}]}]\}x_2(kT) + \{[2 - (t - T_1) - e^{-(t-T_1)}](T_1 - 1 + e^{-T_1}) + (1 - e^{-T_1})[1 - e^{-\alpha T_1}]]\}r(kT) + [(t - T_1) - 1 + e^{-T_1}]r(kT + T_1) \quad (10-138)$$

$$x_2(t) = \{e^{-(t-T_1)}(-1 + e^{-T_1})[2 - T_1 e^{-T_1}][1 - e^{-(t-T_1)}]\}x_1(kT) + \{e^{-[1-e^{-T_1}]} - e^{-[1-e^{-T_1}]}]\}x_2(kT) + \{(1 - e^{-T_1})e^{-[1-e^{-T_1}]} - (T_1 - 1 + e^{-T_1})[1 - e^{-(t-T_1)}]\}r(kT) + [1 - e^{-(t-T_1)}]r(kT + T_1) \quad (10-139)$$

The transition equations describing the state variables at the sampling instants are obtained by substituting $t = kT + T$ into Eqs. (10-138) and (10-139).

Cyclic-rate sampler

R
z. o. h.

S
S

st
x

X5
D(s)

z.o.h. G(s)

Sc
X4

X3
x2

X1
(a)

KT
(k+1)T

(K+2)T
(k+3)T

(K+4)T
Sc

T1
T1

T1
T1

(k+2)T
KT

KT + T1 (k+1)T (k+1)T+T1
(k+2)T+T1

(b)

Fig. 10-25 Cyclic-rate sampled-data system. (a) Block diagram. (b) Sampling schemes.

$G(s)$
 $x_2(1019)$
 s^{-1}
 $X_5(s)$
 $X_4(s)$
 $x_3(s)$
 $X_2(s)$
 $X_1(s)$
 -1
 -1
 (a)
 $x_4(KT)$
 $X_2(KT)0$
 $X_a(KT)$
 $x_a(KT)$
 A_H
 1
 s^{-1}
 $r(KT) X_5(KT+)$
 $X_5(s)$
 $X_4(s) x_3(KT+) X_z(s)$
 $X_2(s) X_1(s)$
 -1
 -1
 (b)

$-(1-e)$
 $1-e^{-T}$
 $X_4(KT)$
 $T_1-1+e^{-T_1}$
 $X_2(KT)$
 (KT)
 e^T
 $1-e^{-T}$
 $-x_1(KT+T_1)$
 $X_2(KT+T_1)$
 $1-e^{-T}$
 $8x_3(KT+T_1)$
 $X_q(KT+T_1*)9$
 5^{-1}
 $\backslash x_q(KT + T_1)$
 $X_2(KT+T_1)$
 $-x_1(KT+T_1+)$
 $7)$
 $X_5(KT+T_1*) X_5(s)$
 $X_4(s)$
 $X_5(KT+T_1)$
 -1
 $X_3(s)$
 $X_2(s) X_1(s)$
 $X_3(KT+T_1*)$
 (c)

Fig. 10-26 Transition flow graphs of cyclic-rate sampled system in Example 10-7.

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Example 10-7

As another example, let us consider the cyclic-rate sampled-data system shown in Fig. 10-25a. The sampling schemes of the samplers are shown in Fig. 10-25b. The transfer functions of the system are given as

$$D(s) = 1/(s + 1) \text{ and } G(s) = 1/s(s + 1)$$

The state variables of the system are assigned as shown in Fig. 10-25a, where variables X_1 , X_2 , and x are state variables of the continuous dynamic elements and x and x are outputs of the sample-and-hold units. The transition flow graphs of the continuous dynamic elements $G(s)$ and $D(s)$ are shown in Fig. 10-26a. At $t = t_0 = kT$, both samplers operate; the following transition equations are written:

$$x(kT^+) = x(kT)$$

$$x_2(kT^+) = x_2(kT)$$

=

$$x_5(kT^+) = x(kT)$$

$$z(kT^+) = x(kT)$$

$$x_5(kT^+) = r(kT) - x_1(kT)$$

-

$$(10-140)$$

$$(10-141)$$

$$(10-142)$$

$$(10-143)$$

$$(10-144)$$

Through the use of these transition equations, the transition flow graph of the over-all system during the time interval $kT < t \leq T_1$ is drawn in Fig. 10-26b. Applying Mason's gain formula to the flow graph of Fig. 10-26b, we can write the following equations:

1

1

$$(AT) + G + I(T)$$

(s

$$X_1(s) = - \frac{x_1(kT)}{s(5^2 + 1)} + \frac{x_2(kT)}{s(5^2 + 1)} + \frac{5^2(5 + 1)x(kT)}{s(5^2 + 1)}$$

$$(10-145)$$

1

1

$$X_2(s) = \frac{5}{s(5 + 1)} + \frac{1}{s(5 + 1)} x_2(kT) + \frac{x(kT)}{s(5 + 1)}$$

$$(10-146)$$

$$X_1(3) = \dots, (AT)$$

$$(10-147)$$

1

$$X_1(5) = - \frac{5(5 + 1)}{Jx_1(kT) + 5 + 1} x(kT)$$

1

+

$$\frac{s(s + 1)}{s(s + 1)} r(kT) \quad (10-148)$$

$$(10-149)$$

S

1

S+

$$Xs(s) = -1 \frac{x_1(kT)}{s} + \frac{1}{s} r(kT)$$

Taking the inverse Laplace transform of the last five equations, we have at $t = kT + T_1$,

-

$$x_1(kT + T_1) = x_1(kT) + (1 - e^{-1}) x_2(kT) + (T_1 - 1 + e^{-1}) x(kT) \quad (10-150)$$

$$x_2(kT + T_1) = e^{-T_1} x_2(kT) + (1 - e^{-T_1}) x(kT)$$

$$x_3(kT + T_1) = x_1(kT)$$

-

-

$$(10-151)$$

$$(10-152)$$

$$x_1(kT + T_1) = - (1 - e^{-1}) x(kT) + e^{-1} x_1(kT) + (1 - e^{-T}) r(kT) \quad (10-153)$$

$$x_5(kT + T_1) = -x_1(kT) + r(kT)$$

$$(10-154)$$

At $t = kT + T_1$, the cyclic-rate sampler closes, but the uniform-rate sampler is open. The following transition equations are written:

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$$x_1(kT + T_1) = x_1(kT + T_1)$$

(10-155)

$$x_2(kT + T_1) = x_2(kT + T_1)$$

(10-156)

$$x_3(kT + T_1) = x_3(kT + T_1)$$

(10-157)

$$x_4(kT + T_1) = x_4(kT + T_1)$$

(10-158)

(10-159)

$$x_5(kT + T_1) = x_5(kT + T_1)$$

Now, the transition flow graph of the system during the time interval $kT + T_1 < t \leq (k + 1)T$ is constructed by use of Eqs. (10-150) through (10-159) and the flow graphs of Fig. 10-26a. The complete transition flow graph is shown in Fig. 10-26c. The transition equations are written by inspection from Fig. 10-26c.

-

e

$$X(s) = -x_1(kT) + [1 - e^{-sT_1}] = (47)$$

+

S

-

$$[T_1 = 1 + z^{-2}]$$

e-T

S

$$X(s) = (1 - e^{-sT_1})x_1(kT) +$$

5+1

-e

$$-T_1$$

S

S

$$s(s + e^{-sT_1})x_2(kT) + [1 - e^{-sT_1}] = (4T) + 1(1 + 1) + (AT)$$

s²(s

XA

1- e-T

s² (s +

" + (1+1)] = (4T) + 1(1 + T)

$$X(s) = -\frac{1 - e^{-sT_1}}{s^2} x_1(kT) + x_1(kT) + 1 - z^{-2} r(kT)$$

+

1

-e

+

1)

+

[

1

1

S+

;-;

=

1

$$s(s + 1)]x_1(kT)$$

-

z. o. h.

S₂

Γ

Fig. 10-27 Sampled-data control system with nonsynchronized samplers.

1

for the time interval $kT < t < kT + T_1$ are written from the transition flow graph. At $t = kT + T_1$, the sampler S_1 operates, but S_2 does not. The same procedure described above is repeated until the final state transition flow graph and equations at $kT + T_1 < t < (k+1)T$ are obtained. The state transition equations at $t = (k+1)T$ are

$$x_1[(k+1)T] = x_1(kT) - T r(kT) + T r(kT)$$

(10-165)

$$x_2[(k+1)T] = [1 - e^{-T-T_1}]x_1(kT) + [e^{-T_1} - T_1(1 - e^{-(T-T_1)})]x_2(kT)$$

(10-166)

$$x_3[(k+1)T] = -x_1(kT) + r(kT)$$

(10-167)

$$x_1[(k+1)T] = x_1(kT) - T_1 x_2(kT) + T_1 r(kT)$$

-

(10-168)

9x₁(10)

0x₁(10)

5-1

생이

(to) 1

x₃(10)

x₃(5)

x₁(5)

-14

1

x₂(10)

x₂(10) X₂(s)

-1

X₁(s)=C(s)

X₄(s)

x₄(10)

xa(10)

to = kT < t < (k + 1)T

Fig. 10-28 State transition flow graph for the sampled-data system with nonsynchronized samplers shown in Fig. 10-27.

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((KT) 1

-14

x₃(KT+T₁)

T

0x₁ (KT)

1

X₁(KT+T₁)

5-1

15-1

5-1

1

X₃(5)

X₁(5)

X₁(s)=C(s)

1

-T₁

5-1

$$\begin{aligned} & 5-1 \\ & X2(KT) \\ & 5-1 \\ & x2(KT+T_1^*) \quad X2(5) \\ & X2(KT+T_1) \\ & X4(s) \quad x4(KT+T_1^*) \\ & -1 \\ & 1-e^{-sT_1} \\ & 1 \\ & X4(KT) \quad X4(KT+T_1) \\ & t > (k+1)T \end{aligned}$$

Fig. 10-29 State transition flow graph for the sampled-data system with nonsynchronized samplers shown in Fig. 10-27.

The details of this illustrative example are left for the readers to carry out as an exercise.

Although only the cyclic sampling and the nonsynchronous sampling are considered in this section, we can see from the techniques demonstrated in the preceding examples that the state transition method can be applied to other sampling schemes such as multirate, multiorder, and noninstantaneous.

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11

STATISTICAL ANALYSIS AND

DESIGN OF SAMPLED-DATA

CONTROL SYSTEMS

11.1 INTRODUCTION

In many practical applications of control systems, the actual signals and disturbances applied to a given control system are random in nature, unlike the deterministic step, ramp, or sinusoidal functions considered in the preceding chapters. These signals and disturbances which are random functions of time can often be described only by their statistical properties. Therefore, the treatment of control systems must be considered with continuous as well as sampled-data for actual random signals and disturbances.

The chief aim of the statistical design of control systems with random inputs is the creation of an optimum system that will render a minimum error between the actual output and the desired output. The design criteria are based mainly on the minimization of the mean square error, a concept which was first used by Wiener. The mean-square-error criterion may not be ideal for all control systems. However, the popularity of the criterion stems from the fact that if the minimization of the mean square error is used as the design criterion, the random signals or disturbances are sufficient. N. Wiener, The Extrapolation, Interpolation, and Smoothing of Stationary Time Series, (New York: John Wiley & Sons, Inc., 1948).

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$n(t)$

$s(t)$

$M(s)$

$c(t)$

Actual system

$e(t) = c(t) - c_a(t)$

$r(t)$

$r(t)$

$c_a(t)$

$M_o(s)$

Desired system

(a)

$M(z)$

$n(t)$

$s(t)$

$C(1)$

$c(t)$

$M(s)$

Actual system

(11-1)

$e(t) = c(t) - c_a(t)$

$r(t)$

$r(t)$

$c_a(t)$

$c_a(t)$

$$e(nT) = c(nT) - ca(nt)$$

$M_o(s)$

Desired system

(b)

Fig. 11-1 (a) Block diagram illustrating the optimum design of a linear continuous-data system. (b) Block diagram illustrating the optimum design of a linear sampled-data system.

efficiently described by their correlation functions and power spectral density functions. Shown in Fig. 11-1a is a block diagram which illustrates the formulation of a linear optimum system with continuous-data, based on the minimum mean square error criterion. The design objective is to select a system with the transfer function $M(s)$ where $M(s)$ may represent an open-loop system or a closed-loop system with constraints on open-loop elements in a manner that would minimize the mean square value of the error $e(t)$. The input signal is assumed to be the sum of a random signal $r(t)$ and a random noise $n(t)$. The error $e(t)$ is defined as the difference between the actual system output $c(t)$ and the desired system output $ca(t)$. The concept of the minimum mean-square-error design of linear continuous-data systems can be extended to the optimum design of systems with sampled-data. Fig. 11-1b illustrates the block diagram of optimum design of a sampled-data system. The design objective is the determination of the z-transfer function $M(z)$ that will minimize the mean square sampled error $e^2(nT)$. In reality, the block diagram of Fig. 11-1a may represent a sampled-data system if the actual system with transfer function $M(s)$ contains samplers or digital elements.

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11.2 STATISTICAL CHARACTERISTICS OF CONTINUOUS

RANDOM SIGNALS

In this section we shall briefly review some of the important properties of continuous random signals so that the concepts of statistical description of random processes may be extended to discrete random signals. Some previous knowledge of basic probability theory and statistical theory is assumed. Therefore, the material presented here consists of merely the definitions and descriptions of some important terms and concepts often used in the analysis of random signals.

Sample Space and Sample Points. The aggregation of all possible outcomes of a random experiment is called a sample space. With each basic outcome of the experiment, a sample point may be assigned. A sample space may consist of a finite number of sample points, and the sample space is said to be discrete. For instance, as a die is thrown, the sample space consists of six discrete sample points; as a coin is thrown, the sample space consists of two discrete points, namely, heads or tails. On the other hand, a discrete sample space may consist of an infinite number of points and is called a continuous sample space. A well-known example of a continuous sample space is a thermal noise voltage caused by thermally excited electrons in a conductor.

Random Variable. The sample points in a sample space are designated a variable that takes on the values of the points. A random variable may be discrete or continuous in correspondence with the discrete sample space or the continuous sample space. If the random variable is denoted by X , then in the coin throwing experiment X is discrete, $X = \text{heads}$ or $X = \text{tails}$, whereas in a thermal noise voltage, X is a continuous random variable and may take on all possible values of the noise voltage.

Probability Density Functions. The probability density function, $p(x)$, describes the statistical characteristics of a random process.

The probability that a random variable X will lie in an infinitesimal range between x and $x + dx$ is

$$P(x < X < x + dx) = p(x) dx$$

(11-1)

The probability that the variable X will lie in the range x to x is

$$P(x_1 < X < x_2) = \int_{x_1}^{x_2} p(x) dx$$

(11-2)

When the random variable represents points of a random signal or a random process that is a function of time, probability density

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functions of various orders may conveniently be defined. The first-order probability density function $p(x)$ can also be written as $p(x, t)$, denoting the probability that the variable X lies between x and $x + dx$ at the time t . The second-order probability density function, $p(x_1, t_1; x_2, t_2)$, denotes the probability that the X lies between x_1 and $x_1 + dx_1$ at t_1 and also lies between x_2 and $x_2 + dx_2$ at time t_2 . Higher order probability density functions may be defined similarly.

Probability Distribution Function. The probability distribution function $P(x, t)$ is defined as the probability that the random variable will be less than some value x at the time t . Therefore,

$$P(x, t) = P[X(t) < x] = \int_{-\infty}^x p(x, t) dx$$

If $P(x, t)$ possesses a first derivative, we have

(11-3)

$$p(x, t)$$

$$= \frac{dP(x, t)}{dx}$$

(11-4)

Therefore, Eqs. (11-3) and (11-4) relate the probability density function and the probability distribution function.

Time Approach and Ensemble Approach. In general, the statistical properties of a random signal may be measured in two ways: the time approach and the ensemble approach. In the time approach measurements of the random signal are made beginning at some time t_1 . An ensemble approach may also be used in making statistical measurements. Assuming that we can collect a large number of identical signal generators, the ensemble approach suggests the measurements of the large number random signals simultaneously at the time t_1 .

Stationary and Ergodic Processes. If in a random process, the statistical properties are constant and are independent of the time origin at which they are measured, the process is said to be stationary with respect to time. In the time approach, if the measurements of a random signal made beginning at time t_1 produce identical results to the measurements made beginning at time t_2 or at any other time, the process is stationary. In the ensemble approach, a random process is defined as stationary in time if the statistical properties measured on a large number of random signals from identical sources are the same, regardless of the time at which the measurements are made.

A random process is said to be ergodic if its statistical properties are identical under the time and the ensemble measurements. The definitions imply that an ergodic process is always stationary, since the stationary process is a presupposition of ergodicity. However, not

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all stationary processes are ergodic. A random process may be shown to be stationary under the time ensemble measurements, but the two results may not be identical.

Moments and Averages. The nth moment of a random variable $x(t)$ is defined to be the statistical average of the nth power of $x(t)$:

$$\overline{x^n} = \text{av}[x^n(t)] = \overline{x^n(t)} = \langle x^n(t) \rangle = \int_{-\infty}^{\infty} x^n p(x, t) dx \quad (11-5)$$

where $p(x, t)$ is the probability density function of x .

The first moment of $x(t)$ is just the average value of $x(t)$;

that is,

$$\overline{x(t)} = \langle x(t) \rangle = \int_{-\infty}^{\infty} x p(x, t) dx \quad (11-6)$$

Equation (11-6) gives the average from the ensemble approach. We could, however, perform a time averaging process. The time average of $x(t)$ is simply

$$\overline{x(t)} = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^{+T} x(t) dt$$

$$\quad (11-7)$$

If $x(t)$ belongs to a random process possessing the ergodic property, the time average is equal to the ensemble average.

The second moments of $x(t)$ in the time average and the ensemble average are given by

and

$$\overline{x^2(t)} = \langle x^2(t) \rangle = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^{+T} x^2(t) dt$$

$$\quad (11-8)$$

$$\quad (11-9)$$

$$\quad (11-10)$$

$$\quad (11-11)$$

$$\overline{x^2(t)} = \langle x^2(t) \rangle = \int_{-\infty}^{\infty} x^2 p(x, t) dx$$

$$\quad (11-8)$$

$$\quad (11-9)$$

$$\quad (11-10)$$

respectively. From Eq. (11-8) we see that $\langle x^2(t) \rangle$ is simply the mean square value of $x(t)$. When $x(t)$ represents the error signal defined in Fig. 11-1a, Eq. (11-8) or Eq. (11-9) gives the mean square error for optimum system design.

The concept of moments may be extended to two random variables. The $(n+m)$ th order joint moment of the joint probability density function of two random variables x and y is

$$\langle x^n(t_1) y^m(t_2) \rangle = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^n(t_1) y^m(t_2) p(x, t_1; y, t_2) dx dy \quad (11-10)$$

In the equation above $p(x, t_1; y, t_2)$ is the probability that x has a value in the range between x and $x + dx$ at time t_1 , and y has a value in the range between y and $y + dy$ at t_2 and is called the joint probability density function of x and y .

Correlation Functions. The correlation functions are measurements of the statistical dependence of one random signal upon another or upon itself. If $x(t)$ and $y(t)$ are two random processes or signals,

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the cross-correlation between x and y is represented by the cross-correlation function

$$R_{xy}(t_1, t_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x y p(x, t_1; y, t_2) dx dy$$

$$\quad (11-11)$$

Equation (11-11) indicates that the cross-correlation function of x and y is equal to the first-order joint moment (or mean) of the joint probability density of x and y . If x and y are stationary processes, the probability density function does not depend on the origin of

time. The cross-correlation function given by Eq. (11-11) is independent of time t_1 and can be written as

$$\phi_{xy}(\tau) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) y(t + \tau) dt \quad (11-12)$$

For a stationary process, the cross-correlation function can be determined from a time average. Thus

$$\phi_{xy}(\tau) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) y(t + \tau) dt \quad (11-13)$$

Furthermore, if the process is ergodic, Eqs. (11-12) and (11-13) are identical.

When $x(t) = y(t)$ or when only one random signal is involved, the correlation is given by the autocorrelation function and is defined to be

$$\phi_{xx}(\tau) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) x(t + \tau) dt \quad (11-14)$$

$$\phi_{xx}(\tau) = \sqrt{\frac{1}{T} \int_0^T x^2(t) dt} \quad (11-15)$$

Some of the important properties of the correlation functions are given below without proof:

(1) $\phi_{xx}(\tau)$ is an even function; i.e., $\phi_{xx}(\tau) = \phi_{xx}(-\tau)$.

(2) $|\phi_{xx}(\tau)| \leq \phi_{xx}(0)$

(3) $\phi_{xy}(\tau) = \phi_{yx}(\tau)$

(4) $\phi_{xx}(0) = \overline{x^2(t)}$ (mean square value of x).

Power Spectral Density Function. Essentially, the correlation functions introduced in the last paragraph give descriptions of random signals in terms of time domain characteristics. This procedure is often desirable. On the other hand, random signals can and often should be described in terms of frequency-domain characteristics. If the correlation functions are looked upon as time functions (of τ), it is natural to take the Laplace transform of $\phi(\tau)$ or $\phi_{xx}(\tau)$ in order to obtain frequency-domain characteristics. Since the correlation functions theoretically extend from $-\infty$ to $+\infty$ in τ , a two-sided

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Laplace transform has to be carried out. Therefore, the autospectral density function is defined to be

$$P_{xx}(s) = \int_{-\infty}^{\infty} \phi_{xx}(\tau) e^{-s\tau} d\tau$$

and the cross-power spectral density function is

$$P_{xy}(s) = \int_{-\infty}^{\infty} \phi_{xy}(\tau) e^{-s\tau} d\tau \quad (11-16)$$

$$(11-17)$$

The correlation functions can be obtained by taking the inverse Laplace transform of the power density functions in the usual manner. Thus

$$\phi_{xx}(\tau) = \frac{1}{2\pi j} \int_{\sigma - j\infty}^{\sigma + j\infty} P_{xx}(s) e^{s\tau} ds \quad (11-18)$$

Some important properties of the power spectral density functions are given below:

$P_{xx}(s)$

(1) $P_{xx}(j\omega)$ is an even function of frequency. Thus $P_{xx}(j\omega) = P_{xx}(-j\omega)$

(2) $P_{zy}(j\omega) = P_{yz}(-j\omega)$

(3) If $x(t)$ and $y(t)$ are the input and output signals, respectively, of a linear system with transfer function $G(s)$, the power spectral densities of $x(t)$ and $y(t)$ are related by

$$P_{yy}(s) = G(s)G(-s)P_{xx}(s)$$

11.3 STATISTICAL CHARACTERISTICS OF SAMPLED RANDOM SIGNALS

(11-19)

In the preceding sections we have defined correlation functions and power spectral density functions which are used for the characterization of continuous random signals. These concepts can also be extended to the statistical descriptions of sampled or discrete random signals. As has been pointed out, there are two ways of determining the correlation functions of random processes: the time average and the ensemble average. However, in sampled-data systems, in addition to the two ways of averaging, there are at least two possible ways of considering a sampled signal. First of all, we can consider the values of a signal at the sampling instants only. In this case, the signal is treated as a sampled sequence or a sequence of numbers. Another way of considering a signal is to treat it as a sampled signal consisting of a train of amplitude modulated pulses. Referring to Fig. 11-2a, $r(t)$ is a member of a stationary random process which

† Van der Pol, Balh, and H. Bremmer, Operational Calculus Based on the Two-sided Laplace Integral, (London: Cambridge University Press, 1950).

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$r(t)$

(a)

塔(1)

$r(t)$

分(1)

-37

-27

-T

Op

T

27

37

(b)

$r(nT)$ $r(1)$

* (1)

-37 -27

-T

0

T

27

37

(c)

Fig. 11-2 (a) Sampler. (b) Continuous and sampled random

signals. (c) Sampled random sequence and impulse train.

is being sampled periodically every T second. The time during which the sampling switch is closed is p and is assumed to be very short compared with the sampling period T . The waveforms of the continuous and the sampled signals are depicted in Fig. 11-2b. If $r(t)$ is a stationary ergodic process, the autocorrelation function of $r(t)$ may be obtained as either a time average or an ensemble average.

Time average: $\overline{r(T)}$

Ensemble average:

=

1

$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T r(t) r(t + \tau) dt$

τ

00

(11-20)

$\overline{r(T)} = \int \int x_1 x_2 \text{pr}(x_1, x_2; \tau) dx_1 dx_2$ (11-21)

where $\text{pr}(x_1, x_2; \tau)$ is the probability that $r(t)$ has a value x_1 at any time and a value x_2 at τ seconds later.

First, let us consider that the signal $r(t)$ is characterized by the values at the sampling instants only. The sampled sequence is denoted by $r(nT)$, where $n = 0, \pm 1, \pm 2, \pm 3, \dots, \pm \infty$. We shall now show that if $r(t)$ is assumed to be a member of a stationary ergodic process, $r(nT)$ is also a stationary ergodic random sequence, and the time averaging and ensemble averaging are equivalent for this discrete case. Furthermore, the sampled sequence averages are equal to the time average of the continuous time function $r(t)$.

Since $r(t)$ and $r(nT)$ are all ergodic, the second probability density

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function of $r(t)$ is equal to the second probability density of the sampled random sequence $r(nT)$.

$\text{Pr}(X_1, X_2; T) = \text{pr}_{rr}[X_1, X_2; kT]$

(11-22)

where $kT = T$, and k is an integer.

The autocorrelation function of $r(nT)$ may be expressed as an ensemble average.

$\overline{r(T)} = \int \int x_1 x_2 \text{pr}(x_1, x_2; kT) dx_1 dx_2$

τ

However, referring to Eq. (11-22), we have

$\text{Pr}_{12}(kT) = \int \int x_1 x_2 \text{pr}(x_1, x_2; kT) dx_1 dx_2$

(11-23)

(11-24)

Comparing Eqs. (11-24) and (11-21), we have

$\text{Pr}_{rr}(kT) = \overline{r(T)}|_{T=kT} = \overline{r(T)}$

(11-25)

proving that when $r(t)$ is an ergodic process, the sampled sequence average and the continuous time average are equivalent. Equation (11-25) also provides an interesting result in that the autocorrelation function of a stationary random sequence $r(nT)$ may be obtained by sampling the autocorrelation function $R(T)$ of the continuous ergodic process $r(t)$. For instance, assuming that the autocorrelation function

$R(T)$

τ

(0)

$r(1)$

(T)

$*(\tau), r(nT)$

(KT)

(b)

P(7)

Proro(T)

Pro(37)

-4T -3T -2T -T

0

T 2T 3T 4T

(c)

T

(11-31)

$$\lim_{N \rightarrow \infty} \sum_{m=-N}^{\infty} \sum_{n=0}^{\infty} \int_0^T S(t - mT)(t - nT + \tau) dt$$

 Letting $\tau = mT$, Eq. (11-31) becomes

$$\lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{m=-N}^{\infty} \sum_{n=0}^{\infty} \int_0^T r(mT)r(nT) dt$$

$$= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \int_{-T/2}^{T/2} r(\tau) r(\tau) d\tau$$

$$= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \int_{-T/2}^{T/2} r(\tau) r(\tau) d\tau$$

$$= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \int_{-T/2}^{T/2} r(\tau) r(\tau) d\tau$$

$$= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \int_{-T/2}^{T/2} r(\tau) r(\tau) d\tau$$

$$= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \int_{-T/2}^{T/2} r(\tau) r(\tau) d\tau$$

$$= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \int_{-T/2}^{T/2} r(\tau) r(\tau) d\tau$$

$$= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \int_{-T/2}^{T/2} r(\tau) r(\tau) d\tau$$

(11-36)
 The last equation shows that the autocorrelation function of a stationary sampled impulse train $r^*(t)$ is an impulse train in τ modulated by $1/T$ times the autocorrelation function of the stationary continuous signal $r(t)$.
 However, the sampled random process $r(t)$ and the impulse modulated signal $r^*(t)$ are not stationary processes (although $r(t)$ is stationary). The correlation functions derived from the ensemble

approach are not equal to those of the time averages. For instance, the autocorrelation function of $r^*(t)$ based on the ensemble average is

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{m=-N}^N \sum_{n=-N}^N r(mT) r(nT) \delta(t - mT) \delta(t - nT + \tau)$$

Simplifying, Eq. (11-37) becomes

$$\Phi_{TOT}(\tau, t) = \sum_{M=-\infty}^{\infty} \sum_{N=1}^{\infty} \text{Pr}\{[(n - m)T] \delta(t - mT) \delta(t - nT + \tau)\} \quad (11-38)$$

In addition to being a function of the shift variable τ , the autocorrelation function determined from the ensemble approach is also a periodic function of time. This fact checks with the physical interpretation of ensemble averaging. Since any number of the ensemble of $r^*(t)$ has values only at the sampling instants $t = kT$, the average of the ensemble has to be evaluated at the sampling instants and has zero values at all other times. The difference between Eqs. (11-36) and (11-38) is illustrated by the typical correlation function of $r^*(t)$ shown in Fig. 11-4. The time-averaged autocorrelation function of $r^*(t)$ is time-invariant and is shown to be a sequence of impulse sheets. The ensemble-averaged autocorrelation functions of $r^*(t)$ appear as impulse trains in a two-dimensional t versus τ plane. In the following sections only the time-invariant correlation functions will be considered. Design of optimum sampled-data systems using the time-invariant correlation functions has been investigated by DeRusso.

In the same manner, we can show that the cross-correlation

+ P. M. DeRusso, "Ultimate Performance Limitations of Linear Sampled-Data Systems," M. I. T. Electronic System Laboratory Report 7843-R-3, June 1959.

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Pr(T)
\$,*(7)
-3T
-2T
-T
2T
3T
(a)
-3T
-2T
,,(T, 1)
(b)
\$(7)
2T
3T

Fig. 11-4 Typical correlation functions showing the difference between the ensemble average and the time average of the non-

stationary time series $r^*(t)$. (a) Autocorrelation function of $r^*(t)$ from the time approach (time-invariant). (b) Autocorrelation function of $r^*(t)$ from the ensemble approach (time-variant). function of two stationary sampled random sequences $r(nT)$ and $c(nT)$ is given by

$$\Pr_{cn}(kT) = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N r(nT) c(nT + kT) \quad (11-39)$$

and the cross-correlation function of two sampled impulse trains $r^*(t)$ and $c^*(t)$ is

$$\Pr_{cc}(\tau) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T r^*(t) c^*(t + \tau) dt$$

$$\Pr_{cc}(kT) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T r^*(t) c^*(t + kT) dt \quad (11-40)$$

where $\Pr_{cc}(kT)$ is given by Eq. (11-39).

As in the continuous case, we can define the sampled power

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spectral density function of a sampled signal $r^*(t)$ as the Laplace transform or the z-transform of the correlation function of $r^*(t)$. Therefore,

$$\Pr_{rr}(s) = \mathcal{L}[\Pr_{rr}(\tau)] = \int_{-\infty}^{\infty} \Pr_{rr}(kT) e^{-skT} dk \quad (11-41)$$

$$\Pr_{rr}(s) = \sum_{k=-\infty}^{\infty} \Pr_{rr}(kT) e^{-skT}$$

Taking the z-transform on both sides of Eq. (11-42) yields

$$\Phi_{rr}(z) = \mathcal{Z}[\Pr_{rr}(kT)] = \sum_{k=-\infty}^{\infty} \Pr_{rr}(kT) z^{-k} \quad (11-43)$$

Since $\Pr_{rr}(kT) = \Pr_{rr}^*(kT)$, Eq. (11-43) becomes

$$\Phi_{rr}(z) = \frac{1}{T} \sum_{k=-\infty}^{\infty} \Pr_{rr}(kT) z^{-k} \quad (11-44)$$

Note that except for the multiplying factor $1/T$ and the limits of summation, Eq. (11-44) is exactly analogous to the relation of the z-transform of the time series $r^*(t)$: that is

$$R(z) = \sum_{k=0}^{\infty} r(kT) z^{-k} \quad (11-45)$$

where it is assumed that $r(t)$ exists only for t greater than zero. Equation (11-44) also implies that whenever the correlation function of a stationary random process $r(t)$ is given, the sampled power spectral density function of $r(t)$ or of $r^*(t)$ can be obtained with the aid of the z-transform table. Equation (11-44) can be written as or

$$P_{rr}(z) = \sum_{k=-\infty}^{\infty} p_{rr}(kT) z^{-k} = p_{rr}(0) + \sum_{k=1}^{\infty} p_{rr}(kT) z^{-k} + \sum_{k=1}^{\infty} p_{rr}(-kT) z^k \quad (11-46)$$

$$P_{rr}(z) = \sum_{k=0}^{\infty} p_{rr}(-kT) z^k + \sum_{k=0}^{\infty} p_{rr}(kT) z^{-k} - p_{rr}(0) \quad (11-47)$$

Since $P_{rr}(T)$ is an even function, $p_{rr}(-kT) = p_{rr}(kT)$, Eq. (11-47) is finally written as

$$P_{rr}(z) = \sum_{k=0}^{\infty} A(z^{-1})^k + A(z) - p_{rr}(0) \quad (11-48)$$

In Eq. (11-48)

$$A(z) = \sum_{k=0}^{\infty} p_{rr}(kT) z^{-k} = Z[p_{rr}(s)]^* \quad (11-49)$$

$$A(z^{-1}) = \sum_{k=0}^{\infty} p_{rr}(kT) z^k = Z[p_{rr}(s)]^- = Z[p_{rr}(s)]_{s=z^{-1}} \quad (11-50)$$

where $Z[p_{rr}(s)]^*$ has all the poles of $P_{rr}(s)$ that lie in the left-half s -plane and $Z[p_{rr}(s)]^-$ has all the poles of $P_{rr}(s)$ that lie in the right-half s -plane and

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$$P_{rr}(s) = Z[p_{rr}(s)]^* + Z[p_{rr}(s)]^- \quad (11-51)$$

$p_{rr}(0)$ is given by $P_{rr}(0)$

$$L^{-1}[P_{rr}(s)]_{t=0} = L^{-1}[P_{rr}(s)]_{t=0} = p_{rr}(0) \quad (11-52)$$

The following example illustrates how the sampled spectral density of a stationary random process may be obtained by use of Eq. (11-48) and the z -transform table.

Example 11-1

Assuming that the auto-power spectral density of a stationary random process $r(t)$ is given by

$$P_{rr}(s) = \frac{36}{-s^2 + 36} \quad (11-53)$$

we wish to find the sampled power spectral density function $P_{rr}(z)$. Using Eq. (11-51), we see that

$$P_{rr}(z) = Z[p_{rr}(s)]^* + Z[p_{rr}(s)]^- = \frac{6+s}{6-s} \quad (11-54)$$

From the z -transform table

$$A(z) = Z[p_{rr}(s)]^*$$

Also

$$P_{rr}(0) = L^{-1}[p, r(s)] + 7 = 0$$

$$A(\approx^1) = [p, , (s)]^- =$$

$$\bullet C + j\omega$$

$$2 - e^{-67}$$

$$32 - 1$$

$$2 - 1 - e$$

$$2\pi j \text{ Sex } [P, , (5)]^* ds =$$

$$c - j\omega$$

$$c - j\omega$$

$$6 + s$$

$$(11-57)$$

Substitution of Eqs. (11-55), (11-56), and (11-57) into Eq. (11-48) yields

$$(11-55)$$

$$(11-56)$$

$$1$$

$$Pc + j\omega$$

$$\text{Sex } 6 \ 3 \ 3 \ ds = 3$$

$$pr \cdot r \bullet (2)$$

$$3$$

$$=$$

$$T$$

$$Z$$

$$-$$

$$Z$$

$$+$$

$$2 - 1$$

$$2 - 1. -6T$$

$$1$$

$$3$$

$$1 - e^{-12T}$$

$$=$$

$$T$$

$$- e^{-6Tz^{-1}}(1 - e^{-\Theta^T z})$$

$$(11-58)$$

$$(11-59)$$

11.4 SAMPLED RANDOM SIGNALS IN SAMPLED-DATA

SYSTEMS

In this section we shall conduct the analysis of sampled-data systems with random signals. Let us consider the open-loop sampled-data system of Fig. 11-5a. The input $r(t)$ is assumed to be a member of a stationary random process. We shall see that the power spectral function and the sampled power spectral function of the system output are given by

and

$$P_{cc}(s) = G(s)G(-s) Pr(s)$$

$$cc$$

$$P_{cc}(z) = G(z) G(z^{-1}) pr \bullet \bullet (2)$$

$$(11-60)$$

$$(11-61)$$

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$$r(t)$$

$$*(t)$$

$$G(s)$$

$$(a)$$

$\rightarrow c^*(1)$
 $-c(1)$
 $4, 0, *(7)$
 $0, *(7)$
 $\$cc(7)$
 $G(s)$
 $G(-s)$
 $\$, *, *(s)$
 $\$, *_c (s)$
 $Dcc(s)$
 (b)
 $0, *, *(z)$
 $\$(K\uparrow)$
 $B, *_c^*(z)$
 $Pc^*_c^*(z)$
 $G(z)$
 $G(z)$
 (KT)
 Γ_{ocn}
 Φ_{cnop}
 (KT)
 (c)

Fig. 11-5 (a) Open-loop sampled-data system with random input.
 (b) Block diagram showing the relationship between power spectral functions in a sampled-data system. (c) Block diagram showing the z-transform relationship between sampled power spectral functions of a sampled-data system.
 respectively. With reference to Fig. 11-5b, Eq. (11-60) implies that if (7) is the input of the system with $G(s)G(-s)$ as the transfer function, the output of the system is $pcc(7)$. Furthermore, when (T) is the input signal, the output of $G(s)$ is $prc(T)$, the cross-correlation function of the sampled signal $r^*(t)$ and continuous output $c(t)$.

By use of the convolutional integral, the output of $G(s)$ is written

$$Sg(o) \lim_{T_o \rightarrow \infty} \frac{1}{T_o} \int_{-T_o}^{T_o} r^*(t) r^*[t+(7-o)] dt = \lim_{T_o \rightarrow \infty} \frac{1}{T_o} \int_{-T_o}^{T_o} g(o) do \quad (11-62)$$

Interchanging the order of integration, Eq. (11-62) becomes

$$\lim_{T_o \rightarrow \infty} \frac{1}{T_o} \int_{-T_o}^{T_o} S'' r^*(t) dt \left[\lim_{T_o \rightarrow \infty} \frac{1}{T_o} \int_{-T_o}^{T_o} g(o) do \right] = \lim_{T_o \rightarrow \infty} \frac{1}{T_o} \int_{-T_o}^{T_o} S'' r^*(t) c(t+7) dt$$

which is the definition of $prc(T)$. Thus

In order to prove that

$$P_{cc}(s) = G(s) Pr(\$) = G(-s) Pr.c(\$) \quad (11-63)$$

(11-64)

(11-65)

we use the properties of $Pee(s)$ and $Pre(s)$ as a basis and write

-

$$Pec(s) = Pec(-s) = G(s) Pr \bullet c(-s) = G(s) Per \bullet (s)$$

-

(11-66)

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Thus when $per(t)$ is the input signal, the output of $G(s)$ is

1

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T c(t) dt = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T c(t+r) dt$$

(11-67)

Interchanging the order of integration in the last equation, we have

1

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T c(t) dt = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T c(t+r) dt$$

$\lim_{T \rightarrow \infty}$

$\lim_{T \rightarrow \infty}$

$\lim_{T \rightarrow \infty}$

$\lim_{T \rightarrow \infty}$

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T c(t) c(t+r) dt$$

-

(11-68)

which is $pec(T)$. Combining Eqs. (11-64) and (11-65), we have Eq.

(11-60).

Equation (11-61) is the counterpart of Eq. (11-60) in the z -domain. The proof of Eq. (11-61) can also be divided into two steps.

First, we shall prove that

$$Pre(z) = G(z) Pr \bullet r \bullet (z)$$

(11-69)

When the input to $G(s)$ is (kT) , the output is written from the superposition principle.

$$\sum_{k=-\infty}^{\infty} pr \bullet (kT) g(mT - kT)$$

$\sum_{k=-\infty}^{\infty}$

$\sum_{k=-\infty}^{\infty}$

$\sum_{k=-\infty}^{\infty}$

$\sum_{k=-\infty}^{\infty}$

$\sum_{k=-\infty}^{\infty}$

$\sum_{k=-\infty}^{\infty}$

$\sum_{k=-\infty}^{\infty}$

$$\sum_{q=-N}^N r(qT) r(qT + kT) g(mT - kT) \quad (11-70)$$

$\sum_{q=-N}^N$

Interchanging the order of summation in the last equation, we have

$\sum_{q=-N}^N$

$\sum_{q=-N}^N$

$$\sum_{k=-\infty}^{\infty} pr \bullet (kT) g(mT - kT)$$

$\sum_{k=-\infty}^{\infty}$

$\sum_{k=-\infty}^{\infty}$

$\sum_{k=-\infty}^{\infty}$

$$\lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{q=-N}^N r(qT) \sum_{k=-\infty}^{\infty} r(qT + kT) g(mT - kT)$$

$\lim_{N \rightarrow \infty}$

$\lim_{N \rightarrow \infty}$

$\lim_{N \rightarrow \infty}$

Letting $qk = n$, Eq. (11-71) reads

$\sum_{k=-\infty}^{\infty}$

$\sum_{k=-\infty}^{\infty}$

$\sum_{k=-\infty}^{\infty}$

$$(11-71)$$

-

$$\sum_{k=-\infty}^{\infty} r(qT) \sum_{k=-\infty}^{\infty} r(nT) g(mT + qT - nT)$$

$$\lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{q=-N}^N c(mT + qT)$$

which is $\text{prc.}(mT)$. Therefore,

$$\text{prc.}(mT) = \sum_{k=-\infty}^{\infty} \text{pr.}(kT) g(mT - kT) \quad (11-72)$$

Multiplying both sides of Eq. (11-73) by z^{-m}/T and taking the summation from $m = -\infty$ to $+\infty$, we have

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$$\sum_{m=-\infty}^{\infty} \text{pr.c.}(mT) z^{-m} = \sum_{m=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} \text{pr.s.}(kT) g(mT - kT) z^{-m}$$

$$= \sum_{k=-\infty}^{\infty} \text{pr.s.}(kT) z^{-k} \sum_{m=k}^{\infty} g(mT - kT) z^{-(m-k)}$$

Therefore, it is proved that

$$\text{Pr.c.}(z) = G(z) \text{Pr.}(z) \quad (11-74)$$

In a similar manner, we can show that

$$\text{Pec}(z) = G(z^{-1}) \text{Pr.}(z) \quad (11-75)$$

Combining Eqs. (11-75) and (11-76) yields

$$\text{Pec}(z) = G(z) G(z^{-1}) \text{Pr.}(z) \quad (11-76)$$

which is the desired result.

The relationship given by Eq. (11-77) is derived with the consideration that only the values of $c(t)$ at the sampling instants are used. The modified z-transform concept can be used in the present studies so that each point of $c(t)$ may be considered. With reference to Fig. 11-6, a fictitious time delay of AT seconds is inserted in the system. By defining $m = 1 - \Delta$, $0 < \Delta < 1$, the modified z-transform of the system output is where

$$C(z, m) = G(z, m) R(z)$$

$$\sum_{k=0}^{\infty} g[(k+m)T]z^{-k}$$

$$\begin{aligned} & (11-78) \\ & (11-79) \\ & c(nT, m) \end{aligned}$$

$$\begin{aligned} & r(t) \\ & * (1) \\ & G(s) \\ & e^{-47} \end{aligned}$$

$$\begin{aligned} & (a) \\ & Orr(KT) \\ & \$, *, *(s) \\ & Prc(KT, m) \\ & G^*(s, m) \\ & G^*(-s, m) \\ & \$, *c^*(s, m) \\ & Occ(KT, m) \\ & Pcc(s, m) \end{aligned}$$

$$(b) \quad c(t-AT)$$

Fig. 11-6 (a) Open-loop sampled-data system with fictitious delay.

(b) Block diagram showing the 2-transform relationship between the modified sampled power spectral functions of a sampled-data system.

† G. W. Johnson, "Statistical Analysis of Sampled-Data Systems," I. R. E. WESCON Convention Records, 1957, pp 187-195; J. T. Tou, "Statistical Design of Digital Control Systems," I. R. E. Trans. on Automatic Control, Sept. 1960, pp. 290-297.

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If $pec(kT, m)$ is defined as the modified autocorrelation function of the random sampled sequence $c(kT + mT - T)$, where $0 < m \leq 1$,
 $pec(kT, m) \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N c(nT+mT-T)c(nT+mT-T+kT)$

$$\begin{aligned} & = \\ & \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N c(nT+mT-T)c(nT+mT-T+kT) \end{aligned}$$

$$(11-80)$$

it can be shown that the modified sampled-spectral density of $c(t)$ is
 $\Phi_{ec}(z, \tau)$

$$\begin{aligned} & = \\ & Pec(z, m) \end{aligned}$$

$$\begin{aligned} & cc \\ & *** \end{aligned}$$

$$\begin{aligned} & -k \\ & 00 \\ & \sum_{k=-\infty}^{\infty} pcc(kT, m) z^{-k} \end{aligned}$$

$$\begin{aligned} & -2^{-1} \\ & T \quad k=-\infty \\ & (11-81) \end{aligned}$$

Then, $Pee(2, m)$ is related to the sampled spectral density of $r(t)$ by
 $G(z, m)G(z^{-1}, m)Pr \cdot r^*(z)$

$$(11-82)$$

The proofs of Eqs. (11-81) and (11-82) are straightforward; therefore, they will not be elaborated here. The reader may derive these

two equations as an exercise.

Equation (11-82) can be expressed in terms of real frequency ω by letting $s = j\omega$ and $z = e^{sT}$:

$$P_{cc}(j\omega, m) = |G^*(j\omega, m)|^2 P_{\mu\mu}(j\omega) \quad (11-83)$$

Similarly, Eq. (11-77) is written as a function of real frequency ω .

$$P_{cc}(j\omega) = |G^*(j\omega)|^2 P_{rr}(j\omega) \quad (11-84)$$

Before entering the discussion of mean square error and the optimum design of sampled-data systems with random signals, a comparison of the statistical descriptions of continuous-data and sampled-data many prove helpful. Corresponding functions and relationships between continuous-data systems are tabulated in Table 11-1.

Thus far, we have limited our illustrations to the derivation of the correlation relations between signals of a single path system. In the design of optimum systems using the mean-square-error criterion, the cross-correlation between signals in two independent systems must be evaluated often. Shown in Fig. 11-7 are two independent open-loop sampled-data systems (in a design problem one may denote the actual system, whereas the other may be the ideal or desired system). The cross-spectral densities $P_{c,c}(z)$ and $P_{c,c}(s)$ are to be expressed as functions of the system transfer functions and the power spectral densities of the other signals. In a manner similar to the method used in proving Eqs. (11-60) and (11-61), we can show that

$$P_{11}C_2(z) = G_1(z^{-1}) P_{r_1c}(z) = G_1(z^{-1}) G_2(z) P_{rr}(z) \quad (11-85)$$

$$P_{c,c}(z) = P_{c,c}(z^{-1}) = G_2(z^{-1}) P_{r,r}(z)$$

and

$$P_{11}C_2(s) = G_1(s^{-1}) P_{r_1c}(s) = G_1(s^{-1}) G_2(s) P_{rr}(s) \quad (11-86)$$

Similarly, the cross-spectral density between the two continuous outputs $c_1(t)$ and $c_2(t)$ may be written as

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TABLE 11-1

CORRESPONDENCE BETWEEN STATISTICAL PROPERTIES OF SAMPLED-DATA SYSTEMS AND CONTINUOUS-DATA SYSTEMS

Stationary

random process

Continuous-data systems

$r(1)$ or $R(s)$

Sampled-data systems

$r(nT)$, $r^*(1)$, $R^*(s)$, or $R(z)$

To

1

Correlation

functions

$$r_{12}(T) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T r_{12}(t) dt$$

$$r_{12}(KT) = \lim_{K \rightarrow \infty} \frac{1}{K} \sum_{k=0}^{K-1} r_{12}(kT)$$

$2N+1$

∞

To 400

1

$$S_{12}(s) =$$

$$= \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T r_{12}(t) e^{-st} dt$$

Spectral

density

Input-output
relations

$$40,0(7) ==$$

$$k = -\infty$$

N

$$\sum_{n=-N}^{\infty} r(nT) r(nT + kT)$$

$$n = -N$$

$$r^*(\frac{1}{T}) r^*(1+7) dt$$

$$, (KT) 8^{(7-KT)}$$

$$, (5) = /00 \$4, , (7) \text{ `ST OT}$$

$$, *, *(5) = =$$

$$(KT)_{ek}$$

$$-kTs$$

$$k = -\infty$$

$$, 0, 0(2) =$$

$$\sum$$

$$k = -\infty$$

$$, , (2, m) ===$$

$$P_1(KT) \tilde{Z}^k$$

$$k = -\infty$$

$$, (KT, m) z^k$$

$$C(s) = G(s)R(s)$$

$$C(z) = G(z)R(z)$$

$$C(jw) = G(jw)R(jw)$$

$$C(s) = G^*(s)R^*(s)$$

$$r(1)$$

$$c(1)$$

$$r(1)$$

$$*(1)$$

$$G(s)$$

$$G(s)$$

$$c^*(1)$$

$$C(1)$$

$$, (7)$$

$$, 10(5)$$

$$C(7)$$

$$, *, *(T)$$

$$G(s)$$

$$G(-s)$$

$$0^\circ C(T)$$

$$G(s)$$

$$, cc(T)$$

$$, (s)$$

$$Pre(s)$$

$$G(-s)$$

$$Pcc(s)$$

$$4, *, *(s)$$

$$, *, c(s)$$

$$cc(s)$$

$$cc(s) = G(s)G(-s), , (s)$$

$$, , (KT)$$

$$Prc(KT)$$

$$G(z)$$

$$cc(AT)$$

$$G(2^{-1})$$

$$, (2)$$

$$Prc(z)$$

$$Pcc(z)$$

$$(jw) = |G(jw)| 20 (jw)$$

$$Pcc(5) G(s)G^*(-s), , (s)$$

$$(z) = G(z)G(z), (z)$$

$$P_{cc}(z, m) = |G(j\omega)|^2 \Phi_{cc}(z, m) = G(z, m)G(z^{-1}, m)\Phi_{cc}(z)$$

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$\Phi_{cc}(1)$

$\Phi_{cc}(1)$

$G_1(s)$

$-C_1(1)$

$\cdot C_1(t)$

$\Phi_{cc}(t)$

$\Phi_{cc}(1)$

$\Phi_{cc}(1)$

$G_2(s)$

$C_2(t)$

Fig. 11-7 Block diagrams illustrating statistical relationships between input and output of sampled-data systems.

$\Phi_{cc}(s)$

=

$G_1(-s) G_2(s) \Phi_{cc}(s)$

(11-87)

Similar relations can also be derived in terms of the modified z-transforms. For instance

$P_{cc}(z, m) = G_1(z^{-1}, m) G_2(z, m) \Phi_{cc}(z)$

(11-88)

where $G_1(z^{-1}, m)$ and $G_2(z, m)$ are the modified z-transforms associated with $G_1(-s)$ and $G_2(s)$, respectively. In fact, these above relationships for sampled-data systems are analogous to those for strictly continuous systems. If one is familiar with the relations between signals for continuous-data systems, Eqs. (11-85) through (11-88) may be written out directly. A comparison of the spectral density relationships for continuous-data and sampled-data systems is made in Fig. 11-8.

11.5 MEAN-SQUARE ERROR

As is well known, when the minimization of the mean-square error is chosen as the performance criterion, the design of optimum linear continuous-data control systems may be carried out using the descriptions of the stationary signals in the system by correlation functions and spectral densities. Evidently, the mean-square error criterion can also be extended to the design of optimum linear sampled-data control systems. In this section, we shall relate the mean-square errors of some typical sampled-data systems to the correlation functions and the spectral densities of the input and the output signals. The systems are assumed to be free from noise and disturbances. Although open-loop configurations are illustrated, the derivations can be extended easily to closed-loop systems. Let us consider the sampled-data system shown in Fig. 11-9. The system is characterized by having sampled inputs and outputs. The

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Continuous-data systems

Sampled-data systems

(1)

$G_1(s)$

$C_1(1)$

(1)

$G_1(s)$

(1)

$\Phi_{cc}(1)$

$G_2(s)$

$C_2(1)$
 $72(1)$
 $G_2(s)$
 $62(1)$
 (7)
 $G_1(s)$
 $P12c(7)$
 (s)
 (s)
 $G_2(-s)$
 $06269(71$
 (7)
 $Proc(T)$
 $\$229,(5)$
 $Br2r,(2)$
 $G_1(z)$
 $PC2C(7)$
 $\$129,(2)$
 $G_2(2-1)$
 (7)
 $\$(7)$
 $Bar2(5)$
 $G_2(s)$
 $PC,C2(7)$
 (7)
 $G_2(-s)$
 $P(7)$
 $\$2,92(5)$
 $G_1(-s)$
 $PCC2(7)$
 $Br_112(2)$
 $G_2(s)$
 $\$c_152(2)$
 $\$121(7)$
 $\$6,02(2)$
 $Pc2(7)$
 $G_2(-s)$
 $\$121(5)$
 $G(s)$
 $\$26,(7)$
 721
 (7)
 $*(T)$
 $\$6254155$
 $\$299(5)$
 $G_2(21)$
 $\$120(7)$
 $Brary(2)$
 $\$6259(2)$
 $G_1(z)$
 $\$112(7)$
 $\$6209(2)$
 $PC(7)$
 $G_2(s)$
 Pc,cz
 $(+)$
 $\$(7)$
 (s)
 $G_1(-s)$
 $Price(T)$
 $Brac_2(5)$
 $G_2(z)$

Pcc (7)

$\$1_1 r_2^2(2)$

$G_1(2)$

(z)

Discrete output signals

** (7)

$\$(s)$

$G_2(s)$

Pric₂

(T)

(7)

$G_1(-s)$

Bric₂(s)

Continuous output signals

$G_2(z, m)$

Pr.co(z, m) $G_1(z, m)$ $\$c, c, (2, m)$

Modified z-transform of output

Fig. 11-8 Block diagrams illustrating the relations between input and output signals of two systems.

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$r(t)$

$* (t)$

$c(1)$

$c^*(t)$

$M(z)$

$r(nT)$

T

$c(nT)$

$e(nT)$

$e(t)$

(1)

$Ca(t)$

$ca(t)$

$Ma(z)$

$r(NT)$

$Ca(nt)$

Fig. 11-9 Sampled-data system to be designed in the sense of minimum mean-square error.

error sequence is defined as the difference between the desired output sequence $ca(nT)$ and the actual output sequence $c(nT)$; that is,

$$e(nT) = ca(nT) - c(nT)$$

The mean-square sampled error is given by

$$e(nT) = c\ll(nT) - c(nT)$$

=

.

(11-89)

$$c\grave{a}(nT) - ca(nT)c(nT) - c(nT)ca(nT) + c^2(nT) \quad (11-90)$$

In view of the relationships

$$e^2(nT) = pee(0) \lim$$

=

-

1

2nj

1

2N + 1

$N \rightarrow \infty$

$$f. pee(2) z^{-1} dz$$

N

$$\sum_{n=-N}^N \tilde{e}^2(nT)$$

$$\frac{1}{2N+1} \sum_{n=-N}^N \tilde{e}^2(nT)$$

$$\text{and } C_a(NT) = c(NT)$$

$$= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N \tilde{e}^2(nT)$$

$$= \lim_{N \rightarrow \infty} \frac{1}{2\pi j} \oint_{\Gamma} p_{ee}(z) z^{-1} dz$$

$$(11-92)$$

the sampled spectral density of $e(nT)$ may be written as

$$P_{ee}(z) = p_{caca}(z) - p_{\zeta\zeta}(z) - p_{\epsilon\epsilon}(z) + p_{cc}(z)$$

From Fig. 11-8 and Table 11-1, the following relationships may be written:

$$P_{caca}(z) =$$

$$= \frac{1}{T} \int_{-T/2}^{T/2} M(z) M(z^{-1}) p_{rr}(z) dz$$

$$= \frac{1}{T} \int_{-T/2}^{T/2} M(z) M(z^{-1}) p_{rr}(z) dz$$

$$= \frac{1}{T} \int_{-T/2}^{T/2} M(z) M(z^{-1}) p_{rr}(z) dz$$

$$P_{cc}(z) = M(z^{-1}) M(z) p_{rr}(z)$$

$$(11-94)$$

$$(11-95)$$

$$(11-96)$$

$$(11-97)$$

Substitution of Eqs. (11-94) through (11-97) into Eq. (11-93) yields

$$P_{ee}(z) = [M(z) M(z^{-1}) - M(z) M(z^{-1})$$

$- M(z) M(z) + M(z) M(z^{-1})] p_{rr}(z)$ (11-98)

Therefore, Eq. (11-98) shows that the sampled spectral density $P_{ee}(z)$ is represented by the system transfer functions $M(z)$, $M(z^{-1})$, and the input sampled spectral density $p_{rr}(z)$. The mean-square sampled

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error is obtained by substituting $P_{ee}(z)$ into Eq. (11-91). The design objective is the determination of the transfer function $M(z)$ in such a way that the mean-square sampled error $e^2(nT)$ is a minimum. In general, most sampled-data systems have sampled as well as continuous signals. Shown in Fig. 11-10 is a sampled-data system

$$r(t)$$

$$* (+)$$

$$c(1)$$

$$M(s)$$

$$r(t)$$

$$Ca(t)$$

$$Mo(s)$$

$$e(1)$$

T
 $e(t)$
 Fig. 11-10 Sampled-data system with continuous output.
 with continuous output signal. The system is compared with a continuous model M_a . The mean-square error of the system is defined as
 $e^2(t) = [c_a(t) - c(t)]^2$

$$= \int_{-T_0}^{T_0} c_a(t)c(t) - c(t)c_a(t) + c^2(t) dt$$

$$\lim_{T_0 \rightarrow \infty} \frac{1}{2T_0} \int_{-T_0}^{T_0} e^2(t) dt = p_{ee}(0)$$

$$= \frac{1}{2\pi j} \int_{-j\infty}^{j\infty} P_{ee}(s) ds \quad (11-99)$$

Similar expressions can also be written for $c_a(t)$, $c_a(t)c(t)$, and the other terms in Eq. (11-99). Therefore, the spectral density of $e(t)$ is written as

$$P_{ee}(s) = P_{aa}(s) - P_{ac}(s) - P_{ca}(s) + P_{cc}(s) \quad (11-101)$$

From Fig. 11-10 the following relationships between the spectral density functions may be written:

$$P_{aa}(s) = M(s)M(-s)P_{rr}(s) \quad (11-102)$$

$$P_{ac}(s) = M(s)M(-s)P_{rr}(s) \quad (11-103)$$

$$P_{ca}(s) = M(s)M(-s)P_{rr}(s) \quad (11-104)$$

$$P_{cc}(s) = M(s)M(-s)P_{rr}(s) \quad (11-105)$$

where $P_{rr}(s)$ and $P(s)$ involve the cross-correlation between a continuous signal and a sampled signal. We can show that

$$P_{rr}(T) = P_{rr}(7) = P_{rr}(7) \quad (11-106)$$

Substitution of Eqs. (11-101) through (11-106) into Eq. (11-100) yields

† J. R. Ragazzini and G. F. Franklin, page 265. op. cit.

$$c^1(t) = 24j \text{ Sex } \{ [M_o(8) Ma(-3) - M(s) Ma(-s)] / c-joo \}$$

$$-M(-s)Me(s)]p_{,,}(s) + M(s)M(-s)p_{,,}(s) \} ds$$

-
-

(11-107)

As is evident now, the mean-square error may be expressed as a function of the power spectral densities of $r(t)$ and $r^*(t)$ and the transfer functions $M(s)$ and $M_a(s)$. In an analysis problem, $e^2(t)$ may be obtained from Eq. (11-107) if $M_a(s)$, $M(s)$ and $r(t)$ are specified. In the design problem, $r(t)$ and $M_a(t)$ are specified, and the design objective is to determine $M(s)$ so that $e^2(t)$ is a minimum. The transfer function $M(s)$ used here is entirely general and could represent a feedback system with or without sampling.

However, the evaluation of the integral of Eq. (11-107) usually may present some difficulties, since $M(s)$ in sampled-data systems usually contains the term e^{-sT} as well as s . If $e(t)$ can be accurately described by the error sequence $e(nT)$, the mean-square sampled error can be used as a close approximate of the true error. In this case, the z -transform method may be applied. The sampled error is not a good measure of $e(t)$ if the system has inter-sampling ripples or hidden oscillations. A compromise may be made by using the modified z -transform technique. The "modified" error sequence may be written as

$$e(nT, m) = ca(nT, m)$$

$$Ca(nT, m) - c(nT, m)$$

(11-108)

where $0 \leq m \leq 1$. Taking the mean-square values on both sides of

Eq. (11-108), we have

$$e^2(nT, m) = ca(nT, m)$$

-

$$Ca(nT, m)c(nT, m)$$

--

$$- c(nT, m)ca(nT, m) + c^2(nT, m) \quad (11-109)$$

In view of the following relationships,

1

$$e^2(nT, m) =$$

$\frac{1}{2\pi j}$

$$\oint_{\Gamma} p_{ee}(z, m) z^{-1} dz$$

(11-110)

1

=

$\frac{1}{2\pi j}$

$$\oint_{\Gamma} p_{cc}(z, m) z^{-1} dz$$

(11-111)

$$Ca(nT, m)c(nT, m)$$

the modified mean-square error is written as

$$e^2(nT, m)$$

=

$\frac{1}{2\pi j}$

$$\oint_{\Gamma} [p_{ce}(z, m) - p_{\infty}(z, m)$$

-

$$- p_{\infty}ca(z, m) + p_{cc}(z, m)] z^{-1} dz$$

(11-112)

From Figs. 11-8 and 11-10, the following relationships may be written:

$$p_{ca}(z, m) = Z_m[M_a(s)M(-s)p_{rr}(s)]$$

$$p_{ec}(z, \tau)$$

=

$$M(z, m)M(z^{-1}, m)p_{,,}(z)$$

(11-113)

(11-114)

$$p_{cc}(z, m) = Z_m[M(s) Ma(-s)p_{rr}(s)]$$

(11-115)

$$p_{\alpha\alpha}(z, m) = Z_m[M(-s) Ma(s)p_{rr}(s)]$$

(11-116)

CCd

TT

Substitution of Eqs. (11-113) through (11-116) into Eq. (11-112)

yields

$$e^2(nT, m)$$

=

-

$$\frac{1}{2\pi j}$$

$$\int_{-1}^1 \{Z_m[Ma(s) Ma(-s)p_{rr}(s) - M(s) Ma(-s) p_{rr}(s)] + M(z, m) M(z^{-1}, m)p_{rr}(z)\} z^{-1} dz$$

$$Ma(s) M(-s) p_{rr}(s)] + M(z, m) M(z^{-1}, m)p_{rr}(z)\} z^{-1} dz$$

(11-117)

In this last equation for each value of m between zero and unity, we get a mean-square value of the error samples for that particular m . A typical $e^2(nT, m)$ versus m curve may be of the form shown in Fig. 11-11. To obtain the mean-square error $e^2(t)$ that is a constant, Eq. (11-117) must be averaged from $m = 0$ to $m = 1$.

Therefore, $e^2(t)$ is given by

Mean-square

error

$$e^2(nT, m)$$

$$e^2(t)$$

$\int_0^1$

$$e^2(t)$$

Fig. 11-11 Mean-square error $e^2(t)$ and modified mean-square error $e^2(nT, m)$.

$\int_0^1$

$$e^2(t)$$

=

-

$$\int_0^1 e^2(nT, m) dm$$

(11-118)

Substituting Eq. (11-110) into Eq. (11-118) yields

$\int_0^1$

$$e'(t) = \int_0^1 [\int_{-1}^1 p_{ee}(x, m) x^{-1} dz] dm$$

(11-119)

Inversion of the order of integration, which is allowable in this case,

$\int_{-1}^1$

$$e^2(t)$$

=

$\int_0^1$

$$\frac{1}{2\pi j}$$

$$\int_{-1}^1 [\int_0^1 p_{ee}(z, m) dm] x^{-1} dz$$

(11-120)

The following example illustrates the mean-square-error method.

Example 11-2

Figure 11-12 shows a sampled-data system whose mean-square error is to be evaluated. The system M is assumed to be a zero-order hold and $M_1(s) = 1$. The power spectral density of the input signal is

† M. Mori, "Statistical Treatment of Sampled-Data Control Systems for Actual Random Inputs," Trans. A. S. M. E., Basic Engineering, Feb. 1958, pp. 444-456.

$\frac{1 - e^{-ST}}{c(t)}$
 $\frac{e(t)}{ca(t)} = r(t)$
 Fig. 11-12 Zero-order hold system for Example 11-2.

$\frac{\Phi_{\tau+}(s)}{200}$
 ω
 (11-121)
 The mean square error of this system will give a measure of the ability of data-reconstruction of the zero-order hold. From Eqs. (11-100) and (11-101), the mean square error is written as

$$e^2(t) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} [P_{ee}(s) - P_{ec}(s) - P_{\epsilon\epsilon}(s) + P_{\epsilon\epsilon}(s)] ds$$

$$\Phi_{caca}(0) - \Phi_{coc}(0) - \Phi_{oca}(0) + \Phi_{cc}(0)$$

(11-122)
 Substituting Eqs. (11-102) through (11-105) into Eq. (11-122), we have

$$e^2(t) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} M(s) - \frac{1}{s} \{ [1 - M(s)] - M(-s) \} \times P_{rr}(s) + M(s) M(-s) P_{rr}(s) \} ds$$

(11-123)
 where $P_r(s)$ is the sampled spectrum of $r(t)$ and can be determined using the method described in Sec. 11.3. Substituting $M(s)$, $M(-s)$, $P_r(s)$, and $P_r(s)$ into Eq. (11-122), we can show by means of contour integration that

$$\Phi_{caca}(0) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} \Phi_{cac}(0) - \Phi_{cca}(0)$$

and

$$\Phi_{cc}(0)$$

Therefore,
 $e^2(t)$

$$\frac{1 - e^{-ST}}{ST}$$

(11-129)

When the impulse (δ) is applied to the zero-order hold, we can show by taking the inverse transform of $M(s)$ $M(-s)$ that the two-sided impulse response of the zero-order hold in the T-domain is as shown in Fig. 11-14. From this,

$$e^{-sT} (1 - e^{-sT})$$

$$-s^2$$

$$-T$$

$$-T$$

$$T$$

$$T$$

Fig. 11-14 Two-sided impulse response of the zero-order hold in the T-domain.

we conclude that the mean-square value of the output of the zero-order hold must equal T times the mean square value of the input. Therefore,

$$\bullet cc(0) = c^2(t) = T[r^*(t)]^2 = r^2(t)$$

(11-130)

or the mean-square value of the output of a sampler-zero-order hold combination is equal to the mean-square value of the input signal. This is also verified by the result of the integral of Eq. (11-125).

11.6 OPTIMUM SYNTHESIS USING MEAN-SQUARE

ERROR CRITERION

The principle objectives of the preceding section are the introduction of the mean-square-error criterion and the establishment of the relationship between the statistical properties of the input and out-

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put signals of sampled-data systems. In this section, we turn our attention from analysis to synthesis. Furthermore, in all the discussions of the previous section, we have neglected noise and interference in control systems. However, noise and interference are present in most practical control systems, and they tend to obscure the signal used for the optimizing control and are the basic restrictions in the design of optimum control systems.

The statistical synthesis of sampled-data systems in the sense of minimum mean-square error can be readily visualized by referring to the block diagram of Fig. 11-15a. $M(z)$ is the closed-loop transfer

$$M(s)$$

$$D(s).$$

$$5(1)$$

$$r(t) = rs + rn.$$

$$D(s)$$

$$H$$

$$c(t)$$

$$G(s)$$

$$e^*(t)$$

$$rs(t)$$

$$e(t)$$

$$Mo(s)$$

$$Ca(t)$$

$$(a)$$

$$r(t) = rs + rn$$

$$c(t) = Cs + Cn$$

$$M(s)$$

$$rs(t)$$

$$Mo(s)$$

$$(b)$$

$$c(t)$$

Fig. 11-15 (a) Sampled-data system to be designed in the sense of minimum mean-square error. (b) An equivalent system block diagram of the system in (a).

function of the actual sampled-data system which is to be optimized in the sense of minimum mean square error, and $M(z)$ is the desired over-all system transfer function. $G(s)$ is the transfer function of the continuous-data portion of the system and is assumed to be known. $D^*(s)$ or $D(z)$ denote the transfer function of the discrete compensator to be designed. The input signal $r(t)$ is assumed to

:

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consist of a noise component $r(t)$ and a signal component $r_s(t)$.

The input signal and the noise are assumed to be functions of stationary random processes. The system output also consists of two components; one due to $r_n(t)$ and one due to $r_s(t)$. The system error is defined as

$$e(t) = c_a(t) - c(t) = c_a(t) - c_s(t) - c_n(t)$$

(11-131)

If the system is well behaved between sampling instants, design may be carried out based on the mean-square-sampled error $e^2(nT)$. Since the signal $r(t) - c(t)$ is sampled in the actual system, we can insert a fictitious sampler at the input to M without affecting anything. Therefore, the entire system can be represented by the equivalent block diagram of Fig. 11-15b. The error sequence of the system is given by

$$e(nT) = c_a(nT) - c(nT) = c_a(nT) - c_s(nT) - c_n(nT) \quad (11-132)$$

Taking the mean-square value on both sides of the last equation, we have

$$e^2(nT) = c_a^2(nT) + c_n^2(nT) + c_s^2(nT) + c_a(nT)c_n(nT) + c_n(nT)c_s(nT) - c_s(nT)c_a(nT)$$

$$- c_a(nT)c_n(nT)$$

$$c_n(nT)c_a(nT)$$

$$c_s(nT)c_a(nT)$$

(11-133)

Making use of the relations between the mean-square sequence and the sampled spectral densities, the mean-square-sampled error may be written as

$$e^2(nT) = \frac{1}{2\pi j} \oint_{\Gamma} [P_{ee}(z) + P_{ca}(z) + P_{cn}(z) - P_{cs}(z) - P_{sc}(z) - P_{cn}(z)] z^{-1} dz \quad (11-134)$$

$$+ P_{en}(z) + P_{c,c}(z) + P_{c,c}(z) - P_{ec}(z) - P_{csca}(z) - P_{scsn}(z) - P_{encs}(z)] z^{-1} dz$$

In view of the relationships given in Fig. 11-8, $e^2(nT)$ becomes

where

$$e^2(nT)$$

$$= \frac{1}{2\pi j} \int_{\Gamma} \{ [M(z)M(z^{-1}) + Ma(z)Ma(z^{-1}) - M(z)Ma(z^{-1}) - M(z^{-1})Ma(z)] p_{r,r}(z) + [M(z)M(z^{-1}) - M(z)Ma(z^{-1})] p_{r,t}(z) + M(z)M(z^{-1}) p_{t,t}(z) \} z^{-1} dz$$

$$\frac{M(z)}{D(z)G(z)} = \frac{1}{1+G(z)D(z)} \quad (11-135)$$

(11-136)
The synthesis objective is to determine the transfer function $M(z)$ so that the mean-square-sampled error of Eq. (11-135) is a minimum. Once the optimum $M(z)$ is determined, the transfer function of the digital compensator is given by

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$$D(z) = \frac{(2) M(2)}{G(z)[1M(2)]} \quad (11-137)$$

The synthesis procedure developed below is essentially an extension of the theory of optimum design of linear filters developed by N. Wiener and A. Kolmogoroff and is also parallel to the work of J. T. Tou. The application of the Wiener-Kolmogoroff criterion to the design of linear continuous-data control systems is well known and is given in the literature. § The theory of the optimum system design by Wiener and Kolmogoroff is based on an application of the calculus of variations to the integral of Eq. (11-135). If $M(z)$ is an optimum transfer function, $e^2(nT)$ must be at a minimum. Replacing the optimum $M(2)$ by a function $M(z) + e_n(z)$ where e is a parameter and (2) is an arbitrary but physically realizable function of z would increase the mean-square-sampled error to $e^2(nT) + \Delta e^2(nT)$ where $\Delta e^2(nT)$ is the change in $e^2(nT)$. Then, a necessary condition that $e^2(nT)$ is a minimum is

$$\frac{\partial [\Delta e^2(nT)]}{\partial e} = 0 \quad (11-138)$$

To insure that the $\Delta e^2(nT)$ is a minimum, the second derivative of $\Delta e^2(nT)$ must be positive with respect to e when $e = 0$. Substituting $M(z) + e_n(z)$ and $M(z^{-1}) + e_n(z^{-1})$ for $M(2)$ and $M(\approx^1)$, respectively, in Eq. (11-135) and simplifying, we have

$$e^2(nT) + \Delta e^2(nT) = e^2(nT) + 27j \int_{\Gamma} e_n(x) \{ [M(z^{-1}) - Ma(z^{-1})] p_{r,r}(z) + [M(z)M(z^{-1}) - M(z)Ma(z^{-1})] p_{r,t}(z) + M(z)M(z^{-1}) p_{t,t}(z) \} z^{-1} dz$$

$$\text{Thus} \quad \times p_{r,r}(z) + [M(\approx^{-1}) - Ma(z^{-1})] p_{r,t}(z) + M(2^{-1}) p_{t,t}(z)$$

$$+ M(\approx^{-1}) p_{r,r}(x) \} x^{-1} dz + 27j \int_{\Gamma} e_n(x^{-1}) \{ [M(2) - Ma(2)] \times p_{r,t}(2) + M(2) p_{r,t}(2) + M(2) p_{t,t}(z) + M(2) p_{r,t}(2) \}$$

$$-xxx \quad 1 \quad - Ma(z) p_{r,t}(2) \} z^{-1} dz + 27j \int_{\Gamma} e_n(z) e_n(z^{-1}) \{ p_{r,r}(2) + p_{r,t}(2) + p_{t,t}(2) \} z^{-1} dz$$

$$\Delta e^2(nT)$$

1

=

 $2\pi j$

$$\oint_{\Gamma} \frac{1}{z} \{M(z^{-1}) [P_{11}(z) + P_{rs}(z) + P_{rv}(z)$$

$$+ P_{rr}(z)] - M_1(z^{-1}) [P_{11}(z) + P_{rr}(z)]\} z^{-1} dz$$

(11-139)

† A. Kolmogoroff, Bull. Acad. Science, U. S. S. R. Ser. Math., 5, 1941, pp. 3-14

‡

J. T. Tou, "Statistical Design of Digital Control Systems," Trans. on Automatic Control, I. R. E., Sept. 1960, pp. 290-297.

§ H. S. Tsien, Engineering Cybernetics, (New York: McGraw-Hill Book Company, Inc., 1954), Chap. 16; J. G. Truxal, Automatic Feedback Control System Synthesis, (New York: McGraw-Hill Book Company, Inc., 1955), Chap. 8.

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1

$$+ \oint_{\Gamma} \frac{1}{z} \{M(z^{-1}) [P_{11}(z) + P_{ran}(z) + P_{rat}(z) + P_{rata}(z)]$$
 $2\pi j$

1

$$M(z) [P_{rr}(z) + P_{rr}(z)] z^{-1} dz + \oint_{\Gamma} \frac{1}{z} \{M(z^{-1}) [P_{11}(z) + P_{rr}(z)]$$
 $2\pi j$

$$\times [P_{rr}(z) + P_{ran}(z) + P_{rw}(z) + P_{rr}(z)] z^{-1} dz$$

(11-140)

Differentiating with respect to e and setting the resultant equal to zero yields

$$\frac{\partial}{\partial e} [Ae^{2(nT)}]$$
 $\frac{de}{de}$

=0

1

=

$$\oint_{\Gamma} \frac{1}{z} \{M(z^{-1}) P_{11}(z) - M(z^{-1})$$

-

$$\times [P_{rr}(z) + P_{rr}(z)]\} z^{-1} dz + 21; \oint_{\Gamma} \frac{1}{z} \{M(z^{-1}) P_{11}(z)$$
 $2\pi j$

$$- M(z) [P_{rr}(z) + P_{rr}(z)]\} z^{-1} dz = 0$$

(11-141)

where $P_1(z) = P_{11}(z) + P_{rr}(z) + P_{rat}(z) + P_{rata}(z)$ (11-142)

When Eq. (11-141) is satisfied, $M(z)$ is an optimum. However, before the optimum $M(z)$ can be obtained from Eq. (11-141), the equation must be modified. It is evident that $P_{11}(z)$ is an even function of z . Therefore, we assume that $P_{11}(z)$ can be factored into two parts, one part having poles and zeros inside the unit circle and the other part having poles and zeros outside the unit circle. $P_r(z)$ can be written as

TT

Equation (11-141) is now written as

1

$$P_1(z) = P(z) P_{rr}(z)$$

$$\oint_{\Gamma} \frac{1}{z} \{M(z^{-1}) P_{rr}(z) - M(z^{-1}) [P_{11}(z) + P_{rr}(z)]$$

$$- M(z) [P_{rr}(z) + P_{rr}(z)]\} z^{-1} dz = 0$$

1

+

 $2\pi j$

$$\oint_{\Gamma} \frac{1}{z} \{M(z^{-1}) P_{rr}(z) - M(z^{-1}) [P_{11}(z) + P_{rr}(z)]$$
 $+ (2)$

(11-143)

$$P_{rr}(z)]\} z^{-1} dz$$

$$- M(z) [P_{rr}(z) + P_{rr}(z)]\} z^{-1} dz = 0 \quad (11-144)$$
 $\oint_{\Gamma} \frac{1}{z}$

We can also divide each of two functions $M(z^{-1}) [P_{rr}(z) + P_{rr}(z)] /$

$P_{\div}(z)$ and $M_d(z)[P_{,,}(z) + P_{rw}(z)]/b_{77}(z)$ into the sum of two parts; that is,

$$M_a(z^{-1})[P_{r,,}(z) + P_{r,tn}(z)]$$

and

$T(z)$

$$= F(z) = [F_1(z)]_+ + [F_-(z)] \quad (11-145)$$

$$M_d(z)[P_{t,r}(z) + P_{r,,}(z)] = F_2(z) = [F_2(\approx)]_+ + [F_2(\approx)]$$

$P(z)$

(11-146)

where $[F(z)]_+$ and $[F_2(z)]_+$ denote the parts with singularities only inside the unit circle and $[F_-(z)]$ and $[F_2(\approx)]_-$ denote the parts with singularities only outside the unit circle. Note that these notations are not to be confused with those of Eq. (11-143). Substitution of Eqs. (11-145) and (11-146) into Eq. (11-144) yields

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$$\oint_{2\pi j} n(x) P_{\div}(x) \{M(z^{-1}) P_{\tilde{n}}(x) - [F_1(x)]_+ - [F_1(z)]_-\} x^{-1} dz$$

1

+

$$\oint_{2\pi j}$$

-

-

-

$$\oint_{2\pi j} n(x^{-1}) P_{\tilde{n}}(x) \{M(x) P_{\div}(x) - [F_+(z)]_+ - [F_-(z)]_-\} x^{-1} dz = 0$$

(11-147)

For reasons of stability $M(z)$ and $\eta(z)$ must have poles only inside the unit circle in the z -plane. Thus $M(z^{-1})$ and $(\cdot)^{-1}$ are functions with poles only outside the unit circle. Therefore, in the integrals of Eq. (11-147),

and

1

$$\oint_{2\pi j}$$

1

$$\cdot \oint_{2\pi j} n(z) P_{\div}(z) [F_1(z)]_+ + z^{-1} dz = 0$$

(11-148)

= 0

(11-149)

$$\oint_{2\pi j} n(z^{-1}) P_{\tilde{n}}(x) [F_-(z)]_+ z^{-1} dz$$

since the integrand of the integral in Eq. (11-148) has poles only inside the unit circle and the contour integral can be taken around the unit circle I in the clockwise direction; and for similar reasons the integral of (11-149) vanishes. Equation (11-147) now reads

1

$$\oint_{2\pi j} n(x) P_{\div}(z) \{M(\approx^{-1}) P_{\tilde{n}}(x) - [F_1(z)]_+ - [F_1(\approx)]_-\} z^{-1} dz$$

+

1

$$\oint_{2\pi j} n(\approx^{-1}) P_{-(z)} \{M(\approx) P_{\div}(z) - [F_2(x)]_+ + [F_2(\approx)]_-\} z^{-1} dz = 0 \quad (11-150)$$

Now if $M(z)$ is indeed the transfer function of the optimum system, Eq. (11-150) must be satisfied for arbitrary $n(z)$. Thus the quantities inside the brackets of Eq. (11-150) must vanish. The optimum transfer function is given by

1

$$M(z) [F_2(z)]_+ = \{M_a(z) [P_{r,,}(x) + P_{,,}(z)]\}_+; (x) \quad (11-151)$$

=

$$\Phi_+(z)$$

Once $M(z)$ is determined, the transfer function of the digital compensator $D(z)$ follows directly from Eq. (11-137).

Clearly the properties of the optimum system are determined entirely by the specified system $M_a(z)$ and the spectra $P_s(z)$, $P_{r1}(z)$, $P_{r2}(z)$, and $P_n(z)$ of the signal and noise. In the event that noise is absent, $P_{r1}(z) = P_{r2}(z) = P_n(z) = 0$, and $P_{T1}(z)$

$$= \frac{\Phi^*(z) \Phi(z)}{(11-152)}$$

Then Eq. (11-151) gives $M(z) = M_a(z)$ as expected, and the error $e(t)$ is always zero.

To illustrate the synthesis of sampled-data control systems by means of the techniques described in the preceding paragraphs, a simple numerical example is presented below.

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Example 11-3

Consider the sampled-data control system shown in Fig. 11-16. Assume that

$$G(s) =$$

$$= \frac{1}{s(s+1)}$$

$$P_{T1}(s) =$$

$$= 1$$

$$P_r(s) = 1 \text{ (white noise) } T = 0.1 \text{ second}$$

and the input and noise are uncorrelated; that is, $P_r(s) = P_n(s) = 0$.

Design a digital compensator to minimize the mean-square-sampled error when

$$[s(t) + r_n(t)]$$

$$D(z)$$

$$D(s)$$

$$G_o(s)$$

$$G(s)$$

$$c(t)$$

Fig. 11-16 Linear sampled-data control system to be designed in

the sense of minimum mean-square sampled error.

The z-transform of $G_o(s)G(s)$ is

$$G_oG(z):$$

$$= \frac{0.005(z + 0.9)}{(z-1)(z \approx 0.905)}$$

The sampled spectrum of the input signal is found to be

$$P_{T1}(z)$$

$$(11-153)$$

$$P_{T1}(z) =$$

$$(\approx$$

$$0.0905) (1.1)$$

$$(11-154)$$

and the sampled spectrum of the noise is $P_r(z) = 1$.

Therefore, $P_{\pi}(z) = P_{T1}(z) + P_{r1}(z)$

$$P_r(z)$$

Dividing $P_{\pi}(z)$ into two parts according to Eq.

In the preceding paragraphs synthesis of linear sampled-data control systems is conducted on the basis of minimization of the mean-square-sampled error. We shall now discuss a synthesis procedure using the modified z-transform to minimize the mean-square error $e(t)$ of the system. Consider the sampled-data system shown in Fig. 11-17. The input signal and noise are assumed to be functions of a

$$r(t) = r_n + r_s$$

$$r_s(t)$$

$$r_s(t)$$

$$D^*(s)$$

$$D(s)$$

$$M_o(s)$$

$$c(t) = C_s + C_n$$

$$G(s)$$

$$e(t)$$

Fig. 11-17 Sampled-data control system to be designed on the basis of minimum mean-square error.
stationary random process. The desired system transfer function is $M_a(s)$. $D(z)$ is the transfer function of the digital compensator to be determined on the basis of minimum mean-square error. The system error is given by

$$e(t) = c_a(t) - c_s(t) - c_n(t)$$

In terms of samples at and in between sampling instants,

$$e(nT, m)$$

$$= c_a(nT, m) - c_s(nT, m) - c_{n,n}(nT, m)$$

$$- c_n(nT, m)$$

$$(11-160)$$

$$(11-161)$$

Taking the mean-square value on both sides of Eq. (11-161), we have

$$e^2(nT, m) = c_a(nT, m) + c_s(nT, m) + c_n(nT, m)$$

$$+ c_s(nT, m) c_n(nT, m) + c_n(nT, m) c_s(nT, m)$$

$$- c_a(nT, m) c_s(nT, m) - c_s(nT, m) c_a(nT, m)$$

$$- c_a(nT, m) c_n(nT, m) - c_n(nT, m) c_a(nT, m)$$

$$(11-162)$$

Through the use of Eq. (11-120), the mean-square error is given by

† J. T. Tou, "Statistical Design of Linear Discrete-Data Control Systems Via The Modified Z-Transform Method," Journal of the Franklin Institute, 271, No. 4, Apr

il
1961, pp. 249-262.

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$$e^2(t)$$

=
where

$$\Phi_{ec}(z, m)$$

$$=$$

$$2\pi j$$

$$\int [S'_{bee}(z, m) dm] z^{-1} dz$$

$$P_{caca}(z, m) + P_{c,,}(2, m) + P_{c,,,}(z, m)$$

$$+ P_{c,c.}(z, m) + P_{c,(2, m)} - P_{c,(z, m)}$$

$$- P_{c,ca}(2, m) - P_{ec.}(z, m) - P_{cca}(2, m)$$

(11-163)

(11-164)

With reference to Fig. 11-17, the modified z-transform of the closed-loop system may be written as

$$M(z, m) =$$

$$G(z, m) D(z)$$

$$1 + D(z) G(z)$$

=

$$G(z, m) W(z) \quad (11-165)$$

in which

$$W(x)$$

==

$$D(z)$$

$$1 + D(z)G(z)$$

(11-166)

In view of the relations of the modified spectral densities given in Eqs. (11-113) through (11-116), the following relationships may be written:

$$P_{\text{caca}}(z, m) = 3m[M_a(s) M_a(-s)P_{r,r}(s)]$$

$$\Phi_{\text{caca}}$$

(11-167)

$$\Phi_{\text{ec}}(z, m)$$

$$= W(z) W(z^{-1})G(z, m)G(z^{-1}, m)P_{r,r}(z)$$

(11-168)

$$P_{\text{cc}}(z, m)$$

$$= W(z)W(z^{-1})G(z, m)G(z^{-1}, m) \quad (2)$$

(11-169)

$$P_{\text{c,c}}(z, m)$$

$$= W(z) W(z^{-1})G(z, m)G(z^{-1}, m)P_{r,r}(z)$$

(11-170)

$$P_{\text{c}}(z, m) =$$

$$\Phi_{\text{cac}}$$

$$W(z) W(z^{-1})G(z, m)G(z^{-1}, m) P_{r,r}(z)$$

$$m)P_{r,r}(z)$$

(11-171)

$$P_{\text{cc}}(z, m) = W(z)G(z, m) Zm[M_a(-s)P_{r,r}(s)]$$

(11-172)

$$\Phi_{\text{eca}}$$

$$P_{\text{c,ca}}(z, m) = W(z^{-1})G(z^{-1}, m) 3m[M_a(s) P_{r,r}(s)]$$

$$Zm[M_a(s)P_{r,r}(s)]$$

(11-173)

$$P_{\text{ec}}(z, m)$$

$$= W(z)G(z, m) 3m[M_a(-s) P_{r,r}(s)]$$

(11-174)

(11-175)

$$P_{\text{cs}}(z, m) = W(z^{-1})G(z^{-1}, m) 3m[M_a(s)P_{r,r}(s)]$$

Substituting Eqs. (11-167) through (11-175) into Eq. (11-163) and simplifying yields

$$1 e^{\circ}(t) = 2; \int_0^{\infty} \{A_1(z) + W(z)W(z^{-1})A_2(z)\}P_{r,r}(z) dz$$

$$= \frac{W(z)[A_3(z^{-1}) + A_1(z^{-1})]}{W(z)[A_3(z) + A_1(z)]} \int_0^{\infty} dz$$

(11-176)

in which

$$A_1(z) = \int_0^{\infty} 3m[M_a(s) M_a(-s)P_{r,r}(s)] dm$$

(11-177)

$$A_2(z) = \int_0^{\infty} G(z, m)G(z^{-1}, m) dm$$

(11-178)
 $A_1(x) = f' G(x^{-1}, m)$
 $S' G(z^{-1}, m) \int m[Md(S)p, \dots, (s)] dm$
 (11-179)
 $A_1(\approx) = f' G(\approx^{-1},$
 $'G(2^{-1}, m) \int m[Ma(s)pr..(\$)] dm$
 (11-180)
 and (2) is given by Eq. (11-142).

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The Wiener-Kolmogoroff theory can now be applied to Eq. (11-176) so that an optimum $D(z)$ may be determined. The condition for minimizing $e^2(t)$ is determined by giving $W(z)$ a small variation $en(z)$. Therefore, when $W(z)$ and $W(z^{-1})$ are replaced by $W(z) + en(z)$ and $W(z^{-1}) + en(z^{-1})$, respectively, the mean-square error will increase to $e(t) + Ae^2(t)$. Then a necessary condition for $e^2(t)$ to be a minimum is

$$\frac{\partial e^2}{\partial W(z)} = 0$$

$$= 0$$

$$E=0$$

$$(11-181)$$

The minimization procedure bears a close resemblance to the minimization of the mean-square-sampled error. The essential steps are to divide (2) into two parts according to Eq. (11-143) and to divide $A_1(z)$ into $A_1(z)$ and $A_2(z)$ in the same manner. The optimum transfer function $W(z)$ when $e^2(t)$ is a minimum is readily obtained as

$$W(x) = [A_1(z) + A_2(z)] \cdot \frac{1}{p(z)} A_1(z)$$

$$+$$

$$(11-182)$$

in which $\{[A_1(z) + A_2(z)]/p(z) A_1(z)\} +$ denotes the part of $[A_1(z) + A_2(z)]/p(z) A_1(z)$ having singularities only inside the unit circle; that is,

$$A_1(z) + A_2(z)$$

$$p(z) A_1(z)$$

$$= \frac{A_1(z) A_2(z) + A_1(z)}{p(z) A_1(z)} + \frac{A_1(z)}{p(z) A_1(z)}$$

$$+$$

$$(2)$$

$$(11-183)$$

The details of the derivation of Eq. (11-182) are left to the reader as an exercise.

To illustrate the synthesis of sampled-data control systems by means of the modified z-transform technique, a single numerical example is given below.

Example 11-4

Assume that for the system of Fig. 11-17,

$$1 - e^{-ST}$$

$$G(s):$$

$$-$$

$$s(s + 1)$$

$$Ma(s) = 1$$

$$\frac{1}{T} = 0.2 \text{ sec.}$$

$$2$$

$$PT, T, (S)$$

$$=$$

$$-$$

$$S$$

$\Pr(s) = 0.1$ (white noise)

the signal and noise are uncorrelated so that $\Pr(s) \phi_{\eta\eta}(5) = 0$. The object is to design an optimum digital compensator so that the mean-square error of the system is at a minimum. The z-transform of $\Pr,7,(s)$ is

$$\Pr..(2) = \frac{-0.2013z}{1.221(z - 0.819)}$$

Therefore,
 $\Pr(2) = P(2) + \Pr(2) =$

$$0.1 \left(\frac{0.261}{1.221(z - 0.819)} + \frac{3.791}{1.221(z - 0.819)} \right)$$

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Dividing (2) into two parts according to Eq. (11-143), we have

$$\Phi\#(2) = 0.316 \left(\frac{0.261}{1.221(z - 0.819)} + \frac{3.791}{1.221(z - 0.819)} \right)$$

$$P(2) = 0.316 \left(\frac{0.261}{1.221(z - 0.819)} + \frac{3.791}{1.221(z - 0.819)} \right)$$

$$\left(\frac{0.261}{1.221(z - 0.819)} + \frac{3.791}{1.221(z - 0.819)} \right)$$

The modified z-transform of $G(s)$ is

$$G(z, m) = \frac{1}{(z - 1)e^{-0.2m}}$$

$$\left(\frac{1}{z - 1} \right) e^{-0.2m}$$

$$\text{Thus } A_z(\approx) = \int_0^1 G(z, m)G(z^{-1}, m) dm = -0.007(z + 0.305) (\approx + 3.274)$$

$$\left(\frac{1}{z - 1} \right) e^{-0.2m}$$

Dividing $A(z)$ into two parts, we have

$$A(z) = 0.0835 \left(\frac{0.305}{1.221(z - 0.819)} + \frac{3.274}{1.221(z - 0.819)} \right)$$

$$\left(\frac{0.305}{1.221(z - 0.819)} + \frac{3.274}{1.221(z - 0.819)} \right)$$

$$A_z(2) = -0.0835 \left(\frac{0.305}{1.221(z - 0.819)} + \frac{3.274}{1.221(z - 0.819)} \right)$$

$$\left(\frac{0.305}{1.221(z - 0.819)} + \frac{3.274}{1.221(z - 0.819)} \right)$$

(11-190)

(11-191)

The modified z-transform of $p_{r,r}(s)$ is found as follows:

1

--?

$$Z_m[p_{r,r}(s)] = p_{r,r}(\approx, m) = + 3m[p_{r,r}(s)] + + + 3m[p_{rr}(s)] -$$

-0.2m

-

T

$$p_{r,r}(0) = 5$$

e

+2

-0.2m

-0.2

2-1] (11-192)

Substituting Eq. (11-192) and $G(z, m)$ into Eq. (11-179), we have

$$A_s(z) = \int_0^1 G(z^{-1}, m) p_{r,r}(z, m) dm = 0.045(\approx + 2.465) (\approx^2 - 1.2\approx + 0.5)$$

0.819) (z

1.221)2

(11-193)

Therefore,

$A_1(z)$

(2) A (2)

$$-1.7(\approx + 2.465)(\approx^2 - 1.2\approx + 0.5)$$

=

-

$$(\approx - 0.819)(\approx - 3.74)(\approx + 3.274)$$

(11-194)

$A(z)$

$p_{rr}(z) A_2(\approx)$

55].

0.0875

=

(11-195)

-

0.819

It follows from Eq. (11-182) that

$W(\approx)$

=

32.2($\approx$

0.819)

(11-196)

$$(\approx - 0.261)(\approx + 0.305)$$

The transfer function of the desired digital compensator is given by

$D(\approx)$

=

$W()$

$$1 - W(\approx)G(\approx)$$

=

$$32.2(\approx 0.819)$$

$$(\approx + 2.45)(\approx - 2.41)$$

-

(11-197)

Although $D(\approx)$ represents an unstable controller, it still can be realized by a feedback-series pulse-data network using the techniques discussed in Sec. 9.2.

The physical realizability and stability of the system transfer function $M(z)$

are ensured when the part of the function in Eq. (11-182) is chosen with poles only inside the unit circle in the z-plane. However, according to Eq.

(11-197), $D(z)$ may still turn out to have poles outside the unit circle.

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12

NONLINEAR SAMPLED-DATA

CONTROL SYSTEMS

12.1 INTRODUCTION

Problems occurring in practice originally stimulated interest in the study of sampled-data systems containing nonlinear elements. As in systems with continuous signals, account must often be taken of physical imperfections such as saturation, backlash, dead zone, and hysteresis of components in sampled-data systems. Most physical systems can be considered linear only within a limited range of their operations. If a sampled-data system is designed according to the linear theories and methods discussed in the preceding chapters, the components used in the system must operate in linear fashions under various operating conditions. If under certain circumstances the components are driven into regions of nonlinear characteristics, linear design theory describes the system performance only approximately or quite often gives a completely erroneous prediction of it. Broadly speaking, a nonlinear system is one for which the principle of superposition is no longer valid. In linear systems mathematical analysis is characterized by the fact that if the separate application of inputs $r_1(t)$ and $r_2(t)$ produces responses $c_1(t)$ and $c_2(t)$ respectively, the simultaneous application of signals $r_1(t)$ and $r_2(t)$ will produce an output $c_1(t) + c_2(t)$. This is why in linear systems, system response need be analyzed only to the deterministic type of signals such as the step function, ramp function, acceleration

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function, and the sine function. These signals can then be combined to approximate signals of almost any kind. Unfortunately, this unique property of superposition does not carry over to nonlinear systems. This fact is also a basic reason why nonlinear differential equations (nonlinear difference equations for sampled-data systems) are more difficult to solve and thus require special attention. At present, no definite set of mathematical tools exists to solve a general class of nonlinear systems with either sampled or continuous data. Furthermore, when sampling and nonlinearity are present in the same system, new phenomena arise which can hardly be described by the theories and techniques of linear sampled-data systems or by that of nonlinear system without sampling. The interaction between sampling and nonlinearity brings about a whole class of complicated and unexpected problems.

Although nonlinearities in feedback control systems generally are due to shortcomings of physical components, in one class of systems nonlinear elements are intentionally introduced in order to improve system performance. In systems with continuous-data, the on-off or relay-type servot such as the automobile voltage regulators and on-off furnace controls are common examples of this class of nonlinear system. The concept of relay control is also extended to sampled-data systems. Actually, with the addition of the sampling switch, nonlinear control of sampled-data systems becomes more versatile; several investigators have recently suggested the pulse-width control which is a modified version of the ordinary relay control in which continuous control is exercised over the duration of the full positive or negative correction applied by the relay to the controlled system

during each sampling period.

Two well-known methods, which have proved both useful and popular in the study of continuous-data nonlinear feedback control systems, are the describing function and the phase-plane methods.

† H. L. Hazen, "Theory of Servomechanism," J. Franklin Inst., September 1934;
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Also, the first published studies of nonlinear control systems with sampled-data are the extensions of these two above mentioned methods. For systems which have low-pass frequency response characteristics, the possible unstable behavior of the systems may be assumed to consist of periodic, quasi-sinusoidal oscillations approximated by their first harmonic component. The method is essentially a steady-state analysis; it does not give any information on the transient behavior of the system. On the other hand, the phase method may be regarded as a direct study of the system difference equations. The phase plane diagram gives indication of both the transient and the steady-state behavior of a system in the time domain. However, as is well known, the major limitations of the phase plane technique are that only second-order systems (or higher-order systems which can be approximated by second-order systems) can be effectively studied by the method, since for high-order systems the phase space becomes difficult to visualize, and it cannot be extended to systems with sinusoidal inputs.

12.2 ANALYSIS OF NONLINEAR SAMPLED-DATA SYSTEMS

BY STATE TRANSITION TECHNIQUE

The state transition technique can also be used for the analysis and synthesis of sampled-data control systems with nonlinear elements. Essentially the technique makes use of the variable gain concept described in Sec. 9.9. Consider the block diagram of the nonlinear system shown in Fig. 12-1. We shall use this basic configuration to describe the development of the analysis procedure of the state transition technique.

As shown in Fig. 12-1. the input to the nonlinear element N is $h(t)$ and the output is $m(t)$. The nonlinearity may be represented-

$r(t)$
 $h(t)$
 $m(t)$
 $x_1(t) = c(t)$
z.o.h.

N
 $G_1(s)$
Nonlinear
Linear
element
controlled
process

Fig. 12-1 Block diagram of sampled-data control system with nonlinear element.

† R. E. Kalman, "Nonlinear Aspects of Sampled-Data Control Systems," Proc. of the Symposium on Nonlinear Circuit Analysis, 6, Brooklyn Polytechnic Institute, New York, 1956.
 C. K. Chow, "Contactor Servomechanisms Employing Sampled-Data," Trans. A. I. E. E., 73, Part 2, 1954, pp. 51-64.

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Variable gain

amplifier

$K(10)$

$r(10)$

$h(t)$

$m(1)$

$M(s)$

$x_1(10)$

$x()$

1

Transition flowi

graph of linear

lements

$X_1(s) = C(s)$

Fig. 12-2 Transition flow graph of the nonlinear sampled-data control system in Fig. 12-1.

ed by a variable gain amplifier with gain $K(kT)=K_1$ during the sampling period $kT < t \leq (k+1)T$. K is defined as

k

Kk

=

$m(kT+)$

$h(kT+)$

(12-1)

and the value of K depends upon the characteristic of the nonlinear element N and will have different values during different sampling periods. Based on the above arrangement, the state transition flow graph of the system in Fig. 12-1, is constructed in Fig. 12-2, and the transition equations are written in the following vector-matrix form:

$V[(t - t_0), K(t_0)]x_1(t_0) + \star[(t - t_0), K(t_0)]r(t_0)$

$x(t)$

=

-

When $t = (k+1)T$ and $t_0 = kT$, Eq. (12-2) becomes

(12-2)

(12-3)

$x[(k+1)T] = \mathcal{Y}(T, Kk)x_1(kT) + \star(T, Kx) r(kT)$

The values of K in the last equation are given by Eq. (12-1), in which $h(kT+)$ is determined from

$h(kT+) = r(kT) - x_1(kT)$

-

(12-4)

and $m(kT+)$ is determined from the nonlinear characteristics once $h(kT+)$ is known. Once the values of K during each sampling period are determined, Eq. (12-3) gives the solution of all the state variables at the sampling instants.

The analysis procedure is best illustrated by a numerical example.

Example 12-1

=

Consider that in the nonlinear sampled-data system of Fig. 12-1, $G_1(s) = 1/s(s+1)$; and the nonlinearity is a relay with dead zone and with its input-output characteristics described by

$m(t) =$

+1

$$\begin{aligned} &1 \\ &h(t) \ 0.1 \ D \\ &\geq \\ &= \\ &h(t) \ -0.1 = -D \\ &(12-5) \\ &0 \ -0.1 < h(t) < 0.1 \end{aligned}$$

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The sampling period is one second. Determine the output response of the system when the input $r(t)$ is zero and the initial conditions are $x_1(0) = 0.138$ and $x_2(0) = 0.762$. The state transition flow graph of the system is drawn in Fig. 12-3. Note that this flow graph is identical to that of Fig. 9-42, which is for a linear system with digital controller. As was mentioned previously, the variable gain concept may be applied to a general class of elements, such as nonlinear elements, time-varying elements, linear digital controllers, etc. Therefore, the transition equations of the nonlinear system are identical to those of Eqs. (9-240) and (9-241); that is,

$$\begin{aligned} &K(10) \\ &1 \\ &r(10) \\ &e^{(*)} \\ &h(t) \ H(s) \\ &8 \times 2(10) \\ &(x_1(10) \\ &8 \times 2(10+) \\ &0x_1 \ (10+) \\ &s-1 \\ &X_2(s) \\ &X_1(s) \\ &-1 \end{aligned}$$

Fig. 12-3 State transition flow graph of the nonlinear system in Example 12-1.

$$x_1[(k+1)T] = [1 - K_1(T - 1 + e^{-T})]x_1(kT) + (1 - e^{-1})x_2(kT)$$

$$\begin{aligned} &+ K(T-1+e^{-T})r(kT) \\ x_2[(k+1)T] &= -Kx_1(1 - e^{-TM})x_1(kT) + e^{-1}x_2(kT) \end{aligned}$$

$$\begin{aligned} &+ Kx_1(1-e)r(kT) \\ \text{For } T &= 1 \text{ second and } r(kT) = 0, \text{ Eqs. (12-6) and (12-7) read} \end{aligned}$$

$$x_1[(k+1)T] = (1 - 0.368K)x_1(kT) + 0.632x_2(kT)$$

$$x_2[(k+1)T] = -0.632Kx_1(kT) + 0.368x_2(kT)$$

(12-6)
(12-7)
(12-8)
(12-9)
For $k = 0$, $h(0+) = -x_1(0) = 0.138$, thus $m(0+) = 1$, and from Eq. (12-1), $K_0 = 7.25$. Substitution of these quantities and the initial conditions into Eq. (12-8) and (12-9) yields

$$x_1(T) = (1 - 0.368K)x_1(0) + 0.632x_2(0) = 0.252$$

$$\begin{aligned} &(12-10) \\ x_2(T) &= \\ &0.632Kx_1(0) + 0.368x_2(0) \\ &= -0.352 \end{aligned}$$

$$\begin{aligned} &(12-11) \\ \text{Therefore,} \\ h(T+) &= -x_1(T) \end{aligned}$$

$$= -0.252$$

-1 and $K_1 = 3.97$. Substitution of k

1 and from which, we have $m(T+)$ the values of K_1 , $x_1(T)$ and $x_2(T)$ into Eqs. (12-8) and (12-9) yields $x_1(2T) = -0.338$ and $x_2(2T) = -0.762$. Following the same procedure, the responses x_1 and x_2 can be calculated through any desired number of sampling periods. The results for the first eight sampling periods are computed and tabulated in the following:

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```

0 1 2 3 4 5 6 7 8
k
(kT)
x2(kT)
0.138 0.252 -0.338 -0.452
0.762 -0.352
ho(kT+) -0.138 -0.252
0.138 0.252 -0.338 -0.452 0.138
-0.762
0.352
0.762 -0.352 -0.762 0.352 0.762
0.333
0.452
-0.138 -0.252 0.333
0.452 -0.138
m(kT+) -1
-1
+1
+1
-1
-1
+1
+1
-1
Kk
7.25
3.97
2.96
2.21
7.25
3.97
2.96
2.21
7.25
m(t)
0.4
0.2
c(1) 0
AAAAAS
AAAAAAA
-0.25
-0.4
-0.6
227
24T
1

```

Fig. 12-4 Output response and relay output signal of nonlinear (relay type with dead zone) sampled-data system shown in Fig. 12-1. $G(s) = 1/s(s + 1)$, relay dead zone $D = 0.1$; initial conditions $(0) = 0.138$, $x(0) = 0.762$. The output response is unstable, and the period of self-sustained oscillation is $T_e = 4T$. The output signal $m(t)$ of the relay and the system output response are plotted in Fig. 12-4. For the given initial conditions, the nonlinear sampled-data system has a periodic steady-state solution of period $4T$. The relay output has two consecutive corrections in the positive direction and two in the negative direction.

It is both interesting and instructive to find the plots of the system output $c(t)$ due to different initial conditions. When the initial conditions of $x_1(0) = -0.1$ and $x_2(0) = 0.6$ are assumed, the nonlinear system has an autonomous periodic response of period $6T$. This is equivalent to saying that the difference equations that describe the nonlinear system have a periodic solution of period $6T$. Similarly, by assigning various other combinations of $x_1(0)$ and $x_2(0)$ as initial conditions of the system, other possible modes of sustained oscillations are discovered. When $x_1(0) = 0.1$, and $x_2(0) = 0.4$, the period of oscillation of $c(t)$ is $4T$, and the relay output signal has one correction in the positive direction and one in the negative direction during each cycle. For $x_1(0) = -0.4$ and $x_2(0) = 1.2$, the sustained oscillation in $c(t)$ has a period of $5T$. We shall see in a later section that with $T = 1$ second and relay dead zone $D = 0.1$, the modes mentioned above are the only possible

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 $m(t)$
 0.4
 0.2
 27
 HAAAAAA
 -0.2
 -0.4
 87
 (a)
 24T
 $m(t)$
 +1
 $C(1)$
 1.0t
 0.8
 0.6
 0.4
 Am
 0.2
 27 4T
 8T 10T 12T
 -0.2
 6T
 -0.4
 -0.6
 14T
 187 207 22T 24T
 167
 (b)

Fig. 12-5 Output response and relay output signal of nonlinear (relay type with dead zone) sampled-data system shown in Fig. 12-1. $G_1(s) = 1/s(s + 1)$, relay dead zone $D = 0.1$; initial conditions: (a) $x_1(0) = 0.1$, $x_2(0) = 0.4$; self-sustained oscillation, period $T = 4T$. (b) $x_1(0) = 0.4$, $x_2(0) = 1.2$; self-sustained oscillation, period $T = 5T$

=
-

modes of self-sustained oscillations that may occur in this particular nonlinear system with sampled-data. The waveforms of $m(t)$ and $c(t)$ corresponding to these various initial conditions are plotted in Fig. 12-5. Stable operation of the system can be obtained by increasing the size of the relay dead zone. For instance, if the relay dead zone is increased to 0.6, we can show that the initial conditions used above will all lead to stable responses. Figure 12-6 shows two typical stable responses and the corresponding relay output signals for the assumed initial conditions.

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$m(1)$
+1
 $c(1)$
1.0
0.8
0.6
0.4
0.2
187
24T
AAAAA
-0.2
-0.4
- 0.6
- 0.8
127
2T 4T
8T 0T
147 16T 20T 22T 26T 28T 1
-1.0†
(c)

Fig. 12-5 (c) Output response and relay output signal of non-linear (relay type with dead zone) sampled-data system shown in Fig. 12-1. $G(s) = 1/s(s + 1)$, relay dead zone $D = 0.1$; initial conditions: $x_1(0) = 0.1$, $x_2(0) = 0.6$, self-sustained oscillation period $T = 6T$.

=

The state transition technique can be used to synthesize certain restricted types of nonlinear sampled-data control systems. In this case, the required digital controller and the nonlinear element are represented by a variable gain amplifier. However, because of the nonlinear aspect of the system the state transition technique cannot be relied upon to provide a design solution for all nonlinear systems.

12.3 THE PHASE PLANE METHOD

Phase plane method is a graphical technique of solving linear and nonlinear differential equations. However, phase plane method has been applied to control systems predominantly in the studies of nonlinear systems simply because the Laplace transform method of solving linear differential equations is so convenient and straightforward. J. T. Tou, B. Vadhanaphuti, op. cit.

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$m(t)$
0
1.5
1.0
 $c(t)$

0.5
 3T
 6T 7T
 0
 T
 2T
 4T 5T
 8T 9T 10T 11T 12T

- 0.5
 -1.0

m(t)

1.0

1.0

0.5

c(t)

0

(a)

T 2T 3T 4T 5T 6T 7T 8T 9T 10T 11T 12T

(b)

t

Fig. 12-6 Stable output responses of nonlinear (relay type with dead zone) sampled-data system shown in Fig. 12-1. $G_1(s) = 1/s(s + 1)$, (a) $x_1(0) = 1.5$, $x_2(0) = 0.5$ (b) $x_1(0) = 0.6$, $x_2(0) = 0.4$.

forward that there seems to be no advantage in applying the graphical method to linear systems.

If a second-order system is characterized by the dependent variable $c(t)$ and its first derivative with respect to time $dc(t)/dt$, the phase plane diagram of the system is a plot of $dc(t)/dt$ as a function of c . At any time t_1 , the coordinate $[c(t_1), dc(t)/dt]$ in the phase plane determines the "state" of the system at t_1 . As time is varied from 0 to ∞ , the representing point (c, c') will trace out a continuous path which indicates the complete history of the system in regard to c and c' at all times; the path itself is called the phase plane trajectory. For instance, in a servo system the phase c' represents the first derivative of $c(t)$ with respect to time.

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$V=C'$

14

12

$\phi(1)=0$

$C=R=1$

Maximum

overshoot

Maximum

overshoot

(a)

13

(b)

Fig. 12-7 Phase plane diagram and the corresponding step response of a second-order servo system.

plane diagram is usually a plot of the output velocity c' as a function of the output displacement c . In Fig. 12-7 the typical phase plane diagram and the corresponding output response of a second-order servo system with unit step function input are shown. The arrows on the phase plane trajectory indicate the increase in time. For an electrical network a plot of current i as a function of electric charge q also forms a phase plane diagram.

The phase plane concept may also be extended to higher order

systems or differential equations. However, the representation of the state of the system for higher order systems requires higher-dimen-

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sional plots. This is usually the main difficulty in applying phase plane technique to systems with order higher than two, since a multidimensional "phase space" diagram is very difficult to visualize and construct. In addition to the above-mentioned limitation, the phase plane technique is normally used to study only transient and steady-state performances of systems subject to only step function and linear time function inputs and/or with initial conditions. For continuous-data systems, we know that the phase plane technique does not solve problems involving systems with sinusoidally excited systems. However, we shall show in this chapter that it is possible to apply the phase plane method to sampled-data systems with sine-wave inputs.

Phase Plane Trajectories of Linear Systems with Continuous-Data

At this point, we shall briefly review the construction of phase plane diagrams for linear second-order systems with continuous-data. Consider that the open-loop transfer function of a linear second-order system with unity-feedback is given by

$$T = \frac{G(s)}{C(s)} = \frac{K_m}{K} \frac{E(s)}{s^2(1 + ST)S(1 + ST)} \quad (12-13)$$

where J is inertia/friction, and $K = K_m/f$. The differential equation that describes the relation between the input and the output variables of the closed-loop system may be written as

$$K_r(t) = T \frac{d^2c(t)}{dt^2} + K_c(t) \frac{dc(t)}{dt}$$

$$+ K_c(t) \frac{dc(t)}{dt} \quad (12-14)$$

Making the substitution of

$$\frac{dc}{dt} = v \quad (12-15)$$

$$\frac{dv}{dt} = \frac{d^2c}{dt^2}$$

and

$$v' = \frac{dv}{dt}$$

$$\frac{dv}{dt} = \frac{d^2c}{dt^2}$$

$$X = v \quad (12-16)$$

$\frac{dv}{dc} = TV$
 into Eq. (12-14) yields
 where r , v , and c , are all functions of time. Solving for $\frac{dv}{dc}$ in Eq.

$\frac{dv}{dc} = -\frac{1}{K(rv)}$
 (12-17), we have

$\frac{dv}{dc} = -\frac{1}{K(rv)}$
 (12-18)
 Eq. (12-18) represents an analytic expression for the slope of a trajectory in the c - v plane. If we consider that the reference input to the system is a unit step function, one trajectory exists in the

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Isocline

$V = C'I$
 (slope = $-K$)

$a = 0$
 Isocline
 (slope = ∞)

Isocline
 (slope < 0)

$v = c'$
 $C = R = 1$

(a)
 $c = R = 1$
 (b)
 $\theta = -1/T$
 Isocline (slope > 0)

$\theta < -1/T$
 Phase plane trajectory
 Isocline (slope = 0)

$a = \frac{dv}{dc} = \infty$
 $\frac{dv}{dc} = 0$
 Phase plane trajectory

$v = c'$
 $\sigma = 1/T$

$\alpha = 0$
 C

$\alpha = \infty$

Phase plane trajectory
 $c = R = 1$

(c)

Fig. 12-8 Phase plane trajectories of a second-order system with a unit-step function input and zero initial conditions. (a) $f > 0$, stable system; (b) $f = 0$, sustained oscillation; (c) $f < 0$, unstable system.

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Characteristic Eq.

root locations

X

X

Ajw

s-plane

Phase plane

trajectory

审

Transient response

れ

4jw

s-plane

2

0

s-plane

(a)

(b)

c(t) 3

c(t)

(c)

Fig. 12-9 Phase plane trajectories of a second-order linear system. phase plane to each set of initial conditions $c(0)$, $v(0)$. Frequently, the phase plane trajectories are sketched by finding the lines on which the trajectories have the same slope dv/dc ; these lines are called the "isoclines" of the phase plane trajectories. For instance, by setting dv/dc to zero in Eq. (12-18), we have

-

$$K(r - c) - v = 0$$

(12-19)

which represents a straight line passing through the real axis in the phase plane at $c = r = 1$ with a slope of $-K$. Therefore, any phase

† A. A. Andronow, and C. E. Chaikin, Theory of Oscillations, trans. Lefschetz, S (Princeton, N. J.: Princeton University Press, 1949).

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Characteristic Eq.

root locations

s-plane

Phase plane

trajectory

AC'

Transient response

F&M

(d)

jw

s-plane

(e)

c(t)

Ajw

s-plane

F

C'

OI

0

3

(f)

2

1

Fig. 12-9 Phase plane trajectories of a second-order linear system.

plane trajectory that intersects this line must have a slope of zero at the intersecting point. Similarly, for any constant slope σ of the trajectory in the phase plane, Eq. (12-18) gives

$$\frac{dv}{dc} = \frac{1}{1 + K(r - c)}$$

After simplification, Eq. (12-20) reads

$$\frac{dv}{dc} = \frac{K(r - c)}{1 + \sigma\tau} \quad (12-20)$$

$$(12-21)$$

Equation (12-21) represents the equation of the isoclines in the phase plane for this particular system. All phase plane trajectories that in-

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$$= 0, \tau_0$$

intersect the same isocline, have the same slope σ at the intersections. Figure 12-8a shows the isoclines for several significant values of σ and the phase plane trajectory when the input is a unit-step function and has zero initial conditions ($c = 0$).

The arrows on the trajectory indicate the increase in time. From this phase plane trajectory, the transient response of the system is clearly oscillatory and finally the response reaches a steady-state value of $c = 1$. If the viscous friction coefficient of the system is made equal to zero, the differential equation of the system becomes

$$K_m(r - c) = \frac{d^2c}{dt^2} \quad (12-22)$$

Then, the isoclines in the phase plane are represented by

$$v = \frac{K_m(r - c)}{J_0} \quad (12-23)$$

Equation (12-23) represents straight lines with slopes that are equal to $-K_m/J_0$ and pass through the point ($c = 1, v = 0$) in the phase plane. For the same value of a but with opposite signs, the isoclines are symmetrical with respect to the isocline for $\sigma = 0$ (the $c = R = 1$ line). The corresponding phase plane trajectory with zero initial conditions is elliptical in form as shown in Fig. 12-8b. For a negative viscous friction coefficient, the system is unstable. Clearly the phase plane trajectory will circle outward and eventually go to infinity (Fig. 12-8c).

Singular Points

The points ($c = 1, v = 0$) in the phase plane diagrams of Fig. 12-18 are called the singular points of the phase plane trajectories. A singular point may generally be classified as a focus point, a saddle point, or a vortex. The classification depends entirely on the variations of the coefficients of the differential equation or the roots of the characteristic equation. For instance, the singular point in Fig. 12-18b is a vortex; and the singular point in Fig. 12-18c is an unstable focus point.

If in Eq. (12-14) $r(t) = 0$,

$$J \frac{dc}{dt} + f \frac{dv}{dt} + K_m c = 0$$

dc
 $dt^2 \quad dt$
 (12-24)
 The origin in the phase plane diagram becomes the only singular point. The relationship among the coefficients and the roots of Eq. (12-24) for each type of singularity is given in Table 12-1.

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$K_m > 0$
 $f > 0$
 $f < 2/JK_m$
 TABLE 12-1
 Two complex conjugate roots
 with negative real parts
 Stable
 focus
 $K_m > 0$
 $f < 0$
 $f < 2/JK_m$
 Two complex conjugate roots
 Unstable
 with positive real parts
 focus
 $K_m > 0$
 $f = 0$
 Two imaginary roots
 Vortex
 $K_m > 0$
 $f > 0$
 $f > 2/JK$
 Two negative real roots
 Stable
 node
 $K_m > 0$
 K_m
 < 0
 $f < 0$
 $f = 0$
 $f > /JK_m$
 Two positive real roots
 Unstable
 node
 Two real roots with opposite
 signs
 Saddle
 point

The phase plane trajectories, location of roots of the characteristic equation in the s -plane, and the corresponding transient responses of the six cases given in Table 12-1 are illustrated in Fig. 12-9.

Phase Plane Trajectories of Relay-Type Control Systems with Sampled-Data

The relay-type control system with sampled-data in Example 12-1 will now be studied by means of the phase plane technique. The analysis is limited to second-order systems only. Therefore, we consider the transfer function of the controlled process as

$$G_1(s) = \frac{1}{s(s+1)} \quad (12-25)$$

$$M(s) = \frac{x_2(t_0)}{s^2} = \frac{v_0}{s^2}$$

$$x_1(1\%) = C_0$$

$$\frac{1}{s^2} x_2(10+) = \frac{1}{s^2} EF$$

$$\ddot{x}_2(s) = V(s)$$

$$\frac{1}{s} X_1(s) = \frac{C(s)}{s}$$

Fig. 12-10 State transition flow graph of $G_1(s)$

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 Although the time constant T and the gain K in Eq. (12-25) are assumed to be unity, extension of the following analysis can be made easily to any second-order system in which T and K are nonunity. With the notations of Fig. 12-1, the transition flow graph of the linear controlled process is shown in Fig. 12-10. The input signal is $m(t_0)$ which is the output of the relay in the nonlinear system. The state transition equations for the open-loop system are:

F. J. Mullin, E. I. Jury. "Phase-Plane Approach to Relay Sampled-Data Systems," Trans. A. I. E. E., Part 2, January 1959, pp. 517-524.
 W. Altar, C. W. Helstrom, "Phase Plane Representation of Sampling Servomechanisms," Research Report R-94410-14-B, Westinghouse Research Lab., Pittsburgh, Pa September 1953.

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Although the time constant T and the gain K in Eq. (12-25) are assumed to be unity, extension of the following analysis can be made easily to any second-order system in which T and K are nonunity. With the notations of Fig. 12-1, the transition flow graph of the linear controlled process is shown in Fig. 12-10. The input signal is $m(t_0)$ which is the output of the relay in the nonlinear system. The state transition equations for the open-loop system are:

$$x(t) = c(t) = [(t - t_0) - 1 + e^{-(t-t_0)}] m(t_0) + x_1(t_0)$$

$$x_2(t) = v(t) = e^{-(t-t_0)} x_2(t_0) + [1 - e^{-(t-t_0)}] m(t_0)$$

(12-26)
 (12-27)
 where $x(t_0) = c_0$ and $x_2(t_0) = v_0$. Equation (12-26) can also be written in terms of $v(t)$. Therefore, Solving for t to in Eq. (12-27) yields

$$x_1(t) = c(t) = (t - t_0) m(t_0) + c_0 + v_0 - v(t)$$

$$= - \ln \left[\frac{m(t_0) - v}{m(t_0) - v_0} \right]$$

(12-29)
 where for simplicity $v(t)$ is denoted by v . Substituting Eq. (12-29) into Eq. (12-28), we have

$$\ln \frac{c - c_0 + v_0 - v}{m(t_0) - v} = - \ln m(t_0)$$

$$\frac{m(t_0) v_0 - m(t_0) v}{(12-30)}$$

$$\begin{aligned} m &= +1 \\ m &= -1 \\ m &= +1 \\ &-1 \end{aligned}$$

$$G(s) = 1/s(s + 1)$$

Fig. 12-11 Phase plane trajectories for relay-type system with $m=+1$ and -1 .
C

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or

$$C_0 = - (v - v_0) - m(t_0) \ln$$

$$\begin{aligned} &[; \\ &\frac{m(t_0)}{m(t_0)} v_0. \end{aligned}$$

(12-31)

Equation (12-31) describes the phase plane trajectory of the relay-type control system between the states of (c_0, v_0) and (c, v) for a given $m(t_0)$. The phase plane trajectories for $m(t) = +1$ and -1 are plotted in Fig. 12-11 for various initial conditions. The corresponding values of v according to Eq. (12-31) when c is varied from 0 to ∞ with $m(t_0) = +1$ and zero initial conditions are tabulated in Table 12-2. The data given in Table 12-2 correspond to the $m(t^*) = +1$ trajectory which passes through the origin in Fig. 12-11. For non-zero initial conditions the trajectory must pass through the point (c_0, v_0) in the phase plane. The phase plane trajectory for $m(t)$

TABLE 12-2

V
C
v
C
v
C
0.1
0.0054
1.05
1.95
-0.2
0.018
0.2
0.023
1.10
1.20
-0.4
0.064
0.3
0.057
1.20
0.41
-0.5
0.095
0.4

0.112
 1.28
 0
 -0.7
 0.170
 0.5
 0.194
 1.30
 -0.10
 -0.8
 0.212
 0.6
 0.317
 1.50
 -0.80
 -0.9
 0.258
 0.7
 0.503
 - 1.0
 0.307
 0.8
 0.810
 -1.2
 0.412
 0.84
 1.00
 - 1.4
 0.525
 0.90
 1.40
 -1.6
 0.644
 0.95
 2.05
 - 1.8
 0.770
 0.98
 2.93
 - 2.0
 0.90
 1.00
 ∞

-3.0
 1.61
 -1 has the same shape as that of the trajectory of $m(t) = +1$ except that it is inverted.

When $m(t) = 0$, the phase plane trajectory is reduced to a simple straight line described by

$$C - C_0 = -(v - V_0) \quad (12-32)$$

Equation (12-31) is actually the equation of the phase plane trajectory for a second-order relay-type system with continuous-data as well as with sampled-data, since up to this point no mention is made of the fact that the derivation of Eq. (12-31) is based on sampled-data only. As is apparent, Eq. (12-31) is valid as long as $m(t_0)$ is constant or zero. The difference between the phase plane trajectory of a sampled-data system and that of a continuous-data system appears to be in the switching time of the relay. For a continuous-

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data system, the switching of the relay depends strictly on the magnitude of the actuating signal $e(t)$. But for a sampled-data system, since $e^*(t)$ is a discrete function of time, the switching not only depends on the magnitude of $e^*(t)$ or $e(t)$ at the sampling instants but can occur only at the sampling instants. The construction of the entire phase-plane trajectory of a relay-type system with sampled data may appear to be difficult, since time does not appear explicitly on the phase plane trajectory. Fortunately, a set of switching lines can be derived in the phase plane; the intersections of the phase plane trajectory and the switching lines indicate the possibility of relay switchings. The details are described as follows:

For a sampled-data system, we let t to T , and the state transition takes place from C_n to C_{n+1} and v_n to U_{n+1} , respectively, where c_n denotes the value of $c(t)$ at $t = nT$ and v_n denotes the value of $v(t)$ at $t = nT$. C_{n+1} and v_{n+1} are the values of $c(t)$ and $v(t)$ at $t = (n+1)T$, respectively. Substituting these quantities into Eq. (12-31) yields.

$$C_{n+1} - C_n = - (V_{n+1} - V_n) - M_2 \ln (M_n - U_{n+1})$$

m_n
 M_n

$-$

V_n

(12-33)

where m_n denotes the fact that $m(t)$ can change values at the sampling instants only. Since the trajectory in any sampling interval starts at c_n , U_n and ends at C_{n+1} , U_{n+1} in the phase plane, given c_n , U_n , we must determine exactly where the point C_{n+1} , U_{n+1} lies on the trajectory. The time for the representing point on the trajectory to travel from (c_n, U_n) to (C_{n+1}, U_{n+1}) is equal to one sampling period T . The relationship between C_n , U_n and C_{n+1} , U_{n+1} can be obtained from Eq. (12-28); that is

$$C_{n+1} - C_n = - (U_{n+1} - U_n) - M_2 \ln (M_n - U_{n+1})$$

(12-34)

Equation (12-34) gives the sum of the coordinates of the switching points as a function of the initial conditions (C_n, v_n) , the relay switching function, and the sampling period T . In general, m , and T are known quantities, so that for given c_n and v_n , the corresponding values of C_{n+1} and U_{n+1} are determined from Eq. (12-34). However, to solve for C_{n+1} and U_{n+1} , another independent relation is desired. Since C_{n+1} , U_{n+1} in Eq. (12-34) must also lie on the phase plane trajectory defined by Eq. (12-33), the exact location of the point (C_{n+1}, U_{n+1}) in the phase plane is determined from the intersection of the line described by Eq. (12-34) and the phase plane trajectory passing through (c_n, v_n) .

Let us assume that c and v are such that their sum is A . Hence the point (C_n, v_n) lies on the line $c_n + U_n = A$ in the phase plane. The point (C_{n+1}, U_{n+1}) is found on the line which is described by

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or

00

or

Slope = -1

Switching lines

$M_{n+1} = +1$

$m = +1$

$(Cov_2)'$

$\lambda(C_2 + 2 + n + 2)$

$+ (C_2 + 11 V_{n+1})$

M_n

$(C_2 + 1, V_{n+1})$.

TT

$C + V = A + 2T$

TTT

$C_0 + v = A - 3T \quad C_2 + v = A - T \quad C_2 + v_2 = A + T$

$C + v = A - 2T \quad C_2 + v_{12} = A$

Fig. 12-12 Determination of C_{n+1} , $n+1$, using the equation

$C_{n+1} + 0_{n+1} = C_n + U_n + m_n T$ and the phase plane trajectory.

$C_{n+1} + U_{n+1} = A T$ for

$M_n = +1$

$C_{n+1} + U_{n+1} =$

A

$-$

T for

$m_n = 1$

$C_{n+1} + U_{n+1} =$

A

for

$m_n =$

0

(12-35)

(12-36)

(12-37)

$=$

$=$

By referring to Fig. 12-12 and assuming that at $t = nT$, we can describe the state of the system by the point (C_n, v_n) in the phase plane, a set of straight lines with slopes equal to -1 and equations of $C_n + U_n = A$, $C_n + U_n = A + T$, $C_n + U_n = A + 2T$, $\dots$, $C_n + V_n = A$

$-T$, $C_n + U_n = A - 2T$, $\dots$, are drawn in the phase plane. The line of $c_n + v_n = mA$ actually passes through the point (c_n, v_n) . Now let us determine the switching points (C_{n+1}, U_{n+1}) when m_n takes on

values of $+1$, -1 , and 0 . When m_{n+1} , the operating point in the phase plane will move from (c_n, v_n) along one of the $m_2 = +1$

trajectories shown in Fig. 12-11 until it reaches a point where the m_{n+1} trajectory intersects the line $c_n + V = A + T$; this inter-

section gives the point (C_{n+1}, U_{n+1}) . At this time $t = (n + 1)T$. If m_{n+1} is again equal to $+1$, the operating point continues to move

on the same trajectory until it reaches the line $c_n + U_n = A + 2T$, and the intersection point gives (C_{n+2}, U_{n+2}) . However, at $t = (n +$

$1)T$, if $M_{n+1} = -1$, the operating point will move from (C_{n+1}, U_{n+1}) along a $m_1 = -1$ trajectory until it reaches the $c_n + U_n = A$ line

and the intersecting point is (C_{n+2}, U_{n+2}) . If originally at $t = nT$, m_n

m_n

V_n

$=$

$:$

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$=$

0 , indicating that the relay has a dead zone, the switching line given by Eq. (12-34) becomes

m_n

$$C_{n+1} + U_{n+1} = C_n + V_n$$

=

(12-38)

which is of a form identical to that of the phase plane trajectory when $m_2 = 0$ (Eq. 12-32). Therefore, when $m_a = 0$, the switching lines and the phase plane trajectories are identical and are straight lines with slope of -1. In Fig. 12-12 when $m_2 = 0$, the operating point will move from (C_n, v_n) along the line $c_n + U_n$. However, since time does not appear explicitly on the phase plane trajectory, the location of the (C_{n+1}, U_{n+1}) point on the $c_n + U_n$ line is uncertain. To find the switching point, we must turn to Eq. (12-27). Substituting $m_n = 0$ into Eq. (12-27) yields

$$U_{n+1} = e^{-T} U_n$$

$$V_n =$$

=

(12-39)

and since T is known, given U_n , U_{n+1} is readily determined from Eq. (12-39). The use of the phase plane method for determining the behavior of relay-type sampled-data control systems is illustrated by the following examples.

Example 12-2

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For the relay-type sampled-data control system of Fig. 12-1, let $G(s) = 1/s(s+1)$ and assuming that the relay is ideal. The sampling period is one second. Plot the phase plane diagram of the system for $r(t) = 1.0$ with $C_0 = 0, V_0 = 0$.

$$m_2 + 1 = -1$$

$$(C_1, V_1)$$

$$(C_2, V_2)$$

$$(C_6, V_6)$$

$$+1$$

Limit cycle

2

$$(C_3, V_3)$$

$$(C_7, V_7)$$

$$(C_{va})$$

$$(C_0, V_0)$$

$$+1$$

Switching

lines

0

T

$$C_2 + V = -T$$

$$C_0 + 1 = 0$$

$$C_2 + V = T \quad C_{+12} = 27$$

$$R = 1.0$$

Fig. 12-13 Phase plane diagram of the relay-type sampled-data system in Example 12-2.

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The first step in the construction of the phase plane diagram involves the drawing of the set of switching lines as shown in Fig. 12-13. The reference switching line, which is described by the equation $C_0 + V = 0$, has a slope of -1, and passes through the origin in the phase plane. All other switching lines have the same slope and are spaced at a horizontal distance of integral multiples of T from the $C_0 + V = 0$ line. At $t = 0$, the error signal is +1; hence $m_0 = +1$. The operating point will move from the origin along one of the $+1.0$ trajectories shown in Fig. 12-11 until the trajectory intersects the switching line $C_n + V_n = T$ and the intersection determines the (C_1, V_1) point.

Since c_1 is less than one, e , is positive, and $m_1 = +1$. The operating point will continue to move along the same trajectory until it intersects the next switching line $c_n + v_n = 2T$ and the intersection point gives (c_2, v_2) . Now C_2 is greater than the magnitude of the reference input; hence e , is negative, and m_2

1. The operating point must now be switched to and must move along the -1 trajectory that passes through the (c_2, v_2) point until the trajectory intersects the line $c + v_n T$ and the point of intersection corresponds to (C_3, v_3) . As is apparent, c_3 is still greater than unity; therefore, $m_3 = -1$ and the operating point will continue to move along the -1 trajectory until it intersects the switching line $\dot{c}_n + v_n 0$, and at the intersection $m_1 = +1$ the operating point is switched to a $+1$ trajectory. From Fig. 12-13 we can see that as the above-mentioned switching process continues, the system will eventually become unstable and exhibit a limit cycle. The period of the limit cycle shown in Fig. 12-13 is 4 seconds, and the magnitude of the limit cycle is approximately 0.85. The output response and the relay switching function $m(t)$ of the system are shown in Fig. 12-14.

=

=

$m(1) 0$

-11

1.5-

$c(1)$

1.0f

0.5-

m_1

$2T$

47 67

87 107 127 147 167 187

Fig. 12-14 Step response of the relay-type sampled-data system in Example 12-2.

Example 12-3

For a nonlinear system, the period of oscillation of the output response usually depends on the choice of the initial conditions. Figure 12-15 shows that by choosing different initial conditions, the unit step response of the system in the preceding example can have oscillations with periods of 2 sec. ($T_s = 2T$), 4 sec. ($T_s = 4T$), and 6 sec. ($T_s = 6T$).

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(c_0, v_0)

$t=0$

$t=T$

1.0

+1

$t=10T$

$t=6T$

+1

-1

Limit cycle

period = 4 sec

$t=9T$ $t=5T$

$t=8T$

$t=4T$

2

3

$t=2T$

$t=7T$

-1

$C_2 + v = A + T$

+1=3T
 $C+v=A-2T$ $C_{12}+v_{12}=A-T$ $C_2+V_2=A$
 (a)

t=8T
 1=7T
 +1
 Limit cycle
 period =
 6 sec
 t=2T
 +1
 +1
 t=T
 t=6T-
 -1
 +1
 +1
 -0.5 0
 0.5
 1.5
 Co
 -1
 2.5
 +1 Co, Vot=2T
 1=3T
 -1
 =4T
 1=5T
 -1
 Limit cycle
 $C+v=A+2T$
 period 2 sec
 $C_2+v=A+T$
 $C_2+v=A-T$
 $C_2+v=A$

(b)
 Fig. 12-15 (a) Phase plane diagram of system in Example 12-3;
 $Co = 0.8, 1.2$ (b) Phase plane diagram of the relay-type
 sampled-data system in Example 12-3; $co = 0.5, vo = 0$; $co = 0.92$,
 $Vo = -0.42$.

Example 12-4

In this example we shall construct the phase plane diagram of the second-order relay-type sampled-data system when the relay has a dead zone. The trajectory in the phase plane, when the relay switches inside the dead zone, has been shown to be a straight line of slope 1. Let us assume that the dead zone D is equal to 0.1; $r(t) = 0$, $T = 1$ sec., and the initial conditions of the system are co

-0.4 and $vo = 1.2$. Notice that these conditions are

=
 -

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(co, vo)
 t=6T-
 +1
 1=5T
 m6=0
 t=T
 t=7T-T

0
 --
 2-
 +1
 $t=2T$
 Limit cycle
 period = 5 sec

-1
 $t=3T$
 $t=4T$
 -1
 $C_2+v=A+T$
 $C_n+v=A$
 $D=0.1$
 $C_2+v=A=T$
 $-D=0.1$
 $R=0$

Fig. 12-16 Phase plane diagram of relay-type sampled-data system with dead zone $D = 0.1$; c_0

=
 =
 =
 $-0.4, v_0$
 $= 1.2$.
 identical to those of the system whose response is obtained by the state transition method and is shown in Fig. 12-5b. One of the purposes of this example is to check the previous results by use of the phase plane method. As shown in Fig. 12-16, the set of switching lines are drawn in the phase plane with the reference line $c_n + U_n A$ passing through the initial condition point $(-0.4, 1.2)$. At $t = 0$, the operating point begins to move along the +1 trajectory that corresponds to > 1 , until it intersects the $c_2+v_n = AT$ switching line. Then, at $t = T$, since $e < 0$, the operating point is shown moving along the -1 trajectory for these sampling periods. At $t = 4T$ the error becomes positive, and the operating point is switched on to the +1 trajectory. After two sampling periods at $t = 6T$, the intersection of the +1 trajectory and the $c_n + U_n A$ line is found inside the dead zone and the operating point will move along this switching line, since it is also the trajectory when $m, 0$. The location of the next switching point is found by use of Eq. (12-39). Since at $t = 6T$, $v_e 0.72$ is read from the phase plane diagram, Eq. (12-39) gives

=
 =
 =

$$V_7 = e^1 \times 0.72 = 0.265$$

 (12-40)

Hence the point which corresponds to $t = 7T$ is found on the line $c_n + v_n = A$. Figure 12-16 shows that the phase plane trajectory finally exhibits a limit cycle with a period of 5 sec. ($T = 5T$). The amplitude of oscillation is approximately 1.12, peak-to-peak.

Note that the system response indicated by the phase plane diagram of Fig. 12-16 checks quite closely with the response of Fig. 12-5b, which is obtained by use of the state transition method. The phase plane method gives the exact continuous response at all times, whereas the difference equation

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method, being an analytical method, produces the response only at the sampling instances, and the actual response in between sampling instants must be estimated from the discrete data. This fact explains why the estimated response of Fig. 12-5b shows an amplitude of oscillation of only 0.52 peak-to-peak, whereas the actual amplitude is 0.56.

The reader may find it interesting and instructive to check the responses

shown in Figs. 12-4 and 12-5a and c using the phase plane method in the manner described above.

A stabilizing effect may be achieved if the relay is allowed to have a dead zone. This fact has been shown previously in Sec. 12.2 by the state transition approach, and two stable responses of the relay-type sampled-data system are shown in Fig. 12-6 when the relay dead zone is 0.6. To illustrate the stabilizing effect of the dead zone, the phase plane diagrams of the two responses of Fig. 12-6 are plotted in Fig. 12-17.

m5=0
t=5T
(co, vo)
(co.vo)

+1
84
-1
0

i
1
-1 2

t=4T

m3=0

t=T

t=T

t=2T

t=3T

D=0.6 D=0.6

Fig. 12-17 Phase plane diagram of relay-type sampled-data system with dead zone $D = 0.6$; $c_0 = 1.5$, $v_0 = 0.5$; $c_0 = 0.6$, v_0

=

= 0.4. Several interesting points are observed from the phase plane diagram of Fig. 12-17. For the initial conditions assumed and with $r(t) = 0$, the only stable steady-state value of $c(t)$ for $T = 1$ sec. is zero because operating point of the system at the sampling instants must lie on one of the switching lines, and in this case the only one of these lines which intersects the horizontal (v_0) axis inside the dead zone is the line $c_n + v_n A$, and the point of intersection is at the origin in the phase plane. Also, for the initial conditions

of c_0

1.5 and $v_0 = 0.5$ with $r(t) = 0$, the system is stable only if the point a on the phase plane trajectory lies inside the dead zone. Therefore, the width of the dead zone required for the system to be stable under the specified initial and input conditions can be determined from Fig. 12-17. In a similar fashion, the width of the dead zone required for stability when the system is subjected to a step input of any magnitude can be determined.

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Phase Plane Diagrams of Systems With Sinusoidal Inputs

The phase plane method can also be applied to a second-order relay-type system which has a sine wave input signal. In fact, the method described in the following paragraphs can be used to approximate the response of a continuous-data system with relay and a sine wave input. In this case the sampler and the hold circuit of the system in Fig. 12-1 become fictitious elements.

When analyzing relay-type sampled-data systems with sinusoidal inputs, the relationship between the signal frequency and the sampling frequency is of great importance. In general, the ratio of the signal frequency to the sampling frequency can be one of the fol-

lowing: (1) an integer, (2) a rational number, and (3) an irrational number. The phase plane analysis is simplest when the frequency ratio is an integer. But the phase plane method can also be applied if the frequency ratio is a rational number or an irrational number. The method is best illustrated by a numerical example.

Example 12-5

Consider the relay-type sampled-data system shown in Fig. 12-1. The sampling period is 1.0 sec., $G(s) = 1/s(s + 1)$, and the relay characteristic is described by Eq. (12-5). The initial conditions are $c_1 = 0$, and

$V_0 = 0$. A

sinusoidal input signal

$$r(t) = \sin \omega t$$

=

(12-41)

where $\omega = \pi/2$ rad. is applied to the system. The signal frequency to sampling frequency ratio is two. For the sampling frequency chosen, the input at the sampling instants, $r(nT)$ has the sequence, 0, +1, 0, -1, 0, +1, 0, -1, . . . Since the switching of the relay depends essentially on the error signal at the sampling instants $e(nT)$, where

=

-

=

$$e(nT) = r(nT) - c(nT)$$

-

(12-42)

or on $r(nT)$ for short, the phase plane diagram of the sinusoidally excited system can be constructed in the usual manner simply by considering the sequential variation of r at the sampling instants. Since $r(nT)$ can take on only three values, +1, 0, and -1 in this system, in the phase plane of Fig. 12-18 three dead zones are located with centers at $c = +1$, 0, and -1. For zero initial conditions and $T = 1.0$ sec., the switching lines are drawn in the usual manner.

Now, at $t = 0$, and =

To 0, the system sees a zero input; hence there is no

response. At $t = T$, r_1

+1, and $m_1 = +1$, the operating point of the system

will move along the +1 trajectory that passes through the origin in the phase

plane. At the intersection of the +1 trajectory and the $c + V_n A + T$ line,

$t = 2T$, and also $r_2 = 0$. The dead zone is effectively centered at the origin

at this instant, and since the operating point is outside the dead zone, and

e is negative the operating point is switched to a -1 trajectory. When

$t = 3T$, $r_3 =$

-1. The dead zone is located at $c = -1$, and c , is negative;

hence the operating point continues to stay on the 1 trajectory until $t = 4T$.

=

=

=

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Dead zone $r=-1$

Dead zone $r=0$

Dead zone

= +1

6T

2

=0

2T

-0.1

+1

0.1

5 +1
 5T
 -0.5
 -0.1 0
 0.1
 0.5
 Limit cycle
 (period = 4T)

4T
 +1
 78=0
 -1
 (a)
 $r_1 = -1$
 3-1
 37
 $C_2 + V_2 = T$

$r(1)$
 -1F
 0.5†
 3
 15
 AAAAAA

1 2
 5 6
 810
 9

$c(t)$
 1214
 13
 4 5

0|
 1 2 3
 89
 6 7

10 11
 -0.5
 12 13
 14 15

A
 Period 4 sec

=
 (b)
 Fig. 12-18 Phase plane diagram and input and output signals of the relay-type sampled-data system in Example 12-5.

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 Dead zone $r=-1$
 Dead zone $r=0$

A_v
 Dead zone - +1
 10-0

+1
 0.4
 0.4
 46
 IOT
 -0.4 0.4
 97
 272-0

```

+1
6T
9 +1
6-0
m2=0
73=-1
37
m=0
71|
77=-1
= +1
-1.4
1=T
-0.6
0.4
0.6
1.4
+1
+1
-1
r(1)
Limit
cycle
(period
= 8 sec)
8T
5 +1
57
-1
4=0
C12+v=T
4T
Cn+v=-T
(a)
C+v=0
-1
AAAAAA
.c(1)
0.5
-0.5
1 2 3 4 5
6
9
7 8
10 11 12 13
14 15
17
16
18 19
Period
= 8 sec
(b)
D = 0.4

```

Fig. 12-19 Phase data diagram and input and output signals of the relay-type sampled-data system in Example 12-5.

At $t = 4T$, the -1 trajectory intersects the $cn + Un = 0$, the operating point of the system at $t = 4T$ lies outside of the dead zone centered at the origin, and the operation point is switched to a $+1$ trajectory. As this process is carried on, Fig. 12-18a shows that the phase plane diagram exhibits a limit cycle after the initial transient. The period of the oscillation is equal to the period of the input signal (4 sec). The input signal, input sequence, and the output signal of the system are plotted in Fig. 12-18b. Fig. 12-19 gives the phase plane diagram and the output response of the same system under the same input and initial conditions, but with the dead zone increased to 0.4 . The output response is again a sustained oscillation with period equal to twice the period of the input sinusoid. The preceding example has demonstrated that the phase plane method can be applied to a relay-type sampled-data system with a sinusoidal input in a straightforward manner. The main essence in the method is the identification of the magnitudes of the input sequence r which must be taken into account when computing switching at the sampling instants. Clearly the method can also be applied to cases where the signal frequency to sampling frequency ratio is a rational number or an irrational number. However, in the case of the irrational ratio of frequencies, the input sequence does not vary periodically, and therefore, the output response is not periodic. Normally, an infinite amount of time would be needed to carry out the entire output response. As is clear, the method described above is not limited to only sampled-data systems with sinusoidal inputs. The method can be applied to a relay-type sampled-data system with almost any input as long as the input sequence r is defined at the outset of the analysis. For a ramp input, the phase plane analysis should be conducted in the e versus e' plane, since the output response c would increase with time.

12.4 THE INCREMENTAL PHASE PLANE METHOD+ Difference Equations

In the preceding section, the idea of the conventional phase plane has been applied to the analysis of nonlinear sampled-data system. The conventional phase plane method consists essentially of writing
 ↑ J. A. Aseltine, "Nonlinear Sampled-Data System Analysis by the Incremental Phase Plane Method," Proc. of the International Federation of Automatic Control, Moscow, 1960. J. A. Aseltine, R. A. Nesbit, "The Incremental Phase Plane for Nonlinear Sampled-Data Systems," I. R. E. Trans. on Automatic Control, Vol. AC-5, August 1960, pp. 159-165.

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the second-order differential equation of a system as a pair of first order equations involving two dependent variables (usually one of the variables is the derivative of the other) and then plotting the paths generated in the phase plane by the two variables with time as a parameter.

We have found that difference equations bear the same relation to sample-data systems as ordinary differential equations do to continuous-data equations. This implies that an "incremental phase plane" may be defined as to consist of the plot of the first difference of a variable versus the variable itself.

-->C(2)
 $E(s) \rightarrow E^*(s)$

↑ E(2)

G(s)

-C(s)

Fig. 12-20 Linear open-loop sampled-data system.

Before entering the discussions of the incremental phase plane, let us define the "differences" of a difference equation. Consider the linear open-loop sampled-data system shown in Fig. 12-20. The z-transform of the output signal is written.

$$C(z) = \sum_{n=0}^{\infty} c(nT)z^{-n}$$

(12-43)

The magnitudes of $c(t)$ at the sampling instants are given by $c(0)$, $c(T)$, $c(2T)$, $c(3T)$, . . . $c(nT)$, . . . The sequence of the first difference is given by

$$\Delta c_0 = c(T) - c(0)$$

$$\Delta c_1 = c(2T) - c(T)$$

Δc

$$\Delta c_2 = c(3T) - c(2T)$$

$$\Delta c_{n-1} = c(nT) - c((n-1)T)$$

$$\Delta c_n = c((n+1)T) - c(nT)$$

(12-44)

where the symbol "A" can be regarded as a difference operator.

Likewise, the differences of the sequence of first differences of Eq. (12-44) are called the second differences of the sequence $c(nT)$. Thus the sequence of second difference of the sequence $c(nT)$ is $A^2 c_0$, $A^2 c_1$, $A^2 c_2$, . . .

where

$$A^2 c_n = \Delta c_{n+1} - \Delta c_n$$

$$A^2 c_n = \Delta c_{n+1} - \Delta c_n$$

(12-45)

and from the definition of the first difference, the last equation is written as

$$A^2 c_n = c((n+2)T) - 2c((n+1)T) + c(nT)$$

(12-46)

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In general, the k-th difference of the sequence $c(nT)$ can be defined as

$$A^k c_n = A^k c_{n+1} - A^{k-1} c_n$$

(12-47)

Now we shall see how a difference equation is written from a z-transform equation. From the definition of the z-transformation and the real translation theorem given by Eq. (4-69), the following inverse z-transforms are written:

and

$$z^{-1} [E(x)] = \sum_{n=0}^{\infty} e(nT) \delta(t - nT)$$

$$z^{-1} [x^m E(x)] = \sum_{n=0}^{\infty} e(nT + mT) \delta(t - nT)$$

$$\sum_{k=0}^{\infty} e(kT) \delta(t - kT + mT)$$

$$z^{-1} [x^m E(x)] = \sum_{n=0}^{\infty} e(nT + mT) \delta(t - nT)$$

$$\sum_{k=0}^{\infty} e(kT) \delta(t - kT + mT)$$

$$\sum_{k=0}^{\infty} e(kT) \delta(t - kT + mT)$$

$$\sum_{k=0}^{\infty} e(kT) \delta(t - kT + mT)$$

$$\sum_{k=0}^{\infty} e(kT) \delta(t - kT + mT)$$

where m is a positive integer and $n = 0, 1, 2, \dots$

(12-48)

(12-49)

If we are interested in only the impulse sequence for t greater than 0, then the last term on the right side of Eq. (12-49) vanishes; Eq. (12-49) becomes

$$Z^{-1}[x^m E(x)] = \sum_{n=0}^{\infty} e(nT + mT) \delta(t - nT)$$

(12-50)

Equation (12-50) generally leads to the formulation of difference equations from z -transform equations. As an illustration, consider that the open-loop transfer function of the sampled-data system shown in Fig. 12-20 is given by

$$C(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2} + \dots + a_r z^{-r}}{b_0 + b_1 z^{-1} + b_2 z^{-2} + \dots + b_q z^{-q}}$$

$$E(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + \dots + b_q z^{-q}}{C(z)}$$

Equation (12-51) can be written as

$$(b_0 z^q + b_1 z^{q-1} + \dots + b_q)C(z) = (a_0 z^r + a_1 z^{r-1} + \dots + a_r)E(z)$$

$$(12-51)$$

(12-52)

If we take the inverse z -transform term by term and use the relation given by Eq. (12-50), Eq. (12-52) becomes

$$\sum_{n=0}^{\infty} b_n e[(n+q)T] \delta(t - nT) + b_{q-1} \sum_{n=0}^{\infty} c[(n+q-1)T] \delta(t - nT) + \dots + b_0 \sum_{n=0}^{\infty} c(nT) \delta(t - nT)$$

$$= \sum_{n=0}^{\infty} a_n e[(n+r)T] \delta(t - nT) + a_{r-1} \sum_{n=0}^{\infty} e[(n+r-1)T] \delta(t - nT) + \dots$$

$$+ a_0 \sum_{n=0}^{\infty} e(nT) \delta(t - nT)$$

(12-53)

$$E(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2} + \dots + a_r z^{-r}}{b_0 + b_1 z^{-1} + b_2 z^{-2} + \dots + b_q z^{-q}}$$

$$(12-54)$$

(12-55)

(12-56)

$$E(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2} + \dots + a_r z^{-r}}{b_0 + b_1 z^{-1} + b_2 z^{-2} + \dots + b_q z^{-q}}$$

(12-57)

(12-58)

Equating the coefficients of the impulses at the same sampling instants yields

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$$b_0 c[(n+q)T] + b_1 c[(n+q-1)T] + \dots + b_q c(nT) = a_0 e[(n+r)T] + a_1 e[(n+r-1)T] + \dots + a_r e(nT)$$

(12-59)

Equation (12-59) is known as the difference-equation description of the sampled-data shown in Fig. 12-20. This is analogous to the differential-equation representation of a continuous-data system. In view of the difference sequence defined previously, the difference

equation of Eq. (12-54) can also be written in terms of the differences Δc_n ($n, k = 0, 1, 2, \dots$).

The following example illustrates the formulation of the difference equation of a sampled-data system.

Example 12-6

Assume that the transfer function of the system in Fig. 12-20 is given by

$$G(s) = K/s^2$$

The z-transfer function of the system is ($T = 1$ sec)

$$(12-55)$$

$$\frac{C(z)}{E(z)}$$

$$=$$

$$= G(z) =$$

$$=$$

$$=$$

$$K\alpha$$

$$(\approx -1)^2$$

If we cross-multiply, Eq. (12-56) becomes

$$(x^2 - 2x + 1)C(z) = KzE(z)$$

$$(12-56)$$

$$(12-57)$$

Using the relationship between Eqs. (12-52) and (12-54), the difference equation of the system is written directly from Eq. (12-57)

$$c[(n+2)T] - 2c[(n+1)T] + c(nT) = Ke[(n+1)T]$$

$$-$$

or, in simplified notations,

$$\text{Since } c_{n+2}$$

$$-$$

be written as

$$c_{n+2}$$

$$--$$

$$2c_{n+1} + c_n$$

$$=$$

$$K\epsilon_{n+1}$$

$$(12-58)$$

$$(12-59)$$

$2c_{n+1} + c_n$ is the second difference $\Delta^2 c_n$ of c_n , Eq. (12-59) can

$$\Delta^2 c_n$$

$$=$$

$$K\epsilon_{n+1}$$

$$(12-60)$$

The Incremental Phase Plane (The A-plane)

The incremental phase plane, or the A-plane, is defined as a plane which has the first difference of a variable as ordinate and the variable itself as abscissa. If a second-order sampled-data system is characterized by the dependent variable $c(t)$, the incremental phase plane diagram of the system is a plot of the first difference Δc_2 of the sequence $c(nT)$ as a function of c_n . Each solution point in the A-plane represents a value of c_n and the first difference Δc_1 for $n = 0, 1, 2, 3, \dots$. Hence the trajectory in the A-plane is a curve that passes through all the solution points $(c_n, \Delta c_n)$ for $n = 0, 1, 2, 3, \dots$.

However, we should point out that the difference equation approach

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gives solutions only at the sampling instants. Therefore, although the trajectory of the A-plane is a continuous curve, only discrete points on the trajectory that correspond to sampling instants are significant.

Let us use the relay-type sampled-data system of Example 12-1 to illustrate the construction of the A-plane trajectory. If assump-

tion is made that the relay contacts are able to open and close instantaneously with zero duration, the positions of the relay and the hold device of the system can be interchanged without affecting the analytical results of the system. The equivalent system, with the relay and the hold device transposed, is shown in Fig. 12-21.

Zero-order

hold

Ideal

relay

Controlled

process

$r(t)$ $e(t)$

$x(t)$

$c(t)$

$G_n(s)$

1

$s(s+1)$

$T=1$ sec

$G(s)=1/s(s+1)$

Fig. 12-21 Second-order relay-type sampled-data system.

The z-transform of the system output $c(t)$ is written as a function of the relay output $X(z)$; hence

$C(z)$

=

$0.264 + 0.368z$

$X(z)$

-

$z - 1)(\approx - 0.368)$

(12-61)

Cross-multiplying and rearranging terms, we see that Eq. (12-61) becomes

-

$(z^2 - 1.368z + 0.368) C(z) = (0.264 + 0.368z) X(z)$ (12-62)

The difference equation is determined directly from Eq. (12-62).

$C_{n+2} - 1.368C_{n+1} + 0.368C_n = 0.264x_n + 0.368x_{n+1}$

In terms of the differences, Eq. (12-63) is written as

$A C_n + 0.632 \Delta C_n = 0.264x_n + 0.368x_{n+1}$

(12-63)

(12-64)

For the system under consideration, since x and x_{n+1} can take on values of only +1 and -1, the right-hand side of Eq. (12-64) is always a constant. Hence, Eq. (12-64) can be expressed as

where

$A' C_n + 0.632 \Delta C_n$

= K

$K = 0.264x_n + 0.368x_{n+1}$

(12-65)

(12-66)

65) is

K

$C_n =$

K

$0.32 - \ln|1 - 0.632 \Delta C_n| + A$

It will be shown in the following that the solution of Eq. (12-

1

ΔC_n

(12-67)

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where A is an arbitrary constant. To prove that Eq. (12-67) is a solution of Eq. (12-65), we take the first difference of both sides of

Eq. (12-67):

$$10.632$$

K

$$A_{cn} = \Delta$$

$$AK - \ln |1 - 0.632 A_{c1}| \}$$

1

$$0.632$$

$$A_{2cn}$$

$$(12-68)$$

=

-

$$(12-69)$$

and since

Eq. (12-68) becomes

K

$$A(\ln c_n) \ln c_{n+1} \ln c_n$$

$$A_{cn} = 0.632 [\ln / 1 - 0.632$$

1

$$-0.632 A_{c...} - \ln |1 - 0.632 A_{c1}| - 0.6324^\circ c.$$

K

K

Substituting the relation.

into Eq. (12-70) yields

$$A_{cn+1} = A' C_n + A_{cn}$$

Cn =

$$(12-70)$$

$$(12-71)$$

-

$$0.632$$

$$K_2 \ln |1$$

$$0.632$$

K

$$(A^2 C_n + A_{cn}) -$$

$$A_{c2}) | - \ln |1 -$$

$$0.632$$

K

$$A_{cn}$$

$$1 A_{2cn}$$

$$(12-72)$$

$$0.632$$

After rearranging terms, Eq. (12-72) reads

$$0.632$$

1

$$(A^2 C_n + A_{cn})$$

K

$$A' c_n' + 0.632 A_{c1} = -K \ln$$

1

$$0.632 A_{cn}$$

$$(12-73)$$

K

Substituting Eq. (12-65) into Eq. (12-73) for K, we very easily obtain

$$A' c_n + 0.632 A_{cn} = A' c_n + 0.632 A_{cn}$$

$$(12-74)$$

Therefore, Eq. (12-67) is proved to be a solution of Eq. (12-65).

When the system is subjected to a unit step input, $r(t) = 1.0$.

The value of K at any sampling instant $t = nT$ will depend upon the magnitudes of c_n and C_{+1} . Determining the value of c_{n+1} at $t = nT$ is not a problem, since C_{n+1} is given by

$$C_{n+1} = A C_n + C_n$$

$$(12-75)$$

Once the trajectory is given, the increment in abscissa to C_{2+1} may be determined by adding the ordinate A_{cn} to the abscissa c_n at

$t = nT$.

For the relay system given, the following values of the relay output are possible at $t = nT$:

$x_n = +1$

$C_n < 1$

x_n

$= -1$

$C_n > 1$

(12-76)

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Similarly, at $t = (n + 1)T$,

$x_{n+1} = +1$

$C_{n+1} < 1$

$x_{n+1} = -1$

$C_{n+1} > 1$

(12-77)

Substitution of Eqs. (12-76) and (12-77) into the expression of Eq. (12-66) yields the following possible values of K and combination of x_n and x_{n+1} :

In

x_{n+1}

$K(= 0.264x + 0.368x_{n+1})$

C_n

C_{n+1}

$+1$

$+1$

0.632

< 1

< 1

$+1$

$- 1$

-0.104

< 1

> 1

$- 1$

$+1$

0.104

> 1

< 1

-1

$- 1$

-0.632

> 1

> 1

As we see, K can have only four possible values: 0.632 , -0.632 , 0.104 , and -0.104 . In Fig. 12-22, the incremental phase plane trajectories described by Eq. (12-67) are plotted for these four

A_{cn}

2.5

2.0

1.5

$K=0.632$

-1.0

0.5

C_n

$-2.0 -1.5 -1.0 -0.5 0$

0.5

$1.0 1.5 2.0 2.5$

3

-0.5

-1.0
 K = -0.632
 --1.5
 - -2.0
 --2.5
 -3.0

(a)
 Fig. 12-22 (a) Δ -plane trajectory for $A_{cn} + 0.632\Delta cn = K$;
 $K = +0.632$ and $K = -0.632$.

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possible values of K with $A = 0$. Note that the trajectory for a negative value of K may be obtained by rotating the corresponding positive K trajectory through 180° about the origin. For any initial condition (c_0, A_{c0}) that does not lie on the trajectories of Fig. 12-22, the particular trajectory is simply shifted horizontally so that it goes through the initial point (c_0, A_{c0}) . Since A is an arbitrary constant in Eq. (12-67), shifting any one of the trajectories of Fig. 12-22 horizontally through any distance still satisfies the solution equation.

Before entering the construction of the over-all A -plane diagram of the system, the technique of finding the successive solution (operating) points along the A -plane trajectory needs further explanations. Let us assume that the system has the initial conditions (c_0, A_{c0}) shown in Fig. 12-23a. Before determining the location of the next point (c_1, A_{c1}) in the A -plane, we must find out which one of the four trajectories in Fig. 12-22 is in effect during the time

A_{cn}
 2.5
 2.0
 1.5
 F1.0
 -2.0 -1.5 -1.0 -0.5

V- ----
 $K = -0.104$
 $-0.5 \quad K = 0.104$

C_n
 0 0.5 1.0 1.5
 2.0 2.5 3

-0.5
 F-1.0
 -1.5
 -2.0
 -2.5
 -3.0

(b)
 Fig. 12-22 (b) A -plane trajectory for $A_{cn} + 0.632A_{cn} = K$;
 $K = +0.104$ and $K = -0.104$.

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A_{cn}
 -1.0
 $b(c_2, A_{c2})$
 - 1.0
 kon A Col
 $K = 0.632$
 (c_1, A_{c1})
 - 1.0
 (c_i, A_{c1})
 $K = -0.104$

A_{c0}
 $a(c_2, A_{c2})$
 $-A_{c1}$
 C_n
 2.0
 A_{c0}
 (a)
 A_{c1}
 1.04
 1.0
 $K=0.632$
 C_n
 0
 1.0
 2.0
 (b)

Fig. 12-23 Location of operating points of the A-plane trajectory. interval between 0 and T. Since the value of K depends upon c_1 as well as c_0 , a knowledge of the initial point (c_0, A_{c0}) alone is insufficient to determine the trajectory. The value of c_1 is easily found from

$$C_1 = \frac{A_{c0} + c_0}{K} \quad (12-78)$$

Assuming that c_0 is less than unity and that c_1 determined from Eq. (12-78) is also less than unity, the $K = 0.632$ trajectory should be drawn through the point (c_0, A_{c0}) . The point (c_1, A_{c1}) is easily found on the trajectory by adding A_{c0} to the abscissa c_0 as shown in Fig. 12-23a. The value of c_2 can be found in a similar manner. Now the effective trajectory during the time intervals T and 2T is determined by the values of c_1 and c_2 . If c_2 is still less than unity, the

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=

operating point at $t = 2T$ is still found on the $K = 0.632$ trajectory, since $c < 1$. However, if $c > 1$, then the $K = -0.104$ trajectory

>1 ,

must be drawn passing through the (c_1, A_{c1}) point; and the next point (c_2, A_{c2}) is found on this trajectory. Due to the multiple-valued property of the trajectory ($K = -0.104$), both point a and point b satisfy the relationship

$$C_2 = A_{c1} + C_1 \quad (12-79)$$

This ambiguity is removed by recognizing the fact that time progresses in the direction of the arrows along the trajectory; thus a change of the operating point in either direction from the point (c_1, A_{c1}) along the trajectory must cause a change in the corresponding direction in the point (c_2, A_{c2}) . As shown in Fig. 12-23a, if the operating point is shifted in the forward direction along the trajectory to (c_1, A_{c1}) , the point at a moves in the same direction to a' , whereas the point at b moves in the opposite direction to b' .

Therefore, a is the correct location for (c_2, A_{c2}) . It can be shown that for the $K = +0.104$ and the $K = -0.104$ trajectories, whenever an ambiguity arises, such as between points a and b in Fig. 12-23a, the nearest point reached along the trajectory is the correct one. In Fig. 12-23b, the $K = +0.632$ trajectory is shown plotted through an initial condition point (c_0, A_{c0}) . With the same reasoning used above, we see that the point b rather than point a is the correct point for the next switching point (c_1, A_{c1}) . In general, we can state that for the $K = +0.632$ or the $K = -0.632$ trajectories, the far-

the point reached along the trajectory is the correct switching point.

To illustrate the construction of the complete A-plane diagram for the system, let us assume the initial conditions $c(0) = 0$ and $c'(0) = 0$. For a unit-step function input $r(t) = 1.0$ and $x(0) = 1.0$, the system output during the time interval $0 < t < T$ is given by

Hence

and

-

$$c(t) = t(1 - e^{-t})$$

$$c(T) = c_1 = e^{-1} = 0.368$$

Also $C_1 =$

-

$$C_0 = 0.368$$

(12-80)

(12-81)

(12-82)

Following the procedure described above, the A-plane trajectory of the relay-type sampled-data system is drawn in Fig. 12-24. A limit cycle of period equal to 4 sec. and centered at $c = 1.0$ results.

An interesting comparison can be made of the results of the A-plane trajectory and the response shown in Fig. 12-14 and obtained by use of the ordinary phase plane method. The system response at the sampling instants c_2 indicated in Fig. 12-24 checks very closely with the response at sampling instants in Fig. 12-14. However,

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Acn

+2.0

F1.0

t=T

$$K = 0.632$$

$$(C_0 - C_1) = (0.368 - 0.368) = 0$$

$$t = 5T$$

$$K = -0.104$$

$$t = 2T$$

$$-1.0$$

OI

1.0

2.0

Cn

14T

$$K = 0.632$$

$$-K = -0.632$$

$$K = 0.104$$

$$t = 3T$$

$$-1.0$$

Fig. 12-24 Incremental phase plane (A-plane) trajectory of relay-type sampled-data control system. $G(s) = 1/s(s + 1)$; $T = 1$ sec; ideal relay.

the diagram in Fig. 12-24 does not carry any information on the response between sampling instants.

Based on the foregoing analysis of the relay-type sampled-data system, the incremental phase plane method would seem to have hardly any advantage over the conventional phase plane method.

The construction of the conventional phase plane diagrams of relay-type sampled-data systems simply relies on the use of simple fixed templates of the +1 and -1 trajectories. The switching lines are derived and are drawn in the phase plane, and the intersections of the switching lines and the phase plane trajectories give the response

quantities at the sampling instants. Moreover, the conventional phase plane method gives the response not only at the sampling instants, but at all times. However, how the conventional phase plane method can be extended to analyze sampled-data systems with nonlinearities other than the relay type is not at all clear. On the other hand, although the incremental phase plane method is more difficult to use in analyzing relay-type systems, it can be used to analyze a variety of other nonlinearities. We shall illustrate this feature by considering a sampled-data system with saturation.

The block diagram of a sampled-data system with saturation is shown in Fig. 12-25.

The equation for $C(z)$ is written

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$r(t)$

$e(t)$

$1+$

$-1+$

*

$x(1)$

$c(t)$

$G(s) = 1/2$

Fig. 12-25 Sampled-data control system with saturation. $T = 1.0$

sec, $r(t) = 1.0$.

$C(x)$

2

$=$

Z

$-$

$1)^2 X(2)$

(12-83)

The difference equation of the open-loop system between X and C is

$A^2 C_n = X_{n+1}$

(12-84)

For the saturation characteristic given and $r(t) = 1.0$, $x(t)$ has the following quantities at $t = (n + 1)T$:

$X_{n+1} = +1$

$X_{n+1} =$

$X_{n+1} =$

1

$-$

$-$

-1

$- C_{n+1}$

$C_{n+1} 0$

$0 \leq C_{n+1} 2$

$C_{n+1} 2$

Thus when $C_{n+1} \leq 0$ and $C_{n+1} \geq 2$, Eq. (12-84) reads

$A C_n = +1 = K$

(12-85)

(12-86)

-2

-1

$K = -1$

$A C_n$

-2

-1

C_n

1

2

K = +1

-2

Fig. 12-26 A-plane trajectories for $Ac_1 = K$.

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and when $0 \leq c_{n+1} \leq 2$, Eq. (12-84) becomes

$$A c_n + A c_n + (c_n - 1) = 0$$

(12-87)

We can show that the solution of Eq. (12-86) is

$$(A c_n)^2$$

$$K(2c_n + A c_n) = A$$

(12-88)

where K = +1 or -1; and the solution of Eq. (12-87) is

$$(A c_n)^2 + A c_n (c_n - 1) + (c_n - 1)^2 = A^2$$

(12-89)

where A is an arbitrary constant.

The trajectories described by Eq. (12-88) are plotted in Fig. 12-26 for K = +1 and -1. The trajectories are parabolas with axes displaced 1/2 units from the c_n axis. The K = -1 trajectory is the same as the K = +1 trajectory rotated through an angle of 180 degrees. The trajectories for Eq. (12-89) are a family of ellipses centered at $c_2 = 1$ and are rotated through an angle of +45 degrees. (Fig. 12-27).

 c_n

To construct the entire A-plane trajectory of the closed-loop system, the same procedure described previously may be used. For the initial condition of $c_0 = 0$ and $A c_0 = 0.2$, the A-plane trajectory of the system with saturation is drawn in Fig. 12-28. The step re-

 $A c_n$

$$A = 1.4$$

$$A = 1.2$$

$$A = 1.0$$

$$A = 0.8$$

$$A = 0.6$$

$$1.6$$

$$2.4$$

 c_n Fig. 12-27 A-plane trajectories for $A^2 c_1 + A c_1 + (c_n - 1) = 0$.

+ To prove that Eqs. (12-88) and (12-89) are, respectively, the solutions of Eqs

(12-86) and (12-87), simply take the first difference of both sides of Eqs. (12-88)

and (12-89) and solve for $A c_n$

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 $A c_n$

$$+2.0$$

Ellipse

$$t = T$$

Saturation

region

$$1.0$$

$$-1.0$$

$$-1.0$$

$$0.2 (c_0, c_0)$$

$$t = 2T$$

Saturation
region
OL 1=5T 1.0
2.0

Cn
11=3T
-1.0
1=4T
Ellipse
-2.0

$G(s) = 1/s^2$ (no hold device), $T = 1$ sec,
 $r(t) = 1.0$, c_0

=
= 0, c_0
= 0.2.

Fig. 12-28 A-plane trajectory of sampled-data system with satu-
ration.

sponse has a sustained oscillation in the steady-state. The period of
oscillation is 6 sec, and the amplitude of oscillation is 2.2 peak-to-
peak.

12.5 THE DESCRIBING FUNCTION METHOD

The describing function method introduced by Goldfarb and
Kochenburger has been a powerful and popular tool in the analysis
and synthesis of nonlinear systems with continuous-data. The des-
cribing function enables one to study the steady-state sinusoidal re-
sponse of a nonlinear system of any finite order. The application of
the describing function method relies on the following assumptions
and considerations:

+ L. C. Goldfarb, "On Some Non-linear Phenomena in Regulatory Systems," *Avto-
matika i Telemekhanika*, 8, No. 5, 1947, pp. 349-83.

R. J. Kochenburger, "A Frequency Response Method for Analyzing and Syn-
thesizing Contactor Servomechanisms," *Trans. A. I. E. E.*, 69, Part 1, 1950, pp.
270-284.

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$r(t)$
 $e(t)$

n
 $G(s)$
 $c(t)$
Nonlinear
Controlled
element
process
(linear)

Fig. 12-29 Nonlinear feedback control system with continuous-
data.

(1) The continuous-data system contains only one nonlinear ele-
ment. A typical form of the nonlinear system for describing func-
tion studies is shown in Fig. 12-29.

(2) The input to the nonlinear element n is assumed to be
sinusoidal. The output of n is generally not a sine wave, but never-
theless, it is a periodic function and can be represented by a Fourier
series. Only the fundamental component of the output of the non-
linear element is assumed to be significant. Actually, if the input to
n is considered to be sinusoidal, the output of n will contain com-
ponents at the fundamental frequency and in general at all higher
harmonic frequencies. Since the harmonics in the outputs of most
of the common types of nonlinearities encountered in servo systems
are often of smaller amplitudes than the amplitudes of the funda-
mental components, most servo systems act as low-pass filters so that

the higher harmonics in the output of the nonlinear elements are attenuated when compared to the fundamental component. When the basic assumptions listed above are satisfied, the nonlinear element n can be represented by an equivalent transfer function called the "describing function," which is defined as the ratio of the fundamental component of the output to the amplitude of the input. Therefore,

$$\text{Describing function } N = \frac{\text{Fundamental component of output from Fourier analysis}}{\text{Amplitude of the sinusoidal input signal}} \quad (12-90)$$

Once the describing function of the nonlinear element of a given system is determined, it can be used to study the stability of the nonlinear system. If the nonlinear element is properly represented by its describing function N , where N is generally a complex number and a function of the input signal amplitude and frequency, it follows that the closed-loop transfer function of the system in Fig. 12-29 is

$$\frac{C(s)}{R(s)} = \frac{NG(s)}{1 + NG(s)} \quad (12-91)$$

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The stability of the closed-loop nonlinear system depends upon the zeros of the equation

$$1 + NG(s) = 0 \quad (12-92)$$

Just as in the case of the linear system, if all the zeros of Eq. (12-92) are located in the left-half of the s -plane, the system is stable; if one or more zeros are found in the right-half of the s -plane, the system is unstable. The roots of Eq. (12-92) may also be interpreted as the condition when

$$G(s) = -1/N \quad (12-93)$$

Normally, the stability analysis can be carried out in either of the following domains: (1) the Nyquist diagram, or (2) the gain-phase plot. In the case of the Nyquist analysis, for instance, the system stability is investigated by plotting the $G(s)$ and the $-1/N$ loci in the polar coordinates. The $G(s)$ plot is the ordinary frequency locus of the linear part of the system, but the $-1/N$ locus can be considered to be the locus of the critical point, which in linear theory is the $(-1, j0)$ point in the complex G -plane. When N is a function of only the input signal magnitude E to the nonlinear element, the critical points of $-1/N$ form a locus which is a function of E . If N is a function of both E and ω , a family of loci must be constructed; on each locus ω is constant, but E is a variable parameter.

The relative position of the $G(s)$ locus and the $-1/N$ plot indicates the stability condition of the nonlinear system. When the critical points of $-1/N$ lie to the left of the $G(s)$ locus or are not enclosed by it (assuming that $G(s)$ is itself a stable system function), the closed-loop system is said to be stable, since any disturbances which appear in the system will tend to die out. Conversely, when any part of the $-1/N$ locus lies to the right of the $G(s)$ locus and thus is enclosed by it, the implication is that any disturbances which are characterized by the values of E corresponding to the enclosed critical points will provide unstable operations. The intersection of the $G(s)$ and $-1/N$ loci corresponds to the possibility of a periodic oscillation characterized by the value of E on the $-1/N$ locus and the value of ω on the $G(s)$ locus. For the situation

in which the $-1/N$ locus is a function of both E and w , periodic oscillation is possible only if the values of w at the intersection are identical on the two loci. The periodic solution described above may be stable or unstable despite slight disturbances in amplitude E or frequency from the equilibrium condition. If the equilibrium state is a stable one, the amplitude of oscillation tends to return to its original value following a disturbance. If the equilibrium is unstable, the oscillation either increases or decreases in amplitude depending

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Linear

process

$r(t)$

$e(t)$

$e(t)$

Zero-order $h(t)$

hold

Relay

$G(s)$

$c(1)$

$m(t)$

Nonlinear element n

Fig. 12-30 Relay-type sampled-data control system.

upon the direction of the disturbance. The oscillation may increase indefinitely or enter into some other equilibrium state.

The system shown in Fig. 12-30 demonstrates the extension of the describing function method to the study of nonlinear sampled-data control systems. Early contributions on the study of relay-type sampled-data systems by means of the describing function technique have been made by Chow† and Russell. In these early investigations the nonlinear element n is considered to include the sampling switch, the zero-order hold, and the on-off relay. Thus the system configuration of the sampled-data system is identical to that of the continuous-data system shown in Fig. 12-29. For a sinusoidal input $e(t)$ whose period is an integral multiple of the sampling period T , the output of the nonlinear element n is a periodic rectangular wave. Therefore, the conventional describing function technique ordinarily used for continuous-data systems can be applied directly. The nonlinear sampled-data system is essentially treated as a nonlinear continuous-data system, since the sampler is absorbed by the nonlinear element.

In this section a new describing function called the "discrete describing function" is derived utilizing the z -transformation. In this case the nonlinear element n in the sampled-data system is considered to include only the on-off relay. The discrete describing function method has the following advantages and features:

(1) The discrete describing function is natural for sampled-data systems; it is analogous to the use of the conventional describing function for continuous-data systems.

(2) The discrete describing function method can be applied to systems with or without zero-order hold devices

(3) Systems with more than one synchronized sampler can be analyzed.

† C. K. Chow, "Contactor Servomechanisms Employing Sampled-Data," Trans. A.I.E.E., 73, 1954, pp. 51-64.

E. A. Russell, "Design Criterion for Stability of Sampled-Data On-Off Servomechanisms," Ph. D. thesis, Columbia University, New York, 1953.

§ B. C. Kuo, "The z -transform-Describing Function for Nonlinear Sampled-Data Control Systems," Proc. of the I. R. E., 48, No. 5, May 1940, pp. 941-942.

(4) The actual output of the nonlinear element, rather than the fundamental component of the Fourier series, is considered.

Stability of the System

The nonlinear sampled-data system under investigation is shown in Fig. 12-30. The sampler is assumed to be ideal, meaning that the sampler output $e^*(t)$ is an impulse train magnitude modulated by the input $e(t)$. The relay characteristic has a dead zone D and was given previously by Eq. (12-1). The stability of the nonlinear sampled-data system is defined as follows: If the system has a self-sustained oscillation in response, the system is said to be unstable; otherwise, it is stable. In other words the present analysis deals mainly with the condition of sustained oscillations when the reference input $r(t)$ becomes zero after some time t .

... In the error signal $e(t)$ is sinusoidal with period T_{enT} , where $n = 2, 3, 4, \dots$ and T is the sampling period in seconds, the sampler output $e^*(t)$ must also be a periodic function, although it may not have the same frequency as $e(t)$. Since $e^*(t)$ can have values only at the sampling instants, the period of $e^*(t)$ must also equal nT . Thus when the characteristics of the zero-order hold are considered, the relay, $G(s)$, the functions $h(t)$, $m(t)$, and $c(t)$ see Fig. 12-30 must also have the same period as $e^*(t)$. This is a very important feature of this system.

Under the condition of self-sustained oscillation, $r(t) = 0$; hence $e(t) = r(t) - c(t) = -c(t)$

(12-94)

which means that $e(t)$ must also have the same period as $c(t)$.

Description and Modification of the System Configuration

The z-transform of the output of the system shown in Fig. 12-30 can be written as

where

$$C(z) = E(z) \times NG(z)$$

$$NG(z) = \mathcal{Z}[N(s)G(s)]$$

(12-95)

(12-96)

$N(s)$ represents the transfer function or the describing function of the combination of the hold device and the relay, excluding the sampler. The closed-loop transfer function of the system is given by

$$\frac{C(z)}{R(z)} = \frac{NG(z)}{1 + NG(z)}$$

$$(12-97)$$

However, the describing function technique for stability studies cannot be effectively applied to Eq. (12-97), since the function

transposed.

$NG(z)$ cannot be separated into two functions N and $G(z)$. A modified block diagram of the system is shown in Fig. 12-31, in which the zero-order hold and the relay are transposed. From the analytical point of view the system behavior is not altered at all by this modification. The nonlinear element in this case is considered to include only the relay whose output signal $v^*(t)$ is an impulse train with constant amplitudes. The input to the controlled process $G(s)$ is not affected by the change.

is

Now, the closed-loop transfer function of the modified system

$$\frac{C(z)}{R(z)} = \frac{N(z) G_1(z)}{1 + N(z) G_1(z)} \quad (12-98)$$

$$=$$

where $N(z)$ is defined as the z -transform discrete describing function of the relay. Now the study of the stability of the nonlinear sampled-data system involves the investigation of the equation

or

$$1 + N(z) G_1(z) = 0$$

$$G_1(z) = -1/N(z) \quad (12-99)$$

$$(12-100)$$

The Discrete Describing Function $N(z)$

The derivation of the discrete describing function of a nonlinear element is based on the assumption that the input signal to the nonlinear element is a sinusoidally modulated impulse train. The discrete describing function is defined as the ratio of the z -transform of the output $v^*(t)$ to the z -transform of the assumed sinusoidally modulated input $e^*(t)$. Thus

$$N(z) = V(z)/E(z)$$

Suppose that the input to the sampler is

$$e(t) = E \cos(at + \phi) \quad (12-101)$$

$$(12-101)$$

$$= E(\cos at + \cos at \sin at \sin at)$$

$$\sin at \sin at \sin at \sin at$$

$$\sin at \sin at \sin at \sin at$$

$$(12-102)$$

$$E(z)$$

$$-$$

$$E x$$

$$2z \cos aT + 1$$

$$[(z$$

$$-$$

$$\cos aT) \cos at - \sin aT \sin at]$$

$$-$$

$$\text{The } z\text{-transform of the last}$$

$$(12-103)$$

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Since the period of the self-sustained oscillation is an integral multiple of the sampling period T , only these periods are considered in the derivation of $N(z)$.

With reference to the system shown in Fig. 12-31, the output of the zero-order hold $m(t)$ is either a constant or zero between two successive sampling instants. For a given period T of $c(t)$, $m(t)$ may have many possible forms. Figure 12-32 illustrates the possible forms of $m(t)$ for $T = 4T$. However, since the error signal $e(t)$ is assumed to be a sinusoid with period $T = 4T$, only a few of the

forms of $m(t)$ illustrated in Fig. 12-32 may actually occur in the system. Figure 12-33 shows all the possible wave forms of $m(t)$ when varies from 0 to 360 degrees and $T_c = 4T$. The periodic functions shown in Fig. 12-33a are actually of the same form as those shown in Fig. 12-32b, except for the phase shift of $\pm n\pi/2$, $n = 0, 1, 2, \dots$ Likewise, the periodic functions shown

27
3T
5T
6T
e(t) 0
47
(a)
m(t)
27
3T
T
4T
5T
6T
(b) 4 = 1
m(t)
3T
T
4T
5T
6T
(c) 4 = 2
m(t)
+1
27
6T
0
t
-1
3T
4T
5T
(d)
m(t)
+1
27
3T
6T
0
-1
4T
5T
(e)
m(t)
2T
37
6T
4T 5T
(f)

Fig. 12-32 Possible configuration of $m(t)$ for $T = 4T$. (a) Sinusoidal error signal, $T_e = 4T$. (b), (c), (d), (e), and (f), output signal of zero-order hold device.

$$=-45^\circ = 45^\circ$$

27

37

27

m(t) 0

-1

0

T

4T

4T

37

m(1)

+

(i) $-45^\circ < \phi < +45^\circ$

$-45^\circ = -135^\circ$

e(t)

37 4T

37

T

27

U

0

4T

27

(ii) $-135^\circ < \phi < -45^\circ$

e(t)

$= -135^\circ = 225^\circ$

T

4T

m(1) 0

0

27 37 4T

27

3T

(iii) $-225^\circ < \phi < -135^\circ$

e(t)

$= -225^\circ = 45^\circ$

T

27

T

m(1) 0

-1

0

37

4T

37

27

4T

(iv) $45^\circ < \phi < -225^\circ$

(0)

T = 4T, and $\Delta = 1$.

Fig. 12-33 (a) Possible configurations of m(t) and e(t) when varies from 0 to 360 degrees.

in Fig. 12-33b and Fig. 12-32c are identical except for a phase shift.

Clearly the waveform given in Fig. 12-32d, e, and f cannot occur in any form of self-sustained oscillations in this system. Thus, if the system is to have a self-sustained oscillation characterized by T. = 4T, only the waveforms shown in Fig. 12-33a and b need be considered. It is important to notice that the functions shown in Fig. 12-33a have one positive and one negative relay correction during each period T. and that these of Fig. 12-33b have two positive and two negative relay corrections during each period. These have to

be considered separately even though they all have the same period. Similarly, we can show that for all $T_s = nT$, $n = \text{even integers}$, the number of positive relay correction signals is equal to that of the negative relay corrections during one period T_e . This number of corrections is designated by A . For $T_s = 2T$, it is apparent that A can

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$e(t)$

$= -90^\circ$

$\emptyset = 0^\circ$

+1

$m(t)$ 0

$\frac{27}{4T}$

$\frac{4T}{2T}$

$\frac{2T}{3T}$

$\frac{T}{3T}$

-1

(i) $-90^\circ \ll 0^\circ$

$e(t)$

$= -90^\circ = -180^\circ$

$\frac{1}{T}$

$\frac{37}{37}$

$\frac{37}{37}$

$m(t)$ 0

$\frac{t}{0}$

$\frac{27}{4T}$

$\frac{4T}{2T}$

$\frac{2T}{-1}$

$\frac{-1}{(ii)}$

$-180^\circ \ll -90^\circ$

+1

$\frac{27}{4T}$

$\frac{4T}{2T}$

$m(t)$ 0

$\frac{T}{37}$

$\frac{37}{-1}$

$\frac{-1}{(iii)}$

$-270^\circ \ll -180^\circ$

$= -180^\circ = -270^\circ$

$\frac{AT}{2T}$

$\frac{2T}{3T}$

$e(t)$

$\emptyset = -270^\circ$

$\frac{27}{4T}$

$\frac{4T}{T}$

$\frac{T}{2T}$

$\frac{2T}{m(1)}$

0

$\frac{37}{3T}$

$\frac{3T}{4T}$

(iv) $-360^\circ \ll -270^\circ$

(b)

$T_e = 4T$, $A = 2$.

Fig. 12-33 (b) Possible configurations of $m(t)$ and $e(t)$ when varies from 0 to 360 degrees.

be only unity. In general, for $T_s = nT$, $n = \text{even integers}$, the values

of A can be 1, 2, 3,

(n = 1).

For a given period of $T_c = nT$, n = even integers, the loci of $-1/N(z)$ form regions in the decibel versus phase shift plot, each for one possible value of A. These regions are later defined as the "critical regions" for the specific T_c and A. The symbols $-1/N(z)|_{\max}$ and $-1/N(z)|_{\min}$ are used to indicate the boundaries of the region.

We can show that all the periodic functions given in Fig. 12-33a have the same critical region, thus indicating that the study of the conditions of self-sustained oscillation is independent of the phase shift of $m(t)$ once T_c and A are given.

For $T_c = nT$, n = odd integers, the number of positive and negative relay corrections may be different. This fact is illustrated in Fig. 12-34b for $T_c = 3T$. In this case, the designation of the type of oscillation by $\Delta = 1, 2, 3, \dots$ is inadequate. This type of oscilla-

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$m(t)$

$e(t)|$

T

2T

3T

$m(t)$

3T

T

2T

$m(1)$

+1

(i) $-60^\circ < < 0^\circ$

-60°

8

Q

2T

3T

$e(t)|$

0

$= -60^\circ = -120^\circ$

2T

3T

(ii) $-120^\circ < -60^\circ$

3T

0

T

2T

-1

$\cdot = -180^\circ$

$= -120^\circ$

$e(t)$

0

2T

(iii) $-180^\circ < 0 < -120^\circ$

$m(t)$

T

0

2T

3T

0

(iv) $-240^\circ < < -180^\circ$

$m(t)$

T
27
3T
-1
m(t)
e(t)|
(v) $-300^\circ < < -240^\circ$

2T
-1
3T
37
-180°
240°
-240°-300°

2T
3T
e(t)
0
\$ = 300° = -360°

T 2T
37
(vi) $-360^\circ < < -300^\circ$

T = 3T, A = 1.

Fig. 12-34 (a) Possible configurations of m(t) and e(t) when varies from 0 to 360 degrees.

tion is designated by A, where i is the number of positive relay corrections and j is the number of negative relay corrections during each period Te. Figures 12-34a and b show that if the self-sustained oscillation has a period of 3T, the oscillation may be any one of the three types: A12, A21, or A (the same as ▲ = 1).

1

=

The typical procedure for the derivation of $-1/N(\approx)$, $-1/N(2)\text{linax}$, and $-1/N\text{min}$ is given in the following for T. 4T. With the same procedure, the expressions for $-1/N(z)$, $-1/N(2)\text{linax}$, and $-1/N(\approx)\text{linin}$, for the relay with dead zone, are derived for T. = 2T, 3T, . . . 8T. The corresponding critical regions are plotted in gain-phase plots in Figs. 12-35 through 12-41. If necessary, the critical regions of $-1/N(2)$ can be derived for Te 9T, 10T, . . . etc.

=

Case (1) Te = 4T, A = 1

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m(t)

+1

e(t)

0 = -30°

0 = 30°

27

27

-1

3T

3T

(i) $-30^\circ << 30^\circ$

m(t)

+1

e(t)|

0 = -30° 0

-90°

0

27
 -1
 T
 27
 3T
 3T
 m(t)
 (ii) - $90^\circ \ll -30^\circ$
 37
 T 2T
 -1
 m(t)
 - 1
 27
 m(t)
 e(t)
 $\emptyset = -90^\circ$
 $= -150^\circ$
 27
 37
 (iii) - $150^\circ < < -90^\circ$
 (iv) - $210^\circ < < -150^\circ$
 e(t)|
 0

(v) - $270^\circ \ll -210^\circ$
 $\emptyset = -210^\circ$
 $= -150^\circ$
 $\emptyset = -210^\circ$

37
 $\emptyset$
 27
 $\emptyset = -270^\circ$
 3T
 27
 e(t)|
 $= -270^\circ = -330^\circ$

T
 0
 27
 3T
 +1
 T

0
 -1
 12T 3T
 m(t)

+1
 27
 -1
 37
 (vi) - $330^\circ \ll -270^\circ$

Te 3T, A12, 421-

Fig. 12-34 (b) Possible configurations of m(t) and e(t) when varies from 0 to 360 degrees.

If the output signal of the zero-order hold in Fig. 12-31 is of the form shown in Fig. 12-33a and i, the Laplace transform of the relay output v*(t) is written

1
 $V^*(s)$
 $= \frac{1}{e}$
 $-2TS + e$

-ATS
 -6T's
 -
 e +
 =
 $1 + e^{-21}$
 (12-104)
 The z-transform of $v^*(t)$ is
 $V(z)$

=
 $\frac{x^2 + 1}{1}$
 Thus from Eq. (12-101)

$\frac{1}{N(z)}$
 =
 $\frac{E(z) - Ex(z \cos \sin)}{V(z)}$

=
 $\frac{2^2 + 1}{2}$
 $\sin)$
 $\frac{+1}{2}$
 $E(z \cos \sin)$
 (12-105)
 (12-106)

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 Decibels

18
 16
 ∞
 $- N_1(z) \mid \text{Max}$

ma
 $T=12-$
 12
 $T=10-$

8
 8
 7
 $N(z)$

6
 0
 Decibels

-4
 -8
 16
 $\cdot G_1(z) = 5$

$T=4-$
 $N(z)$ min
 $G_1(z)$

3-
 -220-210 -200 -190 -180-170-160 -150 -140 -130

Phase shift
 Fig. 12-35 Critical region for relay with dead zone, $T_d = 2T$,
 and $G_1(z)$ plot, $G(s) = 1/s(s + 1)$.

12
 -izmin 412,421)

- Nilma (4=1).

N(2

max

2

-81

-220°

-200°

Asymptotic

to = -180°

∞

-180°

T= 8 sec

7

6

3

-Niz)lmin (4=1)

~G₁(z) (T_c=3T)

Phase shift

-160°

-140°

Fig. 12-36 Critical regions for relay with dead zone, T, = 3T,
and G₁(z) plot, G(s) = 1/s(s + 1).

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Decibels

16

12

Asymptotic

to = -180°

∞

N(z)lmin(41)

-zmin(4=2)

-8

-220°

-200°

∞

(4-1)

N(2)

Asymptotic

to = -135°

T-6 sec

5

3

4

2

G₁(z) (T=4T)

1.5

-180°

-160°

-140°

Phase shift

Fig. 12-37 Critical regions for relay with dead zone, T_c = 4T,
and G₁(z) plot, G(s) = 1/s(s + 1).

∞

16

-Nizmin 423, 432)-

12

Nizil max(4=2)

8

Decibels

D

0

-4

-Nzmin(4=2)

-Nizmin 412 421)

1

N(z) min

Asymptotic

to = 180°

6

-N(2) max(412)

5

2

3

1.5

$G_1(z)$ ($T_c=57$).

$X_T=1\text{sec}$

-8

- 220°

- 200°

- 180°

Phase shift

- 160°

- 140°

Fig. 12-38 Critical regions for relay with dead zone, $T_d = 5T$,
and $G(z)$ plot, $G(s) = 1/s(s + 1)$.

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Decibels

Decibels

16

12

4

4

Asymptotic

to = -210°

$N(z)|_{mi}(4=3)$

1

-Niz)lmax (4=2)

-N(z)lmin (4=1))

izmi4=2)

-8L

- 220°

- 200°

Asymptotic

Asymptotic

to = -180°

to = -150°

7

6

max

5

- 180°

Phase shift

$T=1\text{sec}$

- 160°

2

3.

$G_1(z)$ ($T_2 = 6T$)

- 140°

Fig. 12-39 Critical regions for relay with dead zone, $T_d = 6T$,
and $G_1(z)$ plot, $G(s) = 1/s(s + 1)$.

16
Mizil max (4=3)
(z) min(434)
 $T=6/$
12
51
 ∞
4
0
-N(z) min (423)
3
412
-Nizil min (4=3)→→
N(z)
(4-1):
-Niz)lmo(4=2)-
max
(z)lmi(42)
G(z) ($T=7T$).
-4
-8
-220°
-200°
-180°
Phase shift
-160°
2
-140°
<-120°

Fig. 12-40 Critical regions for relay with dead zone, $T_d = 7T$,
and $G_1(z)$ plot, $G(s) = 1/s(s + 1)$.

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Decibels
16
12
8
4
-4
Asymptotic
to = -202.5°
(4-4)
min
max
1
Nizilmin (4=1)
N(z)
1
-z)(4=3)
N(z) min
Asymptotic
to = -180°
Asymptotic
to = -157.5°
- Nizil max(4=1)
2
3

$(4=2)$
 $N(z)_{\max}$
 $N_{\text{izm}}(4=2)$
 $T=1 \text{ sec}$
 $N(2)$
 $\min$
 $-G_1(z) \quad (T=8T)$
 -8
 -220°
 -200°
 -180°
 Phase shift
 -160°
 6
 -140°
 -120°

When T_e
 Fig. 12-41 Critical regions for relay with dead zone, $T, = 8T$,
 and $G_1(z)$ plot, $G(s) = 1/s(s + 1)$.

$=$
 $= 4T$ and $z = 1/90^\circ j$, Eq. (12-106) becomes
 $\frac{1}{N(z)} E(\sin j \cos)$

$=$
 j
 $= E-180^\circ + \& (12-107)$
 With reference to Figs. 12-33a and i, the limitations on the magnitude of $e(t)$ are:

or
 Hence
 $E \cos > D$ and $E \sin$
 $E \min D/\cos$ and

$=$
 $\&$
 $< D$ for $-45^\circ < 45^\circ$
 $E_{\max} = D / | \sin |$

p
 1
 $N(;$

$\ddot{}$
 and

1
 D
 $\max$

$=$
 $E_{\max} | -180^\circ +$

$=$
 $I \sin \$1$
 $-180^\circ + (12-108)$

D
 $=$
 $E \min -180^\circ \&$

$=$
 $N(2)$

1
 1
 $=$
 $(z) \min$
 $N(z) \min$
 Letting
 $E_{\max}$

$=$

then
 $\frac{1}{E_{\max}/D}$ and $E_{\min}$
 $\max$
 $= -$
 $\frac{1}{V(2)} \mid E=E$
 $N(z)$ FE
 $\max$
 $=$
 $=$
 $\sin$
 $\min \cos$
 The values of $-1/N(z) \mid_{\max}$ and $-1/N(x) \mid_{\min}$ when varies from
 -45 deg. to $+45$ deg. are tabulated below:

$=$
 $\cos$
 -180° (12-109)
 $E_{\min}/D$
 (12-110)
 $\frac{1}{-180^\circ} +$ (12-111)
 $\frac{1}{-180^\circ} +$ \$ (12-112)

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ϕ
 -45°
 -30°
 -20°
 -10°
 0°
 $\frac{1}{N(z)} \max$
 $\frac{1}{1.414-225^\circ}$
 $2/-210^\circ$ $2.92/-200^\circ$ $5.78-190^\circ$
 $N(2) \min$
 $\frac{1}{1.414-225^\circ}$ $1.155-210^\circ$
 $\frac{1}{1.06/-200^\circ}$
 $\frac{1}{1.015} / -190^\circ$
 $\frac{1}{-180^\circ}$
 $+10^\circ$
 $+20^\circ$
 $+30^\circ$
 $+45^\circ$
 $5.78/-170^\circ$
 $2.92-160^\circ$
 $2/-150^\circ$
 $\frac{1}{1.414/-135^\circ}$
 $\frac{1}{1.015/-170^\circ}$
 $\frac{1}{1.06} / -160^\circ$ $1.155/-150^\circ$
 $\frac{1}{1.414/-135^\circ}$
 $-$

The critical region for the condition of self-sustained oscillation characterized by $T = 4T$ and $A = 1$ is the region enclosed by the loci of $-1/N(z) \mid_{\max}$ and $-1/N(z) \mid_{\min}$ and is shown in Fig. 12-37. We can show that the oscillations shown in Fig. 12-33a all have the same critical region.
 Case (2) $T_e = 4T$, $A = 2$

It is shown in Fig. 12-33b that when a system has a sustained oscillation of period $4T$, the output of the hold circuit can also have the form of $\Delta = 2$. The Laplace transform of $v^*(t)$ which corresponds to the waveform shown in Fig. 12-33b and is

$$V^*(s) = 1 + \frac{T_s}{-2T_s} e^{-2Ts} + e^{-4Ts} + \dots$$

Hence

$$\text{Thus } V(z) = \frac{1+e^{-2Ts}}{z^2 + 1}$$

$$= \frac{1+e^{-2Ts}}{z^2 + 1} \quad (12-114)$$

$$= \frac{N(z)}{D(z)}$$

$$= \frac{0.707}{z^2 + 1} E(z \cos \sin)$$

For $z = j$, Eq. (12-115) becomes

$$= \frac{N(j)}{D(j)} = 0.707 E^{-135^\circ} + \dots \quad (12-116)$$

The range of ϕ as indicated in Fig. 12-35b and ϕ is between 0° and -90° . Therefore, the limitations on E are:

$$D(z) = EIDAX$$

$$= \infty \quad E_{\min}$$

$$\text{for } -45^\circ < \phi < 0^\circ$$

$$\sin \phi \quad E_{\max}$$

$$= \infty \quad E_{\min}$$

$$\text{for } 90^\circ < \phi < -45^\circ$$

$$\text{Thus } D(z)$$

$$= \frac{1}{N(z)} \quad \text{max} = \infty \quad \text{for } 0^\circ > \phi > -90^\circ \quad (12-117)$$

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$$= N(2) \min_{-135^\circ + \text{for } -45^\circ \leq \theta < 0^\circ} \sin$$

$$= N(z) \min_{-135^\circ + \text{for } -90^\circ \leq \theta < -45^\circ} \cos \quad (12-119)$$

The critical region of $T_e = 4T$ and $A = 2$ is plotted in Fig. 12-37. Examination of the expression derived for $-1/N(z)$ and the conditions on $e(t)$ that give each value of θ leads to the following conclusions:

- (1) The discrete describing function $N(z)$ of the relay under consideration is a function of the relay dead zone D , and of the frequency ω , the amplitude E , and the phase of the input sinusoid $e(t)$.
- (2) The regions bounded by the loci of $-1/N(z)$ and $-1/N(2) \sin$ in the gain-phase plot for $T_e = nT$ ($n = 2, 3, 4, \dots$) are symmetrical about the -180° deg axis. The maximum widths for the critical regions are $(2\pi/T_c)T$ for $T_e = 4T, 6T, 8T, \dots, nT$, $n = \text{even integers}$, and $(2\pi/T_c)(T/2)$ for $T_e = nT$, $n = \text{odd integers}$. When n is very large, the critical regions are very narrow and finally approach a straight line which lies on the 180° deg axis when n is infinite.

- (3) For certain values of E and θ , $-1/N(z)$ is infinite, and the critical regions extend to infinity (open region). The asymptote of the $-1/N(z)$ in ω boundaries is always the -180° deg. axis, whereas the asymptotes of the $-1/N(z)$ in θ boundaries are $(2\pi/T_c)(T/2)$ for $n = \text{even integers} > 2$, and $\pm(2\pi/T_c)(T/4)$ for $n = \text{odd integers greater than 1}$.

Stability Analysis and Limit Cycles From Gain-Phase Plot and Describing Function

The condition of self-sustained oscillation in the system of Fig. 12-30 is defined as existing when the denominator of Eq. (12-98) is equal to zero; that is,

$$1 + N(z) G_1(z) = 0$$

Equation (12-120) can also be written as

$$G_1(z) = -1/N(z) \quad (12-121)$$

Since the left-hand side of Eq. (12-121) is the z -transfer function of the linear plant and the right-hand side consists of the negative inverse of the z -transform describing function of the nonlinear element, the stability analysis of the nonlinear system with sampled-data follows the well-known procedure of the describing function.

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and gain-phase plane studies of a nonlinear system with continuous-data. In terms of the gain-phase plot of $G_1(z)$, the graphical stability study of the sampled-data system consists of the following steps:

- (1) Plot $G_1(z)$ on gain-phase coordinates for $T_c = 2T, 3T, 4T, \dots$ using T as a parameter on the loci.

- (2) Superpose on the gain-phase coordinates the family of critical regions of $1/N(z)$ for $T_e = 2T, 3T, 4T,$

The following conclusion can be reached from the inspection of

the relative position of the $G_1(z)$ loci and the critical regions:
 (1) If a portion of the $G_1(z)$ locus for some T_e falls within the critical region of this T . for a certain relay dead zone D (when D varies, the critical regions of $-1/N(2)$ simply shift up or down along the 180° degree axis), then a set of E , 6 , A and T_e , exists such that

=
-

$$G_1(2) = -1/N(2)$$

Consequently, for any sampling period T the portion of the $G_1(z)$ locus that lies in the corresponding critical region will produce a self-sustained oscillation characterized by Δ at T_e , p , and E , and the system will be unstable.

(2) If the $G_1(z)$ loci for more than one value of T_e fall within their respective critical regions, for the same value of D the system can have more than one mode of oscillation.

(3) The T 's along the portion of the $G_1(z)$ locus outside the corresponding critical region provide a stable system.

A

One special condition exists when the relay dead zone D becomes zero and the relay is considered to be ideal. From the physical viewpoint the relay will provide a corrective output whenever there is a signal at its input, no matter how small this signal may be. Under this condition some types of oscillation disappear in the nonlinear system, and some can only appear for several discrete values of p . For instance, when $D = 0$, the sustained oscillation of the types, $T_e = 4T$ and $\Delta = 1$ can occur only in the system when the phase angle of $e(t)$ is a multiple of $\pm 7/2$. This fact can be observed by referring to Fig. 12-37 in which the critical regions for $A = 1$ and 2 are shifted to

- along the -180° deg axis when D becomes zero.

Evidently, the critical region for $A = 2$ degenerates into a region which is bounded by two vertical straight lines with the left- and the righthand boundaries located at -225° deg and -135° deg, respectively. The critical region for $A = 1$ degenerates into a single vertical line along the -180° deg axis. For $T_e = 6T$ and $D = 0$, self-sustained oscillation of the type $A = 1$ can no longer exist. This is due to the period of $6T$ of the sinusoidal $e(t)$. This $e(t)$ will produce more than two-relay corrective signals in either direction during

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=

one period, whereas $A = 1$ specifies that only one positive and one negative relay correction can be made during each period T .

Referring to Fig. 12-39, the critical region for $A = 1$ is a closed region; when this closed region is shifted to ∞ along the -180° deg axis, it is no longer expected to enclose any portion of the $G_1(2)$ locus for $T_e = 6T$. Similarly, the critical region for $A = 3$ becomes a vertical strip centered along the -180° deg axis with its left and right-hand boundaries located at the -210° deg and -150° deg lines, respectively. The critical region for $A = 2$ is narrowed down to a vertical line along the -180° deg axis. For $T_e = 8T$, the sustained oscillations are characterized only by $A = 3$ and $A = 4$, and those of $\Delta = 1$ and 2 are eliminated.

=

The following example gives a numerical illustration of the application of the critical regions in the sinusoidal study of a nonlinear sampled-data control system.

Example 12-6

Consider that the open-loop transfer function of the relay type sampled-data system of Fig. 12-30 is given by

$$G(s) = \frac{1}{s(s+1)} \quad (12-122)$$

The sampling period is one second, and the relay has a dead zone of 0.1. With reference to the system configuration of Fig. 12-31, we have

$$G_1(s) = \frac{1}{s} \times \frac{1}{s(s+1)} e^{-Ts}$$

The z-transform of $G_1(s)$ is given by

$$G_1(z) = \frac{1 - e^{-T}}{z(T - 1 + e^{-1})} + \frac{1 - e^{-TM}}{z(T + 1)} \quad (12-123)$$

$$= \frac{1 - e^{-T}}{z(T - 1 + e^{-1})} + \frac{1 - e^{-TM}}{z(T + 1)} \quad (12-124)$$

The loci of $G_1(z)$ with T as a variable parameter are plotted in Fig. 12-42 for $T, 2T, 3T, 4T, \dots, 8T$. For the relay dead zone specified ($D = 0.1$), the normalized critical regions of Fig. 12-35 through 12-41 are shifted down by 20 db. Then superposition of these shifted critical regions on the $G_1(z)$ loci of Fig. 12-42 shows that the following modes of oscillations are possible in this system when $T = 1$ sec.

TABLE 12-3

Mode of Osc. T_e (sec.)

C
Amplitude of oscillation
describing function

1
2
3
4
4456
122
method
0.31
0.438
3
0.572
0.838

For example, when $T_e = 4T$, $z = 190^\circ$; when T_e relationships between T , and z are established.

difference equation
method

0.30
0.352
0.525
0.80

5T, z

172°; etc. Thus the

=27
 16
 $T_o = 4T$
 $T_c = 57$
 $6T \quad 7T$
 $T_c = 37$
 $T = 12\text{sec}$

12
 $T = 8$
 -10-
 -9
 87
 Constant
 T loci
 8
 7
 0
 Decibels

4
 -4
 -8
 -10L
 -200°
 -190°
 -180°
 -6

5
 4
 1.5
 -170°
 -160°

Phase shift
 Constant frequency
 loci $G_1(z)$
 -150°
 $-140^\circ \quad -130^\circ$
 $G_1(s)$

=
 -
 $1(1 - e^{-ST})$ 7
 $s^2(s + 1)$

Fig. 12-42 Frequency loci of $G_1(z)$ in gain-phase plane.

These results are obtained by observing that when $D = 0.1$, the $T = 1$ sec. points on the $G_1(z)$ loci fall inside their respective critical regions for only the four modes of oscillations listed above. When $D = 1$, the critical regions of Figs. 12-35 through 12-41 are not shifted; all the $T = 1$ sec. points on

the $G(z)$ loci fall outside their respective critical regions, and the system is always stable (without sustained oscillation). The amplitude of oscillation of $c(t)$ can be computed directly from the $G_1(z)$ loci. For $r(t) = 0$, the amplitude of $c(t)$ is equal to the amplitude of the error $e(t)$. Therefore,

$$E = |e(t)| = |c(t)|$$

Also, self-sustained oscillations occur when

$$G_1(z) = 1/N(z)$$

(12-125)

(12-126)

For certain T . and T , when z is a point on the unit circle in the z -plane, say, 21, we can write

$$|G_1(z)| = | -1/N(z) | \quad (12-127)$$

The values of $| -1/N(z) |$ suggest a simple method of predicting the amplitude of oscillation of $c(t)$ from the values of $G_1(z)$. We observe that $| -1/N(z) | = k E$, where k is a constant for any T_e . From Eqs. (12-125) and (12-127), we have

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$$E = |G_1(z)| \quad (12-128)$$

Equation (12-128) implies that whenever the $G_1(z)$ locus for a certain T_e is given, the locus of the amplitude of oscillation of $c(t)$ is related to the $G_1(z)$ locus by only a constant factor k . However, we should keep in mind that Eq. (12-128) is valid for only the portion of $G_1(z)$ which lies inside the critical regions for the same T_e . For example, for $T_s = 4T$ and $\Delta = 1$, Therefore, Eq. (12-128) gives

$$E = |G_1(z)| T_s = 4T \quad (12-129)$$

which means that when the system has a self-sustained oscillation which is characterized by $T_s = 4T$ and $\Delta = 1$, every point on the $G_1(z)$ locus inside the critical region in the gain-phase plot represents the amplitude of $c(t)$ for the corresponding sampling period T . In the present example for $T = 1$ sec. the following value of E is obtained from the $G_1(z)$ locus in Fig. 12-37: $E = |G_1(z)| T_s = 4 \text{ sec} = 0.31 \text{ (-10 db)}$

Similarly, for $T_s = 4T$ and $\Delta = 2$, Table 12-3 gives $| -1/N(z) | = 0.707E = |G_1(z)| T_s = 4T$

$E = 0.438$
Therefore, for $T = 1 \text{ sec.}$, $E = 0.31/0.707$

An interesting comparison can be made of the results that are obtained in this example by use of the sinusoidal analysis with describing function to those that are computed in the same system by the state transition method in Sec. 12.2. In Sec. 12.2 the same relay sampled-data system is analyzed by means of the difference equation approach. Figures 12-4 and 12-5 show the four different modes of oscillations of the system when the system is subjected to various initial conditions. The amplitudes of oscillation of $c(t)$ are tabulated in the last column of Table 12-3. Whereas the difference equation approach gives the solution for only a particular set of initial conditions, the determination of the possible modes of oscillations under various initial conditions is extremely difficult. However, the describing function method introduced in this section does point out that for the system under study with $T = 1 \text{ sec.}$ and $D = 0.1$, only the four modes of oscillations listed in Table 12-3 are possible under any initial conditions.

The results in Table 12-3 show that the amplitudes of self-sustained oscillation which are predicted by the two different methods are quite close, except for mode 2 when $T_e = 4 \text{ sec}$ and $\Delta = 2$. Note that both the difference equation and the discrete describing function methods deal with the system response at the sampling instants only. Furthermore, the describing function method assumes that the input to the sampler is sinusoidal and that only the fundamental component of the relay output is considered to be significant. In reality, the oscillations in a nonlinear system are seldom sinusoidal. Therefore, the accuracy of the sinusoidal analysis depends entirely on how

much the output $c(t)$ differs from a sine wave. In fact, the amplitudes of

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oscillation listed in Table 12-3 which are determined by the discrete describing function are extremely close to the results obtained by Chowt using the conventional describing function method.

Nonlinear Systems with More than One Synchronized Sampler

One distinct advantage of the discrete describing function method is that it can be applied to a nonlinear system with more than one synchronized sampler. Figure 12-43 shows the block diagram of a nonlinear sampled-data system with two samplers synchronized to open and close at the same time. The closed-loop transfer function of this multi-sampler system is

$r(t)$

N

Sampler

T

Relay

$H(s)$

Zero-order

hold

$G(s)$

$c(t)$

$G_1(s)$

Sampler T

Fig. 12-43 Nonlinear sampled-data system with two synchronized samplers.

$C(z)$

$R(z)$

-

$N(z) G_1(z)$

$1 + G_1(z) H(z) N(z)$

(12-133)

The stability study of the system now involves the investigation of the following condition:

$G_1(z)H(z) = -1/N(z)$

(12-134)

For the relay-type nonlinearity with dead zone, all the critical regions shown in Figs. 12-35 to 12-41 are still valid for this system.

Nonlinear Systems Without Zero-Order Hold-Finite Pulse Width

Considerations

The nonlinear system studied in the preceding sections is considered to have a zero-order hold following the sampler. If the zero-order hold is absent and if the sampling duration of the sampler can be assumed to be infinitesimal, the describing function technique presented here can still be applied. Instead of plotting $G_1(z)$, which includes the transfer function of the hold, only the locus of $G(z)$ in the gain-phase plot need be plotted using the same critical regions. However, in some cases, without a hold device a finite pulse

+ C. K. Chow, op. cit.

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width has to be considered. In this case, an approximation can be made if the narrow pulses with width p are approximated by flat-topped pulses. The sampler can then be represented by an ideal sampler followed by a fictitious hold device which holds the sampled signal for p second only and then drops to zero instantaneously until the next impulse comes along. The transfer function of $G_1(s)$ is

written

$G_1(s)$

--
The z-transform of $G(s)$ is

Conclusion

1

-

$\tilde{G}(s)$

$G(s)$

S

$G_1(z) = (1 - x^{-1})^3 [G(s)] \quad (p \ll T)$

-

(12-135)

(12-136)

A discrete describing function utilizing the z-transformation has been introduced for a relay-type nonlinearity for sampled-data systems. The critical regions which are defined by the describing function are used to study the condition of self-sustained oscillations by investigating the relative position of the $G_1(z)$ loci and the critical regions. The effect of varying the sampling time T on the system stability is shown clearly on the $G_1(z)$ loci. As has been shown, the method can be applied to multisampler system as well as to systems without hold devices. We might note that the discrete describing function method can also be applied to other types of amplitude dependent nonlinearities such as saturation with dead zone.

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PROBLEMS

Problems for Chapter 2

$x(t)$

K

M

F

Figure 2P-1

2-1 A particle of mass M is connected as shown in Fig. 2P-1 and is at rest in its equilibrium position at $t = 0$. The force is in the form of blows of impulses of strength F which are given to the particle at $t = nT$ sec ($n = 0, 1, 2, 3, \dots$). Write the equation of motion for the system and solve for the displacement $x(t)$ of the particle by the Laplace transform method. K linear spring constant = $M(u^2 + w^2)$; $u = \text{constant}$ and $2\pi/T$; f viscous friction coefficient = $2Mu$.

N elementary pulses of width A , such that $NA = p$, and Δ is very small. Therefore, a practical sampler can be represented by N samplers $S_0, S_1, S_2, \dots, S_{(N-1)}$, connected in parallel as shown in Fig. 2P-7b, implying that sampler S_i samples at instants $0, T, 2T, \dots, nT, \dots$; sampler S_1 is actuated at instants $A, A + T, A + 2T, \dots, A + nT$; etc. Derive the pulsed transform of $e^*(t)$.

Problems for Chapter 3

...9

3-1 Two different schemes of triangular hold are shown in Figs. 3P-1a and 3P-1b, respectively. Write the impulse responses and the transfer functions of the triangular holds. Which of these is physically realizable?

500

PROBLEMS

$e(t)$

$e(t)$

t

or 27 37 47 57 67

or 27 37 47 57 67

(a)

(b)

Figure 3P-1

$e(t)$

0T 2T 3T 4T 5T 6T 7T

Figure 3P-2

3-2 Show that the polygonal approximation of a continuous signal shown in Fig. 3P-2 can be made by use of the two triangular holds of Prob.

3-1. Derive the impulse response and the transfer function of the polygonal hold.

Problems for Chapter 4

4-1 Derive expressions for the z-transform of $1/s^n$ and $1/(s+a)^n$. From these z-transform expressions derive the z-transform of $1/s^3$, $1/s^3$, and $1/(s+a)^3$.

4-2 Obtain the z-transform of the following functions:

1

(a) $5(5^2 + 1)$

4-3 Obtain the inverse z-transform of

1

(b) $5^2(s+1)$

0.52

$G(z)$

=

-

$(z - 1)(z - 0.5)$

4-4 Evaluate the inverse z-transform of

$z(x^2 + 2x + 1)$

$G(z)$

=

$(x^2$

-

$+ 1)(z^2 + z + 2)$

by means of the following methods:

(a) the real inversion formula

(b) the partial fraction expansion

(c) power series expansion.

4-5 Obtain the initial-value and the final-value of the inverse of the following functions:

2

(a) $G(z) =$

(1

$$\approx^{-1})(1 - 0.2\approx^{-1})$$

PROBLEMS

501

1

(b) $G(2)$

=

(1

-

(1 $0.52-1$)

4-6 Find the modified z-transform of the following functions:

(a) $1/s^2$

1

(b)

$52(s + a)$

1

(c)

$s(s + a)(s + b)$

Do the modified z-transforms of these functions check with their z-transforms when $m = 1$?

$e(t)$

(t)

$G(s)$

Figure 4P-7

(1)

$c(t)$

4-7 For the open-loop sampled-data system shown in Fig. 4P-7,

2

$G(s) =$

and T

$= 1 \text{ sec}$

$s(s+2)$

For a unit step input, determine the output response at $t = 0, 1/3, 2/3, 1, 4/3, 5/3, 2$, seconds. Use the submultiple sampling method.

4-8 For the open-loop sampled-data system shown in Fig. 4P-7, given that $s^2 + 3.18s + 13$

$G(s)$

=

$T = 1 \text{ sec } r(t) = \text{unit step}$

$(s + 1)(s^2 + 0.04s + 10)$

(a) Obtain the z-transform of the output. Evaluate $c(nT)$.

(b) Obtain the output between sampling instants.

(c) Determine the initial-value and the final-value of the output response

$c(t)$.

4-9 For the open-loop sampled-data system shown in Fig. 4P-7, given

$G(s) =$

100

$s(s^2 + 100)$

$T = 0.1 \text{ sec } r(t) = \text{unit step}$

(a) Use the z-transform method to evaluate the output response.

(b) Use the submultiple sampling method to evaluate the output response at $t = 0, T/3, 2T/3, T, 4T/3, \dots$

Problems for Chapter 5

5-1 For the following sampled-data systems construct the original signal flow graph, the sampled signal flow graph, and the composite signal flow graph. Evaluate $C(s)$ and $C^*(s)$ from the flow graphs.

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$R(s)$

$G_2(s)$

$G_1(s)$

(a)
 $D^*(s)$
 $+$
 $R(s)$
 $C(s)$
 $G_1(s)$
 $G_2(s)$

$+$
 TX
 $D^*(s)$
 (b)
 $R(s)$
 $C(s)$
 $G(s)$

T
 $H_1(s)$
 $D^*(s)$
 T
 $H_2(s)$

(c)
 $C(s)$

Figure 5P-1

5-2 Draw the state transition flow graphs for the following difference equations. Include all initial conditions on the flow graphs. Solve these equations using the flow graphs and the z-transform.

(a) $y[(n + 2)T] + 5y[(n + 1)T] + 3y(nT) = nT$ $y(0) = 1$, $y(T) = 0$

(b) $y[(n + 2)T] + 3y[(n + 1)T] + 2y(nT) = e^{-1}y(0)$ $y(0) = 0$, $y(T) = 0$

5-3 Draw the state transition flow graph for the simultaneous difference equations

-
 $y_1[(n + 2)T] + 3y_1[(n + 1)T] + y_1(nT) - 2y_2(nT) = r_1(nT)$
 $y_2[(n + 2)T] + y_2[(n + 1)T] - 4y_1(nT) + y_2(nT) = r_2(nT)$

Find $Y_1(z)$ and $Y_2(z)$. The initial conditions are

$y_1(0) = 1$,

$y_2(0) = 2$,

$y_1(T) = 0$

$y_2(T) = 0$

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$R(s)$
 $G_1(s)$
 $G_2(s)$
 $C(s)$
 $H_2(s)$

(d)
 $H(s)$

T
 $.X$

T
 $R(s)$
 $G_1(s)$
 $G_2(s)$
 $G_3(s)$

$C(s)$
 $H(s)$

(e)
 $H(s)$

$G_2(s)$

(f)

$+$
 $.X$

T

$R(s)$
 $G_1(s)$
 T
 $G_2(s)$
 $C(s)$
 $GG(s)$
 $G_5(s)$
 $R(s)$
 $G_2(s)$
 $G_2(s)$
 $G_4(s)$
 $C(s)$
 $r(1)$
 $D(s)$
 $G_2(s)$
 (g)

Figure 5P-1 (continued)

z.o.h.

$G(s)$

$c(1)$

Figure 5P-4

504 PROBLEMS

5-4 For the closed-loop sampled-data control system shown in Fig. 5P-4,

$G(s)$

$=$

1

$s(s+1)(s+2)$

The input signal is a unit step function.

T 1 sec

(a) Draw the state transition flow graph for the system.

(b) Obtain the state transition equations in the Laplace domain for

$X_1(s)$, $X_2(s)$, and $X(s)$, where $X_1(s) = C(s)$.

(c) Obtain the state transition equations for $x_1(t)$, $x_2(t)$, and $x(t)$.

$r(t)$

$e(t)$

$ep(1)$

(p)

$h(t)$

z.o.h

$S+1$

Finite duration

sampler

$ep(t)$

$Ah(t)$

$e(t)$

$c(t)=x_1(t)$

Π

kT $kT+p$

KT $KT+p$ $0p$

T $T+P$

$0p$

T $T+P$

Figure 5P-5

5-5 A sampled-data feedback control system is shown in Fig. 5P-5. The

sampler is considered to have a finite sampling duration p (second),

and the sampling period is T second; ($T > p$). The zero-order hold

holds the output of the sampler at the constant level of $e(kT + p)$

during the time interval $kT + p < t < (k + 1)T$. The output relations

of the sampler and the hold are illustrated in the following.

(a) Write the transition equation (or equations) of the system during

the time interval $kT < t < kT + p$.

(b) Write the transition equation (or equations) of the system during the time interval $kT + p < t < (k + 1)T$ as a function of the initial conditions and the input signal.

Assume that the input signal is a unit step function.

$r(t)$

1

K_k

Variable-gain

amplifier

$s(s+1)$

T

$c(t) = x_1(1)$

Figure 5P-6

5-6 In the feedback control system shown in Fig. 5P-6, K represents the gain of a time-varying amplifier; K changes value with respect to time. However, K can change value only at $t = kT$, where $k = 0, 1, 2, \dots$, and $T = 0.1$ second. Determine K for $k = 0, 1, 2$, so that the system reaches a deadbeat response after 0.2 second. The input is a unit-step function.

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Problems for Chapter 6

6-1 For the error-sampled control system with unity-feedback shown in Fig. 6P-1,

$r(t)$

$G(s)$

$c(t)$

1

Figure 6P-1

$G(s) = 5^2(s+5)$

$T = 1$ sec

$r(t) =$ unit step function

(a) Obtain the z-transform of the output.

(b) Obtain the output response at the sampling instants.

(c) Determine the final-value of the output response.

6-2 The characteristic equations of certain sampled-data control systems are J given as follows. Determine the stability of these systems.

(a) $z^3 + 5z^2 + 3z + 2 = 0$

(b) $3z^5 + 4z^4 + z^3 + 2z^2 + 5z + 1 = 0$

(c) $z^4 + 9z^3 + 3z^2 + 9z + 1 = 0$

(d) $z^3 - 1.522z^2 + 2z + 3 = 0$

(e) $24z^4 + 1.36823z^3 + 0.412622z^2 + 0.08z + 0.0025 = 0$

$r(t)$

$G(s)$

$C(1)$

(a)

$\frac{z}{z-1}$

T

$10 + T$

(c)

Figure 6P-3

Hold

$G(s)$

$-c(t)$

(b)
6-3 The sampled-data system shown in Fig. 6P-3b is used to approximate the continuous system in Fig. 6P-3a. The transfer function of the continuous system is

$$\frac{1}{s(s+1)}$$

The input signal is a unit step function.

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- (a) Evaluate the output of the sampled-data model at the sampling instants when the hold device is a zero-order hold. The sampling period is 0.25 second.
(b) Evaluate the output of the sampled-data model at the sampling instants when the hold device is a triangular hold with impulse response shown in Fig. 6-3c. The sampling period is 0.25 second.
(c) Compare the responses in parts (a) and (b) with the response of the continuous system.
(d) Repeat parts (a), (b), and (c), when the sampling period is 1 second.

6 4 It is convenient to use a sampled-data model to approximate a continuous-data system, especially when the system contains a transportation lag. A continuous-data system with transportation lag is shown in Fig. 6P-4a. Two possible sampled-data models which may be used to approximate the continuous system are shown in Fig. 6P-4b and 6P-4c.

$$\frac{e^{-sT}}{s(s+1)}$$

(b)

$$\frac{e^{-sT}}{s(s+1)}$$

(c)

Figure 6P-4

- (a) Determine the output responses of the sampled models at the sampling instants. The input is a unit step function, and $T = 0.5$ second.
(b) Discuss what factors must be considered in the selection of the sampling period in the two sampled models.

65 Given the following closed-loop transfer functions for sampled-data systems, determine the percent overshoot and peak time of the step response.

$$(a) \frac{0.5}{(z-0.5)(z-0.2)}$$

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$$\frac{0.52}{(z-0.5)(z-0.2)}$$

(c) $R(z)$

$$= \frac{0.388(z+0.002)(z+0.717)}{(z^2 -$$

$$+ 0.632)(z - 0.05)$$

6-6 For the sampled-data control system shown in Fig. 6P-1,

$G(s)$

$$= \frac{K}{s(1+0.2s)}$$

(a) Sketch the root loci for the system without the zero-order hold for $T = 0.1$ second and $T = 1$ second. Determine the marginal value of K for stability in each case.

(b) Repeat part (a) when the system has a zero-order hold.

6-7 The following polynomial in z represents the characteristic equation of a certain sampled-data control system. Sketch the root loci of the system. Determine the marginal value of K for stability.

$$23 + Kz^2 + 1.5Kz(K + 1) = 0$$

6-8 The open-loop transfer function of the error-sampled control system shown in Fig. 6P-1 is given by

$$0.012K(z + 2.3)(z - 0.15)(z - 0.22)$$

$$\text{GhoG}(z) = (z - 1)(z - 0.05)[(z - 0.532)^2 + (0.29)^2]$$

(a) Sketch the root loci in the z -plane as a function of K .

(b) Find the value of K which will give a relative damping ratio of 50 percent to the dominant control poles. What is the velocity error constant?

6-9 Sketch the root loci in the z -plane for the sampled-data system shown in Fig. 6P-9.

$r(t)$

$r(t)$

z , o. h.

$T = 0.1$ sec

$K = 0.5$

$$s(s+1)(s+3)$$

$-c(t)$

Figure 6P-9

$G_1(s)$

$c(1)$

$G_2(s)$

Figure 6P-10

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6-10 For the sampled-data system shown in Fig. 6P-10,

3.14

$$G_2(s) = 0.81s + 13.16$$

s^2

$$G_1(s) =$$

1

$s + 1$

The sampling period is 1 second.

(a) Determine the output at the sampling instants for the unit step input.

(b) Determine the output in between the sampling instants (use any method) for a unit step function input. Is the closed-loop system stable?

Problems for Chapter 7

7-1 Sketch the Nyquist plot for the following sampled-data control systems. The systems are assumed to be error-sampled with a zero-order hold and unity feedback. The sampling period is 1 second. Determine the stability of the closed-loop system.

(a) $G(s)$

=

$$s(1 + 0.2s)$$

(b) $G(s)$:

=

1

$$(s + 1)^2$$

Hold

device

2

$$s(1 + 0.2s)$$

$c(t)$

$T = 1 \text{ sec}$

Figure 7P-2

7-2 The block diagram of a sampled-data system is shown in Fig. 7P-2.

Sketch the Nyquist plots of the open-loop transfer function when

(a) the hold device is a zero-order hold.

(b) the hold device is a first-order hold.

Comment on the results.

7-3 For the system in Fig. 7P-2, Problem 7-2, plot the closed-loop frequency response for the two cases. Determine the values of M_r .

7-4 The open-loop transfer function for an error-sampled control system with zero-order hold and unity-feedback is

$G(s) =$

=

1

$$s(s + 1)(s + 2)$$

The sampling period is one second. Determine the phase margin, gain margin, and M_r for the closed-loop system.

Problems for Chapter 8

8-1 Consider the sampled-data system of Fig. 8P-1. The transfer function of the controlled process is given by

$G_1(s)$

=

3

$$s(1 + 0.2s)$$

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$r(t)$.

z.o.h.

$G_c(s)$

$G_1(s)$

$c(t)$

Figure 8P-1

The sampling period is 1 second. Design a continuous-data compensator so that the resonant peak M_r of the system is less than 1.5. The velocity error constant of the system should be maintained at the same level.

8-2 The system shown in Fig. 8P-1 has a transfer function

$G_1(s)$

=

2.5

$$s(s + 1)(s + 2)$$

and a sampling period of 1 second.

(a) What is the transient peak overshoot of the uncompensated system?

(b) Design the continuous-data compensator $G_c(s)$ so that the phase margin of system is 40 degrees or more. What is the peak overshoot of the compensated system?

8-3 A digital compensator has the transfer function

$G(z) =$

-

-

$$(1 - 0.1z^{-1})(1 - 0.5z^{-1})$$

$$(1 - 0.65z^{-1})(1 - 0.82z^{-1})$$

Determine the transfer function $H(s)$ of the equivalent continuous-data feedback structure shown in Fig. 8-16(b).

8-4 A sampled-data system with feedback compensation is shown in Fig.

8P-4. Given

$$r(t)$$

$$H(s)$$

$$z.o.h.$$

$$G_1(s)$$

$$-c(t)$$

Figure 8P-4

$$1$$

$$G_1(s) =$$

$$s(s+1)$$

$$H(s)$$

$$=$$

$$0.123s^2 + 0.448s + 2.42$$

$$(s + 2.35)(s + 1)$$

The sampling period is 1 second. Determine the pulse transfer function.

$D(z)$ of the equivalent system with cascade digital compensation (Fig.

8-15).

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Problems for Chapter 9

9-1 Synthesize series pulsed-data networks to realize the following pulse transfer functions:

$$(1 - 0.12z^{-1})(1 - 0.52z^{-1})$$

$$(a) D(z) =$$

$$(1 - 0.652z^{-1})(1 - 0.8z^{-1})$$

$$(b) D(z) =$$

$$(1 + 2z^{-1})(1 - z^{-1})$$

$$(10.22z^{-1})$$

9-2 Determine which of the following pulse transfer functions can be realized by series pulsed-data network, by feedback pulsed-data network, by both of these two schemes, by neither of these two schemes.

$$(1 - 0.5z^{-1})$$

$$(a) D(z) =$$

$$=$$

$$1$$

$$(b) D(z) =$$

$$=$$

$$(1 - 2z^{-1})$$

$$0.32z^{-1}$$

$$---$$

$$0.18z^{-2}$$

$$(1 - 0.12z^{-1})$$

$$(c) D(z) = 1 - 32z^{-1} + 2z^{-2}$$

$$1$$

$$-$$

$$(d) D(z) =$$

$$=$$

$$32z^{-1} + 2z^{-2}$$

$$1 + 3z^{-1} + 2z^{-2}$$

$$(e) D(z) = 1 - 2z^{-1} + 2z^{-2}$$

9-3 Synthesize a pulse-data R-C network to realize the following pulse transfer function:

$$D(z) =$$

$$z^{-1}(1 - 0.4z^{-1})(1 - 0.9z^{-1})(1 - z^{-1} + 0.5z^{-2})$$

$$-$$

$$(1 - \approx^{-1})(1 - 0.8\approx^{-1})(1 - 0.12\approx^{-1})(1 - 2\approx^{-1})(1 - 32\approx^{-1})$$

-
-
--

9-4 Realize the pulse transfer functions in Problem 9-1 by pulsed-data processing units such as that shown in Fig. 9-10.

9-5 Represent the pulse transfer functions in Problem 9-1 by a set of first-order difference equations. Draw signal flow graphs to portray these difference equations (see Fig. 5-33a).

$r(1)$

$D(s)$

2.0.h.

$G_1(s)$

$c(1)$

T

$T = 1 \text{ sec}$

$\cdot D(z) \cdot$

Figure 9P-6

9-6 For the sampled-data control system shown in Fig. 9P-6, design the digital controller $D(z)$ so that the system satisfies the following requirements:

$K, 30 \text{ sec}$

phase margin

50 degrees

The transfer function of the controlled process is

PROBLEMS 511

$G_1(s) =$

K

$s(s + 2)(s + 10)$

The sampling period is 1 second.

9-7 An error-sampled control system has the block diagram as shown in Fig. 9P-6. The transfer function of the controlled process is

$G_1(s)$

=

$K(s + 1)$

$s(s^2 + s + 2)(s + 3)$

The sampling period is 0.5 second.

(a) Sketch the root loci in the z -plane as a function of K .

(b) What is the marginal value of K for stability?

(c) When K is set at the marginal value for stability, design a digital compensator so that the damping ratio of the closed-loop control poles is equal to 0.707.

9-8 An error-sampled control system with unity-feedback has the open-loop z -transfer function

$G_h G_1(z)$

=

$K(z+1)$

$(\approx -1)^2 (\approx 0.5)$

(a) Sketch the root loci in the z -plane as a function of K .

(b) Design a cascade digital compensator to stabilize the system.

9-9 For the sampled-data control system shown in Fig. 9P-6, the transfer function of the controlled process is

and $T = 1 \text{ second}$.

K

$G_1(s) =$

$s(s+1)$

(a) Design a digital compensator $D(z)$ so that the following specifications are satisfied:

Damping ratio of control poles = 0.707

Peak time t_p , 3 sec

Overshoot 8 percent

K_2

=

1 sec⁻¹

(b) Realize the digital controller by a pulsed-data R-C network.

(c) Determine the unit step response of the compensated system.

(d) Determine the unit ramp response of the compensated system.

(e) Add a dipole near $z = +1$ to the closed-loop transfer function of the compensated system so that the velocity error constant K_v is increased to 10 sec⁻¹. The added transient term in the step response resulting from the addition of the dipole is to have an initial amplitude less than 1 per cent of the unit step input. Evaluate the step response and the ramp response of the compensated system (with increased K_v).

(f) Discuss what effect may the added dipole have on the bandwidth of the closed-loop sampled-data system.

512

PROBLEMS

9-10 Consider the sampled-data control system of Problem 9-6.

(a) Design a digital compensator so that the system output sequence in response to a step-function input has the shortest settling time (deadbeat response).

(b) Design a digital compensator so that the continuous output of the system in response to a step-function input has the shortest settling time without ripples.

9-11 For the sampled-data control system shown in Fig. 9P-6.

$G_1(s)$

=

1

$s(1 + 0.1s)(1 + 0.02s)$

The sampling period is 0.5 second.

(a) Synthesize a digital compensator so that the continuous output follows the step input signal with zero error after a shortest possible transient time.

(b) Plot the system output as a function of time.

9-12 For the system described in Problem 9-11.

(a) Synthesize a digital compensator so that the continuous output follows a ramp input signal with zero error after a shortest possible transient time.

(b) Plot the step response and the ramp response of the compensated system.

Problems for Chapter 10

ultivate & skip sang by)

10-1 Determine the z-transform of the output of the multirate sampled-data system shown in Fig. 10P-1. The input signal is a unit step function.

Plot the output response of the system.

$e(1) X$

1

$c(t)$

$X_c^* (113$

$s(s+1)$

$T_1 = 2 \text{ sec}$

$T_2 = 3 \text{ sec}$

Figure 10P-1

$r(t)$

$c(t)$

z.o.h.

$D_1(s)$

z.o.h.

$G_1(s)$

TIN

Figure 10P-2

10-2 With reference to Fig. 10P-2, the transfer functions of the system are

$$G_1(s)$$

=

K

$$\frac{s(s+1)}{D_1(s)}$$

$$D_1(s)$$

=

The sampling period is 1 second.

1

$$s + 0.5$$

PROBLEMS 513

(a) Investigate the stability of the system as a function of K and N.

Plot the maximum value of K for stability versus N.

(b) Sketch the root loci for the system as functions of K when N = 1,

2, 3, and 4.

10-3 (a) Draw the state transition flow graph for the system in Problem 10-1.

(b) Write the transition equations of the system.

10-4 (a) Draw the state transition flow graph for the system in Problem

10-2 for N = 1 and N = 2.

(b) Write the transition equations from the transition flow graphs.

C(1)

$$12.0.h$$

$$G(s)$$

Sc

Figure 10P-5

10-5 For the sampled-data system shown in Fig. 10P-5,

$$G(s)$$

=

1

$$\frac{s(s+1)(s+2)}{s(s+1)(s+2)}$$

The sampler closes at instants $t = kT, kT + T_1, (k+1)T, (k+1)T + T_1, \dots$, where k is a positive integer ($k = 0, 1, 2, \dots$); T

and T_1

-

0.8 second.

(a) Draw the state transition flow graph for the system.

(b) Write the state transition equations for the system.

= 1 second

$$r(1)$$

1

$$C(1)$$

$$s + 0.5$$

$$z.o.h.$$

$$0.5 \text{ sec}$$

$$z.o. h.$$

5+1

$$1 \text{ sec}$$

Figure 10P-6

10-6 The block diagram of a multirate sampled-data control system is shown in Fig. 10P-6. It is assumed that the samplers are synchronized.

(a) Draw the state transition flow graph for the system.

(b) Write the state transition equations for the system.

(c) Evaluate and plot the system response to a unit step function input.

10-7 The transfer function of the controlled process is

$$G_1(s)$$

=

1

$$\frac{s(s+0.5)(s+1)}{s(s+0.5)(s+1)}$$

(a) Design a multirate digital controller so that the system will respond

514
PROBLEMS

$r(t)$
 $c(t)$
 $D(s)$
→ z.o.h.

$G_1(s)$
1 sec
0.5 sec

Figure 10P-7

to a unit ramp function input with the shortest settling time and zero steady-state error. There should be no ripples in between sampling instants.

(b) Plot the ramp function response of the designed system.

Problems for Chapter 11

(stulis the anger

and

11-1 The auto-power spectral density of a stationary random process $r(t)$ is given by

$\Phi_{rr}(s)$

=

4

0.04 \$2

-

Find the sampled power spectral density function (2).

11-2 The input signal $r(t)$ to the open-loop sampled-data system shown in Fig. 11P-2 is assumed to be a member of a stationary random process.

The auto-power spectral density of $r(t)$ is

$\Phi_{rr}(s)$

=

4

-

(a) Find the following power spectral density functions

$P_{ec}(s)$ $P_{ec}(2)$ $P_{re}(s)$

(b) Find the autocorrelation function of the input, $\phi_{rr}(7)$.

$r(t)$

1

$G(s) =$

$s(s + 1)$

$\cdot C(t)$

$T=1$ sec

Figure 11P-2

11-3 For the open-loop sampled-data system configuration shown in Fig.

11P-2, show that

$P_{ec}(z) = G(2^{-1}) P_{rc}(2)$

11-4 Give proofs to Eqs. (11-85), (11-86), and (11-87).

11-5 Figure 11P-5 shows a sampled-data system whose mean-square error is to be evaluated. Assume that the system M is a zero-order hold and that $M_a(s) = e^{-sT/2}$. The power spectral density of the input signal is

$\Phi_{rr}(s)$

200

$\bullet Tr(s) = w^3 - s^2$

(a) Evaluate the mean-square error $e(t)$ as a function of w_0 and T .

(b) Plot $e^2(t)$ as a function of $w_0 T$.

PROBLEMS 515

$r(t)$

$1-e$

$C(t)$

$e(t)$

$e^{-sT/2}$

$Ca(t)$

Figure 11P-5

$[s(t) + (t)]$

$D(s)$

$G_{no}(s)$

$G_1(s)$

$c(t)$

$\cdot D(z) \cdot$

$s(t)$

$Mg(s)$

$Ca(t)$

$e(t)$

Figure 11P-6

11-6 Consider the sampled-data control system shown in Fig. 11P-6. Given that

1

$$G_1(s) = \frac{5(s+1)(s+2)}{s(s+1)(s+2)}$$

$$\frac{1}{s(s+1)(s+2)}$$

2

$$\Phi_{\pi\pi\pi}(5)$$

=

=

-

1 \$2

Pr, 1 (white noise)

$$P.T.(S) = P.T., T(S) = 0$$

The sampling period is 0.1 second.

(a) Design a digital compensator to minimize the mean-square-sampled error $e^2(nT)$. The desired system transfer function is $M_1(s) = e^{-sT}$,

(b) Find the minimum value of $e^2(nT)$.

11-7 For the sampled-data system shown in Fig. 11P-6.

$$G_1(s)$$

=

1

$$s(s+1)$$

$$Pr, r.(S) = 1 = 53 \quad Prara(s) = 0.1$$

1

$\Phi_{\pi\eta\pi\alpha}$

$$\Phi_{\pi\pi\pi}(5)$$

=

$$P.T.(S) = 0$$

The sampling period is 0.2 second.

(a) Design a digital compensator to minimize the mean-square error $e^2(t)$. The desired system transfer function is $M_1(s) = 1$.

(b) Find the minimum value of $e^2(t)$.

Problems for Chapter 12

38

L

12-1 For the nonlinear sampled-data system shown in Fig. 12P-1, find the following:

516 PROBLEMS

$r(t)$

X

1

z.o.h.

N

$$s(s+1)$$

$c(t)$

+1-

- 0.4

Nonlinear

element

0.4

X

Figure 12P-1

(a) the output response $c(t)$

(b) peak overshoot

(c) steady-state error.

The sampling period is one second; $r(t) = 0.8u(t)$. Assume zero initial conditions. Use the state transition method and the phase plane method.

12-2 (a) Repeat Problem 12-1 if the step input magnitude is increased to

0.9

(b) Repeat Problem 12-1 if the relay dead zone is increased to 0.6.

12-3 A sampled-data system with relay nonlinearity is shown in Fig. 12P-3. Construct the phase plane trajectory in $\dot{c}$ versus c plane when the initial conditions are $c(0) = 2$, $\dot{c}(0) = 0$; $r(t) = 0$. Sketch the output of the system.

$r(t)$

z.o.h.

$T=1\text{sec}$

Figure 12P-3

$r(t)=0$

2

z.o.h.

$s+2$

-2

$T=1\text{sec}$

$c(t)$

Figure 12P-4

12-4 A nonlinear sampled-data system with a lead compensation is shown in Fig. 12P-4. Construct the phase plane trajectory in $\dot{c}$ versus c plane when the initial conditions are $c(0) = 2$, $\dot{c}(0) = 0$; $r(t) = 0$. Sketch the output $c(t)$.

PROBLEMS 517

1.0-

0.5-

-0.75 -0.2

-X

0.2 0.75

-0.5

-1.0

Figure 12P-5

12-5 The nonlinear element contained in the sampled-data system of Fig. 12P-1 is considered to be a quantizer whose input-output characteristic is shown in Fig. 12P-5.

(a) Evaluate the output response $c(t)$ when $r(t) = 0.5u(t)$.

(b) Determine the steady-state error, if there is any.

(c) Use the state transition approach and check the results by the phase plane method.

$r(1)$

-0.2

z.o.h.

2s

0.2

$s(s+1)$

$c(t)$

Figure 12P-6

12-6 For the nonlinear sampled-data system shown in Fig. 12P-6, sketch the phase-plane diagram in $\dot{c}$ versus c . The input signal is a unit step function. Assume $\dot{c}(0) = 0$, $c(0) = 0$.

Slope = +1

$r(t)$

-0.2

$D(z)$
 $z.o.h.$
 $G(s)$
 0.1
 $T = 1\text{sec}$ Digital
 controller
 $\text{Slope} = +0.5$
 Figure 12P-7
 12-7 (a) Design a digital compensator for the nonlinear sampled-data system shown in Fig. 12P-7. The system is to have an output which will reach its final value in the shortest possible time without overshoot and steady-state error. The input signal is a unit step function. The transfer function of the controlled process is

518
 PROBLEMS

2
 $G(s) =$
 $s(s + 0.5)$
 (b) Repeat the design problem for
 $G(s)$

$=$
 1
 $s(s + 1)(s + 2)$
 12-8 Can the phase plane method of analyzing nonlinear sampled-data systems be applied to systems with ramp inputs? Outline a general procedure and give an example if possible.

$r(t) = Q$
 $1-1 +1$

$c()$
 Figure 12P-9
 12-9 A sampled-data system with saturation is shown in Fig. 12P-9. Sketch the trajectories in the incremental phase plane in Ac versus c for the following initial conditions:

(a) $c(0) = 3$ $Ac(0) = 0$
 (b) $c(0) = 0$ $Ac(0) = 3$

12-10 Sketch the phase plane trajectory in the incremental phase plane of Ac versus c for the system in Problem 12P-4.

12-11 For the nonlinear system shown in Fig. 12P-1, is it possible for the system to break into self-sustained oscillation ($r = 0$)? If so, what are the amplitudes and frequencies of the possible modes of oscillations? Find the smallest size of the relay dead zone so that self-sustained oscillations will not occur.

$e-amr$
 $ae-bmr$
 $(a - b)(x - e-bT)$
 $a(z - e-ar)$
 $a(z - e-ar)$
 $e-amT$
 $-a$

APPENDIX
 Laplace transform

$E(s)$
 Time function

$e(t)$
 TABLE OF TRANSFORMS z -TRANSFORMS AND MODIFIED z -TRANSFORMS

z -transform
 $E(x)$

1

2^{-n}
 Modified z-transform
 $E(z, m)$
 $\emptyset$
 2^{-n-1+m}
 1
 $e^{-nt's}$
 S
 $g(t)$
 $d(t - nT)$
 $u(t)$
 t^2
 e^{-at}
 Z
 2^{-1}
 Tz
 $(x - 1)^2$
 $T^2 z (z+1)$
 $(x-1)^3$
 $\lim (-1)^{-1}$
 $a-\emptyset$
 $a n-i$
 2
 $d a n-1$
 Z
 2
 e^{-ar}
 1
 t
 52
 $\ddot{a}m$
 $2!$
 33
 $(n - 1)!$
 $t n-!$
 $s n$
 1
 $s+a$
 1
 1
 $(s + a) (s + b)$
 $b-$
 a
 1
 1
 1
 1
 mT
 $+$
 $-$
 T
 $(x - 1)^2$
 m^2
 T^2
 $2m + 1$
 2
 $+$
 1
 $\lim (-1)^{-1},$
 $a-o$
 $(x - 1)^2 (x-1)^3$
 $a n-1$
 $J a n-1 z$

e^{-amT}
 $2-e-ar$
 $+$
 $e-amt$
 $s(s+a)$
 1
 $s^2 (s+a)$
 $(s+b)$
 $s^2 (s+a)$
 1
 $s(s+a) (s+b)$
 1
 $(s+a)^2$
 $a-$
 $(e-at-e-bt)$
 $-(u(t)-e-at)$
 a
 $1 (1-1)$
 $(1)+0+ \exists (\emptyset-1)e-at$
 $(u(t))$
 $abbe-at-be-bi)$
 $te-al$
 1
 $a(z e$
 $b-a (2-\emptyset-3-2-\emptyset-17)$
 1
 $a (z$
 Tz
 $b(x-1)^2$
 bTz
 $a (x-1)^2$
 2
 $+$
 $(1-e-ar)z$
 $1)(x-e-ar)$
 $-bT$
 $(1-e-ar)z$
 $a(z-1)(z-e-ar)$
 $(a-b)(1-e-ar)z$
 $a(z-1)(z-e-ar)]$
 bz
 $ab [z = 1 + (a - b) (z - e-ar)$
 $-$
 $-$
 az
 $(a-b)(z-e-ar).$
 $Tze-ar$
 $(x-e-a^7)$
 bT
 $-1)^2$
 $-x)$
 قامة
 $e-amt$
 $e-bmT$
 $e-at$
 2
 $e-br$
 2
 $e-amT$
 $-$
 $e-ar$
 b

$$\begin{aligned}
 &T \\
 &+ \\
 &1) \quad 2 \\
 &a^{mT-1} \\
 &a(x-1) \\
 &+ \\
 &+ \\
 &(b^{mT} + 1 \\
 &b e^{-amr} \\
 &ab \quad z = 1 + (a - b) (x - e^{-ar}) \\
 &T e^{-amT} l e^{-ar+m(z - e^{-ar})} \\
 &(z - e^{-ar})
 \end{aligned}$$

$$\begin{aligned}
 &1 \\
 &s^3 (s + a) \\
 &24 (t^2 - 21 + 2u(t) - e^{-ar})
 \end{aligned}$$

$$\begin{aligned}
 &a \\
 &a \\
 &+ \\
 &s \\
 &s^2 + a^2 z
 \end{aligned}$$

$$\begin{aligned}
 &a \\
 &s \\
 &\sin at \\
 &\cos at \\
 &\sin hat \\
 &\cos hat \\
 &Taz
 \end{aligned}$$

$$\begin{aligned}
 &+ \\
 &(a^{T-2}) Tz \\
 &2a(x-1)
 \end{aligned}$$

$$\begin{aligned}
 &z \\
 &+ \\
 &a^2 (x - 1) \\
 &2 e^{-ar}] \\
 &1) a^2 (z - e^{-aT}) \\
 &z \sin aT \\
 &z^2 - 2z \cos aT + 1 \\
 &z^2 \\
 &z(x - \cos aT) \\
 &2z \cos aT + 1 \\
 &z \sin haT \\
 &2z^2 - 2z \cos haT + \\
 &z(z \\
 &\cos haT) \\
 &z^2 - 2z \cos haT + 1
 \end{aligned}$$

$$\begin{aligned}
 &- \\
 &+ \quad 1 \quad (77 \\
 &1) 3
 \end{aligned}$$

$$\begin{aligned}
 &+ \\
 &1) \\
 &(x \\
 &+ 1 - 1/4
 \end{aligned}$$

$$\begin{aligned}
 &- \\
 &(amT)^{2/2} - amT + 1 \\
 &a^2 (z - 1) \\
 &z \sin amT + \sin (1 - m)aT \\
 &2z^2 - 2x \cos aT + 1 \\
 &z \cos amT
 \end{aligned}$$

$$\begin{aligned}
 &- \\
 &\cos (1 \\
 &-
 \end{aligned}$$

$m)aT$
 $2z^2 - 2z \cos aT + 1$
 $z \sin haT + \sin h(1 - m)aT$
 $-$
 $2z^2 - 2x \cos haT + 1$
 $z \cos hamT$
 $-$
 $\cos h(1$
 J
 a
 $s(s^2 + a^2)$
 $\frac{1}{2} (u(t) -$
 $--$
 $\cos at)$
 2
 $z(z - \cos aT)$
 $2z^2$
 a^2
 $s^2(s^2 + a^2)$
 1
 $s(s + a)^2$
 1
 $s^2 (s + a)^2$
 $-$
 $- 12[u(t) - (1 + at)e^{-at}]$
 $-$
 $2 - Zu(e)$
 $+$
 $($
 $+$
 $1-$
 $\sin at$
 Tz
 $(x - 1)^2$
 1
 $2z \cos aT + 1$
 $z \sin aT$
 $a z^2 - 2z \cos aT + 1$
 z
 2
 $---$
 e^{-ar}
 aTe^{-aTz}
 $(z - e^{-ar})?$
 e^{-at}
 $(aT + 2)z^{-2}$
 $(x - 1)^2$
 $-$
 a
 mT
 $+$
 $2 \frac{1}{2}$
 $21 (2271) 2$
 aT
 $-$
 $-$
 e^{-amr}
 $a^2 (z - e^{-ar})$
 $z^2 - 2z \cos haT + 1$
 $z \cos amT$
 T
 $(x - 1)^2$

$\cos (1$
 $m)aT$
 $n)aT]$
 $22 - 2x \cos aT + 1$
 $z \sin amT + \sin (1 - m)aT$
 $-$
 $a(z^2 - 2z \cos aT + 1)$
 $-$
 $+ amT$
 $e-ar$
 $+$
 $aTe-ar$
 $(z - e-ar)?)$
 $e-amT$
 $amT - 2$
 $2-1$
 $2z$
 $+$
 2
 $e-ar$
 1
 $(s + a)^2 + b^2$
 $s+ a$
 $(s + a)^2 + b^2$
 1
 $e-at \sin bt$
 $\frac{1}{2}($
 $e-at \cos bt$
 22
 $-$
 $ze-ar \cos bT$
 22
 $2ze-ar \cos bT + e-2aT$
 1
 $a^2 + b^2 \} l$
 $--$
 $- e-at \sec \& \cos (bt + 4)] \mid a^2 + b^2 z^2$
 2
 $+$
 $-$
 $s[(s + a)^2 + b^2]$
 $= \tan^{-1}(4)$
 22
 $ze-ar \sec \cos (bT -$
 $22-2ze-ar \cos bT + e-2aT$
 $+$
 $aTe-aTz$
 $(z - e-ar)?$
 $ze-ar \sin bT$
 $-$
 $1)$
 $2ze-ar \cos bT + e-2a1$
 $-$
 $+$
 $-$
 $aTe-ar$
 $e-ar)?$
 amT
 $z-$
 $)e-amr]$
 $1 e-amT[z \sin bmT + e-aT \sin (1 - m)bT]$
 $22 - 2ze-ar \cos bT + e-2aT$
 T

$$e^{-amr}$$

$$z^T [z \cos bmT + e^{-ar} \sin (1 - m)bT]$$

$$z^2 - 2ze^{-ar} \cos bT + e^{-2aT}$$

$$1$$

$$a^2 + b^2 z = 1$$

$$e^{-amT} \sec \{z \cos (bmT + 4)\}$$

$$z^2$$

$$2ze^{-ar} \cos bT + e^{-2aT}$$

$$e^{-ar}$$

$$\cos [(1 - m)bT -$$

$$T = \dots]$$

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T

H

G

(GR)

-H

-(GH)

C(s):

=

$$\frac{R(s)G(s)}{1+HG^*(s)} [1+HG^*(s)] RG(s)H(s)G(s)$$

(GH)

C*(s)

=

$$\frac{RG^*(s)}{1+HG(s)}$$

-H

-H

(GR)

C(z, m)

=

X2

$$\begin{aligned}
& HG(z, m)RG(z) \\
& 1+HG(z) \\
& + RG(z, m) \\
& -GH \\
& H \\
& -H \\
& G_1 \\
& G_2 \\
& C \\
& H \\
& G_1 \\
& G_2 \\
& H \\
& X_2 \\
& -H \\
& -H \\
& X_2 \\
& -H \\
& C(s) \\
& = \\
& 8 \\
& C^*(s) \\
& = \\
& G^*(s)R^*(s) \\
& 1+G^*(s)H^*(s) \\
& G^*(s)R^*(s) \\
& 1+G^*(s)H^*(s) \\
& C(z, m) \\
& = \\
& G(z)R(z) \\
& 1+G(z)R(z) \\
& -(HG, G_2) \\
& C(s) \\
& = \\
& RG_1(s)G_2(s) \\
& 1+HG_1G_2^*(s) \\
& (RG) \\
& C^*(s) \\
& = \\
& -(HG_2) \\
& RG_1^*(s)G_2^*(s) \\
& 1+HG_1G_2^{**}(s) \\
& -H \\
& C(z, m) \\
& -6, (624) \\
& = \\
& RG_1(z)G(z, m) \\
& 1+HG_1G_2(z) \\
& C(s): \\
& C^*(s) \\
& = \\
& = \\
& G_1(s)G_2(s)R^*(s) \\
& 1+G_1(s)HG_2^*(s) \\
& G_1^*(s)G_2^*(s)R^*(s) \\
& 1+G_1(s)HG_2^*(s) \\
& X_2 \\
& C(z, m) \\
& = \\
& -H \\
& G_1(z)G_2(z, m)R(z) \\
& 1+G_1(z)HG_2(z)
\end{aligned}$$

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